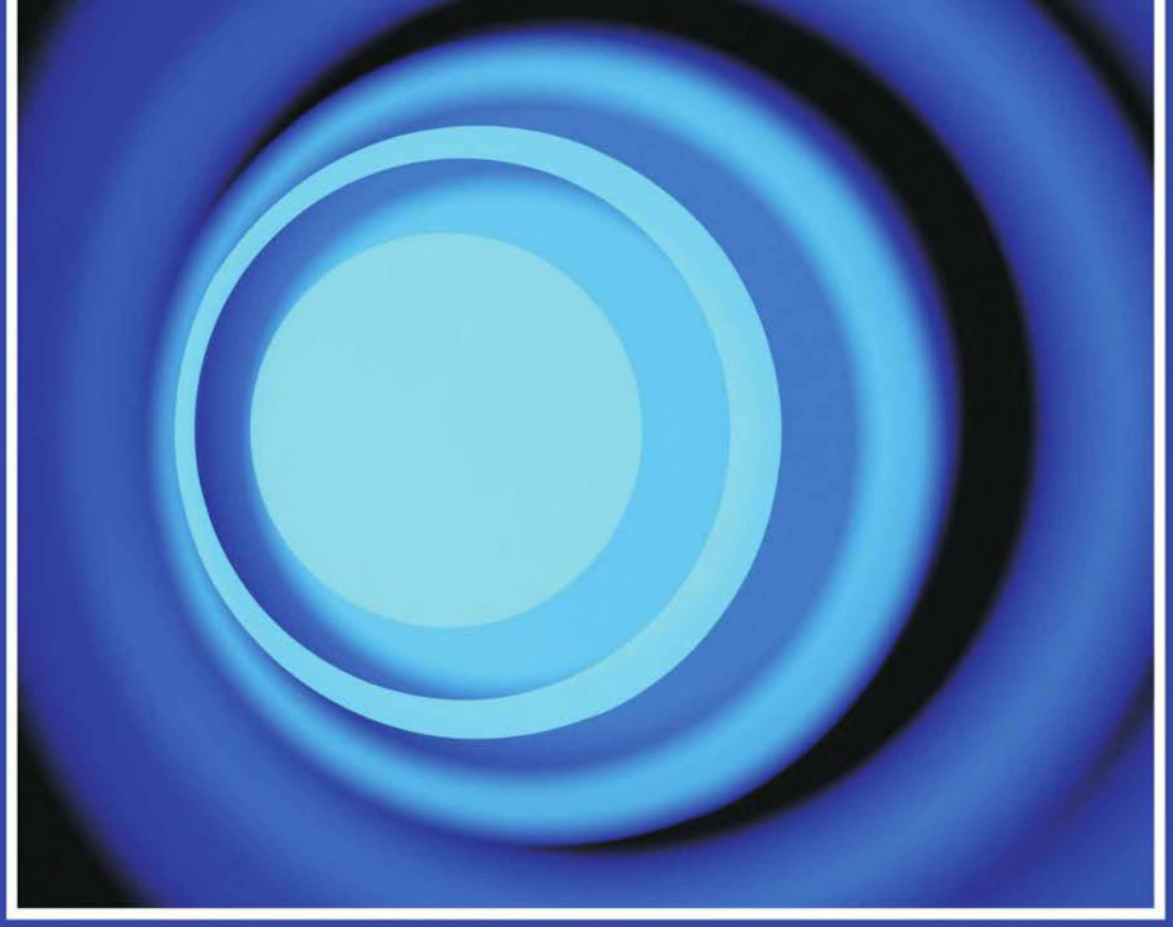
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THE PEARSON GUIDE TO

INORGANIC CHEMISTRY FOR THE IIT JEE

2012

Atul Singhal

ALWAYS LEARNING

PEARSON

The Pearson Guide to

Inorganic Chemistry for the IIT JEE 2012

Dr Atul Singhal

PEARSON

Chandigarh • Delhi • Chennai

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To my grandparents,
parents and teachers

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Preface to the Current Edition

The response to the first edition of the book and positive feedback from teachers and students has motivated me to work upon this revised and thoroughly pruned edition.

It is now power-packed with new questions of the latest IIT-JEE format. I hope that students will find this book useful. All the best!

Dr Atul Singhal
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Preface

The Pearson Guide to Inorganic Chemistry for the IIT JEE is an invaluable book for all the students preparing for the prestigious engineering entrance examination. It provides class-tested course material and problems that will supplement any kind of coaching or resource the students might be using. Because of its comprehensive and in-depth approach, it will be especially helpful for those students who do not have enough time or money to take classroom courses.

- A careful scrutiny of previous years' IIT papers and various other competitive examinations during the last 10 to 12 years was made before writing this book. It is strictly based on the latest IIT syllabus (2009–10) recommended by the executive board. It covers the subject in a structured way and familiarizes students with the trends in these examinations. Not many books in the market can stand up to this material when it comes to the strict alignment with the prescribed syllabus.
- It is written in a lucid manner to assist students to understand the concepts without the help of any guide.
- The objective of this book is to provide this vast subject in a structured and useful manner so as to familiarize the candidates taking the current examinations with the current trends and types of multiple-choice questions asked.
- The multiple-choice questions have been arranged in following categories:

Straight Objective Type Questions (Single Choice), Brainteasers Objective Type Questions (Single Choice), Multiple Correct Answer Type Questions (More than one choice), Linked-Comprehension Type Questions, Assertion and Reasoning Questions, Matrix-Match Type Questions and the IIT JEE Corner.

This book is written to pass on to another generation, my fascination with descriptive inorganic chemistry. Thus, the comments of the readers, both students and instructors, will be sincerely appreciated. Any suggestions for added or updated additional readings would also be welcome.

Dr Atul Singhal
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The contentment and ecstasy that accompany the successful completion of any work would remain essentially incomplete if I fail to mention the people whose constant guidance and support has encouraged me.

I am grateful to all my reverend teachers, especially, the late J. K. Mishra, Dr D. K. Rastogi, the late A. K. Rastogi and my honourable guide, Dr S. K. Agarwala. Their knowledge and wisdom has continued to assist me to present in this work.

I am thankful to my colleagues and friends, Deepak Bhatia, Er Vikas Kaushik, Er A. R. Khan, Vipul Agarwal, Er Ankit Arora, Er Wasim, Manoj Singhal, Vijay Arora, (Director, Dronacharya), Mr. Anupam Shrivastav (Bansal Classes Kota), Mr Rajiv Jain (MVN Faridabad), Mr Ashok Kumar, Mr Sandeep Singhal, Mr N.C. Joshi (Brillant tutorials) and Mr Chandan Kumar (Everon Toppers).

I am indebted to my father, B. K. Singhal, mother Usha Singhal, brothers, Amit Singhal and Karan Singh, and sisters, Ambika and Poonam, who have been my motivation at every step. Their never-ending affection has provided me with moral support and encouragement while writing this book.

Last but not the least, I wish to express my deepest gratitude to my wife Urmila and my little—but witty beyond years, daughters Khushi and Shanvi who always supported me during my work.

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CHEMICAL BONDING

1

Chapter Contents

Orbital overlap and covalent bond; Hybridization involving s, p and d orbitals only; Orbital energy diagrams for homonuclear diatomic species; Hydrogen bond; Polarity in molecules, dipole moment (qualitative aspects only); VSEPR model and shapes of molecules (linear, angular, triangular, square planar, pyramidal, square pyramidal, trigonal bipyramidal, tetrahedral and octahedral) and various levels of multiple-choice questions.

CHEMICAL BOND

Chemical bond is the force of attraction that binds two atoms together. A chemical bond balances the force of attraction and force of repulsion at a particular distance.

A chemical bond is formed to:

- attain the octet state
- minimize energy
- gain stability
- decrease reactivity

When two atoms come close to each other, forces of attraction and repulsion operate between them. The distance at which the attractive forces overcome repulsive forces is called bond distance. Here, the potential energy for the system is lowest, hence the bond is formed.

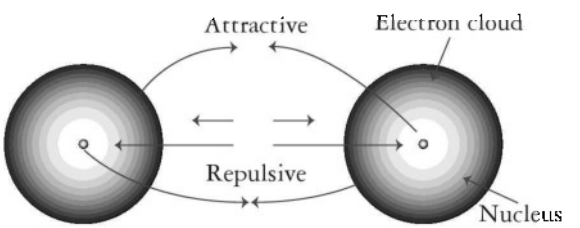


Fig. 1.1

Types of Bonds

Following are the six types of chemical bonds. Here, they are listed in the decreasing order of their respective bond strengths.

1. Ionic bond
2. Covalent bond
3. Coordinate bond
4. Metallic bond
5. Hydrogen bond
6. van der Waals bond

Metallic bond, hydrogen bond and van der Waals bond are interactions.

Octet Rule

It was introduced by Lewis and Kossel. According to this rule, each atom tries to obtain the octet state, that is, a state with eight valence electrons.

Exceptions to the octet rule

- Transition metal ions like Cr^{3+}, Mn^{2+}, Fe^{2+}.
- Pseudo inert gas configuration cations like Zn^{2+}, Cd^{2+}.

Contraction of octet state

- Here, the central atom is electron deficient or does not have an octet state. For example,

$\underset{4}{BeX_2}$ $\underset{6}{BX_3}$ $\underset{6}{AlX_3}$ $\underset{6e^-}{Ge(CH_3)_3}$

Expansion of octet state

- Here, the central atom has more than 8 electrons due to empty d-orbitals. For example, PCl_5, SF_6, OsF_8, ICl_3 etc

$\underset{10}{PCl_5}$ $\underset{12}{SF_6}$ $\underset{16}{OsF_8}$ $\underset{10\ e}{ICl_3}$ etc.

- Odd electronic species like NO, NO_2, ClO_2.
- Interhalogens compounds like IF_7, BrF_3.
- Compounds of xenon such as XeF_2, XeF_4, XeF_6.

IONIC OR KERNEL BOND

An Ionic bond is formed by the complete transfer of valence electrons from a metal to a non-metal. This was first studied by Kossel.

For example,

Na	+	Cl	⟶	Na^-	Cl^-
(2, 8, 1)		(2, 8, 7)		(2, 8)	(2, 8)
Mg	+	O	➤	Mg^{+2}	O^{-2}
(2, 8, 2)		(2, 6)		(2, 8)	(2, 8)

Al + N ➤ Al^{+3} N^{-3}

(2, 8, 3) (2, 5) (2, 8) (2, 8)

- Number of electrons transferred is equal to electrovalency.
- Maximum number of electrons transferred by a metal to non-metal is three, as in the case with AlF_3, (Al metal transfers three electrons to F).
- During electron transfer, the outermost orbit of metal is destroyed. The remaining portion is called core or kernel, hence this bond is also called kernel bond.
- Nature of ionic bond is electrostatic or coloumbic force of attraction.
- It is a non-directional bond.

Conditions for the Formation of an Ionic Bond

The process of bond formation is exothermic where $\Delta H = -Ve$. The essential conditions include

- Metal must have low ionization energy.
- Non-metals must have high electron affinity.
- Ions must have high lattice energy.
- Cation should be large with low electronegativity.
- Anion must be small with high electronegativity.

Born–Haber Cycle

The formation of an ionic compound in terms of energy can be shown by Born–Haber cycle. It is also used to find lattice energy, ionization energy and electron affinity.

For example,

$$M(s) \xrightarrow[+S]{\text{Sublimation}} M(g) \xrightarrow[+I]{\text{Ionization}} M^+(g) + e^-$$

$$\tfrac{1}{2}X_2 \xrightarrow[-1/2\ D]{\text{Decomposition}} X(g) \xrightarrow[-E]{\text{Addition of } e^-} X^-(g)$$

$$M_{(g)} + X\ (g) \xrightarrow[-U]{\text{Crystal formation}} MX(g)$$

$$\Delta H_f = S + \tfrac{1}{2}D + I - E - U$$

Here,

S – Heat of sublimation
D – Heat of dissociation
I – Ionization enthalpy
E – Electron gain enthalpy or electron affinity
U – Lattice energy

For the formation of an ionic solid, energy must be released during its formation, that is, ΔH must be negative.

$$-E - U > S + \tfrac{1}{2} D + I$$

Properties of Ionic Compounds

1. Ionic compounds have solid crystalline structures (flat surfaces), with definite geometry, due to strong electrostatic force of attraction as constituents are arranged in a definite pattern.
2. These compounds are hard in nature.

   Hardness $\propto$ Electrostatic force of attraction

   $\propto$ Charge on ion

   $\propto \frac{1}{\text{Ionic radius}}$
3. Ionic compounds have high value of boiling point, melting point and density due to strong electrostatic force of attraction.

   Boiling point, melting point $\propto$ Electrostatic force of attraction

   Volatile nature $\propto \frac{1}{\text{Electrostatic force of attraction}}$
4. Ionic compounds show isomorphism, that is, they have same crystalline structure. For example, all alums, NaF and MgO, $ZnSO_4.7H_2O$ and $FeSO_4.7H_2O$.
5. These are conductors in fused, molten or aqueous state due to the presence of free ions. In solid state these are non-conductors as no free ions are present.
6. They show fast ionic reactions as activation energy is zero for ions.
7. They do not show space isomerism due to non-directional nature of ionic bond.
8. They have high latice energy (U). It is released during the formation of an ionic solid molecule from its constituent ions.

Lattice Energy

Lattice energy is the energy needed to break an ionic solid molecule into its constitutent ions. It is denoted by U.

$U \propto$ Charge on ion

$\propto \frac{1}{\text{Size of ion}}$

Hence, lattice energy for the following compounds increases in the order shown under:

$NaCl < MgCl_2 < AlCl_3 < SiCl_4$

As charge on a metal atom increases, its size decreases.

In case of univalent and bivalent ionic compounds, lattice energy decreases as follows:

Bi-bi > Uni-bi or Bi-uni > Uni-uni

For example,

$MgO > MgCl_2 > NaCl$.

Some orders of lattice energy

(i) LiX > NaFX > KFX > R6X > CsX
(ii) LiF > LiCl > LiI
(iii) AgF > AgCl > AgI
(iv) BeO > MgO > CaO > SrO

9. Ionic compounds are soluble in polar solvents like water due to high dielectric constant of these solvents. The force of attraction between ions is destroyed and hence they dissolve in the solvent.

Facts Related to Solubility

- If ΔH (hydration) > Lattice energy then the ionic compound is soluble.
- If ΔH (hydration) < Lattice energy then the ionic compound is insoluble
- If ΔH (hydration) = Lattice energy then the compound is at equilibrium state

Some Solubility Orders

a. $LiX < NaX < KX < RbX < CsX$
b. $LiOH < NaOH < KOH < RbOH < CsOH$
c. $BeX_2 < MgX_2 < CaX_2 < BaX_2$
d. $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Ba(OH)_2$
e. $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$
f. $AlF_3 > AlCl_3 > AlBr_3 > AlI_3$

- Crystals of high ionic charges are less soluble. For example, compounds of CO_3^{-2}, SO_4^{-2}, PO_4^{-3} are less soluble.

  Compounds Ba^{+2}, Pb^{+2} are insoluble as lattice energy > ΔH_{hy}.

  Compounds of Ag (salts) are insoluble as lattice energy > ΔH_{hy}.

- Presence of common ions decrease solubility. For example, solubility of AgCl decreases in presence of $AgNO_3$ or KCl, due to the presence of common ions, that is, Ag^+ and Cl^- respectively.

Note: The concept of ionic bond is not a part of IIT-JEE syllabus but has been discussed here for the better understanding of the chapter.

COVALENT BOND

A covalent bond is formed by the equal sharing of electrons between two similar or different atoms.

- If atoms are same or their electronegativity is same the covalent bond between them is non-polar. For example, X – X, O = O, N ≡ N
- If atoms are different or have different value of electronegativity the covalent bond formed between them is polar. For example,

$\overset{+\delta}{H}-\overset{-\delta}{O}-\overset{+\delta}{H}$, $\overset{+\delta}{H}-\overset{-\delta}{X}$

- Here, the number of electrons shared or covalent bonds represent covalency.
- One atom can share maximum three electrons with another atom. For example, in ammonia the covalency of nitrogen atom is three.

Properties of Covalent Compounds

1. Covalent compounds mostly occur in liquid and gaseous state, but if molecular weight of the compound is high they may occur in solid state too.

For example,	F_2,	Cl_2	Br_2	I_2
	'g'	'a'	'l'	's'

	Glucose	Sugar
Molecular wt.	180	342
	(less solid)	(more solid)

2. Solubility of these compounds follows the concepts 'like dissolves like', that is, non-polar solute dissolves in non-polar solvent. For example, CCl_4 dissolves in organic solvents. Similarly, polar solutes dissolves in polar solvent. For example, alcohol and ammonia dissolve in water.
3. Covalent compounds have lower boiling point and melting point values than those of ionic compounds. This is because covalent bond is a weak van der Waals force in nature.

For example,	KOH	>	HX
	Strong ionic force of attraction		Weak van der Waals forces

Boiling point and melting point ∝ Hydrogen bonding
∝ Molecular weight

For example,

H_2O, H_2Te H_2Se H_2S

HF > HI > HBr > HCl

Due to H-bonding; As molecular weight decreases

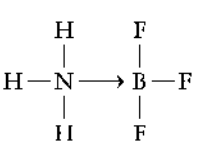
4. Covalent compounds are non-conductors due to absence of free ions, but graphite is a conductor, as the free electrons are available in its hexagonal sheet like structure. In case of diamond, the structure is tetrahedral hence free electrons are not available. Therefore, it is not a conductor.
5. Covalent bond is directional, hence these compounds can show structural and space isomerisms.
6. The reactions involving covalent bond are slow as these need higher activation energy.

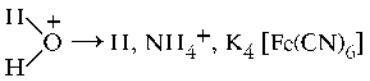
COORDINATE OR DATIVE SEMI-POLAR BOND

Coordinate bond is a special type of bond which is formed by donation of electron pair from the donor to the receiver, that is, it involves partial transfer or unequal sharing of electrons. It is denoted as (⟶) from donor to receiver.

A:	+	B ⟶ (A ⟶ B)
Donor or Lewis base		Receiver Lewis acid

Coordinate bond is intermediate between ionic and covalent bonds, but more closely resembles a covalent bond. The properties of coordinate compounds are more close to covalent compounds. For example,





$\longrightarrow H, NH_4^+, K_4[Fe(CN)_6]$

Sugden Linkage

- Sugden or singlet linkage is formed by donation of one electron and is denoted by (⇀). For example, PCl_5, SF_6, IF_7.
- It was studied for the first time by Lorry and Sidwick.

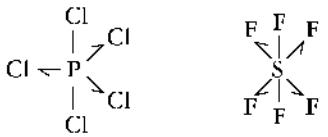


Fig. 1.2

MODERN CONCEPT OF COVALENT BOND

The nature of covalent bond is explained on the basis of Heirler–London's Valence bond theory, Pauling and Slater's overlapping theory and Hund, Mullikan's theory.

Valence Bond Theory or Heitler–London Theory

Orbital concept of covalent bond was introduced by Heitler and London. According to this concept, "A covalent bond is formed due to the half-filled atomic orbitals having electrons with opposite spin to each other".

Features

1. The atoms should have unpaired electrons to form a covalent bond.
2. A covalent bond is formed by the pairing of electrons.
3. The maximum electron density lies between the bonded atoms.
4. There is a tendency to form close shells of the atom, though the octet is not attained in $BeCl_2$, BF_3 etc., and is exceeded in PCl_5, SF_6, IF_7 etc.

- Due to overlapping the potential energy of system decreases.
- The internuclear distance with maximum overlapping and greater decrease of potential energy is known as bond length.

Limitations

1. It cannot explain the formation of odd electron molecule such as ClO_3, NO, H_2^+ etc. In such molecules, electron pairing does not take place.
2. It cannot explain the formation of coordinate bond where only one atom donates a lone pair of electrons to the other atom, molecule or ion.
3. It cannot explain the formation of π-bond.
4. It cannot explain the stereochemistry of the molecules and ions.

Pauling and Slater's Theory

It deals with the directional nature of the bond formed and is simply an extension of Heitler London theory.

1. Greater the overlapping, stronger will be the bond formed. It means bond strength depends upon the overlapping and is directly proportional to the extent of overlapping.
2. A spherically symmetrical orbit, say, s-orbital will not show any preference in direction whereas non-spherical orbitals, say, p- or d-orbitals will tend to form a bond in the direction of maximum electron density with the orbital.

REMEMBER

- The overlapping of the orbitals of only those electrons which take part in the bond formation will occur and not with the electrons of other atoms.
- The wave function of an electron of s-orbital is spherically symmetrical, therefore such an electron exhibits no directional preference in bond formation. The two orbitals having similar energy level, the one which is more directionally concentrated, will form a stronger bond.
- Strongness of overlapping ∝ mode of overlapping.

  For example, linear overlapping is stronger than lateral overlapping.
- Strongness of overlapping ∝ directional nature of orbital.

  For example, p-p > s-p > s-s > p-p
  Linear or axial overlapping — Lateral overlapping

Types of Overlapping

1. **s-s Overlapping:** Overlapping between s-s electrons of two similar or dissimilar atoms is called s-s overlapping and forms a single covalent bond.

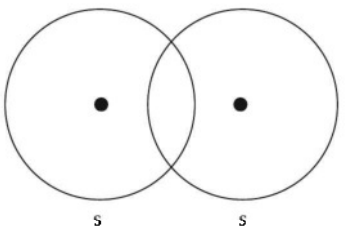


Fig. 1.3 Formation of hydrogen molecule by s-s overlapping.

2. **s-p Overlapping:** Overlapping between s and p electrons is called s-p overlapping. NH_3 is formed by the overlapping between three electrons of nitrogen (px, py and pz) with three electrons of three hydrogen atoms.

$_7N - 1s^2, 2s^2, 2px^1\ py^1\ pz^1$

$_1H = 1s^1$

Strong bond can be formed only when hydrogen electrons approach in the direction of X, Y and Z axis at right angles to each other.

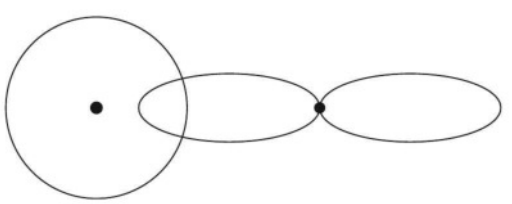
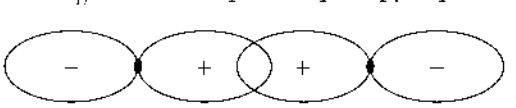


Fig. 1.4

3. **p-p Overlapping:** p-p overlapping is formed by the overlapping of the p-orbitals of the atoms. In case of chlorine molecule, it is formed by the overlapping of the 3pz orbitals of two chlorine atoms.

$_{17}Cl - 1s^2, 2s^2\ 2p^6, 3s^2, 3px^2, 3py^2, 3pz^1$



p – p

Fig. 1.5

Some Important Features of Bond

Bond Length

- Bond length is the average distance between the centers of the nuclei of the two bonded atoms.
- It is determined by X-ray diffraction and spectroscopic methods.
- In case of ionic compounds, it is the sum of ionic radius of cation and anion, while in case of covalent compounds, it is sum of their covalent radius.

Factors affecting bond length

- Bond length ∝ Size of atom. For example,

  HF < HCl < HBr < HI
  (Atomic size)

  Since F < Cl < Br < I

- Bond length $\propto \frac{1}{\text{Bond order or multiplicity}}$

  For example, C C > C = C > C ≡ C
  1.54Å 1.34Å 1.32Å

- Bond length $\propto \frac{1}{s\%}$

  that is, sp^3 > sp^2 > sp
  s% 25% 33% 50%

- Bond length ∝ Electronic repulsion
  For example, H_2 > H_2^+
- Resonance and hyperconjugation also change bond length.
  For example, in benzene, C – C bond length is 1.39 Å, that is, in between C – C and C=C.

Bond Energy

- It is the energy needed to break one mole of bond of a particular type, so as to separate them into gaseous atoms. It is also called bond dissociation energy.
- Bond energy can also be defined as the energy released during the formation of one mole of a particular bond.

Factors affecting bond energy

- Bond energy ∝ Bond order or multiplicity

  For example, C ≡ C > C = C > C – C

- Bond energy $\propto \frac{1}{\text{Bond length or size of atom}}$

  For example,

  HF > HCl > HBr > HI

- Bond energy ∝ s% or s-orbital character involved in hybridisation

  For example,

  sp , sp^2 , sp^3

  50% 33% 25%

- Bond energy $\propto \frac{1}{\text{Lone pair of electrons/ electronic repulsion}}$

  For example, C – C > N – N > O – O > F – F

lp e	0	1	2	3
B.E. (kJ)	247	163	146	138.8

  Some diaromic molecules in order of bond energy are

  C≡O > N≡N > C≡N > C≡C

Bond Angle

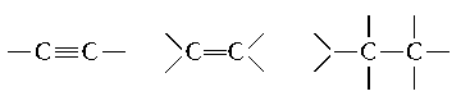
It is the angle between the lines representing the directions of the bonds or the orbitals having bonding pair of electrons.

Factors affecting bond angle

- Bond angle ∝ Bond order ∝ s%

  $\propto \frac{1}{\text{Bond length}}$

  For example,



- Bond angle is also affected by electronic repulsion. (see VSEPR theory)

  For example, NH_4^+ > NH_3 > NH_2^-

  no lp 1 lp e 2 lp e

- Bond angle $\propto \frac{1}{\text{Size of terminal atom}}$

  For example, $I_2O > Br_2O > Cl_2O > OF_2$

- Bond angle $\propto \frac{1}{\text{Size of central atom/electronegativity}}$

  Normally, bond angle decreases when we move down the group, as electronegativity decreases.

  For example, $NH_3 > PH_3 > AsH_3 > BiH_3$

  $H_2O > H_2S > H_2Se > H_2Te$

  $BF_3 > PF_3 > ClF_3$

- Bond angle ∝ Electronegativity of terminal atom

  For example, $PF_3 > PCl_3 < PBr_3 < PI_3$

  PF_3 has more bond angle than PCl_3 due to pπ-dπ bonding.

REMEMBER

PF_3 has greater bond angle when compared to PH_3 due to resonance in PF_3, where a double bond character develops. NH_3 has more bond angle value than NF_3 as F-atom pulls the bpe$^-$ away from N-atom in NF_3.

POLORIZATION AND FAJAN'S RULE

When cation and anion are close to each other, the shape of anion is distorted by the cation. This is known as polarization. Due to this, covalent nature develops in an ionic molecule.

Polarization ∝ Covalent nature $\propto \frac{1}{\text{Ionic nature}}$

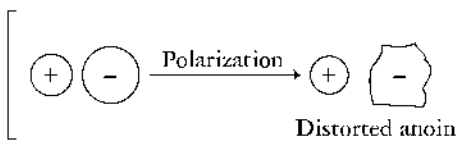


Fig. 1.6

Fajan's Rule

Polarization or covalent nature is explained by the following rules:

Charge on Cation Polarization, covalent nature or polarizing power of a cation ∝ charge on cation. That is greater the charge on cation, greater will be its polarizing power and more will be covalent nature. For example,

$SiCl_4 > AlCl_3 > MgCl_2 > NaCl$

Size of Cation When the charge is same and the anion is common, consider that the covalent nature $\propto \frac{1}{\text{Size of cation}}$

That is, smaller cation has more polarizing power.

For example,

LiCl > NaCl > KCl > RbCl > CsCl

Max. covalent	Max. ionic
Least ionic	Least covalent

$Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

Smallest	Largest
in size	in size

Size of Anion This property is taken into account when the charges are same and the cation is common.

- Polarization or covalent nature ∝ size of anion. Hence, larger anions are more polarized.

  For example, LiF < LiCl < LiBr > LiI

  Since, $F^- < Cl^- < Br^- < I^-$

  Larger the size of anion, easier will be its polarization.
- A cation with 18 valence electrons has more polarizing power than a cation with 8 valence electrons.

For example,

Group IB > Group IA

Cu^+ Na^+

Ag^+ K^+

Group IIB > Group IIA

Zn^{+2} Mg^{+2}

For example,

ZnO > MgO

Zn^{+2} Mg^{+2}

2, 8, 18 2, 8

REMEMBER

As the covalent nature increases, the intensity of the colour increases. For example, $FeCl_3$ is reddish-brown while $FeCl_2$ is greenish-yellow.

Dipole Moment

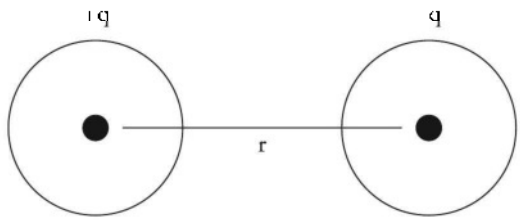


Fig. 1.7

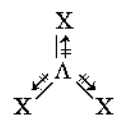
Dipole moment is used to measure the polarity in a molecule. It is denoted by μ. Mathematically, it is given as

$\mu = q \times r$ coulomb metre

$\mu = e \times d$ esu cm

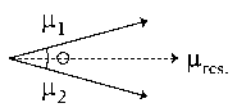
1 debye = 1×10^{-18} esu cm.

- It is represented by (⟞⟶) from electropositive to electronegative species or less electronegative to more electronegative species. For example, AX_3



$\mu = 0$

Non-polar



In case a molecule has more than one polar bonds μ_{net} is given as follows:

$$\mu_{net} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1 \mu_2 \cos\theta}$$

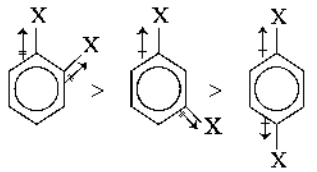
- Dipole moment ∝ Electronegativity difference. For example, HF > HCl > HBr > HI
- Dipole moment ∝ Number of lone pair of electrons.

  For example, HF > H_2O > NH_3

  Fluorine has 3 lone pair, oxygen has 2 lone pair, and ammonia has 1 lone pair of electron.
- Dipole moment ∝ $\frac{1}{\theta}$

  ortho > meta > para

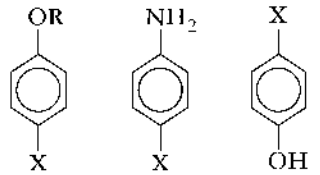
For example,



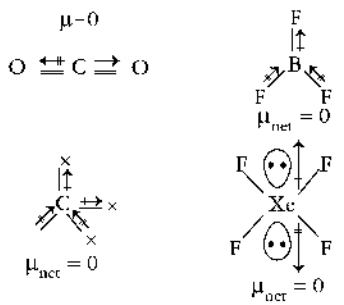
'o' 'm' 'p'

60° 120° 180°

In case of para forms M μ_{net} is positive if both the species are different. For example,

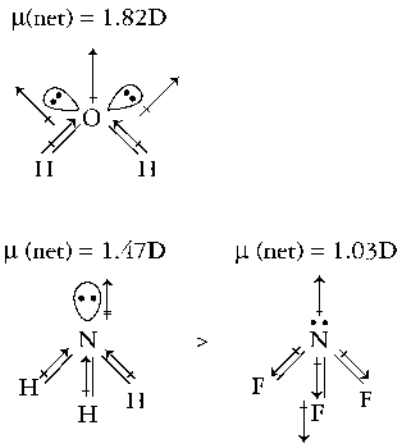


- Homoatomic molecules like X_2, N_2, O_2 and molecules having normal shapes according to hybridization like linear, trigonal, tetrahedral will be nonpolar, as for them, the dipole moment is zero. For example, BX_3, CH_4, CCl_4, $SiCl_4$, PCl_5.



Here, μ (net) = 0 as C = O bonds are in opposing directions.

- Molecules in which the central atom has lone pair of electrons or have distorted shapes, like angular, pyramidal, sea-saw shapes will have some value of dipole moment and will be polar in nature. For example, H_2O, H_2S, OF_2, NH_3, PH_3, PCl_3, SCl_4, SO_2, $SnCl_2$ etc.



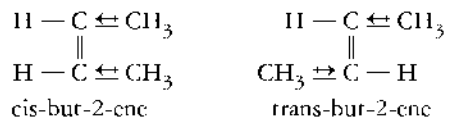
Ammonia has more dipole moment than NF_3 as in ammonia μ (net) is in the direction of lone pair electrons i.e., it is additive while in NF_3 μ (net) is opposite to lone pair i.e., substractive.

- Dipole moment of a cis-alkene is more than trans-alkene. In trans-alkenes, it is zero due to symmetry in most of the cases.

Dipole Moment of Some Common Molecules

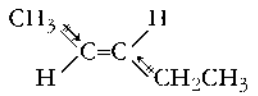
Molecule	H_2	N_2	CH_4	Cl_2	CS_2	C_2H_2	CH_2F_2	O_2	O_3	C_2H_6	SO_2	CO	CsI	NaCl	CH_3OH	C_2H_5OH	H_2O	HF
Dipole moment	0	0	0	0	0	0	1.96D	0	0.52D	0	1.61D	1.12	12.1	8.3	1.69	1.67D	1.84	1.91

Molecule	HCl	NH_3	N_2O	H_2S	H_2O_2	NF_3	$CHCl_3$	PH_3	HCN	$SbCl_3$	CH_3Cl	S_8	PCl_5		Trans-butene	IF_7	PX_3	CX_4
Dipole moment	1.02	1.46	0.17	0.92	1.84	0.55	1.15	0.58	2.93	3.9	1.86	0	0	0	0	0	0	0



Exception: Unsymmetric alkenes with odd number of carbon atoms have some value of dipole moment.

For example, trans-2-penrene



Specific cases of dipole moment

→ $CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4$
Highly polar Non-polar

→ $CH_3Cl > CH_3F > CH_3Br > CH_3I$

Uses

- To find geometry of a complex/molecule etc.
- To find ionic character or nature in a covalent species.

$$\text{Ionic nature } \% = \frac{\mu_{observed}}{\mu_{calculated}\ (q \times r)} \times 100$$

- To distinguish between –cis and –trans alkenes

Cis-but 2-ene > trans bur-2-ene
μ–+ve μ–0

Illustrations

1. **The experimentally determined dipole moment of KF is 2.87 × 10^{-29} cm. The distance between the centers of charge in a KF dipole is 2.66 × 10^{-10} m. Calculate the percentage ionic character of KF.**

Solution

μ = e × d coulombs meter

For KF = 2.66 × 10^{-10} m

For complete separation of a unit charge (electronic charge), e = 1.602 × 10^{-19} C

$$\mu = 1.602 \times 10^{-19} \times 2.66 \times 10^{-10}$$
$$= 4.26 \times 10^{-29} \text{ cm}$$

% ionic character a KCl

$$= \frac{2.87 \times 10^{-29}}{4.26 \times 10^{-29}} \times 100$$
$$= 67.4\ \%.$$

2. The dipole moment of KCl is 3.336 × 10^{-29} coulomb meter which indicates that it is a highly polar molecule. The interatomic distance between K^+ and Cl^- in this molecule is 2.6 × 10^{-10} m. Calculate the dipole moment of KCl molecule if there were opposite charges of one fundamental unit located at each nucleus. Calculate the percentage ionic character of KCl.

Solution

μ = e × d coulombs meter

For KCl d = 2.6 × 10^{-10} m

For complete separation of unit charge (electronic charge), e = 1.602 × 10^{-19} C

$$\mu = 1.602 \times 10^{-19} \times 2.6 \times 10^{-10}$$
$$= 4.1652 \times 10^{-29} \text{ cm}$$
$$\mu\ (KCl) = 3.336 \times 10^{-29} \text{ cm}$$

Per cent ionic character of KCl

$$= \frac{3.336 \times 10^{-29}}{4.1652 \times 10^{-29}} \times 100$$
$$= 80.09\ \%$$

SIGMA AND PI BONDS

Sigma (σ) Bond

Sigma bond is formed by axial or headrohead or linear overlapping between two s–s or s–p or p–p orbitals.

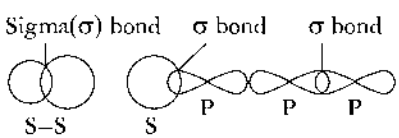


Fig. 1.1

1. Sigma bond is stronger and therefore less reactive, due to more effective and stronger overlapping than bond.
2. The minimum and maximum number of bonds between two bonded atoms is 1.
3. Stability Number of sigma bonds.

4. Reactivity $\propto \frac{1}{\sigma}$
5. In sigma bond, free rotation of the atoms is possible.
6. Sigma bond determines the shape of molecules.

Pi (π) Bond

Pi bond is formed by lateral or sidewise overlapping between the two p orbitals.

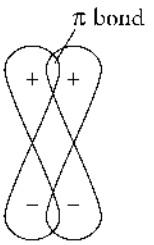


Fig. 1.9

1. It is a weak or less stable bond, and therefore more reactive, due to less effective overlapping.
2. Minimum and maximum number of π bonds between two bonded atoms are 0 and 2, respectively.
3. Stability $\propto \frac{1}{\text{Number of } \pi \text{ bonds}}$
4. Reactivity ∝ Number of π bonds
5. In case of a π bond, free rotation is not possible
6. It does not determine the shape of a molecule but shortens the bond length.

(C−C	>	C=C	>	C≡C)
π-bonds		1		2

Strength of σ- and π-Bonds

The strength of a bond depends upon the extent of overlapping of half-filled atomic orbitals. The extent of overlapping between two atoms is always greater when there is end-to-end overlapping of orbitals. Therefore, a σ-bond is always stronger than π-bond.

REMEMBER

It is the number of unpaired s- or p- electrons present in its atom in the ground state. Thus, covalency of hydrogen atom is 1.

To find sigma and pi bonds in a molecule

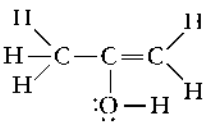
Single bond = 1 σ

Double bond = 1σ, 1π

Triple bond = 1σ, 2π

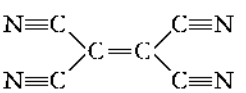
For example,

1. Enolic form of acetone



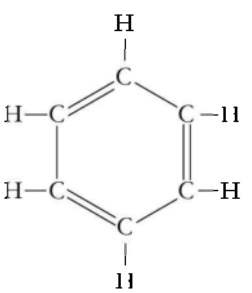
It has 9σ and 1π bond, and 2 lone pairs of electrons.

2. $C_2(CN)_4$



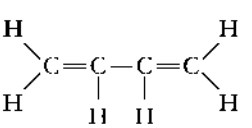
It has 9σ and 9π bond, and 4 lone pairs of electrons.

3.



It has 12σ and 3π bonds.

4. Buta-1, 3-diene



It has 9σ and 2π bonds.

5. $CaC_2 \longrightarrow Ca^{2+} \quad (C \equiv C)^{2-}$

It has 16, 2π bonds.

HYDROGEN BOND

- Hydrogen bond was introduced by Latimer and Rode-bush.
- It is a weak interaction denoted by dotted (...) lines between hydrogen and a highly electronegative and small sized atom like F, O and N. Here, the hydrogen atom is covalently bonded to any of these.
- The nature of a hydrogen bond is either dipole–dipole type, ion–dipole type or dipole–induced dipole type
- HCl has no H-bonding as chlorine is large in size.
- H-bond strength for the following order is 10 kcal per mole, 7 kcal per mole and 2 kcal mole, respectively. HF > H_2O > NH_3

  Hydrogen bonding is of the following two types:

Intermolecular H-bonding

Intermolecular H-bonding is formed between two or more different molecules of the same or different types. For example, HF, H_2O, NH_3, R OH, R COOH, etc.

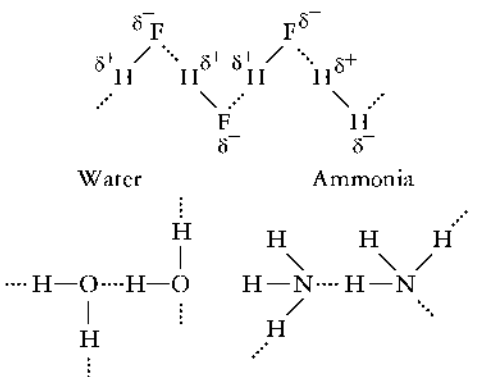


Fig. 1.10

Facts related to intermolecular hydrogen bonding

- One water molecule can form hydrogen bonding with four other water molecules.
- Due to hydrogen bonding in water, the water molecules are closely packed, hence water has less volume but more density than ice where an open cage like structure is observed.
- Water has maximum density at 4°C as above 4°C some hydrogen bonds are broken leading to a decrease in the density.
- Two ice cubes when pressed against each other, form one block due to hydrogen bonding.

Effects of intermolecular H-bonding

- Increase in boiling point, melting point, solubility, thermal srability, viscosity, surface tension and occurence liquid state is observed as molecules get associated more closely due to intermolecular H-bonding.
- HF is a liquid and has a higher boiling point than other HX molecules which are gases at room temperature (Here X = halogens).
- Alcohols are highly soluble in water in any proportion and have higher boiling points than others which are very less soluble in water.
- Glycerol is highly viscous with a high boiling point.

$$\underset{\text{B.P., Viscosity decrease}}{\text{Glycerol} > \text{Glycol} > C_2H_5OH \longrightarrow}$$

- Acids have higher boiling point and solubility than their corresponding acid derivatives.
- In DNA and RNA, the complementary strands are held together by intermolecular H-bonding between the nitrogenous bases of the two strands.
- Nucleic acid and proteins are held together by hydrogen bonds.
- KHF_2 or HF_2^- exists due to hydrogen bonding, but formation of other HX_2^- (for example HCl_2) is not possible, due to absence of hydrogen bonding because of large sizes of the halogen atoms.
- The extent of hydrogen bonding in water is higher than H_2O, hence it has a higher boiling point than HF.

  H_2O > HF > NH_3
- Acids can dimerize due to intermolecular hydrogen bonding. For example, acetic acid dimerizes in benzene.

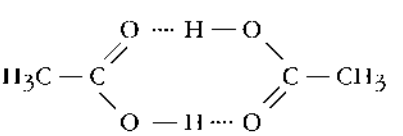


Fig. 1.11

Intramolecular H-bonding or chelation

Intramolecular H-bonding or chelation is formed with in a molecule. For example,

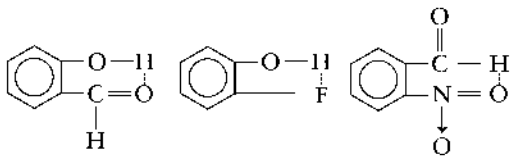


Fig. 1.12

Other examples include pyridine-2-carbonaldoxime and o-hydroxybenzoic acid.

Effects of intramolecular H-bonding

- Due to this bonding the boiling point and acidic nature of the molecule decrease but its volatile nature increases.
- O-nitrophenol has a lower boiling point and reduced acidic nature, but is more volatile than p-nitrophenol. A mixture of both these compounds can be separated by steam distillation method.

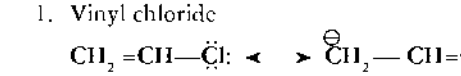
Resonance

- When all the properties of a molecule cannot be explained by a single structural formula, then such molecules are represented by many structural formulas that are canonical structures or contributing or resonating structural representating a single compound.
- It is observed due to the delocalization of π electrons.

Facts about canonical structures

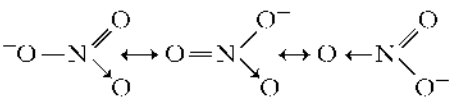
- Canonical structures for a given molecule have the same arrangement of atoms.
- Position and arrangement of atoms are same in canonical structures, they only differ in the distribution of electrons.
- Canonical structures are depicted by the symbol (↔) between them.
- Canonical structures should be planar or nearly planar.
- Total number of paired and unpaired electrons are also same in canonical structures. For example,

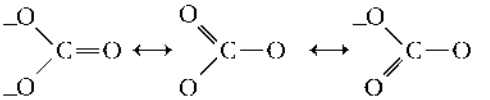
1. Vinyl chloride

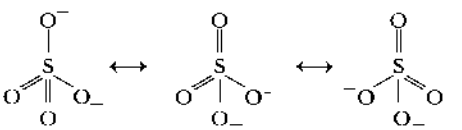


2. $CH_2=CH-CH=CH_2$ ↔ $\overset{\oplus}{C}H_2-CH=CH-CH_2^-$ ↔ $CH_2^--CH=CH-\overset{\oplus}{C}H_2$

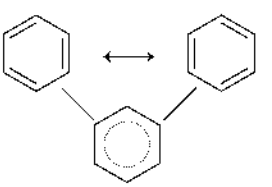
3. $N\equiv N\rightarrow O \leftrightarrow N=\overset{|}{N}=O \leftrightarrow \overset{|}{N}\equiv N-O$

4. 

5. 

6. 

- Resonance changes bond length, for example, in benzene C–C = 1.39 Å, which is an intermediate value between (C C) = 1.54 Å, (C=C) = 1.34 Å



Resonance Energy

- Resonance energy = Energy of most stable canonical structure − Resonance hybrid energy.
- Resonance energy ∝ Number of canonical structure
  Resonance energy ∝ Stability

$$\text{Resonance energy} \propto \frac{1}{\text{Reactivity}}$$

Resonance energy = Expected heat of hydrogenation − Actual heat of hydrogenation.

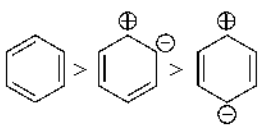
- Due to high resonance energy, benzene is quite stable and undergoes electrophilic substitution reactions. It does not undergo addition reactions,

although it has double bonds (due to delocalization of π electrons or resonance).

- Benzene has 36 kcal/mole of resonance energy.
- Resonance energy of CO_2 is 154.9 kJ.
- In tautomerism, arrangement of atoms is different for its different arrangements but in resonance, the arrangement of atoms is the same.

Stability of Different Canonical Structures

1. A non-polar structure is always more stable than a polar structure. In the following example, the structures are arranged in a decreasing order of stability.

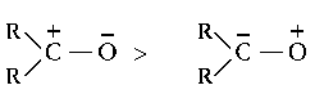


In the last two structures the charges are apart so they are less stable.

2. Greater the number of covalent bonds greater will be the stability. Therefore,

$$CH_3-C\equiv\overset{\oplus}{O} > CH_3-\overset{+}{C}=O$$

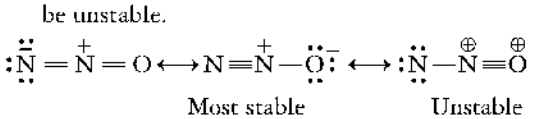
3. The canonical structure in which positive charge in an electro+positive atom and negative charge on the electro–negative atom is more stable. Therefore,



4. The canonical structure in which each atom has an octet state is more stable. Therefore,

$$CH_3-\underset{8e^-}{C}\equiv\overset{\oplus}{O} > CH_3-\underset{6e^-}{\overset{\oplus}{C}}=O$$

5. If like charges are closer then the structure will be unstable.



Most stable Unstable

Types of Resonance

1. **Isovalent resonance** The canonical structures have same number of bonds and same type of charges. For example, SO_2, NO_2, CO_3^{-2}

2. **Heterovalent resonance** Here, the canonical structures have different number of bonds and charges. For example, bura- 1, 3-diene, vinyl cyanide.

Resonance and Bond Order

$$\text{Bond order} = \frac{\text{Total no. of order between two atoms}}{\text{Total no. of major canonical structures}}$$

Example, In SO_3, B.O. = 4/3 = 1.33

In ClO_4^-, B.O. = 7/4 = 1.75

HYBRIDIZATION

- Pauling and Slater introduced this concept to explain the shape of molecules which could not be explained by the valence bond theory.
- It is the intermixing or re-distribution of energy among two or more half-filled, fully filled, incompletely filled or empty orbitals of comparable energy, to form same number of hybrid orbitals. Hybrids have identical energies and similar shapes.

Facts About Hybridization

- Number of atomic orbitals taking part in hybridization is equal to number of hybrids formed.
- Electrons do not undergo hybridization.
- A hybrid bond is always a sigma bond.
- A hybrid bond is always stronger than a non-hybrid bond.
- Hybridization occurs at the time of bond formation.
- Hybridization Overlapping (for enough overlapping, orbitals must be at an appropppriate distance from each other, that is, neither very close nor very far).
- Hybridization increases stability and decreases reactivity and energy of a molecule.
- Hybridization occurs in the central atom in a molecule (NH_3, H_2O, CH_4). Here, the central atoms are N, O, C respectively.
- Hybridization does not occur in isolated atoms but in bonded atoms.

Types of Hybridization

1. **sp hybridization:** Here, one s and one p orbitals form two sp hybrid orbitals after intermixing.

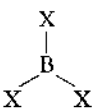
   - Shape of molecule is linear and bond angle is 180°. For example, X–M–X (Mg, Be, Zn, Hg)

     H C ≡ C H
       sp sp

   Some other examples are CO_2, and CS_2.

2. **sp^2 hybridization:** Here, one s and two p orbitals intermix to form three new sp^2 hybrid orbitals.

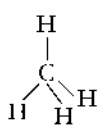
   - Shape of these species is trigonal or coplanar and the bond angle is 120°. For example,



3. **sp^3 hybridization:** Here, one s and three p orbitals intermix to give four new sp^3 hybrid orbitals.



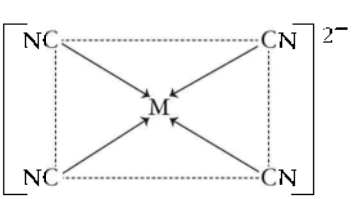
   - Shape of the species is tetrahedral and bond angle is 109° 28′. For example, C_2H_6, CX_4, SiX_4, NH_4^-, BX_4^-, NH_3, PH_3, H_2O, H_2S.



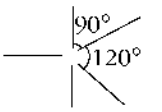
4. **dsp^2 hybridization:** Here, one s, two p and one d orbitals ($d_{x^2-y^2}$) intermix to give four new dsp^2 hybrid orbitals.



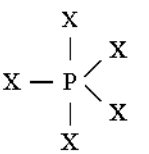
   - Shape of the species is square planar and bond angle is 90°. For example, $[Ni(CN)_4]^{2-}$, $[Cu(CN)_4]^{2-}$



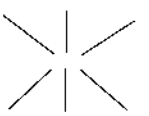
5. **sp^3d hybridization:** Here, one s, three p and one d-orbital (dz^2) intermix to form five new sp^3d hybrid orbitals.



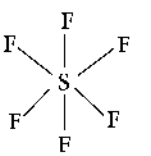
   - Shape of the species is trigonal bipyramidal and bond angles are 90°, 120°. For example, PX_5, XeF_2, I_3^-



6. **sp^3d^2 hybridization:** Here, one s, three p and two d-orbital (d_{z^2} and $d_{x^2-y^2}$) intermix to form six new sp^3d^2 hybrid orbitals.



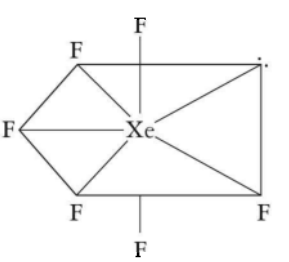
   - Shape of the species is octahedral and bond angle is 90°. For example, SF_6, XeF_4.



7. **sp^3d^3 hybridization:** Here one s, three p and three d orbital (d_{xy}, d_{yz}, d_{xz}) intermix to form seven new sp^3d^3 hybrid orbitals.



   - Shape of the species is pentagonal bipyramidal and bond angle is between 72° to 90°. For example, IF_7, XeF_6.



Rules to find the type of hybridization

For covalent compounds and ions: 1. Count the total number of valence electrons and (±) charge, to find a particular value. For example, in the PO_4^{-3} number of valence electrons is $5 + 4 \times 6 + 3 = 32$. For NH_4^+, this number is $5 + 4 - 1 = 8$

2. Now divide the total value of electrons to get the quotient X (number of bond pair electrons)
   1 If total value of electrons is between 2 to 8, divide it by 2.
   2 If total value is between 10 to 56 divide it by 8.
   3 If total value is 58 or more, divide it by 18.
3. If any remainder is left, divide again as above to get another quotient Y (number of lone pair electrons).
4. If X or X + Y = 2 = sp
   3 = sp^2
   4 = sp^3
   5 = sp^3d
   6 = sp^3d^2
   7 = sp^3d^3

   For example,

   $PCl_5 = 5 + 5 \times 7 = 40/8 = 5$, that is, the hybridization is sp^3d.

   $SF_6 = 6 + 6 \times 7 = 48/8 = 6$, that is, sp^3d^2 hybridization is present.

Rule to find the geometry of covalent compounds: The shape or geometry of a molecule or ion can be obtained by finding the type of hybridization, number of bond pairs and lone pair of electrons using the following relation.

$$P = \tfrac{1}{2}(V + M - C + A)$$

Here, P = total numbers of pairs of electrons around the central atom which gives the present hybridization of the central atom as calculated above.

A = Charge on anion

C = Charge on cation

M = Number of monovalent atoms

V = Number of electrons in the valence shell of the central atom.

To find lone pair of electrons

Lone pair = P − N

Here,

P = Total numbers of pairs of electrons around the central atom which gives hybridization as above.

N = Number of atoms surrounding the central atom or number of bond pairs of electrons.

For example,

1. SO_2

   $P = \tfrac{1}{2}(6 + 0 - 0 + 0) = 3(sp^2)$

   $lp = (3 - 2) = 1$

2. SF_6

   $P = \tfrac{1}{2}(6 + 6 - 0 + 0) = 6(sp^3d^2)$

   $lp = (6 - 6) = 0$

Hybridization in complexes: Coordination number of ligands is used to find the hybridization.

Table 1.1

Coordination Number	*Hybridization*		
2	sp		
3	sp^2		
4	sp^3	or	dsp^2
5	sp^3d	or	dsp^3
6	sp^3d^2	or	d^2sp^3
	When ligands are weak like H_2O, F, Cl, Br, I		When ligands are strong like CO, CN, NH_3

For example, in $[Fe(CN_6)]^{-3}$ the coordination number is 6 and ligand is strong, hence the hybridization is d^2sp^3.

Similary, in $[Fe(H_2O)_6]^{-3}$ the coordination number is 6 and ligand is weak, hence the hybridization is sp^3d^2.

VSEPR (Valence Shell Electron Pair Repulsion Theory)

- Valence shell electron pair repulsion theory was introduced by Nyholm and Gillispie to predict the shape of polyatomic molecules and ions.

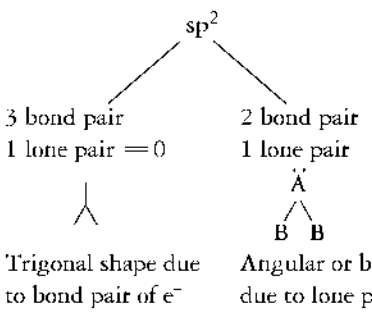
- According to this theory, besides hybridization, the nature of electrons around the central atom also decide the shape of molecule.
- There may be two types of electrons around the central atom, that is, bond pair or lone pair of electrons.
- These electrons undergo electron–electron repulsion and the decreasing order of electronic repulsion follows lp–lp > lp–bp > bp–bp.
- Due to this electronic repulsion, the shape of the molecule becomes distorted and the bond angle changes.

  Distortion in shape $\propto e^-$–e^- repulsion

  Distortion in shape or change in bond angle $\propto$ electronic repulsion.

Geometry of Some Molecules and Ions

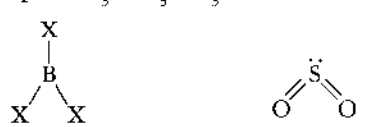
sp^2 hybridization



Trigonal shape due to bond pair of e^-

Angular or bent shape due to lone pair of e^-

For example, BX_3, BH_3, SO_3



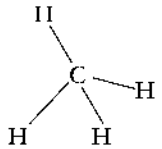
Normal trigonal shape

Bent shape

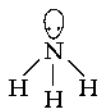
Here, S atom has two bond pairs and one lone pair of electron, so lp–bp type of repulsion distorts the shape, that is, it bends and changes the bond angle and the shape becomes angular. Same holds true for $SnCl_2$ and $PbCl_2$.

sp^3 hybridization

1. When the central atom has four bond pairs of electrons, the shape will be normal with normal bond angle of 109° 28' which The Shape tetrahedral. For example, CH_4, CCl_4, $SiCl_4$, NH_4^+, BX_4.

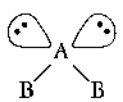


2. When the central atom has 3 bond pairs and 1 lone pair of electron, there will be lp–bp type of repulsion, which distorts the shape and changes the bond angle, that is, the shape becomes pyramidal and the bond angles are less than 109° 28' For example, NH_3, PH_3, NX_3, PX_3



In ammonia, the bond angle is 107°.

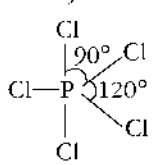
3. When the central atom has 2 lone pair and 2 bond pair of electron, there will be lp–lp type of electronic repulsion, hence the shape will be distorted and it will be angular or bent. For example, H_2O, H_2S, OF_2, SCl_2



4. When the central atom has 3 lone pairs and 1 bond pair of electrons, there will be lp–lp type of electronic repulsion. hence, shape is highly distorted and it becomes linear. For example, I–Cl, HCl

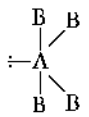
sp^3d hybridization

1. When the central atom has 5 bond pair of electrons, the shape will be normal with normal bond angle, that is, the shape becomes trigonal bipyramidal and bond angle of 90° and 120°. As only bp–bp type of electronic repulsion occurs, hence there is no distortion in shape and no change in bond angle. For example, PCl_5, AsF_5 and PF_5.

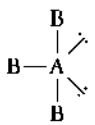


Due to the difference in bond angles here bond length also varies far axial and equitorial bonds.

2. When the central atom has 4 bond pair and 1 lone pair of electrons, the shape will be distorted and it will possess a see-saw like structure. For example, $SeCl_4$, $TeCl_4$, SF_4.

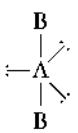


3. When the central atom has 3 bond pairs and 2 lone pair of electrons, the shape will be distorted and it will be a T-shape like structure. For example, ClF_3, BrF_3.



Here, lone pair of electrons occupy equitorial position to minimize e–e repulsion.

4. When the central atom has 2 bond pair and 3 lone pair of electrons, the shape will be distorted and the shape will be linear. For example, XeF_2, I_3^-, ICl_2^-.



sp^3d^2 hybridization

1. When the central atom has 6 bond pairs of electrons, the shape will be normal with normal bond angles that is, octahedral (90°). As only bp–bp type of electronic repulsion occurs, so there is no distortion in shape or change in the bond angle. For example, SF_6, $TeCl_6$





2. When the central atom has 5 bond pair and 1 lone pair of electrons, the shape will be distorted and it will be square pyramidal. For example, BrF_5, IF_5.

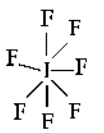


3. When the central atom has 4 bond pair and 2 lone pair of electrons, the shape will be distorted and it will be square planar. For example, XeF_4.



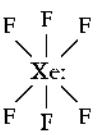
sp^3d^3 hybridization

1. When the central atom has 7 bond pair of electron, the shape will be normal with normal bond angles, that is, pentagonal bipyramidal. As bp–bp type of electronic repulsion occurs, so there is no distortion in shape and no change in bond angle. For example, IF_7.



2. When central atom has 6 bond pair and 1 lone pair of electrons, the shape will be distorted and the shape will be distorted pentagonal bipyramidal. For example, XeF_6.

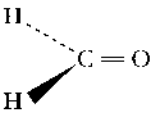
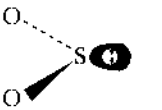
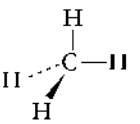
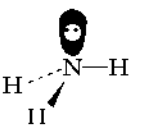
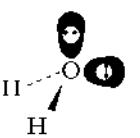
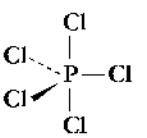
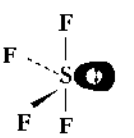
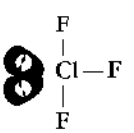
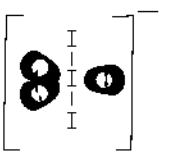


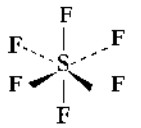
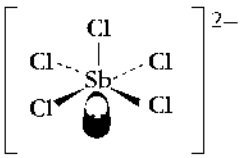
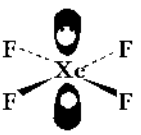


MOLECULAR ORBITAL THEORY

- Molecular orbital theory was given by Hund and Mulliken.
- It is based on LCAO (Linear Combination of Atomic Orbitals) model.
- Atomic orbitals undergo linear combination to form same number of molecular orbitals, if they fulfill the following conditions:
  1. Atomic orbitals must have comparable energies.
  2. Atomic orbitals must overlap linearly for enough and effective overlapping.

Hybridization and Shapes of Some Simple Molecules

Number of Bonds	Number of Lone Pairs	Number of Charge Clouds	Molecular Geometry and Shape	Examples
2	0	2	Linear	O=C=O
3	0	3	Trigonal planar	
2	1		Bent	
4	0	4	Tetrahedral	
3	1		Trigonal pyramidal	
2	2		Bent	
5	0	5	Trigonal bipyramidal	
4	1		See-saw	
3	2		T-shaped	
2	3		Linear	

Number of Bonds	Number of Lone Pairs	Number of Charge Clouds	Molecular Geometry and Shape	Examples
6	0	6	Octahedral	
5	1		Square Pyramidal	
4	2		Square Planar	

3. Atomic orbitals must have same symmetry along with the major molecular axis, for example, if Z axis is the main molecular axis, then only p_z–p_z orbitals will overlap and not p_x or p_y.

- Molecular orbitals are formed due to constructive and destructive interference of atomic orbitals.
- Constructive interaction of orbitals between orbital lobes having same wave function ψ produces bonding molecular orbitals like σ, π and Δ. These are HoMOs (Highest Occupied Molecular Orbitals).

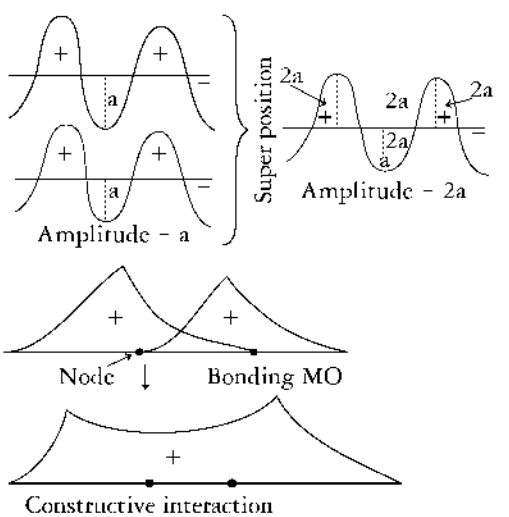


Fig. 1.13

- Destructive interaction between orbitals having different sign of c produces anti-bonding molecular orbitals or LuMOs (Lowest Unoccupied Molecular Orbitals). For example, σ*, π*, d*.

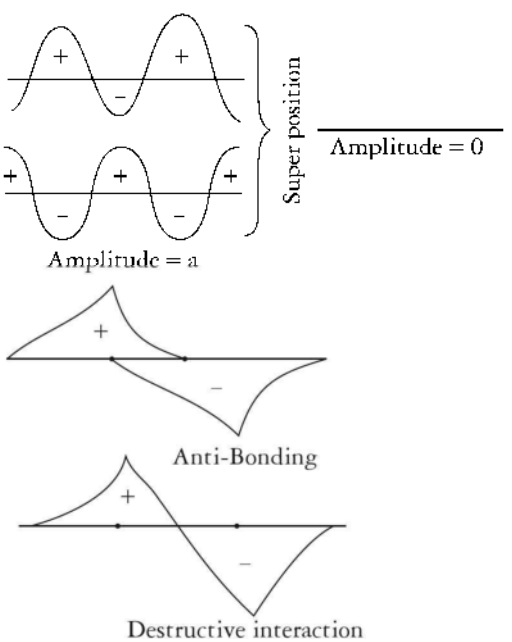


Fig. 1.14

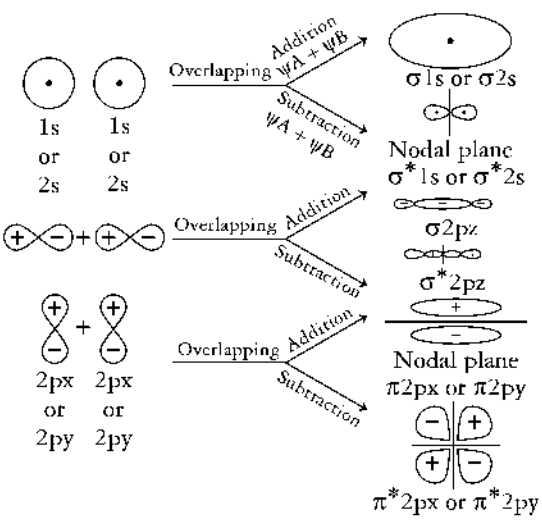
Facts Related to HoMOs and LuMOs

- Energy: LuMOs > HoMOs
- Wavelength: LuMOs < HoMOs

- LuMOs have nodal planes while HoMOs may or may not have nodal planes.
- Electrons contribute force of attraction in HoMOs while they contribute repulsion in LuMOs.
- The shape of the molecular orbitals formed depend upon shape of atomic orbital from which they are formed.
- Like aromic orbitals, molecular orbitals also follow.

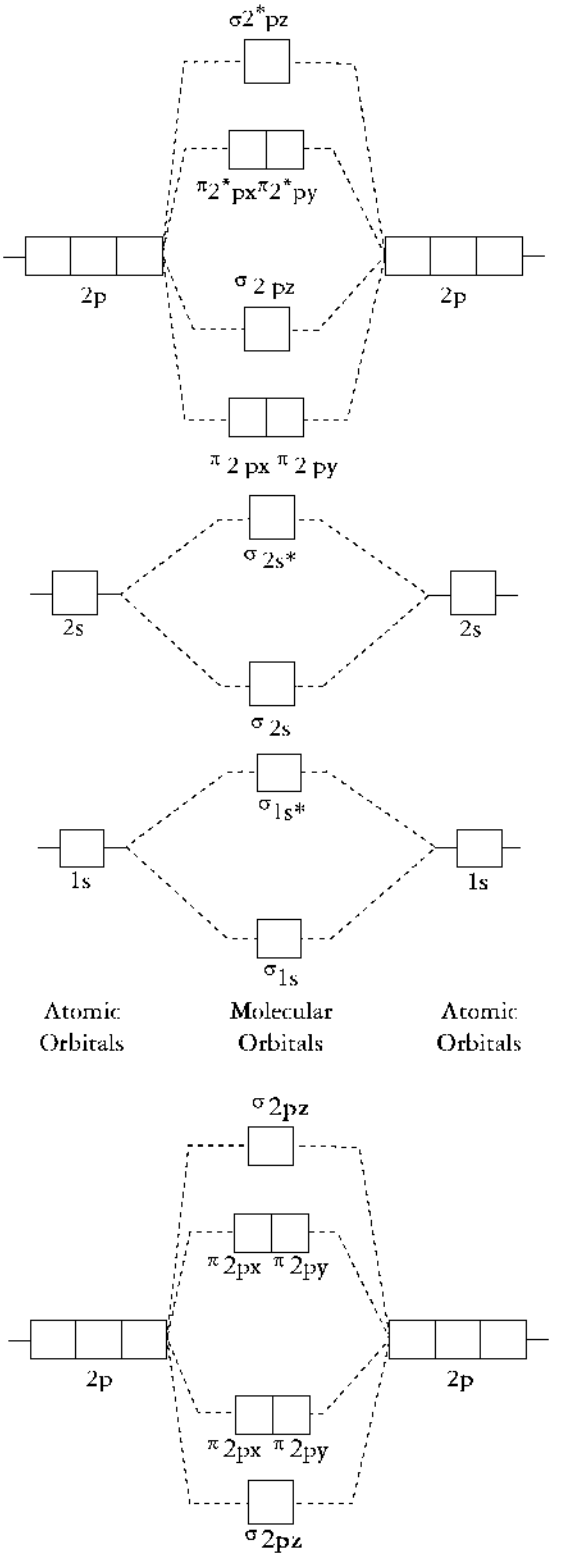
1. Pauli exclusion principle—Any molecular orbital can have a maximum of two electrons with opposite spin.
2. Hund's rule In degenerate molecular orbital, before pairing, each molecular orbital must have one electron.
3. Aufbau principle—Electrons are filled from molecular orbital of lower energy to higher energy.

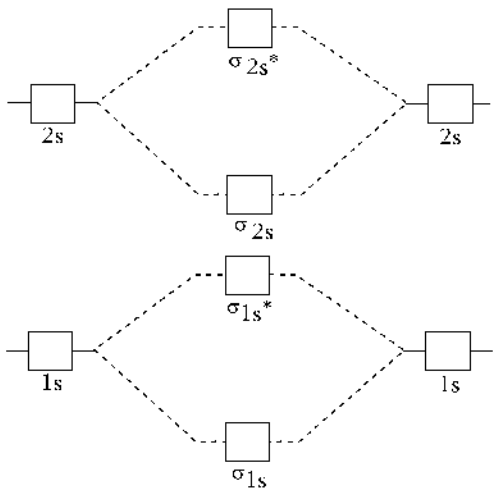
Formation of Various Molecular Orbitals



Order of Filling Electrons in Molecular Orbital

(1) Molecular energy level diagram for diatomic homonuclear molecules such as N_2, C_2 and B_2





2. Molecular orbital energy level diagram for diatomic homonuclear molecules such as O_2, F_2 etc.

(i) Molecules with N_2 configuration or 14 e⁻.
$\sigma 1s\ \sigma^*1s, \sigma 2s\ \sigma^*2s, \pi 2p_x \approx \pi 2p_y, \sigma 2p_z$

(ii) Molecules with O_2 configuration or more than 14 e.
$\sigma 1s\ \sigma^*1s, \sigma 2s\ \sigma^*2s, \sigma 2p_z\ \pi 2p_x = \pi 2p_y, \pi^*2p_x = \pi^*\ 2p_y, \sigma^*2p_z$

- σ 1s is the lowest energy molecular orbital while σ^*2p_z is the highest energy molecular orbital.
- Due to intermixing of 2s and 2p orbitals in cases where the number of elecruons is more than 16, $\sigma 2p_z$ is taken after σ^*2s here.
- Bond order $= \frac{n_b - n_a}{2}$

Here n_b = Number of bonding molecular orbital electrons

n_a = Number of anti-bonding molecular orbital electrons

Bond order ∝ Bond dissociation energy

∝ Bond angle

$\propto \frac{1}{\text{Bond length}}$

- Higher the bond order, higher will be stability and shorter will be the bond length.
- If unpaired electrons (n = 1, 2) are present in a molecule it is paramagnetic.
- If n = 0 that is no unpaired electrons, molecule is diamagnetic.

Examples,

H_2 : $\sigma (1s)^2$

H_2^+ : $\sigma (1s)^1$

H_2^- : $\sigma (1s)^2\ \sigma^* (1s)^1$

N_2^- : KK $\sigma (2s)^2, \sigma^* (2s)^2, \pi(2px)^2 = \pi (2py)^2, \sigma(2pz)^2\ \pi^* (2px)^1$

N_2^{2-} : KK $\sigma (2s)^2, \sigma^* (2s)^2, \pi(2px)^2 - \pi (2py)^2, \sigma 2pz^2\ \pi^* (2px)^1 - \pi^* (2py)^1$

O_2 : KK $\sigma (2s)^2, \sigma^* (2s)^2, \sigma (2pz)^2, \pi (2px)^2 - \pi (2py)^2\ \pi^* (2px)^1 - \pi^* (2py)^1$

O_2^- : KK, $\sigma (2s)^2, \sigma^* (2s)^2, \sigma (2pz)^2, \pi (2px)^2\ \pi (2py)^2 - \pi^* (2px)^1$

O_2^- : KK $\sigma (2s)^2, \sigma^* (2s)^2, \sigma,(2pz)^2, \pi (2px)^2\ \pi (2py)^2, \pi^* (2px)^2 = \pi^* (2py)^1$

O_2^{2-} : KK, $\sigma (2s)^2\ \sigma^* (2s)^2, \sigma (2pz)^2, \pi (2px)^2 = \pi (2py)^2\ \pi^* (2px)^2 = \pi^* (2py)^2$

F_2 : KK, $\sigma (2s)^2\ \sigma^* (2s)^2, \sigma (2pz)^2, \pi (2px)^2 - \pi (2py)^2\ \pi^* (2px)^2 - \pi^* (2py)^2$

K K Srands for $\sigma\ 1s^2\ \sigma^*\ 1s^2$ here.

Some Orders Related to Molecular Orbital Theory

■

	O_2^{2+}	O_2^+	O_2	O_2	O_2^{2-}
Bond order	3	2.5	2	1.5	1

Decreasing order of bond order, bond angle, bond dissociation energy.
Increasing order of bond lengrh.

- Magnetic Nature

	O_2	>	O_2^-	=	O_2	>	O_2^{2+}	=	O_2^{2-}
n	2		1		1		0		0
	Paramagnetic		Weakly Paramagnetic						Diamagnetic

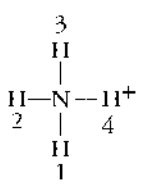
- CO, NO , CN , N_2 (14 e) all have bond order 3 and are diamagneric.
- NO, CN, N_2^-, N_2^- all have a bond order equal to 2.5 and are paramagnetic as n=1.
- H_2, Li_2, B_2 all have a bond order equal to one and are diamagnetic except B_2.
- H_2 , H_2^+, He_2^+ all have a bond equal to order ½ and are paramagneric.
- All molecules with fractional bond order are paramagnetic.
- Molecules with whole number bond order are mostly diamagneric, except O_2, B_2, N_2^{2-}.

REMEMBER

- When z-axis is the major molecular axis, s molecular orbital will be formed only by s–s, p_z–p_z or s–p_z. AO and no MO is formed by p_x or p_y.
- $\sigma 1s$, $\sigma 2s$, $\sigma 2p_z$ do not have any nodal plans.
- σ^*1s, σ^*2s, σ^*2p_z, $\pi 2p_x$, $\pi 2p_y$ all have one nodal plane each.
- π^*2p_x and π^*2p_y have two nodal planes each.

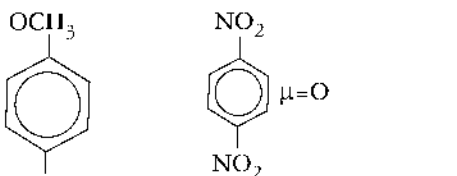
SOME IMPORTANT GUIDELINES

- Formyl Charge: It is equal to $V - N - 1/2\ B$, where
  V = Total number of valence electrons in the free atom.
  N = Total number of lone pair of electrons.
  B = Total number of shared electrons that is, bonded electrons.
  For example, in case of NH_4^+

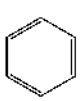


  Here, F.C. on N-atom = $5 - 0 - \frac{1}{2} \times 8 = +1$
  F.C. on 1, 2, 3 H-atoms = $1 - 0 - \frac{1}{2} \times 2 = 0$
  F.C. on H (4 H-atom) = $0 - 0 - \frac{1}{2} \times 2 = -1$
- CO_3^{2-}, NO_3^- involve only pπ–pπ bonding.
- SO_4^{2}, PO_4^{3}, ClO_4 involve pπ–dπ bonding.
- **In a hypervalent,** for example, SO_4^{2-}, PO_4^{-3}, ClO_4^-, at least one atom has more than 8 electrons.
- Molecular solids have low heat of fusion.
- Breaking of covalent bonds occurs during melting of SiO_2.
- **Banana bonding** is shown by boron hydride, that is, diborane B_2H_6.
- **Band Theory of Metallic Bonds** It is based on molecular orbital theory. The highest occupied energy band is called the valence band while the lowest occupied energy band is called as conduction band. The difference in the energy between top of valence bond and bottom of conduction bond is called **energy gap**. For example,

1. When energy gap is very very small, conduction occurs (as in merals).
2. When energy gap is small, less conduction occurs (as in semi-conductor).
3. When energy gap is large, no conduction occurs (as in insulators).



(does not have zero dipole moment,) (does not have zero dipole moment,)

- C_2 molecule has no σ bond but has a π bond.
- Bond energy: $Cl_2 > F_2 > Br_2 > I_2$.
- H_{obs} of CO is greater than $\mu_{expected}$ due to presence of a coordinate bond.
- AgX, $BaSO_4$, $PbSO_4$, are nearly insoluble or insoluble in H_2O as $H_{Hyd} >$ Lattice energy.
- 

  In benzyne, the triple bond (≡) is partial due to less effective overlapping. In it all C atoms are sp^2 hybridized.
- Bond length of CO > CO^+ as bond order of CO is 3 and of CO^+ is 3.5.

  ⟶ Bond energy order
  $2s - 2s < 2s-2p < 2.p-2p.$

  ⟶ p–p < s–s < s–p < p–p { strongness of lateral ← axial overlaping → overlaping}

  ⟶ NO >NO> NO^2 > NO^1 (Bond length)

  ⟶ $NO^+ > NO^{2-} = NO > NO^-$ (Bond order)

Straight Objective Type Questions (Single Choice)

1. KF combines with HF to form KHF_2. The compound contains the species
   (a) K^+, F^- and H^+ (b) K^-, F^- and HF
   (c) K^+ and $[HF_2]^-$ (d) $[KHF]^+$ and F_2

2. The bond order in O_2^- ion is
   (a) 2 (b) 1
   (c) 2.5 (d) 1.5

3. Which of the following molecules has the smallest bond angle?
   (a) NH_3 (b) H_2O
   (c) H_2Se (d) H_2S

4. The number of sigma and pi bonds present in tetracyanoethylene $[(CN)_2C - C(CN)_2]$ molecule are, respectively.
   (a) 5 σ and 9 π (b) 5 σ and 8π
   (c) 9 σ and 9 π (d) 9 σ and 7 π

5. Unusually high boiling point of water is the result of
   (a) Intermolecular hydrogen bonding
   (b) Both intra and inter molecular hydrogen bonding
   (c) High specific heat
   (d) Intramolecular hydrogen bonding

6. Dissolution of ionic solid in water is accompanied by release of energy represented by $\Delta H_{solution}$. This implies that
   (a) $\Delta H_{lattice} > \Delta H_{hydration}$
   (b) $\Delta H_{lattice} = \Delta H_{hydration}$
   (c) $\Delta H_{hydration} > \Delta H_{lattice}$
   (d) $\Delta H_{lattice} : \Delta H_{hydration} = 1 : 2$

7. In which of the following molecules, the central atom does not follow the octet rule?
   (a) H_2O (b) H_2S
   (c) BF_3 (d) CO_2

8. d^2sp^3 hybridization of atomic orbitals gives
   (a) Triangular structure
   (b) Tetragonal structure
   (c) Square planar structure
   (d) Octahedral structure

9. The strength of sigma bonds formed by axial overlap of s or p orbitals of the 2nd shell of the participating atoms decrease as
   (a) s-s > p-s > p-p (b) s-s > p-p > s-p
   (c) p-p > s-p > s-s (d) p-s > s-s > p-p

10. The bond angle and dipole moment of water, respectively are
    (a) 102.5° , 1.56 D
    (b) 107.5°, 1.56 D
    (c) 104.5°, 1.84 D
    (d) 109.5° 1.84 D

11. The sequence of ionic mobility in aqueous solution is
    (a) $Rb^+ > K^+ > Cs^+ > Na^+$
    (b) $Na^- > K^+ > Rb^+ > Cs+$
    (c) $K^+ > Na^+ > Rb^- > Cs^+$
    (d) $Cs^+ > Rb^- > K^+ > Na^+$

12. Which of the following statements is true?
    (a) The dipole moment of NH_3 is zero.
    (b) The dipole moment of NF_3 is equal to NH_3.
    (c) The dipole moment of NF_3 is zero.
    (d) The dipole moment of NF_3 is less than NH_3.

13. Which of the following is correctly based on the molecular orbital theory for peroxide ion?
    (a) Its bond order is two and it is paramagnetic.
    (b) Its bond order is one and it is diamagnetic.
    (c) Its bond order is two and it is diamagnetic.
    (d) Its bond order is one and it is paramagnetic.

14. Bond angle in water is
    (a) 120° (b) 107°
    (c) 104.5° (d) 109.5°

15. The linear structure is assumed by
    I. $SnCl_2$ II. NCO
    III. NO_2^+ IV. CS_2
    (a) I, II and III (b) II, III and IV
    (c) I, III and IV (d) none of the above

16. Which one of the following has zero dipole moment?
(a) ClF (b) PCl_3
(c) SiF_4 (d) $CFCl_3$

17. Which of the following is an electron deficient molecule?
(a) C_2H_6 (b) B_2H_6
(c) SiH_4 (d) PH_3

18. Which of the following is paramagnetic with bond order 0.5?
(a) H_2^+ (b) O_2^-
(c) B_2 (d) F_2

19. Which type of bond is not present in HNO_2 molecule?
(a) Covalent
(b) Coordinate
(c) Ionic
(d) Ionic as well as coordinate

20. The angular shape of ozone molecule (O_3) consist of
(a) 1 sigma and 1 pi bonds
(b) 2 sigma and 1 pi bonds
(c) 1 sigma and 2 pi bonds
(d) 2 sigma and 2 pi bonds

21. Which of the following molecules has trigonal planar geometry?
(a) BF_3 (b) NH_3
(c) PCl_3 (d) IF_3

22. The hybridization of oxygen atom in H_2O_2 is
(a) sp^3 (b) sp^2
(c) sp (d) sp^3d^2

23. Which of the following would have a permanent dipole moment?
(a) SiF_4 (b) SF_4
(c) XeF_4 (d) BF_3

24. In $TeCl_4$, the central atom, tellurium, involves hybridization
(a) dsp^2 (b) sp^3
(c) sp^3d (d) sp^3d^2

25. In which of the following pairs, the two species are isostructural?
(a) SF_4 and XeF_4
(b) SO_3^{2-} and NO_3^-
(c) BF_3 and NF_3
(d) BrO_3^- and XeO_3

26. The correct sequence of increasing covalent character is represented by
(a) LiCl < NaCl < $BeCl_2$
(b) $BeCl_2$ < LiCl < NaCl
(c) NaCl < LiCl < $BeCl_2$
(d) $BeCl_2$ < NaCl < LiCl

27. What is the correct order of the strength of hydrogen bonds?
(a) NH---N < OH---O < FH---F
(b) ClH---Cl > NH---N > OH---O
(c) ClH---Cl < NH---N < OH---O
(d) NH---N > OH---O > FH---F

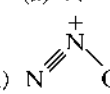
28. Bond angle of 109°28' is found in
(a) NH_3 (b) H_2O
(c) $\overset{\ominus}{C}H_3$ (d) $\overset{\oplus}{N}H_4$

29. The hybrid states of C in CS_2, HCHO and C_{60} molecule respectively, are
(a) sp, sp^2, sp^3 (b) sp, sp^2, sp^2
(c) sp^2, sp^2, sp^2 (d) sp, sp^3, sp^3

30. Which of the following molecules has almost negligible tendency to form hydrogen bonds?
(a) NH_3 (b) H_2O
(c) HF (d) HI

31. Hybridization of the underlined atom changes in
(a) $\underline{Al}H_3$ changes to AlH_4^-
(b) $H_2\underline{O}$ changes to H_3O^+
(c) $\underline{N}H_3$ changes to NH_4^+
(d) all of these

32. Identify the correct sequence of increasing number of π-bonds in the structures of the following molecules.
I. $H_2S_2O_6$ II. H_2SO_3
III. $H_2S_2O_5$
(a) I, II, III (b) II, III, I
(c) II, I, III (d) I, III, II

33. N_2O is isoelectronic with CO_2 and N_3^-, which is the structure of N_2O?
(a) N – O – N (b) N ≡ N ⟶ O
(c)  (d) 

34. The number of lone pairs on Xe in XeF_2, XeF_4 and XeF_6, respectively are
(a) 3, 2, 1 (b) 2, 4, 6
(c) 1, 2, 3 (d) 6, 4, 2

35. The two atoms X and Y lie on the top of group 2 and group 16, respectively. On combination, they form compound of the type
(a) X_2Y_2 (b) XY
(c) X_2Y (d) XY_2

36. Pairs of species having identical shapes for molecules is
(a) BF_3, PCl_3 (b) PF_5, IF_5
(c) CF_4, SF_4 (d) XeF_2, CO_2

37. Shape of ClF_3 is
(a) T-shaped (b) V-shaped
(c) Pyramidal (d) Equilateral triangle

38. A square planar complex is formed by hybridization of the following atomic orbitals
(a) s, p_x, p_y, p_z (b) s, p_x, p_y, p_z, d
(c) $dx^2 - y^2$, s, p_x, p_y (d) s, p_x, p_y, p_z, dz^2

39. Which of the following represents the Lewis structure of N_2 molecule?
(a) $\overset{\cdot\cdot}{\underset{\cdot\cdot}{N}} \equiv \overset{\cdot\cdot}{\underset{\cdot\cdot}{N}}$ (b) $\overset{\cdot}{\underset{\cdot}{N}} \equiv \overset{\cdot\cdot}{\underset{\cdot\cdot}{N}}$
(c) $\overset{\cdot\cdot}{\underset{\cdot\cdot}{N}} \equiv \overset{\cdot\cdot}{\underset{\cdot\cdot}{N}}$ (d) $\overset{\cdot}{\underset{\cdot}{N}} - \overset{\cdot\cdot}{\underset{\cdot\cdot}{N}}$

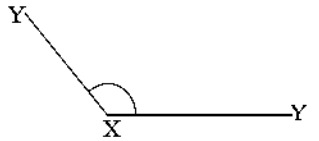
40. Inter-molecular hydrogen bonding exists in
(a) o-nitrophenol
(b) o-chlorophenol
(c) Ammonium chloride
(d) Water

41. The geometrical shape of sp^3d hybridization is
(a) Trigonal bipyramidal
(b) Linear
(c) Tetrahedral
(d) Square planar

42. In which of the following species, intramolecular H-bonding occurs?
I. Acetate ion II. Salicylate ion
III. Propanoic acid IV. O-Nitrophenol
(a) I, III, IV (b) I, II, IV
(c) II, IV (d) IV only

43. Which one of the following compounds has the smallest bond angle in its molecule?
(a) SO_2 (b) OH_2
(c) SH_2 (d) NH_3

44. Which bond angle, 0 would result in the maximum dipole moment for the triatomic molecule XY_2 shown below:



(a) 0 - 120° (b) 0 - 90°
(c) 0 - 145° (d) 0 - 175°

45. Which of the following has the highest bond order?
(a) N_2 (b) O_2
(c) He_2 (d) H_2

46. The energy of hydrogen bond is of the order of
(a) 40 kJ mol^{-1} (b) 140 kJ mol^{-1}
(c) 400 kJ^{-1} (d) 4 kJ mol^{-1}

47. Which one of the following pairs of molecules will have permanent dipole moments for both members?
(a) SiF_4 and NO_2 (b) NO_2 and CO_2
(c) NO_2 and O_3 (d) SiF_4 and CO_2

48. In the formation of N_2^+ from N_2, the electron is removed from a
(a) σ – orbital (b) π – orbital
(c) σ^* – orbital (d) π^* – orbital

49. Which of the following is diamagnetic?
(a) H_2^+ (b) O_2
(c) Li_2 (d) He_2^-

50. How many types of F – S – F bonds are present in SF_4?
(a) 5 (b) 4
(c) 3 (d) 2

51. An octahedral complex is formed when hybrid orbitals of the following type are involved
(a) d^2sp^3 (b) dsp^2
(c) sp^3 (d) sp^2

52. The pair of species having identical shapes for molecules of both species is
(a) CF_4, SF_4 (b) XeF_2, CO_2
(c) BF_3, PCl_3 (d) PF_5, IF_5

53. Among the following compounds, the one that is polar and has the central atom with sp^2 hybridization is

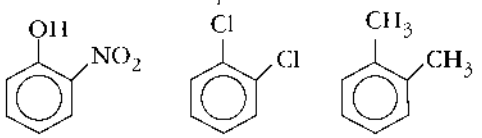
(a) SiF_4 (b) BF_3
(c) $HClO_2$ (d) H_2CO_3

54. The number of σ and π bonds in allyl isocyanide are
(a) 9σ, 3π
(b) 9σ, 9π
(c) 3σ, 4π
(d) 5σ, 7π

55. Iodine pentafluoride undergoes the following hybridization
(a) d^2sp^3 (b) dsp^3
(c) sp^3d (d) sp^3d^2

56. The correct order regarding the electronegativity of hybrid orbitals of carbon is
(a) $sp < sp^2 > sp^3$ (b) $sp < sp^2 < sp^3$
(c) $sp > sp^2 < sp^3$ (d) $sp > sp^2 > sp^3$

57. Correct order of dipole moment is



(1) (2) (3)

(1) (2) (3)
(a) 1 – 2 – 3 (b) 3 < 2 < 1
(c) 1 < 2 < 3 (d) 2 < 3 < 1

58. The maximum number of hydrogen bonds that a molecule of water can have is
(a) 3 (b) 4
(c) 1.3 (d) 2.4

59. The number of lone pairs of electrons present in central atom of ClF_3 is
(a) 0 (b) 1
(c) 2 (d) 3

60. In $[Ag(CN)_2]^-$ the number of π bonds is
(a) 2 (b) 3
(c) 4 (d) 6

61. Which of the following has the regular tetrahedral structure?
(a) BF_4 (b) SF_4
(c) $[Ni(CN)_4]^2$ (d) XeF_4

62. Among the following compounds, the one that is polar and has the central atom with sp^2 hybridization is
(a) SiF_4 (b) BF_3
(c) $HClO_2$ (d) H_2CO_3

63. Malleability and ductility of metals can be accounted due to
(a) the capacity of layers of metal ions to slide over the other.
(b) the crystalline structure in metal.
(c) the presence of electrostatic force.
(d) the interaction of electrons with metal ions in the other.

64. Bond order of nitric oxide is
(a) 1 (b) 2.5
(c) 2 (d) 1.5

65. The maximum number of 90° angles between bond pair-bond pair of electrons is observed in
(a) sp^3d^2 hybridization
(b) sp^3d hybridization
(c) dsp^3 hybridization
(d) dsp^2 hybridization

66. Which combination of atomic orbitals is not allowed according to MO theory?
(a) px – px (b) px – py
(c) py – py (d) pz – pz

67. Which one of the following species is diamagnetic in nature?
(a) H_2^- (b) H_2
(c) H_2^+ (d) He_2^+

68. Bond angle in XeO_3 is
(a) 107° (b) 119°
(c) 92° (d) 103°

69. The electronegativity difference between N and F is greater than that between N and H yet, the dipole moment of NH_3 (1.5 D) is larger than that of NF_3 (0.2 D). This is because
(a) in NH_3 as well as NF_3 the atomic dipole and bond dipole are in opposite directions.
(b) in NH_3 the atomic dipole and bond dipole are in the opposite directions whereas in NF_3 these are in the same direction.
(c) in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in same direction.
(d) in NH_3 the atomic dipole and bond dipole and in the same direction whereas in NF_3 these are in opposite directions.

70. Lattice energy of an ionic compound depends upon
(a) packing of ions only.
(b) size of the ion only.
(c) charge on the ion and size of the ion.
(d) charge on the ion only.

71. Which of the following molecule is planar?
(a) CH_4 (b) NH_3
(c) C_2H_4 (d) $SiCl_4$

72. The number of unpaired electrons in a paramagnetic diatomic molecule of an element with atomic number 16 is
(a) 4 (b) 1
(c) 2 (d) 3

73. The paramagnetism of O_2^+ is due to the presence of an odd electron in the MO
(a) σ^*2s (b) $\pi 2py$
(c) $\pi 2pz$ (d) π^*2py

74. Which of the following is electron deficient molecule?
(a) C_2H_6 (b) PH_5
(c) B_2H_6 (d) SiH_4

75. Which of the following molecules/ions does not contain unpaired electrons?
(a) O_2^{2-} (b) B_2
(c) N_2 (d) O_2

76. Sulphur reacts with chlorine in the 1:2 ratio and forms X. Hydrolysis of X gives a sulphur compound Y. What is the hybridization state of central atom in the anion of Y?
(a) sp^3 (b) sp
(c) sp^2 (d) sp^3d

77. In which of the following molecules, all the bonds are not equal?
(a) AlF_3 (b) NF_3
(c) ClF_3 (d) BF_3

78. PCl_5 in solid state splits into
(a) PCl_6^+ and PCl_4^-
(b) PCl_4^+ and PCl_6^-
(c) PCl_3 and Cl_2
(d) PCl_4^+ and Cl^-

79. Which of the following is not isostructural with $SiCl_4$?
(a) PO_4^{3-} (b) NH_4^+
(c) SCl_4 (d) SO_4^{2-}

80. In which of the following molecules/ions all bonds are not equal?
(a) SF_4 (b) SiF_4
(c) XeF_4 (d) BF_4^-

81. Which of the following species has a linear shape?
(a) NO_2^+ (b) O_3
(c) NO_2^- (d) SO_2

82. How many sigma and pi bonds are present in toluene?
(a) $3\pi + 6\sigma$ (b) $3\pi + 8\sigma$
(c) $6\pi + 6\sigma$ (d) $3\pi + 15\sigma$

Brainteasers Objective Type Questions (Single Choice)

83. The hybrid state of central iodine atom is I_3^+ and I_3^- are, respectively
(a) sp^3, sp^3d (b) sp^3d, sp^3
(c) sp^3d, sp^3d (d) sp^2, sp^3d

84. Arrange the following compounds in order of increasing dipole moment
I. Toluene
II. m-dichlorobenzene
III. o-dichlorobenzene
IV. p-dichlorobenzene
(a) I < IV < II < III
(b) IV < I < II < III
(c) IV < I < III < II
(d) IV < II < I < III

85. The correct order of the lattice energies of the following ionic compounds is

(a) $NaCl > MgBr_2 > CaO > Al_2O_3$

(b) $Al_2O_3 > CaO > MgBr_2 > NaCl$

(c) $NaCl > CaO > MgBr_2 > Al_2O_3$

(d) $MgBr_2 > Al_2O_3 > CaO > NaCl$

86. The following compounds are to be arranged in order of their increasing thermal stabilities. Identify the correct order:

I. K_2CO_3 II. $MgCO_3$

III. $CaCO_3$ IV. $BeCO_3$

(a) I < II < III < IV

(b) IV < II < III < I

(c) IV < II < I < III

(d) II < IV < III < I

87. Which of the following set contains species having same angle around the central atom?

(a) SF_4, CH_4, SeF_4 (b) NF_3, BCl_3, NH_3

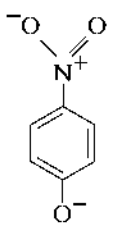
(c) BF_3, NF_3, $AlCl_3$ (d) BF_3, BCl_3, BBr_3

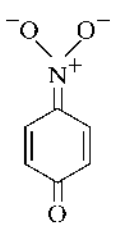
88. The magnetic moment of KO_2 at room temperature is

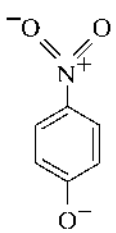
(a) 1.43 BM (b) 2.64 BM

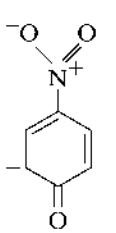
(c) 2.41 BM (d) 1.73 BM

89. The most unlikely representation of resonance structure of p-nitrophenoxide is

(a) 

(b) 

(c) 

(d) 

90. Four elements P, Q, R, S have atomic numbers $Z-1$, Z, $Z+1$ and $Z+2$, respectively. If Z is 9, then bond between which pair of elements will be ionic?

(a) P and R (b) S and Q

(c) S and R (d) Q and R

91. Among the following species, identify the isostructural pairs:

NF_3, NO_3^-, BF_3, H_3O^+, HN_3

(a) [NF_3, NO_3^-] and [BF_3, H_3O^+]

(b) [NF_3, HN_3] and [NO_3^-, BF_3]

(c) [NF_3, H_3O^+] and [NO_3^-, BF_3]

(d) [NF_3, H_3O^+] and [HN_3, BF_3]

92. Pick out the isoelectronic structures from the following:

(I) CH_3, (II) H_3O^+, (III) NH_3, (IV) CH_3

(a) I and II (b) III and IV

(c) I and III (d) II, III and IV

93. Four diatomic species are listed in different sequences. Which of these presents the correct order of their increasing bond order?

(a) $C_2^{2-} < He_2^+ < NO < O_2^-$

(b) $He_2^+ < O_2 < NO < C_2^{2-}$

(c) $O_2 < NO < C_2^{2-} < He_2^+$

(d) $NO < C_2^{2-} < O_2^- < He_2^-$

94. The electric configuration of four elements are as follows:

A : $1s^2\ 2s^2\ 2p^4$ B : $1s^2\ 2s^2\ 2p^5$

C : $1s^2\ 2s^2\ 2p^6\ 3s^1$ D : $1s^2\ 2s^2\ 2p^6\ 3s^2$

Now decide the possible formulae of ionic compounds that could be formed between them.

(a) A_2C, DA, CB, D_2B

(b) AC, DA, CB, CB

(c) C_2A, DA, CB, DB_2

(d) AC, D_2A, C_2B, DB

95. Which of the following pair consists of only paramagnetic species?

(a) [O_2, O_2^{2-}] (b) [O_2, NO]

(c) [NO, NO] (d) [CO, NO]

96. The correct order of increasing bond angles in the following triatomic species is

(a) $NO_2^+ < NO_2 < NO_2^-$

(b) $NO_2^- < NO_2 < NO_2$

(c) $NO_2 < NO_2 < NO_2$

(d) $NO_2^- < NO_2 < NO_2^+$

97. The correct order in which the O – O bond length increases in the following is

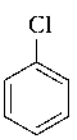
(a) $O_2 < H_2O_2 < O_3$
(b) $O_3 < H_2O_2 < O_2$
(c) $H_2O_2 < O_2 < O_3$
(d) $O_2 < O_3 < H_2O_2$

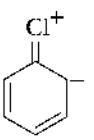
98. In which one of the following pairs, molecules/ions have similar shape?
(a) CCl_4 and $PrCl_4$
(b) NH_3 and BF_3
(c) BF_3 and t-butyl carbonium ion
(d) CO_2 and H_2O

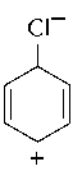
99. How many bonding MOs can be formulated for benzene and how many of these can degenerate?
(a) 3, 6 (b) 3, 2
(c) 6, 3 (d) 2, 3

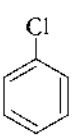
100. The correct order of bond angles (smallest first) in H_2S, NH_3, BF_3 and SiH_4 is
(a) $H_2S < SiH_4 < NH_3 < BF_3$
(b) $NH_3 < H_2S < SiH_4 < BF_3$
(c) $H_2S < NH_3 < SiH_4 < BF_3$
(d) $H_2S < NH_3 < BF_3 < SiH_4$

101. Which of the following is not a resonating form of chlorobenzene?

(a) 

(b) 

(c) 

(d) 

102. The bond order in NO is 2.5 while that in NO^- is 3. Which of the following statements is true for these two species?
(a) bond length in NO^+ is greater than in NO
(b) bond length in NO is greater than in NO
(c) bond length in NO^- is equal than in NO
(d) bond length is unpredictable

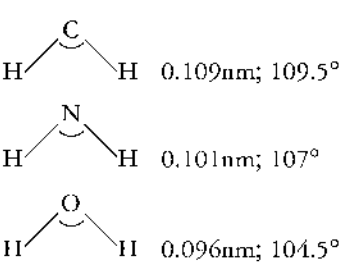
103. The sequence that correctly describes the relative bond strength pertaining to oxygen molecule and its cation or anion is
(a) $O_2^{2-} > O_2^- > O_2 > O_2^-$
(b) $O_2 > O_2^- > O_2 > O_2^{2-}$
(c) $O_2^+ > O_2 > O_2^{2-} > O_2^-$
(d) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

104. The dipole moments of methane and its halogen derivatives are in the order
(a) $CH_4 < CH_2Cl_2 < CHCl_3 < CH_3Cl$
(b) $CH_3Cl < CH_2Cl_2 < CHCl_3 < CH_4$
(c) $CH_4 < CHCl_3 < CH_2Cl_2 < CH_3Cl$
(d) $CH_4 < CH_3Cl < CH_2Cl_2 > CHCl_3$

105. The correct order of increasing bond angles is
(a) $OF_2 < H_2O < Cl_2O < ClO_2$
(b) $ClO_2 < OF_2 < Cl_2O < H_2O$
(c) $ClO_2 < Cl_2O < H_2O < OF_2$
(d) $OF_2 < Cl_2O < H_2O < ClO_2$

106. The correct order of bond energies for the C-H bond is
(a) $(CH_3)_3C\text{-}H < (CH_3)_2CH\text{-}H < H_3C\text{-}H < H_3C.CH_2\text{-}H$
(b) $H_3C\text{-}H < (CH_3)_3C\text{-}H < H_3C.CH_2\text{-}H < (CH_3)_2CH\text{-}H$
(c) $(CH_3)_3C\text{-}H < (CH_3)_2CH\text{-}H < H_3C.CH_2\text{-}H < H_3C\text{-}H$
(d) $H_3C\text{-}H < H_3C.CH_2\text{-}H < (CH_3)_2CH\text{-}H < (CH_3)_3C\text{-}H$

107. The bond lengths and bond angles in the molecules of methane, ammonia and water are given as under:



The variation in bond angle is a result of
I. The increasing repulsion between hydrogen atoms as the bond length decreases.
II. The number of non-bonding electron pairs in the molecule.
III. A non-bonding electron pair having a greater repulsive force than a bonding electron pair.

(a) Only I is correct
(b) Only III is correct
(c) (II) and (III) are correct
(d) (I) and (II) are correct

108. The bond angles of NH_3, NH_4^+ and NH_2^- are in the order
(a) $NH_2^- > NH_3 > NH_4^+$
(b) $NH_4^+ > NH_3 > NH_2^-$
(c) $NH_3 > NH_2^- > NH_4^+$
(d) $NH_3 > NH_4^+ > NH_2^-$

109. The correct order of bond order values among the following is

I. NO^- II. NO_2^+
III. NO IV. NO_2^-
V. NO_2

(a) (II) < (III) < (IV) < (I) < (V)
(b) (V) < (I) < (IV) = (III) < (II)
(c) (V) < (I) < (IV) < (III) < (II)
(d) (I) < (IV) < (III) < (II) < (V)

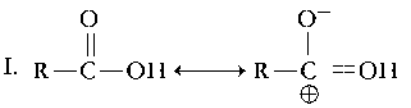
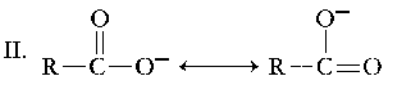
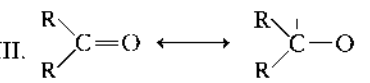
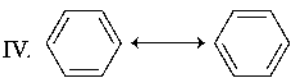
110. Among phosphate, sulphate and chloride of sodium metal the solubility in water increases as
(a) chloride > phosphate > sulphate
(b) sulphate > phosphate > chloride
(c) chloride > sulphate > phosphate
(d) phosphate > chloride > sulphate

111. Bond angle of NH_3, PH_3, AsH_3, and SbH_3 is in the order
(a) $SbH_3 > AsH_3 > PH_3 > NH_3$
(b) $PH_3 > AsH_3 > SbH_3 > NH_3$
(c) $SbH_3 > AsH_3 > NH_3 > PH_3$
(d) $NH_3 > PH_3 > AsH_3 > SbH_3$

112. Shape and hybridization of IF_5, respectively are
(a) Sea-saw, sp^3d
(b) Trigonal bipyramidal, sp^3d
(c) Square pyramidal sp^3d^2
(d) Pentagonal pyramidal, sp^3d^3

113. In terms of polar character, which of the following order is correct?
(a) $NH_3 < H_2O < HF < H_2S$
(b) $H_2S < NH_3 < H_2O < HF$
(c) $H_2O < NH_3 < H_2S < HF$
(d) $HF < H_2O < NH_3 < H_2S$

114. Which among the following pairs represent isovalent resonating forms?

I. 
II. 
III. 
IV. 

(a) I, II, IV (b) II, III, IV
(c) III, IV (d) I, III

115. Consider the following halogen containing compounds:

I. $CHCl_3$ II. CCl_4
III. CH_2Cl_2 IV. CH_3Cl
V. $C_6H_5Cl_2$

The compounds with a net zero dipole moment are:
(a) II and V only
(b) III only
(c) III and IV only
(d) IV only

116. Consider the following molecules or ions:

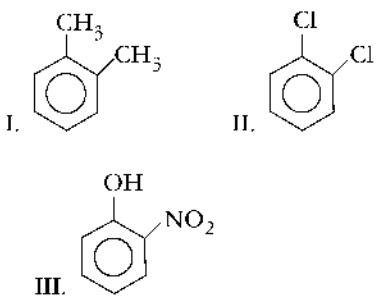
I. CH_2Cl_2 II. NH_4^-
III. SO_4^{2-} IV. ClO_4^-
V. NH_3

sp^3 hybridization is involved in the formation of
(a) I, II, V only
(b) I, II only
(c) I, II, III, IV only
(d) I, II, III, IV, V

117. If the molecule of HCl were totally polar, the expected value of dipole moment is 6.12 D but the experimental value of dipole moment was 1.03 D. Calculate the percentage ionic character.
(a) 17 (b) 83
(c) 50 (d) 90

118. The correct increasing bond angle among BF_3, PF_3 and ClF_3 follows the order
(a) $BF_3 < PF_3 < ClF_3$
(b) $PF_3 < BF_3 < ClF_3$
(c) $ClF_3 < PF_3 < BF_3$
(d) $BF_3 = PF_3 = ClF_3$

119. Correct order of dipole moment in the following structures is



(a) I = II = III (b) I < II < III
(c) II < III < I (d) III < II < I

120. The decreasing order of the boiling points of the following hydrides is

I. NH_3 II. PH_3
III. AsH_3 IV. SbH_3
V. H_2O

(a) (v) > (iv) > (i) > (iii) > (ii)
(b) (v) > (i) > (ii) > (iii) > (iv)
(c) (ii) > (iv) > (iii) > (i) > (v)
(d) (iv) > (iii) > (i) > (ii) > (v)

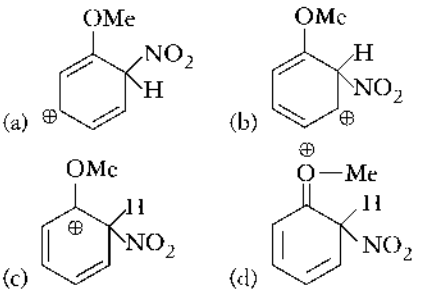
121. Electronic structure of four elements is P, Q, R, S are given as:

P : $1s^2, 2s^2$ Q : $1s^2\ 2s^2\ 2p^2$
R : $1s^2\ 2s^2\ 2p^5$ S : $1s^2\ 2s^2\ 2p^6$

The tendency to form covalent bond is maximum in
(a) P (b) Q
(c) R (d) S

122. The calculated bond order of super oxide ion (O_2^-).
(a) 1.5 (b) 0
(c) 3.5 (d) 2

123. Which of the following canonical forms is the most stable?



124. Among the following, the species having square planar geometry for central atom are

I. XeF_4 II. SF_4
III. $[NiCl_4]^{2-}$ IV. $[PdCl_4]^{2-}$
(a) I and IV (b) I and II
(c) II and III (d) III and IV

125. The lattice enthalpy and hydration enthalpy of four compounds are given as:

Compound	Lattice Enthalpy (in kJmol^{-1})	Hydration enthalpy (in kJ mol^{-1})
P	+ 780	− 920
Q	+ 1012	− 812
R	+ 828	− 878
S	+ 632	− 600

The pair of compounds which is soluble in water is
(a) P and S (b) Q and P
(c) P and R (d) Q and S

126. The formal charges on three oxygen atoms of ozone molecule are, respectively
(a) 0, 0, 0 (b) +1, 0, +1
(c) +1, 0, −1 (d) −1, +1, −1

127. Which of the following is the correct order of electronegativity of Xe in the given compounds?
(a) $XeF_4 < XeO_2F_2 < XeO_3 < XeF_6$
(b) $XeF_4 < XeF_6 < XeO_2F_2 < XeO_3$
(c) $XeF_6 < XeF_4 < XeO_2F_2 < XeO_3$
(d) $XeO_3 < XeO_2F_2 < XeF_4 < XeF_6$

128. Which of the following types of bonds are present in $CuSO_4.5H_2O$?

I. Electrovalent II. Covalent

III. Coordinate

Select the correct answer using the codes given as:

(a) I and II only (b) I and III only

(c) I, II and III (d) II and III only

129. In which of the following set, both the species have pπ-dπ bonding?

(a) CO_3^{2-}, NO_3^- (b) SO_4^{2-}, ClO_3^-

(c) PO_4^{3-}, NO_3^- (d) SO_4^{2-}, CO_3^{2-}

130. Which is the correct order of increasing dipole moment?

(a) ClF < BrCl < NO < HCl < LiI

(b) LiI < BrCl < HCl < ClF < NO

(c) BrCl < NO < HCl < ClF < LiI

(d) NO < BrCl < HCl < LiI < ClF

131. Which of the following species are hypervalent?

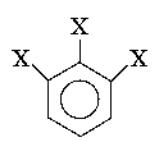
I. ClO_4 II. BF_3

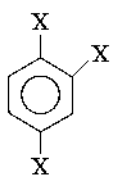
III. SO_4^{2} IV. CO_3^{2}

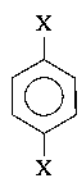
(a) I, II, III (b) I, II

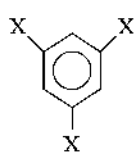
(c) I, III (d) III, IV

132. Which of the following has maximum dipole moment?

(a) 

(b) 

(c) 

(d) 

133. Which of the following order is correct regarding the dipole moment?

(a) $PF_2Cl_3 < PCl_2BrF_2 < PF_3Cl_2$

(b) $PCl_2BrF_2 < PF_3Cl_2 < PF_2Cl_3$

(c) $PF_2Cl_3 < PF_3Cl_2 < PCl_2BrF_2$

(d) $PF_3Cl_2 < PF_2Cl_3 < PCl_2BrF_2$

134. On analysis, a certain compound was found to contain 254 g of (P) and 80 g of (Q). If the atomic weight of (P) is 127 and that of (Q) is 16, then formula of the compound containing (P) and (Q) is

(a) P_2Q_5 (b) P_5Q_2

(c) P_2Q (d) PQ

135. Among N_2, NCl, NCl^+ and NCl^-, the most and the least paramagnetic are, respectively

(a) NCl , N_2 (b) NCl , N_2

(c) NCl , NCl^- (d) NCl, N_2

136. The stability of the species NO_2, NO_2 and NO_2 is in the order

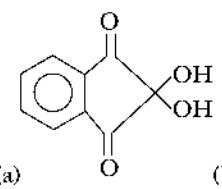
(a) $NO_2^+ > NO_2 > NO_2^-$

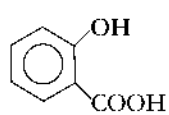
(b) $NO_2 > NO_2 > NO_2$

(c) $NO_2 > NO_2 > NO_2^-$

(d) None of these

137. Intramolecular H-bonding is possible in

(a) 

(b) 

(c) $CCl_3CH(OH)_2$ (d) All of these

138. The interionic distance of K^+ and Cl^- ions in crystalline KCl is 3.14 Å. The screening constant is 10.87 in both ions. Calculate the radius of the K^+ ion.

(a) 2.35 Å (b) 2.70 Å

(c) 1.35 Å (d) 0.675 Å

139. The correct increasing order of dipole moment of the following is

I. cis-pent-2-ene

II. trans-but-2-ene

III. trans-pent-2-ene

IV. p-dichlorobenzene

(a) II < IV < III < I

(b) II = IV < III < I

(c) II = IV < I < III

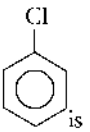
(d) III = II < IV < I

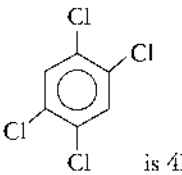
140. The number of sigma and pi-bonds in the compound $(CN)_2C - C(CH_3)(MCO)_3$ is
(a) 16 σ and 10 π bonds
(b) 15 σ and 11 π bonds
(c) 17 σ and 11 π bonds
(d) 16 σ and 11 π bonds

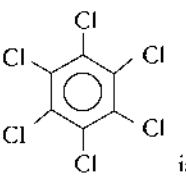
141. Consider the given statements about the molecule: $(H_3C)_2CH - CH - CH - C \equiv C - CH - CH_2$.
I. Three carbon atoms are sp^3 hybridized
II. Three carbon atoms are sp^2 hybridized
III. Two carbon atoms are sp hybridized of three statements:
(a) I and II are correct
(b) I and III are correct
(c) II and III are correct
(d) I, II and III are correct

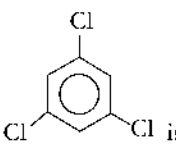
142. The correct decreasing order of bond angles is
(a) $ClF_3 > PF_3 > NF_3 > BF_3$
(b) $BF_3 > PF_3 > NF_3 > ClF_3$
(c) $BF_3 > NF_3 > PF_3 > ClF_3$
(d) $BF_3 > ClF_3 > PF_3 > NF_3$

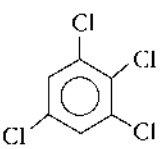
143. The correct order of bond energies for the C-H bond is
(a) $(CH_3)_3C\text{-}H < (CH_3)_2CH\text{-}H < H_3C\text{-}H < H_3C.CH_2\text{-}H$
(b) $H_3C\text{-}H < (CH_3)_3C\text{-}H < H_3C.CH_2\text{-}H < (CH_3)_2CH\text{-}H$
(c) $(CH_3)_3C\text{-}H < (CH_3)_2CH\text{-}H < H_3C.CH_2\text{-}H < H_3C\text{-}H$
(d) $H_3C\text{-}H < H_3C.CH_2\text{-}H < (CH_3)_2CH\text{-}H < (CH_3)_3C\text{-}H$

144. The dipole moment of  is X. The dipole moment of:

(a)  is $4X\sqrt{3}$

(b)  is 3X

(c)  is $6X\sqrt{2}$

(d)  X.

Multiple Correct Answer Type Questions (More Than One Choice)

145. Which of the following statement(s) is/are correct?
(a) The crystal lattice of ice is mostly formed by the covalent as well as hydrogen bonds.
(b) The density of water increases when heated from 0°C to 4°C due to the change in the structure of the cluster of water molecules.
(c) The density of water increases from 0°C to a maximum at 4°C because the entropy of the system increases.
(d) Above 4°C the thermal agitation of water molecules increases. Therefore, intermolecular distance increases and water starts expanding.

146. Which is/are correctly matched?
(a) ClF_3 : T-shape
(b) XeO_3 : Trigonal bipyramidal
(c) XeF_2 : Linear shape
(d) $XeOF_4$: Square planar

147. Which type of chemical bonds are present in N_2O_5?
(a) π-Bond (b) Coordinate bond
(c) Covalent bond (d) Ionic bond

148. Which of the following is/are correct statement(s)?
(a) Dipole moment of CH_3Cl is greater than CH_3F
(b) ICl_2 has a linear shap.
(c) XeF_2 has two lone pair of electrons.
(d) $SnCl_2$ is nonlinear molecule

149. Select the correct statement(s):
(a) $2p_x$ and $2p_y$ orbitals of carbon can be hybridized to yield two new more stable orbitals.
(b) Effective hybridization is not possible with orbitals of widely different energies.
(c) An sp hybrid orbital is not lower in energy than both s- and p- orbitals.
(d) The concept of hybridizarion has a greater significance in the VB theory of localized orbitals than in the MO theory.

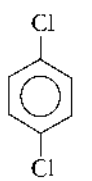
150. CO_2 molecule is isostructural with
(a) $SnCl_2$ (b) $HgCl_2$
(c) NO_2 (d) C_2H_2

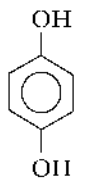
151. Which of the following is/are correct statement(s)?
(a) In $(CH_3)_3N$, C – N – C bond angle is approximately 107° whereas in $(SiH_3)_3N$, Si–N–Si bond angle is approximately 120°.
(b) In dilute solution of H_2CO_3 some C–O bond lengths are equal but in H_2CO_3 (l) all C–O bond lengths are unequal.
(c) CO_2 is a monomer while SiO_2 is a three dimensional giant molecule.
(d) Diamond is the conductor of electricity.

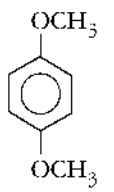
152. The halogens form compound among themselves with the formula AA', AA_3', AA_5' and AA_7', where A is the heavier halogen. Which of the following pairs representing their structures and being polar and non polar are correct?
(a) AA' linear polar
(b) AA_3' T shaped polar
(c) AA_5' – square pyramidal – polar
(d) AA_7' pentagonal bipyramidal non polar

153. Select the incorrect statement(s).
(a) Bond distances are often shortened by ionic–covalent resonance.
(b) Bond order changes linearly with bond length.
(c) Decrease of bond order increases both the bond length and energy.
(d) An increase of the force constant is a consequence of the increase of bond length.

154. The molecule(s) with some finite positive dipole moment is/are:

(a) 

(b) 

(c) 

(d) PF_3Cl_2

155. Bond order decreases in which of the given transitions?
(a) $O_2 \rightarrow O_2^+$ (b) $O_2 \rightarrow O_2^{2-}$
(c) $N_2 \rightarrow N_2^+$ (d) $O_2^- \rightarrow O_2^{2-}$

156. In which case, the hybridization of central atom does not change
(a) NH_3 to NH_4
(b) BCl_3 to BCl_4
(c) SO_2Cl_2 to H_2SO_4
(d) BF_3 to $BF_3.NH_3$

157. Select the correct statement(s).
(a) Acetonitrile has a higher boiling point than HCN.

(b) The hydrogen bond energies of F – H … O, F – H …. F, H – H …. N and N – H … O are all quite similar, being close to the average value which is ~ 7 kcal/mole of a hydrogen bond.

(c) The boiling point of NH_3 is lower than that of $N(CH_3)_3$.

(d) HF has a higher boiling point than CH_3F.

158. Molecule with see-saw shape is
(a) $XeOF_4$ (b) SF_4
(c) XeO_2F_2 (d) XeO_2F_2

159. The pair of species having the same bond order is/are
(a) CN and CO (b) CN and NO
(c) N_2 and NO^+ (d) CN^- and CO

160. The molecule(s) in which the central atom is sp^3d^2 hybridised is/are
(a) XeF_4 (b) XeF_6
(c) SF_6 (d) $I_3^{\oplus}$

161. Which of the following statement is/are correct?
(a) In CCl_4, only dispersive forces exist.
(b) In $CHCl_3$ molecule, both dipole forces as well as dispersive forces exist.
(c) In H_2O molecule, both hydrogen bonds as well as dispersive forces are present
(d) In cyanogen, the number of sigma and pi-bonds are 3 and 4, respectively.

162. Where of these species diamagnetic?
(a) C_2 (b) O_2
(c) O_2^{2+} (d) O_2^{2-}

163. Which of the following is/are incorrect?
(a) The rate of ionic reactions are very slow.
(b) The number of electrons present in the valence shell of S in SF_6 is 12.
(c) According to VSEPR theory, $SnCl_2$ is a linear molecule.
(d) The correct order of stability to form ionic compounds among Na^1, Mg^{21} and Al^3 is $Al^{31} > Mg^2 > Na$.

164. Which compound(s) among the following contain an ionic bond?
(a) NaOH (b) HCl
(c) K_2S (d) LiH

165. Which of the following are characteristics of covalent compounds?
(a) They have low melting and boiling points.
(b) They are formed between two atoms having no or very small electronegativity difference.
(c) They may or may not be insoluble in water.
(d) Their molecules have indefinite geometry.

166. Which of these represents the correct order of the mentioned property?
(a) $PI_3 > PBr_3 > PF_3 > PCl_3$ (Bond angle)
(b) $H_2O > SO_2 > NH_3 > NF_3$ (Dipole moment)
(c) $NO^+ > NO^{2+} > NO > NO^-$ (Bond order)
(d) $O_2 > O_2^+ = O_2^- > O_2^{2-}$ (Paramagnetic nature)

167. The molecule that will have dipole moment is/are
(a) 2, 2-Dimethylpropane
(b) trans – pent – 2 – ene
(c) cis – hex – 3 – ene
(d) 2, 2, 3, 3 – Tetramethylbutane

168. Which of the following statement(s) is/are incorrect?
(a) A sigma bond is weaker than a pi bond.
(b) There are 4 coordinate bonds in the Lewis structure of NH_3^+ ion.
(c) The covalent bond is directional in nature.
(d) pi bond is present in addition to sigma bond.

169. Which of the following is/are nonpolar?
(a) SiF_4 (b) XeF_4
(c) SF_4 (d) BF_3

170. Which among the following are having sp^3d hybridization of the central atom?
(a) XeF_4 (b) ClF_3
(c) BrO_3 (d) XeO_2F_2

171. Which of the following is/are incorrect?
(a) The number of sigma and pi bonds in phenol are 12σ and 3π.
(b) In ClF_3 lone pairs occupy equatorial position.
(c) CO_2 is linear while SO_2 is bent molecule
(d) Intramolecular hydrogen bonding makes removal of H easier.

172. Which of the following order(s) is/are correct here?

(a) Ionic character: $MCl < MCl_2 < MCl_3$

(b) Increasing polarizibility:
$F^- < Cl^- < Br^- < I^-$

(c) Increasing polarizing power:
$Na^1 < Ca^2 < Mg^{21} < Al^3$

(d) Increasing covalent character:
LiF < LiCl < LiBr < LiI

173. Which species among the following have shape similar to sulphate ion?
(a) XeF_4 (b) SF_4
(c) XeO_4 (d) SiF_4

174. Which are the species in which sulphur undergoes sp^3 hybridization?
(a) SF_4 (b) SCl_2
(c) SO_4^{2-} (d) H_2S

175. Which of the following species is/are capable of forming a coordination bond with BF_3?
(a) F^- (b) NH_3
(c) NH_4^1 (d) Ca^{21}

176. Which of the following molecule(s) is/are non-polar?
(a) BF_3 (b) NCl_3
(c) $CHCl_3$ (d) PCl_5

177. Select the incorrect statement(s).
(a) When a covalent bond is formed, transfer of electrons takes place.
(b) Pure H_2O does not contain any ion
(c) A bond is formed when attractive forces overcome repulsive forces.
(d) HF is less polar than HBr.

178. Which one or more among the following involves (s) $p\pi - d\pi$ bonding?
(a) $(SiH_3)_3N$: (b) $(CH_3)_3N$:
(c) $:\overset{\ominus}{C}Cl_3$ (d) $:\overset{\ominus}{C}F_3$

179. In which of these compounds sp^3 hybridization is not used by carbon atom (underlined) for bond formation?
(a) $CH_3.\underline{C}OOH$ (b) $O\underline{C}(NH_2)_2$
(c) $(CH_3)_3\underline{C}.OH$ (d) $C_2H_5.\underline{C}HO$

180. The linear structure is/are assumed by
(a) $SnCl_2$ (b) NCO^-
(c) NO_2^+ (d) CS_2

Linked-Comprehension Type Questions

Comprehension–I

The link formed due to the electrostatic attraction between the positive and negative ions is known as electrovalent bond and compounds having these bonds are called electrovalent or ionic compound. These compounds are characterized by high boiling and melting points, solubility in water, lattice energy etc. The development of covalent nature in these compounds can be expressed by using Fazan's rule.

181. Which is the correct order of increasing ionic nature of following molecules?
I. LiF II. ClF_3
III. K_2O IV. SO_2
(a) II < I < IV < III
(b) II < IV < I < III
(c) II < IV < III < I
(d) IV < II < I < III

182. Which is the correct decreasing order of covalent nature?
(a) $SiCl_4 > AlCl_3 > ZnCl_2 > CaCl_2 > MgCl_2$
(b) $SiCl_4 > AlCl_3 > MgCl_2 > CaCl_2 > ZnCl_2$
(c) $SiCl_4 > AlCl_3 > CaCl_2 > MgCl_2 > ZnCl_2$
(d) $SiCl_4 > AlCl_3 > ZnCl_2 > MgCl_2 > CaCl_2$

183. Which of the following are insoluble in water?
I. AgBr II. $BaSO_4$
III. KOH IV. $Ca_3(PO_4)_2$
(a) I and II (b) I, II and IV
(c) II and IV (d) II, III and IV

184. Which among these does not represent the correct order?
(a) MgO > $MgCl_2$ > NaCl (lattice energy)
(b) $CaF_2 > SrF_2 > BF_2$ (solubility in water)
(c) LiF > NaF > KF (solubility in water)
(d) Both (b) and (c)

Comprehension–2

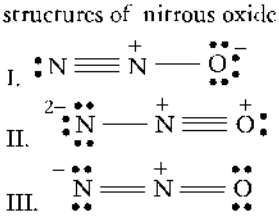
A number of molecules and polyatomic ions cannot be described accurately by a single Lewis structure and a number of descriptions based on the same skeletal structure are written and these taken together represent the molecule or ion. These structures have almost similar energies, same arrangement of atoms and have same number of bonding and non-bonding pair of electrons. These contributing structures or canonical forms taken together constitute the resonance hybrid which represents the molecule or ion.

185. Which of the following is not correct about resonance?
   (a) It averages the bond features as a whole.
   (b) It stabilizes the molecule since energy of the resonance hybrid is lower than that of any of the single canonical structure.
   (c) There is no equilibrium between these canonical structures.
   (d) These canonical structures have real existence also.

186. The value of bond order of CO bond for CO_3^{2-} ion is
   (a) 1.25 (b) 1.33
   (c) 1.5 (d) 1.0

187. Due to resonance in benzene, the value of C-C bond length and bond order are, respectively
   (a) 115 pm and 1.5
   (b) 1.39 Å and 1.33
   (c) 1.39 Å and 1.5
   (d) 115 pm and 2.0

188. The correct order of stability for the resonating structures of nitrous oxide can be given as



   (a) III > II > I (b) I > II > III
   (c) I > III > II (d) I > II = III

Comprehension–3

Pauling introduced the concept of hybridization of atomic orbitals for explaining the characteristic shapes of polyatomic molecules. For example sp, sp^2 and sp^3 hybridizations of atomic orbitals of Be, B, C, N, P and O are used to explain the formation and geometrical shapes of molecules like $BeCl_2$, BF_3, CCl_4, PH_3 and H_2O. They also explain the formation of multiple bonds in molecules like ethyne and ethene.

189. Which is not correct regarding hybridization?
   I Hybrid orbitals are formed when atomic orbitals have comparable energies.
   II. For hybridization atomic orbitals must be fairly apart.
   III. Hybrids always have identical energy and identical shapes.
   IV. The electron waves in hybrid orbitals attract each other.
   (a) I and II (b) II and III
   (c) III and IV (d) I and III

190. Which of there is correctly matched?

Molecule	Hybridization of central atom
I. IF_4^-	sp^3d
II. ICl_4	sp^3d^2
III $BeCl_2$ in solid state	sp^2
(IV) PCl_4^+	dsp^2

   (a) I and II (b) I and III
   (c) II and III (d) I, II and IV

191. In hepta-3-ene and 1-yne, the hybridization state of carbon atom number 2 and 3 are, respectively
   (a) sp^2 and sp, respectively
   (b) sp and sp^2, respectively
   (c) sp only
   (d) sp and sp^3, respectively

192. In which of the following change, the hybridization state of central atom does not change?
   (a) Conversion of $AlCl_3$ into Al_2Cl_6
   (b) PCl_5 into PCl_6^-
   (c) Toluene into benzaldehyde
   (d) NH_3 into NCl_3

Comprehension–4

The VSEPR theory is a simple treatment used for predicting the geometrical shapes of molecules. It is based on the assumption that electron pairs repel each other and, therefore tend to remain as far apart as possible. According to this theory, molecular geometry can be determined by repulsions between lone pairs and lone pairs; lone pairs and bonding pairs and bonding pairs and bonding pairs. The order of these repulsions is as follows:

lp – lp > lp – bp > bp – bp

These repulsion effects result in deviations from idealized shapes and alterations in bond angle in molecules.

193. VSEPR model is remarkably successful in deciding the geometry quite accurately when
   (a) there is no energy difference between possible structures.
   (b) there is very little difference of energy between possible structures.
   (c) there is very large difference of energy between possible structures.
   (d) may be both for (b) and (c)

194. The shape of OSF_4 is
   (a) Square planar
   (b) Trigonal bipyramidal
   (c) See-saw
   (d) Octahedral

195. The shape of $IO(OH)_5$ is supposed to be nearly
   (a) Trigonal bipyramidal
   (b) Square pyramidal
   (c) Octahedral
   (d) Square planar

196. Which is not correctly matched?

Molecule	Shape
(a) $Sb(Ph)_5$	Square pyramidal
(b) $TeCl_6^{2-}$	Octahedral
(c) PCl_6^-	Octahedral
(d) $InCl_4^{2-}$	Square planar

Comprehension–5

Molecular orbital theory describes bonding in terms of the combination and arrangement of atomic orbitals to form molecular orbitals. The electronic configuration of molecules can be written by filling electrons in various MO using Aufbau, Pauli, and Hund's principles. From this configuration a number of properties like bond order, bond length, magnetic nature of diatomic species can be easily explained.

197. σ^*1s has lower energy than $\sigma 2s$ although σ^*1s is antibonding while $\sigma 2s$ is bonding molecular orbital
   (a) σ^*1s is non planar while $\sigma 2s$ is planar
   (b) σ^*1s is more far from $\sigma 2s$ from the nucleus
   (c) σ^*1s is unsymmetrical while $\sigma 2s$ is symmetrical
   (d) σ^*1s originates from lower energy atomic orbital than $\sigma 2s$

198. Find out the correct statements.
   I. Bond length in N_2^- is less than in N_2
   II. NO has more bond length than NO^+
   III. N_2^+ has less dissociation energy than N_2
   IV. O_2 has more dissociation energy than O_2
   (a) I and II (b) II and III
   (c) III and IV (d) II and IV

199. Which of the following have identical bond order?
   I. NO II. O_2^-
   III. CN IV. O_2F_2
   (a) I and III (b) II and III
   (c) I, III and IV (d) II, III and IV

200. Find out the incorrect order.
   (a) $O_2 > O_2^- = O_2^+ > O_2^{2+}$ (magnetic nature)
   (b) $N_2^{2-} > N_2^- > N_2^+ > N_2$ (bond length)
   (c) $CO = N_2 > CO^+ > NO$ (bond order)
   (d) $H_2 > H_2^+ > H_2^-$ (stability)

Assertion and Reasoning Questions

In the following questions two statements (Assertion) (A) and Reason (R) are given. Mark

(a) If both A and R are correct and R is the correct explanation of A
(b) If both A and R are correct but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true
(e) A and R both are false

201. (A): The compound $(CF_3)_3$ N shows almost no basic behaviour even though $(CH_3)_3$ N does.
(R): There is no hydrogen bonding in $(CF_3)_3N$.

202. (A): Water is a good solvent for ionic compounds but poor one for covalent compounds.
(R): Hydration energy of ions released is sufficient to overcome lattice energy and to break hydrogen bonds in water while covalently bonded compounds interact so weakly that even van der Waal's force between molecules of covalent compounds cannot be broken.

203. (A): Dipole moment of cis-pent-2-ene is higher than that of trans-pent-2-ene.
(R): trans-pent-2-ene is more stable than cis-pent-2-ene.

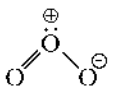
204. (A): All F – S – F angles in SF_4 are greater than 90° but less than 180°.
(R): The lone pair-bond pair repulsion is weaker than bond pair-bond pair repulsion.

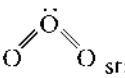
205. (A): In solid state, hybridization of Be atom in $BeCl_2$ is found to be sp^3.
(R): As two chlorine atoms form two coordinate bonds with Be atom which already have two covalent bonds also.

206. (A): Covalent bond in hydrogen molecule is nonpolar.
(R): Most of the covalent bonds are polar.

207. (A): Interelectronic repulsion between bond pair-bond pair, bond pair-lone pair and lone pair-lone pair in a molecule follows the order: bond pair-bond pair < bond-pair-lone pair < lone pair-lone pair.
(R): Bond pair electrons are found in between two nuclei whereas lone pair electrons are attached with only one nucleus and occupy more space.

208. (A): The electronic structure of O_3 is



(R):  structure is not allowed because octet around O cannot be expanded.

209. (A): LiCl is predominantly a covalent compound.
(R): Electronegativity difference between Li and Cl is too small.

210. (A): Ionic compounds tend to be nonvolatile.
(R): Intermolecular forces in these compounds are weak.

211. (A): Bond order in a molecule can assume any value, positive or negative, integral or fractional, including zero.
(R): It depends upon the number of electrons in the bonding and antibonding orbitals.

212. (A): The atoms in a covalent molecule are said to share electrons, yet some covalent molecules are polar.
(R): In polar covalent molecule, the shared electrons spend more time on the average near one of the atoms.

213. (A): Nitrogen is unreactive at room temperature but becomes reactive at elevated temperatures (on heating or in the presence of catalysis).
(R): In nitrogen molecule, there is extensive delocalization of electrons.

214. (A): The solubility of MgF_2 is more than that of $MgCl_2$.
(R): As $MgCl_2$ has less lattice energy than MgF_2.

215. (A): The bond angle of PBr_3 is greater than PH_3 but the bond angle of NBr_3 is less than NH_3.
(R): Electronegativity of phosphorous atom is less than that of nitrogen.

216. (A): Double bonds in C_2 consists of both pi bonds.
(R): The four electrons are present in the 2π molecular orbitals.

217. (A): N_2 and NO^+ both are diamagnetic substances.
(R): NO^+ is isoelectronic with N_2.

218. (A): AlF_3 is a high melting solid while SiF_4 is a gas.
(R): AlF_3 is an ionic compound while SiF_4 is a polar covalent compound.

219. (A): NO_3^- is planar while NH_3 is pyramidal.
(R): N in NO_3^- is sp^2 and in NH_3 it is sp^3 hybridised.

220. (A): Na_2SO_4 is soluble in water while $BaSO_4$ is insoluble.
(R): Lattice energy of $BaSO_4$ exceeds its hydration energy.

221. (A): The energy splitting between $\sigma 2p$ and $\pi 2p$ orbitals is quite large.
(R): The overlap of p-orbitals is more when they are oriented along the axis to form sigma orbitals than when they are oriented to overlap side base to form π-orbitals.

222. (A): Bond order can assume any value including zero.
(R): Higher the bond order, shorter is the bond length and greater is the bond energy.

223. (A): The dipole moment helps to predict whether a molecule is polar or non-polar.
(R): The dipole moment helps of predict the geometry of molecules.

224. (A): p-Hydroxybenzoic acid has lower boiling point than o-hydroxybenzoic acid.
(R): o-Hydroxybenzoic acid has intramolecular hydrogen bonding.

[*IIT 2007*]

225. (A): Boron always forms covalent bond.
(R): The small size of B^{3-} favours formation of covalent bond.

[*IIT 2007*]

Matrix–Match Type Questions

	p	q	r	s
(A)	O	O	O	O
(B)	O	O	O	O
(C)	O	O	O	O
(D)	O	O	O	O

226. Match the following:

Column I	Column II
A. XeF_4	(p) Sea-saw
B. ScF_4	(q) Square planar
C. XeO_3	(r) Pyramidal
D. XeO_4	(s) Tetrahedral
	(t) one lone pair of electron on central atom

227. Match the following:

Column I (Molecule)	Column II (Bond Angle)
A. BF_3	(p) 88°
B. NF_3	(q) 96°
C. PF_3	(r) 103°
D. ClF_3	(s) 120°

228. Match the following:

Column I (Name of the compound)	Column II (Structure/ Geometry)
A. Chlorine trifluoride	(p) Triangular planar
B. Boron trifluoride	(q) Triangular pyramidal
C. Nitrogen trifluoride	(r) T-shaped
D. Sulphur hexafluoride	(s) Regular octahedral
	(t) Central atom with lone pair of electrons

229. Match the following:

Column I	Column II
A. XeF_4	(p) dsp^2
B. H_2O	(q) sp^3

C. PCl_5 (r) sp^3d^2

D. $[Pt(NH_3)_4]^{2+}$ (s) sp^3d

(t) square planar shape

230. Match the following:

Column I	Column II
A. $MgCl_2$	(p) Ionic
B. CaC_2	(q) Covalent
C. $CuSO_4.5H_2O$	(r) Coordinate
D. KHF_2	(s) Hydrogen bond

231. Match the following:

Column I	Column II
A. ICl_2	(p) Linear
B. BrF_2^-	(q) Pyramidal
C. ClF_4^-	(r) Tetrahedral
D. $AlCl_4^-$	(s) Square planar
	(t) Angular

232. March the following:

Column I	Column II
A. N_2	(p) 1.0
B. O_2	(q) 2.0
C. F_2	(r) 2.5
D. O_2	(s) 3.0
	(t) Paramagnetic

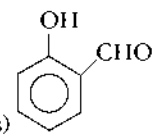
233. Match the following:

Column I	Column II
A. dsp^2	(p) Square planar
B. sp^2	(q) Trigonal
C. d^2sp^3	(r) Octahedral
D. sp^3d	(s) Use of $dx^2 - y^2$
	(t) Trigonal bipyramidal

234. Match the following:

Column I	Column II
A. Zero dipole moment with nonpolar bonds	(p) BF_3
B. Zero dipole moment with polar bonds	(q) O_2
C. Molecule with higher dipole moment	(r) $CHCl_3$
D. Molecule having lower dipole moment	(s) CH_3Cl

235. Match the following:

Column I	Column II
A. Intermolecular H-bonding	(p) Ne
B. Intramolecular H-bonding	(q) NaCl
C. Van der Waal's forces	(r) H_2O
D. Strongest bonding	(s) 
	(t) Chloral hydrate

236. Match column I and column II and choose the correct matching codes from the choices given.

Column I	Column II
A. PCl_5	(p) Pentagonal bipyramidal
B. IF_7	(q) Pyramidal
C. H_3O^+	(r) Trigonal bipyramidal
D. NH_4^+	(s) Terrahedral
	(r) Involvement of d-bital in hybridization

237. Match the following:

Column I (Molecules)	Column II (Bond order)
A. $O_2[SbF_6]$	(p) 1.0
B. RbO_2	(q) 2.0
C. Na_2O_2	(r) 2.5
D. BaO_2	(s) 1.5

238. Match column I and column II and write the correct answer:

Column I (Species)	Column II (Geometry)
A. H_3O^+	(p) Planar
B. $H_2C - NH$	(q) Angular
C. ClO_2	(r) Tetrahedral
D. NH_4^+	(s) Pyramidal

239. Match column I (Molecules) with column II (Boiling points) and select the correct answer.

Column I (Molecules)	Column II (Boiling points)
A. NH_3	(p) 240 K
B. PH_3	(q) 211 K
C. AsH_3	(r) 186 K
D. SbH_3	(s) 264 K

240. Match the compounds in the list I with list II.

Column I	Column II
A. XeO_3	(p) Planar triangular
B. $XeOF_4$	(q) Angular
C. BO_3^{3-}	(r) Trigonal pyramid
D. $SnCl_2$	(s) Square pyramid

The IIT–JEE Corner

241. Among KO_2, AlO_2^-, BaO_2 and NO_2^+, unpaired electron is present in
(a) NO_2^+ and BaO_2 (b) KO_2 and AlO_2^-
(c) KO_2 only (d) BaO_2 only
[*IIT 1997*]

242. Which one of the following compounds has sp^2 hybridization?
(a) CO_2 (b) SO_2
(c) N_2O (d) CO
[*IIT 1997*]

243. Which of these contains both polar and non-polar bonds?
(a) NH_4Cl (b) HCN
(c) H_2O_2 (d) CH_4
[*IIT 1997*]

244. Among the following compounds, the one that is polar and has the central atom with sp^2 hybridization is
(a) H_2CO_3 (b) SiF_4
(c) BF_3 (d) $HClO_2$
[*IIT 1997*]

245. The geometry and the type of hybrid orbitals present about the central atom in BF_3 is
(a) Linear, sp (b) Trigonal planar, sp^2
(c) Tetrahedral, sp^3 (d) Pyramidal, sp^3
[*IIT 1998*]

246. The correct order of increasing C–O bond length of CO, CO_3^{2-}, CO_2 is
(a) $CO_3^{2-} < CO_2 < CO$
(b) $CO_2 < CO_3^{2-} < CO$
(c) $CO < CO_3^{2-} < CO_2$
(d) $CO < CO_2 < CO_3^{2}$
[*IIT 1999*]

247. The hybridization of atomic orbitals of nitrogen in NO_2^+, NO_3^- and NH_4^+ are
(a) sp^2, sp^3 and sp^2, respectively
(b) sp, sp^2 and sp^3, respectively
(c) sp^2, sp and sp^3, respectively
(d) sp^2, sp^3 and sp, respectively
[*IIT 2000*]

248. Amongst H_2O, H_2S, H_2Se and H_2Te, the one with the highest boiling point is
(a) H_2O because of hydrogen bonding.
(b) H_2Te because of higher molecular weight.
(c) H_2S because of hydrogen bonding.
(d) H_2Se because of lower molecular weight.
[*IIT 2000*]

249. The common features among the species CN$^-$, CO and NO$^+$ are
(a) Bond order three and isoelectronic.
(b) Bond order three and weak field ligands.
(c) Bond order two and π-acceptors.
(d) Isoelectronic and weak field ligands.
[*IIT 2001*]

250. The correct order of hybridization of the central atom in the following species NH_3, $PtCl_4^{-2}$, PCl_5 and BCl_3 is
(a) dsp^2, sp^3d, sp^2 and sp^3
(b) sp^3, dsp^2, sp^3d, sp^2
(c) dsp^2, sp^2, sp^3, sp^3d
(d) dsp^2, sp^3, sp^2, sp^3d
[*IIT 2001*]

251. Specify the coordination geometry around and hybridization of N and B atoms in a 1:1 complex of BF_3 and NH_3.
(a) N : tetrahedral, sp^3 ; B : tetrahedral, sp^3
(b) N : pyramidal, sp^3 ; B : pyramidal, sp^3

(c) N : pyramidal, sp^3 ; B : planar, sp^2
(d) N : pyramidal, sp^3 ; B : rerrahedral, sp^3

[*IIT 2002*]

252. The nodal plane in the π–bond of erhene is locared in
(a) rhe molecular plane.
(b) a plane parallel ro rhe molecular plane.
(c) a plane perpendicular ro rhe molecular plane which bisecrs rhe carbon-carbon σ bond ar right angle.
(d) a plane perpendicular ro rhe molecular plane which contains rhe carbon-carbon bond.

[*IIT 2002*]

253. Which of the following are isoelectronic and isostructural?

NO_3^-, CO_3^{-2}, ClO_3^-, SO_3

(a) NO_3^-, CO_3^{-2} (b) SO_3, NO_3^-
(c) ClO_3^-, CO_3^{-2} (d) CO_3^{-2}, SO_3

[*IIT 2003*]

254. Among the following, the molecule with the highest dipole moment is
(a) CH_2Cl_2 (b) CH_3Cl
(c) $CHCl_3$ (d) CCl_4

[*IIT 2003*]

255. The acid having O – O bond is
(a) $H_2S_2O_3$ (b) $H_2S_2O_6$
(c) $H_2S_2O_8$ (d) $H_2S_4O_6$

[*IIT 2003*]

256. Toral number of lone pair of electrons in $XeOF_4$ is
(a) 0 (b) 1
(c) 2 (d) 3

[*IIT 2003*]

257. According to rhe molecular orbital rheory which of rhe following sraremenrs abour rhe magnetic character and bond order is correct regarding O_2 ?
(a) paramagnetic and bond order < O_2
(b) paramagnetic and bond order > O_2
(c) diamagnetic and bond order < O_2
(d) diamagnetic and bond order > O_2

[*IIT 2003*]

258. The compound which has maximum number of lone pairs of electrons on central arom
(a) $[ClO_3]$ (b) XeF_4
(c) SF_4 (d) $[I_3]$

[*IIT 2005*]

259. Among the following, the paramagnetic compound is
(a) O_3 (b) N_2O
(c) Na_2O_2 (d) KO_2

[*IIT 2007*]

260. The speeies having bond order different from rhar in CO is
(a) N_2 (b) CN^-
(c) NO^+ (d) NO^-

[*IIT 2007*]

261. The percentage of p-character in the orbitals forming P – P bonds in P_4 is
(a) 25 (b) 75
(c) 33 (d) 50

[*IIT 2007*]

262. The number of warer molecules directly bonded ro rhe meral centre in $CuSo_4 . 5H_2O$ (Darken rhe correct bubble here)

0	1	2	3	4	5	6	7	8	9

[*IIT 2009*]

263. Match the following:

Column I	Column II
A. B_2P	(p) Paramagnetic
B. N_2	(q) Undergoes oxidation
C. O_2	(r) Undergoes reduction
D. O_2	(s) Bond order ≥ 2
	(t) Mixing of s,p orbitals

[*IIT 2009*]

ANSWERS

Stratght Objective Type Questions

1. (c)	2. (d)	3. (c)	4. (c)
5. (a)	6. (c)	7. (c)	8. (d)
9. (c)	10. (c)	11. (d)	12. (d)
13. (b)	14. (c)	15. (b)	16. (c)
17. (b)	18. (a)	19. (d)	20. (b)
21. (a)	22. (a)	23. (b)	24. (c)
25. (d)	26. (c)	27. (a)	28. (d)
29. (b)	30. (d)	31. (a)	32. (b)
33. (b)	34. (a)	35. (b)	36. (d)
37. (a)	38. (c)	39. (a)	40. (d)
41. (a)	42. (c)	43. (c)	44. (b)
45. (a)	46. (a)	47. (c)	48. (a)
49. (c)	50. (d)	51. (a)	52. (b)
53. (d)	54. (a)	55. (d)	56. (d)
57. (d)	58. (b)	59. (c)	60. (c)
61. (a)	62. (d)	63. (a)	64. (b)
65. (d)	66. (b)	67. (b)	68. (d)
69. (d)	70. (c)	71. (c)	72. (c)
73. (d)	74. (c)	75. (a)	76. (a)
77. (c)	78. (b)	79. (c)	80. (a)
81. (a)	82. (d)		

Brainteasers Objective Type Questions

83. (a)	84. (b)	85. (b)	86. (b)
87. (d)	88. (d)	89. (c)	90. (b)
91. (c)	92. (d)	93. (b)	94. (c)
95. (b)	96. (d)	97. (d)	98. (c)
99. (b)	100. (c)	101. (c)	102. (a)
103. (d)	104. (c)	105. (a)	106. (c)
107. (c)	108. (b)	109. (b)	110. (c)
111. (d)	112. (c)	113. (b)	114. (a)
115. (a)	116. (d)	117. (a)	118. (c)
119. (c)	120. (a)	121. (c)	122. (a)
123. (c)	124. (a)	125. (c)	126. (a)
127. (c)	128. (c)	129. (b)	130. (c)
131. (c)	132. (a)	133. (c)	134. (a)
135. (d)	136. (a)	137. (d)	138. (c)
139. (b)	140. (c)	141. (b)	142. (c)
143. (c)	144. (d)		

Multiple Correct Answer Choice Type Questions

145. (a), (b), (d)	146. (a), (c), ((d))
147. (b), (c)	148. (a), (b), (d)
149. (b), (c), (d)	150. (b), (d)
151. (a), (b), (c)	152. (a), (b), (c), (d)
153. (b), (c), (d)	154. (b), (c), (d)
155. (b), (c), (d)	156. (a), (c)
157. (a), (c), (d)	158. (b), (d)
159. (b), (c), (d)	160. (a), (c)
161. (a), (b), (c), (d)	162. (a), (c), (d)
163. (a), (c), (d)	164. (a), (c), (d)
165. (a), (b), (c)	166. (a), (b), (d)
167. (b), (c)	168. (a), (b), (d)
169. (a), (b), (d)	170. (b), (d)
171. (a), (d)	172. (b), (c) ,(d)
173. (a), (c), (d)	174. (b), (c), (d)
175. (a), (b)	176. (a), (d)
177. (a), (b), (d)	178. (a), (c)
179. (a), (b), (d)	180. (b), (c), (d)

Linked-Comprehension Type Questions

Comprehension–I

181. (b)	182. (d)	183. (b)	184. (d)

Comprehension–2

185. (d) 186. (b) 187. (c) 188. (c)

Comprehension–3

189. (c) 190. (a) 191. (b) 192. (d)

Comprehension–4

193. (b) 194. (b) 195. (c) 196. (d)

Comprehension–5

197. (d) 198. (b) 199. (c) 200. (c)

Assertion and Reasoning Questions

201. (b) 202. (a) 203. (b) 204. (c)
205. (a) 206. (b) 207. (b) 208. (a)
209. (c) 210. (c) 211. (a) 212. (a)
213. (c) 214. (d) 215. (b) 216. (a)
217. (b) 218. (c) 219. (a) 220. (a)
221. (a) 222. (b) 223. (a) 224. (d)
225. (a)

Matrix–Match Type Questions

226. (a)-(q), (b)-(p, r), (c)-(r, r), (d)-(s)
227. (a)-(s), (b)-(r), (c)-(q), (d)-(p)
228. (a)-(r, r), (b)-(p), (c)-(q, r), (d)-(s)
229. (a)-(r, t), (b)-(q), (c)-(s), (d)-(p, t)
230. (a)-(p), (b)-(p, q), (c)-(p, q, r, s), (d)-(p, q, s)
231. (a)-(p), (b)-(r), (c)-(s), (d)-(r)
232. (a)-(s), (b)-(q, t), (c)-(p), (d)-(r, t)
233. (a)-(p, s), (b)-(q), (c)-(r, s), (d)-(t)
234. (a)-(q), (b)-(p), (c)-(s), (d)-(r)
235. (a)-(r), (b)-(s, t), (c)-(p), (d)-(q)
236. (a)-(r, r), (b)-(p, r), (c)-(q), (d)-(s)
237. (a)-(r), (b)-(s), (c)-(p), (d)-(p)
238. (a)-(s), (b)-(p), (c)-(q), (d)-(r)
239. (a)-(p), (b)-(r), (c)-(q), (d)-(s)
240. (a)-(r), (b)-(s), (c)-(p), (d)-(q)

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241. (c) 242. (b) 243. (c) 244. (a)
245. (b) 246. (d) 247. (b) 248. (a)
249. (a) 250. (b) 251. (a) 252. (a)
253. (a) 254. (b) 255. (d) 256. (b)
257. (b) 258. (d) 259. (d) 260. (d)
261. (b) 262. (d) 263. (a)

HINTS AND EXPLANATIONS

Straight Objective Type Questions

1. $KHF_2 \rightarrow K^+ + HF_2^-$

   HF_2^- ion exists due to hydrogen bonding between [H – F .....H].

2. The electronic configuration of O_2^-

   $O_2^- \rightarrow \sigma\ 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \sigma p_z^2\ \pi(2p_x)^2 = \pi 2p_y^2\ \pi^*(2p_x)^2\ \pi^*(2p_y)^1$

   Bond order = ½ [$N_b = N_a$]

   – 1/2 [10 – 7] – 3/2 – 1.5

3. H – Se – H angle in H_2Se is least i.e., 91°.

5. Due to intermolecular hydrogen bond in H_2O, its molecules are associated with each other which is responsible for unusually high boiling point of water.

7. In BF_3, the central atom (boron) has six electrons in the valence shell.

10. Bond angle of H_2O is 104.5°.

11. As smaller the size of cation, higher will be hydration and its effective size will increase hence mobility in aqueous solution will decrease.

12. The dipole moment of NF_3 is less than NH_3.

13. Molecular orbital configuration of peroxide ion, O_2^{2-} is

    $\sigma^2 1s\ \sigma^{*2}1s\ \sigma^2 2s\ \sigma^{*2}2s\ \sigma^2 2p_z\ \pi^2 2p_x\ \pi^2 2p_y\ \pi^{*2}2p_x\ \pi^{*2}2p_y$

    It is diamagnetic.

    Bond order = ½ (10 – 8) = 1.

14. Bond angle of water is 104.5°.

15. $SnCl_2$ has sp^2 hybridization and angular structure. In CS_2, carbon is sp hybridized and is linear. NCO^- and NO_2^+ being isoelectronic with CS_2 have same type of shape.

16. SiF_4 has symmetrical tetrahedral structure so individual bond moments cancel each other resulting in zero dipole moment.

17. The compound of which central atom is octetless is known as electrodeficient compound. Hence, B_2H_6 is electron deficient compound.

19. H – O – N – O. Only covalent bonds are present.

20. The angular shape of ozone molecule consists of 2 sigma and 1 pi bond.

21. BF_3 is sp^2 hybridized. Hence, it is trigonal planner.

23. SF_4 has permanent dipole moment.

26. As difference of electronegativity increases, percentage ionic character increases and covalent character decreases i.e., negativity difference decreases and covalent character increases.

    Na is more +ve than Li and Li is more +ve than Be.

27. Order of strength of hydrogen bonds

    NH---N < OH---O < FH---F

    13 18 40 kJ mol

    As greater the difference in the electronegativity, greater is the strength of the H-bond.

28. As it is sp^3d hybridized with tetrahedral shape.

30. Molecule having almost negligible tendency to form hydrogen bonds is HI as hydrogen bonding depends on two factors:

    (i) Higher electronegativity of X in HX

    (ii) Small size of X

    Electronegativity of I is low and its size is also large. Therefore, both the factors fail here. Hence, no hydrogen bonding is present in HI.

31. Here, hybridization changes from sp^2 to sp^3.

38. As square planar geometry has dsp^2 hybridization.

39. Lewis structure of N_2 molecule is

    $\overset{x}{\underset{x}{N}} \equiv N \overset{x}{\underset{x}{}}$

41. In sp^3d hybridization (one 's' + three 'p' + one dz^2 orbital) the shape of molecule becomes trigonal bipyramid.

43.

Molecule	Hybridization	Repulsion	Bond angle
SO_2	sp^2	lp – bp bp – bp	119°
OH_2	sp^3	lp – lp bp – lp bp – lp	104.5°
SH_2	sp^3	– do –	90°
NH_3	sp^3	lp – bp bp – bp	107°

44. $\mu = \sqrt{(\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos\theta)}$
if $\theta = 90°$, μ is maximum.

45. As bond order for N_2, O_2, He_2, H_2 are 3, 2, 0, 1, respectively

51. Depending upon the nature of ligand, octahedral complex is of two types:
(I) Inner orbital complex (d^2sp^3) → formed under the influence of strong ligand.
(II) Outer orbital complex (sp^3d^2) → formed under the influence of weak ligands.

52. As both are linear in shape.

56. As the s-character increases in hybridized orbitals hence its electronegativity increases.
Hence $sp > sp^2 > sp^3$.

57. In (1) Hydrogen bonding decreases $\theta < 60°$
In (2) dipole-dipole repulsion increases $\theta > 60°$
In (3) θ remains 60°
Hence, the correct order is 2 < 3 < 1.

58. The molecular of water consists of two hydrogen atoms bonded to oxygen atom by covalent bonds. Because of the polar nature of water molecular, the water molecules are held together by intermolecular hydrogen bonds. In this arrangement, each oxygen is tetrahedrally surrounded by four hydrogen atom; two by covalent bond and two hydrogen bonds.

60. As $-C \equiv N -$ two cyanide ions have 4π bonds.

61. Tetrahedral structure is associated with sp^3 hybridized central atom without any lone pair.

63. When a metal is beaten, it does not break but is converted into sheet. It is said to posses the property of malleability. Due to ductile nature, metals can be drawn into wires. These two properties of metals can be accounted due to the capacity of layers of metal ions to slide over one another.

64. Molecular orbital configuration of NO is
$\sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \pi 2px^2\ \pi 2py^2\ \sigma 2pz^2\ \pi^*2px^1$
Bond order = ½ (10 – 5) = 1.5

65. sp^3d^2 hybridization has octahedral structure such that four hybrid orbitals are at 90° w.r.t each other and others hence two at 90° with first four.

67. Since H_2 has no unpaired electron so it is diamagnetic.
$He_2^+ \longrightarrow \sigma(1s)^2\sigma^*(1s)^1$, one unpaired e
$H_2 \longrightarrow \sigma(1s)^2\sigma^*(1s)^0$, no unpaired e
$H_2^+ \longrightarrow \sigma(1s)^1\sigma^*(1s)^0$, no unpaired e^-
$H_2 \longrightarrow \sigma(1s)^2\sigma^*(1s)^1$, one unpaired e .

Due to absence of unpaired electrons, H_2 will be diamagnetic.

70. Lattice energy $\propto$ charge on ion
$\propto$ 1 / size of ion

The value of lattice energy depends on the charges present on the two ions and the distance between them.

71. Ethylene is a planar molecule in which carbon atom is sp^2 hybridized.

72. S_2 molecule is paramagnetic like O_2 as both have two unpaired electrons.

75. O_2 has two unpaired electrons but are paired in O_2^{2-}.

77. Chlorine atom in ClF_3 is sp^3d hybridized. hence bonds are not equal as it has distorted T-shape.

79. Here, SCl_4 is see-saw in shape while rest are tetrahedral.

80. In SF_4 the bonds are not equal as it has See-saw shape and have 2 axial and 2 equatorial bonds.

81.
O – N – O
|
Here N has no lone pair electron.

Brainteasers Objective Type Questions

85. Smaller the size of ions and more the charge, more is the lattice energy.

88. Here number of unpaired electron = 1
Magnetic moment = $\sqrt{n(n+2)}$ B.M.
$= \sqrt{1(1+2)}$
$= \sqrt{3} = 1.73$ B.M.

89. As nitrogen cannot be pentavalent so structure (c) is incorrect.

90. As Q is fluorine and S is sodium hence they form NaF which is an ionic compound.

91. NF_3, H_3O^+ are pyramidal in shape while NO_3^-, BF_3 are planar molecules.

92. I CH_3^+ ; 6 + 3 − 1 = 8 (electrons)

II H_3O^- ; 8 + 3 − 1 = 10

III NH_3 ; 7 + 3 = 10

IV CH_3^- ; 6 + 3 + 1 = 10

Thus, II, III and IV are isoelectronic structures.

93. $He_2^+ < O_2 < NO < C_2^{2-}$
Bond order 0.5 1.5 2.5 3.0

94. Here, A, B, C and D are O, F, Na and Mg, respectively hence the compounds formed by them are C_2A (Na_2O), DA (MgO), CB (NaF) and DB_2 (MgF_2), respectively.

95. Both O_2 and NO are paramagnetic as O_2 contains two unpaired electrons and NO has one unpaired electron.

96. $NO_2^+ > NO_2 > NO_2^-$
132° 130° 115° (Bond angles)

97. Bond length ∝ 1/Bond order
B O of O_2 = 2
B O of O_3 = 1.5
B O of H_2O_2 = 1
So, $O_2 < O_3 < H_2O_2$

98. BF_3 and t-butyl carbonium ion. Both of these have sp^2 hybridization and trigonal planar shape.

100.

Species	lp	bp	VSEPR	Bond angle
H_2S	2	2	lp-lp lp-bp	90°
NH_3	1	3	bp-bp lp-bp bp-bp	107°
BF_3	0	3	bp-bp	120°
SiH_4	0	4	bp-bp	109°28°

Hence, bond angle $H_2S < NH_3 < SiH_4 < BF_3$

102. As bond length ∝ 1 / bond order.
Bond length is inversely proportional to bond-order.

Bond order in NO^+ = 3

NO = 2.5

Hence, bond length in NO > NO

103. As bond strength ∝ bond order.

111. As per VSEPR theory, bond angle decreases with increase in size of the valence shell of the central atom.

112. Count the number of bond pairs and lone-pair around the central atom.

IF_5 : Summation of group number = 42

Bond pairs = $\frac{42}{8}$

= 5 (Residue 2)

Lone pair = $\frac{2}{2}$

= 1

5 bond pairs , 1 lone pair means the geometry is square pyramidal.

117. Per cent ionic character

$$= \frac{\text{obs. dipole moment}}{\text{cal dipole moment}} \times 100$$

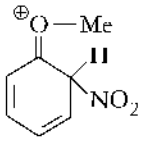
$$= \frac{1.03\ D}{6.12\ D} \times 100 = 17\ \%$$

122. The bond order of superoxide ion O_2^- is

$O_2 \rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \approx \pi 2p_y^2 \pi^* 2p_x^2 \approx \pi^* 2p_y^1$

Bond order = $1/2\ [N_b - N_a]$

= 1/2 [10 − 7] = 1.5

123. 

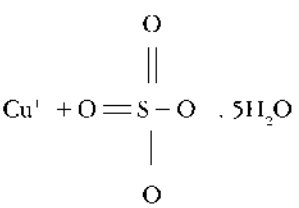
Since in this structure, every atom (except H) has a stable octet of electrons, hence it is the most stable one.

124. In XeF_4: sp^3d^2 hybridization. Shape is square planar instead of octahedral due to presence of two lone pair of electrons on Xe atom.

SF_4: SF_4 molecules shows sp^3d.

125. A compound is soluble in water when its hydration enthalpy is greater than its lattice enthalpy.

128. $CuSO_4.5H_2O$



Here, ionic and covalent bonds are present in $CuSO_4$ while H_2O molecules are attached by coordinate bonds.

132. Here the order of dipole moment will be

(A) > (B) > (C) = (D).

134. Moles of (P) = $\frac{254}{127} = 2$

Moles of (Q) = $\frac{80}{16} = 5$

Hence, the formula is P_2Q_5.

135. As N_2 has no unpaired electron while NCl (isoelectronic with O_2 in terms of valence electrons) has two unpaired electrons.

138. By Pauling's rule, the ionic radius is inversely proportional to the effective nuclear charge. Thus,

$$r_K^- = \frac{\text{const}\,(c)}{19 - 10.87} = \frac{c}{8.13};$$

$$r_{Cl}^- = \frac{c}{17 - 10.87} = \frac{c}{6.13};$$

$$r_K + r_{Cl} = c(1/8.13 + 1/6.13) = 3.14\text{Å}$$

$$r_K / (r_K^+ + r_{Cl}) = \frac{6.13}{6.13 \times 8.13}.$$

$$r_{K^-} = \frac{6.13}{14.26} \times 3.14 = 1.35\ \text{Å}$$

141. This molecule has four carbon atoms (i.e., $-CH - CH-$ and $-CH - CH_2$) sp^2 hybridized.

142. As for them the bond angles are $BF_3 - 120°$, $NF_3 - 106°$, $PF_3 - 101°$, $ClF_3 - 90°$ and 180°.

Multiple Correct Answers Type Questions

148. As XeF_2 has three lone pair of electrons hence it is incorrect.

157. As boiling point of HF is +19.5°C, boiling point of CH_3F – –78°C.

159. As the bond order of these species are

CN = 2.5

N_2 – 3.0

CO = 3.0

CN – 3.0

NO^+ – 3.0

163. As in SF_6 there are 12 electrons in the valence shell of sulphur atom hence it is correct.

166. The bond angle of PF_3 is more than PCl_3 because of $p\pi$-$d\pi$ bonding in PF_3

As bond order of NO^2 and NO is same i.e., 2.5 so the correct order is

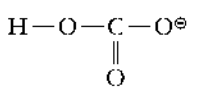
$NO^+ > NO^{2+} - NO > NO$.

169.

SiF_4	sp^3	tetrahedral	non-polar
XeF_4	sp^3d^2	square planar	non-polar
SF_4	sp^3d	square pyramidal	polar
BF_3	sp^2	trigonal planar	non-polar

171. $NaHCO_3 \rightarrow Na^+ + HCO_3^-$

Ionic bond



Covalent bond

180. $SnCl_2$ has sp^2 hybridization and angular structure. In CS_2 carbon is sp hybridized and is linear. NCO and NO_2 being isoelectronic with CS_2 have same type of shape.

Linked-Comprehension Type Questions

Comprehension–1

183. As AgBr, $BaSO_4$ and $Ca_3(PO_4)_2$ have more lattice energy than hydration energy hence they are insoluble in water.

Comprehension–3

189. As hybrids always have identical energy but nonidentical shapes and the electrons waves in hybrid orbitals repel each other.

190. As $BeCl_2$ in solid srare is sp^3 hybridized and PCl_4^+ is also sp^3 hybridized.

Comprehension–5

199. Here rhe bond order of NO^+, CN and O_2F_2 is same i.e., 3, since they are isoelectronic and have same molecular orbiral electronic configuration.

200. As the correct bond order is as follows:

$$CO^+ > CO = N_2 > NO$$

B.O. 3.5 3.0 3.0 2.5

Assertion and Reasoning Questions

203. Dipole momenr of cis-penr-2-ene is more rhan rrans-penr-2-ene as in cis form μ (bond momenr) are additive while in rrans form μ are subtractive.

210. Ionic compounds are nor volarile as they have srrong electrosraric force of arracrion berween rhe ions. This means they have very high boiling poinrs.

217. As both N_2 and NO^+ do not have any unpaired electron hence they are diamagnetic.

218. AlF_3 being ionic is a solid while SiF_4 being a non polar covalent molecule is a gas.

220. Na_2SO_4 is soluble as for ir hydrarion energy is more rhan larrice energy while $BaSO_4$ is insoluble due ro less hydrarion energy than larrice energy.

224. As boiling point will be more in p-hydroxybenzoic acid due to presence of intermolecular hydrogen bonding in it.

225. According ro Fajan's rule small carions having high charge densiry always have tendency ro form covalenr bond.

The IIT–JEE Corner

241. KO_2 is potassium superoxide. Superoxide ion (O_2^{-1}) has one unpaired electron in the antibonding molecular orbital.

242. In SO_2 S-arom is sp^2 hybridized.

243. The O – O bond is non polar while O – H bond is polar.

245. In BF_5, B-arom has sp^2 hybridizarion and molecules have rrigonal bipyramidal geomerry.

246. As bond length $\propto$ 1/bond order

	Bond order
$C^- \equiv O^+$	3
O – C – O	2
CO_3^{-2}	1.33

Since, the bond length increases as the bond order decreases i.e., $CO < CO_2 < CO_3^{-2}$

248. H_2O has highesr boiling poinr due ro H-bonding. The order is $H_2S < H_2Se < H_2Te < H_2O$.

249. All the species are isoelectronic since each one of them has 14 e^- and bond order = 3.

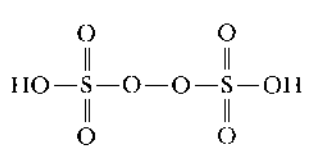
251. In rhe 1 : 1 complex

$H_3N: \rightarrow BF_3$, both N and B have sp^3 hybridized and tetrahedral geometry.

252. The nodal plane in rhe π-bond of erhene is locared in rhe molecular plane.

253. Borh NO_3^- and CO_3^{-2} have same number of elecrrons (32 elecrrons) and sp^2 hybridized cenrral arom (isosrrucrural).

255. Only $H_2S_2O_8$ has O – O bond



257. For O_2

Bond order = $\frac{1}{2}(N_b - N_a) = \frac{1}{2}(10 - 6) = 2$

for O_2^+

Bond order = ½ (10 − 5) = 2.5

Thus, B.O. of O_2^+ > B. O of O_2. Further, as there is one unpaired electron present, hence the ion is paramagnetic.

259. O_2^{2-} – $\sigma 1s^2\ \sigma^*1s^2,\ \sigma 2s^2\ \sigma^*2s^2,\ \sigma 2p_z^2,\ \pi 2p_x^2 - \pi 2p_y^2,\ \pi^*2p_x^2 - \pi^*2p_y^2$

Number of unpaired electrons = 0

$N - N \longrightarrow O$ number of unpaired electrons = 0

$O - O^{\oplus} - O^{\ominus} \leftrightarrow O - O \rightarrow O$

Number of unpaired electrons = 0

O_2^- – $\sigma 1s^2,\ \sigma^*1s^2\ \sigma 2s^2,\ \sigma^*2s^2,\ \sigma 2p_z^2,\ \pi 2p_x^2 - \pi 2p_y^2,\ \pi^*2p_x^2 - \pi^*2p_y^1$

Number of unpaired electrons = 1

So O_2^- is paramagnetic.

260. NO (16 electron system)

Bond order = 2

NO^+, CN^- and N_2 are isoelectronic with CO therefore all have same bond order (i.e., 3)

261. P is sp^3 hybridized in P_4.

Solved Subjective Questions

1. The experimentally determined N–F bond length in NF_3 is greater than the sum of single bond covalent radii of N and F.

[*IIT 1995*]

Solution

The experimentally determined N – F bond length in NF_3 is greater then the sum of single bond covalent radii of N and F. It is due to the smaller sizes of N and F atoms which leads to more repulsion in bonding.

2. $(SiH_3)_3N$ is a weaker base than $(CH_3)_3N$. Why?

[*IIT 1995*]

Solution

It is due to the fact that lone pair of electrons on nitrogen in $(SiH_3)_3N$ are used, in pπ—dπ back bonding by Si atom. On the other hand in $(CH_3)_3N$ such a pπ–dπ bonding is not possible due to absence of vacant d-orbitals in carbon. Hence, $(CH_3)_3N$ is more basic than $(SiH_3)_3N$.

3. In trimethylamine, the nitrogen has a pyramidal geometry whereas in trisilylamine $N(SiH_3)_3$, it has a planar geometry. Why?

Solution

In trisilyamine the pair of electrons occupying the p-orbital of N overlaps with the empty d-orbital on Si which results pπ pπ bonding. Hence, it has a planar structure. Similar pπ pπ bonding is impossible in $(CH_3)_3N$ due to the absence of d orbital in C atom so it has a pyramidal geometry.

4. The dipole moment of NH_3 is more than that of NF_3. Why?

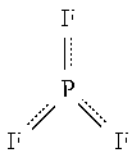
Solution

The dipole moment of NH_3 acts in the directions H → N and thus moment due to unshared pair of electron can naturally increase the moment of the NH_3 molecule on the other hand in the case of NF_3, the dipole moment acts in the direction N → F and thus unshared electron pair can partially neutralize the dipole moment, causing a lower moment of NF_3 w.r.t of NH_3.

5. Explain why PH_3 has less bond angle value than PF_3.

Solution

Although PH_3 and PF_3 are also pyramidal in shape with one lone pair on P-atom, yet PF_3 has greater bond angle than PH_3. It is due to resonance in PF_3 which leads to partial double bond character as shown here.



As a result, repulsions between P F bonds are large and hence the bond angle is large. There is no possibility for the formation of double bonds in PH_3

6. Give reason for the following:
   Ethylene molecules are planar.

Solution

This is because each C-atom involves sp^2 hybridization.

7. Explain the following:
   AlF_3 is a high melting solid whereas SiF_4 is a gas.

Solution

AlF_3 is an ionic solid due to large difference in electronegativities of Al and F whereas of SiF_4 is a covalent compound and hence there are only weak van der Waals forces among their molecules.

8. Explain why CH_3Cl has more dipole moment than CH_3F.

Solution

CH_3Cl has more dipole moment than CH_3F because C–Cl bond length is quite larger than C–F .

9. Give reason for the following:
   The molecule of $MgCl_2$ is linear while that of stannous chloride is angular.

Solution

As electronic configuration of $_{12}Mg$ is $1s^2\ 2s^2\ 2p^6\ 3s^2$ in the ground state. Hence, in the excited state, it is $1s^2\ 2s^2\ 2p^6\ 3s^1\ 3p^1_x$.

It undergoes sp hybridization. Therefore, the shape of $MgCl_2$ is linear.

As electronic configuration of Sn [Kr] $5s^2\ 5p^1_x\ 5p^1_y$.

Hence it undergoes sp^2 hybridization. The two half-filled hybrid orbitals form bonds with Cl atoms while the third is occupied by a lone pair. Hence, $SnCl_2$ is bent or V-shaped.

10. State with reasons:
    Which is more volatile ortho nitrophenol or para nitrophenol?

Solution

Ortho-nitrophenol is more volatile than para nitrophenol because of intramolecular hydrogen bonding or chelation.

11. Hydroxy benzaldehyde is a liquid at room temperature while p-hydroxy benzaldehyde is a high melting solid Why?

[*IIT 1999*]

Solution

As o-Hydroxy benzaldehyde or salycyldehyde shows intramolecular H-bonding or chelation, a weaker one than intermolecular H-bonding in p-hydroxy benzaldehyde.

12. Sodium metal vaporizes on heating and the vapour will have diatomic molecules of sodium (Na_2).
    What type of bonding is present in these molecules? Justify your answer.

Solution

Covalent bonds are present in Na_2. It can be expressed on the basis of molecular orbital theory. The electronic configuration of $_{11}Na = 1s^2\ 2s^2\ 2p^6\ 3s^1$. The 3s atomic orbitals of the two sodium atoms combine to forms σ_{3s} and σ^*_{3s} molecular orbitals. These two electrons enter into σ_{3s}. Hence, bond order is 1. So the two sodium atoms are linked by a linked by a single covalent bond.

13. Indicate the type of bonds present in NH_4NO_3 and state the mode of hybridization of two N atoms.

Solution

In ammonium nitrate all the three types of bonds, that is, ionic, covalent and coordinate bonds are present. In it nitrogen atom of NH_4^+ is sp^3 hybridized while in NO^{3-} it is sp^2 hybridized.

14. The dipole moment of KCl is 30336×10^{29} coulomb metre which indicates that it is a higher polar molecule. The interatomic distance between K and Cl in this molecule is 2.6×10^{10} m. Calculate the dipole moment of KCl molecule, if there were opposite charges of one fundamental unit located at each nucleus, Calculate percentage ionic character of KCl.

Solution

If there were opposite charges of one fundamental unit, i.e.,

$q = 1.602 \times 10^{-19}$ coulombs, then

$\mu = q \times d$

$= (1.602 \times 10^{-19}\text{ coulombs}) \times (2.6 \times 10^{-10}\text{m})$

$= 4.1652 \times 10^{-29}$ coulombs metre

As $\mu_{observed} = 3.336 \times 10^{-29}$ coulomb metre

Hence % ionic character $= \frac{\mu_{observed}}{3.336 \times 10^{-29}} \times 100$

$= \frac{4.1652 \times 10^{-29}}{4.1652 \times 10^{-29}}$

$= 80.09\ \%$

15. Find the dipole moment of HCl molecule if the bond length is 1.2475 (if $D = 3.336 \times 10^{-30}$ CM).

Solution

$\mu = q.r = (1.6023 \times 10^{-19}\text{ C})(1.2476 \times 10^{-10}\text{m})$

$= 1.999 \times 10^{-29}$ C-m

$$\mu = \frac{1.999 \times 10^{-29}\text{ C-m}}{3.336 \times 10^{-30}\text{ C-m-D}^{-1}}$$

$= 5.99D$

16. Explain the difference in the nature of bonding in LiF and LiI.

[*IIT 1996*]

Questions For Self-Assessment

17. Using the VSEPR theory, identify the type of hybridization and draw the structure of OF_2. What are the oxidation states of O and F?

[*IIT 1996*]

18. Interpret the nonlinear shape of H_2S molecule and nonplanar shape of PCl_3 using valence shell electron pair repulsion (VSEPR) theory. (Atomic number H = 1, P = 15, Cl = 17)

[*IIT 1998*]

19. Write the MO electron distribution of O_2. Specify its bond order and magnetic property.

[*IIT 2000*]

20. Using VSEPR theory, draw the shape of PCl_5 and BrF_5.

[*IIT 2003*]

21. Which one is more stable in diethyl ether, anhydrous $AlCl_3$ or hydrous $AlCl_3$? Explain in terms of bonding.

[*IIT 2003*]

22. Draw the shape of XeF_4 and OSF_4 according to VSEPR theory. Show the lone pair of electrons on the central atom.

[*IIT 2003*]

23. On the basis of ground state electronic configuration arrange the following molecules in increasing O O bond length order.

KO_2, O_2, $O_2[AsF_6]$

[*IIT 2003*]

24. Predict whether the following molecules are isostructural or not. Justify your answer.

(i) NMe_3

(ii) $N(SiMe_3)_3$

[*IIT 2005*]

25. What is the effect of the following ionization processes on the bond orders in C_2 and O_2?

26. How does bond energy vary from N^-_2 to N^-_2 and why?

27. What is the hybrid state of $BeCl_2$? What will be the change in the hybrid state of $BeCl_2$ in the solid state?

28. BeF_2 and BF_3 both are stable inspite of contraction of octet rule. Why?

29. PX_5 exists but not PH_5. Explain?

PERIODIC PROPERTIES

2

Chapter Contents

Periodic laws, Type of elements (s, p, d, f), Periodic trend in properties (Atomic and ionic radii, Ionization energy, Electronegativity etc.) and various level multiple-choice questions.

LONG FORM OF PERIODIC TABLE

It is also known as extended form of periodic table or Mosley's periodic table. It was developed by Range, Werner and Burey on the basis of electronic configuration of elements.

Features: The main feature of long form of periodic table are as follows:

(i) It is divided into four blocks i.e., s, p, d and f on the basis of electronic configuration.
(ii) It has seven horizontal rows known as periods.
(iii) Here first period is the shortest period having only two elements (H, He).
(iv) Second and third period are the short periods having eight elements each.
(v) Fourth and fifth periods are the long periods having 18 elements each.
(vi) Sixth period is the longest period having 32 elements.
(vi) Seventh period is incomplete having 26 elements.
(vii) IIIrd B group (IIIrd vertical row) is the largest group having 32 elements including lanthanides and actinides.
(viii) The first element of each group is an alkali metal while the last element is an inert gas.

TYPE OF ELEMENTS

There are four types of known elements

(i) The s-block and p-block elements are called representative elements.
(ii) The d-blocks elements are called transition elements.
(iii) The f-blocks elements are called inner transition elements.

s-Block Elements

(i) The elements having ns^1 and ns^2 electronic configurations in their outermost shell are called s-block elements.
(ii) Elements with ns^1 configuration are called group 1 (alkali elements).
(iii) Elements with ns^2 configuration are called group 2 (alkaline earth elements).
(iv) They are highly reactive and readily form univalent or bivalent positive ions by losing the valence electrons.

(v) The elements of this block are soft, malleable and good conductors of heat and electricity.

(vi) The elements have largest atomic and ionic radii but lowest ionization energies.

(vii) They show fix valency and oxidation states.

(viii) The loss of the outermost electrons(s) occurs readily to form M^{+1} (in case of alkali metals) or M^{+2} ions (in case of alkaline earth elements).

(xi) Except beryllium compounds all other compounds of this block elements are predominantly ionic.

(x) They are soft metals having low melting points and boiling points.

(xi) These metals and their salts impart characteristic colour to the flame.
For example, sodium salt imparts a golden yellow colour to flame.

(xii) The elements of this group have large size, strong reducing nature, high electropositive nature, very low electronegativity values, ionization energy and electron affinity.

p-Block Elements

(i) The elements whose atoms have incomplete p-orbitals in their outermost shell or in which the last electron enters any p-orbital are known as p-block elements.

(ii) The general outer electronic configuration for these elements varies from $ns^2\ np^{1-6}$.

(iii) Elements of groups 13 (IIIrd A), 14 (IVth A), 15 (Vth A), 16 (VIth A), 17 (VIIth A), 18 (VIIIth A) are p-block elements.

(iv) Group 13 (IIIrd A) have one electron in p-orbital whereas group 18 (VIIIth A) (inert gas) have 6 electrons in their outer p-orbitals. The outer p-orbitals in an inert gas are fully-filled with electrons.

Boron	$1s^2\ 2s^2\ 2p^1$
Oxygen	$1s^2\ 2s^2\ 2p^4$
Neon	$1s^2\ 2s^2\ 2p^6$

(v) They include both metals and nonmetals but there is a regular gradation from metallic to nonmetallic character as we move from left to right across the period.

(vi) This block element do not impart colour to the flame test.

(vii) Except F and inert gases all other elements of this block show variable oxidation states.

Gp.	IIIA	IVA	VA	VIA	VIIA
O.S.	+3	+4	+5	+6	+7
		to −4	to −3	to −2	to −1

(viii) They have quite high ionization energies and the values tend to increase as we move from left to right across the period.

(ix) They form covalent compounds mostly like oxides, halides, sulphides, carbonates etc.

(x) Except metals, in the other elements of this block are nonconductors.

(xi) A number of elements of this block show catenation property and allotropy like C, Si, Ge, S, O etc.

(xii) As we move from left to right, there is a gradation from reducing to oxidizing properties.

d-Block Elements

(i) The d-block is in between s and p blocks. These are the elements which have incomplete d-orbital.

(ii) The d-block elements are called transition elements as their properties lies in between s- and p-block elements.

(iii) A transition element is one whose atom or at least one of its ion has incomplete d-orbital or in which the last electron is present in any d-orbital.

(iv) Their general outer electronic configuration is $(n-1)\ d^{1-10}\ ns^{1-2}$

(v) These elements are between 2–13 group in three series of 10 elements each.

(vi) They show variable valency and oxidation state because of the participation of ns- and (n−1)d electrons in their chemical bond formation due to nearly similar energies.

(vii) These are metals with high values of melting points, boiling points, densities, thermal stabilities and hardness etc.

(viii) They are ductile and malleable.

(ix) They good conductors of heat and electricity due to presence of mobile or free electrons.

(x) They form coloured ions and complexes.

(xi) Metals and their ions are generally paramagnetic in nature because of the presence of unpaired electrons.

(xii) These metals form a number of alloys as they have almost similar sizes.

(xiii) These metals and their compounds are widely used as catalysts.

(xiv) These metals also form nonstoichiometric and interstitial compounds with small size atoms like H, C, N, O which can be easily fitted in the vacant sides of the lattices of these metals.

Example, $Fe_{0.93}O$, ZrH_2, WC etc.

f-Blocks Elements

(i) The elements placed in two separate rows at the bottom of the periodic table are f-block elements.

(ii) They have incomplete f-orbitals in their electronic configurations.

(iii) The elements from cerium to lutetium having incomplete 4-f orbitals are lanthanones or lanthanoids.

(iv) The elements from thorium to lawrencium having incomplete 5-f orbitals are actinoids.

(v) In addition to incomplete 5-f orbitals, they also have 1–2 incomplete d-orbitals in their penultimate shells.

(vi) Their general outer configuration is $(n-2)\,f^{1-14}\,(n-1)\,d^{1-2}\,ns^2$.

(vii) Many of actinoid elements have been made only in nanogram quantities or less by nuclear reactions and their chemistry is not fully studied.

(viii) Many of them are synthetic elements.

(ix) The elements coming after uranium are called transuranium elements.

(x) They are metals having high melting and boiling points.

(xi) They show variable oxidation states (variable valency) however their most common and stable oxidation state is +3.

(xii) They form coloured ions and complexes.

(xiii) Actinides are radioactive in nature.

TRENDS IN PERIODIC PROPERTIES OF ELEMENTS

These properties of the elements vary periodically with their atomic numbers when we move from left to right across the period or top to bottom in any group.

Atomic Size or Radius

It is the distance between the centre of nucleus of atom to its outer most shell of electrons i.e., penultimate shell electron.

The absolute value of atomic radius cannot be determined because

- it is not possible to locate the exact position of electrons in an atom as an orbital has no sharp boundaries.
- it is not possible to isolate an individual atom for its size determination due to its small size.
- in a group of atoms, the probability distribution of electrons is influenced by the presence of neighbouring atoms hence size of an atom may change from one environment to another.

The values of atomic radii are derived from bond lengths measured by various techniques such as X-ray diffraction, electron diffraction, infrared spectroscopy, nuclear magnetic resonance spectroscopy etc.

As the absolute value of atomic size cannot be determine so it is expressed in terms of the operational definitions such as ionic radius, covalent radius, van der Waal's radius and metallic radius.

(i) **Covalent Radius:** It is half of the distance between two successive nuclei of two covalent bonded similar atoms in a molecule. It is also called single bond covalent radius.

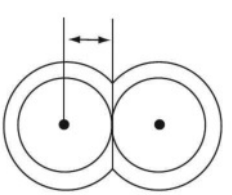


Fig. 2.1

If the bond length in between the two atoms say A – A is 'a', then

Covalent radius (r_{cov})

= ½ [internuclear distance between two covalently bonded similar atoms] = ½ a.

(ii) **van der Waal's Radius ($r_{v.w}$):** It is one half of the distance between the nuclei of two non bonded isolated adjacent atoms belonging to two neighbouring molecules of an element in the solid state.

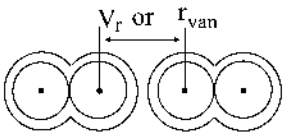


Fig. 2.2

(iii) **Metallic Radius:** It is half of the distance between any two successive nuclei of two adjacent metal atoms in a closely packed crystal lattice.

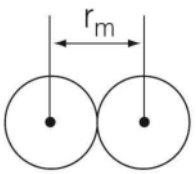


Fig. 2.3

(iv) **Ionic Radius:** It is the effective distance from the centre of the nuclei of an ion up to which it has an influence over electron cloud.

REMEMBER

- van der Waal's radius > metallic radius > anionic radius > covalent radius > cationic radius

Factors Affecting Size

Effective Nuclear Charge and Screening Effect: The force of attraction present between the nucleus and the valence electrons is reduced by the force of repulsion exerted by inner shell electrons on these valence electrons. It is called screening or shielding effect and the net force is called effective nuclear charge. It was shown by Zeff.

$Zeff = Z - \sigma$

$\sigma \propto$ Number of inner shell electrons

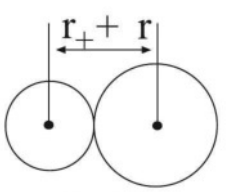


Fig. 2.4

Here Z is atomic number and σ is screening constant.

Size $\propto$ 1/Effective nuclear charge

Size $\propto$ Number of orbits

$\propto$ Number of inner shell electrons (σ)

$\propto$ Electronic repulsion

Size $\propto$ 1/Bond order or multiplicity

Variation in the Value of Radii

In a Period: On the moving from left to right, atomic size decreases as the number of atomic orbitals is the same, while the number of electrons increase as a result effective nuclear charge increases.

IA > IIA > IIIA > IVA > VA ≈ VI > VII < Zero group.

For example,

Li > Be > B > C > N ≈ O > F

Na > Mg > Al > Si > P > S > Cl

In case of zero group, only van der Waal's radius is considered which is always more than the covalent radius.

In a Group: Atomic radius increases from top to bottom as the number of shells or orbitals increase and screening effect increases thus Zeff decreases.

For example,

Li < Na < K < Rb < Cs

F < Cl < Br < I

Size of cation is always smaller than its Atom: As during cation formation outermost orbit is destroyed and number of valence electron decreases thus, Zeff increases and size decreases.

- Size of cation $\propto$ 1/Zeff or magnitude of positive charge
- $M^{+3} < M^{+2} < M^{+} < M$

For example,

$Fe^{+3} < Fe^{+2} < Fe$

Size of Anion is greater than size of its Atom: During anion formation electrons are upraken so Zeff decreases and size increases.

Size of anion $\propto$ magnitude of negative charge

$M^{-4} > M^{-3} > M^{-2} > M^{-} > M$

For example,

$O^{-2} > O > O$

In general, Anion > Atom > Cation

For example,

$X^{-} > X > X^{+}$

In general, for isoelectronic species ionic size decreases as follows:

$M^{-4} > M^{-3} > M^{-2} > M^{-} > M^{+} > M^{+2} > M^{+3} > M^{+4}$

For example,

$C^{-4} > N^{-3} > O^{-2} > F^{-} > Na^{+} > Mg^{-2} > Al^{-3} > Si^{+4} > P^{5-}$

$P^{-3} > S^{-2} > Cl^{-} > K^{+} > Ca^{+2} > Sc^{+3} > Ti^{+4} > Mn^{5+}$

REMEMBER

If the covalent bond present between atoms have different electronegativities, then atomic radius is determined by a formula given by Shoemaker and Stevenson.

$$\Delta A - B = r_A + r_B - 0.09 (X_A - X_B)$$

Here X_A and X_B are electronegativities of A and B, respectively.

Ionization Enthalpy

It is also called ionization energy or ionization potential.

It is defined as The energy required to remove the most loosely bound electron from an isolated atom in the gaseous state resulting in the formation of a positive ion.

$M - 1e^{-} \longrightarrow M^{+} - I_1$

$M^{+} - 1e^{-} \longrightarrow M^{-2} - I_2$

$M^{+2} - 1e \longrightarrow M^{+3} - I_3$

Here, I_1, I_2 and I_3 are the first, second and third ionization enthalpies, respectively.

$$I_1 < I_2 < I_3 < I_4 \longrightarrow$$

The value increases as removal of an electron becomes more and more difficult

Unit: eV/atom or kcal/mole or kJ/mole.

Factors Affecting Ionization Energy

(i) Ionization energy ∝ Effective nuclear charge (Zeff).

(ii) Ionization energy ∝ 1/Atomic size.

(iii) Ionization energy ∝ 1/Screening effect of the inner electrons.
Due to more screening effect, Zeff decreases and removal of valence electron becomes easier.

(iv) Ionization energy ∝ stable electronic configuration.
As completely filled or half-filled orbitals, electronic configurations are stable thus removal of electron is difficult hence more ionization energy is needed.

(v) Ionization energy ∝ 1/Penetration effect of the electrons
For example, more closer the orbital to the nucleus more will be the value of ionization energy due to greater force of attraction between electrons and nucleus.
s > p > d > f

Variation in value of Ionization Energy

In period: On moving from left to right in a period the ionization energy increases as Zeff increases and as size decreases hence the removal of the electron becomes more and more difficult.

Order for First Ionization Energy in any Period:

s^1 s^2 p^1 p^2 p^3 p^4 p^5 p^6

IA < IIA > IIIA < IVA < VA > VIA < VIIA < VIIIA

Maximum

In case of IIA, first ionization energy is more than IIIA as in IIA, ns^2 (fulfilled state) is present but in IIIA np^1 (incomplete) is less stable state.

In case of VA, first ionization energy is more than VIA as in VA ns^2 np^3 (half-filled, more stable state) is present but in VIA ns^2 np^4 (incomplete, less stable state) is present.

In any period an inert gas has maximum value of first ionization energy due to most stable outer state.

The largest jump in between I_1 and I_2 is in case of alkali metals as ns^1 configurations changes into inert gas configuration (ns^2 np^6).

For example,

I_1: Li < Be > B < C < N > O < F < Ne

Na < Mg > Al < Si < P > S < Cl < Ar

I_2: Li > Ne > O > F > N > B > C > Be

Na > Ar > Cl > S > P > Al > Si > Mg

In Group: On moving top to bottom in a group ionization energy decreases as Zeff decreases and size increases so removal of electron becomes more and more easy.

For example,

He > Ne > Ar > Kr > Xe

F > Cl > Br > I

Li > Na > K > Rb > Cs

Period	IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA
I	H							He
	1312							2372
II	Li	Be	B	C	N	O	F	Ne
	520	899	801	1086	1402	1314	1681	2081
III	Na	Mg	Al	Si	P	S	Cl	Ar
	496	737.6	577	786	1011	999	1255	1520
IV	K	Ca	Ga	Ge	As	Se	Br	Kr
	419	590	579	760	946	941	1142	1350
V	Rb	Sr	In	Sn	Sb	Te	I	Xe

	403	549	558	708	884	869	1009	1170
VI	Cs	Ba	Ti	Pb	Bi	Po	At	Rn
	374	502	589	715	703	813	917	1037

REMEMBER

- In periodic table, helium has maximum first ionization energy (1500 eV) while cesium has lowest value.
- The largest jump between I_2 and I_3 is for alkaline earth metals as configuration changes from ns^1 to inert gas configuration ($ns^2\ np^6$).

$$\underset{1s^2\ 2s^2}{Be} \xrightarrow[I_1]{-e^-} \underset{2s^1}{Be^+} \xrightarrow[I_2]{-e^-} \underset{1s^2}{Be^{2+}} \xrightarrow[I_3]{-e^-}$$

Importance of Ionization Energy

The elements having low ionization energies are reducing agents, basic in nature, form cations, ionic compounds and show maximum photoelectric effect.

For example, K, Cs shows maximum photoelectric effect and hence is used in photoelectric cells.

Electron Gain Enthalpy or Electron Affinity (EA)

It is the amount of energy released when a neutral isolated gaseous atom accepts an extra electron to form a gaseous anion.

$$M + 1e^- \longrightarrow M^- + E_1$$

$$M^- + 1e \longrightarrow M^{2-} + E_2$$

$$M^{2-} + 1e \longrightarrow M^{3-} + E_3$$

Here, E_1, E_2 and E_3 are the first, second and third electron gain enthalpy, respectively.

REMEMBER

$E_1 > E_2 > E_3$

The value of electron affinity decreases as the addition of electron becomes more and more difficult and is possible only by absorbing some part of energy. For example, E_2 becomes endothermic in comparison to E_2. (E_1 is exoergonic and E_2 is endoergonic)

For example:

$$O + e^- \xrightarrow{+e^-} O^- \xrightarrow{+e^-} O^{2-}$$

$$E_1 > > E_2$$

Unit: eV/atom or kcal/mole or kJ/mole.

The electron affinity cannot be determined directly, however it is obtained indirectly from Born–Haber cycle.

Factors Affecting Electron Affinity

(i) Electron affinity $\propto$ 1/Atomic size
(ii) Electron affinity $\propto$ Effective nuclear charge
(iii) Electron affinity $\propto$ 1/Screening effect
(iv) Electron affinity $\propto$ 1/Stable electronic configuration

The electron affinity of inert gases is zero this is because of $ns^2\ np^6$ configuration in valency shell hence there is no possibility of adding extra electrons.

Electron affinity of Mg, Be is practically zero due to extra stability of completely filled s-orbitals in them.

If an atom has half-filled orbits, its electron affinity will be very low (in case of Vth group).

Variation in Value of Electron Affinity In Period

On moving from left to right in a period, electron affinity increases as Zeff increases and size decreases.

In general electron affinity follows the following trend:

Halogens > Oxygen family > Carbon family > Nitrogen family > Metals of group I and XIII > Metals of group II > Zero group.

The sequence of electron affinity in IInd period is as follows:

Be < N < Li < B < C < O < F

The sequence of electron affinity in third period is as follows:

Mg < Na < Al < P < Si < S < Cl

In Group:

On moving down the group, electron affinity decreases as Zeff decreases and size increases.

Exceptional Values of Electron Affinity:

EA of F < EA of Cl
EA of C < EA of Si
EA of N < EA of P
EA of O < EA of S

Here, in case of II period elements, electron affinity is less due to their small size and electronic repulsion is more as a result addition of electron becomes difficult.

Period↓ Group →	1	2	13	14	15	16	17	18
1	H							He
	−73							+48
2	Li	Be	B	C	N	O	F	Ne
	−60	+66	−83	−122	+31	−141	−328	+116
3	Na	Mg	Al	Si	P	S	Cl	Ar
	−53	+67	−50	−119	−74	−200	−349	+96
4	K	Ca	Ga	Ge	As	Se	Br	Kr
	−48	–	−36	−116	−77	−195	−325	+96
5	Rb	Sr	In	Sn	Sb	Te	I	Xe
	−47	–	−29	−120	−101	−190	−295	+77
6	Cs	Ba	Tl	Pb	Bi	Po	At	Rn
	−46	–	−30	−101	−110	−174	−270	+68

REMEMBER

- The sequence of electron affinity of VII group is as follows: I < Br < F < Cl
- Oxidizing power of element ∝ Electron affinity
- Reactivity of non metals ∝ Electron affinity

Electronegativity

According to Pauling (1931), it is the power or tendency of an atom in a molecule to attract the shared pair of electrons towards itself.

He considered the reaction of the type

$$\tfrac{1}{2}(A - A) + \tfrac{1}{2}(B - B) \rightarrow A - B$$

The bond dissociation energy of A – B is higher than the mean of bond dissociation energies of (A – A) and (B – B) bonds and their difference, Δ is related to the difference in electronegativities of A and B, respectively.

$$\Delta = E_{A-B} - (E_{A-A} + E_{B-B})^{1/2}$$
$$= 23 (X_A - X_B)^2$$
$$\Delta = 23 (X_A - X_B)^2$$
$$0.208 \sqrt{\Delta} = (X_A - X_B)$$

Here, E_{A-B}, E_{A-A} and E_{B-B} represent bond dissociation energies of A – B, A – A and B – B in kcal, respectively. X_A and X_B are electronegativities of A and B, respectively.

Pauling took geometrical mean instead of arithmetic mean of E_{A-A} and E_{B-B} and introduced the following empirical relation:

$$E_{A-B} - (E_{A-A} \times E_{B-B})^{1/2} = \Delta$$
$$= 30 (X_A - X_B)$$
$$0.182 \sqrt{\Delta} = X_A - X_B$$

- If energies are taken in kJ, the relation is given as:

$$0.088 \sqrt{\Delta} = X_A - X_B$$

For example,

- If we want to calculate electronegativity of fluorine, the bond energies (kJ mole^{-1}) of hydrogen, fluorine and hydrogen fluoride are 436, 153 and 565, respectively.

$$\Delta = E_{H-F} - [E_{H-H} \times E_{F-F}]^{1/2}$$
$$= 565 - [436 \times 153]^{1/2} = 306.7$$
$$X_F - X_H = 0.088 \sqrt{\Delta}$$
$$X_F - 2.2 = 0.088 \sqrt{306.7}$$

On calculating, $X_F = 4$

Allred and Rochow's Method

They proposed following empirical relation for calculating the electronegativity value:

$$X = -0.359 \times \frac{Zeff}{r^2} + 0.744$$

Here, X is the electronegativity and 'r' is covalent radius of the atom.

For example, in case of aluminium

$$Z_{eff} = 13 - 9.5 = 3.5$$

r = 1.25 Å

$$\text{So, } X_{Cl} = -0.359 \times \frac{3.5}{(1.25)^2} + 0.744 = 1.54.$$

Mulliken's Method

$$X = \frac{[IE + EA]}{2}$$

Here, IE = Ionization energy in eV

EA = Electron affinity in eV

When these are in kJ/mol replace 2 by 540.

For example,

$$X_{Cl} = \frac{1251 + 349}{540} \text{ cal} = 2.94.$$

Factors Affecting Electronegativity

(i) Electronegativity ∝ Zeff
∝ 1/Size

(ii) Ionization Energy and Electron Affinity:
Electronegativity ∝ Ionization energy
∝ Electron affinity

(iii) **Charge on Atom:** The cation will be more electronegative than parent atom, which in turn will be more electronegative than its anion. Higher the positive charge (oxidation state) greater will be its electronegativity.

For example, $Fe^{+3} > Fe^{+2}$

(iv) **Effect of Substitution:** The electronegativity of an atom depends upon the nature of substituent attached to that atom.

For example, carbon atom in CF_3I acquires greater positive charge than in CH_3I, therefore C atom in CF_3I is more electronegative than in CH_3I.

(v) The difference in electronegativity of an atom caused by substituents results in different chemical behaviour of that atom.

(vi) Electronegativity ∝ s percentage

So, $sp > sp^2 > sp^3$

For example,

$C \equiv C^- > C = C^- > C - C^-$
↓
more electronegative

Variation in Electronegativity Value

In Period

On moving from left to right in a period the electronegativity increases as Zeff increases and size decreases.

Order for first electronegativity in any Period

IA < IIA > IIIA < IVA < VA > VIA < VIIA
Min. Max.

For example,

Li < Be < B < C < N < O < F
Na < Mg < Al < Si < P < S < Cl

In any period halogens have maximum value of electronegativity while alkali metals have lowest of electronegativity.

Electronegativity of zero group elements is zero. Since, they have stable octet state they have no tendency to attract electrons.

In Group:

On moving from top to bottom in a group electronegativity decreases as Zeff decreases and size increases.

For example,
F > Cl > Br > I
O > S > Se > Te
N > P > As > Sb

Group →	1 IA	2 IIA	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA
I Period	H						
	2.1						
II Period	Li	Be	B	C	N	O	F
	1.0	1.5	2.0	2.5	3.0	3.5	4.0
III Period	Na	Mg	Al	Si	P	S	Cl
	0.9	1.2	1.5	1.8	2.1	2.5	3.0
IV Period	K	Ca	Ga	Ge	As	Se	Br
	0.8	1.0	1.6	1.8	2.0	2.4	2.8
V Period	Rb	Sr	In	Sn	Sb	Te	I
	0.8	1.0	1.7	1.8	1.9	2.01	2.5
VI Period	Cs	Ba	Tl	Pb	Bi	Po	At
	0.7	0.9	1.8	1.9	1.9	1.76	2.2

Decreases ↓
→ Increases

REMEMBER

Decreasing order of electronegativity
- F > O > N > Cl > C > B
- 4 3.5 3 2.97 2.5 2
- Almost all metalloids have nearly 2 value of electronegativity.

Applications of Electronegativity

(1) **Calculation of partial ionic character in a covalent bond:** It depends upon two factors:

(i) The electronegativity difference between two bonded atoms.

(ii) Dipole moment of the compound.
Hannay and Smyth Equation:

Ionic character %
$= 16 (X_A - X_B) + 3.5 (X_A - X_B)^2$

Ionic character %
$= 1 - e^{-1/4} \cdot (X_A - X_B)^2$

When electronegativity difference is greater than 1.7 the compound will be ionic in nature.

CsF is most ionic due to maximum electronegativity difference (3.3).

(2) **Bond Strength:** It is directly proportional to the electronegativity difference between the bonded atoms.

For example, HF > HCl > HBr > HI

(3) **Bond Angle:** Bond angle ∝ Electronegativity of central atom

In case of hydrides of Vth and VIth group bond angle decreases down the group as the electronegativity of central atom decreases the electron pair shift towards hydrogen atom as a result bp-lp repulsion decreases.

For example,

NH_3	>	PH_3	>	AsH_3	>	SbH_3
107°		93°		91.8°		91.3°

Nature of Oxides and Hydroxides

On moving from left to right in a period basic nature decreases while acidic nature increases.

For example,

Na_2O, MgO, Al_2O_3, SiO_2, P_2O_5, SO_2, Cl_2O_7

Most Basic — Most acidic

$NaOH > Mg(OH)_2 > Al(OH)_3 > Si(OH)_4$

Most basic — Most acidic

On moving down the group, basic nature increases. For example, In IA group Cs_2O and CsOH will be most basic (also in periodic table).

Nature of Oxyacids

In a period, the strength of the oxyacids formed by nonmetals increases from left to right.

II Period

H_3BO_3 H_2CO_3 HNO_3 → Strength increases

III Period

H_2SiO_3 H_3PO_4 H_2SO_4 $HClO_4$ → Strength increases

In a group, the strength of the oxyacids of nonmetals decreases.

V Group

HNO_3 H_3PO_4 H_3AsO_4 → Strength increases

VII Group

$HClO_4$ $HBrO_4$ HIO_4 → Strength increases

Nature of Hydrides

The nature of the hydrides changes from basic to acidic on moving from left to right in a period.

NH_3	H_2O	HF
Weak base	Neutral	Weak acid
PH_3	H_2S	HCl
Very weak base	Weak acid	Strong acid

Atomic Volume: It is the volume occupied by one mole atoms of the element in solid state at its melting point. It is obtained by dividing the gram atomic mass with density of the element.

In a period on moving left to right first it decreases to a minimum value and then start increasing.

In group, it normally increases down the group.

Boiling Point, Melting Point and Density: Boiling point, melting point and density increase to a maximum value and then decrease while on moving down the group, it increases.

Hydration Energy (ΔH_{hy}) and Lattice Energy (U)

ΔH_{hy} or U ∝ Charge on ion
∝ 1/Size of ion

i.e., these increase left to right in a period and decrease down the group.

For example,

$Li < Be^{2+} < B^{3+}$

$Li > Na > K^+$

Ionic Mobility

Ionic mobility ∝ 1/Charge on ion
∝ Size

i.e., it deceases left to right in a period and increases down the group.

For example,
$Li^+ > Be^{2-} > B^{3+}$
$Li^+ < Na^+ < K^- < Rb^- < Cs^+$

UNFORGETTABLE GUIDELINES

Diagonal Relationship

Certain II period elements show some similarities with the III period elements which are diagonal to them. It is called diagonal relationship. It is due to the similar ionic sizes, electronegativities and polarizing power.

$$\text{Polarizing power} = \frac{\text{Ionic charge}}{(\text{Ionic radius})^2}$$

I	II	III	IV
Li	Be	B	C
Na	Mg	Al	Si

- Diagonal relationship does not occur after IV group.
- Example of diagonal relationship: Both Li_2CO_3 and $MgCO_3$ on heating form their oxides and CO_2.

$$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$$

$$MgCO_3 \xrightarrow{\Delta} MgO + CO_2$$

- BeO and Al_2O_3 are amphoteric oxides.
- Carbides of Be and Al on hydrolysis give methane gas.

$$Al_4C_3 + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_4$$

$$Be_2C + 2H_2O \longrightarrow 2\,BeO + CH_4$$

- When the difference between ionization energy and electron affinity is high a covalent bond is formed.
- H, Be, Al, Ga, In, Tl, Sn, Pb, Sb, Bi forms amphoteric oxides.
- Fe, CO and Ni have nearly same Zeff value hence have almost same size and ionization energy etc.
- Properties increasing on moving from left to right in a period and decreasing from top to bottom in a group are nonmetallic nature, electronegative nature, oxidizing nature, electronegativity, ionization energy, electron affinity, lattice energy and hydration energy.
- Properties decreasing on moving from left to right in a period and increasing from top to bottom in a group are metallic nature, electropositive nature, reducing nature, basic nature, radius or size, and ionic mobility etc.

Nomenclature of Elements with Atomic Number >100

The IUPAC proposed a system for naming elements with $Z > 100$. By using these rules as follows.

- The names are derived by using roots for the three digits in the atomic number of the element and adding the ending–ium. The roots for the numbers are:

0	1	2	3	4	5	6	7	8	9
nil	un	bi	tri	quad	pent	hex	sept	oct	enn

- In some cases the names are shortened; bi ium and tri ium are shortened to bium and trium, and enn nil is shortened to ennil.
- The symbol for the element is made from the first letters from the roots which make up the name. The strange mixture of Latin and Greek roots has been chosen to ensure that the smbols are all different.

IUPAC nomenclature for the superheavy elements

Atomic No.	Name	Symbol	Atomic No.	Name	Symbol
101	un-nil-unium	Unu	113	un-un-trium	Uut
102	un-nil-bium	Unb	114	un-un-quadium	Uuq
103	un-nil-trium	Unt	115	un-un-pentium	Uup
104	un-nil-quadium	Unq	116	un-un-hexium	Uuh
105	un-nil-pentium	Unp	117	un-un-siptium	Uus
106	un-nil-hexium	Unh	118	un-un-octium	Uuo
107	un-nil-septium	Uns	119	un-un-ennium	Uue
108	un-nil-octium	Uno	120	un-bi-nilium	Ubn
109	un-nil-ennium	Une	130	un-tri-nillium	Utn
110	un-un-nilium	Uun	140	un-quad-nilium	Uqn
111	un-un-unium	Uuu	150	un-pent-nilium	Upn
112	un-un-bium	Uub			

Straight Objective Type Questions (Single Choice)

1. A section of periodic table is given below with elements A, B and X, Y in two groups. Which of the bonds is least polar?
   (a) AY (b) AX
   (c) BY (d) BX

2. The formation of O^{2-} (g) starting from O (g) is endothermic by 603 kJ mol^{-1}. If electron affinity of O (g) is −141 kJ mol^{-1}, the second electron affinity of oxygen would be
   (a) +744 kJ mol^{-1} (b) −744 kJ mol^{-1}
   (c) +462 kJ mol^{-1} (d) −462 kJ mol^{-1}

3. The sum of first three ionization energies of Al is 53.0 eV atom $^{-1}$ and the sum of first two ionization energies of Na is 52.2 eV atom $^{-1}$. Out of Al(III) and Na(II)
   (a) Al (III) is more stable than Na (II)
   (b) Na (II) is more stable than Al (III)
   (c) Both are equally unstable
   (d) Both are equally stable

4. The correct order of first ionization potential is
   (a) F > He > Mg > N > O
   (b) He > F > N > O > Mg
   (c) He > O > F > N > Mg
   (d) N > F > He > O > Mg

5. The element ununoctium belongs to
   (a) p-block (b) noble gases
   (c) d-block (d) s-block

6. The correct sequence of the ionic radii of the following is
   (a) $I^- > S^{2-} > Cl^- > O^{2-} > F^-$
   (b) $S^{2-} > I^- > O^{2-} > Cl^- > F^-$
   (c) $I^- > Cl^- > S^{2-} > O^{2-} > F^-$
   (d) $I^- > S^{2-} > Cl^- > F^- > O^{2-}$

7. Which of the following arrangement shows the correct order of increasing stability?
   (a) $N^{2+} < As^{2+} < Sb^{2+} < Bi^{2+}$
   (b) $Zn^{2+} < Cu^{2+} < As^{2+} < Bi^{2+}$
   (c) $Cu^{2+} < Co^{2+} < P^{2+} < N^{2+}$
   (d) $C^{2+} < Ge^{2+} < Sn^{2+} < Pb^{2+}$

8. The first, second, third and fourth ionization energies of a given element are 0.80, 2.43, 3.66 and 25.03 MJ mol^{-1}, respectively. The element is
   (a) Boron (b) Carbon
   (c) Aluminium (d) Nitrogen

9. Which of the following arrangements show the correct order of increasing lattice energy?
   (a) BaSO4 < SrSO4 < CaSO4 < MgSO4
   (b) MgCO3 < CaCO3 < SrCO3 < BaCO3
   (c) LiF < LiCl < LiBr < LiI
   (d) NaF < KF < RbF < CsF

10. Which one of the following arrangements does not truly represent the property indicated against it?
    (a) Br2 < Cl2 < F2: electronegativity
    (b) Br2 < F2 < Cl2: electron affinity
    (c) Br2 < Cl2 < F2: bond energy
    (d) $Br_2 < Cl_2 < F_2$: oxidizing power

11. The ions O^{2-}, F^-, Na^+, Mg^{2+} and Al^{3+} are isoelectronic. Their ionic radii show
    (a) an increase from O^{2-} to F^- and then decrease from Na^+ to Al^{3+}.
    (b) an decrease from O^{2-} to F^- and then increase from Na^+ to Al^{3+}.
    (c) a significant increase from O^{2-} to Al^{3+}.
    (d) a significant decrease from O^{2-} to Al^{3+}.

12. Which of the following statements are correct?
    (a) HF is a stronger acid than HCl.
    (b) Among halide ions, iodide is the most powerful reducing agent.
    (c) Fluorine is the only halogen that does not show a variable oxidation state.
    (d) HOCl is a stronger acid than HOBr.
    (a) 2 and 4 (b) 2 and 3
    (c) 1, 2 and 3 (d) 2, 3 and 4

13. Identify the correct order in which the ionic radius of the following ions increases:
    (I) F (II) Na
    (III) Na^{3+}
    (a) III, I, II (b) I, II, III
    (c) II, III, I (d) II, I, III

14. Four successive members of the first row transition elements are listed with their atomic numbers. Which one of them is expected to have the highest third ionization energy?
(a) iron ($Z = 26$)
(b) vanadium ($Z = 23$)
(c) manganese ($Z = 25$)
(d) chromium ($Z = 24$)

15. Identify the correct order in which the covalent radius of the following elements increases
(I) Ti (II) Ca
(III) Sc
(a) (I) (II) (III) (b) (II) (I) (III)
(c) (I) (III) (II) (d) (III) (II) (I)

16. Which one of the following arrangements represents the correct order of electron gain enthalpy (with negative sign) of the given atomic species?
(a) S < O < Cl < F (b) Cl < F < S < O
(c) F < Cl < O < S (d) O < S < F < Cl

17. The electronic affinity values (in kJ mol^{-1}) of three halogens X, Y and Z are, respectively −349, −333 and −325. Then X, Y and Z respectively are
(a) F_2, Cl_2 and Br_2 (b) Cl_2, F_2 and Br_2
(c) Cl_2, Br_2 and F_2 (d) Br_2, Cl_2 and F_2

18. Which one of the following orders is not in accordance with the property stated against it?
(a) $F_2 > Cl_2 > Br_2 > I_2$; electronegativity
(b) $F_2 > Cl_2 > Br_2 > I_2$; bond dissociation energy
(c) $F_2 > Cl_2 > Br_2 > I_2$; oxidizing power
(d) HI > HBr > HCl > HF; acidic property in water

19. A sudden large jump between the values of second and third ionization energies of elements would be associated with which of the following electronic configuration?
(a) $1s^2\ 2s^2\ 2p^6\ 3s^1$
(b) $1s^2\ 2s^2\ 2p^6\ 3s^2$
(c) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
(d) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

20. Atomic radii of fluorine and neon in Angstrom units are, respectively given by
(a) 1.60, 1.60 (b) 0.72, 0.72
(c) 0.72, 1.60 (d) 1.60, 0.72

21. Amongst the following elements (whose electronic configurations are given), the correct increasing order of ionization energy is
(I) [Ne] $3s^2\ 3p^1$ (II) [Ne] $3s^2\ 3p^3$
(III) [Ne] $3s^2\ 3p^2$ (IV) [Ar] $3d^{10}\ 4s^2\ 4p^3$
(a) III < I < II < IV (b) IV < II < III < I
(c) I < III < IV < II (d) II < IV < I < III

22. The statement that is not correct for periodic classification of elements is
(a) The properties of elements are a periodic function of their atomic numbers.
(b) Nonmetallic elements are less in number than metallic elements.
(c) The first ionization energies of elements along a period do not vary in a regular manner with increase in atomic number.
(d) For transition elements, the d-subshells are filled with electrons monotonically with increase in atomic number.

23. Sodium forms Na^+ ion but it does not form Na^{2+} because
(a) very low value of $(IE)_1$ and $(IE)_2$.
(b) very high value of $(IE)_1$ and $(IE)_2$.
(c) low value of $(IE)_1$ and low value of $(IE)_2$.
(d) low value of $(IE)_1$ and high value of $(IE)_2$.

24. Correct order of (IE) among the elements Be, B, C, N, O is
(a) B < Be < C < O < N
(b) B < Be < C < N < O
(c) Be < B < C < N < O
(d) Be < B < O < N < C

25. Which of the following order is wrong?
(a) $NH_3 < PH_3 < AsH_3$ acidic
(b) Li < Be < B < C — $(IE)_1$
(c) $Al_2O_3 > MgO < Na_2O < K_2O$ — Basic
(d) $Li^+ < Na^+ < K^+ < Cs^+$ — ionic radius

26. Amongst the following elements (whose electronic configurations are given), the one having the highest ionization energy is
(a) [Ne] $3s^2\ 3p^1$ (b) [Ne] $3s^2\ 3p^3$
(c) [Ne] $3s^2\ 3p^2$ (d) [Ar] $3d^{10}\ 4s^2\ 4p^3$

27. The outermost electronic configuration of the most electronegative element is
(a) $ns^2\ np^5$ (b) $ns^2\ np^4$
(c) $ns^2\ np^3$ (d) $ns^2\ np^6$

28. Which one of the following is the smallest in size?
(a) Na^+ (b) F^-
(c) N^{3-} (d) O^{2-}

29. Which one of the following is the strongest base?
(a) AsH_3 (b) PH_3
(c) NH_3 (d) SbH_3

30. Electronegativity of an element is 1.0 on the Pauling scale. The value on Mulliken scale is
(a) 1.4 (b) 2.8
(c) 3.6 (d) 2.0

31. Which of the following is paramagnetic?
(a) CN (b) O_2
(c) NO^+ (d) CO

32. Which of these is the most basic oxide?
(a) FeO (b) CuO
(c) SnO_2 (d) K_2O

33. The first ionization potential of Na, Mg, Al and Si are in the order
(a) Na > Mg > Al < Si
(b) Na > Mg > Al > Si
(c) Na < Mg < Al < Si
(d) Na < Mg > Al < Si

34. The electronegativity of the following elements increase in the order
(a) Si, P, C, N (b) N, Si, C, P
(c) P, Si, N, C (d) C, N, Si, P

35. Ionic radii are
(a) directly proportional to square of effective nuclear charge.
(b) inversely proportional to square of effective nuclear charge.
(c) directly proportional to effective nuclear charge.
(d) inversely proportional to effective nuclear charge.

36. Atomic radii of fluorine and neon in Angstrom units are, respectively given by
(a) 1.60, 1.60 (b) 0.72, 0.72
(c) 0.72, 1.60 (d) none of these

37. The element with the highest first ionization potential is
(a) nitrogen (b) oxygen
(c) boron (d) carbon

38. The correct order of second ionization potential of carbon, nitrogen, oxygen and fluorine is
(a) O > N > F > C (b) O > F > N > C
(c) F > O > N > C (d) C > N > O > F

39. The electronic configuration of elements A, B and C are [He] $2s^1$, [Ne] $3s^1$ and [Ar] $4s^1$ respectively. Which one of the following order is correct for the first ionization potentials (in kJ mol^{-1}) of A, B and C?
(a) A > B > C (b) C > B > A
(c) B > C > A (d) C > A > B

40. The order of first ionization energies of the elements Li, Be, B, Na is
(a) Be > Li > B > Na
(b) B > Be > Li > Na
(c) Na > Li > B > Be
(d) Be > B > Li > Na

Brainteasers Objective Type Questions (Single Choice)

41. A sudden large jump between the values of second and third ionization energies of elements would be associated with which of the following electronic configuration?
(a) $1s^2\ 2s^2\ 2p^6\ 3s^1$
(b) $1s^2\ 2s^2\ 2p^6\ 3s^2$
(c) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
(d) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

42. The formation of the oxide ion O^{2-} (g) requires first an exothermic and then an endothermic step as shown here:
$O(g) + e \rightarrow O(g);\ \Delta H^\circ = -142\ kJ\ mol^{-1}$
$O^-(g) + e^- \longrightarrow O^{2-}(g);\ \Delta H^\circ = 844\ kJ\ mol^{-1}$
This is because
(a) oxygen is more electronegative.
(b) oxygen has high electron affinity.

(c) O^- ion has comparatively larger size than oxygen atom.

(d) O^- ion will tend to resist the addition of another electron.

43. In which of the following arrangements, the order is according to the property indicated against it?

(I) I < Br < F < Cl
Increasing electron gain enthalpy

(II) Li < Na < K < Rb
Increasing metallic radius

(III) B < C < N < O
Increasing first ionization energy

(IV) $Al^{3-} < Mg^{2+} < Na^- < F^-$
Increasing ionic size

(a) I, II and III (b) I and III only
(c) I, II and IV (d) I and II only

44. Among Al_2O_3, SiO_2, P_2O_5 and SO_2, the correct order of acid strength is

(a) $Al_2O_3 < SiO_2 < P_2O_5 < SO_2$
(b) $SO_2 < P_2O_5 < SiO_2 < Al_2O_3$
(c) $SiO_2 < SO_2 < Al_2O_3 < P_2O_5$
(d) $Al_2O_3 < SiO_2 < SO_2 < P_2O_5$

45. The amount of energy released when 10^6 atoms of iodine in vapour state is converted into I^- ion is 4.8×10^{-13} J. What is the electron affinity of iodine in kJ/mole?

(a) 489 kJ (b) 289 kJ
(c) 259 kJ (d) 389 kJ

46. The electron affinities of N, O, S and Cl are

(a) O = Cl < N = S
(b) O < S < Cl < N
(c) N < O < S < Cl
(d) O < N < Cl < S

47. An element (X) which occurs in the first short period has an outer electronic structure s^2p^1. What is the formula and acid–base character of its oxides?

(a) XO_3, basic (b) X_2O_3, acidic
(c) X_2O_5, basic (d) XO_2 acidic

48. In the descending order of a group in modern periodic table which of this following would be true?

(I) All the atoms have the same number of valence electrons.

(II) Gram atomic volume increases.

(III) Electronegativity decreases.

(IV) Metallic character decreases and the basic nature of their oxides decrease.

Select the correct answer by using the following codes:

(a) I, II and III (b) II, III and IV
(c) II and III (d) I and III

49. The electronic configuration of elements A, B and C are [He] $2s^1$, [Ne] $3s^1$ and [Ar] $4s^1$, respectively. Which one of the following order is correct for the first ionization potentials (in kJ mol^{-1}) of A, B and C?

(a) A > B > C (b) C > B > A
(c) B > C > A (d) C > A > B

50. If for an element (P), the value of successive ionization energies I_1, I_2, I_3, I_4 and I_5 are 800, 2427, 3658, 25024 and 32824 kJ/mole respectively, then the number of valence electron present are

(a) 5 (b) 4
(c) 3 (d) 2

51. Fluorine is a better oxidizing agent than iodine. The most probable reason is

(a) Fluorine is less stable than iodine.
(b) Fluorine has smaller atomic radius than iodine.
(c) Fluorine is more reactive than iodine.
(d) F^- ions have greater electron attracting power than I^- ions.

52. The electronic configurations of four elements are given here. Arrange these elements in the correct order of the magnitude (without sign) of their electron affinity.

(I) $2s^2\,2p^5$ (II) $3s^2\,3p^5$
(III) $2s^2\,2p^4$ (IV) $3s^2\,3p^4$

Select the correct answer using the codes given under:

(a) III < IV < II < I
(b) III < IV < I < II
(c) I < II < IV < III
(d) II < I < IV < III

53. The atomic numbers of V, Cr, Mn and Fe are, respectively 23, 24, 25 and 26. Which one of

these may be expected to have the highest second ionization enthalpy?

(a) V (b) Cr

(c) Mn (d) FE

54. If an element (X) shows two oxidation states +2 and +3 and forms an oxide in such a way that ratio of the element showing +2 and +3 state is 1 : 3 in compound. The formula of this oxide is

(a) X_4O_{11} (b) X_8O_{11}

(c) X_5O_{11} (d) A_9O_{11}

55. Three elements X, Y and Z are present in the III[rd] short period and their oxides are ionic, amphoteric and giant molecule n, respectively. The correct order of the atomic number of X, Y and Z in the order:

(a) Z < Y < X (b) Y < Z < X

(c) X < Z < Y (d) X < Y < Z

56. The density of aluminium is 2.70 gcm^{-3}. Its atomic weight is 26.98. Using the data, calculate the radius of the aluminum atom approximately assuming, it to be spherical.

(a) 2.582 Å (b) 1.356 Å

(c) 1.583 Å (d) 3.158 Å

57. The electronic configuration of the atom having maximum difference in first and second ionization energies is

(a) $1s^2, 2s^2, 2p^1$ (b) $1s^2, 2s^2, 2p^6, 3s^1$

(c) $1s^2, 2s^2, 2p^5$ (d) $1s^2, 2s^2, 2p^6, 2s^2$

58. In which one of the following pairs, the radius of the second species is greater than that of the first?

(a) O^{2-}, N^{3-} (b) Na, Mg

(c) Al, Be (d) Li^-, Be^{2+}

59. Which of the following statement is correct?

(a) The first ionization energy of magnesium is less than that of sodium and aluminium.

(b) The second ionization energy of helium is very nearly double the ionization energy of hydrogen.

(c) From Li to Ne, the first ionization energy values exhibit a smooth increase.

(d) The second ionization energy of magnesium is less than the corresponding values for sodium and aluminium.

60. The electron affinity of chlorine is 3.7 eV. How much energy in kcal is released when 4 g of chlorine is completely converted into Cl$^-$ ion in a gaseous state. (Given: 1 eV = 23.06 kcal mol^{-1})

(a) 9.6 kcal (b) 19.6 kcal

(c) 4.8 kcal (d) 11.6 kcal

61. The difference between atomic radii is maximum in which one of the following pairs.

(a) Rb, Cs (b) K, Rb

(c) Na, K (d) Li, Na

62. The ionization energies (in eV) of a certain element, A are given as under:

I	II	III	IV	V	VI
11.3	24.4	47.9	64.5	392	489.8

The element will form

(a) very few compounds (being an inert element).

(b) both ionic and covalent compounds.

(c) mostly ionic compounds.

(d) mostly covalent compounds.

63. Element X has ionization energies: 7.6, 15.0, 80.1, 109.3, ... (in eV). Element Y has corresponding values: 8.1, 16.3, 33.5, 45.1, 166.7, 205.1, 264.4, ... (in eV). Assuming that one can have a plausible surmise from these values, what would be the likely formula for a compound that may be formed from X and Y?

(a) X_2Y (b) XY

(c) XY_2 (d) X_2Y_3

64. The first and second ionization energies of Mn are in the ratio 0.475 : 1. If the sum of the two energy values is 2226 kJ/mol, calculate the second ionization energy (in kJ/mol).

(a) 1095 (b) 1960

(c) 1069 (d) 1509

65. The second ionization energies of the C, N, O and F atoms are such that

(a) O > N > F > C (b) F > O > N > C

(c) C > O > N > F (d) O > F > N > C

Multiple Correct Answer Type Questions (One or More Than One Choice)

66. Choose the correct statement.
(a) Ce, Gd, U are lanthanoid.
(b) Cu, Ag, Au are known as coinage metal.
(c) Li although the first member of alkali metal but it is strongest reducing agent.
(d) Reducing character decreases down the group.

67. Which of the following represent the incorrect order of ionization energies?
(a) F > N > O > C (b) F > O > N > C
(c) I > Br > F > Cl (d) F > Cl > Br > I

68. Choose the correct statement/s.
(a) $1s^2$ element belongs to p-block.
(b) [Xe] $4f^{14}$ $5d^1$ $6s^2$ element belong to f-block.
(c) [Ar] $3d^5$ $4s^1$ element belong to s-block.
(d) [Ar] $3d^{10}$ $4s^2$ $4d^6$ element is noble gas.

69. Choose the correct order.
(a) $Be^{2+} < Li^+ < Ca^{2-} < K^+$ (ionic radii).
(b) Sulphur has highest electron affinity among chalcogens.
(c) Cl has highest negative electron gain enthalpy.
(d) F is second most electronegative element.

70. Which of the following is/are correct order regarding radius?
(a) $Al^{3+} < Mg^{2+} < Na^+$
(b) $B^{3+} < Ga^{3+} < Al^{3+}$
(c) $Be^{2+} < B^{3} < Li^+$
(d) $Cl^- < S^{2-} < P^{3-}$

71. Which of the following element are artificially made and do not exist in nature?
(a) Bi (b) Ge
(c) Tc (d) At

72. Which of the following pairs contain metalloid?
(a) In, Tl (b) Ge, Ga
(c) As, Te (d) I, Bi

73. Which of the following statement is/are correct?
(a) Fluorine has the highest electron affinity.
(b) Helium has the highest ionization energy.
(c) Alkali metals are the strongest oxidizing agents.
(d) Carbon has the highest melting point.

74. Choose the pair in which IE_1 of first element is greater than IE_1 of second element but in case of IE_2 order is reversed.
(a) P, S (b) F, O
(c) Mg, Al (d) N, O

75. Which of the following pair of oxides are neutral?
(a) Al_2O_3 and B_2O_3
(b) MnO and Mn_2O_7
(c) CO and H_2O
(d) NO and N_2O

76. In halogens, which of the following property increases from iodine to fluorine?
(a) Electronegativity
(b) Bond length
(c) Reducing power
(d) Ionization energy of the element

77. Select the process which is/are endothermic here.
(a) $H + e^- \longrightarrow H^-$
(b) $O^- + e^- \longrightarrow O^{2-}$
(c) $Ar - e \longrightarrow Ar^+$
(d) $X \longrightarrow X + e$

78. Which of the following sets of ions are isoelectronic?
(a) Li^+, Be^{2+}, Be^{3+} (b) Na , Mg^2 , Al^{3+}
(c) P^{3-}, S^{2-}, K^+ (d) Cl^-, Br^-, I^-

79. Which of the following sequence contains atomic number of only representative elements?
(a) 13, 33, 54, 83 (b) 55, 12, 48, 53
(c) 22, 33, 55, 66 (d) 3, 33, 53, 87

80. Which of the following pairs has elements that belong to the same period?
(a) Ca and Zn (b) Mg and As
(c) Ca and Ar (d) Na and Cl

81. Choose the correct order:
(a) Reducing strength of the element depends on the magnitude of the ionization energy.
(b) Mo (IV) > Mo (III) > Mo (II) (Electronegativity order)

(c) Mo II > Mo III > Mo IV
(Electronegativity order)
(d) Fe I < Fe II < Fe III
(Electronegativity order)

82. In which of the following, orders of electron affinity of elements or ions shown here is/are correct?
(a) $O^- > S^-$ (b) $N^- > P$
(c) S > O (d) O > S

83. The correct statement about d-block element is
(a) Ionic radii increases in the series.
(b) All the transition metal ions are coloured.
(c) They are all metals.
(d) They show variable valency.

84. Which of the following is the correct order in accordance to the electropositive nature of metals?
(a) Mg < Ca < Sr
(b) K > Ca > Ga
(c) Fe > Cu > Zn
(d) Fe < Mg < Zn

85. Choose the correct order.
(a) Si < P < Cl < S (2nd ionization energy)
(b) F < O < N < C (2nd ionization energy)
(c) Na and Mg are typical element
(d) $Be^{2+} < Li^+ < Na < K^+$
(Mobility of ion in aqueous solution)

86. Which of the following pairs have nearly same size?
(a) Zr, Hf (b) Nb, Ta
(c) Fe, Zn (d) K, Rb

87. The electronic configurations of four elements are given as under:
(I) $1s^2\ 2s^2\ 2p^5$ (II) $1s^2\ 2s^2\ 2p^4$
(III) $1s^2\ 2s^2\ 2p^3$ (IV) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
Which of the following arrangements gives the correct order in terms of increasing electronegativity of the elements?
(a) III < II < IV < I
(b) II > III > I > IV
(c) IV < III < II < I
(d) I < II < III < IV

88. Which of the following statement is/are incorrect?
(a) All the elements belonging to s-block are metals.
(b) Group 18 elements are known as acrogens.
(c) Halogens are strong reducing agents.
(d) All the elements belonging to d-block are metals.

89. Which of the following is/are correct?
(a) IE_2 (Mg) < IE_2 (Na)
(b) IE_1 (Na) > IE_1 (Mg)
(c) IE_4 (Na) > IE_4 (Mg)
(d) IE_3 (Mg) > IE_3 (Na)

90. The successive ionization energy values for an element X are given below
(I) 1st ionization energy = 410 kJ mol^{-1}
(II) 2nd ionization energy = 820 kJ mol^{-1}
(III) 3rd ionization energy = 1100 kJ mol^{-1}
(IV) 4th ionization energy = 1500 kJ mol^{-1}
(V) 5th ionization energy = 3200 kJ mol^{-1}
Find the number of valence electron for the atom, X.
(a) (IV) (b) (III)
(c) (V) (d) (II)

91. Which of the following trends of atomic size is/are incorrect?
(a) Rb > K > Na > Li
(b) F > O > N > C
(c) Ne > He > Ar > Kr
(d) Be > B > C > N

92. The electronic configuration of four elements are
(I) [Xe] $6s^1$ (II) [Xe] $4f^{14}5d^1\ 6s^2$
(III) [Ar] $4s^2\ 4p^5$ (IV) [Ar] $3d^7\ 4s^2$
Which one of the following statements about these elements is not correct?
(a) (I) is a strong reducing agent.
(b) (II) is a d-block element.
(c) (III) has high electron affinity.
(d) (IV) shows variable oxidation state.

93. Which of the following orders are correct for the property indicated in brackets?
(a) Cl > S > O > N (electron affinity)
(b) Si > Mg > Al > Na (first ionization enthalpy)
(c) $NH_3 > NF_3 > BF_3$ (dipole moment)
(d) $HClO_4 > HBrO_4 > HIO_4$ (pKa values)

94. Which of the following statements is/are not true about the diagonal relationship of Be and Al?

(I) Both react with NaOH to liberate hydrogen.
(II) Their oxides are basic.
(III) They are made passive by nitric acid.
(IV) Their carbides give acetylene on treatment with water.

(a) Only (I) (b) (II) and (IV)
(c) Only (IV) (d) (I) and (IV)

95. Pick the statement(s) which is (are) not true about the diagonal relationship of Li and Mg.
(I) Polarizing powers of Li and Mg^{2+} are almost the same.
(II) Like Li, Mg decomposes water very fast.
(III) LiCl and $MgCl_2$ are deliquescent
(IV) Like Li, Mg readily reacts with liquid bromine at ordinary temperature.

(a) (II) and (IV) (b) (II) and (III)
(c) Only (II) (d) Only (I)

96. The correct statement among the following is
(a) The first ionization potential of Al is less than the first ionization potential of Mg.
(b) The second ionization potential of Mg is greater than the second ionization potential of Na.
(c) The first ionization potential of Na is less than the first ioniation potential of Mg.
(d) The third ionization potential of Mg is greater than the third ionization potential of Al.

97. The statement which is not correct for periodic classification of elements is
(a) The properties of elements are a periodic function of their atomic numbers.
(b) Nonmetallic elements are less in number than metallic elements.
(c) The first ionization energies of elements along a period do not vary in a regular manner with increase in atomic number.
(d) For transition elements, the d-subshells are filled with electrons monotonically with increase in atomic number.

Linked-Comprehension Type Questions

COMPREHENSION–1

Ionization potential is the minimum amount of energy needed to remove the outermost electron from a gaseous isolated atom. Unit for ionization potential is eV/atom or kJ/mole or kcal/mole.

Successive ionization energy is the amount of energy needed to remove electron successively from a gaseous ion may be termed as IE_2, IE_3, IE_4 etc. The difference in the values of IE_1, IE_2 and IE_3 helps to determine electronic configuration of the elements.

Element	IE_1	IE_2	IE_3 (kcal/mol)
P	300	550	920
Q	98	735	1100
R	118	1090	1650
S	496	946	500

98. Which element belongs to group one?
(a) P (b) Q
(c) R (d) S

99. Which element is a noble gas?
(a) S (b) R
(c) P (d) Q

100. Which of these is a nonmetal?
(a) P (b) Q
(c) R (d) S

101. Which element forms srable unipositive ion?
(a) P (b) Q
(c) R (d) S

COMPREHENSION–2

According to Mulliken the electronegativity of an atom can be defined as the arithmetic mean of its ionization energy and electron affinity.

$X_A = \frac{1}{2}$ (I.P. + E.A.)

One more relationship can also be derived if the values are given in eV as follows:

$$X_A = \frac{\text{Ionization potential} - \text{Electron affinity}}{5.6}$$

In case of a pure covalent bond between X – Y

$$\frac{(IA)_X - (EA)_X}{5.6} = \frac{(IP)_Y - (EA)_Y}{5.6} X_X = X_Y$$

102. According to Mulliken, the electronegativity depends on
(a) Electron affinity
(b) Ionization potential
(c) Electron gain enthalpy
(d) Both (a) and (b)

103. Calculation of electronegativity from the 2nd equation given in passage is related to Pauling scale by
(a) 2.8 times lower than the E.N. than Pauling scale.
(b) 2.8 times higher than the E.N. of Pauling value.
(c) Equal to E.N. of Pauling value.
(d) No such a relation is possible.

104. For the formation of $X^- - Y^+$ bond the condition will be

(a) $\frac{(IP)_X - (EA)_Y}{5.6} = \frac{(IP)_X - (EA)_Y}{5.6}$

(b) $\frac{(IP)_X - (EA)_Y}{2.8} = \frac{(IP)_X - (EA)_Y}{2.8}$

(c) $\frac{(IP)_Y - (EA)_Y}{5.6} > \frac{(IP)_X - (EA)_Y}{5.6}$

(d) $\frac{(IP)_X - (EA)_X}{5.6} > \frac{(IP)_Y - (EA)_Y}{5.6}$

COMPREHENSION–3

In the long form of periodic table the elements have been arranged considering their electronic configurations. If we consider elements A, B, C, D and E which have the following electronic configurations:

(a) $1s^2\ 2s^2\ 2p^1$

(b) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

(c) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

(d) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

(e) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$

105. Which of the following elements belong to same group in the periodic table?
(a) C and D (b) A and B
(c) C and D (d) A and E

106. The correct order of atomic radii for these elements will be
(a) D > E > C > B > A
(b) E > D > A > C > B
(c) E > B > C > D > A
(d) E > D > C > B > A

107. The correct order of electron affinity of these elements can be given in increasing order as
(a) E < A < B < C < D
(b) E < B < C < A < D
(c) E < C < B < A < D
(d) E < C < B < A < D

108. The correct order of first ionization energy of these elements can be given in increasing order as
(a) D < A < B < C < E
(b) B < E < C < A < D
(c) B < C < E < A < D
(d) B < E < A < C < D

Assertion and Reasoning Questions

In the following questions two statements (Assertion) A and Reason (R) are given. Mark
(a) If A and R both are correct and R is the correct explanation of A
(b) If A and R both are correct but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

109. (A): Bond dissociation energy of F_2 is less than Cl_2.
(R): Due to smaller size of fluorine there is greater electron repulsion between the F atoms than Cl atoms.

110. (A): Ions K^+, S^{2-}, Sc^{3+} are isoelectronic.
(R): In each ion, the total number of electrons are 18.

111. (A): Ionizarion energy of magnesium is more than thar of aluminium.
(R): In aluminium 3p-orbiral is completely filled whereas in magnesium it is not completely filled.

112. (A): Fluorine is more electronegative than chlorine.
(R): Fluorine is smaller in size than chlorine.

113. (A): Ionization energy for s-electrons is more than the p-electrons for the same shell.
(R): s-electrons are closer to the nucleus than p-electrons hence more tightly attached.

114. (A): Li and Mg show diagonal relationship.
(R): Li and Mg have same atomic radius.

115. (A): He and Be both have the same outer electronic configuration like ns^2 type.
(R): Both are chemically inert.

116. (A): The first ionization energy of N is more stable than that of O.
(R): Oxygen after losing one electron gets a stable electronic configuration.

117. (A): For noble gases in the solid state the crystal radii are actually van der Waal's radii.
(R): In crystals of noble gases no chemical forces operate between the atom.

118. (A): Electron gain enthalpy of oxygen is less than that of fluorine but greater than that of nitrogen.
(R): Ionizarion enthalpy is as follows
N > O > F

119. (A): Second ionization enthalpy will be higher than the first ionization enthalpy.
(R): Ionization enthalpy is a quantitative measure of the tendency of an element to lose electron.

120. (A): Noble gases have large positive electron gain enthalpy.
(R): Electron has to enter the next higher principle quantum level.

Matrix–Match Type Questions

	p	q	r	s	t
(A)	O	O	O	O	O
(B)	O	O	O	O	O
(C)	O	O	O	O	O
(D)	O	O	O	O	O

121. Match the following:

Column I	Column II
A. F	(p) Smallest anion
B. Cl	(q) Most electronegarive
C. Br	(r) Maximum electron affinity
D. I	(s) Metallic lusture
	(t) Pungent liquid

122. March the following:

Column I	Column II
A. Nonmetal	(p) Halogens
B. High ionization energy	(q) Inert gases
C. Low boiling point	(r) Alkali metals
D. Maximum electron affinity	(s) Transition metals

123. Match the following:

Column I	Column II
A. Nitrogen	(p) Metal
B. Oxygen	(q) Electropositive
C. Calcium	(r) Electronegative
D. Cesium	(s) High ionization energies

124. March the following:

Column I	Column II
A. F_2	(p) Gaseous molecules
B. O_2	(q) Highest bond energy
C. N_2	(r) Paramagneric nature
D. Cl_2	(s) Diamagneric

125. March the following:

Column I	Column II
A. Reducrants	(p) Calcium
B. Srrongesr oxidant	(q) Sulphur
C. Basic oxides	(r) Fluorine
D. Acidic oxides	(s) Cesium

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126. Property of alkaline earth metals that increases with their atomic number is
(a) ionization energy
(b) solubility of their hydroxides
(c) solubility of their sulphates
(d) electronegativity
[*IIT 1997*]

127. The correct statement among the following is
(a) The first ionization potential of Al is less than the first ionization potential of Mg.
(b) Second ionization potential of Mg is greater than the second ionization potential of Na.
(c) The first ionization potential of Na is less than the first ionization potential of Mg.
(d) The third ionization potential of Mg is greater than the third ionization potential of Al.
[*IIT 1997*]

128. The correct order of radii is
(a) N < Be < B (b) $F^- < O^{2-} < N^{3-}$
(c) Na < Li < K (d) $Fe^{+3} < Fe^{+2} < Fe^{+4}$
[*IIT Screening 2000*]

129. Identify the least stable in amongst the following
(a) Li (b) Be
(c) B^- (d) C^-
[*IIT Screening 2001*]

130. The set representing the correct order of first ionization potential is
(a) K > Na > Li (b) Be > Mg > Ca
(c) B > C > N (d) Ge > Si > C
[*IIT Screening 2001*]

131. Identify the correct order of acidic strengths of CO_2, CuO, CaO and H_2O
(a) $CaO < CuO < H_2O < CO_2$
(b) $H_2O < CuO < CaO < CO_2$
(c) $CaO < H_2O < CuO < CO_2$
(d) $H_2O < CO_2 < CaO < CuO$
[*IIT 2002*]

ANSWERS

STRAIGHT OBJECTIVE TYPE QUESTIONS

1. (a)	2. (a)	3. (a)	4. (b)
5. (b)	6. (a)	7. (d)	8. (a)
9. (b)	10. (c)	11. (c)	12. (d)
13. (d)	14. (c)	15. (c)	16. (b)
17. (b)	18. (b)	19. (b)	20. (c)
21. (c)	22. (d)	23. (d)	24. (a)
25. (b)	26. (b)	27. (c)	28. (a)
29. (c)	30. (b)	31. (b)	32. (d)
33. (d)	34. (a)	35. (d)	36. (c)
37. (a)	38. (b)	39. (a)	40. (d)

BRAINTEASERS OBJECTIVE TYPE QUESTIONS

41. (b)	42. (d)	43. (c)	44. (a)
45. (b)	46. (c)	47. (b)	48. (a)
49. (a)	50. (c)	51. (b)	52. (b)
53. (b)	54. (b)	55. (d)	56. (c)
57. (b)	58. (a)	59. (d)	60. (a)
61. (c)	62. (d)	63. (a)	64. (d)
65. (d)			

MULTIPLE CORRECT ANSWER TYPE QUESTION

66. (b), (c)
67. (b), (c)
68. (a), (b), (d)
69. (a), (b), (c)
70. (a), (b), (d)
71. (c), (d)
72. (b), (c)
73. (b), (d)
74. (a), (d)
75. (c), (d)
76. (a), (d)
77. (b), (c), (d)
78. (a), (b), (c)
79. (a), (d)
80. (a), (d)
81. (a), (b), (d)
82. (c), (d)
83. (c), (d)
84. (a), (b), (c)
85. (a), (c), (d)
86. (a), (b)
87. (c)
88. (a), (c)
89. (a), (d)
90. (a)
91. (b), (c)
92. (b)
93. (a), (b), (c)
94. (b)
95. (a)
96. (b)
97. (d)

Linked–Comprehesion Type Questions

Comprehension–1

98. (c) 99. (a) 100. (a) 101. (b)

Comprehension–2

102. (d) 103. (b) 104. (d)

Comprehension–3

105. (b) 106. (c) 107. (b) 108. (d)

Assertion and Reasoning Questions

109. (a) 110. (a) 111. (c) 112. (b)
113. (a) 114. (c) 115. (c) 116. (b)
117. (b) 118. (c) 119. (b) 120. (a)

Matrix–Match Type Questions

121. a-(p, q), b-(r), c-(r), d-(s)

122. a-(p, q), b-(p, q), c-(p), d-(q)

123. a-(r, s), b-(r, s), c-(p, q), d-(p, q)

124. a-(p, s), b-(p, r), c-(p, q, s), d-(p, s)

125. a-(p, s), b-(r), c-(p, s), d-(q, r)

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126. (b) 127. (b) 128. (b)
129. (b) 130. (b) 131. (a)

HINTS AND EXPLANATIONS

Straight Objective Type Questions

1. More the ionization energy of one atom and less the electron affinity of the other and less polar (or less ionic) would be bond formed between them. I.E. of A > I.E. of B and the E.A of X > E.A. of Y, therefore A-Y bond is least polar.
2. $O(g) + 2e \rightarrow O^{2-}(g)$, $\Delta H = 603\ kJ\ mol^{-1}$ (i)

   $O(g) + e^- \longrightarrow O^-$, $\Delta H = -141\ kJ\ mol^{-1}$ (ii)

   Equation (i) and (ii) gives:

   $O + e \rightarrow O^{2-}$, $\Delta H = 603 - (-141)$

   $= 744\ kJ\ mol^{-1}$
3. As ionization energy is not the only criteria for the stability of an oxidation state.
5. 118th element: The last element of the actinides, lawrencium (Lr), with atomic number 103, is in the 3rd group. The 118th element is in the 18th group, which is the group of noble gases.
6. Radii of anions carrying same charge decrease from left to right in a period and increase down the group.
12. As HF is not stronger acid than HCl because fluorine is more electronegative than chlorine therefore hydrogen does not donate easily in than in HCl.
14. As it has half-filled 3d-orbital hence it will have more ionization energy.
16. Cl has highest affinity but with negative sing its value is lowest.
17. As in case of halogens the electron affinity decreases as follows:

    $$Cl > F > Br > I$$
18. As bond dissociation energy decreases in the order:

    $$Cl_2 > Br_2 > F_2 > I_2.$$
20. Atomic radius of neon being van der Waal's radius higher than that of fluorine which is infact its covalent radius.
21. As both (2) and (4) have exactly half-filled configuration, but (2) has smaller size than (4), hence it has higher ionization energy.
25. As the correct order of ionization energy is

    $$Li < B < Be < C$$
26. Both B and D have exactly half-filled configuration, but B has smaller size than D, hence it has higher I.E.
28. Number of electrons are same in all but nuclear charge is maximum in Na^-. Hence, it is the smallest.
29. Nitrogen, being smallest in size, can give up its lone pair of electrons most easily.
31. O_2^- has one unpaired electron, so paramagnetic. Other species has no unpaired electron.
32. Alkali metal oxide (K_2O) is most basic in nature.
33. IE_1 of Mg is higher than that of Na because of increased nuclear charge and also that of Al because in Mg a 3 s-electron has to be removed while in Al it is the 3p-electron. The IE_1 of Si is, however, higher than those of Mg and Al because of it increased nuclear charge. Hence, the order is

    $$Na < Mg > Al < Si$$
34. Si and P are in the 3rd period while C and N are in the 2nd period. Elements in 2nd period have higher electronegativities than those in the 3rd period. Since N has smaller size and higher nuclear charge than C, its electronegativity is higher than that of C. similarly the electronegativity of P is higher than that of Si. Thus, the order is Si, P, C, N.
36. Atomic radius of neon being van der Waal's radius higher than that of fluorine which is infact its covalent radius.
37. Amongst B, C, N, O; N has the highest first ionization energy, due to its half-filled 2p orbital which is more stable.
38. The correct order is:

    $O > F > N > C$

Brainteasers Objective Type Questions

41. As ionization energy decreases with the increase in number of orbits or down the group.
42. It is because of the electronic repulsion.
43. As in (III) the correct order of increasing the first ionization enthalpy is B < C < O < N.
44. As acidic nature increases left to right in a period with increase in electronegativity.
45. Electron affinity $= \dfrac{\text{Energy release} \times N_0}{\text{Number of atom ionized}}$

    $$\frac{-4.8 \times 10^{13} \times 6.023 \times 10^{23}}{10^6}$$

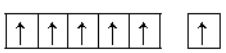
    $= 28.9104 \times 10^4\ J = 289\ kJ.$

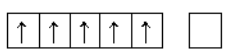
46. Chlorine being the group 17 element has maximum electronegativity. 'N' has zero electron affinity because extra stability is associated with exactly half-filled orbitals. Sulphur has more electron affinity than 'O' because the effect of small size of O atom is more than offset by the repulsion of electrons already present in 2p-orbitlas of O atom.

49. As ionization energy decreases with the increase in number of orbits or down the group.

50. As the difference between I_5 and I_4 is maximum so the element (P) has three valence electrons as after losing three electrons it acquires noble gas configuration.

51. Since fluorine atom has a smaller size than iodine, it has greater electron attracting power and so it is a better oxidizing agent.

53. Cr : [Ar]  $3d^5$ $4s^1$

Cr^+: [Ar]  $3d^5$ $4s^0$

(by first I.P.)

This is stable electronic configuration hence formation of Cr^{2+} by second IP requires maximum enthalpy.

55. Here, X is Mg, Y is Al, Z is Si as MgO is ionic, Al_2O_3 is amphoteric and SiO_2 is a giant molecule.

56. Molar volume $= \frac{26.98\ cm^3}{2.70}$

$= \frac{4}{3}\ \pi r^3 \times 6.02 \times 10^{23}$

$r^3 = \frac{26.98}{2.70} \times \frac{10^{24} \times 3}{4 \cdot 3.14 \times 6.02 \times 10^{23}}\ (Å)^3$

$= \frac{10^2 \times 3}{4 \cdot 3.14 \quad 6.02}$

$r = (300 / 12.56 \times 6.02)^{1/3}$

$= 1.583\ Å$

59. The first ionization energy values of Na, Mg and Al are 5.1, 7.6 and 6.0 eV. The second ionization energy values of Na, Mg and Al are 47.3, 15.03 and 18.82 eV.

60. $Cl + e \rightarrow Cl^- + 3.7$ eV

35.5 $\quad$ 3.7 × 23.06 kcal

As energy released for conversion of 4 g gaseous chlorine into Cl^- ions

$= \frac{3.7 \times 23.06}{35.5} \times 4 = 9.6$ kcal

61. The difference in atomic radii is maximum in Na and K.

62. The fifth ionization energy shows a sudden increase (most likely 1s-orbital)

The likely electron configuration is $1s^2\ 2s^2\ 2p^2$.

63. Element X can lose its first two outermost electrons easily. It is most likely bivalent. By similar reasoning element Y is tetravalent. Therefore, the compound may be X_2Y.

64. $E_1 : E_2 = 0.475 : 1$.

Using the proportionally constant k,

$E_1k + E_2k = 2226$ kJ/mol.

While, $E_1k / E_2k = 0.475/1$.

On solving,

$E_2k = 2226 \times 1/1.475 = 1509$ kJ/mol.

65. Half-filled $2p^3$ subshell of O^+ is more stable than $2p^4$ subshell of F^+ So IE_2 of O will be greater than that of F^-.

Multiple Correct Answer Type Questions

96. IE_2 of Na is greater than Mg because second electron is to be removed from stable noble gas configuration in case of sodium.

Comprehension–3

105. As (a) is boron and (b) is aluminium and both these elements are present in IIIA group.

106. As atomic radii is directly proportional to number of orbits and inner shell electrons while inversely proportional to number of valence electrons. Hence, the correct order of size is

E > B > C > D > A

107. E < B < C < A < D

or

Ca	Al	P	B	Cl
–	−50	−74	−83	−349 kJ mol^{-1}

108. B < E < A < C < D

or

Al	Ca	B	P	Cl
577	590	801	1011	1255 kJ mol^{-1}

Assertion and Reasoning

119. It is difficult to remove an electron from a positively charged ion than a neutral atom.

120. Noble gases have large positive electron gain enthalpies because the electron has to enter the next high principal quantum level leading to a very unstable electronic configuration.

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126. Solubility of hydroxides of group II increases down the group.

127. IE_2 of Na is greater than Mg because second electron is to be removed from stable noble gas configuration in case of sodium.

128. For the isoelectronic ions, the size decreases with increase in nuclear charge.

129. Be ($1s^2\ 2s^2$) due to its completely filled 2s-sub shell has least tendency to take up an electron. As such Be is least stable.

130. Down the group ionization energy decreases.

131. CaO is basic while CO_2 is most acidic of these. The increasing acidic strength order is
$CaO < CuO < H_2O < CO_2$

Solved Subjective Questions

1. Arrange the following in:

   (i) Decreasing order of ionic size:
   Mg^{2-}, O^{2-}, Na^+, F^-
   O^{-2}, F^-, Na^-, Mg^{-2}

   (ii) Increasing order of acidic property:
   ZnO, Na_2O_2, P_2O_5, MgO
   Na_2O_2, MgO, ZnO, P_2O_5

   (iii) Increasing order of first ionization potential:
   Mg, Al, Si, Na
   Na, Al, Mg, Si

   (iv) Increasing order of size:
   Cl^-, S^{2-}, Ca^{2+}
   S^{2-}, Cl^-, Ca^{2+}

   (v) Increasing order of ionic size:
   N^{3-}, Na^-, F^-, O^{2-}, Mg^{2+}
   Mg^{2+}, Na^+, F^-, O^{2-}, N^{3-}

   (vi) Increasing order of basic character:
   MgO, SrO, K_2O, NiO, Cs_2O
   NiO, MgO, SrO, K_2O, Cs_2O

   (vii) Decreasing order of first ionization energy for
   B, Al, Ga, In, Tl
   B, Tl, Ga, Al, In

2. Arrange the following ions in order of their increasing radii:
   Li^-, Mg^{2-}, K^-, Al^{3-}

   Solution *[IIT, July 1997]*

   $Li^- < Al^{3+} < Mg^{2+} < K^+$.

3. Compare qualitatively the first and second ionization potentials of Cu and Zn.

   Solution *[IIT 1996]*

   Cu: $3d^{10}$, $4s^1$

   Zn: $3d^{10}$, $4s^2$

   IP_2 values of Cu shows a jump whereas no such jump is noticed is IP_2 values of Zn.

4. Anhydrous $AlCl_3$ is covalent. From the data given below, predict whether it would remain covalent or become ionic in aqueous solution. (Ionization energy for $AlCl_3$ – 5137 kJ mol^{-1}; $\Delta H_{Hydration}$ for Al^{3+} – –4665 kJ mol^{-1}; $\Delta H_{Hydration}$ for Cl^- – –381 kJ mol^{-1})

   Solution *[IIT July 1997]*

   $AlCl_3 + aq. \rightarrow AlCl_3(aq.)$; ΔH – ?

   ΔH = Energy released during hydration + Energy used during ionization

   $= -4665 - 3 \times 381 + 5137 = -671$

Thus, formation of ions will take place because $\Delta H_h > \Delta H_{ionization}$

5. $BaSO_4$ being an electrovalent compound does not pass into solution state in water. Explain.

Solution

As the hydration energy of $BaSO_4$ being lesser than the lattice energy the ions are not separated hence it does not pass into the solution state.

6. $NaCl_{aq.}$ gives a white precipitate with $AgNO_3$ solution but CCl_4 or $CHCl_3$ does not. Explain.

Solution

As NaCl being an ionic compound can furnish Cl^- and reacts with $AgNO_{3(aq)}$ to give ionic reaction, forming AgCl as white ppt. While CCl_4 and $C1Cl_3$ being covalent cannot furnish Cl^- ions in solution.

7. Why argon (at. mass 39.94) has been placed before potassium (at. mass 39.10) in the periodic table?

Solution

In modern periodic table, elements have been placed in order of their increasing atomic numbers. As the atomic number of argon is 18 and that of potassium is 19. Hence, argon has been placed before potassium.

8. Calculate the electronegativity of fluorine from the following data:

$E_{H-H} = 104.2$ kcal. mol^{-1}

$E_{B\,B} - 36.6$ kcal mol^{-1}

$E_{H\,F} - 134.6$ kcal mol^{-1}

$X_H = 2.1$

Solution

Let the electronegativity of fluorine be X_F and on applying Pauling's equation,

$$X_F - X_H - 0.208\,[E_{H-F} - \tfrac{1}{2}(E_{F-F} + E_{H-H})]^{1/2}$$

So

$$X_F - 2.1 = 0.208\,[134.6 - \tfrac{1}{2}(104.2 + 36.6)]^{1/2} = 3.76$$

9. A monoatomic anion of unit charge contains 45 neutrons and 36 electrons. What is the atomic mass of the element and in which group of the periodic table does it belong to.

Solution

No. of neutrons of the monoromic anion - 45

No. of neutrons of the monoromic anion - 36–1–35

Atomic number of the atom (Z = 35). Therefore, the atom is bromine and it belongs to roup 17, i.e., halogen family.

No. of protons - No. of electrons - 35

Atomic mass - No. of neutrons + No. of protons - 45 + 35 - 80

10. Give the name and atomic number of the inert gas atom in which the total number of d- electrons is equal to the difference in numbers of total p and s-electrons.

Solution

The first inert gas that contains d-electrons is krypton. Its atomic number is 36 and its electronic configuration is as follows:

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6$

Here total number of d-electrons = 10

Total number of p-electrons - 6+6+6 - 18

Total number of s-electrons = 2+2+2+2 = 8

Hence difference in total number of p-and s- electrons - 18 - 8 - 10.

Hence, this inert gas is Krypton.

11. Show by a chemical reaction with water that Na_2O is a basic oxide and Cl_2O_7 is an acidic oxide.

Solution

Na_2O reacts with water to form sodium oxide which turns red litmus blue, hence Na_2O is a basic oxide.

$$Na_2O + H_2O \rightarrow 2NaOH$$

Sodium oxide Sodium hydroxide

Cl_2O_7 reacts with water to form perchloric acid which turns blue litmus red. Hence, Cl_2O_7 is an acidic oxides.

$$Cl_2O_7 + H_2O \rightarrow 211ClO_4$$

Chlorine (VII) oxide Perchloric acid

12. Energy of an electron in the ground state of the hydrogen atom is -2.8×10^{-18} J. Calculate the ionization enthalpy of atomic hydrogen in terms of kJmol^{-1}

Solution

Ionization energy is the amount of energy needed to remove the electron from the ground state to infinity.

Now, energy of the electron in the ground state = -2.8×10^{-15} J

As energy of the electron at affinity = 0.

Hence, the energy required to remove an electron in the ground state of hydrogen atom = 0 −(its energy in the ground state) = −(− 2.18 × 10^{-18} J) = 2.18 × 10^{18} J

Hence ionization enthalpy per mole of hydrogen atoms = $\frac{2.18 \times 10^{18} \times 6.02 \times 1023 \text{ kJ}}{1000}$

= 1323.36 kJmol^{-1} = 1312.36 × 10^3 Jmol^{-1}

Questions for Self Assessment

13. Explain why nitrogen has a less favourable or more positive electron affinity than carbon and oxygen.
14. Which has the more negative electron affinity between Br and Br^-?
15. Predict if the element with atomic number 115 has less or more ionization energy than that of Bi.
16. What is the difference between electron gain enthalpy and electronegativity?
17. Explain why second ionization energy of sodium is more than that of Mg.
18. Explain why the first ionization energy for group 13 elements variate irregularly.
19. Give the reason why oxygen has less favourable electron gain enthalpy than Sulphur.
20. The first ionization energy of carbon atom is greater than that of boron atom whereas, the reverse is true for the second ionization energy.

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PREPARATION AND PROPERTIES OF NONMETALS

3

Chapter Contents

Isolation/preparation and properties of boron, silicon, nitrogen, phosphorous, oxygen, sulphur and halogens, properties of allotropes of carbon (diamond and graphite), phosphorous and sulphur and various levels of multiple-choice questions.

BORON

Occurrence

Boron is not present in free state in nature. In the combined state, it is present in the form of the salts of boric acid. Some of the important minerals of boron are as follows:

1. Borox $Na_2B_4O_7.10H_2O$
2. Colemanite $Ca_2B_6H_{11}.5H_2O$
3. Kernite (Rasorite) — $Na_2B_4O_7.4H_2O$
4. Boracite — $2Mg_3B_8O_{15}.MgCl_2$
5. Boric acid H_3BO_3
6. Boronatrocalcite $CaB_4O_7.NaBO_2.8H_2O$

Extraction of Boron

1. From Borax or Colemanite
   Boron can be extracted from borax or colemanite ores in two steps:

(i) ***Preparation of B_2O_3:***

(a) **From Borax:** First of all, finely powdered borax is converted into B_2O_3 by heating it with conc. HCl or H_2SO_4 as follows:

$$Na_2B_4O_7 + 2HCl \longrightarrow 2NaCl + \underset{\text{Tetraboric acid}}{H_2B_4O_7}$$

$$Na_2B_4O_7 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2B_4O_7$$

$$H_2B_4O_7 + 5H_2O \longrightarrow \underset{\text{Ortho boric acid}}{4H_3BO_3}$$

$$2H_3BO_3 \xrightarrow{\Delta} B_2O_3 + 3H_2O$$

or

$$Na_2[B_4O_5(OH)_4].8H_2O \xrightarrow{\text{acid}} \underset{\text{Orthoboric acid}}{H_3BO_3} \xrightarrow{\text{heat}} B_2O_3$$

(b) **From Colemanite:** The powdered form of colemanite is fused with sodium carbonate and the fused mass is treated with hot water in which calcium carbonate is insoluble and hence gets precipitated. The filtrate solution having borax and

sodium meta borate is crystallized to get the crystals of borax first. In the remaining solution, when CO_2 is passed, sodium meraborate changes into borax. The obtained borax can be converted into B_2O_3 as in the process (a).

$$Ca_2B_6O_{11} + 2Na_2CO_3 \rightarrow 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$$

$$4NaBO_2 + CO_2 \rightarrow Na_2B_4O_7 + Na_2CO_3$$

- Colemanite can be converted into B_2O_3 by adding the mineral in water and passing SO_2 gas in the suspension as follows:

$$Ca_2B_6O_{11} + 11H_2O + 4SO_2 \rightarrow 2Ca(HSO_3)_2 + 6H_3BO_3$$

$$2H_3BO_3 \xrightarrow{\Delta} B_2O_3 + 3H_2O$$

(ii) ***Reduction of B_2O_3:*** Amorphous boron of low purity (called Moissan boron) can be obtained by reducing B_2O_3 with Mg or Na at a high temperature. It is nearly 95–98% pure and is black in colour.

$$B_2O_3 \xrightarrow{Mg/Na/K} B + MgO \text{ or } Na_2O \text{ or } K_2O$$

2. Modern Method (Electrolytic Method)
Boron can be obtained by the electrolysis of a fused mixture having boric anhydride, MgO and MgF_2 at 1373 K. The electrolysis is carried out in a carbon crucible which acts as an anode and an iron rod is used as cathode here. The magnesium that is discharged at the cathode reduces B_2O_3 into boron as follows:

$$2MgO \rightarrow 2Mg + O_2$$

$$B_2O_3 + 3Mg \longrightarrow 2B + 3MgO$$

To remove the impurities, boron is heated in a vacuum at 1373 K electrically where the impurities are removed and pure boron is obtained.

- Crystalline form of boron can be obtained by reducing B_2O_3 with Al powder. This aluminium can be recovered by heating the fused with sodium hydroxide solution.

3. Thermal decomposition of diborane or other boron hydrides also gives boron.

$$B_2H_6 \xrightarrow{Heat} 2B\,(g) + 3H_2\,(g)$$

4. By the reduction of volatile boron compounds with dihydrogen at high temperature (1543°C).

$$2BCl_3\,(g) + 3H_2\,(g) \xrightarrow{1543^\circ C} 2B\,(s) + 6HCl\,(g)$$

5. Pyrolysis of BI_3 (van arkel method) also gives boron.

$$2BI_3\,(g) \xrightarrow{\text{Tungsten, }\Delta} 2B + 3I_2\,(g)$$

6. By heating potassium fluoborate (KBF_4) with magnesium or potassium:

$$2KBF_4 + 3Mg \xrightarrow{\Delta} 2B + 2KF + 3MgF_2$$

$$KBF_4 + 3K \xrightarrow{\Delta} B + 4KF$$

Physical Properties

1. The two allotropic forms of boron are amorphous boron (dark brown) and crystalline boron (black).
2. It has two isotopic forms, namely $_5B^{10}$ and $_5B^{11}$.
3. It is a very hard substance with melting point higher than 2177°C and boiling point is about 2552°C. It is nonfusible under ordinary pressure.
4. It is a nonconductor of electricity.

Chemical Properties

Boron is not much reactive so it cannot react with many chemical reagents at ordinary temperature. Some of the chemical properties of boron are as follows:

1. Action of Air
Amorphous boron burns in air at 973K with a reddish flame giving boron oxide (B_2O_3) and boron nitride (BN).

$$4B + 3O_2 \xrightarrow{973\ K} 2B_2O_3$$

$$2B + N_2 \xrightarrow{973\ K} 2BN$$

2. Reaction with Water
Boron is not attacked by water under normal conditions, however, when steam is passed over red hot boron, hydrogen is liberated as follows:

$$2B + 3H_2O \rightarrow B_2O_3 + 3H_2\uparrow$$

3. Action of Alkalies
Boron reacts with fused KOH and NaOH to give borates.

$$2B + 6KOH \longrightarrow \underset{\text{Potassium borate}}{2K_3BO_3} + 3H_2$$

4. Reducing Nature
It is a powerful reducing agent.

$$3CO_2 + 4B \longrightarrow 2B_2O_3 + 3C$$

$$3SiO_2 + 4B \longrightarrow 2B_2O_3 + 3Si$$

5. Reaction with Acids
Concentrated H_2SO_4 or HNO_3 oxidizes boron into boric acid, however, it does not react with nonoxidizing acids.

$$2B + 3H_2SO_4 \xrightarrow{\Delta} \underset{\text{Boric acid}}{2H_3BO_3} + 3SO_2$$

$$2B + 6HNO_3 \longrightarrow \underset{\text{Conc.}}{2H_3BO_3} + 6NO_2\uparrow$$

6. Reaction with Metals
Boron combines with metals upon heating giving borides which are normally very hard and have high melting points.

$$2Mg + 2B \longrightarrow \underset{\text{Magnesium boride}}{Mg_3B_2}$$

$$Cr + B \longrightarrow CrB$$

7. Reaction with Nonmetals
When it is heated with carbon in an electric furnace it gives an extremely hard substance, boron carbide (B_4C). Boron gives B_2S_3 on heating with sulphur. It also reacts with Cl_2 and Br_2 but at higher temperatures.

$$2B + 3Cl_2 \xrightarrow{\Delta} 2BCl_3$$

Uses

1. It is used as a semiconductor.
2. Boron steel or boron carbide rods can be used for controlling nuclear reactions as boron has a very high cross section for capturing the neutrons. Moreover it can absorb neutrons to create another boron having even number of neutrons.

$${}_5B^{10} + {}_0n^1 \longrightarrow {}_5B^{11}$$

3. It is used in glass industry for making specific types of glass.
4. Boron filaments are used in making light and composite material for aircrafts.

Anomalous Behaviour of Boron

Boron being the first element of its group differs from other elements of its own group in many properties. The anomalous behaviour of boron is due to:

(i) its small size
(ii) high ionization energy
(iii) high electronegativity
(iv) absence of vacant d-orbitals in its valence shell

Some of the important properties in which boron differs from other members of its group are as follows:

1. Boron is a nonmetal while rest of the members of this group are metals.
2. The melting and boiling points of boron are much higher than those of aluminium and the other members of this group.
3. It can show only +3 oxidation state while other elements of the group can show +1 and +3 oxidation states.
4. The trihalides of boron exist as discrete monomeric molecules while those of the other elements of its group can exist as dimmers having halogen-bridged structures.
5. Boron shows allotropy and exists in two forms—crystalline and amorphous, while other elements of its group do not show allotropy.
6. It cannot decompose water or steam like other members of the group which can decompose water.
7. Boron is a nonconductor of electricity while the other elements of its group are good conductors.
8. It can combine with metals to form borides while rest of the elements of the group can not do so.
9. Boron shows a maximum covalence of four in its compounds (e.g., BH_4^-). Aluminium and other members of this family can show a covalence of even six due to presence of d-orbitals in them. Example, $[Al(OH)^6]$.

10. Boron forms only covalent compounds, while aluminium forms both covalent as well as ionic compounds. The other members of this group form mainly ionic compounds.
11. The oxide and hydroxide of boron are weakly acidic, while those of aluminium and the other members of this group are either amphoteric or basic in nature.
12. The oxide of boron i.e., B_2O_3 is soluble in water while the oxides of other elements are insoluble in water.
13. Boron forms a number of series of hydrides called boranes. The boranes are stable covalent compounds and are volatile in nature. The hydrides of aluminium and the other elements of this group are nonvolatile solids and are less stable and decompose on heating.

Diagonal Relationship between Boron and Silicon

Boron (III group) is placed diagonally opposite to silicon (IV group) in the periodic table. Therefore, boron shows diagonal relationship with silicon.

The diagonal relationship between boron and silicon is due to following factors:

(i) The atomic radius of boron (85 pm) is close to that of silicon (118 pm).

(ii) Both have almost similar values of electronegativity (B = 2.0, Si = 1.8) and ionization energy (B = 800 kJ mol^{-1}, Si = 786 kJ mol^{-1}).

Some of the important properties in which boron resembles silicon are as follows:

1. Occurrence

   Both boron and silicon do not occur in free state in nature. But they are always found in the combined state.

2. Allotropy

   Both boron and silicon can show allotropy and exist in two allotropic forms i.e., amorphous and crystalline.

3. Nonmetallic Character

   Both boron and silicon are typical nonmetals with high melting and boiling points. Both are nonconductors of electricity.

4. Semi-conducting Nature

   Both of these are semiconductors at high temperature.

5. Due to high ionization energies both boron and silicon can not form cations easily.

6. Both boron and silicon cannot combine with carbon to form carbides i.e., B_4C, SiC which are very hard in nature and used for cutting and abrasing purposes.

7. Formation of Hydrides

   Both boron and silicon can form a large number of hydrides known as boranes and silanes respectively

   For example, Silane, SiH_4 Disilane, Si_2H_6 etc.

8. Reaction with Alkalies

   Both boron and silicon dissolve in alkalies to form borates and silicates, respectively with the evolution of hydrogen. Boron dissolves in fused alkalies, whereas silicon dissolves both in fused as well as aqueous alkalies.

$$2B + 6NaO \longrightarrow \underset{\text{Sodium borate}}{2Na_2BO_3} + 3H_2$$

$$Si + 2NaOH + 2H_2O \longrightarrow \underset{\text{Sodium silicate}}{Na_2SiO_3} + 2H_2$$

9. Formation of Halides

   These can form halides when heated with respective halogens and halides are covalent in nature and hygroscopic and volatile. They can be easily hydrolyzed by water.

$$SiCl_4 + 4H_2O \longrightarrow \underset{\text{Silicic acid}}{H_4SiO_4} + 4HCl$$

$$BCl_3 + 3H_2O \longrightarrow \underset{\text{Boric acid}}{H_3BO_3} + 3HCl$$

   Due to release of HCl, their chlorides fume in moist air.

10. Formation of Oxides

    These can form oxides when heated in oxygen and their oxides (B_2O_3, SiO_2) are high melting solids and acidic in nature and dissolve in alkalies to form borates and silicates respectively.

$$B_2O_3 + 6NaOH \longrightarrow 2Na_3BO_3 + 3H_2O$$

$$SiO_2 + 2NaOH \longrightarrow Na_2SiO_3 + H_2O$$

11. Formarion of Binary Compounds with Merals

Both can react with many metals forming binary compounds called borides and silicides respectively. For example,

$$2B + 3Mg \xrightarrow{\Delta} \underset{\text{Magnesium boride}}{Mg_3B_2}$$

$$Si + 2Mg \xrightarrow[\text{(In absence of air)}]{\Delta} \underset{\text{Magnesium silicide}}{Mg_2Si}$$

Some of these borides and silicides on hydrolysis results boranes and silanes, respectively.

SILICON

Silicon is the second most abundant element (28 percent by weight) in earth crust. It is widely present as silica (SiO_2).

Three crystalline modifications of SiO_2 are quartz, cristobalite and tridymite of which first two are important. Quartz is used as a piezoelectric material.

Silica occurs as silicates mainly because silica has a great affinity for oxygen. For example, aluminium silicate {Rock clay} (most widely distributed).

Preparation

1. From Silica

Here silica is reduced by coke in an electric furnace to get silicon.

$$SiO_2 + 2C \longrightarrow Si + 2CO$$

$$Si + C \longrightarrow SiC$$

$$2SiC + SiO_2 \rightarrow \underset{\text{Crystalline form}}{3Si} + 2CO$$

2. Silica can also be reduced into silicon by magnesium powder.

$$SiO_2 + 3Mg \rightarrow \underset{\text{Amorphous form}}{Si} + 2MgO$$

3. From Silicon Tetrachloride

Ultra pure silicon is obtained by zone refining and by the reduction of very pure $SiCl_4$.

$$SiCl_4 + 2H_2 \longrightarrow Si + 4HCl$$

4. From Silicon Hydride

$$SiH_4 \xrightarrow{402^\circ C} Si + 2H_2$$

It is purified by zone refining process.

Physiochemical Properties

1. It is a very hard solid element which has two allotropic forms i.e.,

(a) **Amorphous:** It is dark brown powder and a more reactive form.

(b) **Crystalline:** It is yellow crystalline solid, iso structural to diamond and less reactive.

2. Reaction with Merals

It combines with merals giving silicides.

For example,

$$2Mg + Si \rightarrow Mg_2Si$$

3. Solubility

It dissolves in aqua regia and alkalies. It gives a silicate on fusion with alkali.

For example,

$$Si + 2NaOH + H_2O \rightarrow Na_2SiO_3 + 2H_2O$$

4. With Steam

It decomposes steam as follows:

$$Si + 2H_2O \rightarrow SiO_2 + 2H_2$$

5. Ignition

Amorphous silicon can burn with O_2 and F_2.

$$Si + O_2 \rightarrow SiO_2$$

$$Si + 2F_2 \longrightarrow SiF_4$$

Uses

1. Silicon chips doped with P, As, Al or Ga to enhance the semiconductor properties are used for computing devices.
2. It is used in the manufacture of many alloys with high strength, hardness, resistant power against acids like ferrosilicon, manganese silicon, bronze etc. Ferro-silicon is used to prepare acid resistant steel.

NITROGEN (N_2)

It was discovered by Daniel Rutherford who called it foul air or mephitic air (killer of life). Lavoisier established its elemental nature and called it Azote (means without life). The name nitrogen was derived from nitre in which nitrogen element is present.

It exists as a diatomic molecule known as dinitrogen.

Occurrence: It is present in free state as well as in combined state. In combined state it exists in salt peter (KNO_3), Chile saltpetre ($NaNO_3$). It is present 70 percent by volume in air. It is the essential constituent of all living cell of plants and animals.

Preparation

1. By Fractional Distillation of Liquid Air
   It can be obtained by liquefying air (having N_2 and O_2) followed by fractional evaporation as N_2 being more volatile than O_2, boils off more rapidly than oxygen. Nitrogen obtained from here has small amount of O_2 and inert gases as impurities. Here, the apparatus used is Claude's apparatus and it is the commercial method of preparation of N_2.
2. From Nitrogen Containing Compounds

(i) ***From Ammonium Dichromate:***

$$(NH_4)_2Cr_2O_7 \xrightarrow{\text{Heat}} Cr_2O_3 + N_2 + 4H_2O$$

(ii) ***Lab Method:*** In laboratory it is prepared by heating an aqueous solution of ammonium chloride and sodium nitrite as follows:

$$NH_4Cl + NaNO_2 \xrightarrow{\Delta} NH_4NO_2 + NaCl$$

$$NH_4NO_2 \xrightarrow{\text{Heat}} N_2 + 2H_2O$$

(iii) ***From Copper and Nitric Acid:*** When vapours of nitric acid are passed strongly over heated copper, nitrogen is formed.

$$5Cu + 2HNO_3 \longrightarrow 5CuO + N_2 + H_2O$$

(iv) ***By the Oxidation of Ammonia:*** Chlorine or CuO can oxidize ammonia into nitrogen as follows:

$$8NH_3 + 3Cl_2 \longrightarrow 6NH_4Cl + N_2$$

$$2NH_3 + 3CuO \xrightarrow{\text{Heat}} 3Cu + N_2 + 3H_2O$$

3. From Sodium or Barium Azide
   On heating sodium or barium azide in vacuum, very pure form of nitrogen in obtained.

$$2NaN_3 \longrightarrow 2Na + 3N_2$$

$$Ba(N_3)_2 \longrightarrow Ba + 3N_2$$

Physical Properties

1. Nitrogen is a colourless, odourless, tasteless and nonpoisonous gas which can be liquefied.
2. It is slightly lighter than air and has a vapour density of 14.
3. It is partially soluble in water and can be liquefied to a colourless liquid also.
4. It is noncombustible and does not support combustion.
5. It has a melting point of 63.2 K and a boiling point 70.2 K.

CHEMICAL PROPERTIES

1. With Hydrogen
   It reacts with hydrogen at 400–500°C and a pressure of 200 atmosphere in presence of a catalyst (Fe filling and Mo) to give ammonia.

$$N_2 + 3H_2 \xrightarrow[200\ \text{atm}]{400-500^\circ C} 2NH_3$$

2. With Oxygen
   It reacts with oxygen at 3000°C in presence of electric arc to give nitric oxide.

$$N_2 + O_2 \xrightarrow{3000^\circ C} 2NO$$

3. With Metals
   It forms metals nitrides at red hot temperature.
   For example,

$$2Al + N_2 \xrightarrow{1073\ K} 2AlN$$

$$3Mg + N_2 \xrightarrow{723\ K} Mg_3N_2$$

$$6Li + N_2 \xrightarrow{723\ K} 2Li_3N$$

4. With Nonmetals
   It reacts with boron or silicon at bright red heat to form boron and silicon nitrides, respectively.

$$2B + N_2 \longrightarrow 2BN$$

$$3Si + 2N_2 \longrightarrow Si_3N_4$$

5. With Calcium Carbide
   It forms calcium cyanamide (nitrolim) with calcium carbide at 1273 K as follows:

$$\underset{}{CaC_2} + N_2 \longrightarrow \underset{\text{Nitrolim}}{CaCN_2} + C$$

6. With Al_2O_3
   Nitrogen on heating with alumina and carbon gives aluminium nitride as follows:

$$\underset{\text{Alumina}}{Al_2O_3} + N_2 + 3C \xrightarrow{2100\ K} \underset{\text{Aluminium nitride}}{2AlN} + 3CO$$

Uses

1. It is used in the preparation of ammonia and some other chemicals like calcium cyanamide, nitric acid etc.
2. Liquid nitrogen is used as a refrigerant to preserve biological materials and in freezing food materials.
3. It is also used in cryosurgery.
4. It is used for providing inert atmosphere in several metallurgical operations.
5. It is used as an inert diluent for reactive chemicals.
6. It is also used for filling electric bulbs.

- **Active Nitrogen:** Nitrogen gas on electric discharge at low pressure gives active nitrogen which is more reactive but very less stable and changes into normal form again.

PHOSPHOROUS

It was discovered by Brand and Scheeley. It was isolated from bone ash and Lavoisier confirmed its elemental nature. Since it glows in night it is called phosphorous. It is poisonous but essential for growth and maintenance.

Occurrence

Phosphorous is a very reactive element and occurs mainly in the form of phosphate minerals in the earth crust. It is an essential constituent of plants and animals. It is mainly present in bones and teeth in the form of calcium phosphate and also in animals cells (in DNA and RNA). Phosphoprotein is present in brain, milk, eggs.

Ores

Some important ores of phosphorous are as follows:

Phosphorite or rock phosphate: $Ca_3(PO_4)_2$

Chloroapatite: $3Ca_3(PO_4)_2 \cdot CaCl_2$

Fluorapatite: $3Ca_3(PO_4)_2 \cdot CaF_2$

Hydroxyapatite: $Ca_5(PO_4)_3OH$

or

$3Ca_3(PO_4)_2 \cdot Ca(OH)_2$

Extraction

It is extracted mainly from phosphorite ore by using the following methods:

1. Old Process or Retort Process

   Here phosphorous is extracted from bone ash which has mainly calcium phosphate. When bone ash is heated with concentrated sulphuric acid insoluble in calcium sulphate and ortho phosphoric acids are formed.

$$Ca_3(PO_4)_2 + 3H_2SO_4 \longrightarrow 3CaSO_4\downarrow + 2H_3PO_4$$

   From here calcium sulphate is removed by filtration and the syrupy ortho phosphoric acid is evaporated which on decomposition gives meta phosphoric acid.

$$H_3PO_4 \longrightarrow HPO_3 + H_2O$$

   Now meta phosphoric acid is mixed with powdered coke and distilled in fire clay retorts at red hot temperature to get phosphorous which is vapourized and vapours are condensed under water.

$$4HPO_3 + 10C \xrightarrow{\text{Red hot}} P_4 + 10CO + 2H_2O$$

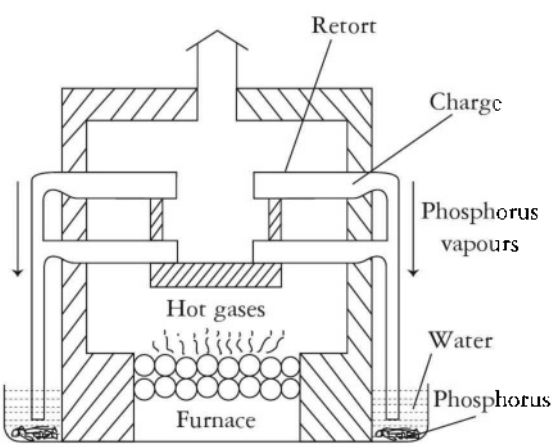


Fig 3.1 Retort process

2. Electrothermic Process

   Here phosphorous is prepared by heating a mixture of phosphate rock, silica and coke in an electric furnace at 1500°C. This is not an electrolytic process. The electricity is used for this purpose.

   The solid mixture is fed into the furnace. The more volatile phosphorous pentaoxide P_4O_{10} is first

displaced from calcium phosphate by nonvolatile silica, SiO_2.

$$Ca_3(PO_4)_2 + 3SiO_2 \xrightarrow[1500°C]{\Delta} 3CaSiO_3 + P_2O_5$$

which is then reduced to phosphorous by coke and carbon monoxide is formed.

$$P_4O_{10} + 10\,C \xrightarrow{\Delta} P_4 + CO\uparrow$$

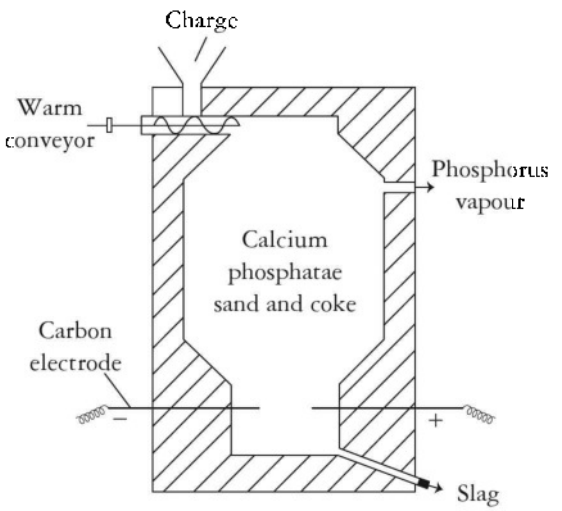


Fig 3.2 Electro thermic process

Purification

The impure phosphorous is first melted under acidified solution of $K_2Cr_2O_7$ so that the impurities get oxidized and can be collected and removed from the surface of the molten phosphorous. Phosphorous is then redistilled in absence of air to get the vapours of pure white phosphorous which are condensed under water.

OXYGEN (O_2)

It was discovered by Priestley and Scheele and called it the vital air or fire air. Lavoisier named it oxygen (*Oxus* means acid and *gennas* means maker).

Occurrence

In freestate oxygen is present in air. Air has 21 percent O_2 by volume or 23 percent by weight. Earth crust has 46.5 percent oxygen (most abundant element in it). In combined state it exists in water, nitrates, sulphates etc. It is an essential constituent of all living beings. The animal and plant tissues have nearly 50–70 percent oxygen.

The dioxygen present in atmosphere is mainly due to photosynthesis of green plants.

$$nH_2O + nCO_2 \xrightarrow{\text{sun light}} (CH_2O)n + nO_2$$

- Naturally occurring oxygen has three isotopes ${}_8O^{16}$, ${}_8O^{17}$, ${}_8O^{18}$ in abundance ratio 99.76 percent, 0.037 percent and 0.204 percent, respectively.

Preparation

1. By Thermal Decompozition of Oxygen Rich Salts
   It can also be prepared by heating oxygen rich salts like $K_2Cr_2O_7$, $KMnO_4$ etc.,

$$2KMnO_4 \xrightarrow{\Delta} \underset{\text{Potassium manganate}}{K_2MnO_4} + MnO_2 + O_2$$

$$2K_2Cr_2O_7 \xrightarrow{\Delta} 2K_2O + 2Cr_2O_3 + 3O_2$$

$$2KClO_4 \xrightarrow{\Delta} 2KCl + 4O_2$$

$$2KNO_3 \xrightarrow{\Delta} 2KNO_2 + O_2$$

2. By Heating Metal Oxides
   Some metal oxides on heating gives oxygen.
   For example,

$$2HgO \xrightarrow{720\ K} 2Hg + O_2$$

$$2Ag_2O \xrightarrow{620\ K} 4Ag + O_2$$

$$2BaO_2 \xrightarrow{\Delta} 2BaO + O_2$$

$$2H_2O_2 \xrightarrow{\Delta} 2H_2O + O_2$$

$$2Na_2O_2 + 2H_2O \longrightarrow 4NaOH + O_2$$

$$2PbO_2 \longrightarrow 2PbO + O_2$$

$$2KBrO_3 \xrightarrow[MnO_2]{\Delta} 2KBr + 3O_2$$

3. By the Action of Water on Acidified KMnO4 and Sodium Peroxide

$$2KMnO_4 + 5Na_2O_2 + 8H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 5Na_2SO_4 + 8H_2O + 5O_2$$

4. Laborarory method
   In Laborarory it is prepared by heating a mixture of potassium chlorate and MnO_2 in 4:1 ratio at 420K.

$$2KClO_3 \xrightarrow[MnO_2]{420\ K} 2KCl + 3O_2$$

- The O_2 formed can be collected by downward displacement of water. Here MnO_2 lowers the decomposition temperature of $KClO_3$ and acts as a catalyst.

Physical Properties

1. It is a colourless, tasteless and odourless gas which is heavier than air.
2. It is partially soluble in water and quite soluble in alkaline pyrogallol which absorbs it.
3. It can be liquefied into a pale blue liquid (90K) by compressing it at low temperature.
4. It can be further solidified into a blue solid (54.4K) by cooling.
5. It is paramagnetic in all physical states.
6. It is a noncombustible gas, however it supports combustion very highly.

CHEMICAL PROPERTIES

1. Reactions with Nonmetals
   Most of the nonmetals burn in oxygen giving acidic oxides.
   For example,

$$C + O_2 \longrightarrow CO_2$$

$$S + O_2 \longrightarrow SO_2$$

$$N_2 + O_2 \xrightarrow{3000^\circ C} 2NO$$

2. **Reaction with Metals**
   Nearly all metals burn in air or oxygen giving basic oxides.
   For example,

$$4Na + O_2 \longrightarrow 2Na_2O$$

$$2Mg + O_2 \longrightarrow 2MgO$$

$$4Fe + 3O_2 \xrightarrow{\Delta} 2Fe_2O_3$$

3. Reaction with Ammonia
   When it is heated with ammonia in presence of red hot platinum gauge catalyst at 800°C, nitric oxide is formed.

$$4NH_3 + 5O_2 \xrightarrow{Pt\ gauge} 4NO + 6H_2O$$

4. With Sulphur Dioxide
   It oxidizes sulphur dioxide into SO_3 as follows:

$$2SO_2 + O_2 \xrightarrow{Pt\ asbestos,\ \Delta} 2SO_3$$

5. With Carbon Disulphide
   It changes CS_2 into CO_2 and SO_2 as follows:

$$CS_2 + 3O_2 \longrightarrow CO_2 + 2SO_2$$

6. With Hydrogen Chloride
   It oxidizes HCl into chlorine at 425°C in presence of cupric chloride.

$$4HCl + O_2 \xrightarrow{Cupric\ chloride,\ \Delta} 2Cl_2 + 2H_2O$$

7. With Metal Sulphides
   It changes metal sulphides into their oxides at high temperature as follows:

$$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$$

$$2HgS + 3O_2 \longrightarrow 2HgO + 2SO_2$$

8. Oxidative Reactions
   It oxidizes a number of organic compounds in presence of catalyst.
   For example,

$$2CH_3OH + O_2 \xrightarrow{Pt} 2HCHO + 2H_2O$$

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$$

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

Uses

1. A mixture of O_2 and He or CO_2 is used in artificial respiration.
2. A mixture of powdered charcoal and liquid oxygen is used an explosive in coal mining.
3. Liquid O_2 is rocket fuel component.
4. It is used in metallurgical process for removing the impurities of metals by oxidation.
5. It is used in the preparation of nitric acid, sulphuric acid, phenol etc.

- Liquid O_2 $\xrightarrow{-218^\circ C}$ Solid O_2
  Pale blue White
- Liquid air can be converted into O_2 by fractional distillation (Cloud method).

Structure

The structure of oxygen molecule is as follows:

$$:\ddot{O}=\ddot{O}:$$

This structure fails to explain the paramagnetic nature of oxygen hence a new structure of O_2 was introduced which is given as

:O:::O:

This structure could not explain high dissociation energy of O_2 molecule and also does not obey octet rule.

Pauling proposed the following structure of oxygen.

This structure explains paramagnetic nature, high dissociation energy of oxygen but does not follow octet rule.

ABNORMAL BEHAVIOUR OF OXYGEN

Oxygen shows some abnormal behaviour to other elements of its group. It is due to its small size, high electronegativity and absence of d-orbital in its valence shell.

For example,

1. Atomicity and Physical State
   Oxygen is a diatomic gas and readily forms multiple bonds while other elements are solids and have puckered eight membered rings (For example, S_8 and Se_8).
2. Oxidation State
   It shows −2 oxidation state in its compounds mainly while the other elements of this group can show +2, +4 and +6 oxidation state also.
3. Magnetic Nature
   Oxygen is paramagnetic while others are diamagnetic.
4. Hydrides
   H_2O, the hydride of oxygen, forms strong hydrogen bonds so it is liquid at room temperature while others hydrides of this group elements do not form hydrogen bonding and are gases.
5. Hydrogen Bonding
   As it is most electronegative element in its group hence its compounds like H_2O, ROH, RCOOH etc., have hydrogen bonding while other elements of the group do not have hydrogen bonding due to their less electronegativity and large size.
6. Nature of Compounds
   Its forms more ionic compounds than the other elements of its group.
7. Formation of Multiple Bonds
   It can easily form multiple bonds with C, N, or elements of comparable size while rest elements of the group have no tendency to form such bonds.

SULPHUR (S)

The term sulphur is derived from Sanskrit word 'sulveri' which means 'killer of copper'. Lavoisier established its elemental nature.

Occurrence

It occurs in free state in volcanic regions. In combined state, it occurs in the form of sulphide and sulphate ores.

Celestine: $SrSO_4$

Gypsum: $CaSO_4.2H_2O$

Copper pyrites: $Cu_2S\ Fe_2S_3$

Galena: PbS

Zinc blende: ZnS

Iron pyrites: FeS_2

EXTRACTION

1. Frasch Method or Louisiana Method
   In Louisiana and Texas, sulphur is found as deposits of thickness of nearly 125 feet present at depth of 500 1000 feet beneath the surface of the earth.
2. From Iron Pyrites
   It is a good source of sulphur. On distillation gives impure sulphur which can be changed into pure form by boiling it at 720°C.

$$3FeS_2 + 5O_2 \xrightarrow[\text{Limited}]{\text{distill}} Fe_3O_4 + 3SO_2 + 3S$$

$$FeS + CO_2 \xrightarrow{1000\,^\circ C} FeO + CO + S$$

3. From Natural Gas
   Natural gas has a considerable amount of H_2S which is separated by absorbing it in mono ethanolamine. From this H_2S, sulphur can be obtained as follows:

$$H_2S + \tfrac{1}{2}\,O_2 \xrightarrow{\text{Low temp.}} 1/8\,S_8 + H_2O$$

$$2H_2S + 3O_2 \rightarrow 2SO_2 + 2H_2O$$

$$2H_2S + SO_2 \xrightarrow{Fe_2O_3,\ 303\ K} 3/8\ S_8 + 2H_2O$$

4. From Spent Oxides of Coal Gas
   It has mainly ferric sulphide and from it sulphur is obtained as follows:

$$Fe_2O_3 + 3H_2S \longrightarrow Fe_2S_3 + 3H_2O$$

$$2Fe_2S_3 + 3O_2 + 2H_2O \rightarrow 2Fe_2O_3 + 6S + 2H_2O$$

5. From Alkali waste of Leblanc Process
   When the alkali waste containing CaS mainly suspended in water and carbon dioxide is circulated, H_2S is formed which on burning in insufficient amount of air or oxygen gives sulphur.

$$CaS + H_2O + CO_2 \longrightarrow H_2S + CaCO_3$$

$$2H_2S + O_2 \rightarrow 2H_2O + 2S$$

Physical Properties

1. It is pale yellow, brittle crystalline solid having a faint smell.
2. It is insoluble in water but soluble in organic solvents like benzene, CS_2, turpentine etc.
3. It is a poor conductor of heat but a bad conductor of electricity.
4. It shows allotropy.

Chemical Properties

1. Burning
   It burns in air with a pale blue flame to give SO_2 and some traces of SO_3.

$$S + O_2 \xrightarrow{\Delta} SO_2$$

$$2SO_2 + O_2 \overset{\Delta}{\rightarrow} 2SO_3$$

2. Reaction with Metals
   It reacts with metals like Cu, Fe, Zn etc., on heating to give their sulphides.

$$Fe + S \rightarrow FeS$$

$$Hg + S \longrightarrow HgS$$

3. Reaction with Nonmetals
   It reacts with nonmetals like C, P, As etc., on heating to give their sulphides.

$$C + 2S \rightarrow CS_2$$

$$2P + 5S \longrightarrow P_2S_5$$

$$2As + 3S \rightarrow As_2S_3$$

When hydrogen and chlorine is passed through boiling sulphur following reactions occur.

$$S + H_2 \rightarrow H_2S$$

$$2S + Cl_2 \longrightarrow S_2Cl_2$$

4. Reaction with Acids
   It reduces hot and concentrated acids into their oxides as follows:

$$S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$$

$$S + 6HNO_3 \longrightarrow H_2SO_4 + 6NO_2 + 2H_2O$$

5. Reaction with Alkalies
   Sulphur dissolves in hot alkali like NaOH to form sulphide and thiosulphates as follows:

$$4S + 6NaOH \xrightarrow{\text{Hot}} 2Na_2S + Na_2S_2O_3 + 3H_2O$$

$$4S + 6KOH \rightarrow 2K_2S + K_2S_2O_3 + 3H_2O$$

If sulphur is in excess Na_2S reacts with it to give sodium penta sulphide.

$$\underset{\text{(Excess)}}{Na_2S} + 4S \rightarrow \underset{\text{Sodium penta sulphide}}{Na_2S_5}$$

6. Formation of Polysulphides
   Many sulphides in their aqueous solutions can further combine with sulphur to give polysulphides of molecular formula M_2S_n. Here, M is a monovalent metal or radical and the value of n may be 2 to 5.

$$Na_2S + 4S \longrightarrow \underset{\text{Sodium pentasulphide}}{Na_2S_5}$$

$$(NH_4)_2S + (n-1)S \rightarrow \underset{\text{Yellow ammonium sulphide}}{(NH_4)_2S_n}$$

Yellow ammonium sulphide is an important analytical reagent which is used for the separation of IIA group radicals from IIIB group radicals in qualitative analysis.

Uses

1. It is used in the manufacture of fire works, gun powder, matches etc.

2. It is also used in the manufacrure of many medicines and dyes.
3. It is used in manufacture of many sulphides like H_2S, CS_2 etc.
4. It is also used in the manufacture of SO_2, SO_3 etc.
5. It is also used in the vulcanizarion of rubber.

HALOGENS FLUORINE (F_2)

It was discovered by Scheele and first prepared by Moissan.

Occurrence

Fluorine does nor occur free in narure as it is highly reactive and presenr in combined srare as ores. Traces of fluoride occur in sea warer, bones, milk, blood, reeth erc. Some importanr ores of fluorine arc as follows:

Fluorspar: CaF_2

Cryorile: Na_3AlF_6

Fluoraparire: $[3Ca_3(PO_4)_2.CaF_2]$

Late Discovery of Fluorine: Fluorine was discovered very lare due ro rhe following reasons:

1. Highly reactive nature (reacts even with glass container).
2. Oxidation potential is very low.
3. HF is very srable, bad conducror poisonous and corrosive.

PREPARATION

1. Moissan Method
   Moissan prepared F_2 from electrolysis of KHF_2 and HF (anhydrous) in 1:12 ratio, in a Pt–Ir–V tube at anode at $-23^{\circ}C$. The HF presenr in fluorine can be removed by passing over NaF as $NaHF_2$.
   Reactions

   $$KHF_2 \rightarrow KF + HF$$

   $$KF \longrightarrow K^+ + F^-$$

   Ar carhode

   $$K^+ + e^- \longrightarrow K$$

   $$2K + 2HF \rightarrow 2KF + H_2$$

   Ar anode

   $$2F^- \longrightarrow F_2 + 2e^-$$

   $$HF + NaF \rightarrow NaHF_2$$

2. Modern Methods
   In these methods fluorine is obtained by the electrolysis of fused fluorides like KHF_2 mainly in electrolytic cells made of Cu, Ni, Monel metal etc.

   Here, it is necessary that the electrolyte and the vessel in which fluorine is to be collected must be dry as fluorine reacts with moisture to give O_2 and O_3 mixture.

   The fluorine obtained must be free from HF before storing by passing it through NaF as HF will attack the vessel due to its more corrosive nature.

(i) **Dennis Method:** Here a V-shaped copper rube is used as the elecrrolyric cell in which graphire elecrrodes are sealed by the caps as shown in Fig. 3.3. When rhe elecrrolyre KHF_2 is made complerely dry at 403 K, elecrrolysis is carried our. The liberated fluorine is collered ar anode which is passed rhrough copper U-rubes ro remove HF vapours.

Reactions: Just like in Moissan method.

$$NaF + HF \rightarrow NaHF_2$$

■ Here, there is always a danger of explosion due to intermixing of H_2 and F_2.

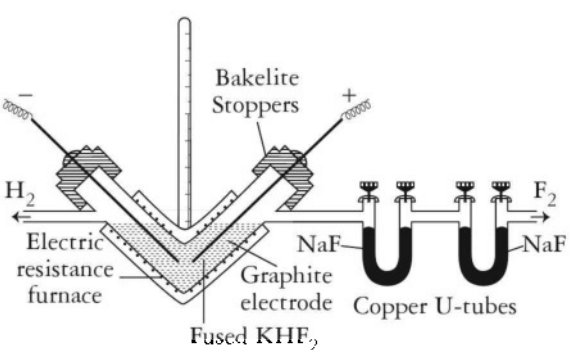


Fig. 3.3 Dennis' Method

(ii) **Whytlay Gray Method:** Whytlay and Gray process is also used ro prepare fluorine. Here rhe elecrrolyre i.e., fused KHF_2 (HF + KF in 2:1 ratio) in elecrrolyric cell (copper vessel) is elecrrolysed ro give fluorine ar anode. Here, ar carhode hydrogen is liberared.

Reactions are similiar as in Moissan method.

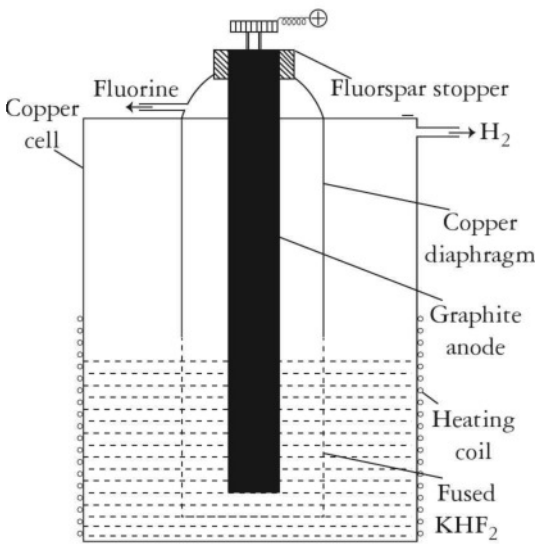


Fig. 3.4 Whytlay Method

Physical Properties

1. It is a paleyellow coloured gas with pungent smell which is heavier than air and poisonous in nature.
2. It is convertible and pale yellow liquid at 86 K and into yellow crystalline solid at 55 K.
3. It is diamagnetic in nature.
4. It is most reactive among all the halogens.
5. It is stored in wax vessel as it can react even with glass vessel.
6. F_2 is the strongest oxidizing agent. It is always reduced therefore does not show disproportionation reaction.

Chemical Properties

1. Reaction with Elements
   Being very highly reactive it reacts with metals as well as nonmetals and even with inert gas like Xe except He, Ar, N_2 and O_2 even under normal conditions.

$$H_2 + F_2 \rightarrow 2HF + 1170 \text{ kJ (violent reaction)}$$

$$2B + 3F_2 \rightarrow 2BF_3$$

$$C + 2F_2 \longrightarrow CF_4$$

$$S + 3F_2 \rightarrow SF_6$$

$$Mg + F_2 \longrightarrow MgF_2$$

$$2Ag + F_2 \rightarrow 2AgF$$

2. With Halogens
   It can form interhalogen compound with all halogens like IF_5, IF_7 etc.

$$I_2 + 5F_2 \xrightarrow{20°C} 2IF_5$$

$$I_2 + 7F_2 \xrightarrow{250-300°C} IF_7$$

$$Cl_2 + 3F_2 \xrightarrow{300°C} 2ClF_3$$

3. With Metal Halides
   It liberates halogens (Cl_2, Br_2, I_2) from metal halides due to its high reactivity and more oxidizing nature.

$$F_2 + 2MX \longrightarrow 2MF + X_2$$

4. Reaction with Alkalies
   It reacts with dilute and concentrated alkali as follows:

$$2F_2 + 2NaOH_{\text{dil.}} \rightarrow 2NaF + OF_2 + H_2O$$

$$2F_2 + 4NaOH \longrightarrow 4NaF + O_2 + 2H_2O$$

5. Oxidizing Properties
   Being an strong oxidant it can oxidize following compounds like chlorates into perchlorates, iodates into periodates etc., as follows:

$$KClO_3 + F_2 + H_2O \longrightarrow KClO_4 + H_2F_2$$

$$KIO_3 + F_2 + H_2O \rightarrow KIO_4 + H_2F_2$$

6. With Oxygen and Xenon
   It forms OF_2, O_2F_2 with oxygen and XeF_2, XeF_4, XeF_6 with xenon.

$$Xe + F_2 \xrightarrow[400°C]{\text{Ni tube}} XeF_2$$

2 : 1

$$Xe + 2F_2 \xrightarrow[400°C]{\text{Ni tube}} XeF_4$$

1 : 5

$$Xe + 3F_2 \xrightarrow[300°C,\ 50-60\ \text{atm}]{\text{Ni tube}} XeF_6$$

1 : 20

7. Fluorinarion of Merhyl Alcohol
   Merhyl alcohol can be fluorinated by fluorine as follows:

$$CH_3OH + 4F_2 \xrightarrow{AgF_2} CF_3OF + 4HF$$

Uses

1. It is used in rockets as a fuel and as an insecticide.
2. It is used to prepare UF_6 which is used for nuclear power generation, which used in separation of uranium isotopes.
3. NaF is used in fluoride rooth pasre.
4. CF_2Cl_2 or freon-12 is used in refrigerator and air conditioner for cooling.
5. CF_3CHBrI is fluothane an anaesthetic agent.
6. Teflon $(C_2F_4)_n$ used in coaring nonsrick pans, insularors.
7. SF_6 is a best insularor and used in X-ray and high volrage machine.

Abnormal Behaviour of Fluorine

It shows variation in properries rhan other halogen and due to rhe following reasons:

(a) Small atomic size

(b) High electronegativity

(c) Low bond energy

(d) Absence of 2d-orbitals

For example, most electronegative, most reactive, maximum ionic character, most stable salts (HF), strongest oxidizing agent, no formation of polyhalides and other than monovalent compounds (F^-_3 not possible).

(i) Reactivity: It is the most reactive halogen due to low dissociation energy of F – F bond.

(ii) Oxidation State: It always shows –1 oxidation state in its compounds as it is most electronegative element and does not have any d-vacant orbital. However, other halogens can also show +1 to +7 oxidation state in their compounds.

(iii) Hydrogen Bonding: It can form hydrogen bonding in its hydrides due to its high electronegarivity and small size while other halogens cannot do so.

(iv) Nature of Compounds: Being most electronegative element, it has a strong tendency to form ionic compounds while other halogens form less ionic compounds and covalent compounds.

(v) Polyhalide Ions: It cannot form X_3^- type of polyhalides due to absence of vacant d-orbitals while other halogens can form such ions like I_3^-, Cl_3^- etc.

(vi) Dissolution in Water: It dissolves in water to give a mixture of O_2 and O_3 while other halogens Cl_2, Br_2 react with water in sunlight and iodine does not react as follows:

$$2F_2 + 2H_2O \rightarrow 4HF + O_2$$

$$3F_2 + 3H_2O \longrightarrow 6HF + O_3$$

$$Cl_2 + H_2O \rightarrow HCl + HClO$$

(vii) Nature of HF: HF shows difference in its properties when compared to other HX due to hydrogen bonding.

For example, it is a liquid while rest HX are gases under ordinary conditions.

It is weak acid due to high dissociation energy of H – F while rest are strong acids due to less dissociation energy of HX bond.

It also exists as H_2F_2 in dimeric form while other HX are always monomers.

(viii) It can form hexachloride with sulphur i.e., SF_6 while other halogens cannot do so.

(ix) It can form any oxyacid while the other halogens forms a number of oxyacids.

It has a strong tendency to form complexes like $[AlF_6]^{3-}$, $[FeF_6]^{3-}$ while other halogens have low tendency to form complexes.

REMEMBER

- F^- does not form precipitate with Ag^+ as AgF is soluble in H_2O due to high hydration energy.
- AgF has more lattice energy than other AgX.
- HF can be oxidized by anode only.
- Dennis and Whytlay–Grey process is also used to prepare F_2.
- H_2F_2 dissolves glass (Etching of glass) forming Hydrofluorosilicic acid (H_2SiF_6).
- CaF_2 is insoluble in water.

CHLORINE

It was discovered by Scheele and named by Davy.

Occurrence

It occurs in combined state mainly in nature in the form of following chloride ores:

Sea water: [NaCl]

Carnalite: $MgCl_2.KCl.6H_2O$

Horn Silver: AgCl

Sylvine: KCl

Chlorapatite: $3Ca_3(PO_4)_2.CaCl_2$

Preparation

1. By the Oxidation of Hydrochloric Acid
   Hydrochloric acid can be oxidized into chlorine by using any of these oxidants MnO_2, $KMnO_4$, $K_2Cr_2O_7$, O_3, Pb_3O_4, PbO_2, etc.

   $$HCl + \text{oxidizing agent} \rightarrow Cl_2\uparrow$$

   For example,

   $$K_2Cr_2O_7 + 14HCl \rightarrow 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$$

   $$2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$$

   $$Pb_3O_4 + 8HCl \longrightarrow 2PbCl_2 + 4H_2O + Cl_2$$

   $$NaClO + 2HCl \rightarrow NaCl + H_2O + Cl_2$$

2. From Metal Chlorides
   When any metal chloride is heated with conc. H_2SO_4 in presence of MnO_2, chlorine is formed as follows:

   $$MCl + MnO_2 + \text{conc. } H_2SO_4 \xrightarrow{\Delta} Cl_2\uparrow$$

   For example,

   $$2NaCl + MnO_2 + 3H_2SO_4 \rightarrow 2NaHSO_4 + MnSO_4 + 2H_2O + Cl_2\uparrow$$

3. Laboratory Method
   In lab, chlorine is formed as follows:

(i) By the oxidation of HCl using MnO2: When concentrated HCl is heated with MnO_2 in a round bottom flask, chlorine is formed.

$$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$$

(ii) By the oxidation of HCl using $KMnO_4$: When cold and concentrated HCl is treated with $KMnO_4$, chlorine is formed.

$$2KMnO_4 + 16HCl \longrightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$$

In both these methods, chlorine is first passed through water and then through concentrated H_2SO_4 to remove HCl gas and water vapours, respectively. Here, chlorine is collected by upward displacement of air method.

4. From Platinic Chloride or Gold Chloride
   When these are heated in a hard glass tube, pure chlorine is obtained as follows:

   $$PtCl_4 \xrightarrow{647\ K} PtCl_2 + Cl_2 \xrightarrow{855\ K} Pt + 2Cl_2$$

   $$2AuCl_3 \xrightarrow{448\ K} 2AuCl + 2Cl_2 \xrightarrow{458\ K} 2Au + 3Cl_2$$

5. Deacon's Method
   Here chlorine can be obtained by the oxidation of HCl by air in presence of catalyst cupric chloride at 400–450°C.

   $$4HCl + \underset{\substack{\text{air}\\ 4:1}}{O_2} \xrightarrow{CuCl_2} Cl_2 + 2H_2O$$

   Here, the catalytic action of cupric chloride can be explained by the mechanism given as under.

   $$2CuCl_2 \xrightarrow{\text{High temp.}} Cu_2Cl_2 + Cl_2$$

   $$2Cu_2Cl_2 + O_2\ (\text{air}) \longrightarrow \underset{\text{Copper oxychloride}}{2Cu_2OCl_2}$$

   $$Cu_2OCl_2 + 2HCl \rightarrow 2CuCl_2 + H_2O$$

   Here, chlorine is associated with HCl, air (O_2, N_2) and steam. From it HCl and water can be removed by passing it through washing tower and drying tower, respectively. Now chlorine has impurities of O_2 and N_2. Cl_2 is made moisture free by conc. H_2SO_4.

6. By the Electrolysis of Brine Solution
   Cl_2 is also prepared by the electrolysis of brine (aq NaCl) at anode by using Nelson, Castner-Kelner or Salvey cell etc.

   $$NaCl \rightleftharpoons Na^+ + Cl^-$$

   $$H_2O \rightleftharpoons H^+ + OH^-$$

At Cathode

$H+ + e^- \longrightarrow H$

$H + H \rightarrow H_2$

At Anode

$Cl \rightarrow Cl + e$

$Cl + Cl \longrightarrow Cl_2$

PHYSICAL PROPERTIES

1. It is a greenish yellow gas having a pungent suffocating smell and poisonous in nature.
2. It is soluble in water and its aqueous solution is known as chlorine water.
3. It is 2.5 times heavier than air.
4. It can be easily liquefied and solidified.

Chemical Properties

1. With Hydrogen
   It has a great affinity with hydrogen and reacts with it in dark also.

   $H_2 + Cl_2 \longrightarrow 2HCl$

2. Reaction with Metals and NonMetals
   It combines with metals as well as nonmetals to form their chlorides at ordinary temperature except with N_2, O_2 and noble gases.

   $2B + 3Cl_2 \longrightarrow 2BCl_3$

   $2P + 3Cl_2 \rightarrow 2PCl_3$

   $2As + 3Cl_2 \longrightarrow 2AsCl_3$

   $2S + Cl_2 \rightarrow S_2Cl_2$

   $2Na + Cl_2 \longrightarrow 2NaCl$

   $2Al + 3Cl_2 \rightarrow 2AlCl_3$ (Bright sparks)

   $2Fe + 3Cl_2 \longrightarrow 2FeCl_3$

3. Reaction with Alkalies
   It can react with dilute as well as concentrated NaOH to give following products as follows:

   $Cl_2 + 2NaOH \longrightarrow NaClO + NaCl + H_2O$
   dil. and cold — Sodium hypochlorite

   $Cl_2 + 2KOH \rightarrow KClO + KCl + H_2O$
   dil. and cold — Potassium hypochlorite

   $3Cl_2 + 6NaOH \rightarrow NaClO_3 + 5NaCl + 3H_2O$
   conc. and hot — Sodiumchlorate

   $3Cl_2 + 6KOH \longrightarrow KClO_3 + 5KCl + 3H_2O$
   conc. and hot — Potassium chlorate

   $2Cl_2 + 2Ca(OH)_2 \rightarrow Ca(ClO)_2 + CaCl_2 + 2H_2O$
   cold and dil. — Calcium hypochlorite

   $6Cl_2 + 6Ca(OH)_2 \longrightarrow Ca(ClO_3)_2 + 5CaCl_2 + 6H_2O$
   Hot and conc. — Calcium hypochlorate

   It reacts with dry slaked lime to give bleaching powder.

   $Cl_2 + Ca(OH)_2 \rightarrow CaOCl_2.H_2O$
   Slaked lime — Bleaching powder

4. Oxidizing and Bleaching Nature
   It decomposes water forming HCl and HClO which is unstable and decomposes giving nascent oxygen which is responsible for the oxidizing and bleaching action of chlorine.

   $Cl_2 + H_2O \rightarrow HCl + HOCl$

   $HOCl \xrightarrow{\Delta} HCl + [O]$

   Coloured substance + [O] → Colourless substance

   The bleaching action of chlorine is due to oxidation hence it is permanent.

   It oxidizes potassium bromide and iodide into bromine and iodine as follows:

   $2KBr + Cl_2 \rightarrow 2KCl + Br_2$

   $2KI + Cl_2 \longrightarrow 2KCl + I_2$

   It oxidizes ferrous salts into ferric salts as follows:

   $2FeCl_2 + Cl_2 \longrightarrow 2FeCl_3$

   $2FeSO_4 + H_2SO_4 + Cl_2 \rightarrow Fe_2(SO_4)_3 + 2HCl$

   It oxidizes sulphites into sulphates as follows:

   $Na_2SO_3 + H_2O + Cl_2 \rightarrow Na_2SO_4 + 2HCl$

   It can oxidize hypo into sodium thio sulphate.

   $Na_2S_2O_3 + H_2O + Cl_2 \rightarrow Na_2SO_4 + 2HCl + S$

   It oxidizes moist SO_2 into H_2SO_4 as follows:

   $SO_2 + 2H_2O + Cl_2 \rightarrow Na_2SO_4 + 2HCl$

5. With Ammonia
With ammonia it reacts as follows:

$$NH_3 + 3Cl_2 \longrightarrow \underset{\text{Explosive}}{NCl} + 3HCl$$
(Excess)

$$8NH_3 + 3Cl_2 \rightarrow 6NH_4Cl + N_2\uparrow$$
(Excess)

6. Addition Reaction
It undergoes addition reaction with CO, SO_2 and unsaturated hydrocarbons like ethene as follows:

$$CO + Cl_2 \rightarrow COCl_2$$

$$SO_2 + Cl_2 \longrightarrow \underset{\text{Sulphuryl chloride}}{SO_2Cl_2}$$

$$C_2H_4 + Cl_2 \rightarrow \underset{\text{Ethylene chloride}}{C_2H_4Cl_2}$$

Uses

1. Cl_2 is used as a bleaching agent and a water purifier.
2. It is used to prepare bleaching powder, D.D.T, $CHCl_3$ etc.
3. It is used to manufacture war gas like $COCl_2$ (phosgene), $C_2H_4Cl.S.C_2H_4Cl$ (mustard gas), $CCl_3.NO_2$ (tear gas).
4. It is used in the extraction of gold and platinum also.
5. It is used to bleach vegetable, coloured matter, paper textiles but not for silk and wool (destroyable by chlorine).
6. It is also used in the preparation of insecticides like B.H.C. (gammaxene) and D.D.T. (di-chloro diphenyl trichloroethane)

BROMINE

It was discovered by Ballard.

Occurrence

It is found in combined state in the forms of ores like

Carnalite: $KCl.MgCl_2.6H_2O + MgBr_2$ (.01 – 0.1%)

Bromocarnalite: $KBr.MgCl_2.6H_2O$

Bromagyrite: AgBr

In sea water many bromides like NaBr, KBr, $MgBr_2$ etc., are present in small amounts. These bromides are also present in mineral springs and salt lakes.

Preparation

1. Laboratory Method
In laboratory, bromine can be prepared by treating metal bromides with concentration H_2SO_4 in presence of MnO_2 as follows:

$$M - Br + H_2SO_4 + MnO_2 \rightarrow Br_2\uparrow$$

For example,

$$2KBr + 3H_2SO_4 + MnO_2 \rightarrow 2KHSO_4 + MnSO_4 + 2H_2O + Br_2$$

2. By Passing Chlorine in Meral Bromides
When chlorine is passed through metals bromides, bromine is released as chlorine is more reactive so replaces bromine from bromides.

$$2M - Br + Cl_2 \longrightarrow 2MCl + Br_2\uparrow$$

For example,

$$2KBr + Cl_2 \longrightarrow 2KCl + Br_2\uparrow$$

$$MgBr_2 + Cl_2 \rightarrow MgCl_2 + Br_2\uparrow$$

3. From a mixture of Potassium Bromide and Potassium Bromate
When HCl is added to this mixture bromine is formed as follows:

$$5\,KBr + KBrO_3 + 6HCl \rightarrow 3KCl + 3Br_2 + 3H_2O$$

4. From Carnalite
First remove KCl, $MgCl_2$ and other halides (except $MgBr_2$) from carnalite mother liquor by fractional crystallization. Now, chlorine is passed through the solution having $MgBr_2$ mainly to obtained bromine as follows:

$$MgBr_2 + Cl_2 \longrightarrow MgCl_2 + Br_2\uparrow$$

Those vapours which can not be condensed are passed over moist iron filling to get iron bromides as follows:

$$Fe + Br_2 \longrightarrow FeBr_2$$

$$3FeBr_2 + Br_2 \rightarrow \underset{\text{Ferroso-ferric bromide}}{Fe_3Br_8}$$

Bromine obrained by this method has the impurities of Cl_2 and I_2 which can be removed by distilling it with KBr (to remove chlorine) and with zinc oxide to remove iodine.

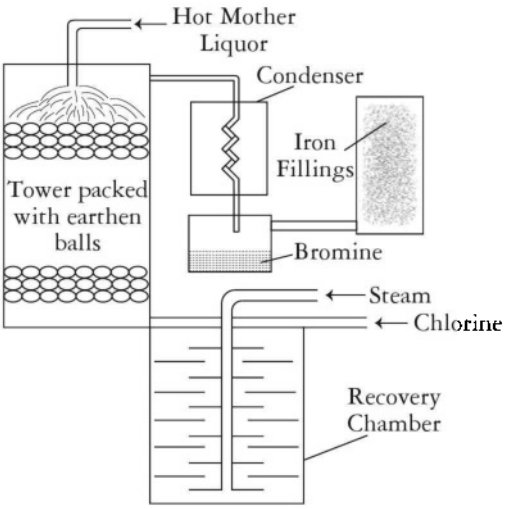


Fig. 3.5 Carnalite

5. From Sea Water
Sea water, the major source of bromine is first of all slightly acidified and then treated with chlorine to obtain bromine as follows:

$$Cl_2 + \underset{\text{(sea water)}}{\text{Bromides}} \rightarrow \text{Chlorides} + \text{Bromine}$$

When the bromine vapours are passed through absorption towers having sodium carbonate solution following reactions occurs:

$$3Na_2CO_3 + 3Br_2 \rightarrow NaBrO_3 + 5NaBr + 3CO_2$$

$$10\ NaBr + 2NaBrO_3 + 6H_2SO_4 \longrightarrow 6Na_2SO_4 + 6Br_2 + 6H_2O$$

or

$$5\ NaBr + NaBrO_3 + 3H_2SO_4 \rightarrow 3Na_2SO_4 + 3Br_2 + 3H_2O$$

From here, the vapours of bromine are obtain and condensed.

Physical Properties

1. It is a reddish brown, heavy and poisonous mobile liquid.
2. It has a boiling point of 331.5 K, freezing point of 265.8 K and a density of 3.2 g/cm^3.
3. It has a bad irritating smell which effects the eyes, nose and throat.
4. It is quite soluble in water but highly soluble in organic solvents like chloroform, benzene etc. giving a reddish brown solution.

Chemical Properties

1. Reaction with Hydrogen
It reacts with hydrogen at 473 K or at room temperature in presence of catalyst like platinum.

$$H_2 + Br_2 \rightleftharpoons 2HBr$$

2. Reaction with Elements
It can react with nonmetals and metals directly as follows:

$$2B + 3Br_2 \xrightarrow{\text{Red heat}} 2BBr_3$$

$$2P + 3Br_2 \longrightarrow 2PBr_3$$

$$2As + 3Br_2 \rightarrow 2AsBr_3$$

$$2Na + Br_2 \longrightarrow 2NaBr$$

$$2K + Br_2 \rightarrow 2KBr$$

$$2Al + 3Br_2 \longrightarrow 2AlBr_3$$

$$Zn + Br_2 \rightarrow ZnBr_2$$

- Br_2 does not combine with N, O, S directly.

3. Reaction with Alkalies
It can react with dilute as well as concentrated NaOH to give following products as follows:

$$\underset{\text{Cold and dil.}}{2NaOH + Br_2} \longrightarrow \underset{\text{Sodium hypobromite}}{NaBr + NaBrO} + H_2O$$

$$\underset{\text{Hot and conc.}}{6NaOH + 3Br_2} \rightarrow \underset{\text{Sodium bromate}}{5NaBr + NaBrO_3} + 3H_2O$$

$$\underset{\text{Hot and conc.}}{3Na_2CO_3 + 3Br_2} \longrightarrow 5NaBr + NaBrO_3 + 3CO_2$$

$$\underset{\text{Hot and conc.}}{6Ba(OH)_2 + 6Br_2} \rightarrow \underset{\text{Barium bromate}}{5BaBr_2 + Ba(BrO_3)_2} + 6H_2O$$

With dry slaked like it gives a bleaching powder like compound.

$$Ca(OH)_2 + 3Br_2 \rightarrow CaOBr_2 + H_2O$$

4. Oxidizing and Bleaching Agent
   Br_2 acts as an oxidizing agent and a bleaching agent as it generates nascent oxygen as follows:

   $$Br_2 + H_2O \longrightarrow HBr + HOBr \xrightarrow{\Delta} HBr + [O]$$

   It can oxidize KI into iodine.

   $$2KI + Br_2 \longrightarrow 2KBr + I_2$$

   It can oxidize sulphites into sulphates

   $$Na_2SO_3 + Br_2 + H_2O \rightarrow Na_2SO_4 + 2HBr$$

   It can oxidize thiosulphate into sulphate.

   $$Na_2S_2O_3 + Br_2 + H_2O \rightarrow Na_2SO_4 + S + 2HBr$$

   It can oxidize arsenites into arsenates.

   $$Na_3AsO_3 + Br_2 + H_2O \rightarrow Na_3AsO_4 + 2HBr$$

   It can oxidize hydrogen sulphide into sulphur.

   $$H_2S + Br_2 \rightarrow 2HBr + S$$

   It can oxidize SO_2 into H_2SO_4.

   $$SO_2 + Br_2 + 2H_2O \rightarrow 2HBr + H_2SO_4$$

5. With Ammonia
   Bromine reacts with ammonia as follows:

   $$NH_3 + 3Br_2 \longrightarrow \underset{\text{Excess}}{NBr_3} + 3HBr$$

   $$8NH_3 + 3Br_2 \rightarrow \underset{\text{Excess}}{6NH_4Br} + N_2$$

6. With Mercuric Oxide
   It reacts with mercuric oxide to give mercury oxybromide. However, if bromine vapours are passed over dry HgO at 333 K, Br_2O is formed.

   $$2HgO + 2Br_2 + H_2O \longrightarrow \underset{\text{Mercury oxybromide}}{HgBr_2.HgO} + 2HBrO$$

   $$HgO + 2Br_2 \rightarrow Br_2O + \underset{\text{Bromine monoxide}}{HgBr_2}$$

7. With Organic Compounds
   It gives addition and substitution reactions with organic compounds (hydrocarbons) as follows:

   $$C_2H_4 + Br_2 \longrightarrow C_2H_4Br_2$$

   $$C_2H_6 + Br_2 \rightarrow C_2H_5Br + HBr$$

Uses

1. It is used to prepare tetra ethyl lead (TEL) a famous antiknocking substance.

   $$C_2H_6 + Br_2 \xrightarrow{h\nu} C_2H_5Br + HBr$$

   $$4C_2H_5Br + 4Na - Pb \longrightarrow (C_2H_5)_4Pb + 4NaBr + 3Pb$$

2. AgBr is used in phorography.
3. The major use of bromine is in the manufacture of ethylene bromide which is used as an additive to leaded petrol (antiknock gasoline component).
4. It is also used to prepare bromine water, dyes, drugs, AgBr, benzyl bromide (tear gas) etc.
5. It is used as a germicide, oxidizing and bleaching agent.

IODINE (I_2)

It was discovered by Courtios and named by Gay Lussac (iodi means violet). It is the rarest halogen with metallic lustre.

Occurrence

It is present in combined state in the form of salts like iodides, iodates etc.

Main Source

The main sources of iodine are

(i) Kelp (ash of sea weeds like Laminaria) 1% metal iodide

(ii) Caliche or Crude chile salt petre: $NaIO_3$ (0.2 %)

Preparation

1. From Kelp
   On commercial scale iodine is prepared from kelp which are the ashes of sea weeds like laminaria. Kelp has many salts of Na and K like chlorides, carbonates, sulphates and iodides. Here, iodides are in solution. When this solution (mother liquor) is heated with concentrated sulphuric acid in presence of MnO_2 in iron retorts, I_2 is formed which is sublimated in pure form.

   $$2NaI + 3H_2SO_4 + MnO_2 \xrightarrow{\Delta} 2NaHSO_4 + MnSO_4 + 2H_2O + I_2 \uparrow$$

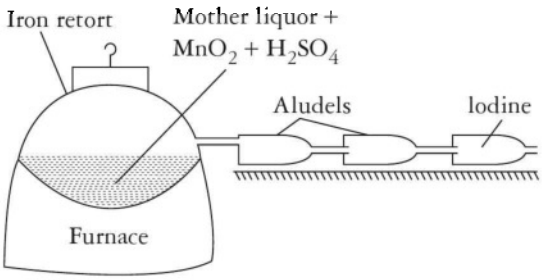


Fig. 3.6 I_2 Preparation from Kelp

2. From Caliche
The aqueous solution of chile salrpetre is concentrated and cooled down to remove sodium nitrate crystals. When the remaining solution (mother liquor) is treated with sodium hydrogen sulphite (in a calculated amount) iodine is precipitated as follows:

$$2NaIO_3 + 5NaHSO_3 \xrightarrow{\Delta} 3NaHSO_4 + 2Na_2SO_4 + H_2O + I_2\uparrow$$

■ If $NaHSO_3$ is in excess, HI will be produced here. Impurities of Cl_2, Br_2 are present as ICl, IBr and these are removed by KI as follows:

$$KI + ICl \longrightarrow KCl + I_2$$

$$KI + IBr \longrightarrow KBr + I_2$$

3. From KI
When chlorine or bromine is passed through KI solution iodine is formed.

$$2KI + X_2 \longrightarrow 2KX + I_2\uparrow$$

$(X_2 = Br_2, Cl_2)$

4. From Metal Iodides
When metal iodides are treated with concentrated H_2SO_4 in presence of MnO_2 iodine is formed (Lab Method).

$$2MI + H_2SO_4 + MnO_2 \longrightarrow I_2\uparrow$$

For example,

$$2KI + 3H_2SO_4 + MnO_2 \xrightarrow{\Delta} 2KHSO_4 + MnSO_4 + 2H_2O + I_2\uparrow$$

5. From a Mixture of Sodium Iodide and Sodium Iodate
When the mixture is treated with concentrated sulphuric acid iodine is formed as follows:

$$5\,NaI + NaIO_3 + 3H_2SO_4 \longrightarrow 3Na_2SO_4 + 3I_2 + 3H_2O$$

Physical Properties

1. It is a dark violet shining solid which sublimes on heating.
2. It has a melting point of 387K and a boiling point of 457K.
3. Its vapours are highly pungent and irritating.
4. It is least soluble in water among halogens and its aqueous solution is brownish in colour. Its solubility increases in presence of KI due to the formation of KI_3.
5. It exists as a diatomic molecule however on heating at 1973 K it changes into atomic form.

CHEMICAL PROPERTIES

1. Reaction with KI
It reacts with KI giving potassium tri-iodide as follows:

$$KI + I_2 \longrightarrow 2KI_3$$

KI_3 decomposes easily into iodine.

■ The solubility of I_2 in water increases in presence of KI due to the formation of KI_3.

2. Reaction with Hypo Solution
When it reacts with hypo solution, sodium tetra thionate is formed and the colour of I_2 disappears.

$$2Na_2S_2O_3 + I_2 \longrightarrow \underset{\text{Sodium tetrathionate}}{Na_2S_4O_6} + 2NaI$$

3. Reaction with Nitric Acid
It reacts with nitric acid to give iodic acid.

$$I_2 + 10HNO_3 \longrightarrow \underset{\text{Iodic acid}}{2HIO_3} + 10NO_2 + 4H_2O$$

■ Cl_2, Br_2 does not give this reaction.

4. Reaction with Elements
It can combine only with H, P, As, Sb in presence of platinum catalyst as follows:

$$H_2 + I_2 \xrightarrow{\Delta, Pt} 2HI$$

$$P_4 + 6I_2 \xrightarrow{\Delta} 4PI_3$$

$$2K + I_2 \longrightarrow 2KI$$

$$2Fe + 3I_2 \xrightarrow{\Delta} 2FeI_3$$

5. Oxidizing Properties
It behaves like a weak oxidizing agent and oxidizes following compounds as follows:
It oxidizes H_2S into sulphur.

$$H_2S + I_2 \longrightarrow 2HI + S$$

It oxidizes ferrous sulphate into ferric sulphate.

$$2FeSO_4 + H_2SO_4 + I_2 \longrightarrow Fe_2(SO_4)_3 + 2HI$$

It oxidizes SO_2 into H_2SO_4.

$$SO_2 + I_2 + 2H_2O \longrightarrow H_2SO_4 + 2HI$$

It oxidizes sodium sulphite into sodium sulphate.

$$Na_2SO_3 + I_2 + H_2O \longrightarrow Na_2SO_4 + 2HI$$

It oxidizes sodium arsenite into sodium arsenate.

$$Na_3AsO_3 + I_2 + H_2O \longrightarrow Na_3AsO_4 + 2HI$$

6. With Alkalies
When it reacts with cold and dilute NaOH solution, sodium hypoiodite is formed which undergoes hydrolysis to give hypoiodous acid.

$$2NaOH + I_2 \rightarrow NaIO + NaI + H_2O$$

$$NaIO + H_2O \longrightarrow HIO + NaOH$$

or

$$NaOH + I_2 \longrightarrow \underset{\text{Hypoiodous acid}}{NaI + HIO}$$

When it reacts with hot and concentrated NaOH solution sodium, iodate is formed.

$$6NaOH + 3I_2 \longrightarrow 5NaI + \underset{\text{Sodium iodate}}{NaIO_3} + 3H_2O$$

7. Displacement of Chlorine and Bromine
Iodine can displace Cl_2 and Br_2 from their oxy salts as follows:

$$2KClO_3 + I_2 \rightarrow 2KIO_3 + Cl_2$$

$$2KBrO_3 + I_2 \longrightarrow 2KIO_3 + Br_2$$

8. With Ammonia
It gives a mild explosive with ammonia as follows:

$$2NH_3 + 3I_2 \rightarrow \underset{\text{Explosive}}{NI_3.NH_3} + 3HI$$

$$8NI_3.NH_3 \longrightarrow 5N_2 + 9I_2 + 6NH_4I$$

9. With Mercuric Oxide
When it reacts with fresh precipitate of mercuric oxide, hypoiodous acid is formed as follows:

$$2HgO + 2I_2 + H_2O \rightarrow HgI_2.HgO + 2HIO$$

10. Iodoform Reaction
When it reacts with ethyl alcohol and alkali KOH, a yellow precipitate of iodoform is formed.

$$C_2H_5OH + 6KOH + 4I_2 \longrightarrow CHI_3 + HCOOK + 5KI + 5H_2O$$

Uses

1. I_2 is used to prepare tincture (2% solution of I_2 in alcohol), iodex, iodoform and to increase production of eggs.
2. It is used in the manufacture of photosensitive papers, films etc.
3. Solution of I_2 in KI can be used in goitre's treatment.

REMEMBER

- Insufficient iodine in the diet leads to goiter (enlargement of the thyroid gland) in humans beings.
- Organic solvents dissolve I_2 to form brown solution due to formation of charge transfer complex.
- I_2 dissolves in organic solvents due to free I_2 molecules.
- Iodine forms I^+ (iodinium ion). It shows basic nature of iodine. For example, ICl.

ALLOTROPIC FORMS OF CARBON

Occurrence

In occurs in freestate in the form of coal, diamond and graphite and in combined state in the form of CO_2, carbonates and hydrocarbons. It exists in two allotropic forms: crystalline and amorphous.

1. Crystalline Form
   It is of three types as follows:

A. Diamond
   It is the hardest known substance. It is the purest form of carbon with high density (3.5 g/ml), melting point (7350°C) and refractive index (2.45).

   Structure: In diamond carbon atom is sp^3 hybridized with tetrahedral structure. Here, one carbon atom is attached with four other carbon atoms by covalent bonds and this gives diamond a three-dimensional polymeric structure. Here C – C bond length is 1.54 Å and bond angle is 109° 28'.

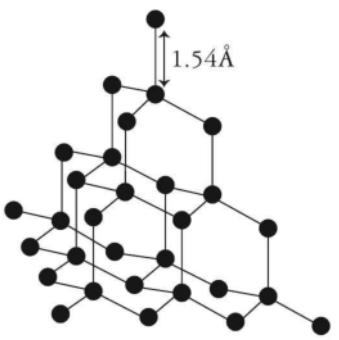


Fig. 3.7 Diamond

Properties

(i) The high value of boiling point, melting point, density are due to its three-dimensional structure which needs high energy for bond breaking. Light rays, X-rays (highest) can pass through it.

(ii) It is transparent solid with a specific gravity of 3.52.

(iii) Due to high refractive index on proper cutting it produces maximum total internal reflection.

(iv) It is bad conductor due to absence of free electrons.

(v) It is insoluble in all solvents.

(vi) Pure diamonds are colourless while impure diamond may be coloured. Kohinoor diamond is of 186 carat while Pit diamond is of 136 carat.

(vii) Diamond on heating in vacuum at 1800 2000°C converts into graphite.

(viii) Being chemically inert it does not react with acids, bases etc.

(ix) When it is heated in air upto 1173 K, CO_2 is formed and on heating with fluorine upto 973 K it gives CF_4.

$$C + O_2 \longrightarrow CO_2$$

$$C + 2F_2 \longrightarrow CF_4$$

Uses

Diamond dyes are often used in making cutting tools. It is used as an abrasive.

B. Graphite
   It is a gray coloured substance with metallic lustre and more stable and reactive than diamond.

   Structure: It has a two-dimensional sheet like hexagonal structure in which each carbon atom is sp^2 hybridized and it is attached with three other carbon atoms in a hexagonal planar structure method. Here C – C bond length is 1.42 Å and the distance between two successive layer is 3.35 Å. These layers are held together by weak van der Waal's forces and slippery in nature hence graphite is soft and have low density. hence C – C bond length is shorter than that of diamond because a π bond is formed by the fourth valence electron present in each carbon atom which is free.

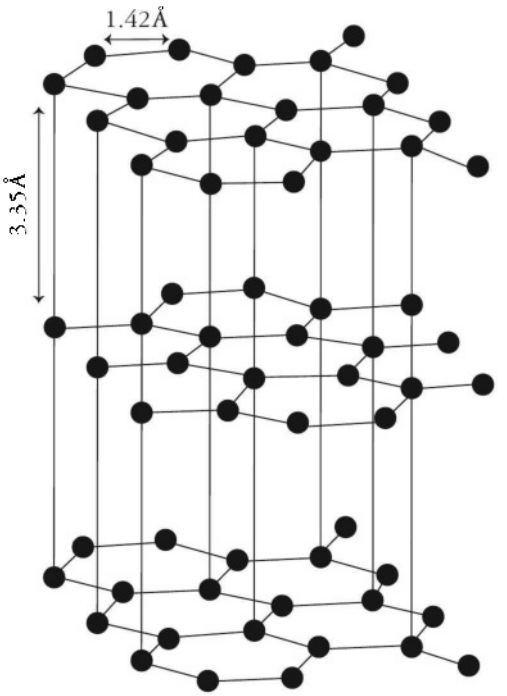


Fig 3.8. Graphite

Due to these π-electrons which are able to move in these slippery structures graphite can conduct electricity.

Preparation: Acheson method is used to prepare graphite.

$$SiO_2 + 3C \xrightarrow{\Delta} SiC + 2CO$$

$$SiC \xrightarrow{\Delta} C + Si$$

Properties

(i) It is a soft, grayish, greasy crystalline solid with a density of 2.5 g/ml.

(ii) Graphite has lowest energy among carbon allotropes.

(iii) It is a conductor of electricity due to the presence of free electrons in its slippery hexagonal sheets. Its conductivity increases with increase of temperature.

(iv) It is called plumbago or black lead as it leaves a black mark on paper.

(v) It is thermodynamically more stable than diamond and it has 1.9 kJ less free energy of formation than diamond.

(vi) On heating at 1873 K and at a very-very high pressure (50,000 60,000 atm) it changes into diamond.

(vii) It is more reactive form than diamond, however it does not react with dilute acids or alkalies.

(viii) Graphite on heating with conc. HNO_3 gives graphitic acid ($C_{11}H_4O_5$).

(ix) Graphite on oxidation with alkaline $KMnO_4$ gives mellitic acid or benzene hexacarboxylic acid [$C_6(COOH)_6$] and oxalic acid.

(x) On heating graphite with vapours some metals like K, Rb, Cs at nearly 600 K, a paramagnetic compound C_8M is obtained (M = K, Rb, Cs).

Uses

(i) It is used in pencil leads (graphite and clay) and as a lubricant in making electrodes.

(ii) It is also used in electroplating, electrotyping, painting, stoves and in making refractory crucibles.

(iii) In nuclear reactor it is used as moderator.

(iv) It is a better lubricant on earth than moon.

ALLOTROPIC FORMS OF PHOSPHOROUS

1. White Phosphorous

Preparation: White phosphorous is formed as a soft waxy solid whenever phosphorous vapours are condensed.

Physical Properties

1. It is extremely reactive due to strain in the P_4 molecule.
2. It is soft, waxy solid having low melting Point (317 K) and a characteristic garlic smell.
3. Its vapours are highly poisonous and it causes a disease known as 'Phossy jaw' due to which decay of jaw bones occurs.
4. It is very slightly soluble in water, but is soluble in benzene, carbon disulphide and sulphur monochloride.
5. It turns yellow on exposure to light, therefore it is also known as yellow phosphorous.
6. **Phosphorescence**
   On exposure to air, white phosphorous emits a faint green light visible in the dark. The phenomenon of the emission of light is called phosphorescence.
7. It has very low ignition temperature, hence it is kept under water.

CHEMICAL PROPERTIES

1. Heating Effect
   On heating in an inert atmosphere at 240°C, it changes into red phosphorous.

$$\text{Yellow P} \xrightarrow[\text{Inert atm.}]{240^\circ C} \text{Red P}$$

2. Oxidation
   It inflames in dry air at about 50°C and forms P_4O_{10}.

$$P_4 + 5O_2 \longrightarrow P_4O_{10}$$

3. Reaction with Halogens
   White phosphorous ignites spontaneously in chlorine and reacts violently with the halogens.

$$P_4 + 6Cl_2 \longrightarrow 4PCl_3$$

$$P_4 + 10Cl_2 \longrightarrow 4PCl_5$$

4. Reducing Nature
It can reduce oxidizing agents like conc. HNO_3, conc. H_2SO_4 into their oxides and get oxidized itself into phosphoric acid

$$P_4 + 20\,HNO_3 \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$$
Phosphoric acid

$$P_4 + 10\,H_2SO_4 \longrightarrow 4H_3PO_4 + 10SO_2 + 4H_2O$$

5. Reaction with Strong Alkali
Phosphine gas is obtained when white phosphorous is treated with a strong alkali in an inert atmosphere.

$$P_4 + 3KOH + 3H_2O \longrightarrow 3KH_2PO_2 + PH_3$$
Potassium hypophosphite

■ This reaction is not given by red phosphorous.

6. Reaction with Alkali Salts
Phosphorous reacts with solutions of easily reducable metal salt. For example, those of Cu, Ag and Au to give the metal phosphate. For example:

$$3P_4 + 12CuSO_4 + 24H_2O \longrightarrow 4Cu_3P + 8\,H_3PO_3 + 12\,H_2SO_4$$

$$P_4 + 10\,CuSO_4 + 16H_2O \longrightarrow 10\,Cu + 4H_3PO_4 + 10H_2SO_4$$

7. Reaction with Metals
It reacts with a number of metals forming their phosphides For example:

$$6Mg + P_4 \longrightarrow 2\,Mg_3P_2$$
Magnesium phosphide

$$6Ca + P_4 \longrightarrow 2\,Ca_3P_2$$
Calcium phosphide

8. Reaction with Sulphur
It violently explodes with sulphur forming a number of sulphide such as P_2S_3, P_2S_5, P_4S_3 and P_4S_7.

$$8P_4 + 3S_8 \longrightarrow 8P_4S_3$$
Tetra phosphorous trisulphide

$$P_4 + 10\,S \longrightarrow 2P_2S_5$$
Phosphorous pentasulphide

Uses: It is used as a rat poison and for making phosphorous bronze, tracer bullets, smoke screens etc.

Structure: As the vapour density of white phosphorous is 62 hence it leads to the molecular formula P_4. In its structure the four phosphorous atoms lie at the corners of a regular tetrahedron. Each phosphorous atom is linked to each of the other three atoms by covalent bonds. The P – P bond length is 2.21 Å and the ∠PPP bond angle is 60°. It suggests that the molecule is under strain hence very reactive in nature.

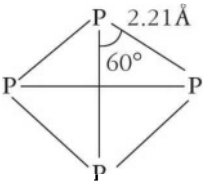


Fig. 3.9 Structure of White Phosphorous

2. Red Phosphorous

Preparation: This allotrope is made from white phosphorous by heating it in the absence of air in an inert atmosphere of CO_2 or coal gas to 270°C for several days in an egg shaped iron vessel as shown under:

$$\text{White 'P'} \xrightarrow[CO_2 \text{ or coal gas}]{540\text{–}570\ K} \text{Red 'P'}$$

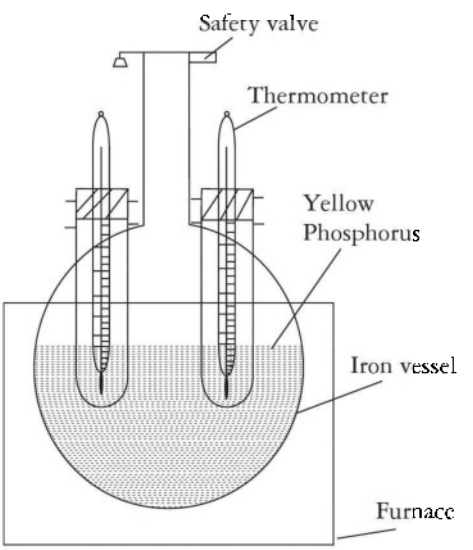


Fig. 3.10 Preparation of Red Phosphorous

Physical Properties

1. It is dark red powder with specific gravity 2.1 g/ml and a stable variety of phosphorous.
2. It is hard, solid, odourless and nonpoisonous in nature and is insoluble in water as well as in organic solvents like carbon disulphide etc.

3. It is much less reactive than white phosphorous.
4. It does not show phosphorescence.
5. As its ignition temperature is much higher than that of white phosphorous and thus does not catch fire easily i.e., can be kept is open air.
6. It changes ro white phosphorous when it is vapourized by sublimarion and the vapours are condensed (P_4).

Chemical Properties

1. Heating Effect
   Red phosphorous burns only when heated above 260°C to form phosphorous pentaoxide.

$$P_4 + 5O_2 \xrightarrow{260^\circ C} P_4O_{10}$$

2. Reaction with Nonmetals and Metals
   It can reacts with halogens, sulphur and metals when heated as follows:

$$2P + 5Cl_2 \xrightarrow{\Delta} 2PCl_5$$

$$2P + 3S \xrightarrow{\Delta} P_2S_3$$

$$P + 3Na \xrightarrow{\Delta} Na_3P$$

Uses: Red phosphorous is largely used in the match industry and it is preferred to yellow phosphorous because of its nonpoisonous nature.

Structure: Red phosphorous exists as chains of P_4 linked togerher by covalent bonds to give a polymeric amorphous srructure.

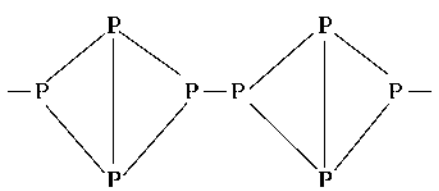


Fig. 3.11 Srructure of Red Phosphorous

Difference between White and Red Phosphorous

White Phosphorous	Red Phosphorous
1. It is a yellowish soft waxy olid.	It is red hard brittle powder.
2. Its molecular formula is P4.	It is a complex polymer.
3. It's density is 1.8 g/cm^3	Its density is 2.1 g/cm^3.
4. Its M.P. is 44°C (under water).	Its M.P. is 590°C (under pressure).
5. Its B.P. is 280°C.	It sublimates at 400°C (1 atm).
6. It is toxic in nature.	It is nontoxic in nature.
7. It shows phosphorescence.	It does not show phosphorescence.
8. It is soluble in benzene, CS_2 etc.	It is insoluble in solvents.
9. It ignites in air at 50°C	It ignites in air at 260°C
10. It ignites in chlorine	It requires heat before it burns in chlorine.
11. It reacts with hot alkali to give phosphorine.	It does not react with alkalies.
12. Phosphorescence and oxidation on exposure to air.	There is little or no oxidation.

3. Black Phosphorous

Preparation: It is obtained by heating white phosphorous at 470 K under very high pressure as follows:

$$\text{White 'P'} \xrightarrow[1000 - 10000 \text{ atm. pressure}]{470 \text{ K}} \text{Black P'}$$

Physical Properties

1. It is most inactive but most stable allotrope of phosphorous.
2. It is a good conductor of heat and electricity resembling graphite in this respect.
3. Its density is higher than that of red phosphorous.

Structure: Black phosphorous is also a polymer of high molecular weight. It has a double layered crystal lattice in which each layer is made up of zig-zag chains linking phosphorous atoms.

REMEMBER

(i) There are three distinct allotropes with acid that cast into sticks, and must be kept under water as white phosphorous inflames in contact with air at quite low temperature.

(ii) The other forms of phosphorous are obtained from white phosphorous.

(iii) Radioactive phosphorous (P^{32}) is used in the treatment of leukemia and other blood disorders and in the preparation of its compounds.

ALLOTROPIC FORMS OF SULPHUR

It is present in these two allotropic forms which are as follows:

1. Crystalline
   There are two crystalline allotrope of sulphur as follows:

(i) **Rhombic (S_R) or Octahedral or α-Sulphur:** It is the most common, most stable, allotrope of sulphur. It is a pale yellow crystalline solid having a ring puckered structure (S_8). It is insoluble in water but soluble in benzene, CS_2, turpentine etc. It has a melting point of 112.8°C and a specific gravity of 2.06 g cm^{-3}.

This form has S_8 units at room temperature and the eight sulphur atoms are arranged in a ring puckered structure.

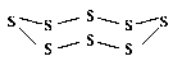


Fig. 3.12 Structure of Rhombic Sulphur

It can be prepared by evaporating the solution of roll sulphur in CS_2.

(ii) **Monoclinic (S_M) or Prismatic or β-Sulphur:** It is obtained by melting rhombic sulphur at above 95.6°C.

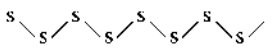
$$\alpha - \text{Sulphur} \xrightarrow{95.6^\circ C} \beta - \text{Sulphur}$$

It has a melting point of 19.2°C and a gravity of 1.98 g cm^{-3}. It is insoluble in water but soluble in CS_2.

It is stable only above 96°C and below this temperature it changes into rhombic sulphur. It has also ring puckered structure like rhombic sulphur but different in symmetry.

2. Amorphous
   Amorphous allotropes of sulphur are as follows:

(i) **Plastic or γ-Sulphur:** It is a super cooled liquid which is obtained by sudden cooling of boiling sulphur by a stream of cold water. It is dark in colour and insoluble in carbon disulphide and water. It has no sharp melting point but its specific gravity is 1.95 g cm^{-3}. It has an open chain structure in which each sulphur atom is linked to two other sulphur atoms by covalent bonds.



An open chain structure of S_8 molecule of γ-sulphur

(ii) **Colloidal Sulphur:** It can be prepared by passing H_2S through nitric acid as follows:

$$H_2S + 2HNO_3 \rightarrow 2H_2O + 2NO_2 + S$$

It can also be prepared by the reaction of H_2S with SO_2 as follows:

$$SO_2 + 2H_2S \longrightarrow 2H_2O + 3S\downarrow$$

It is also prepared by the reaction of sodium thiosulphate and dilute HCl as follows:

$$Na_2S_2O_3 + 2HCl \longrightarrow 2NaCl + SO_2 + H_2O + S\downarrow$$

■ On heating or keeping for a long time colloidal sulphur changes into ordinary sulphur.

(iii) **Milk of Sulphur:** When milk of lime and flower of sulphur are boiled together a mixture of CaS_5 and CaS_2O_3 is formed. When this mixture is HCl a white amorphous precipitate of milk of sulphur is obtained as follows:

$$3Ca(OH)_2 + 12\,S \longrightarrow 2CaS_5 + CaS_2O_3 + 3H_2O$$

$$2CaS_5 + CaS_2O_3 + 6HCl \rightarrow 3CaCl_2 + 3H_2O + 12\,S$$

Milk of Sulphur

On standing for a long time, it changes into rhombic sulphur. It is soluble in CS_2.

Straight Objective Type Questions (Single Choice)

1. Amorphous boron is prepared by heating B_2O_3 with
   (a) Mg (b) Mn
   (c) SiO_2 (d) Hg
2. Method used for obtaining highly pure silicon used as a semiconductor material is
   (a) electrochemical (b) oxidation
   (c) zone refining (d) crystallization
3. In graphite, electrons are
   (a) spread out between the structure
   (b) localized on earth C-atom
   (c) present in antibonding orbital
   (d) localized on every third C-atom
4. The halogen which is most easily reduced is
   (a) F_2 (b) Cl_2
   (c) Br_2 (d) I_2
5. Elemental silicon to be used as semiconductor is purified by
   (a) floatation
   (b) electrolysis
   (c) zone refining
   (d) heating under vacuum
6. The number of covalent bonds in fluorine molecules is
   (a) 1 (b) 2
   (c) 3 (d) 4
7. Boron when heated with carbon forms
   (a) B_4C (b) B_2C_3
   (c) BC_2 (d) B_4C_3
8. Chlorine acts as a bleaching agent only in presence of
   (a) sunlight (b) pure oxygen
   (c) dry air (d) moisture
9. On boiling an aqueous solution of $KClO_3$ with I_2, which reaction takes place?
   (a) $2KClO_3 + I_2 \rightarrow 2KIO_3 + Cl_2$
   (b) $5KClO_3 + 3I_2 + 3H_2O \longrightarrow 5KCl + 6HIO_3$
   (c) $KClO_3 + I_2 + H_2O \longrightarrow KCl + 2HIO_2$
   (d) $KClO_3 + I_2 + H_2O \rightarrow KClO_4 + 2HI$
10. Nitrogen is liberated by the thermal decomposition of only
    (a) NH_4NO_2 (b) NaN_3
    (c) $(NH_4)_2Cr_2O_7$ (d) all the three
11. Which of the following is chemically inactive allotropic form of carbon?
    (a) coal (b) diamond
    (c) animal charcoal (d) charcoal
12. Which bonding feature about white phosphorous is not correct?
    (a) Each P atom contains a lone pair of electrons.
    (b) The molecule contains 6P – P bonds.
    (c) The P – P – P bond angles in the molecules are 109.5°.
    (d) It consists of P_4 tetrahedral molecules.
13. Silicon shows a diagonal relation with
    (a) phosphorous (b) magnesium
    (c) boron (d) carbon
14. The use of diamond as a gem depends on its
    (a) chemical inertness
    (b) hardness
    (c) purest form of carbon
    (d) high refractive index
15. Which one of the following allotropic forms of carbon is isomorphous with crystalline silicon?
    (a) coal (b) diamond
    (c) coke (d) graphite
16. Active nitrogen can be made by passing an electric spark through N_2 gas at
    (a) high pressure
    (b) very low pressure (2 mm of Hg)
    (c) ordinary pressure
    (d) very low temperature
17. Each of the following is true of white and red phosphorous except that they
    (a) can be oxidized by heating in air.
    (b) can be converted into one another.
    (c) both are soluble in CS_2.
    (d) consist of the same kind of atoms.

18. What is the number of free electrons present on each carbon atom in graphite?
(a) zero (b) 1
(c) 2 (d) 3

19. The bleaching action of chlorine is due to the liberation of which the following?
(a) [O] (b) O_2
(c) HOCl (d) HCl

20. Among the following, the properties of which pair of halogens are more similar than those of the other pairs?
(a) Fluorine and astatine
(b) Chlorine and bromine
(c) Fluorine and bromine
(d) Fluorine and chlorine

21. Graphite has a layered structure. The distance between the layers is
(a) 115 pm (b) 375 pm
(d) 335 pm (d) 235 pm

22. Which of the following is a super fluid?
(a) helium I (b) helium II
(c) krypton I (d) argon II

23. Which of the following pairs of halogens have approximately identical bond energy?
(a) F_2 and Cl_2 (b) Cl_2 and I_2
(c) F_2 and Br_2 (d) F_2 and I_2

24. N_2 is prepared commercially by
(a) heating a mixture of ammonium chloride and sodium nitrite.
(b) the serpeck process.
(c) the fractional distillation of liquefied air.
(d) heating ammonium dichromate.

25. Silicon has the characteristics of a
(a) nonmetal (b) metalloid
(c) metal (d) none of these

26. Carbon forms a large number of compounds due to its
(a) property of catenation
(b) variable valency
(c) tetra valency
(d) large chemical affinity

27. The bleaching action of chlorine occurs in the presence of
(a) sunlight
(b) moisture
(c) pure sulphur dioxide
(d) pure oxygen

28. It is possible to obtain oxygen from air by fractional distillation because
(a) oxygen has a lower density than nitrogen.
(b) oxygen has high b.p. than nitrogen.
(c) oxygen is more reactive than nitrogen.
(d) oxygen is in a different group of the periodic table from nitrogen.

29. Nitrogen is a relatively inactive element because
(a) dissociation energy of its molecule is fairly high
(b) its electronegativity is fairly high
(c) it has low atomic radius
(d) nitrogen atom has a stable electronic configuration

30. One can obtain a silica garden, if
(a) silicon salts are grown in garden
(b) crystals of coloured cations are added to a strong solution of sodium silicate
(c) silicon tetrafluoride is hydrolyzed
(d) sodium silicate solution is heated with base

31. Which of the following, when heated give nitrogen gas?
(a) $(NH_4)_2Cr_2O_7$ (b) $Ba(N_3)_2$
(c) NH_4NO_3 (d) both (a) and (b)

32. Silicon is an important constituent of
(a) amalgams (b) chlorophyll
(c) haemoglobin (d) rocks

33. Which of the following allotropic forms of sulphur is the most stable thermodynamically?
(a) β-monoclinic
(b) γ-monoclinic
(c) plastic sulphur
(d) orthorhombic sulphur

34. Which of the following is not correct?
(a) The hybridization of C in graphite is sp^2.
(b) SiO_2 reacts with Na_2CO_3 and liberates CO.
(c) SiO_2 is used as acid flux.
(d) The distance between the layers in graphite is 3.35×10^{-3} cm.

35. Fluorine is prepared by the electrolysis of:
(a) A solution of KHF_2 in HF
(b) Molten NaF

(c) A solution of KHF_2 in KF
(d) A molten mixture of $Ca_3(PO_4)_2.CaF_2$ and cryolite

36. Boro and silicon resemble in all respects except
(a) both form halides which are Lewis acids.
(b) their chlorides hydrolyze to their respective acids.
(c) both form acidic oxides.
(d) their hydrates are stable.

37. Fluorine shows only one oxidation state (−1) because it has
(a) a small covalent radius
(b) a low bond energy
(c) a high electronegativity
(d) no d-orbital available for bonding

38. Which of the following properties is not exibited by nitrogen?
(a) Supporter of life
(b) Catenation
(c) Low boiling point
(d) Hydrogen bonding

39. Which of the following halogens is extracted from sea-weeds?
(a) I_2 (b) Cl_2
(c) F_2 (d) Br_2

40. $Na_2S_2O_3$ is reduced by I_2 to:
(a) Na_2SO_4 (b) $NaHSO_3$
(c) Na_2S (d) $Na_2S_4O_6$

41. Fluorine reacts with H_2S to produce:
(a) S and HF (b) SF_2
(c) SF_6 (d) SF_4

42. Which of the following has least bond energy?
(a) H_2 (b) F_2
(c) O_2 (d) N_2

43. The element evolving two different gases on reaction with conc. sulphuric acid is
(a) S (b) C
(c) Sn (d) P

44. Chlorine acts as an oxidizing agent when it reacts with:
(a) $Fe_2(SO_4)_3$ (b) O_3
(c) $KMnO_4$ (d) $FeSO_4$

45. The most reactive and least reactive forms of phosphorous are respectively:
(a) White and black phosphorous
(b) White and red phosphorous
(c) Red and white phosphorous
(d) Scarlet and red phosphorous

46. Which of the following noble gases is the most polarized?
(a) Xenon (b) Radon
(c) Helium (d) Krypton

47. The electrolysis of brine produces:
(a) NaOH and $NaClO_3$
(b) NaCl and NaClO
(c) Only Cl_2
(d) Cl_2 and NaOH

48. Diamond is hard because
(a) All the four valence electrons are bonded to each carbon atoms by covalent bonds.
(b) It is a giant molecule.
(c) It is made up of carbon atoms.
(d) It cannot be burnt.

49. Which of the following noble gases is the least polarized?
(a) Krypton (b) Radon
(c) Helium (d) Xenon

50. The high oxidizing power of fluorine is due to
(a) low heat of dissociation and high heat of hydration.
(b) high heat of dissociation and high heat of hydration.
(c) high electron affinity.
(d) high heat of dissociation and low heat of hydration.

51. Fluorine reacts with aqueous $KClO_3$ to produce
(a) KCl (b) $KClO_4$
(c) KClO (d) $KClO_2$

52. Deep-sea divers breathe using a mixture of
(a) O_2 and He (b) O_2 and H_2
(c) O_2 and Ar (d) O_2 and Kr

53. Which one of the following pairs is obtained on heating ammonium dichromate?
(a) N_2 and H_2O (b) NO and NO_2
(c) N_2O and H_2O (d) NO_2 and H_2O

54. In the manufacture of bromine from sea water, the mother liquor containing bromides is treated with
(a) chlorine (b) iodine
(c) sulphur dioxide (d) carbon dioxide

Brainteasers Objective Type Questions (Single Choice)

55. An orange solid (A) on heating gives a green residue (B), a colourless gas (C) and water vapour. The dry gas (C) on passing over heated magnesium gave a white solid (D). (D) on reaction with water gives a gas (E) which gives dense white fumes with HCl. Here, (E) will be
    (a) N_2 (b) NO_2
    (c) NH_3 (d) N_2O

56. Which of the following statements are correct?
    (I) The hybridization of C in graphite is sp^2.
    (II) SiO_2 reacts with Na_2CO_3 and liberates CO.
    (III) SiO_2 is used as acid flux.
    (IV) The distance between the layers in graphite is 3.35×10^{-3} cm.
    (a) 1, 2 and 3 (b) 1, 2 and 4
    (c) 1, 3 and 4 (d) 2, 3 and 4

57. A greenish yellow gas reacts with an alkali metal hydroxide to form a halate which can be used in fire works and safety matches. The gas and halate, respectively are
    (a) Br_2, $KBrO_3$ (b) Cl_2, $KClO_3$
    (c) I_2, $NaIO_3$ (d) Cl_2, $NaClO_3$

58. Graphite is a soft, solid, lubricant, extremely difficult to melt. The reason for this anomalous behaviour is that graphite
    (a) has molecules of variable molecular masses like polymers.
    (b) has carbon atoms arranged in large plates of rings of strongly bound carbon atoms with weak interplate bonds.
    (c) is an allotropic form of diamond.
    (d) is a noncrystalline substance.

59. Boron has an extremely high melting point because of
    (a) the strong binding forces in the covalent polymer
    (b) the strong van der Waals forces between its atoms
    (c) allotropy
    (d) its ionic crystal structure

60. The colour of halogens progressively deepens from fluorine to iodine because
    (a) Fluorescence and phosphorescence become more intense as the atomic numbers of halogen increases.
    (b) Halogens of higher atomic number absorb light of longer wavelength since the difference in energy between the ground state and excited state decreases as the atomic number increases.
    (c) Halogens of higher atomic number absorb light of shorter wavelength since the difference in energy between the ground state and excited state increases as the atomic number increases.
    (d) The standard electrode potential increases from I_2 to F_2.

61. Carbon shows strong catenation while silicon shows little or no catenation because
    (a) The Si–Si bond is stronger than the C–C bond.
    (b) The C–C bond is stronger than the Si–Si bond
    (c) Silicon is a metalloid and carbon is a nonmetal.
    (d) Silicon forms ionic compounds whereas carbon forms covalent compounds.

62. Which of the following has been arranged in order of increasing bond energy?
    (a) $I_2 < F_2 < Br_2 < Cl_2$
    (b) $F_2 < Br_2 < Cl_2 < I_2$
    (c) $F_2 < I_2 < Cl_2 < Br_2$
    (d) $Cl_2 < I_2 < Br_2 < F_2$

63. The interlayer distance in graphite is
    (a) The same as the covalent radius of carbon.
    (b) More than twice the covalent radius of carbon.
    (c) Very small since the layers are tightly packed.
    (d) Many times larger than the covalent radius of carbon.

64. Helium II is the most extraordinary liquid with
    (a) Very high viscosity and zero heat conductivity.
    (b) Very high viscosity and very high heat conductivity.
    (c) Zero viscosity and very high heat conductivity.
    (d) Zero viscosity and low heat conductivity.

65. The F–F bond is weak because
    (a) The length of the F – F bond much larger than the bond lengths in other halogen molecules.
    (b) The F – F bond distance is small and hence the internuclear repulsion between the two F-atoms is very low.
    (c) The repulsion between the nonbonding pairs of electrons of two fluorine atoms is large.
    (d) The ionization energy of the fluorine atom is very low.

66. In which of the following reactions is bromine liberated?
    (a) $KBr\,(aq) + SO_2 \xrightarrow{\Delta}$
    (b) $KBr\,(aq) + F_2 \xrightarrow{\Delta}$
    (c) $KBr\,(aq) + I_2 \xrightarrow{\Delta}$
    (d) $HI\,(aq) + KBr\,(aq) \xrightarrow{\Delta}$

67. Compounds formed when noble gases gets trapped in the cavities of the crystal lattices of certain organic and inorganic compounds are known as
    (a) Clathrates
    (b) Stoichiometric compounds
    (c) Polyhydrates
    (d) Supercooled solids

68. Which of the following statements is correct for silicon?
    (a) It forms an oxide (SiO_2) that is amphoteric and has a Gaint structure
    (b) It does not undergo coordination number expansion.
    (c) It forms strong but unconjugated multiple bonds of the $p\pi$-$d\pi$ variety, especially with O and N.
    (d) It forms molecular halides that are not hydrolyzed.

69. Which of the following allotropic forms of phosphorous is the most stable, least reactive, has graphite like structure and is a good conductor of electricity?
    (a) Red phosphorous
    (b) White phosphorous
    (c) Scarlet phosphorous
    (d) Black phosphorous

70. Fluorine may be prepared by the electrolysis of a molten mixture of KHF_2 and KF. The anode is made of
    (a) Iron (b) Ungraphitized carbon
    (c) Copper (d) Graphite

71. Which of the following statement is incorrect for graphite?
    (a) Its layers are very tightly packed, almost without any space between them.
    (b) It cleaves easily between the layers which accounts for the remarkable softness of the crystals.
    (c) Its density is lower than that of diamond.
    (d) It has a layered structure and the bonding between the layers is very weak.

72. Sulphur reacts with chlorine in 1:2 ratio and forms (X). (X) on hydrolysis gives a sulphur compound (Y). What is the hybridized state of central atom in the anion of (Y)?
    (a) sp^3d (b) sp^2
    (c) sp^3 (d) sp

73. Which of the following reactions does not liberate iodine?
    (a) $CuSO_4 + KI \longrightarrow$
    (b) $NaIO_3 + NaHSO_4 + Cl_2 \longrightarrow$
    (c) $KI + MnO_2 + H_2SO_4 \longrightarrow$
    (d) $NaIO_3 + NaHSO_3 \longrightarrow$

74. In the reaction, $P_4 + 3KOH + 3H_2O \rightarrow 3KH_2PO_2 + PH_3$, phosphorous is
    (a) Only oxidized.
    (b) Oxidized as well as reduced.
    (c) Only reduced.
    (d) Neither oxidized nor reduced.

75. Which of the following statement is correct?
    (a) Graphite has such a high thermodynamical stability that diamond spontaneously changes into graphite in ordinary conditions.
    (b) Graphite and diamond have equal thermodynamic stability.
    (c) Graphite is thermodynamically more stable than diamond.
    (d) Diamond is thermodynamically more stable than graphite.

76. Bromine reacts with hot and concentrated Na_2CO_3 to produce
    (a) $NaBrO + NaBrO_3 + CO_2$
    (b) $NaBr + NaBrO_3 + CO_2$
    (c) $NaBr + NaBrO + CO_2$
    (d) $NaBr + NaBrO_4 + CO_2$

77. White phosphorous is more reactive than the nitrogen molecule because the
(a) P–P bond in phosphorous is weaker than the N≡N bond in nitrogen.
(b) P–P–P bond angle is 120° whereas N_2 is linear.
(c) Electronegativity of phosphorous is low.
(d) Ionization energy of phosphorous is greater than that of N_2.

78. Fluorine reacts with water to produce
(a) F_2O and F_2O_2 (b) F , O_2 and H
(c) HF and H_2O_2 (d) HF, O_2 and F_2O_2

79. In the cyclo-S_8 molecule of rhombic sulphur, all the S–S bond lengths and all the S–S–S bond angles are, respectively (give approximate values)
(a) 214 pm and 108°
(b) 204 pm and 105°
(c) 104 pm and 120°
(d) 102 pm and 125°

80. In the equilibrium, C (s, diamond) ⇌ C (s, graphite) + Heat (density of diamond and graphite are 3.5 and 2.3 g/cm^3, respectively), the equilibrium will be shifted to the left at
(a) High temperature and low pressure.
(b) Low temperature and very high pressure.
(c) Low temperature and low pressure.
(d) High temperature and very high pressure.

81. In clathrates atoms or molecules the bond formed is
(a) Covalent (b) Ionic
(c) Metallic (d) They do not form bonds

82. Which of the following reactions does not produce bromine?
(a) $NaBr + I_2 \rightarrow$
(b) $MgBr_2 + Cl_2 \rightarrow$
(c) $NaBrO_3 + NaBr + H_2SO_4 \longrightarrow$
(d) $NaBr + MnO_2 + H_2SO_4 \longrightarrow$

83. Carbon atoms in diamond are bonded to each other in a
(a) Tetrahedral configuration
(b) Octahedral configuration
(c) Linear configuration
(d) Planar configuration

84. Fluorine reacts with dilute NaOH and concentrated NaOH to, respectively produce
(a) F_2O_2 and NaF (b) OF_2 and NaF
(c) OF_2 and O_2 (d) HF and O_2

85. Which of the following statements is incorrect for helium?
(a) It has a negative Joule Thomson coefficient above 40 K.
(b) It has to be compressed before it can liquefy.
(c) It has a positive Joule–Thomson coefficient above 40 K.
(d) Its spontaneous expansion causes it to warm up.

86. In which of the following reactions is HF liberated?
(a) $F_2 + KCl \xrightarrow{\Delta}$
(b) $F_2 + NaOH \xrightarrow{\Delta}$
(c) $KHF_2 \xrightarrow{\Delta}$
(d) $PH_4F \xrightarrow{\Delta}$

87. In carbon-60, all carbon atoms are
(a) sp^2-hybridized with a graphite like shape.
(b) sp^2-hybridized with a diamond shape.
(c) sp^3-hybridized with a square antiprism shape.
(d) sp^2-hybridized with a truncated icosahedron shape.

88. In which of the following properties does white phosphorous resemble red phosphorous?
(a) Reaction with concentrated NaOH to produce PH_3.
(b) Fluorescence and phosphorescence.
(c) Burning in the presence of air.
(d) Solubility in an organic solvent.

89. Which of the following mixtures of noble gases are used to produce laser beams?
(a) Ar and Rn (b) He and Kr
(c) He and Ne (d) Kr and Ar

90. The standard reduction potentials of the halogens are in the order
(a) $Cl_2 > F_2 > I_2 > Br_2$
(b) $F_2 > Cl_2 > Br_2 > I_2$
(c) $F_2 > Cl_2 > I_2 > Br_2$
(d) $I_2 > Br_2 > Cl_2 > F_2$

91. Carbon-60 contains
(a) 12 pentagons and 20 hexagons.
(b) 20 pentagons and 12 hexagons.

(c) 30 pentagons and 20 hexagons.
(d) 24 pentagons and 36 hexagons.

92. In which of the following reactions is Cl_2 (gas) produced?
(a) $KCl + Br_2 \longrightarrow$
(b) $NaCl + K_2Cr_2O_7 + \text{conc. } H_2SO_4 \longrightarrow$
(c) $NaOCl + NH_3 \longrightarrow$
(d) $Ca(OCl)Cl + H_2O \longrightarrow$

93. Helium is suitable for low temperature gas thermometry because of its
(a) High boiling point and high polarizability.
(b) Low boiling point and near-ideal behaviour.
(c) High transition temperature.
(d) Real behaviour.

94. Phosphorous is obtained by the reduction of phosphate rock using
(a) Al at high temperature.
(b) Coke and silica at high temperature.
(c) Silica at high temperature.
(d) Fe_2O_3 and coke at high temperature.

95. Very pure silicon is prepared by
(a) Heating SiO_2 with KF.
(b) The electrolysis of SiO_2 in SiF_4.
(c) Reducing pure silicon tetrachloride with magnesium.
(d) Decomposing $K_2[SiF_6]$.

96. $NaHSO_4$ reacts with F_2 to produce mainly
(a) $Na_2S_2O_5$ (b) $Na_2S_2O_8$
(c) $Na_2S_2O_3$ (d) $Na_2S_2O_7$

97. Very pure silicon is an insulator, but becomes a p-type or an n-type semiconductor when doped with a
(a) Group 12 and a group 16 element, respectively
(b) Group 4 and group 6 element, respectively
(c) Group 1 and a group 12 element, respectively
(d) Group 13 and a group 15 element, respectively

98. Chlorine can be manufactured by the oxidation of HCl in air in the presence of a $CuCl_2$ catalyst at 450°C. The process is known as the
(a) Nelson process
(b) Deacon process
(c) Solvay process
(d) Chloride process

99. White phosphorous reacts with calcium to form a certain compound which, on hydrolysis, produces
(a) P_4O_6 (b) P_4O_{10}
(c) PH_3 (d) P_2H_4

100. Cl_2 reacts with dilute NaOH and concentrated NaOH to, respectively produce
(a) $NaClO_3$ and NaClO
(b) NaClO and $NaClO_2$
(c) NaCl and $NaClO_4$
(d) NaClO and $NaClO_3$

Multiple Correct Answer Type Question (More Than One Choice)

101. Which of the following halogens does not turn starch iodine paper blue?
(a) I_2 (b) F_2
(c) Cl_2 (d) Br_2

102. Iodine reacts with hypo to give
(a) Na_2S (b) NaI
(c) Na_2SO_3 (d) $Na_2S_4O_6$

103. Electrolysis of aqueous solution of Brine (NaCl) gives
(a) O_2 (b) NaOH
(c) H_2 (d) Cl_2

104. Cl_2 reacts with hot aqueous NaOH to give
(a) NaClO (b) $NaClO_4$
(c) NaCl (d) $NaClO_3$

105. The halogens, which are not attacked by conc. HNO_3 are
(a) Br_2 (b) F_2
(c) I_2 (d) Cl_2

106. Select the incorrect statement about Buckyball or Buckminister fullerene?
(a) It is an allotrope of carbon.
(b) It is referred as C-60 and C-70.
(c) In it carbon atoms are sp^3 hybridized.
(d) It is inert like diamond.

107. Which one of the following arrangements do not truely represent the property indicated against it?
(a) $Br_2 < Cl_2 < F_2$ Electronegativity
(b) $Br_2 < Cl_2 < F_2$ — Bond energy
(c) $Br_2 < Cl_2 < F_2$ — Electron affinity
(d) $Br_2 < Cl_2 < F_2$ oxidizing power

108. Which of the following noble gases do not form any compound?
(a) Kr (b) He
(c) Xe (d) Ne

109. Which of the following names can be used for group VIIIA elements?
(a) Rare earths
(b) Inert gases
(c) Rare gases of armosphere
(d) Noble gases

110. Which of the following compounds contain boron?
(a) Colemanite (b) Kernite
(c) Borax (d) Cristoballite

111. Which of the following will displace the halogen from the solution of the halide?
(a) Cl_2 added to a KBr solution
(b) Br_2 added to an NaCl solution
(c) Br_2 added to a KI solution
(d) Cl_2 added to an NaF solution

112. Which of the following pair of reactants give oxygen on reaction with each other?
(a) Cl_2, NaOH (dilute and cold)
(b) F_2, H_2O
(c) F_2, NaOH (hot and conc.)
(d) $CaOCl_2$, H_2SO_4 (dil. and less amount)

113. Nitrogen is prepared by heating
(a) Microcosmic salt, $NaNH_4HPO_4.4H_2O$
(b) A mixture of CuO and NH_3
(c) A mixture of NH_4Cl and $NaNO_3$
(d) Barium azide

114. Chlorine is produced by the
(a) Evaporation of sea water .
(b) Action of concentrated H_2SO_4 on NaCl in the presence of MnO_2.
(c) Electrolysis of an aqueous solution of NaCl.
(d) Action of concentrated HCl on MnO_2.

115. Graphite is a
(a) Good conductor of electricity.
(b) Good insulator.
(c) Bad conductor of heat.
(d) Good conductor of heat.

116. Chlorine behaves as an oxidizing agent upon reaction with
(a) $Na_2S_2O_3$ (b) $NaNO_2$
(c) $Fe_2(SO_4)_3$ (d) O_3

117. Which of the following allotropes of sulphur exist as S_8 molecules with a puckered-ring structure assuming a crown conformation?
(a) α-Rhombic (b) β-Monoclinic
(c) γ-Rhombic (d) γ-Monoclinic

118. Which of the following statements are correct for the nitrogen molecule?
(a) It is a good ligand.
(b) The bond order is 2.2.
(c) It easily reacts with magnesium even at room temperature.
(d) It is used in the Serpeck's process.

119. Select the incorrect statement.
(a) Fluorine cannot show oxidation number greater than zero.
(b) Halogens do not directly combine with oxygen.
(c) Iodine cannot displace chlorine from $KClO_3$.
(d) Euchlorine is a mixture of Cl_2 and Cl_2O_7.

120. Which of the following compounds react with fluorine?
(a) KF (b) NaCl
(c) Al_2O_3 (d) B_2O_3

121. Which of the following are bent?
(a) Br_2^- (b) I_3^-
(c) I_3^+ (d) F_3^-

122. Which of the following statements are correct for the P_4 molecule?
(a) It reacts with nitrogen to form a phosphorous nitrogen polymer.
(b) The electron affinity of phosphorous is negative.
(c) The P–P bond distance are equal.
(d) The P–P–P bond angles are 60°.

123. Which of the following statement is/are correct?
(a) C_{60} has a geodesic dome structure.
(b) Solid C_{60} has a cubic close-packed structure.
(c) The name buckminsterfullerene was given to C_{60}.
(d) The common name for C_{60} is 'bucky ball'.

124. Buckminsterfullerene is prepared by
(a) Vapourization of carbon by resistive heating.
(b) Passing an arc discharge between carbon electrodes in a tube containing helium at 100 torr.
(c) The pulsed laser vapourization of graphite.
(d) None of these.

125. Which of the following reactions are possible?
(a) $Ba(N_3)_2 \xrightarrow{\Delta} Ba + 3N_2$ (Pure form)
(b) $12C + 9H_2SO_4 \longrightarrow C_6(COOH)_6 + 6H_2O + 9SO_2$ (Mellitic acid)
(c) $CaC_2 + N_2 \xrightarrow{100^\circ C} CaCN_2 + C$
(d) $KIO_3 + F_2 + H_2O \rightarrow KIO_4 + H_2F_2$

Linked–Comprehension Type Questions

Comprehension-1

If we are given some elements without mentioning their names and considering them as (P), (Q), (R), (S) and (T). Here, we have been given some properties regarding these elements.

Elements	I_1(kJmol^{-1})	Electronegativity
(P)	1012	2.1
(Q)	1314	3.5
(R)	1402	3.0
(S)	1680	4.0
(T)	1256	3.2

126. Which of these elements has maximum number of allotropes?
(a) R (b) P
(c) Q (d) S

127. For these elements the decreasing order of electron affinity is
(a) S > T > Q > P > R
(b) S > T > P > Q > R
(c) T > S > Q > P > R
(d) T > S > P > Q > R

128. Here lowest and highest possible oxidation state are shown by
(a) Lowest by P and highest by T.
(b) Lowest by S and highest by T.
(c) Lowest by P, R and highest by S.
(d) Lowest by P, R and highest by T.

Comprehension-2

A black powder (P) when heated with a light metal chloride and conc. sulphuric acid gives out a greenish yellow gas (Q). This gas (Q) on passing through liquor ammonia liberates another gas (R) and on passing through boiling KOH gives compounds, one of which (X) on heating with the black powder (P) gives another gas (S).

129. Here the black powder (P) is
(a) Cr_2O_3 (b) HgS
(c) MnO_2 (d) PbS

130. Here greenish yellow gas is
(a) Oxygen (b) SO_2
(c) Fluorine (d) Chlorine

131. The gas (R) has a bond order of
(a) 2 (b) 1
(c) 3 (d) 2.5

132. Here the compound (X) and the gas (S) are respectively
(a) KCl and O_2 (b) $KClO_3$ and O_2
(c) $KClO_3$ and SO_2 (d) KCl and F_2

Comprehension-3

Three elements (P), (Q), (R) are taken. Both (P) and (Q) can form compounds with (R). (P) forms five compounds with (R) in which its oxidation states are +1 to +5 while (Q) can form mainly two compounds with it, with oxidation number +3 and +5.

133. Here (P), (Q) and (R)
(a) P, N, S (b) P, O, N
(c) N, P, O (d) N, P, S

134. The number of sigma bonds in the compounds of (Q) and (R) in +3 and +5 oxidation states are, respectively
(a) 12, 14 (b) 12, 16
(c) 14, 16 (d) 16, 18

135. The formula of compounds of (P) and (R) having brown and blue colour are, respectively
(a) PR_2 and P_2R_5 (b) P_2R_3 and PR_2
(c) P_2R and PR_2 (d) PR_2 and P_2R_3

Assertion and Reasoning Questions

In the following questions two statements (Assertion) (A) and Reason (R) are given Mark
(a) If both A and R are correct and R is the correct explanation of A
(b) If both A and R are correct but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true
(e) A and R both are false

136. (A): White phosphorous is less stable than red phosphorous.
(R): White phosphorous exists as P_4 molecules.

137. (A): Among elements of group 13, boron has the highest melting point.
(R): Boron is metallic in nature.

138. (A): Diamond is harder than graphite.
(R): Graphite is more stable than diamond.

139. (A): P_4 is more reactive than N_2.
(R): P–P bonds are relatively weaker than N≡N bond.

140. (A): Trihalides of boron family fume in moist air.
(R): Trihalides of group 13 elements are hygroscopic.

141. (A): Al forms $[AlF_6]^{3-}$ but B does not form $[BF_6]^{3-}$.
(R): B does not react with F_2.

142. (A): When a mixture of potassium chlorate and MnO_2 in 4:1 ratio is heated at 375°C, O_2 is formed.
(R): Here, MnO_2 must be free from carbon.

143. (A): White phosphorous is stored under water.
(R): White phosphorous is highly reactive and catches fire spontaneously in air.

144. (A): Si–Si bonds are much stronger then Si–O bonds.
(R): Silicon does not form double bonds with itself.

145. (A): Bond energy of chlorine is more than bond energy of fluorine.
(R): Bond energy of X_2 (X ➤ halogen) decreases down the group.

146. (A): Phosphorite mineral cannot be used as a fertilizer of phosphorous.
(R): Super phosphate of lime is used as a fertilizer of phosphorous.

147. (A): The stronger oxidizing property of F is due to the smaller the value of dissociation energy of F_2 molecule.

(R): There is a repulsion operating between the nonbonding 2p electrons on the bonded F-atoms.

148. (A): In presence of moisture, Cl_2 can act as an oxidant and bleaching agent.

(R): Chlorine reacts with moisture to give HCl and HClO. HClO being less stable decomposes to give nascent oxygen.

149. (A): Deep sea divers use heliumoxygen mixture for respiration.

(R): Helium is inert in nature.

150. (A): Iodine is liberated when KI is added to a solution of Cu^{2+} ions but Cl_2 is not liberated when KCl is added to a solution of Cu^{2+} ions.

(R): I^- ions are strong reducing agent whereas as Cl^- ions does not act as reducing agent.

151. (A): Bond strengths in the nitrogen, oxygen and fluorine molecules follow the order $N_2 > O_2 > F_2$.

(R): The electronegativity increases in the order N < O < F.

152. (A): Fluorine acts as a stronger oxidizing agent than oxygen.

(R): Fluorine is more electronegative than oxygen.

153. (A): Iodine is less soluble in water.

(R): It becomes more soluble in presence of KI due to the formation of KI_3.

154. (A): In S_8 molecule, each S atom is bonded to two S atoms.

(R): Each S atom is S_8 molecules is sp^3 hybridized, having two lone pairs of electrons and bonded to two other S-atoms.

155. (A): Iodine shows oxidation state of +1 and +3 in the compounds ICl and ICl_3, respectively.

(R): Iodine coming below the halogens F, Cl and Br in the halogen group of elements in the periodic table shows a higher degree of electropositive nature.

Matrix–Match Type Questions

	p	q	r	s
(A)	O	O	O	O
(B)	O	O	O	O
(C)	O	O	O	O
(D)	O	O	O	O

156. Match the following:

Column I (Allotropic form)	Column II (Structure)
A. Engel's sulphur	(p) Crystalline form yellow crystals
B. χ-sulphur	(q) Fibrous or rubber like
C. Rhombic sulphur	(r) Puckered S_8 rings (crown configuration)
D. γ-monoclinic	(s) S_4 rings, chair conformation, unstable

157. Match the following:

Column I	Column II
A. Boron	(p) Amphoteric oxide
B. Carbon	(q) Acidic oxide
C. Silicon	(r) Catenation
D. Phosphorous	(s) Allotropy

158. Match the following:

Column I	Column II
A. F_2	(p) Metallic lusture
B. Cl_2	(q) Most electronegative
C. Br_2	(r) Highest bond energy
D. I_2	(s) Reddish liquid

159. March the following:

Column I	List II
A. Diamond	(p) sp^2 hybridization
B. Graphite	(q) sp^3 hybridization
C. Fullerene	(r) Tetrahedral
D. Norbide	(s) Carbide

160. Match the following:

Column I	Column II
A. F_2	(p) pale yellow coloured gas
B. Cl_2	(q) violet colour solid
C. Br_2	(r) orange red liquid
D. I_2	(s) greenish coloured gas

161. March the following:

Column I	Column II
A. F_2	(p) Oxidizing nature
B. Cl_2	(q) Reaction with dil or conc. NaOH
C. Br_2	(r) Pale greenish yellow
D. I_2	(s) No bleaching action

162. Match the following:

Column I	Column II
A. Diamond	(p) Amorphous allotrope
B. Lubricant	(q) Crystalline allotrope
C. Sugar charcoal	(r) SiO_2
D. Acid flux	(s) Graphite

163. Match the following:

Column I	Column II
A. Oxygen	(p) Octa atomic
B. Sulphur	(q) Diatomic
C. Nitrogen	(r) Paramagnetic
D. Chlorine	(s) Diamagnetic

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164. White phosphorus (P_4) has
I. six P-P sigma bonds
II. four P-P single bonds
III. four lone pairs of electrons
IV. PPP angle of 60°
(a) 1, 2, 3 (b) 2, 3, 4
(c) 1, 3, 4 (d) all are correct
[*IIT 1998*]

165. Sodium nitrate decomposes above 800°C to give
(a) N_2 (b) O_2
(c) NO_2 (d) Na_2O
[*IIT 1998*]

166. A liquid (A) is treated with Na_2CO_3 solution. A mixture of two salts (B) and (C) are produced in the solution. The mixture on acidification with sulphuric acid and distillation produces the liquid (A) again. Identify (A) here.
(a) Cl_2 (b) O_2
(c) Br_2 (d) N_2
[*IIT 2000*]

167. $(NH_4)_2Cr_2O_7$ on heating liberates a gas. The same gas will be obtained by
(a) heating NH_4NO_2
(b) heating NH_4NO_3
(c) treating H_2O_2 with $NaNO_2$
(d) treating Mg_3N_2 with H_2O
[*IIT 2004*]

168. The gas which is liberated when PbO_2 reacts with conc. HNO_3
(a) NO_2 (b) O_2
(c) N_2O (d) N_2
[*IIT 2005*]

169. Which allotrope of phosphorus is thermodynamically most stable?
(a) Red (b) Black
(c) White (d) Yellow
[*IIT 2005*]

170. The reaction of P_4 with X leads selectively to P_4O_6. Then 'X' is
(a) Dry O_2 (b) A mixture of O_2, N_2
(c) Moist O_2 (d) O_2 with Aq NaOH
[*IIT 2009*]

ANSWERS

Straight Objective Type Questions

1. (a)	2. (c)	3. (d)	4. (a)
5. (c)	6. (a)	7. (a)	8. (d)
9. (a)	10. (d)	11. (b)	12. (b)
13. (c)	14. (d)	15. (b)	16. (b)
17. (c)	18. (b)	19. (a)	20. (b)
21. (c)	22. (b)	23. (d)	24. (c)
25. (b)	26. (a)	27. (b)	28. (b)
29. (a)	30. (b)	31. (d)	32. (d)
33. (d)	34. (b)	35. (a)	36. (a)
37. (d)	38. (a)	39. (a)	40. (d)
41. (a)	42. (b)	43. (b)	44. (d)
45. (b)	46. (a)	47. (d)	48. (a)
49. (c)	50. (a)	51. (b)	52. (a)
53. (a)	54. (a)		

Brainteasers Objective Type Questions

55. (c)	56. (b)	57. (b)	58. (b)
59. (a)	60. (b)	61. (b)	62. (a)
63. (b)	64. (c)	65. (c)	66. (b)
67. (a)	68. (c)	69. (d)	70. (b)
71. (a)	72. (c)	73. (b)	74. (b)
75. (c)	76. (b)	77. (a)	78. (b)
79. (b)	80. (d)	81. (d)	82. (a)
83. (a)	84. (b)	85. (c)	86. (c)
87. (d)	88. (a)	89. (c)	90. (b)
91. (a)	92. (d)	93. (b)	94. (b)
95. (c)	96. (b)	97. (d)	98. (b)
99. (c)	100. (d)		

Multiple Correct Answers Type Questions

101. (a), (b)	102. (b), (d)
103. (b), (c), (d)	104. (c), (d)
105. (a), (b), (d)	106. (c), (d)
107. (b), (c)	108. (b), (d)
109. (b), (c), (d)	110. (a), (b), (c)
111. (a), (c)	112. (b), (c), (d)
113. (b), (d)	114. (b), (c), (d)
115. (a), (d)	116. (a), (b)
117. (a), (b), (d)	118. (a), (d)
119. (c), (d)	120. (a), (b), (c)
121. (a), (c)	122. (c), (d)
123. (a), (b), (c), (d)	124. (a), (b), (c)
125. (a), (b), (d)	

Linked–Comprehension Type Questions

Comprehension-1

126. (b) 127. (c) 128. (d)

Comprehension-2

129. (c) 130. (d) 131. (c) 132. (b)

Comprehension-3

133. (c) 134. (b) 135. (d)

Assertion and Reasoning Questions

136. (b)	137. (c)	138. (b)	139. (a)
140. (c)	141. (b)	142. (b)	143. (a)

144. (d) 145. (b) 146. (b) 147. (a)

148. (a) 149. (b) 150. (a) 151. (b)

152. (a) 153. (b) 154. (a) 155. (a)

Matrix–Match Type Questions

156. (a)-(s), (b)-(p), (d)-(r)

157. (a)-(p, (s), (b)-(q), (r), (s), (c)-(q), (r), (s), (d)-(q), (r), (s)

158. (a)-(q), (b)-(r), (c)-(s), (d)-(p)

159. (a)-(q), (r), (b)-(p), (c)-(p), (d)-(s)

160. (a)-(p), (b)-(s), (c)-(r), (d)-(q)

161. (a)-(p), (q), (r), (b)-(p), (q), (r), (c)-(p), (q) (d)-(p), (q), (s)

162. (a)-(q), (b)-(s), (c)-(p), (d)-(r)

163. (a)-(q), (r), (b)-(p), (c)-(q), (s), (d)-(q), (s)

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164. (c) 165. (b) 166. (c) 167. (a)

168. (b) 169. (b)

HINTS AND EXPLANATIONS

Straight Objective Type Questions

2. Si obtained by reduction of $SiCl_4$ with H_2 is further purified by zone refining method to obtain Si of very high purity. Silicon is purified by zone-refining process because the impurities present in it are more soluble in the liquid phase than in the solid phase.

3. In graphite, each carbon is sp^2 hybridized and the single occupied unhybridized p-orbitals of C-atoms overlap sidewise to give π-electron cloud which is delocalized and thus the electrons are spread between the structure.

4. F_2 is the strongest oxidizing agent and hence is most easily reduced.

5. Zone refining method is the best method for purification of semi-conductor.

7. $4B + C \xrightarrow{\text{Heated}} B_4C$

   The molecular formula of boron carbide is $B_{12}C_3$.

8. Bleaching action of chlorine is only in the presence of moisture where nascent oxygen is displaced from H_2O.

   $Cl_2 + H_2O \longrightarrow HCl + HClO$ (unstable)

   $HClO \longrightarrow HCl + [O]$

10. $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

    $2NaN_3 \xrightarrow{\Delta} 2Na + 3N_2$
    Sodium azide

    $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

11. Diamond is chemically inactive allotrope of carbon because its compact structure and saturated nature.

14. Diamond has high refractive index. The value of $\mu = 2$, only some synthetic compound having such a high value of refractive index.

17. Both white and red phosphorous are not soluble in CS_2, only white P is soluble.

19. $Cl_2 + H_2O \xrightarrow{\Delta} HCl + HClO$

    $HClO \longrightarrow HCl + [O]$

    $Cl_2 + H_2O \longrightarrow 2HCl + [O]$

    The bleaching action of chlorine is due to the liberation of nascent oxygen [O].

28. Air is liquefied by making use of the Joule–Thomson effect (cooling by expansion of the gas). Water vapour and CO_2 are removed by solidification. The remaining constituents of liquid air i.e., liquid oxygen and liquid nitrogen are separated by means of fractional distillation (b.p. of $O_2 = -183^\circ C$; b.p. of $N_2 = -195.8^\circ C$).

29. N_2 molecule contains triple bond between N atoms having very high dissociation energy (946 $kJmol^{-1}$) due to which it is relatively inactive.

30. This solution gives a coloured growth of crystals after sometime known as silica garden.

31. $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

    $Ba(N_3)_2 \xrightarrow{\Delta} 3N_2 + Ba$

    $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

32. Rocks are mainly made up of silicates.

36. Silicon due to its larger size form hydrides which are not so stable. However, hydrides of boron are stable.

40. This is a very familiar redox reaction.

$$S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + I$$

42. In F_2 due to greater inter electronic repulsions.

43. $C + 2H_2SO_4 \rightarrow CO_2 + 2H_2O + 2SO_2$

Here, carbon is oxidized to CO_2 and H_2SO_4 is reduced to SO_2.

53. Nitrogen in the laboratory can be obtained by heating ammonium dichromate.

$$(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_2 + 4H_2O$$

Nitrogen is collected by downward displacement of water.

54. Bromine in the mother liquor is oxidized to Br_2 by Cl_2 which is a stronger oxidizing agent.

$$2Br^- + Cl_2 \rightarrow Br_2 + 2Cl^-$$

Brainteasers Objective Type Questions

55. $\underset{(A)}{(NH_4)_2Cr_2O_7} \xrightarrow{\Delta} \underset{(C)}{N_2} + \underset{\substack{(B)\\ \text{Green}}}{Cr_2O_3} + 4H_2O$

$$N_2 + 3Mg \xrightarrow{\Delta} Mg_3N_2$$

$$\underset{(D)}{Mg_3N_2} + 6H_2O \xrightarrow{\Delta} 3Mg(OH)_2 + \underset{(E)}{2NH_3}$$

$$\underset{(E)}{NH_3} + HCl \longrightarrow \underset{\text{(White fumes)}}{NH_4Cl}$$

57. The gas is Cl_2 and the halate used in fire works and safety matches is $KClO_3$.

$$\underset{\substack{\text{Greenish}\\ \text{yellow gas}}}{3Cl_2} + 6KOH \longrightarrow \underset{\substack{\text{Potassium chlorate}\\ \text{(Halate)}}}{KClO_3} + 5KCl + 3H_2O$$

72. $S + 2Cl_2 \longrightarrow \underset{(X)}{SCl_4}$

$$SCl_4 + 4H_2O \rightarrow S(OH)_4 + 4HCl$$

$$\downarrow$$

$$\underset{(Y)}{H_2SO_3} + H_2O$$

The hybridized state of S in (Y) is sp^3.

Multiple Correct Answer Type Questions

106. As in ir carbon atoms are sp^2 hybridized and it is reacting just like graphite.

112. $Cl_2 + 2NaOH \longrightarrow NaCl + NaClO + H_2O$

$$2F_2 + 2H_2O \longrightarrow 4HF + O_2$$

$$2F_2 + 4NaOH \longrightarrow 4NaF + 2H_2O + O_2$$

$$2CaOCl_2 + H_2SO_4 \rightarrow CaCl_2 + CaSO_4 + 2HCl + O_2$$

119. As iodine can displace chlorine from $KClO_3$ and Euchlorine is a mixture of Cl_2 and ClO_2.

125. As the reaction (C) occurs at 1000°C.

$$CaC_2 + N_2 \xrightarrow{1000^\circ C} CaCN_2 + C$$

Linked–Comprehension Type Questions

Comprehension-1

126. Here, electronegativity and I_1 of P suggests that it is phosphorous so it will have maximum number of allotropes.

127. As the values of I_1 and electronegativity suggests that P, Q, R, S, T are, respectively phosphorous, oxygen, nitrogen, fluorine and chlorine and for them electron affinity decreases as

$$Cl > F > O > P > N$$

For example, T > S > Q > P > R.

128. As P, R are phosphorous and nitrogen which may have −3 oxidation state while T is chlorine has highest +7 oxidation state also.

Comprehension-2

129. $\underset{(P)}{MnO_2} + 2NaCl + 3H_2SO_4 \rightarrow 2NaHSO_4 + MnSO_4 + \underset{\substack{(Q)\\ \text{Greenish yellow}}}{Cl_2} + 2H_2O$

131. $3Cl_2 + 8NH_3 \rightarrow \underset{(R)}{N_2}\uparrow + 6NH_4Cl$

It is nitrogen which has a bond order 3.

132. $3Cl_2 + 6KOH \rightarrow \underset{(X)}{KClO_3} + 5KCl + 3H_2O$

$\underset{(X)}{2KClO_3} + MnO_2 \rightarrow 2KCl + 3O_2 + \underset{(S)}{MnO_2}$

Comprehension-3

134. Here, compounds are P_4O_6 and P_4O_{10}, respectively in which the number of sigma bonds are 12 and 16, respectively.

135. Since PR_2 is NO_2 which is a brown colour oxide and P_2R_3 is N_2O_3 which has light blue colour.

Assertion and Reasoning Questions

140. Trihalides of boron family undergo hydrolysis producing hydrogen halides due to which they fume in moist air.

141. B does not have vacant d-orbitals as second shell is the outermost shell.

142. As in presence of carbon an explosive mixture of $KClO_3$ and carbon is obtained so in order to avoid any vigorous reaction there should be no carbon.

143. The ignition temperature of white P is low (about 30°C). In air it readily catches fire giving dense fumes of phosphorous pentoxide. It is therefore, kept in water.

144. Si–Si bonds are weaker than Si–O bonds Si has no tendency to form double bonds with itself.

149. Unlike nitrogen, helium is not soluble in blood even under high pressure , that is why it is used by seadivers.

151. Bond strength not only depends on the electronegativity of the elements.

155. Oxidation states of iodine in ICl and ICl_3 are +1 and +3, respectively as Cl is more electronegative than I because down the group electropositive nature increases.

The IIT-JEE Corner

164. Six P–P bonds four lone pairs bond angle is 60°.

166. The liquid (A) is bromine which on rearment with sodium carbonate forms a mixture of NaBr and $NaBrO_3$ (sodium bromate). The mixture with conc. H_2SO_4 on distillation gives the liquid bromine again.

$3Br_2 + 3Na_2CO_3 \rightarrow 5NaBr + NaBrO_3 + 3CO_2$

$5NaBr + NaBrO_3 + 3H_2SO_4 \longrightarrow 3Na_2SO_4 + 3Br_2 + 3H_2O$

170. Here N_2 and O_2 mixture is used.

$P_4 + 3O_2 \rightarrow P_4O_6$

Here, N_2 reacts in further oxidation.

Solved Subjective Questions

1. State with balanced equations, what happens when:

(i) Nitrogen is obtained in the reaction of aqueous ammonia with potassium permanganate.

Solution

$4KMnO_4 + 2H_2O \longrightarrow 4MnO_2 + 4KOH + 3O_2$

$4NH_3 + 3O_2 \longrightarrow 2N_2 + 6H_2O$

(ii) Elemental phosphorous reacts with conc. HNO_3 to give phosphoric acid.

Solution

$P_4 + 20HNO_3 \xrightarrow{I_2 \text{ as catalyst}} 4H_3PO_4 + 20NO_2 + 4H_2O$

(iii) Sulphur is precipitated in the reaction of hydrogen sulphide with sodium bisulphite solution.

Solution

$2H_2S + NaHSO_3 + H \longrightarrow 3S\downarrow + 3H_2O + Na^+$

(iv) Phosphorous is treated with concentrated nitric acid.

or

Manufacture of phosphoric acid from phosphorous.

[*IIT 1997*]

Solution

$$[2HNO_3 \longrightarrow H_2O + 2NO_2 + [O]] \times 10$$

$$P_4 + 10\,[O] \longrightarrow P_4O_{10}$$

$$P_4O_{10} + 6H_2O \longrightarrow 4H_3PO_4$$

$$P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$$

2. State with balanced equations what happens when:

(i) Sodium salt of an acid (A) is formed on boiling white phosphorous with NaOH solution.

(ii) On passing chlorine through phosphorous kept fused under water, another acid (B) is formed which on strong heating gives metaphosphorous acid.

(iii) Phosphorous on treatment with conc. HNO_3 gives an acid (C) which is also formed by the action of dilute H_2SO_4 on powdered phosphorite rock.

(iv) (A) on treatment with a solution of $HgCl_2$ first gives a white precipitate of compound (D) and then a grey precipitate of (E).

Identify (A) to (E) and write balanced equations.

Solution

(i) $P_4 + 3NaOH + 3H_2O \longrightarrow$

$3NaH_2PO_2 + PH_3$

Sodium hypophosphite

Hence, acid (A) is H_3PO_2.

(ii) $2P + 3Cl_2 \longrightarrow 2PCl_3$

$2PCl_3 + 6H_2O \longrightarrow 2H_3PO_3 + 6HCl$

(B)

Phosphorous acid

$H_3PO_3 \xrightarrow{\text{Heat}} HPO_2 + H_2O$

(iii) $P_4 + 20\,HNO_3 \longrightarrow 4H_3PO_4 + 3CaSO_4$

(C)

Orthophosphoric acid

$Ca_3(PO_4)_2 + 3H_2SO_4 \longrightarrow$

Phosphorite rock

$2H_3PO_4 + 3CaSO_4$

(iv) $H_3PO_2 + 2H_2O \longrightarrow H_3PO_4 + 4H$

$2HgCl_2 + 2H \longrightarrow Hg_2Cl_2 + 2HCl$

(D)

White ppt.

$Hg_2Cl_2 + 2H \longrightarrow 2Hg + 2HCl$

(E)

Grey ppt.

3. Identify (P) to (S) in the following scheme of reaction.

$$KO_2 + S \xrightarrow{\Delta} (P) \xrightarrow{BaCl_2} \underset{\text{White ppt.}}{(Q)}$$

$$\text{Carbon,} \xrightarrow{\Delta} (R) \longrightarrow HCl \xrightarrow{(S)} \underset{\text{Gas}}{}$$

Solution

$$KO_2 + S \xrightarrow{\Delta} K_2SO_4 \xrightarrow{BaCl_2} \underset{\text{White ppt.}}{BaSO_4}$$

$$\text{Carbon, } \Delta \longrightarrow BaS \xrightarrow{HCl} \underset{\text{Gas}}{H_2S}$$

4. State with balanced equations what happens when:

(i) White phosphorous (P_4) is boiled with a strong solution of sodium hydroxide in an inert atmosphere.

(ii) Sodium iodate is treated with sodium bisulphide solution.

When white phosphorous is boiled with aqueous NaOH or alcoholic solution of potassium hydroxide, phosphine gas (PH_3) is liberated.

$$P_4 + 3NaOH + 3H_2O \longrightarrow \underset{\text{Sodium hypophosphite}}{NaH_2PO_2} + PH_3 \uparrow$$

(ii) $5NaHSO_3 + 2NaIO_3 \rightarrow 3NaHSO_4 + 2Na_2SO_4 + I_2 + H_2O$

(iii) Phosphorous reacts with nitric acid to give equimolar ratio of nitric oxide and nitrogen dioxide.

$4P + 10HNO_3 + H_2O \rightarrow 5NO + 5NO_2 + 4H_3PO_4$

(iv) Sodium bromate reacts with fluorine in presence of alkali.

Solution

$NaBrO_3 + F_2 + 2NaOH \longrightarrow NaBrO_4 + 2NaF + H_2O$

5. A soluble compound of a poisonous element M, when heated with Zn/H_2SO_4 gives a colourless and extremely poisonous gaseous compound N, which on passing through a heated tube gives a silvery mirror of element M. Identify M and N.

[*IIT 1997*]

Solution

M is As;

$AsCl_3 + 6H \xrightarrow{Zn/H_2SO_4} AsH_3 + 3HCl$

$2AsH_3 \xrightarrow{\Delta} 2As + 3H_2$

6. A colourless inorganic salt decomposes completely at about 250° C to give only two products, (B) and (C), leaving no residue. The oxide (C) is a liquid at room temperature and nautral to moist litmus paper while the gas (B) is a neutral oxide. White phophorus burns in excess of (B) to produce a strong white dehydrating agent. Write balanced equation for the reactions involved in the above process.

[*IIT 1996*]

Solution

$NH_4NO_3 \rightarrow N_2O + 2H_2O$

$10N_2O + P_4 \longrightarrow \underset{\text{(Dehydrating agent)}}{P_4O_{10}} + 10N_2$

7. Complete and balance the following chemical reactions

(i) Red phosphorous is reacted with iodine in presence of water.

$P + I_2 + H_2O \longrightarrow \ldots\ldots + \ldots\ldots$

(ii) Anhydrous potassium nitrate is heated with excess of metallic potassium.

$KNO_3 (s) + K (s) \rightarrow$

$NH_3 + NaOCl \longrightarrow$

Solution

(i) $2P + 3I_2 + 6H_2O \rightarrow 2H_3PO_4 + 6HI$

(ii) $2KNO_3 + 10\,K \longrightarrow 6K_2O + N_2$

(iii) $2NH_3 + NaOCl \rightarrow \underset{\text{Hydrazine}}{H_2N.NH_2} + NaCl + H_2O$

8. How is boron obtained from borax? Give chemical equations with reaction conditions. Write the structure of B_2H_6 and its reaction with HCl.

[*IIT 2002*]

Solution

When hot and concentrated HCl is mixed with borax ($Na_2B_4O_7.10\,H_2O$), the partially soluble H_3BO_3 is formed which on subsequent heating gives B_2O_3 which is reduced to boron on heating with Mg, Na or K.

$\underset{\text{Anhydrous}}{Na_2B_4O_7} + 2HCl \longrightarrow 2NaCl + \underset{\text{Hot \& conc.}}{H_2B_4O_7}$

$H_2B_4O_7 + 5H_2O \longrightarrow 4H_3BO_3 \downarrow$

$2H_3BO_3 \xrightarrow{\text{Strong heating}} B_2O_3 + 3H_2O$

$B_2O_3 + 6K \rightarrow 2B + 3K_2O$

or

$B_2O_3 + 6Na \longrightarrow 2B + 3Na_2O$

or

$B_2O_3 + 3Mg \rightarrow 2B + 3MgO$

$B_2H_6 + HCl \longrightarrow B_2H_5Cl + H_2$

Generally, this reaction occurs in the presence of Lewis acid ($AlCl_3$).

9. Mention the products formed in the following:

(i) Chlorine gas is bubbled through a solution of ferrous bromide.

(ii) Iodine is added to a solution of stannous chloride.

(iii) Aluminium sulphide gives a foul odour when it becomes damp. Write a balanced chemical equation for the reaction.

Solution

(i) $2FeBr_2 + 3Cl_2 \longrightarrow 2FeCl_3 + 2Br_2$

(ii) $SnCl_2 + I_2 \longrightarrow SnCl_2I_2$

$2SnCl_2 + 2I_2 \longrightarrow SnCl_4 + SnI_4$

[*IIT 1997*]

(iii) $Al_2S_3 + 6H_2O \longrightarrow 2Al(OH)_3\downarrow + 3H_2S\uparrow$ (foul odour)

Foul odour, on damping of Al_2S_3 is due to formation of H_2S gas.

10. (i) The first ionization energy of B is less then that of C whereas the second ionization energy of B is more than that of C.

Solution

Boron has one electron on its outermost orbit while carbon has two. When the first electron is taken out energy required in boron is less. But for 2nd IE, the 2nd electron in case of boron is to be removed from a completely fulfilled shell whereas there is still one electron in the outermost shell of carbon. Hence 1st IE of B is less than that of carbon while reverse is true for 2nd IE.

(ii) Gallium has a higher ionization energy than aluminium.

Solution

Gallium has higher IE than that of aluminium because of higher effective nuclear charge for gallium which is due to the fact that additional $3d^{10}$ electrons do not shield the nuclear charge effectively so that the outer electrons are more strongly held.

11. (i) Bond dissociation energy of $N_2 > N_2^+$ but that of $O_2 < O_2^+$

Solution

Bond order of N_2 is 3.0 while that of N_2^+ is 2.5 hence bond dissociation energy of $N_2 > N_2^+$. In O_2 molecule of the 16 electrons 10 are present in bonding and 6 in antibonding orbitals, the bond order is2.0. One of the anti-bonding electron is lost to give O_2^+ ion; the bond order changes to 2.5 and hence the bond dissociation energy of $O_2^+ > O_2$.

(ii) Boron has a high melting point.

Solution

Melting point of boron is very high because it has a giant three dimensional structure in which the boron atoms are held together by strong force of attraction, which means more energy is needed to break the structure.

(iii) Boron cannot from B^{3+} ion.

Solution

As The sum of the first IE for boron is very high hence the total energy required to produces B^{3+} ions is much more than would be compensated by lattice energies of ionic compounds or by hydration energy of B^{3+} ions in solution. As a result boron can not form B^{3+} ions.

12. (i) Diamond is covalent yet its melting point is very high.

Solution

In diamond, the carbon atoms are arranged in cubic closed packed pattern, i.e., they have a face-centered cubic structure. Each carbon atom is sp^3 hybridized and is bound to four other C atoms. This results a three dimensional network of strong covalent bonds. This causes the melting point of diamond to be unusually high.

(ii) Elemental silicon does not form graphite like structure as carbon does.

Solution

Carbon forms graphite like structure. In this structure each carbon in a layer is covalently bonded to three other carbon atoms by overlap of sp^2 hybrid orbitals in a hexagonal pattern.

Two adjacent layers are bonded by p electron interactions.

On the other hand, Si does not form graphite like structure because

(i) The size of Si atom is larger than that of carbon. The overlap of two 3p orbitals results in poor overlap and hence a weaker bond

(ii) The catenation tendency to form ring structure by Si atom is very small because Si–Si bond energy is lower than C–C bond energy.

Questions For Self Assessment

13. Given reasons for the following:

(i) Carbons acts as an abrasive and also as a lubricant.

[*IIT 1981*]

(ii) Sulphur melts to a clear mobile liquid at 119°C, but on further heating above 160°C, it becomes viscous.

[*IIT 1981*]

(iii) Graphite is used as a solid lubricant.

[*IIT 1985*]

14. Give reasons for the following:

(i) Fluorine cannot be prepared from fluorides by chemical oxidation.

[*IIT 1985*]

(ii) Valency of oxygen is generally two, whereas sulphur shows valency of two, four and six.

[*IIT 1988*]

(iii) Bond dissociation energy of F_2 is less than that of Cl_2.

[*IIT 1992*]

15. (i) Give reasons why elemental nitrogen exists as a diatomic molecule whereas elemental phosphorous as a tetraatomic molecule.

[*IIT 2000*]

(ii) The O–O bond energy is less than the S–S bond energy. Explain.

16. (i) Write the two resonance structures of ozone which satisfy the octet rule.

(ii) O_2 is inert room temperature. Why?

(iii) Red phosphorus is denser and chemically less reactive than white phosphorus. Why?

COMPOUNDS OF LIGHTER METALS–I

4

Chapter Contents

Oxides, peroxides, hydroxides, carbonates, bicarbonates, chlorides and sulphates of sodium, potassium, magnesium, calcium and Compounds of aluminium (like alumina, aluminium chlorides and alums) and various levels of multiple-choice questions.

COMPOUNDS OF ALKALI METALS

These metals mainly form ionic compounds as they have a strong tendency to loose one valence electron to form M^+ type of cation. This tendency increases on moving from Li to Cs, that is, on moving down the group.

Action of Air

Alkali metals (except lithium) in air get tarnished at once and give oxides, hydroxides and carbonates so they are kept in inert solvents like kerosene oil, paraffin oil to prevent their reaction with air.

$$M \xrightarrow{O_2} M_2O \xrightarrow{H_2O} MOH \xrightarrow{CO_2} M_2CO_3$$

$$M_2O \xrightarrow{O_2} M_2O_2$$

- Reactivity of IA metals towards air is

  Cs > Rb > K > Na > Li

Oxides of Alkali Metals

Alkali metals on combustion with oxygen give oxides (M_2O), peroxide (M_2O_2) and superoxide (MO_2).

$$M \xrightarrow{O_2} M_2O \xrightarrow{O_2} M_2O_2 \xrightarrow{O_2} MO_2$$

The formation and stability of these type of oxides by alkali metals can be expressed on the basis of their ionic radii and lattice energy. Li^+ being the smallest cation, has a strong positive field around it thus it can stabilize only a smaller anion while Na^+ being larger in size can stabilize a more large anion and so on. The strong positive field around Li^+ can attract the negative charge very firmly hence O^{2-} ion cannot form O_2^{2-} ion. That is the reason why Li_2O is possible but in case of other M^+, K^-, Rb^+, Cs^-, O^{2-} can further combine with oxygen so other oxides are also possible as follows:

$$\underset{\text{Monoxide ion}}{O^{2-}} \xrightarrow{|O|} \underset{\text{Peroxide ion}}{O_2^{2-}} \xrightarrow{O_2} \underset{\text{Super oxide ion}}{2O_2}$$

- Li gives only Li_2O, Na forms true peroxide and K, Rb, Cs form MO_2 (superoxides).

Li_2O	Na_2O	K_2O	Rb_2O	Cs_2O
White	Pale Yellow	Bright yellow	Orange	

Li_2O, Na_2O, K_2O and Rb_2O have anti-fluorite crystal structures while CS_2O has an anti $CdCl_2$ layer structure. It undergoes sublimation and gives white sublimates like NH_4X, $HgCl_2$, Hg_2Cl_2, $AlCl_3$, As_2O_3 and Sb_2O_3.

On moving from Li_2O to Cs_2O, basic nature increases.

M_2O_2 are colourless, diamagnetic in nature and strong oxidizing agents. MO_2 are coloured, paramagnetic and stronger oxidizing agents than M_2O_2.

$$Na_2O_2 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O_2$$

$$4KO_2 + 2H_2SO_4 \longrightarrow 2K_2SO_4 + 2H_2O + 3O_2$$

$$2KO_2 + 2H_2O \rightarrow 2KOH + H_2O_2 + O_2$$

KO_2 is used in space capsules for artificial respiration.

$$KO_2 + 2CO_2 \rightarrow 2K_2CO_3 + 3O_2 \uparrow$$

Hydroxides of Alkali Metals (MOH)

Alkali metals react with water to form metal hydroxide which are strongly alkaline in nature as these metals are highly electropositive.

$$2M + 2H_2O \rightarrow 2MOH + H_2 \uparrow$$

Reactivity order of alkali towards water is

Cs > Rb > K > Na > Li

These metal hydroxides are white crystalline solids having high solubility in water and alcohol.

LiOH , NaOH , KOH , RbOH, CsOH ➤

Basic nature, solubility in H_2O, thermal stability increase from LiOH to CsOH.

Halides of Alkali Metals (MX)

Alkali metals react with halogens to form metal halides, which are ionic in nature.

$$2M + X_2 \rightarrow 2MX + \text{heat}$$

Reactivity order of alkali metals: Reactivity order towards halogens is

Cs > Rb > K > Na > Li

LiX , NaX , KX , RbX, CsX ➤

Ionic nature, solubility in H_2O increases while lattice energy decreases.

For example,

LiF < NaF < KF < RbF < CsF

KF < KCl < KBr < KI

These halides are colourless, however on heating they become coloured due to nonstoichiometry and crystal defects.

REMEMBER

- LiF is ionic but insoluble in H_2O due to very high lattice energy while LiCl, LiBr, LiI are covalent less soluble or insoluble as their hydration energy is less than lattice energy.
- Other M X are crystalline solids with high melting and boiling points.
- Halides of K, Rb, Cs can also form polyhalides.

  For example,

  $KI + I_2 \longrightarrow KI_3$

Carbonates of Alkali Metals (M_2CO_3)

Alkali metals form M_2CO_3 type of carbonates which are thermally very stable and readily soluble in water. They are solubility in water and thermal stability increases from Li_2CO_3 to Cs_2CO_3 as electropositive nature increases down the group.

$$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$$

These carbonates do not decompose on heating except Li_2CO_3 which decompose on heating.

$$M_2CO_3 \xrightarrow{\Delta} XXX$$

$$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2 \uparrow$$

Li_2CO_3 is insoluble in water and decompose on heating as Li^+ ion being smallest cation

has maximum polarizing power so it distorts the electron cloud of the neighbour oxygen atoms of the carbonate ion as a result C–O becomes weak and Li O becomes stronger so Li_2O is formed on decomposition.

Bicarbonates of Alkali Metals ($MHCO_3$)

Alkali metals form $MHCO_3$ type of bicarbonates and their solubility in H_2O and thermal stability increases from $NaHCO_3$ to $CsHCO_3$. While $LiHCO_3$ is insoluble in water.

On heating, bicarbonates decompose as follows:

$$2MHCO_3 \xrightarrow{\Delta} M_2CO_3 + CO_2 + H_2O$$

- Only alkali metals form stable solid bicarbonates due to their highly electropositive nature (except lithium).

Sulphates of Alkali Metals (M_2SO_4)

These elements form M_2SO_4 type of sulphates and these are soluble in water except Li_2SO_4. These sulphates on fusion with carbon form sulphides as follows:

$$M_2SO_4 + 4C \rightarrow M_2S + 4CO$$

COMPOUNDS OF SODIUM

Sodium oxide (Na_2O)

Preparation

1. By adding sodium nitrate or sodium nitrite with sodium

$$2NaNO_3 + 10Na \longrightarrow 6Na_2O + N_2$$

$$2NaNO_3 + 6Na \rightarrow 4Na_2O + N_2$$

2. By adding sodium azide (NaN_3) with $NaNO_2$

$$3NaNO_3 + NaNO_2 \rightarrow 2Na_2O + 5N_2$$

(Pure)

Physiochemical Properties

1. It is white amorphous powder.
2. Reaction with Water

   It dissolves in water with evolution of much heat.

$$Na_2O + H_2O \rightarrow 2NaOH$$

3. Effect of Heat

$$2Na_2O \xrightarrow{400^\circ C} Na_2O_2 + 2Na$$

4. Reaction with Liquid Ammonia

$$Na_2O + NH_3 \rightarrow \underset{\text{Sodamide}}{NaOH + NaNH_2}$$

Sodium Peroxide (Na_2O_2)

Preparation

It is prepared by heating sodium with excess of dry air, free from CO_2 on aluminium tray as follows:

$$2Na + O_2 \xrightarrow{350^\circ C} Na_2O_2$$

Physiochemical Properties

1. It is slightly yellow powder due to the presence of small amount of sodium oxide but in pure form it is colourless.

2. Effect of Air

It turns white on exposure to air due to the formation of NaOH and Na_2CO_3.

$$2Na_2O_2 + 2H_2O \rightarrow 4NaOH + O_2$$

$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

3. Reaction with Water

When it is treated with water it gives NaOH and oxygen.

$$2Na_2O_2 + 2H_2O \longrightarrow 4NaOH + O_2$$

4. With dilute Acids

It reacts with cold and dilute acids to give hydrogen peroxide while on heating it gives water and oxygen as follows:

$$Na_2O_2 + H_2SO_4 \longrightarrow Na_2SO + H_2O_2$$

$$2Na_2O_2 + 2H_2SO_4 \xrightarrow{\Delta} 2Na_2SO_4 + 2H_2O + O_2$$

5. As a Powerful Oxidizing Agent

Being a strong oxidizing agent it oxidizes chromic compounds into chromates and sulphides into sulphates etc.

$$3Na_2O_2 + 3H_2O_2 \longrightarrow 6NaOH + 3[O]$$

$$2Cr(OH)_3 + 4NaOH + 3[O] \longrightarrow 2Na_2CrO_4 + 5H_2O$$

$$2Cr(OH)_3 + 3Na_2O_2 \longrightarrow 2Na_2CrO_4 + 2NaOH + 2H_2O$$

$$2C_6H_5COCl + Na_2O_2 \longrightarrow (C_6H_5CO)_2O_2 + 2NaCl$$

Benzoyl chloride — Benzoyl peroxide

6. Action of CO and CO_2

It reacts with carbon monoxide and carbon dioxide to form sodium carbonate as follows:

$$Na_2O_2 + CO \longrightarrow Na_2CO_3$$

$$2Na_2O_2 + 2CO_2 \longrightarrow 2Na_2CO_3 + O_2$$

Uses

1. As it readily combines with CO and CO_2, it is used for the purification of air in submarines, diving bells and other confined spaces.
2. It is used as an oxidizing agent.
3. It is also used for bleaching straw, silk etc., in the form of soda bleach (Na_2O_2 + dil. HCl) i.e., oxone.
4. It is used in the preparation of dyes and some other chemicals like benzoyl peroxide, sodium per borate etc.

Caustic Soda or Sodium Hydroxide (NaOH)

Methods of Preparation

It is prepared by the following methods:

Gossage or Causticization Method

In this method a suspension of lime [$CaO + Ca(OH)_2$] is treated with sodium carbonate to obtain NaOH as follows:

$$Na_2CO_3 + Ca(OH)_2 \longrightarrow 2NaOH + CaCO_3$$

(aq.) (aq.) (aq.)

From here, calcium carbonate can be easily separated and caustic soda solution can be easily drained out and evaporated to dryness to obtain in crystalline form. This NaOH is not pure and has impurities like $CaCO_3$, Na_2CO_3, $Ca(OH)_2$ etc.

Lowig's Method

Here a mixture of sodium carbonate and ferric oxide is heated in a revolving furnace upto redness to get sodium ferrite which is first of all cooled and hydrolyzed by hot water into NaOH solution and in soluble ferric oxide. The solution is filtered and evaporated upto dryness to get the flacks of NaOH.

$$Na_2CO_3 + Fe_2O_3 \xrightarrow[CO_2\uparrow]{\text{Fusion}} 2NaFeO_2$$

$$2NaFeO_2 \xrightarrow[\Delta]{H_2O} 2\,NaOH + Fe_2O_3$$

- In both Gossage and Lowig's method, the initial material is sodium carbonate.

Modern Method or By Electrolysis of Brine Solution (aq. NaCl)

For the electrolysis of aqueous NaCl electrolytic cells like Nelson, Castner Kelner, Salvay cells are used. Here NaOH, Cl_2, H_2 are formed.

Reactions

$$NaCl \rightleftharpoons Na^+ + Cl^-$$

$$H_2O \rightleftharpoons H^+ + OH$$

At Anode

$$2Cl^- \rightleftharpoons Cl_2 + 2e^-$$

At Cathode

$$2Na^+ + 2e \longrightarrow 2Na$$

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2\uparrow$$

- As here chlorine can react with NaOH solution even in cold hence it is necessary that it must be kept away from NaOH by using a porous diaphragm or by using a mercury cathode so that this reaction can be checked.

$$2NaOH + Cl_2 \longrightarrow NaClO + NaCl + H_2O$$

Physical Properties

1. It is a white crystalline, deliquescent soapy solid with a meting point of 591.4 K.
2. It is highly soluble in water but less soluble in alcohol.
3. It is corrosive in nature and bitter in taste.

Chemical Properties

1. Effect of Atmosphere

It absorbs moisture and CO_2 from atmosphere and changes into Na_2CO_3, hence it cannot be kept in atmosphere.

$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

2. Basic Nature

It is a strong base hence it reacts with acid or acidic oxide to give salts as follows:

$$NaOH + HNO_3 \longrightarrow NaNO_3 + H_2O$$

$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

$$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$$

It can also react with amphoteric oxides like Al_2O_3, ZnO, SnO, PbO etc., to form sodium metalates as follows:

$$Al_2O_3 + 2NaOH \longrightarrow \underset{\text{Sodium meta aluminate}}{2NaAlO_2} + H_2O$$

$$ZnO + 2NaOH \longrightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2} + H_2O$$

$$SnO + 2NaOH \longrightarrow \underset{\text{Sodium stannite}}{Na_2SnO_2} + H_2O$$

$$PbO + 2NaOH \longrightarrow \underset{\text{Sodium plumbite}}{Na_2PbO_2} + H_2O$$

3. With Metals

It can react with metals like Zn, Sn, Pb and Al to give sodium metalates but it cannot react with alkali metals.

$$\text{IA metal} + NaOH \longrightarrow \text{No reaction}$$

$$M(\text{Rest metals}) + 2NaOH \longrightarrow \underset{\text{Sodium metalate}}{Na_2MO_2} + H_2\uparrow$$

M may be Zn, Sn, Pb, Be.

For example,

$$2NaOH + Zn \longrightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2} + H_2$$

4. Reactions with Non-metals

It can react with nonmetals like B, Si, P, S and halogens as follows:

With Boron

$$2B + 6NaOH \longrightarrow \underset{\text{Sodium borate}}{2Na_3BO_3} + 3H_2$$

With Silicon

$$2NaOH + Si + H_2O \longrightarrow \underset{\text{Sodium silicate}}{Na_2SiO_3} + 2H_2$$

With White Phosphorous

$$P_4 + 3NaOH + 3H_2O \longrightarrow \underset{\text{Sodium hypophosphite}}{3NaH_2PO_2} + PH_3$$

With Sulphur

$$4S + 6NaOH \longrightarrow \underset{\text{Hypo}}{Na_2S_2O_3} + 2Na_2S + 3H_2O$$

With Halogens

$$\underset{\text{Dil and cold}}{2NaOH + X_2} \longrightarrow \underset{\text{Sodium hypo halite}}{NaXO} + NaX + H_2O$$

For example,

$$2NaOH + Cl_2 \longrightarrow NaOCl + NaCl + H_2O$$

$$\underset{\text{Hot and conc.}}{6NaOH + 3X_2} \longrightarrow 5NaX + \underset{\text{Sodium halate}}{NaXO_3} + 3H_2O$$

For example,

$$6NaOH + 3Br_2 \longrightarrow 5NaBr + NaBrO_3 + 3H_2O$$

5. With Salts

It can react with metallic salts to form metallic hydroxides which may be either insoluble or further soluble in NaOH to give oxyacid salts.

(a) Salts of Fe, Cr, Cu form insoluble hydroxides with NaOH as follows:

$$FeCl_3 + 3NaOH \longrightarrow \underset{\text{Red ppt.}}{Fe(OH)_3} + 3NaCl$$

$$CrCl_3 + 3NaOH \longrightarrow \underset{\text{Green ppt.}}{Cr(OH)_3} + 3NaCl$$

$$CuSO_4 + 2NaOH \longrightarrow \underset{\text{Blue colour ppt.}}{Cu(OH)_2} + Na_2SO_4$$

(b) With those salts whose insoluble hydroxides further dissolve in excess of NaOH to give sodium metalates:

For example, when NaOH is added in $AlCl_3$, sodium meta aluminate is formed as follows:

$$3NaOH + AlCl_3 \longrightarrow \underset{\text{White ppt.}}{Al(OH)_3} + 3NaCl$$

$$NaOH + Al(OH)_3 \rightarrow \underset{\text{Sodium metaaluminate}}{2NaAlO} + 2H_2O$$

When NaOH is added in the given compounds, their sodium maralates are formed.

$$SnCl_2 + 2NaOH \rightarrow Sn(OH)_2 + 2NaCl$$

$$Sn(OH)_2 + 2NaOH \longrightarrow Na_2SnO_2 + 2H_2O$$

$$ZnCl_2 + 2NaOH \rightarrow Zn(OH)_2 + 2NaCl$$

$$Zn(OH)_2 + 2NaOH \longrightarrow Na_2ZnO_2 + 2H_2O$$

(c) **Unstable Hydroxides:** Some salts give stable hydroxides with NaOH.

For example:

$$2AgNO_3 + 2NaOH \rightarrow 2AgOH + 2NaNO_3$$

$$2AgOH \longrightarrow \underset{\text{Brown}}{Ag_2O} + H_2O$$

$$HgCl_2 + 2NaOH \rightarrow Hg(OH)_2 + 2NaCl$$

$$Hg(OH)_2 \longrightarrow \underset{\text{Yellow}}{HgO} + H_2O$$

With Ammonium Salts: It liberates ammonia.

$$NH_4Cl + NaOH \xrightarrow{\Delta} NH_3\uparrow + HCl$$

Formation of HCOOH: It reacts with carbon monoxides at 150–200°C under pressure to form sodium formate which undergoes hydrolysis to give formic acid.

$$NaOH + CO \xrightarrow[5\text{–}10\text{ atm 'P'}]{150\text{–}2000} HCOONa \xrightarrow{H.OH} HCOOH$$

Uses of NaOH

1. It is used in the hydrolysis of ester and saponification.
2. It is used in the preparation of paper, soap, rayon, dyes, drugs and a number of chemicals.
3. It is used as a reagent in the laboratory.
4. It is used in the refining of Bauxite (by Bacyer's method), petroleum and vegetable oils etc.
5. It is used to prepare H_2 gas [Ujeno method]

$$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2\uparrow$$

Sodium Carbonate or Washing Soda ($Na_2CO_3.10\ H_2O$)

- Na_2CO_3: soda-ash
- $Na_2CO_3.H_2O$: crystal carbonate

Methods of Preparation

It is prepared by the following methods:

Le–Blanc Process

Here, the raw material used to prepare Na_2CO_3 are NaCl, $CaCO_3$, coke and H_2SO_4.

$$2NaCl + H_2SO_4 \rightarrow \underset{\text{Salt cake}}{Na_2SO_4} + 2HCl$$

$$Na_2SO_4 + 4C \rightarrow \underset{\text{Black salt}}{Na_2S} + 4CO\uparrow$$

$$Na_2S + CaCO_3 \rightarrow Na_2CO_3 + \underset{\text{Black ash}}{CaS}$$

- This method is used to prepare washing soda, salt cake, black salt, black ash.

Solvay Ammonia Soda Process

Na_2CO_3 is industrially prepared by Solvay method using NaCl, $CaCO_3$ and NH_3 as follows:

$$2NH_3 + CO_2 + H_2O \rightarrow (NH_4)_2CO_3$$

$$2NaCl + (NH_4)_2CO_3 \rightarrow Na_2CO_3 + 2NH_4Cl$$

$$Na_2CO_3 + CO_2 + H_2O \rightarrow 2NaHCO_3$$

$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$$

- Here, Na_2CO_3, $NaHCO_3$, $CaCl_2$ are obtained as byproducts. $NaHCO_3$ is precipitated due to its low solubility product. $KHCO_3$ cannot be precipitated due to its high solubility product hence it cannot be prepared by this method.

Details of the Process: This process is carried out in following steps, onebyone.

(i) ***Saturation of Brine Solution by Ammonia:*** It is done in a saturated tank (ammonia absorber) by passing ammonia gas through 30% brine solution. Here, the possible impurities of calcium, magnesium and iron salts present in brine can be removed as carbonates by carbon dioxide present in ammonia. These impurities can be easily removed by filtration in the form of precipitates and the clear liquid is allowed to enter the carbonation tower.

$$2NH_3 + CO_2 + H_2O \longrightarrow (NH_4)_2CO_3$$

$$CaCl_2 + 2(NH_4)_2CO_3 \rightarrow CaCO_3 \downarrow + 2NH_4Cl$$

$$MgCl_2 + 2(NH_4)_2CO_3 \rightarrow MgCO_3 \downarrow + 2NH_4Cl$$

(ii) ***Carbonation*:** The clear ammonical brine solution is cooled and then allowed to enter the carbonation tower. It flows down slowly in the tower and when it comes in contact with the carbon dioxide stream, the sparingly soluble sodium bicarbonate is formed along with ammonium chloride.

$$NH_3 + H_2O + CO_2 \rightarrow NH_4HCO_3$$

$$NaCl + NH_4HCO_3 \rightarrow NaHCO_3 + NH_4Cl$$

Here, CO_2 is prepared by heating lime stone in a lime kiln and the lime formed in the kiln is dissolved in water to get $Ca(OH)_2$ which is transferred to ammonia recovery tower.

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

$$CaO + H_2O \rightarrow Ca(OH)_2$$

(iii) ***Filtration*:** The milky solution having $NaHCO_3$ as a fine suspension obtained after carbonation is filtered with the help of a rotary vacuum filter. From here the filtrate of $NaHCO_3$ is separated and the remaining solution having NH_4Cl (with little NH_4HCO_3) is pumped into ammonia recovery tower.

(iv) ***Calcination*:** Sodium bicarbonate obtained after filtration is strongly heated in specially designed cylindrical vessels for ignition to obtain sodium carbonate.

$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$$

(v) ***Recovery of Ammonia*:** The filtrate from the vacuum filter having NH_4Cl is mixed with slaked lime and steam is passed. Here ammonia is formed again as follows:

$$2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + H_2O + CaCl_2$$

$$NH_4HCO_3 \xrightarrow{\Delta} NH_3 + H_2O + CO_2$$

From here the mixture of ammonia and carbon dioxide is again passed into the saturating tower so that the process goes on repeating.

- Sodium carbonate obtained by this process is quite pure and the byproduct obtained in this process is calcium chloride. It is a very cheap process as well.

Physical Properties

1. It is a white crystalline solid partially soluble in water and its aqueous solution is basic.

2. Efflorescence

It is an efflorescent substance and looses its water of crystallization to form its monohydrate in open air as follows:

$$Na_2CO_3.10H_2O \xrightarrow{Air} Na_2CO_3.H_2O + 9H_2O$$

'Powder form' (loss of weight)

3. Heating Effect

On heating it changes into anhydrous sodium carbonate as follows:

$$Na_2CO_3.10H_2O \xrightarrow{10\,H_2O} Na_2CO_3 \xrightarrow[\text{Strong}]{\Delta} X$$

4. With Acids

It is easily decomposed by acids as follows:

$$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + NaCl$$

$$NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2 \uparrow$$

5. With CO_2

When CO_2 is passed through the concentrated solution of sodium carbonate, sodium bicarbonate gets precipitated.

$$Na_2CO_3 + H_2O + CO_2 \longrightarrow 2NaHCO_3$$

6. With Silica

When it is treated with silica it gives sodium silicate as follows:

$$Na_2CO_3 + SiO_2 \longrightarrow Na_2SiO_3 + CO_2 \uparrow$$

Sodium silicate is known as water glass or soluble glass as it dissolves in water.

7. With Sulphur and Sulphur dioxide

When aqueous solution of sodium carbonate is treated with sulphur dioxide and sulphur, sodium thiosulphate is formed.

$$Na_2CO_3\,(aq) + SO_2 \longrightarrow Na_2SO_3 + CO_2$$

$$Na_2SO_3 + S \longrightarrow Na_2S_2O_3$$

8. With Salts of NonAlkali Metals

It reacts with salts of nonalkali metals to form insoluble normal or basic carbonates as follows:

(a) **With Lead Acetate:** When it is treated with lead acetate it gives basic lead carbonate as follows:

$$3(CH_3COO)_2Pb + 3Na_2CO_3 + H_2O \longrightarrow 2PbCO_3Pb(OH)_2 + CO_2 + 6CH_3COONa$$

Basic lead carbonate

(b) **Reaction with Copper Sulphate:** When it is treated with copper sulphate it gives basic copper carbonate as follows:

$$2CuSO_4 + 2Na_2CO_3 + H_2O \longrightarrow Cu(OH)_2CuCO_3 + CO_2 + Na_2SO_4$$

Basic copper carbonate

(c) **With Zinc Sulphate:** When it is treated with zinc sulphate it gives basic zinc carbonate as follows:

$$5ZnSO_4 + 5Na_2CO_3 + 4H_2O \longrightarrow [3Zn(OH)_2 2ZnCO_3].H_2O + 3CO_2 + 5Na_2SO_4$$

Basic zinc carbonate

(d) Carbonates of some metals like Fe, Al, Sn undergo hydrolysis at once into hydroxides as follows:

$$Fe_2(SO_4)_3 + 3Na_2CO_3 \longrightarrow Fe_2(CO_3)_3 + 3Na_2SO_4$$

$$Fe_2(CO_3)_3 + 3H_2O \longrightarrow 2Fe(OH)_3 + 3CO_2$$

Uses

1. It is used for softening hard water, refining of petrol and in the manufacture of glass, borax etc.
2. $Na_2CO_3 + K_2CO_3$ is a fusion mixture hence it is used in quantitative and qualitative analysis.
3. It is used for washing purpose in laundry.
4. It is used in paper, textile and paint industries.
5. Na_2CO_3 gives pink colour with HPH, yellow colour with MeOH or methyl red and blue colour with red litmus.

Sodium Bicarbonate or Baking Soda ($NaHCO_3$)

Preparation

It is prepared by passing carbon dioxide through a saturated solution of sodium carbonate as follows:

$$Na_2CO_3 + CO_2 + H_2O \longrightarrow 2NaHCO_3$$

Sodium bicarbonate can be easily precipitated from here as it is less soluble in water.

Physiochemical Properties

1. It is a white crystalline solid partially soluble in water and its aqueous solution is basic. $NaHCO_3$ gives yellow colour with methyl red or methyl orange but no colour with HPH i.e., phenolphthalein.

$$HCO_3^- + H_2O \rightarrow H_2O + CO_2 + OH^-$$

$$NaHCO_3 + H_2O \longrightarrow H_2CO_3 + NaOH$$

2. **Heating Effect**

On heating at 100 °C, it decomposes into Na_2CO_3 with the evolution of CO_2.

$$2NaHCO_3 \xrightarrow{100°C} Na_2CO_3 + CO_2 + H_2O$$

3. With zinc sulphate it forms normal zinc carbonate.

$$ZnSO_4 + 2NaHCO_3 \longrightarrow ZnCO_3 + Na_2SO_4 + H_2O + CO_2$$

4. **With Dilute Acids**

It reacts with dilute acids as follows:

$$NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$$

Uses

1. It is used to remove acidity in stomach and in the making of baking powder (Sodium bicarbonate + potassium hydrogen tartarate).
2. It is used in fire extinguishers.
3. It is used in medicines as a mild antiseptic for skin infections.
4. It is used in making effervescent drinks.

REMEMBER

- On dissolving equimolar amount of Na_2CO_3 and $NaHCO_3$ and cooling the solution, crystals of sodium sesquicarbonate ($Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$) are obtained which are used for wool washing.

SODIUM CHLORIDE (NaCl)

It is called common salt or rock salt or sea salt or table salt. Sea water has nearly 2.95% NaCl. 28% aqueous NaCl solution is called brine.

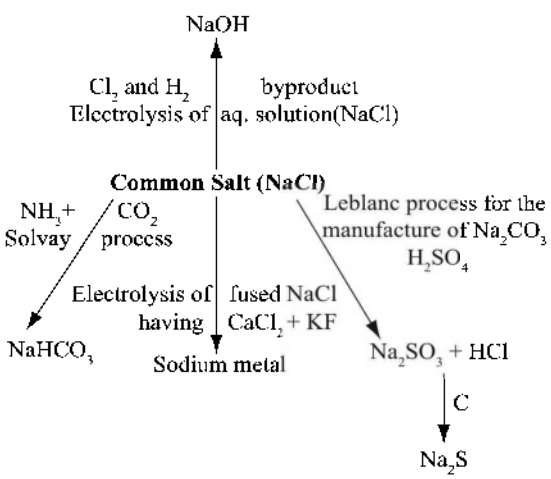
Preparation

It is mainly manufactured from sea water by evaporation in sun. As it contains impurities of $MgCl_2$, $CaCl_2$, $CaSO_4$ hence it is further purified by passing HCl gas where due to common ion effect pure NaCl gets precipitated.

Physical Properties

1. It is a white crystalline solid and hygroscopic in nature.
2. It melts at 1081K and boils at 1713K.
3. It dissolves in water and the process of dissolution is endothermic.
4. Its solubility is 36 g per 100 g of water at 273K. The solubility does not increase much with increase in temperature.

Summary of Some Important Reactions



Uses

1. It is an essential constituent of our food.

2. It is used in the manufacture of sodium, sodium hydroxide, washing soda, hydrogen chloride, chlorine etc.
3. It is used in freezing mixture.
4. It is used for the preparation of soap.
5. It is used for regenerating ion exchange resins.

Sodium Sulphate or Glauber's Salt ($Na_2SO_4 .10H_2O$)

Preparation

1. Anhydrous sodium sulphate or salt cake is obtained during the manufacture of sodium carbonate when NaCl and H_2SO_4 are heated together.

$$2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$$

2. **From Chile Salt Petre ($NaNO_3$)**

When chile salt petre is treated with H_2SO_4, nitre cake ($NaHSO_4$) is formed which on heating with NaCl gives salt cake.

$$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$$

$$NaCl + NaHSO_4 \longrightarrow Na_2SO_4 + HCl$$

- Salt cake on dissolving in water followed by crystallization below 305K gives the crystals of Glauber's salt.

3. **Hargreaves Process**

It is prepared by passing SO_2 mixed with air and water vapours over heated NaCl in iron retort.

$$4NaCl + 2SO_2 + 2H_2O + O_2 \rightarrow 2Na_2SO_4 + 4HCl$$

Physiochemical Properties

1. Anhydrous form is crystalline solid and soluble in water.
2. **Reduction**

On heating with carbon it reduced into Na_2S with concentrated H_2SO_4

$$Na_2SO_4 + 4C \rightarrow Na_2S + 4CO$$

3. **With Concentrated H_2SO_4**

When equimolar amount of anhydrous Na_2SO_4 and concentrated H_2SO_4 is cooled prismatic crystals of sodium bisulphite are formed.

$$Na_2SO_4 + H_2SO_4 \longrightarrow 2NaHSO_4$$

4. **Reaction with Lead and Barium Salts**

It forms their insoluble sulphates as follows:

$$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$$

$$PbCl_2 + Na_2SO_4 \longrightarrow PbSO_4 + 2NaCl$$

Uses

1. It is used in the manufacture of craft paper, paper board, window glass etc.
2. It is also used as a mild laxative in medicine.
3. It is also used in the preparation of hypo, Na_2S, $NaHSO_4$.

COMPOUNDS OF POTASSIUM

Potassium forms oxides, hydroxides, carbonate and halides as follows:

Oxides

Potassium forms following three oxides:

K_2O: (Potassium mono oxide)

KO_2: (Potassium super oxide)

K_2O_3: (Potassium sesquioxide)

Potassium superoxide (KO_2)

Preparation

1. KO_2 is prepared by burning potassium with oxygen (free from moisture) as follows:

$$K + O_2 \longrightarrow KO_2$$

2. KO_2 can also be prepared by treating KOH with ozone as follows:

$$2KOH + O_3 \longrightarrow 2KO_2 + H_2O$$

Physiochemical Properties

1. It is a chrome yellow powder which easily dissolves in water and gives H_2O_2 as follows:

$$2KO_2 + 2H_2O \longrightarrow 2KOH + H_2O_2 + O_2$$

2. It can react with CO and CO_2 as follows:

$$2KO_2 + CO \rightarrow K_2CO_3 + O_2$$

$$2KO_2 + CO_2 \longrightarrow K_2CO_3 + 3/2\ O_2$$

3. It reacts with sulphur on heating to form K_2SO_4.

$$2KO_2 + S \longrightarrow K_2SO_4$$

Uses

It is used as air purifier in space capsules and breathing mask as it not only produces oxygen but also removes CO_2. It is also used as an oxidizing agent.

- K_2O_3 is prepared by passing oxygen through potassium dissolved in liquid ammonia.

$$4K + 3O_2 \rightarrow 2K_2O_3$$

- K_2O is a white hygroscopic solid while KO_2 is chrome yellow powder.
- K_2O and KO_2 on hydrolysis forms KOH.

$$K_2O + H_2O \rightarrow 2KOH$$

Potassium Hydroxide (KOH)

Preparation

1. It is mainly obtained by the electrolysis of an aqueous solution of KCl just like NaOH.
2. It is also obtained by the action of soda lime on potassium carbonate.

Physiochemical Properties

1. It has similar properties to NaOH but it is more stronger and have more solubility in alcohols than NaOH.
2. Its aqueous solution is known as potash lye.
3. It is a better absorber of CO_2 than NaOH as K_2CO_3 being more soluble does not separate salt.

Uses

1. Alcoholic KOH is used as a dehydrating agent for the dehydration of alkyl halides in organic chemistry.

$$C_2H_5Br + \text{alc. KOH} \longrightarrow C_2H_4 + KBr + H_2O$$

2. It is used for the absorption of gases like CO_2, SO_2 etc.
3. It is also used in making soaps.

Potassium Carbonate (K_2CO_3)

It is also called pearl ash.

Preparation

It is prepared by Leblanc process.

Leblanc Process

Here, KCl is treated with H_2SO_4 to form K_2SO_4 which on further heating with $CaCO_3$ and carbon gives K_2CO_3 as follows:

$$KCl + H_2SO_4 \rightarrow KHSO_4 + HCl$$

$$KCl + KHSO_4 \rightarrow K_2SO_4 + HCl$$

$$K_2SO_4 + 2C + CaCO_3 \longrightarrow K_2CO_3 + CaS + 2CO_2$$

Black ash

Precht Process

$$2KCl + 3[MgCO_3.3H_2O] + CO_2 \xrightarrow{100°C} 2[KHCO_3.MgCO_3.4H_2O] + MgCl_2$$

Potassium magnesium hydrogen carbonate

$$2\ [KHCO_3.MgCO_3.4H_2O] \xrightarrow{140°C} K_2CO_3 + 2MgCO_3 + CO_2 + 9H_2O$$

Physiochemical Properties

1. It is a white deliquescent solid.
2. It is highly soluble in water and due to hydrolysis, it forms an alkaline solution.

$$K_2CO_3 + H_2O \rightleftharpoons KHCO_3 + K^+ + OH^-$$

- Due to its high solubility in water, it cannot be precipitated hence cannot be formed by Solvay method.

3. Melts at lower temperature (700°C) when mixed with Na_2CO_3, thus it forms fusion mixture ($Na_2CO_3 + K_2CO_3$).
4. Action of CO_2

$$K_2CO_3 + H_2O + CO_2 \rightarrow 2KHCO_3$$

Uses

1. It is used in the manufacturing of hard glass and soft soap.

2. Its mixture with Na_2CO_3 can be used as a fusion mixture in laboratory.

Potassium Bicarbonates ($KHCO_3$)

Preparation

It is prepared by passing CO_2 through cold saturated solution of K_2CO_3.

$$K_2CO_3 + H_2O + CO_2 \longrightarrow 2KHCO_3$$

Physiochemical Properties

1. It is in the form of white powder.
2. Chemically, it resembles $NaHCO_3$ except that it is more soluble in water.

Uses

It is used in making powder and medicines.

Potassium Chloride (KCl)

Occurrence: It occurs as sylvine (KCl) and as carnalite ($KCl.MgCl_2.6H_2O$).

Preparation

From Sylvine (KCl)

It is a mixture of KCl and NaCl. When the boiling hot saturated solution of the mixed salts in water is cooled, KCl separates out and NaCl is left behind in the solution.

From Carnalite ($KCl.MgCl_2.6H_2O$)

It is always found mixed with NaCl and $MgSO_4$. To remove NaCl and $MgSO_4$, the ore is grounded and extracted with a hot 20% solution of $MgCl_2$. Carnalite dissolves while NaCl and $MgSO_4$ remain undissolved. These are filtered off and the solution which contains only carnalite is crystallized so that cubic crystals of KCl separate out leaving behind $MgCl_2$ in the solution.

Physiochemical Properties

1. It is a colourless crystalline solid having cubic crystals.
2. Its melting point and boiling point are 768°C and 1411°C, respectively.
3. It is extremely soluble in water and closely resembles NaCl in most of its properties, except that its solubility increases rapidly with the temperature and is more readily fusible.

Uses

1. It is used in the manufacture of potassium and its compounds.
2. It is used as a potassium fertilizer (K-type fertiliser), since it supplies potassium (as K_2O) to the soil.

Potassium Iodide KI

Preparation

1. When ferroso enferric iodide and potassium carbonate are treated as follows, potassium iodide is formed.

$$4K_2CO_3 + Fe_3I_8 + 4H_2O \rightarrow 8KI + 4CO_2 + \underset{\text{ppt.}}{Fe(OH)_2.2Fe(OH)_3}$$

From here the precipitate can be easily filtered off and the solution on crystallization gives the crystals of KI.

2. When KOH or K_2CO_3 are treated with HI, KI is formed as follows:

$$KOH + HI \rightarrow KI + H_2O$$

$$K_2CO_3 + 2HI \longrightarrow 2KI + CO_2 + H_2O$$

3. When iodine is heated with hot and conc. solution of KOH, KI and KIO_3 solution is formed which is evaporated to dryness and the obtained solid residue is ignited with powdered charcoal to obtained KI as follows:

$$3I_2 + 6KOH \longrightarrow 5KI + KIO_3 + 3H_2O$$

$$KIO_3 + 3C \rightarrow KI + 3CO$$

Physiochemical Properties

1. It is a white crystalline solid which is highly soluble in water and alcohol.
2. It dissolves free iodine and forms KI_3.

$$KI + I_2 \rightarrow \underset{\text{Unstable}}{KI_3}$$

3. With H_2SO_4

Here, sulphuric acid decomposes KI as follows:

$$2KI + 2H_2SO_4 \rightarrow K_2SO_4 + I_2 + SO_2 + 2H_2O$$

4. As a Reducing Agent

Being a strong reducing agent it can reduce $KMnO_4$, $K_2Cr_2O_7$, $CuSO_4$, etc. as follows:

It reduces $KMnO_4$ into $MnSO_4$ as follows:

$$2KMnO_4 + 10KI + 8H_2SO_4 \rightarrow 6K_2SO_4 + 2Mn(SO_4) + 5I_2 + 8H_2O$$

It reduces $K_2Cr_2O_7$ into chromium sulphate as follows:

$$6KI + K_2Cr_2O_7 + 7H_2SO_4 \rightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 3I_2 + 7H_2O$$

It reduces HNO_3 into nitrogen dioxide as follows:

$$2KI + 4HNO_3 \rightarrow 2KNO_3 + 2NO_2 + I_2 + 2H_2O$$

It reduces copper sulphate into cuprous iodide as follows:

$$2CuSO_4 + 4KI \rightarrow \underset{\text{Cuprous iodide}}{Cu_2I_2} + 2K_2SO_4 + I_2$$

5. Formation of Insoluble Iodides

It forms insoluble iodides with lead acetate and silver nitrate etc. as follows:

$$Pb(CH_3COO)_2 + 2KI \longrightarrow \underset{\text{Yellow}}{PbI_2} + 2CH_3COOK$$

$$AgNO_3 + KI \rightarrow \underset{\text{Yellow}}{AgI} + KNO_3$$

6. With Mercurous Chloride

When it is treated with $HgCl_2$, a red precipitate of HgI_2 is formed which further dissolves in excess of KI giving potassium mercuric iodide. Its alkaline solution is called Nessler's reagent which is used for the detection and estimation of ammonium ions.

$$HgCl_2 + 2KI \rightarrow HgI_2 + 2KCl$$

$$HgI_2 + 2KI \longrightarrow K_2HgI_4$$

Uses

1. It is used for making Nessler's reagent.
2. It is used as a solvent of iodine as a reagent in laboratory.
3. It is used in medicine and photography.

Potassium Sulphate (K_2SO_4)

Preparation

1. By heating naturally occurring mineral, schonite:

$$K_2SO_4.MgSO_4.6H_2O + 2KCl \xrightarrow{\Delta} 2K_2SO_4 + MgCl_2 + 6H_2O$$

2. By treating KCl with H_2SO_4:

$$2KCl + H_2SO_4 \longrightarrow K_2SO_4 + 2HCl$$

3. By heating dry KO_2 with sulphur:

$$2KO_2 + S \xrightarrow{\Delta} K_2SO_4$$

Physiochemical Properties

1. It is a white crystalline solid.
2. It does not form hydrates.
3. When heated with carbon, it is reduced to potassium sulphide.
4. It resembles Na_2SO_4.

GENERAL REVIEW OF COMPOUNDS OF ALKALINE EARTH METALS

These metals form many compounds like oxides, hydroxides, halides, sulphates, carbonates etc., as follows:

1. Oxides of Alkaline Earth Metals (MO)

These metals react slowly with oxygen to give MO type oxides.

$$2M + O_2 \longrightarrow 2MO$$

For example

BeO	MgO	CaO	BaO
Covalent	magnesia	quicklime	baryta
Amphoteric		Basic	

- These oxides are white crystalline solids and quite stable, except BeO (covalent) rest are ionic oxides

having face-centered cubic structure. Except, BeO rest oxides react with water to form basic hydroxides as follows:

$$MO + H_2O \longrightarrow M(OH)_2 + \text{Heat}$$

From BeO to BaO: Ionic nature, solubility in water and basic nature increase while lattice energy and stability decrease.

2. Peroxides of Alkaline Earth Metals (MO_2)

Ba, Sr also form peroxides due to their high electropositivity as follows:

$$2BaO + O_2 \xrightarrow{\Delta} 2BaO_2$$

$$2SrO + O_2 \xrightarrow{\Delta} 2SrO_2$$

All MO_2 are white solid ionic compounds.

- Be metal is relatively less reactive and does not react below 880K while magnesium burns with white dazzling light.

3. Hydroxides of Alkaline Earth Metals $M(OH)_2$

Except Be, all metals react with water to give hydroxides and the decreasing order of reactivity towards water is as follows:

Ba > Sr > Ca > Mg

$$M + 2H_2O \longrightarrow M(OH)_2 + H_2\uparrow$$

Beryllium does not react with water even at higher temperature due to its lower oxidation potential while magnesium reacts with hot water.

$Be(OH)_2$, $Mg(OH)_2$, $Ca(OH)_2$ $Sr(OH)_2$, $Ba(OH)_2$

→ Basic nature and solubility in water increase

$Be(OH)_2$ is amphoteric in nature while rest are basic and their basicity is less than that of alkali metal hydroxides as in these metal hydroxides the M–O bonds are stronger than in alkali metal hydroxides.

Solubility increases from $Be(OH)_2$ to $Ba(OH)_2$ or down the group as hydration energy becomes more than the lattice energy as lattice energy decreases much more than hydration energy. It is confirmed by the increase in solubility product from $Be(OH)_2$ to $Ba(OH)_2$.

- Aq. solution of $Ca(OH)_2$ is lime water.
- Aq. solution of $Ba(OH)_2$ is Baryta water.

4. Halides of Alkaline Earth Metals (MX_2)

All these metals combine directly with halogens at higher temperature to form MX_2 type of halides.

$$M + X_2 \xrightarrow{\Delta} MX_2$$

These halides can also be obtained by the action of HX on MO or $M(OH)_2$ or MCO_3.

$$MO + 2HX \longrightarrow MX_2 + H_2O$$

$$M(OH)_2 + 2HX \longrightarrow MX_2 + 2H_2O$$

$$MCO_3 + 2HX \longrightarrow MX_2 + H_2O + CO_2$$

These halides are ionic, hygroscopic in nature, high melting, nonvolatile solids and form hydrates. For example, $CaCl_2.6H_2O$.

BeX_2 MgX_2 CaX_2 SrX_2 BaX_2

→ Ionic nature increases

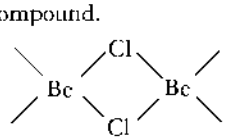
→ Solubility decreases
(Except in fluorides which are insoluble)

$BeCl_2$ is covalent polymeric halide due to small size of Be^{2+} and more polarizing power. It is a low melting volatile solid.

$BeCl_2$ fumes in moist air as it is very easily hydrolyzed by water as follows:

$$BeCl_2 + 2H_2O \longrightarrow Be(OH)_2 + 2HCl$$

Structure of $BeCl_2$: The structure is just like BeH_2 which is polymeric as $BeCl_2$ is electron deficient compound.



5. Carbonates of Alkaline Earth Metals (MCO_3)

These metals form MCO_3 type of carbonates which are insoluble in water but dissolve in $CO_2 + H_2O$ and their solubility decreases down the group.

$$MCO_3 + H_2O + CO_2 \longrightarrow M(HCO_3)_2$$

Thermal srability and ionic narure increases from $BeCO_3$ ro $BaCO_3$ as the polarizing power of M^{2} ion decreases with rhe increase in size from Be^{21} to Ba^{21}.

The increase of thermal stability can also be explained by the decomposition temperatures of these carbonates.

Decomposition Temp (K)	$BeCO_3$ (< 300); $MgCO_3$ (813) $CaCO_3$ (1173); $SrCO_3$ (1563); $BaCO_3$ (1633)

- Group IIA metals do nor form solid bicarbonates.
- $SrCO_3$ is used in the manufacture of glass of the picrure tube for colour relevision.

6. Sulphates of Alkaline Earth Metals (MSO_4)

These merals form MSO_4 rype of sulphares by rhe action of dilure H_2SO_4 as follows:

$$M + (dil.)H_2SO_4 \longrightarrow MSO_4 + H_2 \uparrow$$

$$MO + H_2SO_4 \rightarrow MSO_4 + H_2O$$

$$M(OH)_2 + H_2SO_4 \longrightarrow MSO_4 + 2H_2O$$

$$MCO_3 + H_2SO_4 \rightarrow MSO_4 + H_2O + CO_2$$

The solubiliry of rhese sulphares in water decreases from $BeSO_4$ ro $BaSO_4$ as rheir solubiliry product and heat of hydrarion of solvarion decreases.

	$BeSO_4$	$MgSO_4$	$CaSO_4$
Ksp	very high	10	2.4×10^{5}

$SrSO_4$	$BaSO_4$
7.6×10^{7}	1.5×10^{9}

These sulphares are rhermally very srable due ro rheir high larrice energies bur on srrong heating give meral oxides as follows:

$$2MSO_4 \xrightarrow{Heat} 2MO + 2SO_2 + O_2$$

Their srability of rhe sulphares increases down rhe group or from $BeSO_4$ ro $MgSO_4$ as rhe rhermal decomposition and basic narure of rhe meral increases.

Decomposition Temp (K)	$BeCO_3$ (< 300); $MgCO_3$ (813) $CaCO_3$ (1173); $SrCO_3$ (1563); $BaCO_3$ (1633)

- $BaSO_4$ is insoluble in H_2O hence used to detect stomach problem (barium meal).
- $MgSO_4 .7H_2O$ or $[Mg(H_2O)_6 SO_4 .H_2O]$ is Epsom salt (mild laxarive).
- $K_2SO_4 . MgSO_4 . 6H_2O$ is potash magnesis (a fertilizer).

Compounds of Magnesia (MgO)

Preparation

It is prepared by heating following compounds as follows:

$$Mg(OH)_2 \xrightarrow{\Delta} MgO + H_2O$$

$$MgCO_3 \xrightarrow{\Delta} MgO + CO_2$$

$$2Mg(NO_3)_2 \xrightarrow{\Delta} 2MgO + 4NO_2 + O_2$$

$$2Mg\,(s) + O_2 \xrightarrow{\Delta} MgO\,(s)$$

$$MgCl_2 .6H_2O \xrightarrow[Heating]{Strong} MgO\,(s)$$

Physiochemical Properties

1. It is an amorphous white powder and slighrly soluble in warer as follows:

$$MgO + H_2O \longrightarrow Mg(OH)_2$$

2. It is quite stable and fuses at 2800 K hence it is used as a refractory material for lining electric furnaces.
3. It is a basic oxide and forms salt with acids.

$$MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$$

$$MgO + 2HCl \rightarrow MgCl_2 + H_2O$$

Uses

1. It is used as a refractory material in furnace due ro irs high melring point and basic flux narure.
2. It is used in medicine as an antacid.
3. It is also used in making soral cement.

4. It is used for filing rubber.
5. When it is mixed with asbestos it can be used as an insulator for steam pipes and boilers.

Magnesium Hydroxide $Mg(OH)_2$

$Mg(OH)_2$ occurs in nature as brucite.

Preparation

1. **From MgO**

   It is prepared by dissolving MgO into water as follows:

   $$MgO + H_2O \longrightarrow Mg(OH)_2$$

2. **By Treating Magnesium Salt with Alkali**

   $$MgCl_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2 + CaCl_2$$

   $$MgCl_2 + 2NaOH \rightarrow Mg(OH)_2 + 2NaCl$$

Physiochemical Properties

1. It is a white powder and partially soluble in water.
2. It is a basic oxide and forms salts with acids.
3. On heating, $Mg(OH)_2$ decomposes to form MgO.
4. It dissolves in ammonium chloride solution forming a complex ion. It partly explains why magnesium does not precipitate with the hydroxides of Al, Cr and Fe in IIIrd group of qualitative analysis.

   $$Mg(OH)_2 + 2NH_4Cl \rightarrow MgCl_2 + 2NH_4OH$$

Uses

1. $Mg(OH)_2$ is an aqueous suspension used in medicines as an antacid, called 'Milk of Magnesia'.

Magnesium Carbonate ($MgCO_3$)

$MgCO_3$ occurs in nature as dolomite, $MgCO_3.CaCO_3$ and as magnesite $MgCO_3$.

Preparation

1. **By Treating an aqueous solution of magnesium salt with $NaHCO_3$**

   $$MgSO_4 + 2NaHCO_3 \rightarrow MgCO_3 + Na_2SO_4 + H_2O + CO_2$$

2. **By passing CO_2 through suspension of MgO in water**

   $$MgO + H_2O + 2CO_2 \longrightarrow Mg(HCO_3)_2$$

   $$Mg(HCO_3)_2 + MgO \rightarrow 2MgCO_3 + H_2O$$

3. **From Magnesium Sulphate and Sodium Carbonate**

   It cannot be directly obtained by the reaction of these two as a white precipitate of basic magnesium carbonate or magnesium alba is obtained first which is suspended in water and CO_2 is passed into it to obtain magnesium bicarbonate (fluid magnesium) whose solution on boiling gives magnesium carbonate.

   $$2MgSO_4 + 2Na_2CO_3 + H_2O \rightarrow MgCO_3 \cdot Mg(OH)_2 + 2Na_2SO_4 + CO_2$$

   $$MgCO_3.Mg(OH)_2 + 3CO_2 + H_2O \longrightarrow 2Mg(HCO_3)_2$$

   $$Mg(HCO_3)_2 \rightarrow MgCO_3 + CO_2 + H_2O$$

Physiochemical Properties

1. It is a white solid powder which is insoluble in water.
2. On suspension in water and by passing CO_2 it forms $Mg(HCO_3)_2$.

   $$MgCO_3 + H_2O + CO_2 \longrightarrow \underset{\text{Magnesium bicarbonate}}{Mg(HCO_3)_2}$$

3. **With Acids**

   It dissolves in acids giving salts as follows:

   $$MgCO_3 + H_2SO_4 \longrightarrow MgSO_4 + H_2O + CO_2$$

   $$MgCO_3 + 2HCl \rightarrow MgCl_2 + H_2O + CO_2$$

4. **Heating Effect**

   $$MgCO_3 \xrightarrow{\Delta} MgO + CO_2$$

Uses

1. $(MgCO_3)_x.[Mg(OH)_2].3H_2O$ (magnesium alba) is used in tooth powder as an antacid and laxative.
2. It is also used in the manufacturing of glass, ceramics etc.

3. It is also used as a filler for paper rubber and pigments.

Magnesium Bicarbonate $Mg(HCO_3)_2$

Preparation

By passing CO_2 when $MgCO_3$ is suspended in water.

$$MgCO_3 + H_2O + CO_2 \longrightarrow Mg(HCO_3)_2$$

Physiochemical Properties

1. It is found only in liquid form.
2. On Boiling:

$$Mg(HCO_3)_2 \longrightarrow MgCO_3 + H_2O + CO_2$$

3. A solution having 12 grams of $MgCO_3$ per 100 cc of water conraining dissolved CO_2 is known as 'fluid magnesia'.
4. The precipitate obtained by reacting epsom salt ($MgSO_4.7H_2O$) with sodium carbonate is known as magnesia alba. It has got a variable composition $Mg(CO_3)_X\,[Mg(OH)_2]_Y.2H_2O$.

Magnesium Chloride ($MgCl_2$)

Occurrence: It occurs as carnalite [$KCl.MgCl_2.6H_2O$] and bischofite ($MgCl_2.H_2O$).

It is found in sea water and mineral springs etc.

Preparation

1. From Carnalite

It is extracted from carnalite by fractional crystallization. Here the mineral is fused and cooled to 175°C, when practically all KCl is separated out and only $MgCl_2$ is remains in the fused state. This fused mass on cooling gives the crystals of $MgCl_2.6H_2O$ as follows:

$$MgCl_2 + 6H_2O \xrightarrow[\text{in vaccum}]{} MgCl_2.6H_2O$$

2. From Sea Water

When sea water is treated with lime, the magnesium ions present in it can be precipitated in the form of magnesium hydroxide. When the precipitate of $Mg(OH)_2$ is dissolved in HCl and the solution formed is crystallized the crystals of $MgCl_2.6H_2O$ is obrained.

$$Mg^{2-} + Ca(OH)_2 \longrightarrow \underset{\text{(from sea water)}}{Mg(OH)_2\downarrow} + Ca^{2-}$$

$$Mg(OH)_2 + 2HCl \longrightarrow MgCl_2 + H_2O$$

3. Laboratory Method

It can be prepared by dissolving magnesium oxide or magnesium carbonate in dilute hydrochloric acid.

$$MgO + 2HCl \longrightarrow MgCl_2 + H_2O$$

$$MgCO_3 + 2HCl \longrightarrow MgCl_2 + H_2O + CO_2$$

This solution on concentration and cooling gives crystals of $MgCl_2.6H_2O$.

4. Anhydrous $MgCl_2$ cannot be obtained by heating hydrated form of $MgCl_2$ hence it is prepared by heating magnesium metal or hydrated magnesium chloride in a current of dry HCl or chlorine gas.

$$Mg + Cl_2 \longrightarrow \underset{\text{(anhydrous)}}{MgCl_2}$$

Physiochemical Properties

1. It is a colourless, crystalline, deliquescent solid which is highly soluble in water.
2. Heating Effect: On heating it decomposes into magnesium oxide as follows:

$$MgCl_2.6H_2O \xrightarrow{\Delta} Mg(OH)Cl + HCl + 5H_2O$$

$$Mg(OH)Cl \xrightarrow{\Delta} MgO + HCl$$

Uses

1. It is used for the preparation of many magnesium compounds like $Mg(OH)_2$, MgO, $MgCO_3$ etc.
2. When hydrated magnesium chloride is mixed with magnesium oxide it becomes quite hard mass paste known as soral magnesia i.e., $MgCl_2.5MgO \times H_2O$ which is used as a cement in dental filling.
3. It is used in lubricating cotton thread in spinning.

Magnesium Sulphate or Epsom Salt or Epsomite [$MgSO_4.7H_2O$]

It occurs in the form of minerals like epsom salt ($MgSO_4.7H_2O$), kieserite ($MgSO_4.H_2O$), kainite ($KCl.MgSO_4.3H_2O$)

Preparation

1. From Magnesite

It is prepared from magnesite ore by reacting them with dil. H_2SO_4 as follows:

$$MgCO_3 + H_2SO_4 \rightarrow MgSO_4 + H_2O + CO_2$$

2. From Dolomite

When powdered dolomite is boiled with sulphuric acid, it is obtained as calcium sulphate which being less soluble can be easily precipitated while $MgSO_4$ remains in the solution.

$$MgCO_3.CaCO_3 + 2H_2SO_4 \rightarrow MgSO_4 + CaSO_4\downarrow + 2CO_2 + 2H_2O$$

3. From Keiserite ($MgSO_4.H_2O$)

When the powdered ore is dissolved in water, a solution is obtained. When the solution is concentrated and cooled the crystals of magnesium sulphate are obtained.

$$MgSO_4.H_2O + 6H_2O \rightarrow MgSO_4.7H_2O$$

4. Laboratory method

It can be prepared by dissolving MgO or $MgCO_3$ in dilute H_2SO_4 and evaporating the solution to get the crystals of magnesium sulphate.

$$MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$$

$$MgCO_3 + H_2SO_4 \longrightarrow MgSO_4 + H_2O + CO_2$$

Physiochemical Properties

1. It is a colourless crystalline solid which is soluble in water. It is isomorphous with $ZnSO_4.7H_2O$
2. It is an efflorescent compound and loses water of crystallization on exposure to air.
3. Heating effect: It undergo decomposition on heating.

$$MgSO_4.7H_2O \xrightarrow[-6H_2O]{150^\circ C/\Delta} MgSO_4.H_2O$$

$$\underset{\text{Kieserite}}{MgSO_4.H_2O} \xrightarrow[-H_2O]{200^\circ C/\Delta} MgSO_4$$

$$MgSO_4 \xrightarrow{\Delta} MgO + SO_3$$

4. Reduction by Lamp Black:

$$2MgSO_4 + C \longrightarrow 2MgO + 2SO_2 + CO_2$$

Uses

1. It is used as a purgative (medicine) and in dying tanning and ceremic cement.
2. $Mg(ClO_4)_2$ is anhydrous magnesium per chlorate, a famous dehydrating agent.
3. It is used in manufacturing of paints, fire proofing fabrics etc.

REMEMBER

- Magnesia mixture is a solution of $MgCl_2$ and NH_4Cl in ammonia which is used to detect phosphate or arsenate ions.

COMPOUNDS OF CALCIUM

Calcium Oxide or Quick-Lime (CaO)

It is called burnt lime or lime also.

Preparation

1. Industrial Preparation

It is prepared by heating lime stone in a rotatory kiln at 800–1000°C.

Since it is a reversible reaction the evolved CO_2 must be removed so that the equilibrium may shift in the forward direction.

$$CaCO_3 \xrightarrow[1000^\circ C]{\Delta} \underset{\uparrow}{CO_2} + CaO \xrightarrow{H_2O} \underset{\text{Slaked lime}}{Ca(OH)_2}$$

[Slaking of lime]

Physical Properties

1. It is a white amorphous solid with a melting point of 287 K.
2. On exposure to atmosphere, it absorbs moisture and CO_2.
3. On heating in an oxyhydrogen flame, it becomes incandescent and releases a bright white light called lime light.

Chemical Properties

1. Basic Nature

Being a basic oxide it react with acidic oxides to form salts as follows:

$$CaO + CO_2 \longrightarrow CaCO_3$$

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

2. Reaction with Water

On reaction with water it forms calcium hydroxide with a hissing sound. It is an exothermic reaction and known as slaking of lime.

$$CaO \xrightarrow{H_2O} Ca(OH)_2 + 15 \text{ kcal}$$

$[Ca(OH)_2 + H_2O]$ is milk of lime

3. With Carbon

When it is heated with carbon at 3000°C, calcium carbide is formed.

Calcium carbide on reaction with water gives acetylene and with nitrogen gives $CaCN_2$.

$$CaO \xrightarrow[3000°C]{3C} CaC_2 + CO\uparrow$$

Calcium carbide

H_2O → $C_2H_2 + Ca(OH)_2$

$N_2(\Delta)$ → $CaCN_2 + C$

Nitrolim (fertilizer)

Uses

1. It is used in the manufacture of glass, cement, bleaching powder, soda-lime etc.
2. It is used as a basic flux.
3. It is used as a drying agent for gases and alcohols (NH_3, C_2H_5OH).
4. It is used in the refining of sugar.
5. It is also used as a disinfectant and germicide.
6. It is used for white washing.

Calcium Hydroxide or Slaked lime $Ca(OH)_2$

Preparation

It is prepared by the reaction of lime with water as follows:

$$CaO + H_2O \longrightarrow Ca(OH)_2 + 15 \text{ kcal}$$

Physical Properties

- It is a white amorphous powder and sparingly soluble in water. Its suspension in water is called milk of lime while the clear solution is called lime water but chemically both are same.

Chemical Properties

1. Reaction with CO_2

Its slowly absorbs CO_2 from air forming insoluble milky solution of calcium carbonate. If more CO_2 is passed in this milky solution soluble calcium bicarbonate is formed and milkyness gets disappeared.

$$Ca(OH)_2(aq) + CO_2 \longrightarrow CaCO_3 + H_2O$$

$$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$$

Calcium bicarbonate (soluble)

2. Reaction with Chlorine

When it is treated with chlorine in cold milk of lime form, calcium hypochlorite is formed but when it is hot, calcium chlorate is formed.

$$2Ca(OH)_2 + 2Cl_2 \longrightarrow CaCl_2 + 2H_2O + Ca(OCl)_2$$

Calcium hypochlorite

$$6Ca(OH)_2 + 6Cl_2 \longrightarrow 5CaCl_2 + 6H_2O + Ca(ClO_3)_2$$

Calcium chlorate

On heating with chlorine, slaked lime gives bleaching powder.

$$Ca(OH)_2 \xrightarrow[30^\circ C]{Cl_2} \underset{\text{Bleaching powder}}{CaOCl_2}$$

When it is heated with chlorine upto redness or with NH_4Cl, calcium chloride is formed.

$$Ca(OH)_2 \xrightarrow{Cl_2,\ \text{red hot}} CaCl_2$$

$$Ca(OH)_2 \xrightarrow{NH_4Cl} CaCl_2 + NH_3$$

Uses

1. It is used in softening of water, purification of coal gas, sugar etc.
2. It is used in the manufacturing of bleaching powder, sodium carbonate etc.
3. It is also used in making of mortar and plaster used as building materials.

Calcium Oxide or Marble or Lime Stone ($CaCO_3$)

In nature it occurs as lime stone, ice land spar, marble and shells of sea animals.

Preparation

Laboratory Method

In laboratory it is prepared by passing CO_2 through lime water or by adding sodium carbonate solution into calcium chloride as follows:

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$$

Physiochemical Properties

1. It is a white solid and which is almost insoluble in water.
2. Heating effect:

$$CaCO_3 \xrightarrow{1000^\circ C} CaO + CO_2$$

3. With dilute acids:

$$CaCO_3 + H_2SO_4 \longrightarrow CaSO_4 + H_2O + CO_2 \uparrow$$

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2 \uparrow$$

Uses

1. It is used in the preparation of cement, washing soda ($NaHCO_3$ by Solvay method).
2. In the extraction of many metals like iron.
3. Marble in used as a building material.
4. Precipitated chalk is used in the manufacture of paints, medicines and tooth paste etc.

Calcium Sulphate Dihydrate or Gypsum ($CaSO_4.2H_2O$)

Calcium sulphate occurs as anhydride ($CaSO_4$) and gypsum ($CaSO_4.2H_2O$). Naturally occurring calcium sulphate is called alabaster.

Preparation

In laboratory it is prepared by the reaction of calcium carbonate and calcium chloride with dilute acids as follows:

$$CaCO_3 + H_2SO_4 \longrightarrow CaSO_4 + H_2O + CO_2 \uparrow$$

$$CaCl_2 + H_2SO_4 \longrightarrow CaSO_4 + 2HCl$$

$$CaCl_2 + Na_2SO_4 \longrightarrow CaSO_4 + 2NaCl$$

Physiochemical Properties

1. It is a white crystalline solid which is partially soluble in water and its dissolution in water is exothermic.
2. It also dissolves in dilute acids, ammonium sulphate etc.
3. Heating Effect: On heating, gypsum gives plaster of paris (Calcium sulphate hemihydrate). Plaster of Paris when mixed with water, gives a hard mass with light expansion. Gypsum on heating at 200°C gives anhydrous calcium sulphate known as dead burnt plaster.

$$2CaSO_4.2H_2O \underset{\text{Setting } (\Delta H = -Ve)}{\overset{120^\circ C}{\rightleftharpoons}} \underset{\text{Plaster of Paris}}{(CaSO_4)_2.H_2O} + 3H_2O$$

$$\text{Plaster of Paris} \xrightarrow[200^\circ C]{\Delta} \underset{\text{Dead burnt plast}}{2CaSO_4 + H_2O}$$

On very strong hearing, gypsum decomposes into CaO.

$$2CaSO_4 \xrightarrow{\text{Strong heating}} 2CaO + 2SO_2 + O_2$$

4. With Carbon: On strong heating with carbon it forms calcium sulphide.

$$CaSO_4 + 4C \rightarrow CaS + 4CO$$

Uses

1. Gypsum is used in the manufacture of plaster of paris, cement, black board chalk etc.
2. It is also used in manufacturing of mortar and cement.

Plaster of Paris ($CaSO_4 . ½ H_2O$) or [$(CaSO_4)_2 H_2O$]

It is known as calcium sulphate hemihydrate.

Preparation: On heating gypsum at 120°C Plaster of Paris is formed as follows:

$$2CaSO_4.2H_2O \underset{\text{Setting } (\Delta H = -Ve)}{\overset{120^\circ C}{\rightleftharpoons}} \underset{\text{Plaster of Paris}}{(CaSO_4)_2 H_2O} + 3H_2O$$

Physiochemical Properties

1. It is a white powder.
2. Setting of Plaster of Paris: When plaster of paris is mixed with a sufficient amount of water a paste is formed which on standing for some time changes into a hard mass with more volume. Here, gypsum is formed with the evolution of heat.

$$(CaSO_4)_2.H_2O \xrightarrow[H_2O]{\text{Setting}} \underset{\text{Gypsum (orthorhombic)}}{CaSO_4.2H_2O}$$

$$\xrightarrow{\text{Hardening}} \underset{\text{Gypsum (monoclinic)}}{CaSO_4.2H_2O}$$

Setting of plaster of paris may be catalyzed with the help of NaCl while retarded by adding borax or alum.

Plaster of paris with alum is called Keene cement.

3. Heating effect: When plaster of paris is heated at 200°C it gives anhydrous calcium sulphate (dead plaster or dead burnt) which does not have any setting property.

$$(CaSO_4)_2 H_2O + 3H_2O \xrightarrow[200^\circ C]{\Delta} 2CaSO_4 + H_2O$$

Dead burnt plaster

Uses

1. Plaster of paris is used for plaster of broken bones, making statues, toys, chalks etc.
2. It is also used as building material.
3. It is used for making moulds for casting.

COMPOUNDS OF ALUMINIUM

Aluminium oxide or Alumina Al_2O_3

Alumina (aluminium oxide, Al_2O_3) is most stable compound and exists both in the anhydrous and hydrated forms:

(i) Anhydrous Al_2O_3 occurs in nature as colourless crystals in corundum and in combination with different coloured oxides such as ruby (red), sapphire (blue), emerald (green), amethyst (violet) and topaz (yellow).

(ii) Hydrate alumina occurs as bauxite.

Preparation

Al_2O_3 can be prepared by igniting aluminium hydroxide, aluminium nitrate, aluminium sulphate or ammonium alum as follows:

1. From Aluminium Chloride

$$AlCl_3 + 3NH_4OH \rightarrow Al_2O_3 + 3NH_4Cl$$

$$2Al(OH)_3 \longrightarrow Al_2O_3 + 3H_2O$$

2. By Heating Aluminium Sulphate or Ammonium Salt

$$Al_2(SO_4)_3 \xrightarrow{\Delta} Al_2O_3 + 3SO_3$$

$$\underset{\text{Ammonium salt}}{(NH_4)_2SO_4.Al_2(SO_4)_3.24H_2O} \xrightarrow{\Delta}$$

$$Al_2O_3 + 2NH_3 + 4SO_3 + 25H_2O$$

3. By Heating mixture of AlF_3 and B_2O_3

$$2AlF_3 + B_2O_3 \longrightarrow Al_2O_3 + 2BF_3$$

4. Al_2O_3 can also be prepared by **H**. Gold Schmidt's alumino-thermic process. In this process, Al powder is mixed with ferric oxide (thermite) in the ratio of 1:3, respectively and ignited oxides are reduced to the metal and heat is liberated.

$$Fe_2O_3 + 2Al \longrightarrow 2Fe + Al_2O_3$$

5. $4Al + 3O_2 \longrightarrow 2Al_2O_3$

This reaction involves oxidation and the process of anodizing will favour formation of Al_2O_3.

Physiochemical Properties

1. It is a white crystalline, very stable and infusible compound.
2. It is insoluble in water and almost unreactive.
3. Al_2O_3 is not a good conductor of electricity. It is dissolved in cryolite to make it a good conductor.
4. It has a melting point of 2323K and a boiling point of 2523K (due to high affinity of Al for oxygen, the meeting point of aluminium oxide is very high).
5. Electrolytic reduction of pure alumina is not possible because it melts at very high temperature.
6. Amphoteric Nature: Al_2O_3 is an amphoteric oxide because it reacts with acids as well as bases to form salts.

$$Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$$

$$Al_2O_3 + 2NaOH \xrightarrow{\Delta} 2NaAlO_2 + H_2O$$

7. Heating Effect: When alumina is heated above 800°C, an exothermal change takes place, after which it becomes almost insoluble in acids.
8. On hearing with carbon at high temperature:

$$2Al_2O_3 + 9C \xrightarrow{2000°C} Al_4C_3 + 6CO$$

$$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$$

Uses

1. Alumina is used as refractory material, medium in chromatographic separation, and bauxite cement (bauxite + lime).
2. Fused bauxite called aluminium, is used as an abrasive, in the manufacture of Al and its salts and in the preparation of artificial precious gems.
3. Alumina has different capacities for adsorbing other substances on the surface from solution. This property is used in separating mixtures into their components by chromatography.
4. Artificial gems are prepared by directing oxyhydrogen flame on a rod of Al and finely powdered mixture of alumina, fluorspar and little colouring matter. For example, Cr_2O_3 for rubies, 5% TiO_2 and 1.5% Fe_2O_3 for sapphires.

Aluminium Chloride $AlCl_3$ or Al_2Cl_6

Preparation

It is prepared as follows:

1. Anhydrous $AlCl_3$ can be obtained by passing dry HCl or Cl_2 over heated aluminium in absence of air.

$$2Al + 3Cl_2 \longrightarrow 2AlCl_3$$

$$2Al + \underset{(dry)}{6HCl} \longrightarrow 2AlCl_3 + 3H_2$$

2. Macah's Method

Anhydrous $AlCl_3$ can also be obtained when a mixture of alumina and carbon is heated at 1273K in an atmosphere of chlorine. The vapours of $AlCl_3$ on cooling gives solid anhydrous $AlCl_3$.

$$Al_2O_3 + 3C + \underset{dry}{3Cl_2} \xrightarrow{\Delta} 2AlCl_3 + 3CO\uparrow$$

3. Hydrated $AlCl_3$

When aluminium or aluminium hydroxide is dissolved in dilute HCl followed by crystallization hydrated aluminium chloride is obtained.

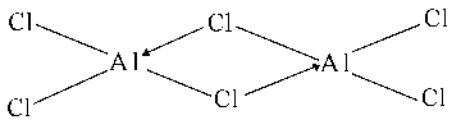
$$2Al + \underset{dry}{6HCl} \xrightarrow{\Delta} 2AlCl_3 + 3H_2$$

$$Al(OH)_3 + 3HCl \longrightarrow AlCl_3 + 3H_2O$$

Physical Properties

1. Anhydrous $AlCl_3$ is covalent and lewis acid in nature.
2. It is a white deliquescent solid and it is soluble in organic solvents like alcohol, ether etc.

3. Dimer of $AlCl_3$ i.e., Al_2Cl_6 has both covalent and coordinate bonds.



4. In $AlCl_3$, Al is sp^2 while in Al_2Cl_6 Al is sp^3 hybridized.

Chemical Properties

1. Hydrolysis

During hydrolysis in moist air, $AlCl_3$ gives fumes of HCl and on dissolving in water it changes into hydrated form.

$$2AlCl_3 + 6H_2O \xrightarrow[\text{Excess}]{} 2Al(OH)_3 + 6HCl$$

$$2AlCl_3 + 12H_2O \rightarrow 2[Al(H_2O)_6]^{3+} \cdot 3Cl^-$$

2. With NaOH

When sodium hydroxide is added slowly and gently into $AlCl_3$ solution a white gelatinous precipitate is formed which dissolves in excess of sodium hydroxide to give sodium meta aluminate.

$$AlCl_3 + 3NaOH \longrightarrow Al(OH)_3 + 3NaCl$$

$$Al(OH)_3 + NaOH \rightarrow NaAlO_2 + 2H_2O$$

3. With NH_4OH

When ammonium hydroxide is added in a solution of $AlCl_3$, a white gelatinous precipitate of $Al(OH)_3$ is formed which does not dissolve in NH_4OH.

$$AlCl_3 + 3NH_4OH \longrightarrow Al(OH)_3 + 3NH_4Cl$$

4. With Ammonia: Anhydrous $AlCl_3$ forms an adduct with gaseous ammonia.

$$Al_2Cl_6 + 12NH_3 \rightarrow 2[AlCl_3.6NH_3]$$

Uses

1. $AlCl_3$ is used as a catalyst to generate electrophile in Friedal Craft's reaction (also in gasoline).
2. It is used as a mordant in dyeing.
3. It can also be used in the manufacture of gasoline (by cracking of high boiling fractions of petroleum).

Alums

These are double salts of M^- and M^{+3} having a general formula:

$$M_2^I SO_4, M_2^{III}(SO_4)_3 .24H_2O$$

or

$$M^I M^{III}(SO_4)_2 .12H_2O$$

Here, M^I may be K^+, Na^+, Cs^-, NH_4^-, Rb^-, Ag (Li^- is not possible due to its small size it can not have coordination number 6) and M^{III} may be Fe^{+3}, Cr^{-3}, Mn^{-3}

Common alum is potash alum

i.e., $K_2SO_4 Al_2(SO_4)_3 24H_2O$

Naming of Alums

If M^{+3} is not Al^{-3}, name of alum is given according to M^{+3}.

Example, $Cs_2SO_4 . Cr_2(SO_4)_3 . 24H_2O$ i.e., chrome alum.

If M^{-3} is Al^{+3}, name of alum is given according to M^- (e.g., rubidium alum) but not with K^+.

If neither K^- nor Al^{+3} are present than name of alum is given according to both the cations.

Example, $(NH_4)_2SO_4 Fe_2(SO_4)_3 24H_2O$
Ferric ammonium alum

Pseudo Alums: When in an alum monovalent element (M^I) is substituted by a bivalent element (M^{II}) it is called pseudo alum. It is not isomorphous with an alum.

Example, $MgSO_4 . Al_2(SO_4)_3 . 24H_2O$

$FeSO_4 . Al_2(SO_4)_3 . 24H_2O$

Some Facts About Alums

1. These can be obtained by mixing equimolar solutions of the constituent sulphates followed by crystallization.
2. These are crystalline compounds and all alums are isomorphous.

3. In alum crystals, 6 water molecules are held by monovalent ion, 6 water molecules are held by trivalent ion, 12 water molecules are held in the crystal structure.
4. Aqueous solutions of alum are acidic due to cationic hydrolysis of trivalent cation.
5. Feature alum or Hair-salt $Al_2SO_4.18H_2O$ is a native form of aluminium sulphate.
6. On heating an alum at high temperature the alum swells up into a porous mass which is called burnt alum.
7. Alums act as coagulants as they are effective in precipitating colloids.
8. Alums have germicide properties.

Potash Alum $K_2SO_4.Al_2(SO_4)_3.24\ H_2O$

Preparation

1. From Bauxite or Aluminium Sulphate

   When bauxite is boiled with sulphuric acid solution of aluminium sulphate is obtained. In it K_2SO_4 is added in a calculated amount followed by concentration and cooling to get the crystals of alum.

$$Al_2O_3 + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2O$$

$$Al_2(SO_4)_3 + K_2SO_4 + 24H_2O \longrightarrow \underset{\text{Potash alum}}{K_2SO_4.Al_2(SO_4)_3.24H_2O}$$

2. From Alunite or Alum Stone

   It is boiled with dilute sulphuric acid and a calculated amount of K_2SO_4 is added in the boiled solution, the resultant solution on cooling gives the crystals of alum.

$$\underset{\text{Alum stone}}{K_2SO_4.Al_2(SO_4)_3.4Al(OH)_3} + 6H_2SO_4 \longrightarrow K_2SO_4 + 3Al_2(SO_4)_3 + 12H_2O$$

$$K_2SO_4 + Al_2(SO_4)_3 + 24H_2O \longrightarrow K_2SO_4.Al_2(SO_4)_3.24H_2O$$

Physiochemical Properties

1. It is a white crystalline solid compound.
2. It is soluble in water and the aqueous solution is acidic due to cationic hydrolysis of $Al_2(SO_4)_3$. The aqueous solution has K^+, Al^{3-} and SO_4^{2-}.
3. Heating Effect: On heating it swells up due to elimination of water molecules as follows:

$$K_2SO_4.Al_2(SO_4)_3.14H_2O \xrightarrow{473\ K} K_2SO_4.Al_2(SO_4)_3 + 24H_2O$$

$$K_2SO_4.Al_2(SO_4)_3 \xrightarrow{\text{Red hot}} K_2SO_4 + Al_2O_3 + 3SO_3$$

Uses of Alums

1. As mordent in dyeing.
2. As a syptic to stop bleeding.
3. In tanning of leather.
4. In purification of water (negative impurities in water).
5. It is used in sizing of cheap quality paper.

Straight Objective Type Questions (Single Choice)

1. Molecular formula of Glauber's salt is
   (a) $FeSO_4.7H_2O$ (b) $Na_2SO_4.10H_2O$
   (c) $MgSO_4.7H_2O$ (d) $CuSO_4.5H_2O$

2. Which one of the following processes will produce hard water?
   (a) Saturation of water with $CaCO_3$
   (b) Addition of Na_2SO_4 to water
   (c) Saturation of water with $MgCO_3$
   (d) Saturation of water with $CaSO_4$

3. Some large white transparent crystals are left out in a bowl for several days. They are then observed to have changed their form into white powder. The crystals may have been of
   (a) Calcium oxide
   (b) Sodium carbonate
   (c) Ammonium chloride
   (d) Sodium chloride

4. Aluminium is not present in which of the following minerals?
   (a) cryolite (b) mica
   (c) feldspar (d) fluorspar

5. Which one of the following processes is used for the manufacturing of calcium?
   (a) Reduction with CaO with hydrogen
   (b) Electrolysis of molten $Ca(OH)_2$
   (c) Electrolysis of a mixture of anhydrous $CaCl_2$ and KCl
   (d) Reduction of CaO with carbon

6. Which of the following statements about anhydrous aluminium chloride is correct?
   (a) It sublimes at 100°C under vacuum.
   (b) It exists as $AlCl_3$ molecules.
   (c) It is a strong Lewis base.
   (d) It is not easily hydrolyzed.

7. The solubility of alkali metal hydroxide follows the order:
   (a) LiOH > CsOH > RbOH > NaOH > KOH
   (b) LiOH > NaOH > KOH > RbOH > CsOH
   (c) LiOH < NaOH < KOH < RbOH < CsOH
   (d) None of these

8. The metallic lusture exhibited by sodium is explained by
   (a) existence of body centred cubic lattice.
   (b) excitation of free proton.
   (c) oscillations of loose electrons.
   (d) diffusion of sodium ions.

9. Which pair of the following chlorides do not impart colour to the flame?
   (a) $MgCl_2$ and $CaCl_2$
   (b) $BeCl_2$ and $MgCl_2$
   (c) $BeCl_2$ and $SrCl_2$
   (d) $CaCl_2$ and $BaCl_2$

10. KO_2 (potassium superoxide) is used in oxygen cylinders in space and submarines because it
    (a) produces ozone
    (b) absorbs CO_2
    (c) absorbs CO_2 and increase O_2 contents
    (d) eliminates moisture

11. Sodium hydroxide being hygroscopic absorbs moisture when exposed to the atmosphere. A student placed a pellet of NaOH on a watch glass. A few days later, he noticed that the pellet was covered with a white solid. Identify this white solid.
    (a) Na metal (b) Na_2SO_3
    (c) Na_2CO_3 (d) Na_2S

12. The oxide that gives hydrogen peroxide on treatment with a dilute acids is
    (a) TiO_2 (b) MnO_2
    (c) PbO_2 (d) Na_2O_2

13. Which of the following compound transforms baking soda into baking powder?
    (a) $KHCO_3$ (b) $NaHCO_3$
    (c) $KHC_4H_4O_6$ (d) KCl

14. When a standard solution of NaOH is left in air for a few hours,
   (a) the concentration of Na^+ ion in solution will remains same.
   (b) a precipitate will form.
   (c) the strength of solution will increase.
   (d) strength of solution will decrease.

15. Baking powder used to make cake is a mixture of starch, $NaHCO_3$ and $Ca(H_2PO_4)_2$. The function of $Ca(H_2PO_4)_2$ is
   (a) Being acidic in nature and gives CO_2 when moistened with $NaHCO_3$.
   (b) To slow down the release of CO_2 gas.
   (c) To act as a filler.
   (d) None of these.

16. When calcium sulphate is mixed with conc. HCl and the paste is formed. What colour is obtained when a pinch of this paste is brought near the flame?
   (a) apple green (b) crimson red
   (c) brick red (d) golden yellow

17. A solution of sodium metal in liquid ammonia is strongly reducing due to the presence of
   (a) sodium hydride (b) sodium amide
   (c) sodium atoms (d) solvated electrons

18. The alkali metals form salt-like hydrides by the direct synthesis at elevated temperature. The thermal stability of these hydrides decreases in which of the following orders?
   (a) NaH > LiH > KH > RbH > CsH
   (b) LiH > NaH > KH > RbH > CsH
   (c) CsH > RbH > KH > NaH > LiH
   (d) KH > NaH > LiH > CsH > RbH

19. Sodium peroxide which is a yellow solid, when exposed to air becomes white due to the formation of
   (a) Na_2O
   (b) H_2O_2
   (c) NaOH and H_2O_2
   (d) NaOH and Na_2CO_3

20. When Na and Li are placed in dry air, one gets:
   (a) Na_2O, Li_2O
   (b) Na_2O, Li_3N, Li_2O
   (c) NaOH, Na_2O, Li_2O
   (d) Na_2O, Li_2O, Li_3N, NH_3

21. Which of these is the main constituent of eggshell?
   (a) $CaSiO_3$ (b) $CaSO_4.2H_2O$
   (c) $CaCO_3$ (d) $CaSO_4.\frac{1}{2}H_2O$

22. In curing cement plasters, water is sprinkled from time to time. This helps in
   (a) keeping sand gravel mixed and cement hydrated.
   (b) keeping it cool.
   (c) converting sand into silicic acid.
   (d) developing interlocking needle like crystals of hydrated silicates.

23. The one which is deliquescent among the following is
   (a) $CuSO_4.5H_2O$ (b) $BaCl_2.2H_2O$
   (c) $FeSO_4.7H_2O$ (d) $CaCl_2$

24. Which one of the following reactions is not associated with the Solvay process of manufacture of sodium carbonate?
   (a) $NaCl + NH_4HCO_3 \rightarrow NaHCO_3 + NH_4Cl$
   (b) $2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$
   (c) $2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$
   (d) $NH_3 + H_2CO_3 \longrightarrow NH_4HCO_3$

25. Baking powder contains
   (a) $NaHCO_3$, starch
   (b) $NaHCO_3$, $Ca(H_2PO_2)_2$ and starch
   (c) $NaHCO_3$
   (d) $NaHCO_3$, $Ca(H_2PO_2)_2$

26. The common name cream of tartar refers to
   (a) $K(SbO)C_4H_4O_6$
   (b) $Mg_2(OH)_2(C_4H_4O_6).H_2O$
   (c) $KHC_4H_4O_6$
   (d) $KNaC_4H_4O_6$

27. In electrolysis of NaCl when Pt electrode is taken then H_2 is liberated at cathode while with Hg cathode, it forms sodium amalgam. The reason for this is
   (a) more voltage is required to reduce H^+ at Hg than at Pt.
   (b) concentration of H^+ ions is larger when Pt electrode is taken.

(c) Na is dissolved in Hg while it does not dissolve in Pt.
(d) Hg is more inert than Pt.

28. Which of these is used to treat acid indigestion?
(a) $Ca(OH)_2$ (b) $Mg(OH)_2$
(c) $Be(OH)_2$ (d) KOH

29. A certain metal M is used to prepare an antacid, which is used as a medicine for acidity. This metal accidentally catches fire, which cannot be extinguished by using CO_2 based extinguishers. The metal M is
(a) Mg (b) Ba
(c) Be (d) C

30. Potassium when heated strongly in oxygen forms
(a) K_2O_2 (b) K_2O
(c) KO_3 (d) KO_2

31. An important ore of magnesium is
(a) malachite (b) cassiterite
(c) carnalite (D) galena

32. When a concentrated solution of ammonia is saturated with sodium chloride in the presence of pieces of dry ice, a white cloud is formed. This is due to the
(a) Precipitation of ammonium carbonate.
(b) Precipitation of ammonium hydrogen carbonate from the mixture.
(c) Precipitation of sodium hydrogen carbonate from the reaction mixture.
(d) Precipitation of sodium carbonate from the reaction mixture.

33. The oxidation states of the most electronegative element in the products of the reaction, BaO_2 with dil. H_2SO_4 are
(a) 0 and –1 (b) –1 and –2
(c) –2 and 0 (d) –2 and +1

34. When KI is added to acidified solution of sodium nitrite
(a) NO gas is liberated and I_2 is set free.
(b) N_2 gas is liberated and HOI is produced.
(c) N_2 gas is liberated and HI is produced.
(d) N_2O gas is liberated and I_2 is set free.

35. Sodium peroxide is used to purify the air in submarines and confined spaces because
(a) It decomposes to form Na_2O.
(b) It reacts with oxygen to form sodium superoxide.
(c) It removes CO_2 and produces O_2.
(d) Both (a) and (b).

36. The metallic lustre exhibited by sodium is explained by
(a) diffusion of sodium ions.
(b) oscillation of loose electrons.
(c) excitation of free electrons.
(d) existence of body centred cubic lattice.

37. Potassium is produced by electrolyzing fused KCl in a cell similar to one used for Na but the cell must be operated at a higher temperature because:
(a) KCl has a higher melting point than NaCl.
(b) K is a stronger reducing agent than Na.
(c) K is more electropositive than Na.
(d) K has higher melting point than Na.

38. When a standard solution of NaOH is left in air for hours then
(a) the strength of solution will increase.
(b) the concentration of Na^+ ion in solution will remains same.
(c) a precipitate will form.
(d) strength of solution will decrease.

39. The compound insoluble in acetic acid is
(a) calcium oxide
(b) calcium carbonate
(c) calcium oxalate
(d) calcium hydroxide

40. KOH is preferably used to absorb CO_2 gas because
(a) $KHCO_3$ is soluble in water and $NaHCO_3$ is sparingly soluble in water
(b) KOH is cheaper than NaOH
(c) KOH is more soluble than NaOH in water
(d) KOH is stronger base than NaOH

41. RbO_2 is
(a) peroxide and diamagnetic.
(b) peroxide and paramagnetic.
(c) superoxide and paramagnetic.
(d) superoxide and diamagnetic.

42. The hydration energy of Mg^{2+} ions is higher than that of
(a) Al^{3+} (b) Be^{2+}
(c) Na^{+} (d) None of these

43. Plaster of Paris hardens by
(a) changing into CaO
(b) giving out water
(c) utilizing water
(d) giving off CO_2

44. An $FeCl_3$ solution reacts with sodium hydroxide to produce
(a) Fe_3O_4 (b) $Fe_2O_3.nH_2O$
(c) Fe_2O_3 and FeO (d) FeO and $FeCl_3$

45. Which of the following set of raw materials are used in the manufacturing of Na_2CO_3 by Solvay process?
(a) $CaCl_2$, NH_3, CO_2
(b) NaOH, NH_3, CO_2
(c) NaCl, NH_3, CO_2
(d) $Ca(OH)_2$, NH_3, CO_2

46. In which of these pairs both species are soluble in NaOH solution?
(a) $Pb(OH)_2$, $Sn(OH)_4$
(b) $Al(OH)_3$, $Fe(OH)_3$
(c) $Zn(OH)_2$, $Cu(OH)_2$
(d) $Cr(OH)_2$, $Sn(OH)_2$

47. A burning strip of magnesium is introduced into a jar containing a gas. After sometime, the walls of the container is coated with carbon. The gas in the container is
(a) H_2O (b) CO_2
(c) O_2 (d) N_2

48. In which of the following pairs, both species on heating do not undergo any chemical change?
(a) $MgCO_3$, $KHCO_3$
(b) Cs_2CO_3, KNO_2
(c) Na_2CO_3, $NaNO_3$
(d) Li_2CO_3, KNO_2

49. Cl_2O_6 on reaction with $Ba(OH)_2$ gives
(a) $BaCl_2O_7$
(b) $Ba(ClO_3)_2 + Ba(ClO_2)_2$
(c) $Ba(Cl_2O_7)_2 + Ba(ClO_2)_2$
(d) $Ba(ClO_3)_2 + Ba(ClO_4)_2$

50. Halides of alkaline earth metals form hydrates such as $MgCl_2.6H_2O$, $CaCl_2.6H_2O$, $BaCl_2.2H_2O$ and $SrCl_2.2H_2O$. This shows that halides of group 2 elements:
(a) can absorb moisture form air.
(b) act as dehydrating agents.
(c) are hydroscopic in nature.
(d) all of the above.

51. Which of the following is correct?
(a) Sodium reduces CO_2 to carbon.
(b) In the Castner's process of sodium extraction, NaCl is used as an electrolyte.
(c) Magnalium is an alloy of Mg and Zn.
(d) Mg reacts with cold water and liberates hydrogen gas.

52. Which hydroxide is insoluble in sodium hydroxide?
(a) $Fe(OH)_3$ (b) $Al(OH)_3$
(c) $Cr(OH)_3$ (d) Both (a) and (c)

53. The metal (M) is prepared by the electrolysis of fused chloride. It reacts with hydrogen to form a colourless solid from which hydrogen is released on treatment with water. The metal (M) is
(a) Ca (b) Al
(c) Zn (d) Cu

54. The magnetic moment of KO_2 at room temperature is
(a) 1.43 BM (d) 2.64 BM
(c) 2.41 BM (d) 1.73 BM

55. When washing soda is heated
(a) CO_2 is released
(b) Water vapour is released
(c) CO_2 is released
(d) $CO + CO_2$ is released

56. Sodium peroxo borate, $Na_2[B_2(O_2)_2(OH)_4]\,6H_2O$ is a constituent of
(a) Bleaching powder
(b) Rocket propellants
(c) Washing powder
(d) Baking powder

57. Solvay process converts which of the following into soda ash?

(a) Caustic soda (b) Brine
(c) Sodium oxide (d) Sodium bicarbonate

58. One mol of calcium hypophosphite on heating gives
(a) 1 mol of calcium pyrophosphate and 3 mols phosphine.
(b) ½ mol of calcium pyrophosphate and 1 mol phosphine.
(c) 1 mol calcium pyrophosphate.
(d) ½ mol calcium phosphate and ½ mol phosphine.

59. A substance which gives a brick red flame and decomposes on heating into brown gas is
(a) $MgCO_3$ (b) $CaCO_3$
(c) $Ca(NO_3)_2$ (d) KNO_3

60. A metal (A) readily forms water soluble sulphate ASO_4, water insoluble hydroxide $A(OH)_2$ and oxide AO. The hydroxide is soluble in NaOH. Then (A) is
(a) Ca (b) Sr
(c) Be (d) Ba

Brainteasers Objective Type Questions (Single Choice)

61. In the following reactions, (X) and (Y), are respectively
$BaC_2 + N_2 \xrightarrow{\Delta} (X)$
$CaC_2 + N_2 \xrightarrow{\Delta} (Y)$
(a) $Ba(CN)_2$ and $Ca(CN)_2$
(b) $Ba(CN)_2$ and $CaCN_2$
(c) $BaCN_2$ and $CaCN_2$
(d) None of these

62. Which of the following is correct statement?
(I) Ca^{2+} ions are important in blood clotting.
(II) $Ca_3(PO_4)_2$ is a part of bones.
(III) $3Ca_3(PO_4).CaF_2$ is a part of enamel of teeth.
(a) (I) and (II) (b) (II) and (III)
(c) (I) and (III) (d) (I), (II) and (III)

63. The dissolution of $Al(OH)_3$ by a solution of NaOH results in the formation of
(a) $[Al(H_2O)_6(OH)_3]$
(b) $[Al(H_2O)_4(OH)]^{2+}$
(c) $[Al(H_2O)_2(OH)_4]^-$
(d) $[Al(H_2O)_3(OH)_3]$

64. Calcium imide on hydrolysis gives gas (P) which on oxidation by bleaching powder gives gas (Q). Gas (Q) on reaction with magnesium gives compound (R) which on hydrolysis again gives gas (P). Identify (P), (Q) and (R).
(a) N_2, NH_3, MgNH
(b) NH_3, N_2, Mg_3N_2
(c) NH_3, NO_2, $Mg(NO_2)_2$
(d) N_2, N_2O_5, $Mg(NO_3)_2$

65. A solution when diluted with H_2O and boiled, gives a white precipitate. On addition of excess NH_4Cl/NH_4OH, the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in NH_4OH/NH_4Cl.
(a) $Al(OH)_3$ (b) $Ca(OH)_2$
(c) $Mg(OH)_2$ (d) $Zn(OH)_2$

66. In the electrolysis of aqueous NaCl solution, side reactions taking place are
(I) $2OH^- + Cl_2 \rightarrow 2OCl^- + H_2$
(II) $2Na + 2H_2O \longrightarrow 2NaOH + H_2$
(III) $4OH^- \longrightarrow O_2 + 2H_2 + 4e^-$
Select the correct alternate:
(a) (I) and (III) (b) (II) and (III)
(c) (I) and (II) (d) (I), (II) and (III)

67. Consider the following statements:
(I). Cs ion is more highly hydrated than other alkali metal ions.
(II). Among the alkali metals Li, Na, K and Rb, Li has the higher melting point.

(III). Among the alkali metals, only Li forms a stable nitride by direct combination.

(a) (I) and (III) are correct

(b) (II) and (III) are correct

(c) (I), (II) and (III) are correct

(d) (I) and (II) are correct

68. A compound (A) on heating gives a colourless gas. The residue is dissolved in water to obtain (B). Excess CO_2 is bubbled through aqueous solution of (B) and (C) is formed. (C) on gentle heating gives black (A). The compound (A) is

(a) $Ca(HCO_3)_2$ (b) $NaHCO_3$

(c) $CaCO_3$ (d) Na_2CO_3

69. The pair of compounds which can exist together in aqueous solution is

(I) NaH_2PO_4 and $NaHCO_3$

(II) Na_2CO_3 and $NaHCO_3$

(III) NaOH and NaH_2PO_2

(IV) $NaHCO_3$ and NaOH

(a) II and III (b) I, II and III

(c) I and IV (d) Only IV

70. Which of the following changes occurs when excess of CO_2 gas is passed into a clear solution of lime water?

(a) A white precipitate containing both $CaCO_3$ and $Ca(HCO_3)_2$ is formed.

(b) Initially, a white precipitate of $CaCO_3$ is formed which changes into soluble $Ca(HCO_3)_2$ on passing excess CO_2 gas.

(c) A white precipitate of $Ca(HCO_3)_2$ is formed.

(d) A white precipitate of $CaCO_3$ is formed.

71. Salt (P) + (Q) $\longrightarrow$ (R) $\xrightarrow[\text{White ppt.}]{BaCl_2}$

(P) is paramagnetic in nature and contains about 55% K, hence (P) is

(a) KO_2 (b) K_2O

(c) K_2SO_4 (d) K_2O_2

72. If NaOH is added to an aqueous solution of Zn^{2+} ions, a white precipitate appears and on adding excess NaOH, the precipitate dissolves. In this solution zinc exists in the

(a) Anionic part

(b) Cationic part

(c) Both in anionic and cationic parts

(d) There is no zinc left in the solution

73. Which of the following compounds is consumed during the preparation of Na_2CO_3 by Solvay's process?

(a) $NaCl + NH_4HCO_3$

(b) $CaCO_3 + NaCl$

(c) $NH_4Cl + CaO + NaCl$

(d) $NH_3 + CaCO_3 + NaCl$

74. (P) + $H_2O \longrightarrow$ NaOH

(P) $\xrightarrow{O_2,\ 400°C}$ (Q) $\xrightarrow{H_2O,\ \text{at } 25°C}$ NaOH + O_2

(Q) is used for oxygenating in submarine.

(P) and (Q) are

(A) Na_2O_2 and O_2

(b) Na_2O and O_2

(c) Na_2O_2 and Na_2O

(d) Na_2O and Na_2O_2

75. When sodium chloride is electrolyzed in Nelson's cell, hydrogen is liberated at cathode and not sodium, because

(a) H^+ does not react with water.

(b) H has lower electrode potential.

(c) H^+ has greater mobility than Na^+.

(d) H is smaller than Na.

76. Which of the following equation is correct?

(a) $3LiNO_3 \xrightarrow{heat} 2LiNO_2 + O_2$

(b) $NaNO_3 + NaNH_2 \xrightarrow{80° - 90°C} 2NaOH + N_2O$

(c) Potassium formate is heated with free exposure to air.

$2HCOOK + O_2 \longrightarrow K_2CO_3 + H_2O + CO_2$

(d) Solid $KBrO_3$ is heated with powdered charcoal.

$2KBrO_3 + 3C \longrightarrow 2KBr + 3CO_2$

77. 2g of aluminium is treated separately with excess of dilute H_2SO_4 and excess of NaOH. The ratio of the volumes of hydrogen evolved is

(a) 1 : 1 (b) 1 : 2

(c) 2 : 1 (d) 2 : 3

78. In the reaction: $K_4[Fe(CN)_6] + K_2CO_3 \longrightarrow$ the product formed is

(a) $FeCO_3$ (b) KCN

(c) $KFeCO_3$ (d) $K_3[Fe(CN)_4]$

79. The major role of fluorspar (CaF_2) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is

(I) as a catalyst.

(II) to make the fused mixture very conducting.

(III) to lower the temperature of melt.

(IV) to decrease the rate of oxidation of carbon at the anode.

(a) (II), (III) (b) (I), (II)

(c) (II), (III), (IV) (d) (III), (IV)

80. In the reaction: $Al_2(SO_4)_3.18H_2O \xrightarrow[-18H_2O]{\text{Heat}} A \xrightarrow{800^\circ C} B + C$. The product A, B and C are, respectively

(a) $Al_2(SO_4)_3$, Al_2O_3, SO_3

(b) Al_2O_3, $Al_2(SO_4)_3$, SO_3

(c) Al_2SO_4, Al_2O_3, SO_3

(d) $Al_2(SO_4)_3$, Al_2O_3, SO_2

81. Which of the following reaction/s are correct here?

(I) $B + NaOH \rightarrow 2Na_3BO_3 + H_2$

(II) $P_4 + NaOH + H_2O \rightarrow NaH_2PO_2 + PH_3$

(III) $S + NaOH \longrightarrow Na_2S_2O_3 + Na_2S + H_2O$

(a) I only (b) III only

(c) II and III (d) I, II and III

82. Gas 'A' is bubbled through slaked lime when a white precipitate is formed. On prolonged bubbling, the precipitate dissolves. On heating the resultant solution, the white precipitate reappears with solution of gas 'B'. The gases 'A' and 'B' respectively are

(a) CO and CO_2 (b) CO_2 and CO_2

(c) CO and CO (d) CO_2 and CO

83. $CO_2 + NH_3 + H_2O \longrightarrow X$

$X + NaCl \rightarrow Y + NH_4Cl$

$2Y \longrightarrow Z + H_2O + CO_2$. 'Z' is

(a) Na_2CO_3 (b) $(NH_4)_2CO_3$

(c) NH_4HCO_3 (d) $NaHCO_3$

84. In the following sequence of reactions, identify the end product (D).

$Na_2CO_3 \xrightarrow{SO_2}$ (A) $\xrightarrow{Na_2CO_3}$ (B)

Elemental S, A $\rightarrow$ (C) $\xrightarrow{I_2}$ (D)

(a) Na_2SO_4 (b) $Na_2S_4O_6$

(c) Na_2S (d) $Na_2S_2O_3$

85. A deliquescent white crystalline hydroxide X reacts with a nitrate Y to form another hydroxide which decomposes to give a insoluble brown layer of its oxide. X is a powerful cautery and breaks down the proteins of skin flesh to a pasty mass. X and Y are

(a) NaOH, $Zn(NO_3)_2$

(b) NaOH, $AgNO_3$

(c) $Ca(OH)_2.HgNO_3$

(d) NaOH, $Al(NO_3)_3$

86. 1.04 g of bleaching powder was made into a paste with water and then made upto 200 ml. 25 ml of this solution was found to oxidize 13.4 ml of a standard solution of arsenic containing 4.3 g Al_2O_3 per litre. Find the available percentage of chlorine in the sample.

(a) 31.7% (b) 36.7%

(c) 41.7% (d) 63.6%

87. Solution of sodium metal in liquid ammonia is strongly reducing due to the presence of the following:

(a) solvated electrons

(b) sodium atoms

(c) sodium hydride

(d) sodium amide

88. Consider the following reactions:

$X + HCl \xrightarrow[\text{(addition)}]{\text{Anhy. } AlCl_3} C_2H_5Cl$

$\xrightarrow[\text{(substitution)}]{\text{Anhy. } ZnCl_2/HCl} Y$

Y can be converted to X on heating with ..... at ..... temperature.

(a) Cu, 300°C

(b) Al_2O_3, 350°C

(c) NaOH/I_2, 60°C

(d) $Ca(OH)_2 + CaOCl_2$, 60°C

89. When zeolite, which is hydrated sodium aluminium silicate, is treated with hard water the sodium ions are exchanged with

(I) H^- ions (II) Ca^{2-} ions
(III) SO_4^{-2} ions (IV) Mg^{2-} ions
(a) (II), (IV) (b) (I), (II), (III)
(c) (II), (IV), (V) (d) all are correct

90. Metal X on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour. Y is
(a) MgO (b) $Mg(NO_3)_2$
(c) Mg_3N_2 (d) NH_3

91. The aqueous solution of an inorganic compound (X) gives white precipitate with NH_4OH which does not dissolve in excess of NH_4OH. This aqueous solution also gives white precipitate with $AgNO_3$ and the precipitate is soluble in dilute HNO_3. Here, (X) is
(a) $AlCl_3$ (b) $AlBr_3$
(c) AlN (d) $Al_2(SO_4)_3$

92. A metal (A) heating in nitrogen gas gives (B). (B) on treatment with H_2O gives a colourless gas which on passing through $CuSO_4$ solution turns it deep blue. Here, (B) is
(a) NH_3 (b) MgO
(c) $Mg(NO_3)_2$ (d) Mg_3N_2

93. Which of the following pairs can be distinguished by the action of heat?
I Na_2CO_3 and $CaCO_3$
II $MgCl_2.6H_2O$ and $CaCl_2.6H_2O$
III $Ca(NO_3)_2$ and $NaNO_3$
(a) I and II (b) I II and III
(C) I and IIII (D) I only

94. Out of the following, metals that cannot be obtained by electrolysis of the aqueous solution of their salts are
I. Ag II. Mg
III. Cu IV. Al
(a) 2, 4 (b) 2, 3
(c) 1, 4 (d) All of these

95. Consider the following reactions:
I. $2Na + 2H_2O \longrightarrow 2NaOH + H_2$
II. $2NaOH + Cl_2 \rightarrow NaCl + NaOCl + H_2O$
III. $4OH \rightarrow O_2 + 2H_2O + 4e$
IV. $2Cl \rightarrow Cl_2 + 2e$
In the diaphragm cell used for the electrolysis of brine, the reactions that occur would include
(a) 2, 3, 4 (b) 1, 3, 4
(c) 1, 2, 3 (d) 1, 2, 4

96. In the following sequence of reactions. Identify (E).

$$Na_2CO_3 + H_2O + CO_2 \longrightarrow (A) \xrightarrow{A,\ ZnCl_2}$$

$$(B) \xrightarrow{\Delta} (C) + (D)\uparrow \xrightarrow{NaOH} (E)$$

(a) $NaHCO_3$ (b) Na_2O_2
(c) Na_2ZnO_2 (d) $ZnCO_3$

97. When brine solution is saturated with NH_3 and CO_2 a slightly alkaline white sodium salt (A) is formed which has pH of nearly 8.4. (A) on heating liberates a gas (B) leaving a highly alkaline residue (C) of pH nearly 10–11. Gas (B) is colourless and turns a solution of $Ca(OH)_2$ milky. Identify (B).
(a) Na_2CO_3 (b) $NaHCO_3$
(c) Na_2S (d) Na_2SO_4

98. The decomposition temperatures of alkaline earth metal carbonates are given as under:

$BeCO_3$	$MgCO_3$	$CaCO_3$
< 100°C	<.....	< 900°C

$SrCO_3$	$BaCO_3$
<.....	< 1300°C

The decomposition temperatures of $MgCO_3$ and $SrCO_3$ are, respectively
(a) 1290°C, 1200°C
(b) 1290°C, 540°C
(c) 540°C, 1290°C
(d) 540°C, 800°C

99. A soft metal (A) gives violet flame test. When burnt in O_2, (A) gives chrome yellow powder (B) which on reaction with water, gives alkaline solution (C) and O_2. Identify the metal.
(a) Na (b) Mg
(c) K (d) Ca

100. Aluminium is more reactive than iron because its standard reduction potential is higher. Still aluminium is less easily corroded than iron because
(a) Al reacts with atmospheric carbon dioxide to form a self-protective layer of Al_2O_3.

(b) It has higher reducing power and forms a self protective layer of Al_2O_3.

(c) It has higher reducing power and does not react with oxygen so easily.

(d) Both (a) and (b)

101. A compound (X) imparts a golden yellow flame and shows the following reactions:

Zn powder when boiled with a concentrated aqueous solution of (X) dissolves and H_2 is evolved. What is X?

(a) KOH (b) NaOH

(c) $Mg(OH)_2$ (d) $Ca(OH)_2$

102. A hydrated colourless solid (A) is water soluble and finds use in medicine as a purgative. When a solution of (A) is treated with ammonium phosphate, a white precipitate is formed. (A) gives a pink mass is cobalt nitrate test. What is (A)?

(a) $FeSO_4.7H_2O$ (b) $Na_2SO_4.10H_2O$

(c) $MgSO_4.7H_2O$ (d) $CaSO_4.2H_2O$

Multiple Correct Answer Type Questions (More Than One Choice)

103. Select the correct statement(s).

(a) $BaSO_4$ is soluble in water.

(b) $Ba(OH)_2$ is soluble in water.

(c) $MgSO_4$ is soluble in water.

(d) CaF_2 is soluble in water.

104. Nitrogen dioxide cannot be obtained by heating

(a) $NaNO_3$ (b) $Pb(NO_3)_2$

(c) KNO_3 (d) $LiNO_3$

105. Select the incorrect statement(s):

(a) Milk of magnesia is an aqueous solution of $Mg(OH)_2$.

(b) KOH is a weaker base than NaOH.

(c) CaO_2 is less stable than MgO_2.

(d) Mg^{2+} ions are precipitated with the addition of NH_4OH in the presence of NH_4Cl.

106. Select the correct statement(s).

(a) $Na_2CO_3.NaHCO_3.2H_2O$ is a mineral called trona.

(b) Li_2CO_3 and $MgCO_3$ both are thermally stable.

(c) Li_2CO_3 is only sparingly soluble in water and no $LiHCO_3$ has been isolated.

(d) K_2CO_3 cannot be prepared by a method similar to the ammonia-soda process.

107. White phosphorous on reaction with $Ca(OH)_2$ gives

(a) $Ca_3(PO_4)_2$ (b) PH_3

(c) $Ca(H_2PO_2)_2$ (d) $Ca(H_2PO_4)_2$

108. Which of the following statement(s) is/are true?

(a) Stability of alkali metal peroxide increases with increase in atomic number.

(b) Hydration energy of AgF is higher than its lattice energy.

(c) Anhydrous $MgCl_2$ cannot be prepared by direct heating of $MgCl_2.6H_2O$.

(d) The milk of magnesia used as antacid is chemically $MgO + MgCl_2$.

109. Which is/are correct statement(s)?

(a) Hydration of alkali metal ion is less than that of IIA group.

(b) Alkaline earth metal ions, because of their much larger charge to size ratio, exert a much stronger electrostatic attraction on the oxygen of water molecule surrounding them.

(c) $NaNO_3$ forms Na_2O on heating.

(d) The heats of hydration of the dipositive alkaline earth metal ions decrease with an increase in their ionic size.

110. Which of the following metals are extracted by using Al as a reducing agent?

(a) W from WO_3 (b) Mg from MgO

(c) Na from Na_2O (d) Cr from Cr_2O_3

111. Nitrogen dioxide can be obtained from
(a) $NaNO_3$ (b) $Cu(NO_3)_2$
(c) $AgNO_3$ (d) $Hg(NO_3)_2$

112. Aluminium becomes passive in
(a) $HClO_4$ (b) Conc. HCl
(c) Conc. HNO_3 (d) H_2CrO_4

113. Select the correct statement(s).
(a) Be dissolves in alkali forming $[Be(OH)_4]^{2-}$.
(b) BeF_2 forms complex ion with NaF in which Be goes with cation.
(c) $BeCO_3$ is kept in the atmosphere of CO_2 since, it is least thermally stable.
(d) BeF_2 forms complex ion with NaF in which Be goes with anion.

114. Which of the following oxides is/are amphoteric?
(a) Na_2O (b) CaO
(c) Al_2O_3 (d) SnO_2

115. Alkali metals are characterized as
(a) Good conductors of heat and electricity.
(b) High reducing nature.
(c) High melting point.
(d) Solubility in liquid ammonia.

116. Which is true about beryllium?
(a) $Be(OH)_2$ is basic in nature only.
(b) Beryllium halides are electron deficient.
(c) Aqueous solution of $BeCl_2$ is acidic.
(d) It forms unusual carbide Be_2C.

117. The pair of compounds which cannot exist together in aqueous solution is
(a) NaH_2PO_4 and Na_2HCO_3
(b) Na_2CO_3 and $NaHCO_3$
(c) NaOH and NaH_2PO_4
(d) $NaHCO_3$ and NaOH

118. Chemical change can take place in
(a) $K_2CO_3 \xrightarrow{\Delta}$
(b) $Li_2CO_3 \xrightarrow{\Delta}$
(c) $NaHCO_3 \xrightarrow{\Delta}$
(d) $Mg(NO_3)_2 \xrightarrow{\Delta}$

119. Select the correct statements about alkali metals.
(a) All form ionic hydrides (MH).
(b) All form (MNH_2) amide.
(c) All form nitrides.
(d) All form superoxides (MO_2).

120. Which among the following compounds are not paramagnetic?
(a) K_2O_2 (b) NO_2
(c) KO_2 (d) K_2O

121. Which of the following statement is/are correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
(a) The vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4.
(b) A deep red vapour is evolved.
(c) Chlorine gas is evolved.
(d) Chromyl chloride is formed.

122. Nitrate of which of the following elements can be converted into their oxides on heating?
(a) Na (b) Li
(c) Mg (d) Rb

123. Which of the following carbonates can evolve CO_2 on heating?
(a) Na_2CO_3 (b) Rb_2CO_3
(c) Li_2CO_3 (d) $MgCO_3$

124. Sodium sulphate is soluble in water whereas barium sulphate is sparingly soluble because
(a) the hydration energy of sodium sulphate is more than its lattice energy.
(b) the lattice energy of barium sulphate is more than its hydration energy.
(c) the lattice energy has no role to play in solubility.
(d) the hydration energy of sodium sulphate is less than its lattice energy.

125. Which of the following oxides have rock salt structure with coordination number 6:6?
(a) MgO (b) CaO
(c) SrO (d) B_2O_3

126. Incorrect match is/are
(a) Soda ash: Na_2CO_3
(b) Pearl ash: $CuCO_3$
(c) Bone ash: K_2CO_3
(d) Baking soda: $NaHCO_3$

127. Mg and Zn have following resemblance
(a) Both of them can be used as electrodes.
(b) MgO and ZnO are amphoteric.
(c) Both of them can be used to prevent corrosion.
(d) $MgCO_3$ and $ZnCO_3$ both on hearing give their oxides.

128. Out of the following the correct statement is/are
(a) $BeCl_2$ is an electron deficient molecule.
(b) $BeCl_2$ is a covalent compound.
(c) The hybridization state of Be in $BeCl_2$ is p^3.
(d) $BeCl_2$ can form dimer.

129. The compound of alkaline earth metals which are amphoteric in nature are
(a) MgO (b) BeO
(c) CaO (d) $Be(OH)_2$

130. Gypsum on heating gives
(a) $CaS + O_2$ (b) $CaO + SO_3$
(c) $CaSO_4 \cdot \frac{1}{2} H_2O$ (d) $CaSO_4$

Linked–Comprehension Type Questions

Comprehension–I

Consider the reactions given below and answer the questions.

(A) $\xrightarrow{\Delta}$ (B) (oxide) + CO_2

(B) + $H_2O \longrightarrow$ (C)

(C) + CO_2 $\rightarrow$ (A) (milky)

(C) + $NH_4Cl \xrightarrow{\Delta}$ (D) gas

(D) + H_2O + CO_2 $\rightarrow$ (E)

(E) + NaCl $\longrightarrow$ (F)

(F) $\xrightarrow{\Delta}$ $Na_2CO_3 + CO_2 + H_2O$

131. Here compound (A) is
(a) CaO (b) Na_2CO_3
(c) $CaCO_3$ (d) $Ca(HCO_3)_2$

132. Here compound (B) and (C) are, respectively
(a) $Ca(OH)_2$ and $CaCO_3$
(b) CaO and $Ca(OH)_2$
(c) $Ca(OH)_2$, CaO
(d) $CaCO_3$ and $Ca(OH)_2$

133. Here (D), (E) and (F) are, respectively
(a) (D) – NH_3, (E) – NH_4HCO_3, (F) – $NaHCO_3$
(b) (D) – NH_3, (E) – NH_4Cl, (F) – NH_4HCO_3
(c) (D) = NH_4HCO_3, (E) = Na_2CO_3, (F) = $NaHCO_3$
(d) None of these

Comprehension–2

Na $\xrightarrow{H_2O}$ (P) $\xrightarrow{CO_2}$ (Q)

$\xrightarrow{SO_2}$ (R) $\xrightarrow{A,\ Na_2S,\ I_2}$ (S)

Ag^+/Salt $\rightarrow$ (T) complex

134. Here, the compound (Q) and (R) are, respectively
(a) $NaHCO_3$, Na_2SO_4
(b) Na_2CO_3, Na_2SO_3
(c) Na_2CO_3, Na_2SO_4
(d) $NaHSO_3$, Na_2SO_3

135. The compound (S) is
(a) $Na_2S_2O_3$ (b) $Na_2S_2O_5$
(c) Na_2S (d) $Na_2S_4O_6$

136. Oxidation number of each sulphur atom in compound (S) are
(a) +6, 0 (b) +5, –1
(c) +5, 2 (d) +6, 2

137. Which statement about (S) is not correct here?
(a) It can be used in photography.
(b) Here (T) can be sodium argento thiosulphate.
(c) In it each sulphur atom is sp hybridized.
(d) It is also called hypo.

Comprehension–3

Metal nitrate (P) on heating decomposes, leaving a solid residue (Q) which goes into solution with dilute HCl. The solution of (Q) gives white precipitate with ammonium carbonate solution. The precipitate (R) is dissolved in dilute HCl and treated with potassium chromate solution to get yellow precipitate (S).

138. Here, metal nitrate (P) and residue (Q) are, respectively
   (a) $NaNO_3$ and Na_2O
   (b) $Ba(NO_3)_2$ and BaO
   (c) $Mg(NO_3)_2$ and MgO
   (d) $LiNO_3$ and Li_2O

139. Here, (R) is given as
   (a) Na_2CO_3 (b) $NaHCO_3$
   (c) $BaCO_3$ (d) $MgCO_3$

140. Here, yellow precipitate (S) is of
   (a) Na_2CrO_4 (b) $BaCr_2O_7$
   (c) $Na_2Cr_2O_7$ (d) $BaCrO_4$

Comprehension–4

Consider the following sequence of reactions and answer the questions given below.

$$(A) + NaOH \xrightarrow{Heat} NaCl + NH_3 + H_2O$$
$$NH_3 + CO_2 + H_2O \longrightarrow (B)$$

$$(B) + NaCl \longrightarrow (C) + NH_4Cl$$

$$(C) \longrightarrow Na_2CO_3 + H_2O + (D)$$

141. Here, the compound (A) is
   (a) NH_4NO_3 (b) NH_4Cl
   (c) NH_4NO_2 (d) Any of these

142. Here the compound (B) and (C) are, respectively
   (a) NH_4HCO_3 and Na_2CO_3
   (b) $NaHCO_3$ and NH_4HCO_3
   (c) NH_4HCO_3 and $NaHCO_3$
   (d) $NaHCO_3$ and Na_2CO_3

143. Which is correct about (D) here?
   (a) It is CO_2 (acidic oxide).
   (b) It is an oxidant.
   (c) In it oxidation number of carbon is +2.
   (d) All of these.

Comprehension–5

Element A burns in nitrogen to give an ionic compound B. Compound B reacts with water to give C and D. A solution of C becomes 'milky' on bubbling carbon dioxide.

144. Here, the element (A) is
   (a) Mg (b) Na
   (c) Ca (d) K

145. Here compound (B) and (C) are, respectively
   (a) Calcium carbide and acetylene
   (b) Calcium nitride and calcium hydroxide
   (c) Magnesium nitride and magnesium hydroxide
   (d) Magnesium carbide and propyne

146. Which is incorrect about this element?
   (a) It forms basic oxide and basic hydroxide.
   (b) Its carbonate decomposes on heating.
   (c) It can form stable complexes even with weak complexing agents also.
   (d) It is essential constituent of bones and teeth.

Comprehension–6

When 16.8 g of white solid X was heated, 4.4 g of acid gas A that turned lime water milky was driven off, together with 1.8 g of a gas B which condensed to a colourless liquid. The solid that remained, Y, dissolved in water to give an alkaline solution, which with excess barium chloride solution gave a white precipitate Z. The precipitate effervesced with the acid giving off carbon dioxide.

147. Here, (X) is
   (a) $KHCO_3$ (b) Na_2CO_3
   (c) NH_4HCO_3 (d) $NaHCO_3$

148. Here, (A) and (B) are, respectively
   (a) CO_2 and N_2
   (b) CO_2 and H_2
   (c) CO_2 and H_2O
   (d) CO_2 and NH_3

149. Here, (Y) and (Z) are, respectively
   (a) $NaHCO_3$ and $BaHCO_3$
   (b) Na_2CO_3 and $BaCO_3$
   (c) NaOH and $BaCO_3$
   (d) K_2CO_3 and $BaCO_3$

Assertion and Reasoning Questions

In the following questions two statements (Assertion) A and Reason (R) are given. Mark.
   (a) If A and R both are correct and R is the correct explanation of A.
   (b) If A and R both are correct but R is not the correct explanation of A.
   (c) A is true but R is false.
   (d) A is false but R is true.

150. (A): Sodium metal can not be obtained by the electrolysis is of its salt in aqueous solution.
   (R): Sodium is above hydrogen in electrochemical series and it reacts with water to produce sodium hydroxide and hydrogen.

151. (A): LiF is practically insoluble in water.
   (R): LiF has very high lattice energy.

152. (A): $BaSO_4$ is highly insoluble in water but readily dissolves in a solution of the sodium salt of ethylene diamine-terra acetic acid.
   (R): Ba^{2-} forms a very stable complex with the anion of the sodium salt of ethylene diamine tetra acetic acid which is soluble in water.

153. (A): $Al(OH)_3$ is amphoteric in nature.
   (R): Al O and O H bonds can be broken with equal ease in $Al(OH)_3$

154. (A): Among the alkali metals, lithium salts exhibit the least electrical conductance in aqueous solutions.
   (R): Smaller the radius of the hydrated cation, lower is the electrical conductance in aqueous solution.

155. (A): Al becomes passive in conc. HNO_3.
   (R): Conc. HNO_3 has no action on aluminium metal.

156. (A): The stability of alkali metal peroxides increases with increasing atomic number.
   (R): Bigger cations form more more stable lattice with bigger anions.

157. (A): $AlCl_3$ forms dimmer Al_2Cl_6 but it dissolves in water forming $[Al(H_2O)_6]^{3+}$ and Cl^- ions.
   (R): Aqueous solution of $AlCl_3$ is acidic due to hydrolysis.

158. (A): Sodium ions are discharged in preference to hydrogen ions at a mercury cathode.
   (R): The nature of cathode can affect the order of discharge of cations.

159. (A): Mg gets oxidized when heating in CO_2 atmosphere.
   (R): Mg has a strong affinity for oxygen.

160. (A): Anhydrous magnesium chloride cannot be obtained by heating $MgCl_2.6H_2O$.
   (R): $MgCl_2.6H_2O$ is highly stable and is unaffected by heat.

161. (A): $BaCO_3$ is more soluble in HNO_3 than in plain water.
   (R): Carbonate is a weak base and reacts with the H^- from the strong acid causing the barium salt to dissociate.

162. (A): The alkali metals can form ionic hydrides which contains the hydride ion.

(R): The alkali metals have low electronegativity, their hydrides conduct electricity when fused and liberate hydrogen at the anode.

163. (A): Sodium reacts with oxygen to form Na_2O_2 whereas potassium reacts with oxygen to form KO_2.

(R): Potassium is more reactive than sodium.

164. (A): Crystals of $NaHCO_3$ and $KHCO_3$ show hydrogen bonds of different kinds.

(R): In $NaHCO_3$, the bicarbonate ions are linked in an infinite chain while in $KHCO_3$, a dimeric chain is formed.

165. (A): MgO is used for lining of steel making furnace.

(R): It is a acidic flux and helps in removing basic impurities.

166. (A): When hot and concentrated NaOH reacts with chlorine, NaCl and NaClO are formed.

(R): It is a case of disproportion or autoredox reaction.

167. (A): $BeCl_2$ fumes in moist air.

(R): $BeCl_2$ reacts with moisture to form HCl gas.

168. (A): Magnesium metal burns in air to give a white ash and this ash gives smell of ammonia in contact with water.

(R): The ash contains magnesium nitride also which is hydrolyzed by water and ammonia is evolved.

169. (A): Magnesium is extracted by the electrolysis of fused mixture of $MgCl_2$, NaCl and $CaCl_2$.

(R): Calcium chloride acts as a reducing agent.

170. (A): Alkali metals dissolve in liquid ammonia to give blue solutions.

(R): Alkali metals in liquid ammonia give solvated species of the type

$[M(NH_3)_n]^+$ (M – alkali metals)

[IIT 2007]

Matrix–Match Type Questions

	p	q	r	s
(A)	O	O	O	O
(B)	O	O	O	O
(C)	O	O	O	O
(D)	O	O	O	O

171. Match the following:

Column I	Column II
A. Sorel's cement	(p) $MgCl_2$
B. Albite	(q) MgO
C. A salt of carnalite	(r) $NaAlSi_3O_8$
D. Glauber's salt	(s) $Na_2SO_4.10H_2O$

172. Match the following:

Column I	Column II
A. Effloresent	(p) NaOH
B. Deliquescent	(q) KOH
C. Fusion mixture	(r) Na_2CO_3 and K_2CO_3
D. Washing soda	(s) $Na_2CO_3.10H_2O$

173. Match the following:

Column I	Column II
A. Beryl	(p) $KCl.MgCl_2.6H_2O$
B. Carnalite	(q) $MgCO_3$
C. Asbestos	(r) $3BeO.Al_2O_3.6SiO_2$
D. Magnesite	(s) $Ca_2Mg_5Si_8O_{22}(OH)_2$

174. Match the following:

Column I	Column II
A. Na_2CO_3	(p) SO_2 absorber
B. Na_2SO_3	(q) Detergent
C. NaOH	(r) Glass
D. NaOCl	(s) Bleach

175. Match the following:

Column I (Compounds)	Column II (Use of Compounds)
A. Magnesium hydroxide	(p) As a fertilizer
B. Barium sulphate	(q) As a purgative
C. Magnesium sulphate	(r) As a constituent of lithopone
D. Calcium cyanamide	(s) As a constituent of sorrel cement

176. Match the following:

Column I	Column II
A. Duralumin	(p) Calcium
B. Magnalium	(q) Magnesium
C. Electron metal	(r) Aluminium
D. Dolomite	(s) Zinc

177. Match the following:

Column I (Compounds)	Column II (Composition)
A. Black ash	(p) K_2CO_3
B. Washing soda	(q) CaS
C. Nitrolim	(r) Na_2CO_3
D. Pearl ash	(s) $CaCN_2 + C$

178. Match the following:

Column I	Column II
A. Castner Kellner cell	(p) $MgCl_2.5MgO.x\ H_2O$
B. Pearl ash	(q) K_2CO_3
C. Solvay's process	(r) Manufacture of $NaHCO_3$
D. Sorel's cement	(s) Manufacture of NaOH

179. Match the following:

Column I	Column II
A. Magnesia	(p) $MgSO_4.7H_2O$
B. Epsom salt	(q) MgO
C. Anhydrone	(r) CaH_2
D. Hydrolith	(s) $Mg(ClO_4)_2$

180. Match the following:

Column I	Column II
A. Plaster of Paris	(p) KCl
B. Sorrel's cement	(q) MgO
C. Component of carnalite	(r) $MgCl_2$
D. Sylvine	(s) $CaSO_4.\ ½\ H_2O$

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181. Which of the following statement is correct for $CsBr_3$?
(a) It is a covalent compound
(b) It contains Cs^{3+} and Br^- ions
(c) It contains Cs^+ and Br_3^- ions
(d) It contains Cs^-, Br^- and lattice Br_2 molecule
[*IIT 1996*]

182. Highly pure dilute solution of sodium in liquid ammonia
I. shows blue colour
II. exhibits electrical conductivity
III. produces sodium amide
IV. produces hydrogen gas
(a) 1, 2 (b) 1, 2, 3
(c) 2, 3, 4 (d) All of these
[*IIT 1998*]

183. Sodium nitrate decomposes above ~ 800°C to give
(a) N_2 (b) O_2
(c) NO_2 (d) Na_2O
[*IIT 1998*]

184. Polyphosphates are used as water softening agents because they
(a) form soluble complexes with anionic species.
(b) precipitate anionic species.
(c) form soluble complexes with cationic species.
(d) precipitate cationic species.
[*IIT 2002*]

185. $MgSO_4$ on reaction with NH_4OH and Na_2HPO_4 forms a white crystalline precipitate. What is its formula?
(a) $Mg(NH_4)PO_4$ (b) $Mg_3(PO_4)_2$
(c) $MgCl_2.MgSO_4$ (d) $MgSO_4$
[*IIT 2006*]

186. White phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a
(a) Dimerization reaction
(b) Disproportion reaction
(c) Condensation reaction
(d) Precipitation reaction
[*IIT 2008*]

187. The compounds formed upon combustion of Na-metal in excess air is/are
(a) Na_2O_2 (b) Na_2O
(c) NaO_2 (d) NaOH
[*IIT 2009*]

ANSWERS

Straight Objective Type Questions

1. (b)	2. (d)	3. (b)	4. (d)
5. (c)	6. (a)	7. (c)	8. (c)
9. (b)	10. (c)	11. (c)	12. (d)
13. (c)	14. (d)	15. (a)	16. (c)
17. (d)	18. (b)	19. (d)	20. (b)
21. (c)	22. (d)	23. (d)	24. (b)
25. (b)	26. (a)	27. (d)	28. (b)
29. (a)	30. (d)	31. (c)	32. (c)
33. (b)	34. (a)	35. (c)	36. (b)
37. (a)	38. (d)	39. (c)	40. (a)
41. (c)	42. (c)	43. (c)	44. (b)
45. (c)	46. (a)	47. (b)	48. (b)
49. (d)	50. (d)	51. (a)	52. (d)

53. (a) 54. (d) 55. (b) 56. (c)
57. (b) 58. (b) 59. (c) 60. (c)

Brainteasers Objective Type Questions

61. (b) 62. (d) 63. (c) 64. (b)
65. (a) 66. (a) 67. (b) 68. (c)
69. (b) 70. (b) 71. (a) 72. (a)
73. (b) 74. (d) 75. (c) 76. (c)
77. (a) 78. (b) 79. (a) 80. (a)
81. (d) 82. (b) 83. (a) 84. (b)
85. (b) 86. (a) 87. (a) 88. (b)
89. (a) 90. (c) 91. (a) 92. (d)
93. (b) 94. (a) 95. (b) 96. (c)
97. (a) 98. (c) 99. (c) 100. (b)
101. (b) 102. (c)

Multiple Correct Answer Type Questions

103. (b), (c) 104. (a), (c)
105. (b), (d) 106. (a), (c), (d)
107. (b), (c) 108. (a), (b), (c)
109. (a), (b), (d) 110. (a), (d)
111. (b), (c), (d) 112. (c), (d)
113. (a), (c), (d) 114. (c), (d)
115. (a), (b), (d) 116. (b), (c), (d)
117. (c), (d) 118. (b), (c), (d)
119. (a), (b) 120. (a), (d)
121. (a), (b), (d) 122. (b), (c)
123. (b), (c), (d) 124. (a), (b)
125. (a), (b), (c) 126. (b), (c)
127. (a), (c), (d) 128. (a), (b), (d)
129. (b), (d) 130. (b), (c), (d)

Linked–Comprehension Type Questions

Comprehension–1

131. (c) 132. (b) 133. (a)

Comprehension–2

134. (b) 135. (a) 136. (d) 137. (c)

Comprehension–3

138. (b) 139. (c) 140. (d)

Comprehension–4

141. (b) 142. (c) 143. (d)

Comprehension–5

144. (c) 145. (b) 146. (c)

Comprehension–6

147. (d) 148. (c) 149. (b)

Assertion and Reasoning Questions

150. (d) 151. (a) 152. (a) 153. (a)
154. (a) 155. (c) 156. (a) 157. (b)
158. (a) 159. (a) 160. (c) 161. (a)
162. (a) 163. (b) 164. (a) 165. (c)
166. (d) 167. (a) 168. (a) 169. (c)
170. (b)

Matrix–Match Type Questions

171. (a)-(q), (b)-(r), (c)-(p), (d)-(s)

172. (a)-(q), (b)-(p), (c)-(r), (d)-(s)

173. (a)-(r), (b)-(p), (c)-(s), (d)-(q)

174. (a)-(q), (b)-(r), (c)-(p), (d)-(s)

175. (a)-(s), (b)-(r), (c)-(q), (d)-(p)

176. (a)-(q), (r), (b)-(q), (r), (c)-(q), (s), (d)-(p), (q)

177. (a)-(q), (b)-(r), (c)-(s), (d)-(p)

178. (a)-(s), (b)-(q), (c)-(r), (d)-(p)

179. (a)-(q), (b)-(p), (c-) (s), (d)-(r)

180. (a)-(s), (b)-(q), (r), (c)-(r), (d)-(p)

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181. (c) 182. (a) 183. (b) 184. (c)

185. (a) 186. (b) 187. (a,b)

HINTS AND EXPLANATIONS

Straight Objective Type Questions

1. Glauber's salt is $Na_2SO_4.10H_2O$.

2. $MgCO_3$ and $CaCO_3$ are water insoluble. $CaSO_4$ dissolves in water adding Ca^{2+} ions which are responsible for producing hardness in water.

3. Due to efflorescence (to give out H_2O) nature of $Na_2CO_3.10H_2O$.

4. Fluorspar is CaF_2. It does not contain aluminium. Feldspar is $KAlSi_3O_8$.

5. Calcium is obtained by electrolysis of a used mass consisting 6 parts $CaCl_2$ and 1 part CaF_2 at about 700°C in an electrolytic cell made of graphite which acts as anode and a water cooled cathode of iron.

   $CaCl_2 \rightleftharpoons Ca^{2+} + 2Cl^-$

   At anode: $2Cl^- \longrightarrow Cl_2 + 2e^-$

   At cathode: $Ca^{2+} + 2e \longrightarrow Ca$

6. $AlCl_3$ exists as a dimmer (Al_2Cl_6). It is a strong Lewis acid as Al has an incomplete octet and has a tendency to gain electrons. $AlCl_3$ undergoes hydrolysis easily and forms an acidic solution.

   $AlCl_3 + 3H_2O \longrightarrow Al(OH)_3 + 3HCl$

   Example, $AlCl_3$ sublimes at 100°C under vaccum.

9. Be and Mg atoms are smaller. The electrons in these atoms, are more strongly bound and hence these are not excited by the energy of flame to higher energy states. The chlorides of these elements therefore, do not give any colour in flame.

10. $4KO_2 + 2CO_2 \longrightarrow 2K_2CO_3 + 3O_2$

11. $2NaOH + CO_2 \longrightarrow \underset{\text{(Solid)}}{Na_2CO_3}$

12. $Na_2O_2 + H_2SO_4 \underset{\text{(20\% ice cold)}}{\longrightarrow} Na_2SO_4 + H_2O_2$

13. $KHC_4H_4O_6$ is potassium hydrogen tartarate and is used to transform baking soda into baking powder.

17. The free ammoniated electrons make the solution of Na in liquid NH_3 a very powerful reducing agent. The ammonical solution of an alkali metal is rather favoured as a reducing agent than its aqueous solution because in aqueous solution, the alkali metal being highly electropositive evolves hydrogen from water (thus H_2O acts as an oxidizing agent) while its solution in ammonia is quite stable, provided no catalyst (transition metal) is present.

18. The ionic character of the bonds in hydrides increase from LiH to CsH hence thermal stability of these hydrides decreases as follows:

LiH > NaH > KH > RbH > CsH

19. Sodium peroxide. [Na_2O_2] is a yellow solid which becomes white when it is exposed to air because Na_2O_2 forms NaOH with moisture and with CO_2 it forms Na_2CO_3.

24. $2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$

This reaction is not associated with Solvay process.

27. H^- ions are discharged at a higher potential when Hg cathode is used than that using Pt cathode.

31. Carnalite is $KCl.MgCl_2.6H_2O$. It is an ore of Mg.

33. $BaO_2 + H_2SO_4 \rightarrow BaSO_4 \downarrow + H_2O_2$

34. $2I^- + 4H^+ + 2NO_2^- \longrightarrow 2NO + I_2 + 2H_2O$

36. These loose electrons absorb radiation and then radiate it back.

38. NaOH being deliquescent absorbs water from atmosphere therefore strength of solution will decrease.

39. Calcium oxalate does not dissolve in acetic acid. All others [CaO, $CaCO_3$ and $Ca(OH)_2$] are bases and hence dissolve in acetic acid.

41. RbO_2 means Rb and O_2^-. O_2^- is the superoxide ion and contains 17 electrons. Thus, paramagnetic in nature.

42. Mg^{2+} has smaller size than Na^+ and thus has higher hydration energy than Na^+.

47. Magnesium wire reduce CO_2 to C

$2Mg + CO_2 \rightarrow 2MgO + C.$

50. Halides of Group 2 elements are hygroscopic and act as dehydrating agents.

54. Here number of unpaired electron = 1

Magnetic moment = $\sqrt{n(n+2)}$ B.M.

= $\sqrt{1(1+2)}$

= $\sqrt{3}$ = 1.73 BM

55. $Na_2CO_3.10H_2O \xrightarrow[-9H_2O]{\Delta} Na_2CO_3.H_2O \xrightarrow{\Delta} Na_2CO_3 + H_2O$

57. Solvay process converts brine into soda ash. Reactions involved in Solvay ammonia soda process are as follows:

$NH_3 + H_2O + CO_2 \longrightarrow NH_4HCO_3$

$NaCl + NH_4HCO_3 \rightarrow NaHCO_3 + NH_4Cl$

$NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$

59. $2Ca(NO_3)_2 \xrightarrow{\Delta} 2CaO + \underset{\text{Brown gas}}{4NO_2\uparrow} + O_2$

Being a calcium compound it imparts a brick red flame.

Brainteasers Objective Type Qusetions

63. $Al(OH)_3 + OH^- \longrightarrow [Al(OH)_4^-]$

Coordination number is 6, thus it exists as $[Al(H_2O)_2(OH)_4]^-$.

64. $\underset{(P)}{Ca(NH)_2} + 2H_2O \longrightarrow Ca(OH)_2 + \underset{(P)}{NH_3}(g)$

$\underset{(P)}{2NH_3} + 3CaOCl_2 \rightarrow \underset{(Q)}{N_2}(g) + 3CaCl_2 + 3H_2O$

$\underset{(Q)}{N_2}(g) + 3Mg \rightarrow \underset{(R)}{Mg_3N_2}$

$\underset{(R)}{Mg_3N_2} + 6H_2O \longrightarrow 3Mg(OH)_2 + \underset{(P)}{2NH_3}$

65. $Al(OH)_3$ is a gelatinous white precipitate, formed by the reaction of Al^{3+} with NH_4OH in the presence of NH_4Cl.

72. $Zn^{2+} + 2NaOH \rightarrow 2Na^+ + Zn(OH)_2$

$Zn(OH)_2 + 2NaOH \longrightarrow Na_2ZnO_2 + 2H_2O$

Thus, Na_2ZnO_2 forms $2Na^+$ and $[ZnO_2]^{2-}$ ions.

75. H^+ has greater mobility than Na^+, because of its smaller size as compared with Na, therefore electrode potential (or reduction potential) of H^+ (0.00 V) is greater than that of Na^+ (–2.71 V). Hence H^+ is reduced in preference to Na.

Cathode

$2H^+(aq) + 2e \rightarrow H_2(g)$

77. $2Al + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2$

$2Al + 2NaOH + 2H_2O \longrightarrow 2NaAlO_2 + 3H_2$

Hence, the ratio of volumes of hydrogen evolved is 1:1.

78. Here KCN is formed as follows:

$K_4[Fe(CN)_6] + K_2CO_3 \rightarrow 5\,KCN + KCNO + Fe + CO_2$

79. Fluorspar is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6).

(i) to make the fused mixture more conducting so alumina in a bad conductor of electricity.

(ii) to lower the melting point of fused mixture to 1140 K, alumina has a high melting point of 2323 K.

80. $Al_2(SO_4)_3.18H_2O \xrightarrow[18H_2O]{Heat} \underset{A}{Al_2(SO_4)_3}$

$\xrightarrow{800°C} \underset{B}{Al_2O_3} + \underset{C}{3SO_3}$

82. 'A' and 'B' are CO_2 and CO_2.

$Ca(OH)_2 \xrightarrow[\Delta]{(A)\ CO_2} CaCO_3 \downarrow \xrightarrow[H_2O]{CO_2}$

$\underset{(soluble)}{Ca(HCO_3)_2} \xrightarrow{\Delta} \underset{(ppt.)}{CaCO_3} + \underset{(B)}{CO_2}\uparrow$

83. $CO_2 + NH_3 + H_2O \rightarrow NH_4HCO_3\ (X)$

$NH_4HCO_3 + NaCl \longrightarrow \underset{(Y)}{NaHCO_3} + NH_4Cl$

$2NaHCO_3 \rightarrow Na_2CO_3\ (Z) + H_2O + CO_2$

84. $Na_2CO_3 + SO_2 \xrightarrow{H_2O} \underset{(A)}{2NaHSO_3} + CO_2$

$\underset{(A)}{2NaHSO_3} + Na_2CO_3 \longrightarrow$

$\underset{(B)}{2Na_2SO_3} + H_2O + CO_2$

$\underset{(B)}{Na_2SO_3} + S \xrightarrow{\Delta} \underset{(C)}{Na_2S_2O_3}$

$2Na_2S_2O_3 + I_2 \longrightarrow \underset{(D)}{Na_2S_4O_6} + 2NaI$

86. $CaOCl_2 + H_2O \rightarrow Ca(OH)_2 + Cl_2$

$As_2O_2 + 2Cl_2 + 2H_2O \longrightarrow As_2O_3 + 4HCl$

Meq = Meq Cl_2

Eq. wt. of $As_2O_3 = \frac{\text{mol. wt.}}{4} = \frac{198}{4} = 49.5$

$N = 4.3/49.5$

Total meq of arsenite used

$= \frac{(13.4 \times 4.3) \times 200}{49.5 \quad 25} = \text{meq } Cl_2$

Meq $(Cl_2) = \frac{9 \times 1000}{35.5}$

g of $Cl_2 = \frac{13.4 \times 4.3 \times 8 \times 35.5}{49.5 \qquad 1000}$

$= 0.33$

Percentage of $Cl_2 = \frac{0.33 \times 100}{1.04} = 31.7\ \%$

87. $Na + (x + y)\,NH_3 \longrightarrow Na(NH_3)x' + e\,(NH_3)_{-y}$. Thus, due to solvated (ammoniated) electron, solution of Na metal in liquid ammonia is highly reducing in nature.

89. $Na_2Al_2Si_2O_8.xH_2O$ is written as Na_2Z, Na^+ ions exchange Ca^{-} and Mg^{2-} ions of hard water.

$Na_2Z + Mg^2 \longrightarrow 2Na^1 + MgZ$

$Na_2Z + Ca^{2-} \rightarrow 2Na^+ + CaZ$

90. $\underset{(X)}{3Mg} + N_2 \xrightarrow{\Delta} \underset{(Y)}{Mg_3N_2}$

$\underset{(Y)}{Mg_3N_2} + 6H_2O \rightarrow 3Mg(OH)_2 + \underset{\text{colourless}}{2NH_3}$

$CuSO_4 + 4NH_3 \rightarrow \underset{\text{Blue complex}}{[Cu(NH_3)_4]SO_4}$

91. $AlCl_3 + 3NH_4OH \rightarrow Al(OH)_3 + 3NH_4Cl$

$Al(OH)_3 + NH_4OH \longrightarrow$ No reaction

$AlCl_3 + 3AgNO_3 \rightarrow \underset{\text{White ppt.}}{3AgCl} + Al(NO_3)_3$

93. (i) $Na_2CO_3 \xrightarrow{\Delta}$ No effect

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

(ii) $MgCl_2.6H_2O \xrightarrow{\Delta} Mg(OH)Cl.5H_2O + HCl$

$CaCl_2.6H_2O \xrightarrow{\Delta} 2CaCl_2 + 6H_2O$

(iii) $2Ca(NO_3)_2 \xrightarrow{\Delta} 2CaO + 2NO_2 + 3O_2$

$2NaNO_3 \xrightarrow{\Delta} 2NaNO_2 + O_2$

94. Mg and Al are more electropositive than hydrogen and hence cannot be obtained by electrolysis of the aqueous solution of their salts.

97. $NaCl + 2NH_3 + CO_2 + H_2O \longrightarrow \underset{(A)}{NaHCO_3} + NH_4Cl$

$2NaHCO_3 \xrightarrow{\Delta} \underset{(B)}{Na_2CO_3} + H_2O + CO_2$

$CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$

98. The decomposition temperature increases down the group.

99. Here (A) is potassium.

$K + O_2 \rightarrow \underset{\text{Superoxide}}{KO_2}$

$4KO_2 + 2H_2O \rightarrow 4KOH + O_2$

Superoxide is paramagnetic and contains a three electron bond.

101. Here (X) is NaOH.

As sodium imparts a golden yellow flame and NaOH also evolves hydrogen and zinc.

$Zn + 2NaOH \longrightarrow Na_2ZnO_2 + H_2$

102. Here (A) is Epsom salt ($MgSO_4.7H_2O$)

$Mg^{2+} + NH_4^+ + PO_4^{3-} \rightarrow \underset{\text{White ppt.}}{Mg(NH_4)PO_4}\downarrow$

$MgSO_4 + Na_2CO_3 \rightarrow MgCO_3 + Na_2SO_4$

$MgCO_3 \xrightarrow{[O]} MgO + CO_2$

$3Co(NO_3)_2 \longrightarrow 2CoO + 4NO_2 + O_2$

$CoO + MgO \rightarrow CoO.MgO$

Multiple Correct Answer Type Questions

114. Na_2O and CaO are basic while Al_2O_3 and SnO_2 are amphoteric.

115. Melting point of alkali metals are lower.

117. Acidic and basic salts do not exist together in aqueous solution because they react with each other.

$2NaOH + NaH_2PO_4 \rightarrow Na_3PO_4 + 2H_2O$

$NaHCO_3 + NaOH \rightarrow Na_2CO_3 + H_2O$

124. When H.E. > L.E. the salt is soluble in water.

Comprehension–3

138. $\underset{(P)}{Ba(NO_3)_2} \rightarrow \underset{(Q)}{BaO} + 2NO_2 + \tfrac{1}{2}O_2$

139. $\underset{(Q)}{BaO} \xrightarrow{HCl} BaCl_2$

$BaCl_2 \xrightarrow{(NH_4)_2CO_3} \underset{(R)}{BaCO_3}$

140. $\underset{(R)}{BaCO_3} \xrightarrow{HCl} BaCl_2$

$BaCl_2 \xrightarrow{K_2CrO_4} \underset{\text{Yellow ppt.}}{\underset{(S)}{BaCrO_4}}$

Comprehension–4

141. $NH_4Cl + NaOH \xrightarrow{\text{Heat}} NaCl + NH_3 + H_2O$
(A)

142. $NH_3 + CO_2 + H_2O \rightarrow \underset{\substack{(B)\\ \text{Ammonium bicarbonate}}}{NH_4HCO_3}$

$\underset{(B)}{NH_4HCO_3} + NaCl \longrightarrow \underset{\substack{(C)\\ \text{Sodium bicarbonate}}}{NaHCO_3} + NH_4Cl$

143. $\underset{(C)}{NaHCO_3} \longrightarrow \underset{(D)}{Na_2CO_3} + H_2O + CO_2$

Comprehension–5

145. $3Ca + N_2 \rightarrow Ca_3N_2$
(A) (B), Calcium nitride (An ionic compound)

$Ca_3N_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2NH_3$
(C) Calcium hydroxide (D)

146. As calcium can form complexes only with strong complexing agents like EDTA.

Comprehension–6

147. (X) is $NaHCO_3$ (molecular wt. = 84)

148. $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3(s) + CO_2(g) + H_2O(g)$
(X) (Y) (A) (B)

2 x 84 = 168g	106 g	44 g	18 g
= 16.8 g	10.6 g	4.4 g	1.8 g

$CO_2 + Ca(OH)_2 \rightarrow CaCO_3\downarrow + H_2O$
Lime water White ppt.

H_2O (g) is condensed to liquid water.

149. $Na_2CO_3 + BaCl_2 \longrightarrow BaCO_3 + NaCl$
(Y) (Z)

$BaCO_3 + 2HCl \longrightarrow BaCl_2 + H_2O + CO_2$
(Z)

Assertion and Reasoning Questions

150. Sodium metal can be obtained by electrolysis of fused salt.

153. As the size of the ion increases, the tendency to rupture the O H bond decreases and hence acidic nature decreases i.e., basic nature increases.

154. Hydration energy decreases down the group i.e., Li is most hydrated the least conducted in aqueous solution.

155. Al metal is rendered passive when treated with conc. HNO_3.

163. K^+ being larger in size than Na^+ has a weaker positive field around it which cannot prevent the conversion of peroxide ion (O_2^{2-}) to superoxide ion $(O)_2^-$.

165. As it is a basic flux hence removes acidic impurities.

167. $BeCl_2 + 2H_2O \longrightarrow Be(OH)_2 + 2HCl$

169. NaCl and $CaCl_2$ are added to provide conductivity to the electrolyte and also to lower the fusion temperature of anhydrous $MgCl_2$.

170. Blue colour is due to solvated electrons.

The IIT–JEE Corner

181. $CsBr_3$ contains Cs^+ and Br_3^- ($Br^- + Br_2$) ions.

182. Highly pure dilute solution of sodium in liquid ammonia is blue in colour due to ammoniated electron. This solution is conducting due to both ammoniated cation and ammoniated electron.

$Na + (x + y)\,NH_3 \rightarrow [Na(NH_3)x]^+ + [e(NH_3)y]^-$

183. $2NaNO_3 \xrightarrow{\Delta} 2NaNO_2 + O_2$

184. Polyphosphates are used as water softener because these form soluble complexes with cationic species (Ca^{+2} and Mg^{+2}) present in hard water. The complex calcium and magnesium ions do not form any precipitate with soap and hence water readily produce lather with soap solution.

$2Ca^{+2} + Na_2[Na_4(PO_3)_6] \longrightarrow Na_2[Ca_2(PO_3)_6] + 4Na^+$

$2Mg^{+2} + Na_2[Na_4(PO_3)_6] \rightarrow Na_2[Mg_2(PO_3)_6] + 4Na^+$
$Na_2[Mg_2(PO_3)_6]$
Soluble complex

185. Test of Mg^{2+} ion

$Mg^{2-} + NH_4OH + Na_2HPO_4 \longrightarrow Mg(NH_4)PO_4$

186. It is a case of disproportion reaction in which white phosphorus on reaction with NaOH gives PH_3 and sodium hydrogen phosphite as follows:

$P_4 + 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3$

187. Here Both Na_2O and Na_2O_2 are formed.

$2Na \xrightarrow[O_2]{Air} Na_2O \xrightarrow{O_2} Na_2O_2$

Solved Subjective Questions

1. The hydroxide of aluminium and iron are insoluble in water. However, NaOH is used to separate one from the other. Why?

Solution

As aluminium hydroxide dissolves in NaOH forming soluble $NaAlO_2$ and can be separated from ferric hydroxide.

$$Al(OH)_3 + NaOH \rightarrow \underset{\text{(Soluble)}}{NaAlO_2} + 3H_2O$$

2. (A) is a binary compound of a univalent metal. 1.422 g of (A) reacts completely with 0.321 g of sulphur in an evacuated and sealed tube to give 1.743 g of a white crystalline solid (B) that formed a hydrated double salt (C) with $Al_2(SO_4)_3$. Identify (A), (B) and (C).

[*IIT 1994*]

(i) (B) forms a double salt with $Al_2(SO_4)_3$ and so it may be K_2SO_4.

(ii) $(A) + S \rightarrow \underset{(B)}{K_2SO_4}$

As 1.743 g K_2SO_4 is obtained by 1.422g (A)

So 174 g K_2SO_4 is obtained by =

$$\frac{1.422 \times 174}{1.743} = 142\text{ g (A)}$$

As 174 g K_2SO_4 requires 32 g S

So 1.743 g K_2SO_4 requires

$$\frac{32 \times 1.743}{174} = 0.321\text{ g S}$$

Hence the given data confirms that (B) is K_2SO_4.

$2(A) + S \rightarrow K_2SO_4$

(A potassium salt)

Mol. wt. of (A) x 2 – 142

Mol. wt. of (A) = 71

As (A) is potassium salt

Hence, mol. wt. of left component

= 71 – 39 = 32

Hence, the salt is KO_2.

$$\underset{(A)}{2KO_2} + S \longrightarrow \underset{(B)}{K_2SO_4} \longrightarrow \underset{(aq)}{Al_2(SO_4)_3}$$

$$\underset{(C)}{K_2SO_4.Al_2(SO_4)_3.24H_2O}$$

3. When gas (A) is passed through dry KOH at low temperature, a deep red coloured compound (B) and a gas (C) are obtained. The gas (A) on reaction with but-2-ene followed by treatment with Zn/H_2O yields acetaldehyde. Identify (A), (B) and (C).

[*IIT 1994*]

Solution

The gas (A) on reaction with but-2-ene followed by reaction with Zn/H_2O gives acetaldehyde and thus (A) is ozone

(i) $O_3 + CH_3 - CH = CH - CH_3 \rightarrow$

$CH_3 - CH - O - CH - CH_3$

O O

Mono ozonide

$\downarrow H_2O, Zn$

$2CH_3CHO$

Acetaldehyde

(ii) $\underset{(A)}{5O_3} + \underset{(B)}{2KOH} \rightarrow \underset{\text{Potassium ozonide (Deep red)}}{2KO_3} + \underset{(C)}{H_2O\,(g) + 5O_2}$

4. Mg_3N_2 when reacted with water gives off NH_3 but HCl is not obtained from $MgCl_2$ on reaction with water at room temperature.

[*IIT 1995*]

Solution

Mg_3N_2 is a salt of a strong base, $Mg(OH)_2$ and a weak acid (NH_3). While, $MgCl_2$ is a salt of a strong base, $Mg(OH)_2$ and a strong acid, HCl and hence it cannot undergo hydrolysis to give HCl.

5. Calcium burns in nitrogen to produce a white powder which dissolves in sufficient water to produce a gas (A) and an alkaline solution. The solution on exposure to air produces a thin solid layer of (B) on the surface. Identify the compounds A and B.

[*IIT 1996*]

Solution

Ca burns in air to form CaO and Ca_3N_2.

$2Ca+O_2 \rightarrow 2CaO$

$3Ca + N_2 \rightarrow Ca_3N_2$

Calcium nitride on hydrolysis with H_2O gives ammonia (A)

$Ca_3N_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2NH_3$

The alkaline solution of $Ca(OH)_2$ formed reacts with CO_2 present in the air to give $CaCO_3$ (B).

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

Hence, A is NH_3 and B is $CaCO_3$.

6. Gradual addition of KI solution to $Bi(NO_3)_3$ solution initially produces a dark brown precipitate which dissolves in excess of KI to give a clear yellow solution. Write chemical equations for the above reactions.

[*IIT 1996*]

Solution

$Bi(NO_3)_3 + H_2O \rightarrow$

$Bi[(OH)(NO_3)_2] + HNO_3$

$NO_3^- + 4H^+ + 3e \rightarrow NO + 2H_2O$

$2I^- \rightarrow I_2 + 2e^-$

$I_2 + KI \rightarrow KI_3$ (Yellow solution)

7. The crystalline salts of alkaline earth metals contain more water of crystallization than the corresponding alkali metal salts. Why?

[*IIT 1997*]

Solution

Because of smaller size and higher nuclear charge, alkaline earth metals have a higher tendency than alkali metals to attract H_2O molecules and thus contain more water of crystallization than alkali metals. For example, $MgCl_2.6H_2O$.

8. Element A burns in nitrogen to give an ionic compound B. Compound B reacts with water to give C and D. A solution of C becomes 'milky' on bubbling carbon dioxide. Identify A, B, C and D.

[*IIT 1997*]

Solution

The element (A) may be calcium.

$3Ca + N_2 \rightarrow Ca_3N_2$

(A) (B), Calcium nitride (An ionic compound)

$Ca_3N_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2NH_3$

(C) (D) Calcium hydroxide

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

It brings milkiness to the solution.

9. Chlorination of calcium hydroxide produces bleaching powder. Write its chemical equation.

[*IIT 1998*]

Solution

Bleaching powder can be obtained by passing Cl_2 into $Ca(OH)_2$. Although bleaching powder is written as $Ca(OCl)_2$, it is actually a mixture.

$3Ca(OH)_2 + 2Cl_2 \rightarrow Ca(OCl)_2.Ca(OH)_2.CaCl_2.2H_2O$

10. White solid is either Na_2O or Na_2O_2. A piece of red litmus paper turns white when it is dipped into a freshly made aqueous solution of the white solid.

(i) Identify the substance and explain with balanced equation.

(ii) Explain what would happen to the red litmus if the white solid were the other compound.

[*IIT 1999*]

Solution

Na_2O_2 is a powerful oxidant and a bleaching agent so it bleaches red litmus paper to white in aqueous solution state.

$$Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O + [O]$$

$$[O] + \underset{\text{red}}{\text{Litmus}} \rightarrow \text{White (bleaching)}$$

The other compound Na_2O gives NaOH on dissolution in water and the red litmus turns to blue.

11. Identify the following:

$$Na_2CO_3 \xrightarrow{SO_2} (A) \xrightarrow{Na_2CO_3} (B) \xrightarrow[\Delta]{S} (C) \xrightarrow{I_2} (D)$$

Also, mention the oxidation state of S in all compound.

(IIT2003)

Solution

$$Na_2CO_3 + SO_2 \xrightarrow{H_2O} \underset{(A)}{2NaHSO_3} + CO_2$$

$$NaHSO_3 + Na_2CO_3 \rightarrow \underset{(B)}{2Na_2SO_3} + H_2O + CO_2$$

$$\underset{(B)}{Na_2SO_3} + S \rightarrow \underset{(C)}{Na_2SO_3}$$

$$\underset{(C)}{2NaS_2O_3} + I_2 \rightarrow \underset{(D)}{Na_2S_4O_6} + 2NaI$$

Oxidation states of S are +4 in $NaHSO_3$ and Na_2SO_3 +6 snd-2 (average+2) in $Na_2S_2O_3$ and+5 and 0 (an average +5/2) in $Na_2S_4O_6$.

12. AlF_3 is not soluble in anhydrous HF but soluble in KF:

(a) Explain this observation.

(b) When BF_3 is added to the above solution, AlF_3 is precipitated. Write the balanced chemical equation.

[IIT 2004]

Solution

(a) AlF_3 dissolves in ionic KF due to the formation of K_3AlF_6.

$$AlF_3 + 3KF \rightarrow K_3[AlF_6]$$

soluble

On the other hand, anhydrous HF being weak acid does not dissociate to appreciable extent and AlF_3 does not form $[AlF_4]$ or $H[AlF_4]$.

(b) Addition of BF_3 in $K_3[AlF_6]$ forms AlF_3 as insoluble mass as BF_3 is more acidic than AlF_3.

$$K_3[AlF_6] + BF_3 \rightarrow AlF_3 + 3K[BF_4]$$

13. When 16.8 g of white solid X were heated, 4.4 g of acid gas A that turned lime water milky was driven off together with 1.8 g of a gas B which condensed to a colourless liquid. The solid that remained, Y, dissolved in water to give an alkaline solution, which with excess barium chloride solution gave a white precipitate Z. The precipitate effervesced with acid giving off carbon dioxide. Identify A, B and Y and write down the equation for the thermal decomposition of X.

Solution

(X) is $NaHCO_3$ (molecular wt. = 84)

Reactions involved are as follows:

$$\underset{(X)}{2NaHCO_3} \xrightarrow{\Delta} \underset{(Y)}{Na_2CO_3(s)} + \underset{(A)}{CO_2(g)} + \underset{(B)}{H_2O(g)}$$

2 x 84 = 168 g	106 g	44 g	18 g
= 16.8 g	0.6 g	4.4 g	1.8 g

$$CO_2 + Ca(OH)_2 \rightarrow \underset{\text{White ppt.}}{CaCO_3\downarrow} + H_2O$$

Lime water

H_2O (g) is condensed to liquid water

$$\underset{(Y)}{Na_2CO_3} + BaCl_2 \rightarrow \underset{(Z)}{BaCO_3} + NaCl$$

$$\underset{(Z)}{BaCO_3} + 2HCl \rightarrow BaCl_2 + H_2O + CO_2$$

14. Why does the solubility of alkaline earth metal hydroxides increase down the group?

Solution

Among alkaline earth metal hydroxides, the anion, i.e., OH^- ion is common; therefore, the lattice enthalpy depends upon the size of the cation. Since, lattice enthalpy decreases much more than the hydration enthalpy with

increasing ionic size, the solubility of the metal hydroxides increases as we move down the group.

15. Why does the solubility of alkaline earth metal carbonates and sulphates decrease down the group?

Solution

The size of CO^{2}_{3} and SO_4^{2} ions are much bigger than those of the cations. Hence, the lattice energies of carbonates and sulphates remain almost constant as we move down the group. Since, hydration enthalpies decrease as the size of the cation increases down the group, the solubility of carbonates and sulphates of alkaline earth metals decreases down the group.

16. Why is Li_2CO_3 decomposed at a lower temperature whereas Na_2CO_3 at higher temperature?

Solution

Li_2CO_3 is a salt of a weak acid (CO_2) with a weak base (LiOH). Since the weak base cannot attrack CO_2 strongly, so Li_2CO_3 decomposes at lower temperature. On the other hand, NaOH is a much stronger base than LiOH and hence can attract CO_2 more strongly, thus Na_2CO_3 is much more stable than Li_2CO_3 and hence decomposes at much higher temperature than Li_2CO_3.

Questions For Self Assessment

17. Why is LiF almost insoluble in water whereas LiCl is soluble not only in water but also in acetone?

18. Explain the following:

    (a) A solution of Na_2CO_3 is alkaline.

    (b) Alkali metals are prepared by electrolysis of their fused chlorides.

    (c) Sodium is found to be more useful than potassium.

19. How would you explain?

    (i) BeO is insoluble but $BeSO_4$ is soluble in water.

    (ii) BaO is soluble but $BaSO_4$ in insoluble in water.

    (iii) LiI is more soluble than KI in ethanol.

20. Arrange the following sulphates of alkaline earth metals in order of decreasing thermal stability.

    $BeSO_4$, $MgSO_4$, $CaSO_4$, $SrSO_4$

    [*IIT 1997*]

21. Give reason in one or two sentences for the following: "The hydroxides of aluminium and iron are insoluble in water. However, NaOH is used to separate one from the other.

22. A metal chloride (P) gives white precipitate (Q) in presence of NH_4OH and $(NH_4)_3PO_4$. Q on heating gives R and a pungent smell gas (S) which turns red litmus blue. Identify P to S and give reactions.

    $P = MgCl_2$ $Q = MgNH_4PO_4.6H_2O$
    $R = Mg_2P_2O_7$ $S = NH_3$

23. An unknown solid mixture contains one or two of the following: CaO, $BaCl_2$, $AgNO_3$, Na_2SO_4, $ZnSO_4$ and NaOH. The mixture is completely soluble in water and the solution gives pink colour with phenolphthalein, when 0.1N HCl solution is gradually added to the above solution, a precipitate is produced which dissolves with further addition of the acid, what is/are present in the solid ? Give equation to explain the appearance of the precipitate and its dissolution.

Solution

NaOH, $ZnSO_4$

24. Complete and balance the following reactions:

    $$Ca_5(PO_4)_3F + H_2SO_4 + H_2O \xrightarrow{Heat} \ldots\ldots + 5CaSO_4.2H_2O + \ldots\ldots$$

    [*IIT 1994*]

25. Write balanced equation for the reactions between (a) Na_2O_2 and water (b) KO_2 and water (c) Na_2O and CO_2

26. An unknown solid mixture contains one or two of the following: $CaCO_3$, $BaCl_2$, $AgNO_3$, Na_2SO_4, $ZnSO_4$ and NaOH. The mixture is completely soluble in water and the solution gives pink colour with phenolphthalein. When dilute hydrochloric acid is gradually added to this solution, a precipitate is produced which dissolves with further addition of the acid. What is present in solid mixture? Give equations to explain the appearance of the precipitate and its dissolution.

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COMPOUNDS OF p–BLOCK ELEMENTS –I

5

Chapter Contents

Boron: diborane, boric acid and borax; Carbon: oxides and oxyacid (carbonic acid); Silicon: silicones, silicates and silicon carbide; Nitrogen: oxides, oxyacids and ammonia; Phosphorous: oxides, oxyacids (phosphorous acid and phosphoric acid) and phosphine and various levels of multiple-choice questions.

COMPOUNDS OF BORON

Hydrides of Boron

Boron reacts with dihydrogen to give a number of hydrides. These hydrides are known as boranes and have a general formula B_nH_{n+4} (Example: B_2H_6) and B_nH_{n+6} (Example: B_4H_{10})

Diborane (B_2H_6)

Preparation

B_2H_6 is prepared as follows:

1. **Laboratory Method**: In laboratory it is prepared by the reaction of sodium borohydride with iodine in a high boiling polyether solvent $(CH_3OCH_2CH_2)_2O$.

$$2NaBH_4 + I_2 \xrightarrow{\text{Polyether}} B_2H_6 + 2NaI + H_2$$

2. **Industrial Method**

$$2BF_3 + 6LiH \xrightarrow{450K} B_2H_6 + LiF$$

3. **From Boron Halides**

$$4BX_3 + 3LiAlH_4 \rightarrow 2B_2H_6 + 3LiX + 3AlX_3$$

For example,

$$4BCl_3 + 3LiAlH_4 \rightarrow 2B_2H_6 + 3LiCl + 3AlCl_3$$

4. When boron halides and excess of hydrogen undergo reaction by passing silent electric discharge at low pressure, B_2H_6 is formed.

$$2BX_3 + 6H_2 \xrightarrow[\text{discharge}]{\text{Silent}} B_2H_6 + 6HCl$$

Physiochemical Properties

1. It is a colourless gas which is stable only at low temperatures.
2. **Heating Effect**: When it is heated in between 100–300°C, higher boranes are formed like B_3H_9, B_4H_{10}, B_5H_{11} etc. while at red hot temperature it decomposes into boron.

$$B_2H_6 \xrightarrow{\text{Red hot}} 2B + 3H_2$$

3. **With Oxygen:** All boranes catch fire in the presence of oxygen and liberate a lot of heat energy so they can be used as high energy fuels also.

$$B_2H_6 + 3O_2 \longrightarrow 2B_2O_3 + 3H_2O$$
$$\Delta H = -2008 \text{ kJ/mol}$$

4. **With Water:** Boranes are readily hydrolyzed by water.

$$B_2H_6 + 6H_2O \longrightarrow 2H_3BO_3 + 6H_2$$

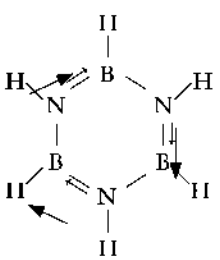
5. **Formation of Borohydrides:** Boranes are used in the formation of hydroborates or borohydrides like $LiBH_4$ or $NaBH_4$, which are extensively used as reducing agents in many organic synthesis.

$$2LiH + B_2H_6 \longrightarrow Li^+[BH_4]$$
$$2NaH + B_2H_6 \longrightarrow 2Na^+[BH_4]^-$$

6. **Formation of Borazine:** Diborane (1 vol) on reaction with NH_3 (2 vol) at 450K gives borazine or borazol or inorganic benzene.

$$3B_2H_6 + 6NH_3 \xrightarrow{450K} 2B_3N_3H_6 + 12H_2$$

or



Borazine
(Isoelectronic structural to benzene)

Fig. 5.1

Structure of Diborane

It is an electron deficient molecule as it does not have enough number of electrons needed for the formation of normal covalent bonds. Its structure was proposed by Dilthey as a bridge structure shown in the figure given below. It is also confirmed by electron diffraction studies.

In this structure of diborane, boron atom is sp^3 hybridized having one empty and three half-filled hybrid orbitals i.e., four hybrid orbitals of equal energy.

$$B^* \; 1s^2 \; 2s^1 \; 2p^1_x \; 2p^1_y \; 2p^0_z$$

In diborane there are not enough electrons (there are only twelve electrons) 3 from each boron atom and 6 from 6 hydrogen atom to form the 7 covalent bonds that an ordinary Lewis structure would required $H_3B.BH_3$.

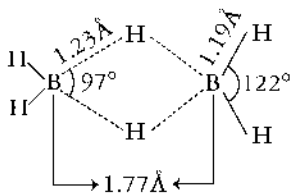


Fig. 5.2 St. of Diborane

These 12 valence electron of boron forms 6 covalent bonds with these 6 hydrogen. Amongst these 6 hydrogen atoms, 2H are termed as bridging hydrogen atoms whereas 4H are termed as terminal hydrogen atoms.

When diborane is methylated then only 4 hydrogen atoms are susceptible to methylation and these 4H atoms are entirely different from remaining 2H atoms. These 2H atoms are known as bridge hydrogen. Hydrogen bridge is also known as Banana bond or tau bond or 3 centre 2 electron (3C – 2e) bond.

Bond lengths: B – B = 1.77 Å
B – H = 1.19 Å
(Normal covalent bond)
B – H – 1.33 Å
3 centre 2 electron bond.

H–B–H bond angle = 121.5° ± 7.5°

H–H–B bond angle (3C – 2e) – 100°

Diborane has two type of BH bonds:

(i) **B–Ht:** (Here H_t = terminal hydrogen) It is a normal covalent bond (two centre electron pair bond i.e., 2c–2e bond).

(ii) **B–Hb:** (Here H_b = bridge hydrogen). It is a bond between three atoms, **B–H_b–B**, (three centre-two electron pair bond i.e., 3c – 3e or banana bond). Such a bond is formed by the overlaping of empty sp^3 hybridized orbital of one boron atom, 1s orbital of hydrogen atom (H_b) and half filled sp^3 hybridized orbital of the other boron atom.

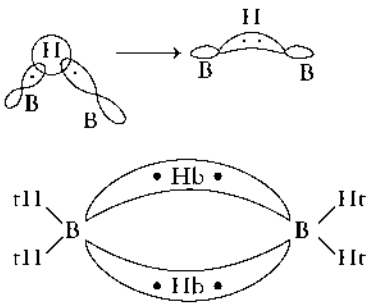


Fig. 5.3

Uses

1. It is used for making high energy fuels and propellants.
2. It can be used as a catalyst in many polymerization reactions.
3. It can be used as a reductant in many organic reactions like hydroboration.

Boric Acid or Ortho Boric Acid (H_3BO_3 or $B(OH)_3$)

Preparation

1. **From Borax**: When aqueous solution of borax is acidified with conc. HCl or H_2SO_4, boric acid solution is formed which can be easily crystallized into crystals of boric acid as follows.

$$Na_2B_4O_7 + 2HCl + 5H_2O \rightarrow 4H_3BO_3 + 2NaCl$$

$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow 4H_3BO_3 + Na_2SO_4$$

2. **From Colemanite**: When SO_2 is passed through a suspension of colemanite in boiling water, boric acid is formed which can be easily crystallized.

$$Ca_2B_6O_{11} + 4SO_2 + 11H_2O \rightarrow 6H_3BO_3 + 2Ca(HSO_3)_2$$

3. **From Boron trichloride**:

$$BCl_3 + 3H_2O \longrightarrow H_3BO_3 + 3HCl$$

Physiochemical Properties

1. It is white, soft, needle like crystalline solid with a soapy touch.
2. **Acidic Nature:** It behaves like a weak monobasic acid. It also behaves like a Lewis acid.

$$H_3BO_3 + H_2O \longrightarrow H_3O + H_2BO_3$$

$$H_3BO_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$$

3. **Solubility:** It is less soluble in cold water but readily soluble in hot water.

Solubility of boric acid $\propto$ temperature

$$H_3BO_3 + H_2O \rightarrow H_3BO_3\,aq. - Heat$$

4. **Heating Effect:** On heating it gives different products at different temperatures as follows:

$$H_3BO_3 \xrightarrow[-H_2O]{373K} \underset{\text{Metaboric acid}}{HBO_2} \xrightarrow[H_2O]{433K}$$

$$\underset{\text{Tetraboric acid}}{H_2B_4O_7} \xrightarrow[H_2O]{\text{strongly heating}} \underset{\text{Boron trioxide}}{B_2O_3}$$

5. **With Ethyl Alcohol:** When it is heated with ethyl alcohol it gives volatile vapours of triethyl borate which burn with a green edged flame.

$$H_3BO_3 + 3C_2H_5OH \rightarrow \underset{\text{Triethyl borate}}{B(OC_2H_5)_3} + 3H_2O$$

6. **With NaOH:**

$$H_3BO_3 + NaOH \rightarrow NaBO_2 + 2H_2O$$

7. **With NaOOH** (Sodium Hydroperoxide)

$$2H_3BO_3 + 2NaOOH \rightarrow \underset{\text{Dihydrated sodium peroxyborate (Crystals)}}{Na_2B_2O_6.2H_2O} + 2H_2O$$

Structure: It has a slippery layer structure in which planar BO_3 units are linked together by hydrogen bonds. In this structure each hydrogen atom can act as a bridge between the two oxygen atoms of the two adjacent BO_3 units. It forms a covalent bond with the oxygen atom of one BO_3 unit and a hydrogen bond with the oxygen atom of another BO_3 unit. In boric acid, boron atom is sp^2 hybridized with planar trigonal shape.

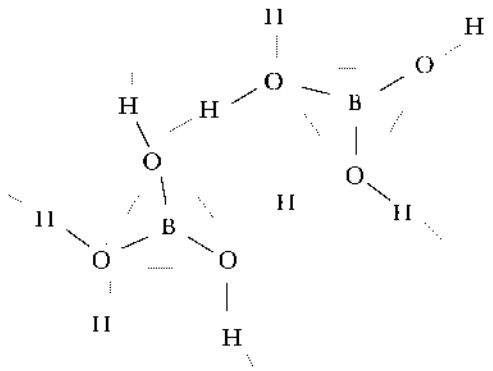


Fig. 5.4 St. of Boric acid

Uses

1. It is used as a antiseptic and a eye lotion (boric lotion).
2. It is also used as a food preservative.
3. H_3BO_3 decreases thermal expansion of glass.
4. It is used in the manufacture of enamels and glazes for pottery.

Borax or Tincal ($Na_2B_4O_7.10H_2O$)

It is called sodium tetraborate decahydrate.

It is also called tincal or suhaga as tincal has nearly 45% borax.

Preparation

1. **From Colemanite**: It is prepared from powdered ore of colemanite [$Ca_2B_6O_{11}$] by boiling it with sodium carbonate followed by filtration as follows:

$$Ca_2B_6O_{11} + 2Na_2CO_3 \rightarrow Na_2B_4O_7 + 2NaBO_2 + 2CaCO_3$$

Sodium metaborate can be further converted into borax by passing CO_2 through it.

$$4NaBO_2 + CO_2 \rightarrow Na_2B_4O_7 + Na_2CO_3$$

2. **From Boric Acid**: Boric acid on treatment with sodium carbonate gives borax as follows:

$$4H_3BO_3 + Na_2CO_3 \rightarrow Na_2B_4O_7 + 6H_2O + CO_2$$

Physiochemical Properties

1. It exists in three forms:

(i) ***Prismatic Borax*** i.e., decahydrate form ($Na_2B_4O_7.10H_2O$): It can be obtained by the crystallization of borax solution at coordinate temperature. It is less soluble in cold water but highly soluble in hot water.

(ii) ***Octahedral Borax*** i.e., pentahydrate form ($Na_2B_4O_7.5H_2O$): It can be obtained by crystallizing borax solution at 333K.

(iii) ***Anhydrous Borax or Borax Glass*** ($Na_2B_4O_7$): It can be obtained heating borax above its melting point (1013K). This type of borax is colourless glassy mass which absorbs moisture and readily changes into decahydrate form.

2. **Hydrolysis**: On hydrolysis it gives basic taqueous solution as follows:

$$Na_2B_4O_7 + 7H_2O \xrightarrow{\Delta} 2Na^+ + 2OH^- + 4H_3BO_3$$

3. **Heating Effect of Borax**: Borax on heating swells up as water molecules are eliminated. When it is further heated at a high temperature a glassy transparent solid mass is obtained.

$$\underset{\text{Prismatic Borax}}{Na_2B_4O_7.10H_2O} \xrightarrow[-10H_2O]{\Delta} \underset{\text{Anhydrous borax}}{Na_2B_4O_7}$$

$$Na_2B_4O_7 \xrightarrow[740°C]{\Delta} \underset{\text{Sodium metaborate (Glassy mass)}}{2NaBO_2} + B_2O_3$$

4. B_2O_3 (acidic, glass like) is used in borax bead test for detecting the presence of basic radicals like

$[CO^{+3}, Cu^{+2}, Cr^{+2}, Ni^{+2}, Fe^{+2}]$

Blue ↓ Green, brown ↓ yellow

$$B_2O_3 + \text{Pt-wire} + \text{Metal peroxide} \rightarrow \underset{\text{Coloured Bead}}{\text{Metaborates}}$$

$$CuSO_4 + B_2O_3 \rightarrow \underset{\text{Copper metaborate}}{CuO.B_2O_3} + SO_3$$

5. **With Ethyl Alcohol and conc. H_2SO_4**: When it is heated with ethyl alcohol and conc. H_2SO_4 volatile vapours of triethyl borate are produced which burn with a green edged flame (test of borate ions).

$$Na_2B_4O_7 + 11_2SO_4 + 511_2O \longrightarrow Na_2SO_4 + 4H_3BO_3$$

$$H_3BO_3 + 3C_2H_5OH \longrightarrow \underset{\text{Tri ethylborate}}{B(OC_2H_5)_3} + 3H_2O$$

Structure of Borax: It can be shown as | Na_2 |B_4O_7 $2H_2O$|. $8H_2O$ |. It has two tetrahedral and two triangular units attached to each other as shown in the Figure 5.5.

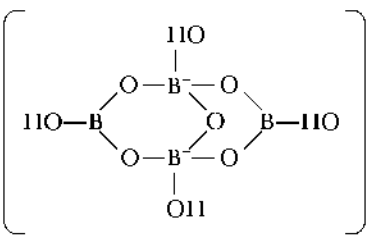


Fig. 5.5 St. of Borax

Uses

1. It is used in borax bead test for detecting basic radicals.
2. It can be used in the softening of water as it can form insoluble calcium and magnesium borates.
3. It is used in making optical and hard borosilicate glasses.
4. It is used to make enamels and glazes.
5. It can be used as a flux and as an antiseptic too.

COMPOUNDS OF CARBON

Carbon Monoxide : C ≺ O :

It is present in some quantity in volcanic gases, exhaust gases of internal combustion engines and chimney gases.

Preparation

It can be prepared as follows:

1. It is obtained by the incomplete combustion of carbon as follows:

$$\underset{\text{Less air}}{2C + O_2} \longrightarrow 2CO$$

2. **From HCOOH:** When formic acid or sodium formate is heated with concentrated H_2SO_4, pure carbon monoxide is formed as follows:

$$HCOOH \xrightarrow{\text{conc. } H_2SO_4} H_2O + CO\uparrow$$

$$2HCOONa + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O + 2CO$$

3. **From Oxalic Acid:** When oxalic acid is treated with concentrated sulphuric acid, carbon monoxide is formed as follows:

$$\begin{matrix} COOH \\ | \\ COOH \end{matrix} \xrightarrow{\text{conc. } H_2SO_4} CO\uparrow + CO_2\uparrow + H_2O$$

4. **From Potassium Ferrocyanide:** When potassium ferrocyanide is treated with sulphuric acid, carbon monoxide is formed as follows:

$$K_4Fe(CN)_6 + 6H_2SO_4 + 6H_2O \longrightarrow 2K_2SO_4 + FeSO_4 + 3(NH_4)_2SO_4 + 6CO$$

5. **By Passing Steam over Red Hot Coke:** When steam is passed over red hot coke, a mixture of CO and H_2 is formed.

$$C + H_2O \longrightarrow \underset{\text{Water gas}}{CO + H_2}$$

6. **By Heating Metal Carbonates:** Carbonates of Ca, Ba, Mg on heating with zinc gives CO as follows:

$$CaCO_3 + Zn \longrightarrow CaO + ZnO + CO$$

7. **By the Reduction of Carbon dioxide:** It can be reduced into CO by passing it over red hot zinc as follows:

$$Zn + CO_2 \xrightarrow{\text{heat}} ZnO + CO$$

8. **By the Reduction of Heavy Metal Oxide:** These oxides on heating with carbon give CO as follows:

$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$$

$$ZnO + C \longrightarrow Zn + CO$$

Physical Properties

1. It is a neutral, highly poisonous, colourless, odourless gas and burns with blue flame.

$$2CO + O_2 \longrightarrow 2CO_2 + \text{heat}$$

2. It is less soluble in water.
3. Its density is nearly equal to that of air.
4. It is a reducing agent in Mond's process (Ni).

5. It is a linear molecule in which carbon atom is sp hybridized.
6. When it mixes with heamoglobin of blood it forms carboxy-haemoglobin which destroys oxygen carrying capacity of blood as it can absorb oxygen.

$$\text{Haemoglobin} + CO \rightarrow \underset{\text{(stable)}}{\text{Carboxy-haemoglobin}}$$

■ Here, due to suffocation, death may occur.

Chemical Properties

1. **Combustion:** Carbon monoxide burns in air with blue flame with an exothermic reaction.

$$2CO + O_2 \longrightarrow 2CO_2$$

2. **Reducing Properties:** It is a strong reducing agent so it is widely used for the extraction of metals from their oxides.

For example,

$$Fe_2O_3 + 3CO \xrightarrow{600\text{–}900^\circ C} 2Fe + 2CO_2$$

$$ZnO + CO \rightarrow Zn + CO_2$$

$$2Cu(OH)_2 + CO \longrightarrow Cu_2O + 2H_2O + CO_2$$

$$I_2O_5 + 5CO \longrightarrow I_2 + 5CO_2$$

This iodine turns $CHCl_3$ and CCl_4 layer violet hence it is a test of CO.

3. **Reaction with Chlorine:** It combines with chlorine in presence of sunlight to form phosgene as follows:

$$CO + Cl_2 \rightarrow \underset{\text{Phosgene}}{COCl_2}$$

4. **Reaction with Metals:** Carbon monoxide reacts with metals to form metal carbonyls.

$$M + xCO \longrightarrow M[CO]_x$$

Here, M may be Ni, Cr, Mo, Fe, Co etc.

For example.,

$$Ni + 4CO \xrightarrow{325\text{–}345K} \underset{\text{Nickelcarbonyl}}{Ni(CO)_4}$$

$$Fe + 5CO \xrightarrow{473K} \underset{\text{Iron pentacarbonyl}}{Fe(CO)_5}$$

5. **Reaction with Dihydrogen:** When it reacts with dihydrogen at 425–675K in presence of a catalyst, methanol is formed as follows:

$$CO + 2H_2 \xrightarrow[ZnO + C]{425\text{–}675K} \underset{\text{Methanol}}{CH_3OH}$$

Uses

1. It is used as a reducing agent and in the preparation of metal carbonyls.
2. It is used in war gas preparation (phosgene)
3. It is used in the refining of nickel in Mond's process
4. It is used in the preparation of methanol, phosgene, synthetic petrol etc.
5. It is used in gaseous fuel as water and producer gas.

REMEMBER

■ Carbogen (mixture of O_2 + 5–10% CO_2) is used for artificial respiration for victims of CO poisoning.

Structure: The structure of CO is represented as follows:

$$:C^{-}\equiv O^{+} \quad \text{or} \quad :C^{-}\equiv O^{+} \quad \text{or} \quad :C \equiv O:$$

Here, carbon atom is sp hybridized and the carbon–oxygen bond length is 1.13 Å. It has lone pair of electrons on carbon atom hence it can act as a ligand and can form a coordinate bond with metal atoms during complex formation.

Carbon Dioxide O = C = O or CO_2

It is present 0.03–0.05 % by volume in air. It is a linear nonpolar molecule having zero value of dipole moment.

Preparation

It is prepared by following methods:

1. **From Carbonates and Bicarbonates:** Metal carbonates and bicarbonates on heating give carbon dioxide as follows:

$$MCO_3 \xrightarrow{\Delta} CO_2 \uparrow + MO$$

For example,

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

$$MHCO_3 \xrightarrow{\Delta} MCO_3 + CO_2 \uparrow + H_2O$$

For example,

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaO + 2CO_2 + H_2O$$

2. **Lab Method:** When metal carbonates or bicarbonates are treated with mineral acids, CO_2 is formed as follows:

$$MCO_3 + HCl \longrightarrow CO_2 + MCl + H_2O$$

$$MHCO_3 + HCl \longrightarrow MCl + H_2O + CO_2$$

For example,

When $CaCO_3$ is treated with dilute HCl, carbon dioxide is formed.

$$CaCO_3 + 2HCl \xrightarrow{\Delta} CaCl_2 + CO_2 + H_2O$$

3. **By Alcoholic Fermentation:** During the fermentation of molasses into alcohol, CO_2 is also formed in good amount.

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{Enzyme}} \underset{\text{Ethanol}}{2C_2H_5OH} + \underset{\text{byproduct}}{2CO_2}$$

4. **By the Complete combustion of Carbon:**

$$C + O_2 \longrightarrow CO_2$$

$$2C + O_2 \longrightarrow 2CO$$

$$2CO + O_2 \longrightarrow 2CO_2$$

5. **From Fuel Gases:** Fuel gases produced during the combustion of coal or coke have a good amount of CO_2 mixed with N_2, O_2 and CO. On passing the gaseous mixture through K_2CO_3, CO_2 is absorbed and potassium bicarbonate is formed which on heating gives CO_2 as follows:

$$K_2CO_3 + CO_2 + H_2O \longrightarrow 2KHCO_3$$

$$2KHCO_3 \longrightarrow K_2CO_3 + H_2O + CO_2$$

Physiochemical Properties

1. It is a colourless, odourless, tasteless gas which is heavier than air.
2. It is partially soluble in water under normal pressure but more soluble at high pressure.
3. It is nonpoisonous but it does not support life.
4. CO_2 is acidic in nature and called carbonic anhydride.

$$H_2O + CO_2 \longrightarrow H_2CO_3$$

5. **Dry Ice:** CO_2 gas can be easily liquefied into a mobile liquid which can be further changed into a white solid mass of CO_2. Solid carbon dioxide is known as dry ice since it does not wet the surface on which it melts. It is used as a coolant for preserving fresh articles in food industry and for making cold baths in the laboratory. Under the name dry cold it is used as a refrigerant. It can provide not only cold but also an inert atmosphere which can help in the killing of unwanted bacteria.

Table 5.1 Comparision of Properties of CO and CO_2

Property	CO	CO_2
B.P. (K)	81.5	194.5
C–O Bond length (pm)	112.8	116.3
Density (g/l) at 273 K	1.250	1.977
ΔH_f (kJ mol^{-1})	110.5	−393.5
M.P. (K)	68	216.4 (at 5.2 atm)
Structure	:C ≡ O:	O = C = O

Chemical Properties

1. **NonCombustible Nature and Nonsupporter of Combustion:** It is neither combustible nor supports combustion, however active metals like Na, K, Mg etc. continue to burn in the atmosphere of CO_2.

$$CO_2 + 4Na \longrightarrow 2Na_2O + C$$

$$CO_2 + 2Mg \longrightarrow 2MgO + C$$

CO_2 can be used as a fire extinguisher except in case of active metals like Na, Mg etc.

2. **Reduction:** When carbon dioxide is passed over red hot coke, it gets reduced to carbon monoxide.

$$\underset{\text{Red hot coke}}{CO_2 + C} \longrightarrow 2CO$$

3. **Acidic Nature:** It is an acidic oxide as it forms carbonic acid on dissolving in water and can form salts with bases as follows:

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

$$Na_2CO_3 + CO_2 + H_2O \longrightarrow 2NaHCO_3$$

4. **Action on Lime Water:** CO_2 turns limewater into milky in soluble calcium carbonate.

$$\underset{\text{Milky}}{Ca(OH)_2} + CO_2 \longrightarrow CaCO_3\downarrow + H_2O$$

This milkiness disappears when more CO_2 is passed as soluble calcium bicarbonate is formed.

$$CaCO_3 + H_2O + CO_2 \longrightarrow \underset{\text{Soluble}}{Ca(HCO_3)_2}$$

$$Ca(HCO_3)_2 \longrightarrow CaCO_3 + CO_2 + H_2O$$

5. **Formation of Carbonates:** It reacts with some basic oxides like Na_2O, K_2O to form their carbonates.

$$K_2O + CO_2 \longrightarrow K_2CO_3$$

$$Na_2O + CO_2 \longrightarrow Na_2CO_3$$

6. **Photosynthesis:** Here, CO_2 is changed into carbohydrates (glucose) etc. by plants in presence of sunlight and chlorophyll

$$6CO_2 + 6H_2O \xrightarrow[\text{Sun light}]{\text{Chlorophyll}} C_6H_{12}O_6 + 6O_2$$

$$6xCO_2 + 5xH_2O \longrightarrow (C_6H_{10}O_5)x + 6xO_2$$

Uses

1. It is used in cold drinks and aerated water.
2. CO_2 (15%) is used as fire extinguisher (except Mg-fire).
3. O_2 + 5–10% CO_2 is carbogen which is used for artificial respiration in case of CO poisoning and pneumonia patients.
4. It is used in the manufacture of sodium carbonate by solvay method.
5. Dry fire extinguisher is $SiO_2 + NaHCO_3$.
6. Foamite fire extinguisher is $NaHCO_3 + Al_2(SO_4)_3$.

Structure

It is a linear molecule with zero dipole moment. Here C–O bond length is 1.15 Å (less than C=O bond) as it is resonance hybrid of following structures.

$$O{=}C{=}O \longleftrightarrow O{\equiv}C{-}O \longleftrightarrow O{-}C{\equiv}O$$

- Carbon can also form some other oxides like C_3O_2, C_5O_2 etc., which are less stable. Some graphite oxides like C_2O and C_4O_3 are also formed which are very unstable.

Carbon Suboxide (C_3O_2) (O=C=C=C=O)

It is obtained by the dehydration of malonic acid.

$$CH_2\begin{matrix}COOH\\COOH\end{matrix} \xrightarrow{\Delta,\ P_4O_{10}} O=C=C=C=O + 2H_2O$$

Some of the chemical reactions of C_3O_2 are as follows:

$$C_3O_2 + 2H_2O \longrightarrow CH_2\begin{matrix}COOH\\COOH\end{matrix}$$
Malonic acid

$$C_3O_2 + 2HCl \longrightarrow CH_2\begin{matrix}COCl\\COCl\end{matrix}$$
Malonyl chloride

$$C_3O_2 + 2NH_3 \longrightarrow CH_2\begin{matrix}CONH_2\\CONH_2\end{matrix}$$
Malonyl amide

COMPOUNDS OF SILICON

Silicones

These are organosilicon polymeric compounds having a general formula $(R_2SiO)_n$. They have Si–O–Si linkages.

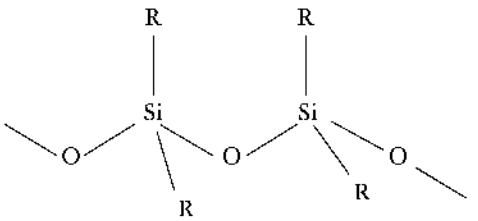


Fig. 5.6

Preparation

The alkyl or aryl substituted chlorosilanes are prepared by the reaction of Grignard reagent and silicon tetrachloride.

$$RMgCl + SiCl_4 \rightarrow RSiCl_3 + MgCl_2$$

Grignard reagent

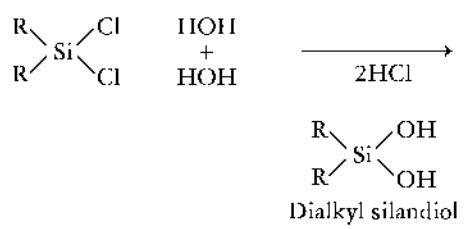
$$2RMgCl + SiCl_4 \longrightarrow R_2SiCl_2 + 2MgCl_2$$

$$2RCl + Si \xrightarrow{Cu,\ 570K} R_2SiCl_2$$

$$3RMgCl + SiCl_4 \longrightarrow R_3SiCl + 3MgCl_2$$

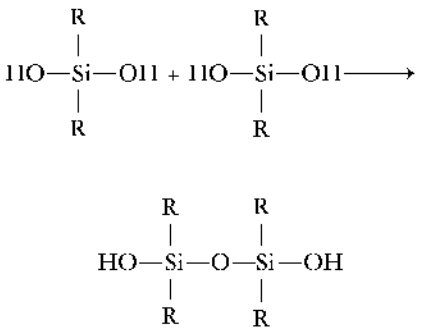
Here, R may be $-CH_3$, $-C_2H_5$ or $-C_6H_5$ etc.,

Dialkyl dichloro silanes on hydrolysis gives silanols as follows:



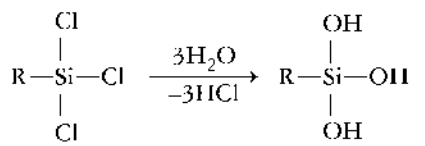
Dialkyl silandiol

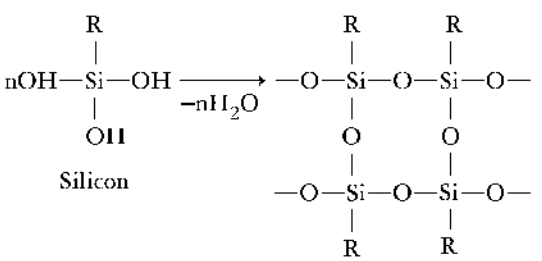
Polymerization of dialkyl silandiol yields linear thermoplastic polymer.



Polymerization continues on both the ends and thus chain increases in length.

$RSiCl_3$ on hydrolysis gives a cross linked silicone. The formation can be explained in three steps:





Silicon

Cyclic (ring) silicones are formed when water is eliminated from the terminal –OH groups of linear silicones.

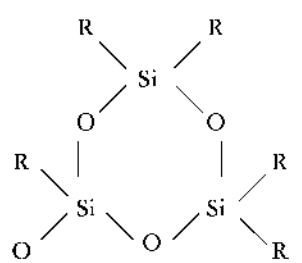


Fig. 5.7 Cyclic silicone

R_3SiCl on hydrolysis forms only a dimer.

$R_3SiOH + OHSiR_3 \rightarrow R_3Si{-}O{-}SiR_3$

Properties

1. The lower silicones are oily liquids, however higher silicons having long chains or ring structures are waxy and rubber like solids.
2. These are nontoxic in nature.
3. These have very high thermal stability.
4. Some low molecular weight silicones dissolve in organic solvents like benzene, ether etc.
5. These are chemically inert, water repellent, insulators, lubricants and antifoaming agents etc.
6. Viscosity of silicone oils remains the same at different temperatures.

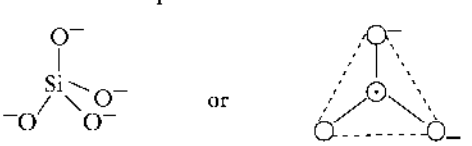
Uses

1. Silicones oils are used for high temperature oil baths, high vacuum pumps and low temperature lubrication.
2. These are mixed with paints and enamels to make them resistant to the effects of high temperature , sunlight, chemicals and damp.
3. These are used for making water proof papers, all weather lubricants etc.
4. They are used as insulating material for electric motors etc.
5. These are used in making vaseline like greases which can be used as lubricants in aeroplanes.
6. Silicone-rubbers are useful as they retain their elasticity over a range of temperatures.
7. Silicones, resins, silicon fluids and silicon rubbers are also widely used.

Silicates

These are alumino silicates of magnesium, calcium etc. For example, Quartz, feldspar, mica and zeolites. These are present in almost all rocks, clays etc. Glass and cement are man made and highly useful silicates.

The silicates are complex network solids having the basic structural unit as silicate ion $(SiO_4)^{4-}$ which is tetrahedral in shape.



On the basis of the manner in which different $(SiO_4)^{4-}$ units are linked together silicates are divided into following types:

(i) ***Orthosilicates:*** These are simple silicates and in such silicates discrete SiO_4^{4-} tetrahedral are present and there is no sharing of oxygen atoms between adjacent tetrahedral.

For example, Zircon $(ZrSiO_4)$
Forsterite or Olivine (Mg_2SiO_4)
Willemite (Zn_2SiO_4)
Phenacite (Be_2SiO_4) etc.

(ii) ***Pyrosilicates or Islands:*** In such silicates the two tetrahedral units share one oxygen atom (corner) between them containing basic unit of $Si_2O_7^{6-}$ anion. Here, each silica atom is surrounded by 3.5 oxygen atoms.

For example, Thortveitite $Sc_2(Si_2O_7)$, Hemimorphite $Zn_4(OH)_2Si_2O_7.H_2O$ etc.

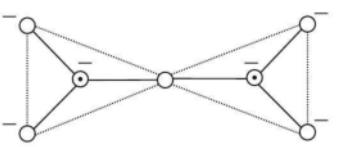
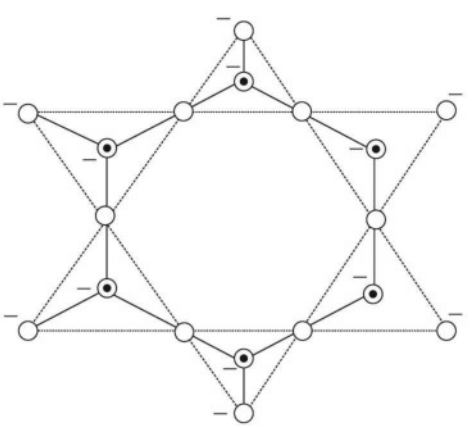


Fig. 5.8 St. of Pyrosilicates

(iii) ***Cyclic or Ring Silicates:*** In such silicates the two tetrahedral units share two oxygen atoms (two corners) per tetrahedron to form a close ring containing basic unit of $(SiO_3)_2^{2n-}$ or $(SiO_3^{2-})_n$.

For example, wollastonite $Ca_3(Si_3O_9)$
Benitoite $BaTi(Si_3O_9)$
Beryl $Be_3Al_2(Si_6O_{18})$
Catapleiite $Na_2ZrSi_3O_9.2H_2O$ etc.



or

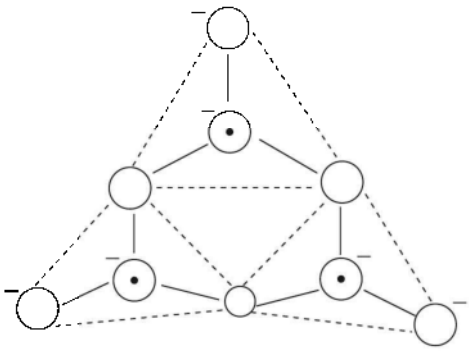


Fig. 5.9 St. of Ring Silicates

(iv) ***Chain Silicates:*** The sharing of two oxygen atoms (two corners) per tetrahedron leads to the formation of a long chain. Anions of these silicates have two general formulas $(SiO_3)_n^{2n-}$ and $(Si_4O_{11})_n^{6n-}$.

For example, pyroxenes and asbestos

Jadeire $NaAl(SiO_3)_2$

Spodumene $LiAl(SiO_3)_2$

Diopside $CaMg(SiO_3)_2$

Tremolire $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$

Enstarire $MgSiO_3$ etc.

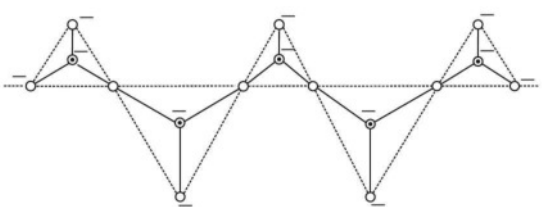


Fig. 5.10 St. of Chain Silicates

(v) ***Two Dimensional Sheet Silicates:*** In such silicates, sharing of three oxygen atoms (three corners) by each tetrahedron unit results in an infinite two dimensional sheet of primary unit $(Si_2O_5)_n^{2n-}$. The sheets are held together by electrostatic force of the cations that lie between them.

Micas: Example, Muscovite: $KAl_2(OH)_2(Si_3AlO_{10})$
Clays: Example, Talc: $Mg_3(OH)_2(Si_4O_{10})$
Kaolin: Example, $Al_2(OH)_4(Si_2O_5)$ etc.

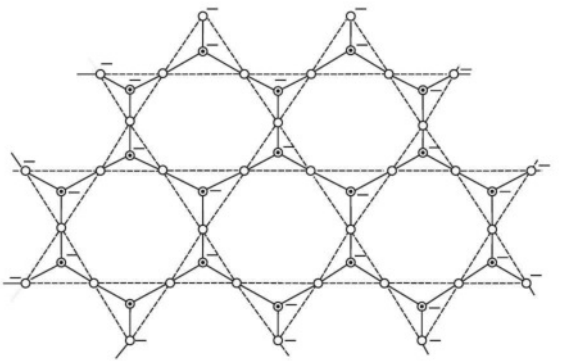


Fig. 5.11 St.of Two Dimensional Sheet Silicates

(vi) ***Three Dimensional or Frame Work Silicates:*** In such silicates all the four oxygen atoms (four corners) of SiO_4^{4-} tetrahedral are shared with other tetrahedral, resulting in a three dimensional network with the general formula $(SiO_2)n$.

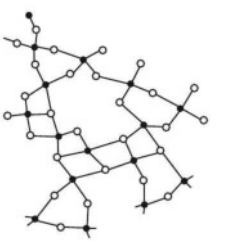


Fig. 5.12 St. of 3-dimenensional Sheet Silicates

For example,

Zeolite: These are hydrated aluminosilicates having a three dimensional structure (honey comb like) with a general formula

$$M_x^{+n}[(AlO_2)_x(SiO_2)_y]_n^{x-}.zH_2O.$$

Here, **M** – Na, K, Ca, and

'n' represents the charge on the metal ion and 'z' is the number of moles of water of hydration. Some important examples of zeolites are as follows:

Erionite: $Na_2K_2CaMg(AlO_2)_2(SiO_2)_6.6H_2O$

Gemelinite: $Na_2Ca(AlO_2)_2(SiO_2)_4 . 6H_2O$

Chabazite: $Na_2Ca(AlO_2)_2(SiO_2)_4.(6H_2O)$

Zeolites are characterized by their open structure which consists of channels and cavities of different sizes ranging from 200 1100 pm in diameter. This open structure enables them to trap water molecules

and other small molecules such as NH_3, CO_2, C_2H_5OH etc., but not the bigger molecules. Thus, zeolites act as molecular sieves.

In zeolites, cations are free to move through the open pore structure and can be exchanged reversibly with other cations. Therefore, they act as ion-exchangers.

Zeolites act as shape-selective catalysts due to presence of cavities. They are widely used ion-exchangers in water softening operations.

Feldspar: These are aluminosilicates of two types:

(i) Orthoclase feldspar: Example, orthoclase $KAlSi_3O_8$

(ii) Plagioclase feldspar: Example, Albite $NaAlSi_3O_8$

Ultramarines: These are also aluminosilicates, but do not contain water of hydration. Most of the ultra marines are coloured and are used as pigments and in calico printing.

For example, Sodalite $Na_8(AlO_2)_6(SiO_2)_6Cl_2$

REMEMBER

- SiO_2 is silica gel (a semipermeable membrane).
- Being adsorbent, it is used to remove sulphur.
- Na_2SiO_3 is used a filter in saponification to remove excess of water.

Carborundum (Silicon Carbide) (SiC)

Preparation

It can be obtained by heating a mixture of sand, carbon, common salt and saw dust in an electric furnace.

$$SiO_2 + 3C \longrightarrow SiC + 2CO$$

Here, salt added acts as flux and saw dust makes the mass porous while two carbon rods connected by a thin carbon core act as electrodes in the furnace. Carborundum (SiC) is obtained round the central core of carbon. It is crushed, washed with H_2SO_4, NaOH, H_2O and finally dried.

Physiochemical Properties

1. In pure form it is colourless but in the commercial samples it is yellow, green or blue coloured.
2. It is a very hard mass but less harder than diamond.
3. It is chemically inert hence it resists the attack of almost all the reagents except fused caustic soda.
4. It dissolves in the fused alkali in presence of air to give sodium silicate as follows:

$$4NaOH + SiC + 2O_2 \longrightarrow Na_2SiO_3 + Na_2CO_3 + 2H_2O$$

Uses

It is used as an abrasive in place of emery. It is used in the manufacturing of grind-stones, knife sharpeners etc.

COMPOUNDS OF NITROGEN

Oxides of Nitrogen

On combining with oxygen, nitrogen forms five oxides For example, nitrous oxide (N_2O), nitric oxide (NO), nitrogen trioxide (N_2O_3), nitrogen dioxide (NO_2) or dinitrogen tetraoxide (N_2O_4) and dinitrogen penta oxide (N_2O_5). These are given as follows:

N_2O	NO	N_2O_3	N_2O	N_2O_5
Colourless Gas	Colourless gas	Bluish liquid	Brown gas	Colourless solid

Acidic nature increases →

Preparation

These oxides are prepared as follows:

(i) ***Preparation of N_2O***

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$$

$$2NO + H_2O + Fe \xrightarrow{\Delta} N_2O + Fe(OH)_2$$

(ii) ***Preparation of NO***

$$N_2 + O_2 \xleftrightarrow{3000^\circ C} 2NO$$

$$3Cu + 8HNO_3 \xrightarrow{\text{dil. \& cold}} 3Cu(NO_3)_2 + 2NO + 4H_2O$$

$$FeSO_4 + NO \longrightarrow \underset{\text{Dark brown}}{FeSO_4.NO} \xrightarrow{\Delta} FeSO_4 + \underset{\text{Pure gas}}{NO}$$

$$[Fe(H_2O)_6]SO_4 + NO \longrightarrow [Fe(H_2O)_5NO]SO_4$$

$$\xrightarrow{\Delta} FeSO_4 + NO + 5H_2O$$

(iii) *Preparation of N_2O_3*

$$NO + NO_2 \xrightarrow{-20^\circ C} \underset{\text{Pale blue solid}}{N_2O_3}$$

(iv) *Preparation of NO_2*

$$Cu + 4HNO_3 \xrightarrow{\text{Conc. \& cold}} Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

$$2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$$

$$2NO + O_2 \longrightarrow 2NO_2$$

(v) *Preparation of N_2O_5*

$$4HNO_3 + P_4O_{10} \longrightarrow 2N_2O_5 + 4HPO_3$$

$$N_2O_4 + O_3 \longrightarrow N_2O_5 + O_2$$

$$4AgNO_3 + 2Cl_2 \xrightarrow{\text{Heat}} 4AgCl + 2N_2O_5 + O_2$$

Shapes of Nitrogen Oxides

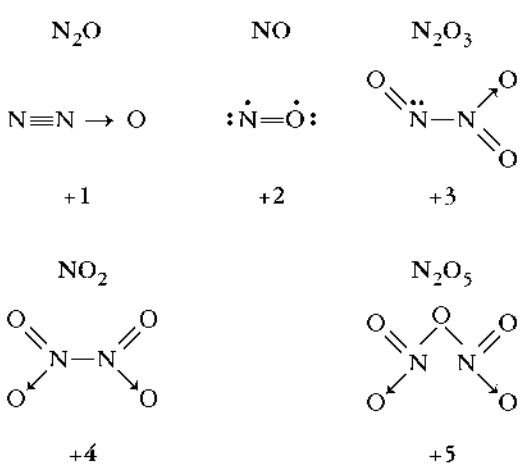
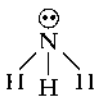


Fig 5.13 Shapes of N-oxides

Some Facts about Nitrogen Oxides

- N_2O is called laughing gas and used as an anesthetic (mixture of $N_2O + O_2$).
- NO is an odd electron molecule having one odd electron i.e., it is paramagnetic in nature.
- NO combines with Fe (II) to form a brown nitrosyl complex of iron (I) i.e., $[Fe(H_2O)_5NO]^{2+}$. It shows reducing nature of NO (detection test of nitrite and nitrate).
- NO_2 is paramagnetic and brown coloured while N_2O_4 is diamagnetic and colourless.
- According to X-ray diffraction the ionic structure of N_2O_5 is $NO_2^+ NO_3^-$ that is why it is called nitronium nitrate.

Ammonia (NH_3)



It was isolated by Priestley. Berthelot suggested that it is a compound of nitrogen and hydrogen and Davy introduced its molecular formula. It is the most important compound of nitrogen which is also used as a starting material for the preparation of some other useful compounds.

Occurrence: It is present in traces in atmosphere.

Preparation

1. **From Ammonium Salts:** Ammonium salts on heating with metal oxide or hydroxide gives ammonia.

$$\text{Ammonium Salt} + \underset{\text{or Hydroxide}}{\text{Metal oxide}} \xrightarrow{\Delta} NH_3$$

For example,

$$2NH_4Cl + Ca(OH)_2 \xrightarrow{\Delta} 2NH_3 + 2H_2O + CaCl_2$$

It is laboratory method for the preparation of ammonia.

2. **From Nitrides:** Nitrides on hydrolysis give ammonia.

$$\text{Any Nitride} + H_2O \longrightarrow NH_3$$

For example,

$$AlN + 3H_2O \longrightarrow Al(OH)_3 + NH_3$$

$$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$$

3. **Cyanamide Process:** Here ammonia is formed when coal and lime are heated at 1000°C and nitrogen is passed as follows:

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2.$$

$$CaO + 3C \xrightarrow{\Delta} CaC_2 + CO$$

$$CaC_2 + N_2 \xrightarrow{\Delta} \underset{\text{Nitrolim}}{CaCN_2} + C \text{ (Graphite)}$$

$$CaCN_2 + 3H_2O \xrightarrow{453\ k} CaCO_3 + 2NH_3$$

- Here anhydrous calcium chloride or CaF_2 is used as a catalyst.

3. **Haber's Process**: When nitrogen and hydrogen in 1:3 ratio are heated at 400–500°C, a pressure of 200 atmosphere and in presence of a catalyst (finely divided Fe filling and molybdenum (promoter)), ammonia is formed as follows:

$$\underset{1:3}{N_2 + 3H_2} \xrightarrow[\text{200–900 atm 'P'}]{\text{Fe/Mo, 500°C}} 2NH_3 + 22400 \text{ cal.}$$

This is an exothermic and reversible reaction.

Favourable Conditions for the formation of Ammonia

Low temperature, high pressure, high concentration of N_2 and H_2 are favourable conditions for the formation of ammonia

(i) ***Low Temperature:*** As the reaction is exothermic hence low temperature is favourable for the formation of ammonia. It is in between 400–500°C.

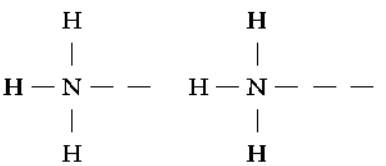
(ii) ***High Pressure:*** As the reaction proceeds with a decrease in volume ($\Delta_n = -2$) hence high pressure is favourable for the formation of more ammonia. It is 200–900 atmosphere.

(iii) ***More Concentration of N_2 and H_2:*** As these are reactants hence more concentration of the reactants favours reaction in forward direction i.e. formation of more ammonia.

(iv) ***Continuous Removal of Ammonia:*** It also favours its formation.

Physical Properties

1. It is a colourless gas having a pungent smell which causes tears in eyes.
2. It is lighter than air hence can be collected by downward displacement of air.
3. Liquid NH_3 is a polar solvent.
4. On cooling it is easily liquefied under pressure and finally freezes into a white snowy crystal.
5. Reducing nature of IA metals can be increased by dissolving them in liquid NH_3 as solvated electrons are formed.
6. NH_3 is highly soluble in water so collected over CaO or Hg. Its high solubility in water is because of hydrogen bonding and it is directly proportional to pressure while inversely proportional to temperature.
7. Ammonia molecules can link together to form associated molecules by hydrogen bonding.



8. NH_3 has high vapour density hence its bottle is not completed filled and cooled before opening to avoid any accident.
9. NH_3 cannot be dried with concentrated H_2SO_4, anhydrous $CaCl_2$, P_2O_5 as it reacts with them.
10. NH_3 has high latent heat of vapourization (Jgm^{-1}), hence it is used in refrigeration.

Chemical Properties

1. **Decomposition**: It is quite stable and can decomposed only at red hot temperature or by electric sparks as follows:

$$NH_3 \xrightarrow{\Delta \text{ Air}} N_2 + H_2O$$

$$NH_3 \xrightarrow{> 500^\circ} N_2 + H_2O$$

2. **Combustion**: It is neither combustionable nor help in combustion, however it burns in an atmosphere of O_2 or air as follows:

$$4NH_3 + 3O_2 \longrightarrow 2N_2 + 6H_2O$$

3. **Oxidation**: On passing over heated copper oxide, it gets oxidized to nitrogen and water.

$$3CuO + 2NH_3 \longrightarrow 3Cu + N_2 + 3H_2O$$

Ammonia is also oxidized by passing it along with O_2 over red hot platinum gauge at 800°C.

$$4NH_3 + 5O_2 \xrightarrow[800^\circ C]{\text{Pt gauge}} 4NO + 6H_2O + \text{Heat}$$

4. **Basic Nature:** Aqueous solution of ammonia is weakly basic in nature (Bronsted base) as following equilibrium is achieved.

$$NH_3(aq) + H_2O \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

Being a weak bronsted base it can turn red litmus blue, phenolphthalein pink and it can react with acids to form salts as follows:

$$NH_3 + HCl \longrightarrow \underset{\text{(White fumes)}}{NH_4Cl}$$

$$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

5. **Formation of Complex Ions or Lewis Basic Nature:** In ammonia, nitrogen atom has one lone pair of electrons hence it can behave like a Lewis base and can form complexes by donating this lone pair to transition metal cations for coordinate bond formation.

$$Cu^{2+} + 4NH_3 \longrightarrow \underset{\text{Tetraamine copper (II) ion}}{[Cu(NH_3)_4]^{2+}}$$

$$Co^{2+} + 6NH_3 \longrightarrow \underset{\text{Hexaamine cobalt (II) ion}}{[Co(NH_3)_6]^{2+}}$$

$$Ag^+ + 2NH_3 \longrightarrow \underset{\text{Diammine silver (I) ion}}{[Ag(NH_3)_2]^+}$$

6. **With Halogens:** Chlorine or bromine can easily oxidized ammonia into nitrogen as follows:

$$8NH_3 + 3X_2 \longrightarrow \underset{\text{Excess}}{6NH_4X} + N_2\uparrow$$

For example,

$$8NH_3 + 3Cl_2 \longrightarrow 6NH_4Cl + N_2$$

$$8NH_3 + 3Br_2 \longrightarrow 6NH_4Cl + N_2$$

If halogen is in excess, reaction is as follows:

$$NH_3 \xrightarrow{3X_2} NX_3 + 3HX$$

For example,

$$NH_3 + \underset{\text{Excess}}{3Cl_2} \longrightarrow \underset{\text{Explosive}}{NCl_3} + 3HCl$$

$$NH_3 + \underset{\text{Excess}}{3Br_2} \longrightarrow NBr_3 + 3HBr$$

When iodine flakes are rubbed with liquor ammonia, a dark brown precipitate of ammoniated nitrogen iodide is formed as follows:

$$2NH_3 + 3I_2 \longrightarrow \underset{\substack{\text{Ammoniated}\\ \text{nitrogen iodide}}}{NH_3 . NI_3} + 3HI$$

7. **Reducing Nature:** Ammonia can also act as a reducing agent as it can be easily oxidized by hypochlorites or bleaching powder into nitrogen and it reduces them.

$$2NH_3 + 3NaClO \longrightarrow N_2 + 3NaCl + 3H_2O$$

$$2NH_3 + 3CaOCl_2 \longrightarrow N_2 + 3CaCl_2 + 3H_2O$$

8. **Formation of Amide:** On passing dry ammonia over heated sodium or potassium metal amides are formed.

$$2Na + 2NH_3 \longrightarrow \underset{\text{Sodamide}}{2NaNH_2} + H_2$$

$$2K + 2NH_3 \longrightarrow \underset{\text{Potassium amide}}{2KNH_2} + H_2$$

9. **Precipitatation of heavy metal ions as hydroxides using aqueous ammonia:** Aqueous ammonia or ammonium hydroxide can be used to precipitate many heavy metal ions like Fe^{3+}, Ar^{3-}, Cr^3, Cu^{2+}, Zn^{2+} etc, from their aqueous salt solutions as follows:

$$FeCl_3 + 3NH_4OH \longrightarrow \underset{\substack{\text{Ferric hydroxide}\\ \text{(Brown)}}}{Fe(OH)_3\downarrow} + 3NH_4Cl$$

$$AlCl_3 + 3NH_4OH \longrightarrow \underset{\substack{\text{Aluminium hydroxide}\\ \text{(White)}}}{Al(OH)_3\downarrow} + 3NH_4Cl$$

$$CrCl_3 + 3NH_4OH \longrightarrow \underset{\substack{\text{Chromium hydroxide}\\ \text{(Green)}}}{Cr(OH)_3\downarrow} + 3NH_4Cl$$

$$CuSO_4 + 2NH_4OH \longrightarrow \underset{\substack{\text{Cupric hydroxide}\\ \text{(Blue)}}}{Cu(OH)_2\downarrow} + (NH_4)_2SO_4$$

$$ZnSO_4 + 2NH_4OH \longrightarrow \underset{\substack{\text{Zinc hydroxide}\\ \text{(White)}}}{Zn(OH)_2\downarrow} + (NH_4)_2SO_4$$

10. **Formation of Complexes using Aqueous Ammonia:** Some salts dissolve in excess of aqueous ammonia (bronsted base) and form soluble complexes.

For example,

$$2NH_4OH \xrightarrow{AgCl} \underset{\text{Diamine silver chloride}}{Ag(NH_3)_2Cl} + H_2O$$

$$2NH_4OH \xrightarrow{HgCl_2} Hg(NH_2)Cl + H_2O + NH_4Cl$$
Mercuric aminochloride (White ppt.)

$$2NH_4OH \xrightarrow{Hg_2Cl_2} Hg + Hg(NH_2)Cl + NH_4Cl + H_2O$$
Mercury mercuric amino chloride (Black ppt.)

$$4NH_4OH \xrightarrow{CuSO_4} Cu(NH_3)_4SO_4 + 4H_2O$$
Tetraamine copper sulphate (Deep blue colour)

$$4NH_4OH \xrightarrow{ZnCl_2} Zn(NH_3)_4Cl_2 + 4H_2O$$

$$4NH_4OH \xrightarrow{ZnSO_4} Zn(NH_3)_4SO_4 + 4H_2O$$
Tetraamine zinc sulphate (water soluble compound)

11. **With Nesseler's Reagent:** Here a red brown precipitate of millon's base of iodide is formed when ammonia is treated with alkaline solution of K_2HgI_4 (Nesseler's reagent) as follows:

$$2K_2HgI_4 + NH_3 + 3KOH \rightarrow H_2N.HgO.HgI\downarrow + 7KI + 2H_2O$$
Brown ppt.

Uses

1. It is used in the preparation of nitric acid and other nitrogen compounds.
2. It is widely used in the preparation of nitrogenous fertilizers e.g., urea, ammonium nitrate, ammonium phosphate, ammonium sulphate etc.
3. It is used for making artificial silk and as a cleansing agent for removing grease in dry cleaning.
4. It is used in the preparation of sodium bicarbonate by Solvay's process.
5. Liquid ammonia is used as a refrigerant in ice plants.

Nitrous Acid (HNO_2)

Preparation

1. **From Barium Nitrate:** By adding calculated amount of ice, cold sulphuric acid to a well cooled solution of barium nitrate solution, nitrous acid is formed as follows:

$$Ba(NO_2)_2 + H_2SO_4 \rightarrow BaSO_4 + 2HNO_2$$

2. **By the action of mineral acids on nitrites:**

$$NaNO_2 + HCl \longrightarrow NaCl + HNO_2$$

$$2KNO_2 + H_2SO_4 \rightarrow K_2SO_4 + 2HNO_2$$

3. **By the oxidation of ammonia with H_2O_2:**

$$NH_3 + 2H_2O_2 \longrightarrow HNO_2 + 4H_2O$$

Physiochemical Properties

1. It has a slightly bluish colour in solution which is believed to be due to the anhydride N_2O_3.
2. It is a weak acid ($Ka = 4.5 \times 10^{-3}$) and reacts with alkalies to form salts.

$$NaOH + HNO_2 \rightarrow NaNO_2 + H_2O$$

3. **Decomposition:** It is very unstable and undergoes autooxidation even on standing. On boiling it decomposes rapidly giving acid.

$$3HNO_2 \xrightarrow{Boil} HNO_3 + H_2O + 2NO$$
Brown fumes

4. **Oxidizing Property:** It acts as an oxidizing agent.

$$H_2S + 2HNO_2 \rightarrow S + 2NO + 2H_2O$$

$$SO_2 + 2HNO_2 \rightarrow H_2SO_4 + 2NO$$

$$KI + 2HNO_2 \longrightarrow 2KOH + 2NO + I_2$$

$$2FeSO_4 + H_2SO_4 + 2HNO_2 \longrightarrow Fe_2(SO_4)_3 + 2NO + 2H_2O$$

$$SnCl_2 + 2HCl + 2HNO_2 \rightarrow SnCl_4 + 2NO + 2H_2$$

5. **Reducing Property:** It acts as a reducing agent towards strong oxidizing agent.

$$2KMnO_4 + 3H_2SO_4 + 5HNO_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5HNO_2$$

$$K_2Cr_2O_7 + 4H_2SO_4 + 3HNO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3HNO_2$$

$$Br_2 + H_2O + HNO_2 \longrightarrow HNO_3 + 2HBr$$

6. **With Ammonia:** It reacts with ammonia gives nitrogen and water.

$$NH_3 + HNO_2 \rightarrow [NH_4NO_2] \rightarrow N_2 + 2H_2O$$

7. It decomposes urea and other aliphatic amino compounds to give nitrogen.

$$\underset{\text{Urea}}{NH_2.CO.NH_2} + 2HNO_2 \longrightarrow CO_2 + 2N_2 + 3H_2O$$

$$\underset{\text{Ethyl amine}}{C_2H_5NH_2} + HNO_2 \longrightarrow C_2H_5OH + N_2 + H_2O$$

8. **Diazotization:** It reacts with aromatic amines to give benzene diazonium chloride.

$$\underset{\text{Aniline hydrochloride}}{C_6H_5NH_2HCl} + HNO_2 \longrightarrow \underset{\text{Benzene diazonium chloride}}{C_6H_5N{=}NCl} + 2H_2O$$

9. **With Sulphuric Acid**

$$2HNO_2 + H_2SO_4 \longrightarrow 2H_2O + SO_2 + 2NO_2$$

Uses

1. It is used as an oxidizing and reducing agent.
2. It is used in the preparation of diazo compounds which are used for making aniline dyes.
3. It is used for the replacement of NH_2 group by one group in aliphatic amines.

Structure

Since nitrous acid forms two series of organic derivatives the nitrites (R-ONO) and nitro compounds (R-NO_2) it is therefore considered to be a tautomeric mixture of two forms.

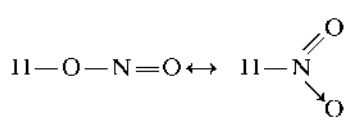


Fig. 5.14 St. of HNO_2

Nitric Acid (HNO_3)

It is the most important oxyacids of nitrogen which is widely used in chemical industries for a number of purposes. It is called aqua fortis (strong water) as it reacts with nearly all metals.

Preparation

1. **Retort Method or Laboratory Method:** In laboratory, it is prepared by heating a mixture of MNO_3 (M = Na, K) and concentrated sulphuric acid in a glass retort as follows:

$$MNO_3 + \underset{\text{Conc.}}{H_2SO_4} \xrightarrow{\Delta} MHSO_4 + HNO_3$$

For example,

$$KNO_3 + \underset{\text{Conc.}}{H_2SO_4} \xrightarrow{\Delta} KHSO_4 + HNO_3$$

The vapours of nitric acid obtained from here are condensed in a glass receiver into liquid form of nitric acid. It has impurities of nitrogen oxides which can be removed by further distillation or by blowing a current of CO_2 through this acid in warm state.

2. **Birkeland–Eyde Process or Arc Method:** In this process first of all nitrogen and oxygen are treated to get nitric oxide at 300°C as follows:

$$N_2 + O_2 \underset{300°C}{\longleftrightarrow} 2NO - \text{energy}$$

According to Le-Chatelier's principle this is an endothermic reaction hence it is favoured by high temperature, as volume is constant ($\Delta_n = 0$). Thus, pressure is kept constant to increase the formation of nitric oxide.

Nitric oxide further combines with oxygen to give nitrogen dioxide. Nitrogen dioxide on absorbing in water gives nitric acid having 30–40% strength as follows:

$$2NO + O_2 \longrightarrow 2NO_2$$

$$NO_2 + H_2O \longrightarrow HNO_3 + HNO_2$$

$$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$$

3. **Ostwald Process:** Here, nitric acid is formed by the oxidation of ammonia by air over platinum gauge at 1025 1175 K as follows:

$$\underset{1:8}{4NH_3 + 5O_2} \xrightarrow[800°C]{Pt} 4NO + 6H_2O + 21.6\ \text{kcal}$$

$$2NO + O_2 \longrightarrow 2NO_2$$

$$3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$$

or

$$4NO_2 + 2H_2O + O_2 \longrightarrow 4HNO_3$$

According to Le-Chatelier's principle , formation of nitric acid is favoured by low temperature, as it is exothermic, high pressure as volume is decreasing during the reaction.

Process: The whole process of manufacture of HNO_3 by this method can be described as follows:

(i) ***Catalyst Chamber or Converter:*** It has a platinum gauge which is initially heated upto 1175 K. When a mixture of ammonia and pure air in 1 : 8 –10 ratio is passed in it, ammonia (upto 95 %) gets oxidized into nitric oxide as follows:

$$4NH_3 + 5O_2 \xrightarrow[800°C]{Pt} 4NO + 6H_2O + 21.6\,kcal$$
1 : 8

(ii) ***Cooling Vessels:*** Here, gaseous mixture of NO and O_2 etc, coming from catalyst chamber are cooled in an aluminium or chromium steel vessel.

(iii) ***Oxidizing Chamber:*** Here, cooled gaseous mixture is oxidized by air as a result NO_2 is formed.

$$2NO + O_2 \longrightarrow 2NO_2$$

(iv) ***Absorption Tower:*** In this tower quartz pieces or acid proof flint are filled and water is spread from the top. Here, NO_2 coming from oxidizing chamber is absorbed by water and dilute nitric acid is formed as follows:

$$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$$

or

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

Concentration of Nitric Acid: Dilute nitric acid obtained from absorbing tower is distilled till a constant mixture is formed to get 68% concentrated HNO_3. It on further distillation with concentrated sulphuric acid, becomes 98% concentrated HNO_3 (fuming nitric acid). On cooling, fuming nitric acid in a freezing mixture crystals of 100% of pure nitric acid is obtained.

Physical Properties

1. It is a colourless pungent smelling liquid in anhydrous form but when it is impure it looks yellowish due its decomposition into NO_2.
2. It is miscible with water and forms a constant boiling mixture (azeotropic mixture) which has 68 percent of HNO_3 and boils at 394K. Due to the formation of this mixture, dilute nitric acid cannot be concentrated beyond 68 percent by boiling.
3. In pure form it has a boiling point of 355.6K, melting point of 231.4K and a specific gravity of 1.504 (at 298K).
4. Fuming nitric acid ($HNO_3 + NO_2$) gives brown fumes due to decomposition into NO_2 hence it is kept in black bottle to avoid decomposition.

$$4HNO_3 \rightarrow 4NO_2 + 2H_2O + O_2$$

5. It has an extremely corrosive action on the skin and causes painful blisters or sores.

Chemical Properties

1. **Acidic Nature:** It is a strong monobasic acid and reacts with basic hydroxides, oxides, carbonates etc. to form salts as follows:

$$HNO_3(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + NO_3^-(aq)$$

$$NaOH + HNO_3 \longrightarrow NaNO_3 + H_2O$$

$$Ca(OH)_2 + 2HNO_3 \longrightarrow Ca(NO_3)_2 + 2H_2O$$

$$MgO + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2O$$

$$Na_2CO_3 + 2HNO_3 \longrightarrow 2NaNO_3 + H_2O + CO_2$$

2. **Heating Effect:** In presence of sunlight it decomposes slowly and looks yellowish due to the formation of NO_2.

$$4HNO_3 \xrightarrow{\Delta} 4NO_2 + 2H_2O + O_2$$

■ Yellow colour can be removed by warming the acid at 60–80°C and bubbling dry air through it.

3. **Oxidizing Nature of HNO_3:** It is a powerful oxidizing agent as it easily gives nascent oxygen on decomposition under different conditions as follows:

Condition I

$$\underset{\text{dil.}}{2HNO_3} \longrightarrow H_2O + 2NO + 3[O]$$

Condition II

$$\underset{\text{conc.}}{4HNO_3} \rightarrow 2H_2O + 4NO_2 + O_2\uparrow$$

Condition III (Oxidation of Metals)

$$Al, Fe, Ni, Cr, Co + \underset{\text{Conc.}}{HNO_3} \rightarrow X\ X$$

■ Reason: It is due to the formation of a stable oxide layer over metal surface (that is why HNO_3 is stored in Al container).

(a) **Oxidation of Nonmetals:** It can oxidize many nonmetals into their oxyacids For example,

(i) It can oxidize boron into orthoboric acid.

$$B + 3HNO_3 \rightarrow H_3BO_3 + 3NO_2$$

(ii) It can oxidize carbon into carbon dioxide.

$$C + 4HNO_3 \rightarrow CO_2 + 4NO_2 + 2H_2O$$

(iii) It can oxidize phosphorous into orthophosphoric acid.

$$2P + 10HNO_3 \rightarrow 2H_3PO_4 + 10NO_2 + 2H_2O$$

(iv) It can oxidize sulphur into sulphuric acid.

$$S + 6HNO_3 \longrightarrow H_2SO_4 + 6NO_2 + 2H_2O$$

(v) It can oxidize iodine into iodic acid.

$$I_2 + 10HNO_3 \rightarrow \underset{\text{Iodic acid}}{2HIO_3} + 10NO_2 + 4H_2O$$

(b) Oxidation of Metalloids: It can oxidize many meralloids into their oxyacids For example,

(i) It can oxidize arsenic into arsenic acid.

$$2As + 10HNO_3 \longrightarrow 2H_3AsO_4 + 10NO_2 + 2H_2O$$

or

$$As + 5HNO_3 \longrightarrow H_3AsO_4 + 5NO_2 + H_2O$$

(ii) It can oxidize antimony into antimonic acid.

$$2Sb + 10HNO_3 \longrightarrow 2H_3SbO_4 + 10NO_2 + 2H_2O$$

or

$$Sb + 5HNO_3 \longrightarrow H_3SbO_4 + 5NO_2 + H_2O$$

(iii) It can oxidize selenium into selenious acid.

$$Se + 4HNO_3 \longrightarrow H_2SeO_3 + 4NO_2 + H_2O$$

(iv) It can oxidize tin into meta stannic acid.

$$Sn + 4HNO_3 \rightarrow H_2SnO_3 + 4NO_2 + H_2O$$

(c) Oxidation of Metals: It can react with nearly all the metals except Au, Pt to give a number of products. The product formation depends upon nature of metal, concentration of HNO_3 and temperature.

■ **Oxidation of Metals above Hydrogen in Electrochemical Series:** Such metals are more electropositive than hydrogen hence it can easily liberate nascent hydrogen from HNO_3. This nascent hydrogen can reduce nitric acid into many products like NO_2, NO, N_2O, N_2 and ammonia as follows:

$$Metal + HNO_3 \rightarrow Metal\ nitrate + H$$

$$2HNO_3 + 2H \rightarrow 2NO_2 + 2H_2O$$

$$2HNO_3 + 6H \longrightarrow 2NO + 4H_2O$$

$$2HNO_3 + 8H \rightarrow N_2O + 5H_2O$$

$$2HNO_3 + 10H \longrightarrow N_2 + 6H_2O$$

$$2HNO_3 + 16H \longrightarrow 2NH_3 + 6H_2O$$

(i) *In Case of Iron*

Very dilute nitric acid and iron give ferrous nitrate and ammonium nitrate

$$4Fe + 10HNO_3 \rightarrow NH_4NO_3 + 4Fe(NO_3)_2 + 3H_2O$$

Dilute nitric acid and iron give ferrous nitrate and nitrous oxide.

$$\underset{\text{dil moderate}}{4Fe + 10HNO_3} \rightarrow N_2O + 4Fe(NO_3)_2 + 5H_2O$$

Concentrated nitric acid and iron give ferrous nitrate and nitrogen dioxide.

$$\underset{\text{conc. \& cold}}{Fe + 6HNO_3} \rightarrow 3NO_2 + Fe(NO_3)_3 + 3H_2O$$

■ With highly concentrated HNO_3 (> 80%) iron becomes passive due to formation of stable oxide layer on its surface.

(ii) *In Case of Zinc*

Very dilute nitric acid (6 percent) and zinc give zinc nitrate and ammonium nitrate

$$\underset{\text{Very dil and cold}}{4Zn + 10HNO_3} \rightarrow NH_4NO_3 + 4Zn(NO_3)_2 + 3H_2O$$

Dilute nitric acid (20%) and zinc give zinc nitrate and nitrous oxide.

$$\underset{\text{dil and cold}}{4Zn + 10HNO_3} \rightarrow 2N_2O + 4Zn(NO_3)_2 + 5H_2O$$

Cold and moderate nitric acid and zinc give zinc nitrate and nitric oxide.

$$\underset{\substack{\text{(conc.)}\\ \text{cold and moderate}}}{3Zn + 8HNO_3} \rightarrow 2NO + 3Zn(NO_3)_2 + 4H_2O$$

Concentrated and cold nitric acid (70%) and zinc give zinc nitrate and nitrogen dioxide.

$$\underset{\text{conc. and cold}}{Zn + 4HNO_3} \longrightarrow 2NO_2 + Zn(NO_3)_2 + 2H_2O$$

(iii) *In Case of Tin*

Dilute nitric acid and tin give tin nitrate and ammonium nitrate

$$4Sn + 10HNO_3 \text{ (dil.)} \longrightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$$

Hot and concentrated nitric acid and tin give meta stannic acid and nitrogen dioxide.

$$Sn + 4HNO_3 \text{ (hot and conc.)} \longrightarrow \underset{\text{Metastannic acid}}{H_2SnO_3} + 4NO_2 + H_2O$$

(iv) *In Case of Lead*

Dilute nitric acid and lead give lead nitrate and nitric oxide.

$$3Pb + 8HNO_3 \text{ (dil.)} \longrightarrow 3Pb(NO_3)_2 + 2NO + 4H_2O$$

Concentrated nitric acid and lead give lead nitrate and nitrogen dioxide.

$$Pb + 4HNO_3 \text{ (conc.)} \longrightarrow Pb(NO_3)_2 + 2NO_2 + 2H_2O$$

- **Oxidation of Metals below Hydrogen in Electrochemical Series**: As these metals are not only less electropositive but also less reactive than hydrogen hence cannot displace nascent hydrogen from nitric acid. Such metals can be oxidize into their oxides by nitric acid. These oxides dissolve in nitric acid to form nitrates as follows:

$$HNO_3 \longrightarrow \text{Reduction product} + H_2O + [O]$$

$$\text{Metal} + [O] \longrightarrow \text{Metal oxide}$$

$$\text{Metal oxide} + HNO_3 \longrightarrow \text{Metal nitrate} + H_2O$$

(i) *In Case of Copper*

Cold and very dilute nitric acid and copper give copper nitrate and nitrous oxide.

$$4Cu + 10HNO_3 \text{ (cold and dil.)} \longrightarrow 4Cu(NO_3)_2 + 5H_2O + N_2O$$

Cold and dilute nitric acid and copper give copper nitrate and nitric oxide.

$$3Cu + 8HNO_3 \text{ (cold and moderately conc.)} \longrightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$$

Hot and concentrated nitric acid and copper give copper nitrate and nitrogen dioxide.

$$Cu + 4HNO_3 \text{ (hot and conc.)} \longrightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$$

$$5Cu + 12HNO_3 \text{ (cold and dil.)} \longrightarrow 5Cu(NO_3)_2 + 6H_2O + N_2$$

(ii) *In Case of Silver*

Dilute nitric acid and silver give silver nitrate and nitric oxide.

$$3Ag + 4HNO_3 \text{ (dil.)} \longrightarrow 3AgNO_3 + NO + 2H_2O$$

Concentrated nitric acid and silver give silver nitrate and nitrogen dioxide.

$$Ag + 2HNO_3 \text{ (conc.)} \longrightarrow AgNO_3 + NO_2 + H_2O$$

In Case of Mercury

Dilute nitric acid and mercury give mercurous nitrate and nitric oxide.

$$6Hg + 8HNO_3 \text{ (dil.)} \longrightarrow \underset{\text{Mercurous nitrate}}{3Hg_2(NO_3)_2} + 2NO + 4H_2O$$

Concentrated nitric acid and mercury give mercuric nitrate and nitrogen dioxide.

$$Hg + 4HNO_3 \text{ (conc.)} \longrightarrow \underset{\text{Mercuric nitrate}}{Hg(NO_3)_2} + 2NO_2 + 2H_2O$$

- **In case of Noble Metals**: Nobles metals like Au, Pt, Rh, Ir etc., are not effected by nitric acid but in aqua regia these metals dissolve as follows:

Gold dissolves in aqua regia to give chloro auric acid and nitrosyl chloride as follows:

$$2Au + 3HNO_3 + 11HCl \longrightarrow \underset{\substack{\text{Chloro auric acid}\\ \text{(Hydrogen tetrachloro aurate)}}}{2HAuCl_4} + 3NOCl + 6H_2O$$

Platinum dissolves in aqua regia to give chloro platinic acid and nitrosyl chloride as follows:

$$Pt + 2HNO_3 + 8HCl \longrightarrow \underset{\text{Chloroplatinic acid}}{H_2PtCl_6} + 2NOCl + 4H_2O$$

- **In Case of Mg and Mn**: Both these metals on reaction with dilute nitric acid form their nitrates and liberate hydrogen as follows:

$$Mg + 2HNO_3 \text{ (dil.)} \longrightarrow Mg(NO_3)_2 + H_2\uparrow$$

$$Mn + 2HNO_3 \text{ (dil.)} \longrightarrow Mn(NO_3)_2 + H_2\uparrow$$

(d) **Oxidation of Compounds:** Nitric acid can oxidize many compounds.

For example,

(i) It can oxidize KI into iodine.

$$6KI + 8HNO_3 \longrightarrow 6KNO_3 + 2NO + 3I_2 + 4H_2O$$

(ii) It can oxidize stannous chloride into stannic chloride.

$$3SnCl_2 + 6HCl + 2HNO_3 \longrightarrow 3SnCl_4 + 2NO + 4H_2O$$

(iii) It can oxidize sulphur dioxide into sulphuric acid.

$$3SO_2 + 2HNO_3 + 2H_2O \longrightarrow 3H_2SO_4 + 2NO$$

(iv) It can oxidize ferrous sulphate into ferric sulphate.

$$6FeSO_4 + 2HNO_3 + 3H_2SO_4 \longrightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$$

(v) It can oxidize hydrogen sulphide into sulphur.

$$3H_2S + 2HNO_3 \longrightarrow 3S + 2NO + 4H_2O$$

Reaction with Organic compounds

(a) ***Oxidation:*** It can oxidize sugar into oxalic acid.

$$\underset{\text{Cane sugar}}{C_{12}H_{22}O_{11}} + 36\,HNO_3 \longrightarrow 6\underset{\text{Oxalic acid}}{(COOH)_2} + 36\,NO_2 + 23\,H_2O$$

(b) ***Nitration:*** It can be used for the nitration of many organic compounds as follows:

$$\underset{\text{Benzene (conc.)}}{C_6H_6} + HNO_3 \xrightarrow[323K]{\text{Conc. } H_2SO_4} \underset{\text{Nitrobenzene}}{C_6H_5NO_2} + H_2O$$

$$\begin{array}{l} CH_2OH \\ | \\ CHOH \\ | \\ CH_2OH \\ \text{Glycerol} \end{array} + 3HNO_3 \xrightarrow[\text{Below } 298K]{\text{Conc. } H_2SO_4} \begin{array}{l} CH_2ONO_2 \\ | \\ CHONO_2 \\ | \\ CH_2ONO_2 \\ \text{Glycerol trinitrate} \end{array} + 3H_2O$$

REMEMBER

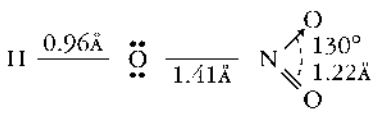
Finger + HNO_3 (conc.) ⟶ Yellow (Xanthoprotein)

Wood + HNO_3 ⟶ Yellow (Nitrocellulose)

Uses

1. It is used as an oxidizing agent not only in inorganic chemistry but also in case of organic chemistry.
2. It is used along with conc. H_2SO_4 as nitrating mixture for the nitration of aromatic compounds.
3. In the manufacture of fertilizers, explosives like T.N.T., nitroglycerine, gun cotton, picric acid etc.
4. It is used in the manufacturing of artificial silk, drugs and dyes etc.
5. It is used in the pickling of stainless steel and etching of metals.
6. In the purification of silver and gold.
7. It is used as an oxidizer in rocket fuels.

Structure: It is a planar molecule as shown in Fig 5.15. In it N–OH, N–O and O–H bond lengths are 1.41, 1.22 and 0.96 Å, respectively.



It has following two resonating structures.



Fig. 5.15 (St. of HNO_3)

OXIDES OF PHOSPHOROUS

Phosphorous forms many oxides, however the main ones are: phosphorous trioxide (P_4O_6), phosphorous tetra oxide (P_4O_8), phosphorous pentaoxide (P_4O_{10}) etc.

Phosphorous Trioxide (P_4O_6)

It is called phosphorous trioxide, however from vapour density measurements the formula obtained is P_4O_6.

Preparation

1. It is obtained by burning phosphorous in a limited supply of air as follows:

$$P_4 + 3O_2 \text{ (limited)} \longrightarrow P_4O_6$$

2. When vapours of phosphorous react with N_2O at low pressure and 600°C, P_2O_3 is formed as follows:

$$P_4 + 6N_2O \rightarrow P_4O_6 + 6N_2$$

Physiochemical Properties

1. It is a white crystalline, highly poisonous, waxy solid with garlic smell which is soluble in CS_2, ether and chloroform.
2. In it phosphorous has +3 oxidation state so it is called phosphorous (III) oxide.
3. **Decomposition:** It can be decomposed on heating at 483K into red phosphorous and phosphorous tetra oxide.

$$4P_4O_6 \xrightarrow{\Delta} 3P_4O_8 + P_4$$

Red phosphorous

■ If temperature is above 713K it dissociates into phosphorous dioxide.

$$2P_2O_3 \longrightarrow 3PO_2 + P$$

4. **Action of Air or O_2:** It can be oxidized by air into phosphorous penta oxide.

$$P_4O_6 + 2O_2 \xrightarrow{\Delta} P_4O_{10}$$

5. **With Chlorine:** It burns in chlorine giving oxy chlorides vigorously.

$$P_4O_6 + 4Cl_2 \rightarrow \underset{\text{Phosphorous oxychloride}}{2POCl_3} + \underset{\text{Meta phosphorous oxychloride}}{2PO_2Cl}$$

6. **Reaction with Water:** It dissolves in cold water slowly giving phosphorous acid while in hot water it gives orthophosphoric acid and phosphine in a violent manner.

$$P_4O_6 + 6H_2O \underset{\text{(Cold)}}{\rightarrow} 4H_3PO_3$$

$$P_4O_6 + 6H_2O \underset{\text{(Hot)}}{\rightarrow} 3H_3PO_4 + PH_3$$

Structure: Its structure is just like that of P_4 molecule i.e, each phosphorous atom is present at the corner of a tetrahedron and is covalently bonded to three other phosphorous atoms through oxygen atoms as shown in the Figure 5.16. It has 6 P – O – P bonds, 12 sigma bonds, and 16 lone pairs of electrons. In it the bond length of P – O is 1.63 Å.

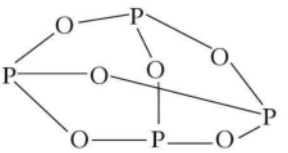


Fig. 5.16 St.of P_4O_6

Phosphorous Penta oxide (P_4O_{10})

It is called phosphorous penta oxide (P_2O_5) however, it is written as P_4O_{10}.

Preparation

It is obtained by heating phosphorous in excess or free supply of air or CO_2 as follows:

$$P_4 + 5O_2 \text{ (excess)} \rightarrow P_4O_{10}$$

$$2P + 5CO_2 \xrightarrow{\Delta} P_2O_5 + 5CO$$

It can be purified by sublimation.

Physiochemical Properties

1. It is a snowy white solid which sublimes on heating.
2. In pure form it is odourless but it gives garlic smell in presence of impurity of P_4O_6.
3. **Reaction with Water:** It reacts with cold water to form metaphosphoric acid and with hot water it gives phosphoric acid.

$$P_4O_{10} + 2H_2O \underset{\text{(Cold)}}{\rightarrow} \underset{\text{Metaphosphoric acid}}{4HPO_3}$$

$$P_4O_{10} + 6H_2O \underset{\text{(Hot)}}{\rightarrow} 4H_3PO_4$$

■ As it forms orthophosphoric acid with water hence it is called phosphoric anhydride.

4. **As Dehydrating Agent:** Below 100°C it is the most effective dehydrating agent as it has a great affinity towards water.

For example,

$$2H_2SO_4 + P_4O_{10} \rightarrow 2SO_3 + 4HPO_3$$

$$4HNO_3 + P_4O_{10} \rightarrow 2N_2O_5 + 4HPO_3$$

$$2CH_3CONH_2 + P_4O_{10} \rightarrow 2CH_3CN + 4HPO_3$$

5. **With Carbon:** On strong heating with carbon it reduced to red phosphorous.

$$P_4O_{10} + 10\,C \longrightarrow P_4 + 10\,CO$$

Uses

It is used as a dehydrating and drying agent.

Structure

Its structure is just like that of P_4O_6 but in it each phosphorous atom is attached to an additional oxygen atom with a coordinate linkage involving a lone pair of electrons present on each phosphorous atom. It has 16 sigma bonds, 4 pi bonds and 24 lone pairs of electrons on oxygen atoms.

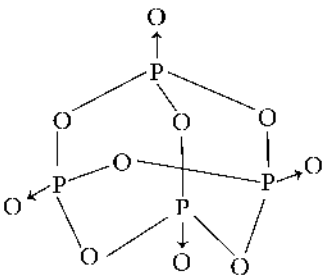


Fig. 5.17 St.of P_4O_{10}

OXYACIDS OF PHOSPHOROUS

Phosphorous forms a number of oxy acids. Some of the important oxyacids are given in Table 5.2:

Table 5.2 Oxy Acids of Phosphorous

Name of oxyacid	Formula	Basicity	Oxidation state of 'P'
Hypophosphorous acid	H_3PO_2	1	+1
Phosphorous acid	H_3PO_3	2	+3
Orthophosphoric acid	H_3PO_4	3	+5
Pyrophosphoric acid	$H_4P_2O_7$	4	+5
Metaphosphoric acid	HPO_3	1	+5
Hypophosphoric acid	$H_4P_2O_6$	4	+4

Hypophosphorous Acid (H_3PO_2)

Preparation

It can be prepared by heating barium hydroxide (baryte) solution with white phosphorous followed by reaction of barium hypophosphite crystals with dilute sulphuric acid as follows:

$$8P + 3Ba(OH)_2 + 6H_2O \rightarrow \underset{\text{Barium hypophosphite}}{3Ba(H_2PO_2)_2} + 2PH_3$$

$$Ba(H_2PO_2)_2 + H_2SO_4 \longrightarrow 2H_3PO_2 + BaSO_4\downarrow$$

Physiochemical Properties

1. It is a colourless crystalline solid with a melting point of 26.3°C which is soluble in water.
2. **Acidic Nature:** It is a monobasic acid and ionizes as

$$H_3PO_2 \rightleftharpoons \underset{\text{Hypophosphite ion}}{H^+ + H_2PO_2^-}$$

3. **Heating Effect:** On heating, it decomposes to give phosphine.

$$2H_3PO_2 \xrightarrow{433\ K} H_3PO_4 + PH_3$$

4. **Reducing Nature:** It acts as a reducing agent.

For example,
It reduces silver nitrate into silver.

$$4AgNO_3 + 2H_2O + H_3PO_2 \rightarrow 4Ag + 4HNO_3 + H_3PO_4$$

- It reduces auric chloride into gold.

$$4AuCl_3 + 6H_2O + 2H_3PO_2 \rightarrow 4Au + 12HCl + 2H_3PO_4$$

- It reduces copper sulphate into cuprous hydride.

$$4CuSO_4 + 3H_2PO_2 + 6H_2O \rightarrow 2Cu_2H_2 + 3H_3PO_4 + 4H_2SO_4$$

- It reduces mercuric chloride into mercury.

$$2HgCl_2 + H_3PO_2 + 2H_2O \longrightarrow 2Hg + H_3PO_4 + 4HCl$$

- It reduces chlorine into hydrogen chloride.

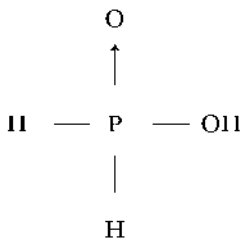
$$H_3PO_2 + 2H_2O + 2Cl_2 \rightarrow H_3PO_4 + 4HCl$$

Uses

Sodium, potassium and calcium hypophosphites are used as nerve tonics medicines.

Structure

In it phosphorous atom is sp^3 hybridized.



Phosphorous Acid (H_3PO_3)

Preparation

1. It is prepared by the action of water upon phosphorous oxide, or by the action of water on phosphorous trichloride.

$$P_4O_6 + 6H_2O \longrightarrow 4H_3PO_3$$

$$PCl_3 + 3H_2O \longrightarrow H_3PO_3 + 3HCl$$

2. It can be prepared by the hydrolysis of PCl_3 by oxalic acid.

$$PCl_3 + 3H_2C_2O_4 \rightarrow H_3PO_3 + 3CO + 3CO_2 + 3HCl$$

Physiochemical Properties

1. It is a white deliquescent crystal with a melting point of 73.6°C and highly soluble in water.
2. It is a dibasic acid which ionizes as follows:

$$H_3PO_3 \rightarrow H_2PO_3^- + H^+$$

$$H_2PO_3^- \rightarrow HPO_3^{2-} + H^+$$

As it is a weak dibasic acid hence it forms two series of salts, e.g, sodium dihydrogen phosphite NaH_2PO_3 and disodium hydrogen phosphite, Na_2HPO_3.

3. **Decomposition:** It decomposes on heating at 200°C into phosphoric acid and phosphine.

$$4H_3PO_3 \longrightarrow 3H_3PO_4 + PH_3$$

Orthophosphoric acid

4. **Reducing Nature:** It can act as a strong reducing agent For example:

- It reduces silver nitrate to metallic silver.

$$2AgNO_3 + H_3PO_3 + H_2O \longrightarrow 2Ag\downarrow + 2HNO_3 + H_3PO_4$$

- It reacts with $HgCl_2$ to form a white precipitate of Hg_2Cl_2 which subsequently turns black due to separation of metallic mercury.

$$2HgCl_2 + H_3PO_3 + H_2O \longrightarrow Hg_2Cl_2\downarrow + 2HCl + H_3PO_4$$

$$Hg_2Cl_2 + H_3PO_3 + H_2O \longrightarrow 2Hg\downarrow + 2HCl + H_3PO_4$$

- It reduces copper sulphate into copper.

$$H_3PO_3 + CuSO_4 + H_2O \rightarrow H_3PO_4 + H_2SO_4 + Cu\downarrow$$

- It reduces auric chloride into gold.

$$3H_3PO_3 + 2AuCl_3 + 3H_2O \longrightarrow 3H_3PO_4 + 6HCl + 2Au\downarrow$$

- It reduces sulphur dioxide into sulphur.

$$SO_2 + 2H_3PO_3 \longrightarrow 2H_3PO_4 + S\downarrow$$

- It reduces PCl_5 into PCl_3.

$$H_3PO_3 + 3PCl_5 \longrightarrow PCl_3 + 3POCl_3 + 3HCl$$

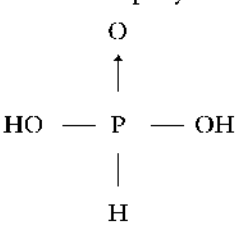
- It decolourizes the solution of iodine and potassium permanganate by reducing them as follows:

$$I_2 + H_2O + H_3PO_3 \longrightarrow H_3PO_4 + 2HI$$

$$2KMnO_4 + 3H_2SO_4 + 5H_3PO_3 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5H_3PO_4$$

Structure

Here, phosphorous atom is sp^3 hybridized.



Orthophosphoric Acid (H_3PO_4)

Preparation

1. It can be conveniently prepared by dissolving P_2O_5 in water and the solution is boiled to form thick syrup.

$$P_2O_5 + 3H_2O \rightarrow 2H_3PO_4$$

2. Red phosphorous when heated with conc. HNO_3, orthophosphoric acid is obtained.

$$P + 5HNO_3 \longrightarrow H_3PO_4 + H_2O + 5NO_2$$

3. On large scale it is prepared by treating phosphorite rock with dil. H_2SO_4

$$Ca_3(PO_4)_2 + 3H_2SO_4 \longrightarrow 3CaSO_4 + 2H_3PO_4$$

4. It can be obtained by heating red phosphorous with 50% HNO_3 on a water bath.

$$P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$$

Physiochemical Properties

1. It is a colourless syrupy liquid which on concentration gives transparent prismatic crystals having melting point of 42.3°C.
2. It is nonvolatile, but dissolves readily in water. It is miscible in all proportions in water.
3. **Heating Effect or Decomposition:** On heating, it decomposes to form pyrophosphoric acid at about 250°C, which, on further heating (316°C), gives metaphosphoric acid as follows:

$$2H_3PO_4 \xrightarrow{250^\circ C} H_4P_2O_7 + H_2O$$

$$H_4P_2O_7 \xrightarrow{>316^\circ C} 2HPO_3 + H_2O$$

On very strong heating at red hot it gives P_4O_{10}.

$$4H_3PO_4 \xrightarrow{\Delta} P_4O_{10} + 6H_2O$$

4. **Acidic Nature:** It is a tribasic acid, because it ionizes in three stages as represented below:

$$H_3PO_4 \longrightarrow H^+ + H_2PO_4^- \text{ (Readily ionized)}$$

Dihydrogen phosphate ion

$$H_2PO_4^- \longrightarrow H^+ + HPO_4^{2-} \text{ (Weak ionization)}$$

Monohydrogen phosphate ion

$$HPO_4^{2-} \longrightarrow H^+ + PO_4^{3-} \text{ (Very weak ionization)}$$

Phosphate ion

Thus, it gives three series of salts

- Sodium dihydrogen phosphate $NaH_2PO_4.H_2O$
- Disodium hydrogen phosphate $Na_2HPO_4.12H_2O$
- Tri-sodium or normal sodium phosphate $Na_3PO_4.12H_2O$

5. **With $MgSO_4$:** When it reacts with magnesium sulphate in presence of ammonium chloride and ammonium hydroxide, a white precipitate of magnesium ammonium phosphate is obtained (test of magnesium ions).

$$MgSO_4 + NH_4Cl + H_3PO_4 \longrightarrow Mg(NH_4)PO_4 + H_2SO_4 + HCl$$

6. **With $AgNO_3$:** With $AgNO_3$ it gives a yellow precipitate of silver phosphate.

$$3AgNO_3 + H_3PO_4 \longrightarrow \underset{\text{Yellow ppt.}}{Ag_3PO_4} + 3HNO_3$$

7. **With $BaCl_2$:** With $BaCl_2$ it gives a white precipitate of barium phosphate.

$$3BaCl_2 + 2H_3PO_4 \longrightarrow \underset{\text{White ppt.}}{Ba_3(PO_4)_2} + 6HCl$$

8. **With Ammonium Molybdate:** When it is heated with ammonium molybdate in presence of nitric acid, a canary yellow coloured precipitate of ammonium phosphomolybdate is obtained (test of PO_4^{3-} ion).

$$H_3PO_4 + 21\,HNO_3 + 12(NH_4)_2MoO_4 \longrightarrow \underset{\text{Ammonium phosphomolybdate}}{(NH_4)_3PO_4.12MoO_3} + 21\,NH_4NO_3 + 12H_2O$$

9. **Preparation of HBr and HI:**

$$3NaBr + H_3PO_4 \longrightarrow 3HBr + Na_3PO_4$$

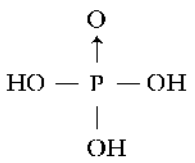
$$3NaI + H_3PO_4 \longrightarrow 3HI + Na_3PO_4$$

Uses

1. It is used for making phosphorous, meta phosphoric acid, pyrophosphoric acid, phosphates and phosphorous fertilizers.
2. It is used to stabilize H_2O_2 or to avoid its decomposition.
3. In the lab method preparation of HBr and HI.

Structure

In it phosphorous atom is sp^3 hybridized.



Phosphine (PH_3)

It is an important hydride of phosphorous which was discovered by Gengembre.

Preparation

1. **From Metal Phosphides**: When water is passed over metal phosphides like sodium phosphide or calcium phosphide, phosphine is obtained.

   Metal Phosphide + H_2O ➤ $PH_3\uparrow$

   For example,

   $Na_3P + 3H_2O$ ➤ $PH_3 + 3NaOH$

   $Ca_3P_2 + 6H_2O$ ➤ $2PH_3 + 3Ca(OH)_2$

2. **From Phosphorous Acid**: When phosphorous acid is heated pure form of phosphine is obtained.

   $4H_3PO_3 \xrightarrow{\Delta\ (In\ aq\ sol)} 3H_3PO_4 + PH_3\uparrow$

3. **From Aluminium Phosphide**: Aluminium phosphide on reaction with dilute sulphuric acid gives phosphine.

   $2AlP + 3H_2SO_4$ ➤ $2PH_3 + Al_2(SO_4)_3$

4. **From Phosphonium Iodide**: When it is heated with caustic soda solution, pure phosphine is obtained as follows:

   $PH_4I + NaOH$ ➤ $PH_3 + NaI + H_2O$

5. **Laboratory Method**: In laboratory phosphine is prepared by heating white phosphorous with concentrated NaOH solution in an inert atmosphere of CO_2 or oil gas as follows:

   $4P + 3NaOH + 3H_2O \xrightarrow{Inert\ gas\ (Ar)} 3NaH_2PO_2 + PH_3$

   White or Yellow

   Phosphine obtained from here is highly inflammable due to the presence of impurity of phosphorous dihydride and hydrogen.

   $P_4 + 4NaOH + 4H_2O$ ➤ $4NaH_2PO_2 + 2H_2$

   $3P_4 + 8NaOH + 8H_2O$ ➤ $8NaH_2PO_2 + 2P_2H_4$

   Here, vortex rings are also due to the formation of P_2H_4. If reaction is carried out in alcoholic medium, P_2H_4 is removed.

   $2P_2H_4 + 7O_2$ ➤ $4H PO_3 + 2H_2O$

- From phosphine P_2H_4 can be removed by passing the gaseous mixture through a freezing mixture where P_2H_4 gets condensed.
- When the gaseous mixture is passed through HI, phosphine is absorbed by forming PH_4 which on reaction with KOH gives pure phosphine.

  $PH_3 + HI$ ➤ PH_4I.

  $PH_4I + KOH \xrightarrow[(30\ \%)]{} PH_3 + HI + H_2O$

Physical Properties

1. It is a highly poisonous, colourless gas having rotten fish like odour.
2. It can be condensed to a colourless liquid having a boiling point of –85°C and can be frozen to a white solid having a melting point of –134°C.

Chemical Properties

1. **Combustion**: It burns in presence of O_2 to give phosphorous pentaoxide.

   $4PH_3 + 8O_2$ ➤ $P_4O_{10} + 6H_2O$

2. **Decomposition**: On heating in absence of air at 45°C it decomposes as follows:

   $4PH_3$ ➤ $P_4 + 6H_2$

3. **Basic Nature**: It is weaker base than ammonia and its aqueous solution is neutral towards litmus. When it reacts with halogen acids phosphonium halides are formed.

   $PH_3 + HI$ ➤ PH_4I

4. **Reaction with Chlorine**: It burns in chlorine to form PCl_3.

   $PH_3 + 3Cl_2$ ➤ $PCl_3 + 3HCl$

5. **Reaction with Copper Sulphate**: When it is passed through copper sulphate solution a black precipitate of copper phosphide is obtained.

$$3CuSO_4 + 2PH_3 \longrightarrow Cu_3P_2 + 3H_2SO_4$$

6. **With Silver Nitrate:** When it is passed through silver nitrate solution a black precipitate of silver phosphide is obtained.

$$3AgNO_3 + PH_3 \longrightarrow Ag_3P + 3HNO_3$$

7. **With Nitric Acid:** With nitric acid it gives P_2O_5 as follows:

$$2PH_3 + 16HNO_3 \longrightarrow P_2O_5 + 16\,NO_2 + 11H_2O$$

8. **Formation of Addition Compounds:** It reacts with anhydrous $AlCl_3$ or $SnCl_4$ or Cu_2Cl_2 to form adducts as follows:

$$AlCl_3 + 2PH_3 \longrightarrow AlCl_3.2PH_3$$

$$SnCl_4 + 2PH_3 \longrightarrow SnCl_4.2PH_3$$

$$Cu_2Cl_2 + 2PH_3 \longrightarrow Cu_2Cl_2.2PH_3$$

9. **Lewis Basic Nature:** In it phosphorous atom has one lone pair of electron thus it can act like a lewis base and form coordinate compounds with lewis acids.

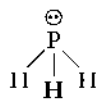
$$BF_3 + :PH_3 \rightarrow Cl-\underset{Cl}{\overset{Cl}{B}}\leftarrow \underset{H}{\overset{H}{P}}-H$$

Uses

1. For making holme signals ($PH_3 + C_2H_2$) and smoke screens.
2. It is used for making many metal phosphides like silver phosphide, copper phosphide etc.

Structure

It has a pyramidal structure in which phosphorous atom is sp^3 hybridized and has a lone pair of electron. Here, H P H bond angle is 93°.



Straight Objective Type Questions (Single Choice)

1. In diborane the two H – B – H angles are nearly
   (a) 60°, 120° (b) 95°, 150°
   (c) 95°, 120° (d) 120°, 180°

2. Which of the following reacts with BCl_3 to form diborane?
   (a) $K_2Cr_2O_7$ (b) NH_3
   (c) NaHg (d) $LiAlH_4$

3. In the reaction

   $B_2H_6 + 2KOH + 2(A) \longrightarrow 2(B) + 6H_2$

   (a) and (b) are, respectively
   (a) HCl, KBO_3 (b) H_2, H_3BO_3
   (c) H_2O, KBO_2 (d) H_2O, KBO_3

4. BCl_3 does not exist as a dimmer but BH_3 exists as B_2H_6 because
   (a) Cl_2 is more electronegative than hydrogen.
   (b) large size of the chlorine atom does not fit between the small sized boron atoms, while small sized hydrogen atoms occupy the space between boron atoms.
   (c) there is $p\pi$=$d\pi$ back bonding in BCl_3.
   (d) both (b) and (c)

5. The structure of diborane (B_2H_6) contains
   (a) four 2c-2e bonds and two 3c-2e bonds
   (b) two 2c-2e bonds and four 3c-2e bonds
   (c) two 2c-2e bonds and two 3c-3e bonds
   (d) four 2c-2e bonds and four 3c-2e bonds

6. In silicon dioxide
   (a) there are double bonds between silicon and oxygen atoms.
   (b) silicon is bonded to two silicon atoms.
   (c) each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms.
   (d) each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bounded to two silicon atoms.

7. The correct order of acidic nature of oxides is in the order
   (a) $N_2O_5 < N_2O_3 < NO_2 < NO < N_2O$
   (b) $N_2O < NO < N_2O_3 < NO_2 < N_2O_5$
   (c) $N_2O_5 < N_2O < N_2O_5 < NO < N_2O$
   (d) $NO < N_2O < N_2O_3 < NO_2 < N_2O_5$

8. H_3PO_3 has ... nonionizable P–H bonds.
   (a) 2 (b) 1
   (c) None (d) 5

9. Phosphine, acetylene and ammonia can be formed by treating water with
   (a) Mg_3P_2, Al_4C_3, Li_3N
   (b) Ca_3P_2, Mg_2C, NH_4NO_3
   (c) Ca_3P_2, CaC_2, $CaCN_2$
   (d) Ca_3P_2, CaC_2, Mg_3N_2

10. SiO_2 is reacted with sodium carbonate. Which gas is liberated?
    (a) O_2 (b) O_3
    (c) CO (d) CO_2

11. NH_3 is converted to hydrazine generally by reaction with which of the following?
    (a) CO (b) $SnCl_2$
    (c) NaOCl (d) NaOH

12. Cl – P – Cl bond angles in PCl_3 molecule are
    (a) 120° and 90° (b) 60° and 90°
    (c) 60° and 120° (d) 120° and 30°

13. When concentrated HNO_3 decomposes slowly in sunlight. It gives
    (a) Brown colour (b) Green colour
    (c) Yellow colour (d) Blue colour

14. The number of hydrogen atom(s) attached to phosphorous atom in hypophosphorous acid is
    (a) zero (b) two
    (c) one (d) three

15. Which of the following compound is used in Dunstan's test?
    (a) Alumina (b) Boric acid
    (c) Potash alum (d) Borax

16. The number of P–O–P and P–O–H bonds present, respectively in pyrophosphoric acid molecule are

(a) 2, 2 (b) 1, 8
(c) 1, 2 (d) 1, 4

17. The equivalent mass of phosphoric acid (H_3PO_4) in the reaction,

$NaOH + H_3PO_4 \longrightarrow NaH_2PO_4 + H_2O$ is

(a) 98 (b) 89
(c) 49 (d) 58

18. Correct order of boiling points of hydrides of nitrogen family is

(a) $NH_3 < PH_3 < AsH_3 < SbH_3$

(b) $PH_3 < AsH_3 < NH_3 < SbH_3$

(c) $NH_3 < SbH_3 < PH_3 < AsH_3$

(d) $PH_3 < NH_3 < SbH_3 < AsH_3$

19. Ammonium compound which on heating does not give NH_3 is

(a) $(NH_4)_2SO_4$ (b) $(NH_4)_2CO_3$
(c) NH_4NO_2 (d) NH_4Cl

20. Which of the following oxides of nitrogen is a coloured gas?

(a) N_2O (b) NO
(c) N_2O_4 (d) NO_2

21. The number of P–O–P bridges in the structure of phosphorous penroxide and phosphorous trioxide are, respectively

(a) 5, 5 (b) 5, 6
(c) 6, 5 (d) 6, 6

22. Addition of concentrated HNO_3 to concentrated H_2SO_4 gives

(a) SO_4^{2-} (b) SO_3
(c) NO_2 (d) NO_3

23. H_3PO_2 is the molecular formula of an acid of phosphorous. Its name and basicity are, respectively

(a) Hypophosphoric acid and two

(b) Hypophosphorous acid and one

(c) Hypophosphorous acid and two

(d) Phosphorous acid and two

24. The strongest base is

(a) NH_3 (b) PH_3
(c) AsH_3 (d) SbH_3

25. Ionization of boric acid in aqueous medium gives, which one of the following?

(a) $[BO_3]^{3-}$ (b) $[B(OH)_4]^-$
(c) $[B(OH)_2O]^-$ (d) $[B(OH)O_2]^{2-}$

26. Which of the following halide of carbon is used as refrigerant?

(a) CCl_4 (b) CF_4
(c) CH_2Cl_2 (d) CF_2Cl_2

27. In $[B_4O_5(OH)_4]^{2-}$ anion, what is the number of B–O–B bridges?

(a) 2 (b) 3
(c) 4 (d) 5

28. Which of these is most explosive?

(a) NCl_3 (b) PCl_3
(c) $AsCl_3$ (d) All of these

29. Two oxides of nitrogen, NO and NO_2 react together at 253°K and form a compound of nitrogen X. X reacts with water to yield another compound of nitrogen Y. The shape of the anion of Y molecule is

(a) triangular planar

(b) pyramidal

(c) tetrahedral

(d) square planar

30. Concentrated HNO_3 reacts with iodine to give:

(a) HI (b) HOI
(c) $HOIO_2$ (d) $HOIO_3$

31. The reaction between NH_2^- and N_2O gives

(a) NO (b) N_2O_5
(c) N_3 (d) NH_2NH_2

32. The lightening bolts in the atmosphere cause the formation of

(a) NO (b) NH_3
(c) NH_4OH (d) NH_2OH

33. Which of the following statement is correct?

(a) NO is acidic colourless and gaseous oxide.

(b) N_2O is angular in shape.

(c) NO_2 is angular in shape with a sweet smell.

(d) NO_2 reacts with NaOH to give a mixture of two salts.

34. The acid having maximum basicity is
(a) H_3PO_3 (b) H_3PO_2
(c) $H_4P_2O_5$ (d) H_3PO_4

35. Cane sugar reacts with concentrated HNO_3 to give
(a) Formic acid
(b) Oxalic acid
(c) CO_2 and water
(d) CO and water

36. One mole of magnesium nitride on the reaction with an excess of water gives
(a) two moles of ammonia
(b) two moles of nitric acid
(c) one mole of ammonia
(d) one mole of nitric acid

37. The bonds present in N_2O_5 are
(a) only ionic
(b) covalent and coordinate
(c) only covalent
(d) covalent and ionic

38. N forms NCl_3 whereas P can form both PCl_3 and PCl_5. Why?
(a) N atoms is larger than P in size.
(b) P has low lying 3d orbitals, which can be used for bonding but N does not have 3d orbitals in its valence shell.
(c) P is more reactive towards Cl than N.
(d) None of these.

39. A gas that cannot be collected over water is
(a) PH_3 (b) O_2
(c) SO_2 (d) N_2

40. Which of the following oxide of nitrogen is most thermally stable?
(a) N_2O (b) NO
(c) N^2O_3 (d) N_2O_5

41. Nitrogen dioxide cannot be prepared by heating
(a) KNO_3 (b) $Pb(NO_3)_2$
(c) $Cu(NO_3)_2$ (d) $AgNO_3$

42. Number of P P bonds in $(HPO_3)_4$ are
(a) zero (b) 1
(c) 2 (d) 3

43. Moderate electrical conductivity is shown by
(a) diamond (b) carborundum
(c) graphite (d) silica

44. The formula of calcium cyanamide is
(a) $Ca(CN)_2$ (b) CaC_2N
(c) CaNCN (d) $CaCHNH_2$

45. On the addition of mineral acid to an aqueous solution of borax, the following compound is
(a) Orthoboric acid
(b) Boron hydride
(c) Pyroboric acid
(d) Metaboric acid

46. Which of the following compounds are formed when boron trichloride is treated with water
(a) B_2H_6 + HCl (b) B_2O_3 + HCl
(c) H_3BO_3 + HCl (d) None of these

47. In diborane
(a) All the atoms are in one plane.
(b) Structure is just like ethane.
(c) Presence of B B bond is observed.
(d) Boron atoms are linked together by hydrogen bridges.

48. Which statement about nitric acid is not true?
(a) It often has a yellow colour due to the formation of NO_2.
(b) Its anhydride is N_2O_3.
(c) It is a strong oxidizing agent.
(d) It is one of the more common strong acids and gets essentially 100% dissociated in water.

49. PCl_3 reacts with water to form
(a) H_3PO_4 (b) PH_3
(c) $POCl_3$ (d) H_3PO_3, HCl

50. PH_4I + NaOH forms
(a) NH_3 (b) P_4O_6
(c) PH_3 (d) P_4O_{10}

Brainteaser Objective Type Questions (Single Choice)

51. What is responsible for the brownish coloured gas when copper dissolves in nitric acid?
(a) $2NO_2(g) \rightarrow N_2O_4(g)$
(b) $NH_4NO_3(l) \rightarrow N_2O(g) + 2H_2O(g)$
(c) $2N_2(g) + O_2(g) \rightarrow 2NO(g)$
(d) $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

52. Which is not an appropriate method of making phosphoric acid?
(a) $P_4(s) + 16H_2O(l) \longrightarrow 4H_3PO_4(aq) + 2H_2(g)$
(b) $P_4O_{10}(s) + 6H_2O(l) \longrightarrow 4H_3PO_4(aq)$
(c) $PH_3(g) + 2O_2(g) \longrightarrow H_3PO_4(l)$
(d) $PCl_5(s) + 4H_2O(l) \longrightarrow H_3PO_4(aq) + 5HCl(aq)$

53. PCl_3 and PCl_5 both exists; NCl_3 exists but NCl_5 does not exist. It is due to
(a) lower electronegativity of P than N.
(b) lower tendency of N to form covalent bond.
(c) availability of vacant d-orbital in P but not in N.
(d) statement is itself incorrect.

54. Identify the oxide of nitrogen in the following reactions.
I $_____ + 3SnCl_2 + 6HCl \rightarrow 3SnCl_4 + 2NH_4OH$
II $4AgNO_3 + 2Cl_2 \rightarrow 4AgCl + ____ + O_2$
(a) NO_2, N_2O_5 (b) NO, N_2O_5
(c) NO_2, N_2O_3 (d) NO, N_2O_3

55. $HNO_2 + H_2SO_3 + H_2O \rightarrow H_2SO_4 + [A]$. A is
(a) NH_3 (b) N_2
(c) NH_2OH (d) NH_4OH

56. Which is correct statement?
(a) The decreasing order of melting point is $NH_3 > (CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$.
(b) The decreasing order of bond angle is $BF_3 > PF_3 > ClF_3$.
(c) The decreasing order of bond dissociation energy is $Cl_2 > Br_2 > F_2 > I_2$.
(d) Both (b) and (c).

57. White phosphorous on reaction with lime water gives calcium salt of an acid (P) along with a gas (Q). Which of the following is correct?
(a) (Q) is more basic than ammonia.
(b) (P) is dibasic acid.
(c) The bond angle in (Q) is less than that in case of ammonia.
(d) (P) on heating gives (Q) and O_2.

58. Which of the following is correct?
(a) P–O bonds in P_4O_6 molecule are 6.
(b) P–P bonds in P_4 molecule are 6.
(c) P O P bonds in P_4O_8 molecule are 8.
(d) Both (a) and (c).

59. A colourless poisonous gas (A) which burns with blue flame is passed through aqueous NaOH at high pressure and temperature to give a compound (B). (B) on heating gives (C) and (C) gives white precipitate (D) with $CaCl_2$. Both (C) and (D) decolourize acidified $KMnO_4$. Here (A) can be
(a) CO (b) CO_2
(c) SO_2 (d) H_2S

60. If O_2 is removed from the formula of anhydride of HNO_2 then the formula of the resulting compound satisfies which of the following properties
(a) It supports combustion.
(b) It cannot react with red hot copper.
(c) It produces tears in eyes.
(d) It is paramagnetic.

61. The correct increasing order of acidity is
(a) $H_2O_3 > CO_2 > H_2O$
(b) $H_2O < H_2O_2 > CO_2$
(c) $H_2O < H_2O_2 < CO_2$
(d) $CO_2 > H_2O_2 > H_2O$

62. When a mixture of NO and NO_2 is passed through an aqueous solution of ammonium sulphate, we get
(a) a dibasic acid whose dinegative anion has all bonds equal with bond angle 90°.
(b) a dibasic acid whose anhydride in solid state form a cyclic trimer which contain three A A bonds.
(c) a diatomic gas which on reaction with acetylene under electric spark gives a monobasic acid whose anion is pseudohalide.
(d) a dibasic acid which has not dehydrating property.

63. Nitrogen is liberated by the thermal decomposition of only
(a) NH_4NO_2 (b) NaN_3
(c) $(NH_4)_2Cr_2O_7$ (d) All of the above

64. H_3PO_4 will be obtained in
(a) Red phosphorous + conc. HNO_3 ⟶
(b) White phosphorous + conc. H_2SO_4 ⟶
(c) PCl_5 + oxalic acid followed by heat
(d) All of these

65. P_4O_{10} on reaction with PCl_5 gives
(a) $POCl_3$ and O_2 (b) $POCl_3$ only
(c) PO_2Cl only (d) $POCl_3$ and Cl_2

66. $H_2C_2O_4 \xrightarrow{\Delta}$ Gas (P) + gas (Q) + liquid (R) Gas (P) burns with a blue flame and is oxidized to gas (Q)

$$\text{Gas (P)} + Cl_2 \longrightarrow (S) \xrightarrow{NH_3,\ \Delta} (T)$$

(P), (Q), (R) and (T) are, respectively
(a) CO, CO_2, $COCl_2$, $HCONH_2$
(b) CO, CO_2, H_2O, $COCl_2$
(c) CO_2, CO, H_2O, $HCONH_2$
(d) CO, CO_2, H_2O, NH_2CONH_2

67. During the preparation of N_2O by heating a mixture of $NaNO_3$ and $(NH_4)_2SO_4$, which of the following is correct?
(a) The evolved gas is passed through $FeSO_4$ solution to remove nitric oxide.
(b) The evolved gas is collected over cold water as it is fairly soluble in hot water.
(c) Both are correct.
(d) None is correct.

68. What may be expected to happen if phosphine gas is mixed with chlorine gas?
(a) PCl_5 and HCl are formed and the mixture cools down.
(b) $PH_3.Cl_2$ is formed with warming up.
(c) PCl_3 and HCl are formed and the mixture warms up
(d) The mixture only cools down

69. Amongst the trihalides of nitrogen, which one is least basic?
(a) NF_3 (b) NCl_3
(c) NBr_3 (d) NI_3

70. Hydrolysis of PI_3 yields
(a) Monobasic acid and a dibasic acid.
(b) Monobasic acid and a salt.
(c) A monobasic acid and tribasic acid.
(d) A monoacidic base and a dibasic acid.

71. NaOH (aq.) on reaction with N_2O_5 gives (X). NaCl (aq.) on reaction with N_2O_5 gives (X) and (Y). (Y) is
(a) $NaNO_2$ (b) NOCl
(c) $NaNO_3$ (d) NO_2Cl

72. Which of the following statements are correct about the reaction between the copper metal and dilute HNO_3?
I The principle reducing product is NO gas.
II Cu metal is oxidized to Cu^{2+} (aq.) ion which is blue in colour.
III NO is paramagnetic and has one unpaired electron in antibonding molecular orbital.
IV NO reacts with O_2 to produce NO_2 which is linear in shape.
Choose the correct statements.
(a) I and III (b) I, II and III
(c) II and III (d) II, III and IV

73. HBr and HI reduce sulphuric acid, HCl can reduce $KMnO_4$ and HF can reduce
(a) $K_2Cr_2O_7$ (b) $KMnO_4$
(c) H_2SO_4 (d) None of the above

74. When HNO_2 reacts with conc. H_2SO_4 two gases (P) and (Q) are produced. The correct statement is

(a) Both are triatomic and colourless.
(b) Both are triatomic and diamagnetic.
(c) Both are odourless.
(d) None of these.

75. For $P_4 + Cl_2$ (excess) $\rightarrow$ (A) $\xrightarrow{H_2O}$ (B) + (C)
Any alkene can react with (C) to give a compound which can perform Wurtz reaction. Here (B) is
(a) HCl (b) H_3PO_2
(c) H_3PO_4 (d) H_3PO_3

76. Consider the following statements for diborane.
I Boron is approximately sp^3 hybridized.
II B H B angle is 180°.
III There are two terminal B–H bonds for each boron atom.
IV There are only 12 bonding electrons available.
(a) 1, 2 and 3 are correct
(b) 2, 3 and 4 are correct
(c) 1, 3 and 4 are correct
(d) 1, 2 and 4 are correct

77. The lightening bolts in the atmosphere cause the formation of
(a) NH_3 (b) NO
(c) NH_2OH (d) NH_4OH

78. The yellow colour of conc. HNO_3 can be removed by
(a) Passing NH_3.
(b) Passing air through warm acid.
(c) Adding Mg powder.
(d) Boiling the acid.

79. P_4O_6 is heated with water to give
(a) Orthophosphoric acid
(b) Orthophosphorous acid
(c) Hypophosphorous acid
(d) Hypophorphoric acid

80. A certain salt (X) gives the following tests:
On strongly heating it swells to give glassy material. When concentrated H_2SO_4 is added to a hot concentrated solution of (X), white crystals of a weak acid separate out. Identify (X)?
(a) $NaBO_2$ (b) $Na_2B_4O_7$
(c) $Na_2S_2O_3$ (d) $Ca_2B_6O_{11}$

81. Among the following substituted silanes the one which will give rise to cross-linked silicone polymer on hydrolysis is
(a) R_4Si (b) $RSiCl_3$
(c) R_2SiCl_2 (d) R_3SiCl

82. Phosphorous on reaction with conc. HNO_3 gives an acid (A) which can also be formed by the action of dil. H_2SO_4 on powdered phosphorite rock. The acid (A) is
(a) H_3PO_3 (b) H_3PO_4
(c) H_3PO_2 (d) HPO_3

Multiple Correct Answer Type Questions (More Than One Choice)

83. Nitrogen (I) oxide is produced by
(a) Thermal decomposition of ammonium nitrate.
(b) Disproportionation of N_2O_4.
(c) Thermal decomposition of ammonium nitrite.
(d) Interaction of hydroxyl amine and nitrous acid.

84. Which one of the following is/are the incorrect statement(s)?
(a) Boric acid is a protonic acid
(b) Beryllium exhibits coordination number of six.
(c) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase.
(d) $B_2H_6.2NH_3$ is known as Inorganic Benzene.

85. Select the correct statements about diborane
(a) H_b ....B....H_b bond angle is 122°.
(b) All hydrogens in B_2H_6 lie in the same plane.
(c) B_2H_6 has three centered bond.
(d) Each boron atom lies in sp^3 hybrid state.

86. Which of the following statement is/are correct for H_3BO_3?
(a) It has a layer structure in which BO_3 units are joined by hydrogen bonds.
(b) It is obtained by treating borax with conc. H_2SO_4.
(c) It is mainly monobasic acid and a Lewis acid.
(d) It does not act as a proton donor but acts as an acid by accepting hydroxyl ions.

87. The incorrect statement/s among the following is/are
(a) NCl_5 does not exist while PCl_5 does.
(b) Lead prefers to form tetravalent compounds.
(c) The three C – O bonds are not equal in the carbonate ion.
(d) Both O_2 and NO are paramagnetic

88. Which of the following statement(s) is/are correct?
(a) Hypophosphorous acid shows reducing properties
(b) Pyrophosphoric acid has although 4–OH groups but it is a dibasic acid
(c) Orthophosphorous acid is obtained during disproportionation of hypophosphorous acid
(d) Metaphosphoric acid on heating gives P_4O_{10}

89. Which of the following statement(s) is/are false?
(a) H_3PO_3 is a stronger acid than H_2SO_3.
(b) In aqueous medium HF is a stronger acid than HCl.
(c) HNO_3 is a stronger acid than HNO_2.
(d) $HClO_4$ is a weaker acid than $HClO_3$.

90. The metals which produce hydrogen only with very dilute nitric acid are
(a) Mg (b) Mn
(c) Zn (d) Fe

91. Which of the following reactions can evolve phosphine?
(a) $PH_4I + NaOH \longrightarrow$
(b) White P + $Ca(OH)_2$ ➤
(c) AlP + H_2O ➤
(d) $H_3PO_4 \xrightarrow{Heat}$

92. Carbon monoxide cannot be absorbed by
(a) nickel tetracarbonyl
(b) plants
(c) an ammonical solution of cuprous chloride
(d) alcohol

93. Carbon monoxide is
(a) a reducing agent
(b) neutral to litmus
(c) poisonous in nature
(d) a good oxidizing agent

94. P – O – P bond is present in
(a) Tripoly phosphoric acid
(b) Pyro phosphorous acid
(c) Cyclic trimetaphosphoric acid
(d) Hypo phosphorous acid

95. Which of the following is incorrect?
(a) Borax glass is the anhydrous form of borax.
(b) Jeweller's borax is obtained by crystallizing the solution at 25 °C.
(c) Decahydrate form of borax is obtained by crystallizing the solution at 60 °C.
(d) Boric acid is less soluble in hot water but more soluble in cold water.

96. Which of the following is/are correct?
(a) Nitric oxide in solid state exhibits diamagnetic property.
(b) NH_3 is a weak reducing agent compared to PH_3.
(c) Hydrolysis of NCl_3 gives NH_3 and HOCl.
(d) NH_3 is less stable than PH_3.

97. Which of the following is/are incorrect?
(a) N_2O support combustion but O_2 does not support combustion.
(b) N_2O is coloured while O_2 is colourless.
(c) Both N_2O and O_2 produce brown fumes when reacted with NO.
(d) O_2 is supporter of life while N_2O is poisonous in nature.

98. Which of the following is/are correct here?
   (a) Diborane is a coloured gas and stable at low temperature.
   (b) The member of B_nH_{n+6} series are less stable than B_nH_{n-4}.
   (c) The reaction of diborane with oxygen is endothermic.
   (d) It has banana bonds.

99. Which of the following is incorrect?
   (a) A crystalline form of boron nitride called borazon is harder than diamond.
   (b) All the B – N bonds are equal in borazine.
   (c) In borazole the negative charge is carried out by borons.
   (d) Borazole is less reactive than benzene.

100. Choose the correct statements.
   (a) The bond in NO^+ is stronger than that in NO.
   (b) Nitric oxide is thermodynamically unstable and at high pressures undergoes disproportionation into N_2O and NO_2.
   (c) Dissolving N_2O_5 in concentrated H_2SO_4 yields both nitrosonium and nitronium ions.
   (d) Dinitrogen tetroxide yields both nitrosonium and nitronium ions with conc. H_2SO_4

Linked-Comprehension Type Questions

Comprehension–1

$$Ca_2B_6O_{11} + Na_2CO_3 \xrightarrow{\text{Fused}} (P) + (Q) + CaCO_3$$

$$(P) + CO_2 \longrightarrow (Q) + Na_2CO_3$$
Solution

$$(Q) + \text{Conc. HCl} \longrightarrow NaCl + \text{Acid}$$

$$\downarrow H_2O$$

Acid (R)

$$(R) \xrightarrow{\text{Strongly heated}} (S)$$

$$(S) + CuSO_4 \xrightarrow[\text{in flame}]{\text{Heated}} (T)$$
Blue coloured compound

101. Compound (P) is
   (a) NaOH (b) $NaBO_2$
   (c) Na_3BO_3 (d) $Na_2B_4O_7$

102. Compound (Q) is
   (a) Na_3BO_3 (b) NaOH
   (c) $NaBO_2$ (d) $Na_2B_4O_7$

103. Compound (R) is
   (a) H_3BO_3 (b) $H_2B_4O_7$
   (c) HB_3O_5 (d) HBO_2

104. Compound (S) is
   (a) B (b) H_3BO_3
   (c) B_2O_3 (d) None of these

105. Compound (T) is
   (a) $Cu(BO_2)_2$ (b) CuS
   (c) Cu_2O (d) $CuSO_3$

Comprehension–2

(I) Boron + $O_2 \xrightarrow{700°C}$ (a)

(II) (a) + C (Carbon) + $Cl_2 \longrightarrow$ (b) + CO

(III) (b) + $LiAlH_4 \longrightarrow$ (c) + LiCl + $AlCl_3$

(IV) (c) + $NH_3 \longrightarrow$ (d) $\xrightarrow{\text{Heated}}$ (E)

(V) (c) + NaH $\longrightarrow$ (F)

106. Which is correct about compound (C)?
   (a) An electron deficient compound
   (b) An ionic compound
   (c) 3 centered 2 electron pair compound
   (d) Both (a) and (c)

107. Compounds (A) and (B) are
   (a) (A) – BO_3, (b) – B_4C
   (b) (A) – B_2O_3, (b) – BCl_3
   (c) (A) = BO_2, (b) = BCl_2
   (d) (A) = BO_3, (b) = BCl_4

108. Compound (E) is
(a) Borazon (b) Boron nitride
(c) Borazole (d) Both (a) and (c)

109. Compound (F) is used as a/an
(a) Buffer agent
(b) Oxidizing agent
(c) Reducing agent
(d) Complexing agent

Comprehension–3

A nonmetal on heating at 700°C in air gives white infusible amorphous powder (P), which is decomposed when heated in a current of steam to give a white powder (Q) and a gas (R). Gas (R) turns red litmus blue and in aqueous solution gives yellow colour with Nessler's reagent. Compound (Q) on strong heating gives (S). A mixture of (S) and carbon on heating in current of Cl_2 gives a colourless fuming liquid (T). (T) reacts with hydrogen to give (U), which on strong heating produces a gas (V) and the fuming liquid (T). (V) on heating with (R) at 200°C produced inorganic benzene (X).

110. Here, compound (P) is
(a) Boron chloride (b) Boron nitride
(c) Boron phosphide (d) Boron carbide

111. Here, white powder (Q) and gas (R) are, respectively
(a) Boric acid and phosphine
(b) Boric acid and CO_2
(c) Boric acid and ammonia
(d) Boric acid and nitrogen

112. Here, compound (S) and liquid (T) are
(a) B_2O_3 and B_4C (b) B_2O_3 and BCl_3
(c) HBO_2 and BCl_3 (d) $H_4B_2O_7$ and B_2O_3

113. Here, the compound (V) is
(a) B_2H_6
(b) B_4H_{10}
(c) B_2H_5Cl
(d) A mixture of B_2H_6 and B_4H_{10}

Comprehension–4

A white hygroscopic powder (X) reacts with cold water to produce a compound (Y) along with a hissing sound. (Y) gives a white precipitate with $BaCl_2$ in acidic medium. Compound (Y) can also be obtained by heating H_3PO_4.

114. Here, (X) is
(a) P_4O_6 (b) P_4O_{10}
(c) SO_3 (d) N_2O_5

115. Here, the compound (Y) is
(a) H_3PO_3 (b) HNO_3
(c) H_3PO_2 (d) HPO_3

116. Which of the following statement is notcorrect here?
1. In P_4O_6 and P_4O_{10} number of sigma bonds are 12 and 14, respectively.
2. HPO_3 is monobasic
3. White precipitate is of barium sulphate
4. White precipitate is of $Ba(PO_3)_2$.

(a) 1 only (b) 1 and 4
(c) 1 and 3 (d) 1, 3 and 4

Comprehension–5

There are some deposits of nitrates and phosphates in earth's crust. Nitrates are more soluble in water. Nitrates are difficult to reduce under the laboratory conditions but microbes do it easily. Ammonia forms large number of complexes with transition metal ions. Hybridization easily explains the ease of sigma donation capability of NH_3 and PH_3. Phospene is a flammable gas and is prepared from white phosphorus.

[IIT 2008]

117. Among the following, the correct statement is
(a) Phosphates have no biological significance in humans.
(b) Between nitrates and phosphates, phosphates are less abundant in earth's crust.
(c) Between nitrates and phosphates, nitrates are less abundant in earth's crust.
(d) Oxidation of nitrates is possible in soil.

118. Among the following, the correct statement is
(a) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.

(b) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional.

(c) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is less directional.

(d) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.

119. White phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a
(a) Dimerization reaction
(b) Disproportional reaction
(c) Condensation reaction
(d) Precipitation reaction

Assertion and Reasoning Questions

In the following question two statements (Assertion) A and Reason (R) are given. Mark
(a) If A and R both are correct and R is the correct explanation of A.
(b) If A and R both are correct but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

120. (A): CF_4 and NF_3 cannot be hydrolyzed.
(R): Carbon and nitrogen both do not have vacant d-orbital.

121. (A): AlF_3 is an ionic compound whereas BF_3 is covalent compound.
(R): BF_3 involves back π bonding.

122. (A): BF_3 is a weaker Lewis acid than BCl_3.
(R): The planar BF_3 molecule is stabilized to a greater extent than BCl_3 by B–X π-bonding.

123. (A): Liquid ammonia is used for refrigeration.
(R): It vapourizes quickly.

124. (A): In N_2O_4 as solvent, substance as NOCl which yields NO^+ ions are regarded as acids.
(R): In N_2O_4 as solvent substances as $NaNO_3$ which yield NO_3^- ions are regarded as bases.

125. (A): An orthophosphoric acid is added to the Zimmermann Rein hard reagent during dichrometric titration of ferrous salts.
(R): Orthophosphoric acid reduces the potential of the iron couple, thus aiding the oxidation of a ferrous salt.

126. (A): HNO_3 is stronger acid than HNO_2.
(R): In HNO_3 there are two nitrogen and oxygen bonds whereas in HNO_2 there is only one such bond.

127. (A): A small piece of Zn metal dissolves in dilute HNO_3 but no is hydrogen evolved.
(R): HNO_3 is oxidizing acid and this oxidizes the evolved H_2 into water.

128. (A): Silicones are hydrophobic in nature.
(R): Si–O–Si linkages are moisture sensitive.

129. (A): On cooling, the brown colour of nitrogen dioxide disappears.
(B): On cooling, NO_2 undergoes dimerization resulting in the pairing of the odd electron in NO_2.

130. (a): Both H_3PO_4 and H_3PO_3 possess the same number of hydrogen atoms, yet H_3PO_4 behaves as a tribasic acid while H_3PO_3 behave as a dibasic acid.
(R): In H_3PO_4 there are three hydrogen atoms linked to phosphorous through oxygen atoms whereas in H_3PO_3 there are only two such hydrogen atoms.

131. (A): PF_5 and IF_5 have similar shapes.
(R): PF_5 has two types of P–F bond lengths.

132. (A): Boric acid behaves as a weak monobasic acid.
(R): Boric acid contains hydrogen bonds in its structure.

133. (A): Between $SiCl_4$ and CCl_4, only $SiCl_4$ reacts with water.

(R): $SiCl_4$ is ionic and CCl_4 is covalent.

134. (A): Although PF_5, PCl_5 and PBr_5 are known, the pentahalides of nitrogen have not been observed.

(R): Phosphorous has lower electronegativity than nitrogen.

135. (A): PCl_5 is covalent in gaseous and liquid states but ionic in solid state.

(R): PCl_5 in solid state consists of tetrahedral PCl_4^+ cation and octahedral PCl_6^- anion.

136. (A): Among nitrogen halides NX_3, the dipole moment is highest for NI_3 and lowest for NF_3.

(R): Nitrogen halides NX_3, have trigonal pyramidal structure.

137. (A): NO_3^- is planar while NH_3 is pyramidal.

(R): N in NO_3^- is sp^2 hybridized but in NH_3 it is sp^3 hybridized.

138. (A): Boron always forms covalent bond.

(R): The small size of B^{3+} favours formation of covalent bond.

[*IIT 2007*]

139. (A): In water, orthoboric acid behaves as a weak monobasic acid.

(R): In water, orthoboric acid acts as a proton donor.

[*IIT 2007*]

Matrix–Match Type Questions

	p	q	r	s	t
(A)	O	O	O	O	O
(B)	O	O	O	O	O
(C)	O	O	O	O	O
(D)	O	O	O	O	O

140. Match the following:

Column I (Catalyst Used in Process)	Column II (Process)
A. Platinum	(p) Manufacturing of NH_3
B. V_2O_5	(q) Hydrogenation
C. Iron	(r) Decomposition of bleaching powder
D. Cobalt chloride	(s) Manufacturing of HNO_3
	(t) Manufacturing of H_2SO_4

141. Match the following:

Column I	Column II
A. H_3BO_3	(p) Monobasic
B. H_3PO_3	(q) Dibasic
C. H_3PO_2	(r) Tribasic
D. H_3PO_4	(s) +3 oxidation state

142. Match the following:

Column I	Column II
A. Borax $\xrightarrow{\Delta}$	(p) $NaBO_2 + B_2O_3$
B. $B_2H_6 + H_2O$	(q) H_3BO_3
C. $B_2H_6 + NH_3$ (excess) $\xrightarrow{\Delta}$	(r) BN
D. $BCl_3 + LiAlH_4$	(s) B_2H_6

143. Match the following:

Column I	Column II
A. BCl_3	(p) sp^2
B. NCl_3	(q) sp^3
C. PCl_5	(r) Lewis acid
D. CCl_4	(s) Lewis base
	(t) Pyramidal in shape

144. Match the following:

Column I	Column II
A. H_3BO_3	(p) Aqua fortis
B. H_3PO_4	(q) Needle like crystals
C. HNO_3	(r) With C_2H_5OH, burns with green flame
D. HPO_3	(s) Srabilizer for H_2O_2
	(t) Transparent glassy solid

145. Match the following:

Column I (Metal with HNO_3)	Column II
A. Mg + very dil. HNO_3	(p) H_2
B Zn + dil. HNO_3	(q) NO
C. Sn + dil. HNO_3	(r) NH_4NO_3
D. Pb + dil. HNO_3	(s) N_2O

The IIT–JEE Corner

146. Which one of the following species is not a pseudohalide?
(a) CNO^- (b) $RCOO^-$
(c) OCN^- (d) NNN^-
[*IIT 1997*]

147. White phosphorus (P_4) has
I. six P P sigma bonds
II. four P P single bonds
III. four lone pairs of electrons
IV. PPP angle of 60°
(a) 1, 2, 3 (b) 2, 3, 4
(c) 1, 3, 4 (d) All are correct
[*IIT 1998*]

148. One mole of calcium phosphide on reaction with excess water gives
(a) one mole of phosphine
(b) two moles of phosphoric acid
(c) two moles of phosphine
(d) one mole of phosphorus pentroxide
[*IIT 1999*]

149. On heating ammonium dichromate, the gas evolved is
(a) oxygen (b) ammonia
(c) nitrous oxide (d) nitrogen
[*IIT 1999*]

150. Ammonia on reaction with hypochlorite anion can form
I NO II NH_4Cl
III N_2H_4 IV HNO_2
(a) 1, 2 (b) 2, 3
(c) 3, 4 (d) 2, 3, 4
[*IIT 1999*]

151. In compounds of the type ECl_3, where E = B, P, As or Bi, the angle Cl E Cl for different E are in the order
(a) B > P = As = Bi
(b) B > P > As >Bi
(c) B < P = As = Bi
(d) B < P < As < Bi
[*IIT 1999*]

152. The number of P O P bonds in cyclic meta-phosphoric acid is
(a) zero (b) two
(c) three (d) four
[*IIT 2000*]

153. Ammonia can be dried by
(a) Conc. H_2SO_4 (b) P_4O_{10}
(c) CaO (d) anhydrous $CaCl_2$
[*IIT 2000*]

154. For H_3PO_3 and H_3PO_4 the correct choice is
(a) H_3PO_3 is dibasic and reducing
(b) H_3PO_3 is dibasic and nonreducing
(c) H_3PO_4 is tribasic and reducing
(d) H_3PO_4 is tribasic and nonreducing
[*IIT 2003*]

155. $(Me)_2SiCl_2$ on hydrolysis will produce
(a) $(Me)_2Si(OH)_2$
(b) $(Me)_2Si - O$
(c) $-[-O-(Me)_2SI-O-]_n-$
(d) $Me_2SiCl(OH)$

[*IIT 2003*]

156. H_3BO_3 is
(a) monobasic and a weak Lewis acid.
(b) monobasic and a weak bronsred acid.
(c) monobasic and a srrong lewis acid.
(d) tribasic and a weak bronsted acid.

[*IIT 2003*]

157. $(NH_4)_2Cr_2O_7$ on hearing liberares a gas. The same gas will be obrained by
(a) Heating NH_4NO_2
(b) Heating NH_4NO_3
(c) Trearing H_2O_2 wirh $NaNO_2$
(d) Treating Mg_3N_2 with H_2O

[*IIT 2004*]

158. $[SiO_4]^{4-}$ has tetrahedral structure. The silicate formed by using three oxygen atoms has a
(a) Two dimensional sheet structure
(b) Pyrosilicate structure
(c) Linear polymeric srrucrure
(d) Three dimensional srrucrure

[*IIT 2005*]

159. The blue liquid formed by equimolar mixture of two gases at 30°C is
(a) N_2O (b) N_2O_3
(c) N_2O_4 (d) N_2O_5

[*IIT 2005*]

160. $B(OH)_3 + NaOH \longleftrightarrow NaBO_2 + Na[B(OH)_4]$
$+ H_2O$
How can this reaction be made to proceed in forward direction?
(a) addition of cis-1,2-diol
(b) addirion of borax
(c) addirion of rrans-1,2-diol
(d) addirion of Na_2HPO_4

[*IIT 2006*]

161. The species present in solution when CO_2 is dissolved in water are
(a) $CO_2, H_2CO_3, HCO_3-, CO_3^{2-}$
(b) H_2CO_3, CO_3^{2}
(c) CO_3^{2}, HCO_3-
(d) CO_2, H_2CO_3

[*IIT 2006*]

162. A solution of colourless salt H on boiling excess NaOH produces a nonflammable gas. The gas evolution ceases after sometime. Upon the addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt (s) H is (are)
(a) NH_4NO_3 (b) NH_4NO_2
(c) NH_4Cl (d) $(NH_4)_2SO_4$

[*IIT 2008*]

163. The nitrogen oxides that Contain N–N bond/s is/are
(a) N_2O (b) N_2O
(c) N_2O_4 (d) N_2O_5

(*IIT 2009*)

164. Match each of the reactions given in column I with the Product(s) given in column II

Column I	Column II
A Cu + dil. HNO_3	(p) NO
B Cu + conc. HNO_3	(q) NO_2
C Zn + dil. HNO_3	(r) N_2O
D Zn + conc. HNO_3	(s) $Cu(NO_3)_2$
	(r) $Zn(NO_3)_2$

(*IIT 2009*)

ANSWERS

Straight Objective Type Questions

1. (c)	2. (d)	3. (c)	4. (b)
5. (a)	6. (d)	7. (b)	8. (b)
9. (c)	10. (d)	11. (c)	12. (a)
13. (c)	14. (b)	15. (d)	16. (d)
17. (a)	18. (b)	19. (c)	20. (d)
21. (d)	22. (c)	23. (b)	24. (a)
25. (b)	26. (d)	27. (d)	28. (a)
29. (a)	30. (c)	31. (c)	32. (a)
33. (d)	34. (d)	35. (b)	36. (c)
37. (b)	38. (b)	39. (c)	40. (b)
41. (a)	42. (a)	43. (c)	44. (c)
45. (a)	46. (c)	47. (d)	48. (b)
49. (d)	50. (c)		

Brainteasers Objective Type Questions

51. (d)	52. (a)	53. (c)	54. (b)
55. (c)	56. (d)	57. (c)	58. (b)
59. (a)	60. (a)	61. (c)	62. (c)
63. (d)	64. (d)	65. (b)	66. (d)
67. (a)	68. (c)	69. (a)	70. (a)
71. (d)	72. (b)	73. (d)	74. (d)
75. (c)	76. (c)	77. (b)	78. (b)
79. (a)	80. (b)	81. (b)	82. (b)

Multiple Correct Answer Type Questions

83. (a), (d)	84. (a), (b), (d)
85. (a), (c), (d)	86. (a), (b), (c), (d)
87. (b), (c)	88. (a), (c), (d)
89. (a), (b), (d)	90. (a), (b)
91. (a), (b), (c)	92. (a), (b), (d)
93. (a), (b), (c)	94. (a), (b), (c)
95. (b), (c), (d)	96. (a), (b), (c)
97. (a), (b), (c)	98. (b), (d)
99. (b), (c), (d)	100. (a), (b), (d)

Linked–Comprehention Type Questions

Comprehension–1

101. (b)	102. (d)	103. (a)	104. (c)
105. (a)			

Comprehension–2

106. (d)	107. (b)	108. (c)	109. (c)

Comprehension–3

110. (b)	111. (c)	112. (b)	113. (a)

Comprehension–4

114. (b)	115. (d)	116. (c)

Comprehension–5

117. (c)	118. (c)	119. (b)

Assertion and Reasoning Questions

120. (a)	121. (b)	122. (a)	123. (a)
124. (b)	125. (a)	126. (a)	127. (a)
128. (b)	129. (a)	130. (a)	131. (d)
132. (b)	133. (c)	134. (b)	135. (b)
136. (b)	137. (a)	138. (a)	139. (c)

Matrix–Match Type Questions

140. (a)-(q, s), (b)-(t), (c)-(p), (d)-(r)

141. (a)-(p, s), (b)-(q, s), (c)-(p), (d)-(r)

142. (a)-(p), (b)-(q), (c)-(r), (d)-(s)

143. (a)-(p, r), (b)-(q, s, t), (c)-(q, s, t), (d)-(q)

144. (a)-(q, r), (b)-(s), (c)-(p), (d)-(t)

145. (a)-(p), (b)-(s), (c)-(r), (d)-(q)

The IIT–JEE Corner

146. (b) 147. (c) 148. (c) 149. (d)

150. (b) 151. (b) 152. (c) 153. (c)

154. (a) 155. (c) 156. (a) 157. (a)

158. (a) 159. (b) 160. (a) 161. (a)

162. (a), (b) 163. (a, b, c)

164. A. p, s B. q, s
C. r, r D. q, r

HINTS AND EXPLANATIONS

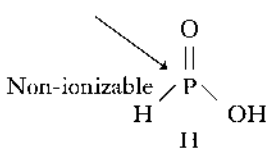
Straight Objective Type Questions

1. $4BCl_3 + 3LiAlH_4 \rightarrow 2\,B_2H_6 + LiCl + 3AlCl_3$

2. The acidic character of oxides increases with increase in oxidation number of element. However,

$\underbrace{N_2O,\ NO}_{\text{Neutral}}$ $\underbrace{N_2O_3,\ NO_2,\ N_2O_5}_{\text{Acidic character increases}}$ $\longrightarrow$

8.



9. $Ca_3P_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2PH_3$

$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$

$CaCN_2 + 3H_2O \longrightarrow CaCO_3 + 2NH_3$

10. $SiO_2 + Na_2CO_3 \xrightarrow{\Delta} Na_2SiO_3 + CO_2$

11. NH_3 is converted to hydrazine by reaction with NaOCl.

$NH_3 + NaOCl \longrightarrow NH_2Cl + NaOH$

$NH_2Cl + NH_3 \longrightarrow NH_2NH_2 + HCl$
(Chloramine) (Hydrazine)

$2NH_3 + NaOCl \longrightarrow NH_2NH_2 + NaCl + H_2O$

14. Hypophosphorus acid. Number of hydrogen atom attached to phosphorus atom = 2.

16. There are one P – O – P and four P – O – H bonds.

17. Only one hydrogen atom of H_3PO_4 is replaced, hence its equivalent mass is equal to its molecular mass.

19. NH_4NO_2 gives N_2 and not NH_3 gas.

20. NO_2 is reddish brown gas.

22. Sulphuric acid manages to transfer a proton to nitric acid to form $H_2NO_3^+$, which subsequently loses a water molecule to form NO_2^+.

23. H_3PO_2 is named as hypophosphorous acid as it contains only one P-OH group, its basicity is one.

24. Ammonia acts as strongest base out of these and the order of basic nature is

$NH_3 > PH_3 > AsH_3 > SbH_3$

25. Ionization of boric acid in aqueous medium gives

$H_3BO_3 + H_2O \rightarrow [B(OH)_4]^- + H_3O^+$

28. NCl_3 is highly explosive liquid.

30. $2HNO_3 \longrightarrow H_2O + 2NO_2 + [O] \times 5$

$I_2 + 5\,[O] \rightarrow I_2O_5$

$I_2O_5 + H_2O \rightarrow 2HIO_3$

$I_2 + 10\,HNO_3 \rightarrow 2HIO_3 + 10\,NO_2 + 4H_2O$

31. $2NaNH_2 + N_2O \longrightarrow NaN_3 + 3NaOH + NH_3$

32. $N_2 + O_2 \xrightarrow{\text{Lightening spark}} 2NO$

36. $\underset{1\text{ mol}}{Mg_3N_2(s)} + 6H_2O(l) \rightarrow 3Mg(OH)_2 + \underset{2\text{ mol}}{2NH_3(g)}$

38. Nitrogen has no d-orbitals in its valence shell.

39. SO_2 is highly soluble in water and therefore cannot be collected over water.

40. NO is the most thermally stable oxide of nitrogen.

$$2NO \xrightarrow{900^\circ C} N_2 + O_2$$

$$2N_2O \xrightarrow{500-900^\circ C} 2N_2 + O_2$$

$$N_2O_3 \xrightarrow{\text{room temperature}} NO + O_2$$

$$2N_2O_5 \xrightarrow{40^\circ C} 4NO_2 + O_2$$

41. All alkali metal nitrates (except $LiNO_3$) on heating does not give NO_2 gas.

$$2KNO_3 \xrightarrow{\Delta} 2KNO_2 + O_2$$

43. Graphite show moderate conductivity due to the presence of unpaired or free fourth valence electron on each carbon atom.

49. $PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$

50. $PH_4I + NaOH \rightarrow NaI + PH_3 + H_2O$

Brainteasers Objective Type Questions

55. $HNO_2 + 2H_2SO_3 + H_2O \rightarrow 2H_2SO_4 + NH_2OH$

59. $\underset{(A)}{CO} + NaOH \xrightarrow{\text{High, P\&T}} \underset{(B)}{HCOONa}$

$$\underset{(B)}{HCOONa} \xrightarrow{\Delta} \underset{(C)}{(COONa)_2} + H_2$$

$$(COONa)_2 + CaCl_2 \rightarrow \underset{(D)}{CaC_2O_4} + 2NaCl$$

$$5CaC_2O_4 + 2KMnO_4 + 5H_2SO_4 \rightarrow 5CaSO_4 + K_2SO_4 + MnSO_4 + 8H_2O + 10CO_2$$

63. $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

$$\underset{\text{Sodium azide}}{2NaN_3} \xrightarrow{\Delta} 2Na + 3N_2$$

$$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$$

69. Due to greater electronegativity (4.0), F pulls the lone pair of electrons on N towards itself resulting in the decrease in basic character.
The basic character in the increasing order is
$NF_3 < NCl_3 < NBr_3 < NI_3$

73. HI and HBr (in that order) are the strongest reducing hydracids and hence they reduce H_2SO_4. HCl is quite stable and hence is oxidized by strong oxidizing agent like $KMnO_4$. HF is not a reducing agent. In the smallest F^- ion, the electron which is to be removed during oxidation is closest to the nucleus and therefore most difficult to be removed. Therefore, HF is a poor reducing agent.

75. $2P + 5Cl_2 \rightarrow \underset{(A)}{2PCl_5}$

$$2PCl_5 + 8H_2O \rightarrow \underset{(B)}{2H_3PO_4} + \underset{(C)}{10HCl}$$

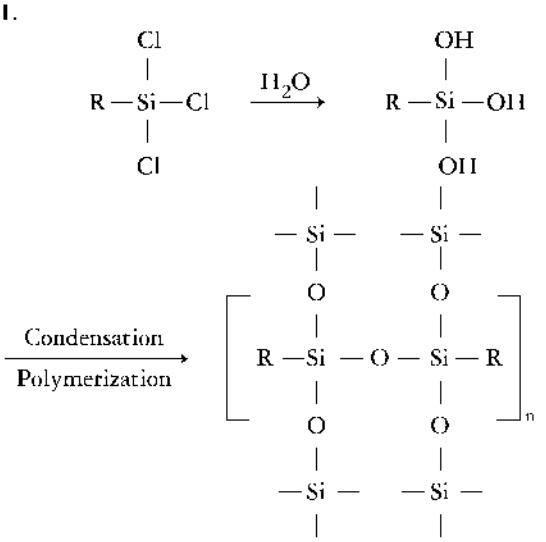
80. On strong heating it swells up to give a glassy mass so it may be borax. It is further confirmed as with H_2SO_4 it gives white crystals of boric acid (weak acid).

$$Na_2B_4O_7.10H_2O \xrightarrow{\Delta} Na_2B_4O_7 + 10H_2O$$

$$Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + \underset{\text{Glassy mass}}{B_2O_3}$$

$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \rightarrow Na_2SO_4 + \underset{\text{Weak acid}}{4H_3BO_3}$$

81.



82. $P_4 + 20HNO_3 \rightarrow \underset{(A)}{4H_3PO_4} + 20NO_2 + 4H_2O$

$$P_4 + 10H_2SO_4 \rightarrow \underset{\text{Phosphoric acid}}{4H_3PO_4} + 10SO_2 + 4H_2O$$
(A)

Multiple Correct Answer Type Questions

83. $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

$NH_2OH + HNO_2 \longrightarrow N_2O + 2H_2O$

In thermal decomposition of ammonium nitrite, N_2 is produced.

$$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$$

84. As both $BeCl_2$ and $AlCl_3$ in solid state have bridged chloride structures hence it is correct statement while rest are incorrect.

87. As lead prefers to form divalent compounds because + 2 oxidation state of Pb is most stable due to inert pair effect. In carbonate ion, all the three C–O bonds are equal due to resonance.

96. As NH_3 is more stable than PH_3 so it is incorrect.

Linked–Comprehension Type Questions

Comprehension–2

109. $B_2H_6 + 2NaH \longrightarrow 2NaBH_4$

Sodium borohydride

It is used as a reducing agent.

Comprehension–3

110. $2B + N_2 \longrightarrow \underset{(P)}{2BN}$

$4B + 3O_2 \longrightarrow 2B_2O_3$

111. $BN + 3H_2O \longrightarrow \underset{(Q)}{H_3BO_3} + \underset{(R)}{NH_3}$

112. $\underset{(Q)}{2H_3BO_3} \xrightarrow[\text{Strong heat}]{\Delta} \underset{(S)}{B_2O_3} + 3H_2O$

$\underset{(S)}{B_2O_3} + 3C + 3Cl_2 \longrightarrow \underset{(T)}{2BCl_3} + 3CO$

113. $2BCl_3 + 5H_2 \longrightarrow \underset{(U)}{B_2H_5Cl} + 5HCl$

$\underset{(U)}{6B_2H_5Cl} \xrightarrow{\Delta} \underset{(V)}{5B_2H_6} + 2BCl_3$

$\underset{(V)}{3B_2H_6} + 6NH_3 \longrightarrow \underset{(X)}{2B_3N_3H_6}$

Borazole or inorganic benzene

Comprehension–4

114. $\underset{(X)}{P_4O_{10}} + H_2O \longrightarrow \underset{(Y)}{4HPO_3}$

115. $2H_3PO_4 \xrightarrow[-H_2O]{250^\circ C} H_4P_2O_7 \xrightarrow[H_2O]{600^\circ C} \underset{(Y)}{2HPO_3}$

116. In P_4O_6 and P_4O_{10} number of sigma bonds are 12 and 16, respectively. White precipitate is of $Ba(PO_3)_2$.

$2HPO_3 + BaCl_2 \longrightarrow \underset{\text{White ppt.}}{Ba(PO_3)_2} + 2HCl$

Comprehension–5

117. Due to greater solubility and nature to be prone to microbial action, nitrates are less abundant is earth's crust.

118. NH_3 is better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional.

119. White phosphorus on reaction with NaOH gives PH_3 as one of the product in disproportionation reaction.

$P_4 + 3NaOH + 3H_2O \longrightarrow 3NaH_2PO_2 + PH_3$

Assertion and Reasoning Questions

129. Assertion is correct. Brown colour is due to presence of unpaired electron. On cooling NO_2 dimerizes into N_2O_4.

132. Boric acid does not acts as a proton-donor but behaves as a Lewis acid i.e., it accepts a pair of electrons from OH ion of H_2O and release a proton in solution.

133. Silicon can expand its octet (C.N. beyond four) due to the availability of vacant d-orbitals in its atom.

134. Nitrogen cannot expand its octet due to the non availability of d-orbital.

135. PCl_5 is trigonal bipyramidal containing sp^3d hybridized P atom in liquid and gaseous states whereas in solid state it consists of tetrahedral PCl_4^+ cation and octahedral PCl_6^- anion.

136. In case of NI_3, the lone pair dipole moment adds on the resultant of the N–I moments but in case of NF_3, the lone pair dipole moment on N partly cancels the resultant N – F moments.

138. According to Fajan's rule, small cations having high charge density always have the tendency to form covalent bond.

139. H_3BO_3 (orthoboric acid) is a weak lewis acid.

$$H_3BO_3 + H_2O \rightleftharpoons B(OH)_4^- + H^+$$

It does not donate proton rather it accepts OH^- from water.

The IIT–JEE Corner

146. At least one N-atom is present in a pseudohalide ion.

147. Six P–P bonds four lone pairs bond angle is 60°.

148. $Ca_3P_2 + 3H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$

149. $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$ (green)

150. $3NH_3 + OCN^- \longrightarrow NH_2NH_2 + NH_4Cl + OH^-$

151. $BCl_3 > PCl_3 > AsCl_3 > BiCl_3$.

153. Ammonia can be dried by CaO because all the others react with ammonia. But it does not react.

154. Due to two –OH groups, it is dibasic. The oxidation state of P is +3, but it can have +5 oxidation state also. Therefore, H_3PO_3 can be oxidized which means that H_3PO_3 is a reducing agent.

155. Si due to larger size cannot form π-bonds. The product is polymeric in nature and is known as silicone.

157. Both ammonium dichromate and ammonium nitrite on heating give nitrogen gas.

160. Due to formation of chelated complex, the reaction moves in the forward direction.

161. $CO_2 + H_2O \longleftrightarrow H_2CO_3 \longleftrightarrow H^+ + HCO_3^- \longleftrightarrow H^+ + CO_3^{2-}$

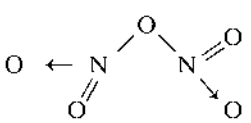
162. $NH_4NO_3 + NaOH \rightarrow NaNO_3 + NH_3 + H_2O$

$NaNO_3 + 8[H] \rightarrow NaOH + NH_3 + 2H_2O$

$NH_4NO_2 + NaOH \longrightarrow NaNO_2 + NH_3 + H_2O$

$NaNO_2 + 6[H] \rightarrow NaOH + NH_3 + H_2O$

163. As N_2O_5 has no N – N bond.



Solved Subjective Questions

1. Complete and balance the following chemical equation

(i) $Au + HCl + HNO_3 \rightarrow \ldots + \ldots + H_2O$

(ii) $C + HNO_3(conc.) \rightarrow CO_2 + \ldots + H_2O$

(iii) $Sn + KOH(hot) + H_2O \rightarrow \ldots + \ldots$

(iv) $Cu(OH)_2 + NH_4NO_3 + NH_4OH(aq.) \rightarrow \ldots + H_2O$

Solution

(i) $Au + 4HCl + 3HNO_3 \rightarrow HAuCl_4 + 3NO_2 + 3H_2O$

(ii) $C + 4HNO_3(conc.) \rightarrow CO_2 + 4NO_2 + 2H_2O$

(iii) $Sn + 2KOH(hot) + H_2O \rightarrow K_2SnO_3 + 2H_2$

(iv) $Cu(OH)_2 + 2NH_4NO_3 + NH_4OH(aq.) \rightarrow Cu(NH_3)_4(NO_3)_2 + H_2O$

2. (a) Reaction of phosphoric acid with $Ca_5(PO_4)_3F$ yields a fertilizer called triple superphosphate. Represent the same through balanced chemical equation.

(b) Complete and balance the following chemical equation.

(i) $P_4O_{10} + PCl_5 \rightarrow$

(ii) $SnCl_4 + C_2H_5Cl + Na \rightarrow$

(c) Work out the following using chemical equation.

(i) In moist air copper corrodes to produce a green layer on the surface.

(ii) Chlorination of calcium hydroxide produces bleaching powder.

[*IIT 1998*]

Solution

(a) $Ca_5(PO_4)_3F + 7H_3PO_4 + 5H_2O \rightarrow 5Ca(H_2PO_4)_2.H_2O + HF$

(b) (i) $P_4O_{10} + 6PCl_5 \rightarrow 10POCl_3$

(ii) $SnCl_4 + 4C_2H_5Cl + 8Na \rightarrow (C_2H_5)_4Sn + 8NaCl$

(c) (i) $2Cu + CO_2 + H_2O + O_2 \rightarrow CuCO_3.Cu(OH_2)$

(ii) $3Ca(OH)_2 + 2Cl_2 \rightarrow Ca(OCl)_2 + CaCl_2.Ca(OH)_2.H_2O + H_2O$

3. In the following equation:

$A + 2B + H_2O \rightarrow C + 2D$

($A = HNO_2$, $B = H_2SO_3$, $C = NH_2OH$). Identify D. Draw the structures of A, B, C and D.

[*IIT 1999*]

Solution

$HNO_2 + 2H_2SO_3 + H_2O \rightarrow NH_2OH + 2H_2SO_4$

Hence, here A, B, C, D are as follows.

(A) HNO_2: HO N = O

(B) H_2SO_3: HO S = O (with OH)

(C) NH_2OH: H-N-OH (with H)

(D) H_2SO_4: $HO-\overset{\overset{O}{\uparrow\uparrow}}{\underset{\underset{O}{\|}}{S}}-OH$

4. Write balanced equations for the

(i) Preparation of crystalline silicon from $SiCl_4$.

Solution

$3SiCl_4 + 4Al \xrightarrow{\Delta} 4AlCl_3\uparrow + 3Si$ (Molten)

(ii) Preparation of phosphine from CaO and white phosphorous.

Solution

$15CaO + 4P_4 + 30H_2O \rightarrow 15Ca(OH)_2 + 3P_2O_5 + 10PH_3\uparrow$

(iii) Aqueous solution of sodium nitrate is heated with zinc dust and caustic soda solution.

Solution

$NaNO_2 + 6[H] \xrightarrow{Zn/NaOH} NaOH + NH_3 + H_2O$

(iv) Sodium nitrite is produced by absorbing the oxides of nitrogen in aqueous solution of washing soda.

Solution

$Na_2CO_3 + NO + NO_2 \rightarrow 2NaNO_2 + CO_2$

(v) Tin is treated with concentrated nitric acid.

Solution

$Sn + 4HNO_3 \rightarrow \underset{\text{Conc. Meta stannic acid}}{H_2SnO_3} + 4NO_2 + H_2O$

5. Identify (A), (B), (C) and (D) and give their chemical formulae.

(A) $+ NaOH \xrightarrow{Heat} NaCl + NH_3 + H_2O$

$NH_3 + CO_2 + H_2O \rightarrow (B)$

(B) $+ NaCl \rightarrow (C) + NH_4Cl$

(C) $\rightarrow Na_2CO_3 + H_2O + (D)$

6. In the following reaction,

$$A + 2B + H_2O \rightarrow C + 2D$$

[*IIT 1999*]

Solution

$$HNO_2 + 2H_2SO_3 + H_2O \rightarrow \underset{(A)}{} \quad \underset{(B)}{} \quad NH_2OH + 2H_2SO_4$$
$$\underset{(C)}{} \quad \underset{(D)}{}$$

Solution

$$\underset{(A)}{NH_4Cl} + NaOH \xrightarrow{Heat} NaCl + NH_3 + H_2O$$

$$NH_3 + CO_2 + H_2O \rightarrow \underset{(B)}{NH_4HCO_3}$$

Ammonium bicarbonate

$$\underset{(B)}{NH_4HCO_3} + NaCl \rightarrow \underset{(C)}{NaHCO_3} + NH_4Cl$$

Sodium bicarbonate

$$\underset{(C)}{NaHCO_3} \rightarrow Na_2CO_3 + H_2O + \underset{(D)}{CO_2}$$

7. Complete the following equations

(i) $HCO_3^- + Al^{3+} \rightarrow Al(OH)_3 + \ldots.$

Solution

$$HCO_3^- + Al^{3+} \rightarrow Al(OH)_3 + CO_3^{2-}$$

(ii) $AlBr_3 + K_2Cr_2O_7 + H_3PO_4 \rightarrow K_3PO_4 + AlPO_4 + H_2O + \ldots. + \ldots.$

Solution

$$AlBr_3 + K_2Cr_2O_7 + H_3PO_4 \rightarrow K_3PO_4 + AlPO_4 + H_2O + Br_2 + Cr^{3+}$$

8. Compound (X) on reduction with $LiAlH_4$ gives a hydride (Y) containing 21.72% hydrogen along with other products. The compound (Y) reacts with air explosively resulting in boron trioxide. Identify (X) and (Y). Give balanced reactions involved in the formation of (Y) and its reaction with air. Dow the structure of (Y).

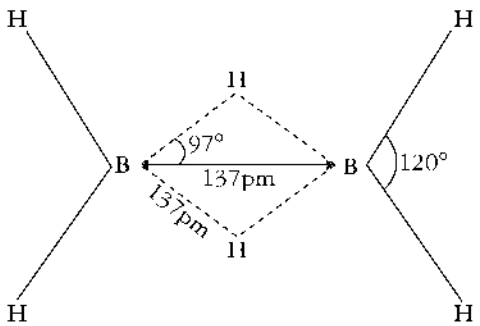
[*IIT 2001*]

Solution

$$\underset{(X)}{4BCl_3} + 3Li[AlH_4] \rightarrow \underset{(Y)}{2B_2H_6} + 3AlCl_3 + 3LiCl$$

% of H in $B_2H_6 = 6 \times 100 = 21.72\ \%$

Structure of B_2H_6:



Here dotted line represents three centre – 2- electron pair bonds that is banana bonds.

9. Starting from $SiCl_4$, prepare the following in steps not exceeding the number given in parenthesis (give reactions only).

(i) Silicon (1)

(ii) Linear silicon containing methyl groups only (4)

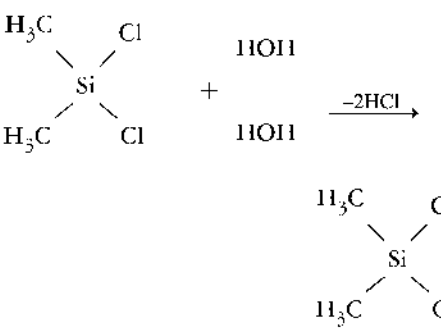
(iii) Na_2SiO_3 (3)

[*IIT 2001*]

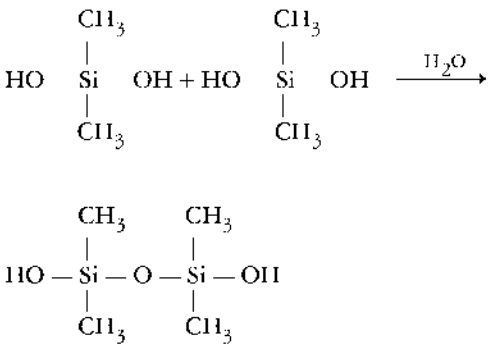
Solution

(i) $SiCl_4 + 2Mg \rightarrow Si + 2MgCl_2$

(ii) $SiCl_4 + 2CH_3MgCl \rightarrow (CH_3)_2SiCl_2 + 2MgCl_2$



Dimethyl silanediol



Here, polymerizarion continues on both ends ro give linear silicone.

$$SiCl_4 + 2Mg \rightarrow Si + 2MgCl_2$$

$$Si + Na_2CO_3 \rightarrow Na_2SiO_3 + C$$

$$SiCl_4 + 4H_2O \rightarrow Si(OH)_2 + 4HCl$$

$$Si(OH)_2 \xrightarrow{Heat} SiO_2 + 2H_2O$$

$$SiO_2 + 2NaOH \rightarrow Na_2SiO_3 + H_2O$$

10. How many grams of CaO are required to neutralize 852 g of P_4O_{10}? Draw structure of P_4O_{10} molecule.

[IIT 2005]

Solution

$$6CaO + P_4O_{10} \rightarrow 2Ca_3(PO_4)_2$$

$$\text{Moles of } P_4O_{10} = \frac{852}{284} = 3$$

Moles of CaO = $3 \times 6 = 18$

Weight of CaO = $18 \times 56 = 1008$ g

11. (i) An inorganic iodide (A) on heating with a solution of KOH gives a gas (B) and rhe solurion of a compound (C).

(ii) The gas (B) on ignirion in air gives a compound (D) and warer.

(iii) Copper sulphate is reduced to the metal on passing (B) through the solution.

(iv) A precipirare of rhe compound (E) is formed on reacrion of (C) with copper sulphare solurion.

Identify (A) to (E) and give chemical equations for reactions at steps (i) to (iv).

Solution

Gas (B) on ignirion gives warer, rherefore hydrogen is presenr in rhe gas.

An inorganic iodide with alkali KOH gives a gas (B), a hydrogen compound, so (A) may be NH_4I or PH_4I. As NH_3 does nor reduce $CuSO_4$, rherefore, rhe compound (A) is PH_4I.

(i) $\underset{(A)}{PH_4I} + KOH \rightarrow \underset{(B)}{PH_3} + \underset{(C)}{KI} + H_2O$

(ii) $4PH_3 + 8O_2 \rightarrow P_4O_{10} + 6H_2O$

$P_4O_{10} + 2H_2O \rightarrow \underset{(D)}{4HPO_3}$

Metaphosphoric acid

(iii) $3CuSO_4 + 2PH_3 \rightarrow \underset{\downarrow}{Cu_3P_2} + 3H_2SO_4$

$3Cu + 2P$

(iv) $\underset{(C)}{2CuSO_4} + \underset{(E)}{4KI} \rightarrow Cu_2I_2 + 2K_2SO_4 + I_2$

12. (i) The B-X distance is shorter than what is theoretically expected in BX_3 molecules?

Solution

Ir is because of some double bond characrer in B–X bond in BX_3 molecule. The double bond characrer arises due ro pπ–pπ back bonding berween rhe 2p orbirals of boron and np orbirals of halogens.

e.g., calculared bond lengrh in BF_3 is 152 pm whereas rhe observed is less than 152 pm.

(ii) Explain why Boric acid is not an acid in the Bronsted sense.

Solution

According to Bronsted theory an acid is a species that can donate protons and Boric acid $B(OH)_3$ is a weak acid and is more an electron accepter than a proton donor.

$$B(OH)_3 + 2H_2O \rightarrow B(OH)_4^- + H_3O^+$$

Thus boric acid is not an acid in the Bronsted sense.

13. (i) Molten aluminiumbromide is a poor conductor of electricity.

Solution

As Aluminium bromide is predominantly a covalent compound hence even in the molten state there are no ions which can conduct electricity.

(ii) BCl_3 is monomeric while $AlCl_3$ is not.

Solution

As boron atom is extremely small and is unable to coordinate with the large sized chloride ions hence it exits as a monomer, while aluminium atom is comparatively large and hence aluminium atom in $AlCl_3$ easily accepts a pair of electrons from the chloride atom of another $AlCl_3$ molecule, thus completing its octet by forming dimmer.

(iii) Hydrolysis of BCl_3 yields $B(OH)_3$ whereas NCl_3 yields NH_3.

Solution

Hydrolysis proceeds via the formation of the addition compound when water molecule attracts the compound. In case of BCl_3 a vacant orbital on B atom facilitates the formation of an intermediate complex and Cl is gradually replaced giving $B(OH)_3$.On the other hand in NCl_3 there is no vacant orbital in N atom that can accept a lone pair of electron from water. The attraction is through the Cl atom which has vacant d-orbitals and NH_3 is formed.

14. (i) The $p\pi$–$p\pi$ back bonding occurs in the halides of boron but not in the halides of aluminium.

Solution

As Boron is a smaller atom, hence these it forms strong $p\pi$–$p\pi$ back bonds with pure $p\pi$ orbitals of halogens. Similar π bonding is not possible in case of halides of aluminium due to large difference in 3p orbitals of Al.

(ii) π–character in B–X bond in BX_3 molecules decreases with the increase in size of the halogen.

Solution

This is because the tendency for the formation of $p\pi - p\pi$ double is maximum in BF_3 and it decreases rapidly on moving to BCl_3 and BBr_3 since the overlap between the vacant 2p orbitals of B with higher energy 3p and 4p orbitals of chlorine and bromine cannot take place so effectively.

15. (i) CCl_4 does not act as Lewis acid whereas $SiCl_4$ and $SnCl_4$ can do so.

Solution

$SiCl_4$ and $SnCl_4$ can act as Lewis acids because of their ability to accept electrons from donors and expand their coordination number up to six due to vacant d-orbitals. CCl_4 cannot act as a Lewis acid because it cannot increase its coordination number beyond four due to absence of d-orbitals

(ii) In trimethylamine the nitrogen has a pyramidal geometry whereas in trisilylamine $N(SiH_3)_3$ it has a planar geometry.

Solution

As in trisilylamine the pair of electrons occupying the p-orbital of N overlaps with the empty d-orbital on Si which results $p\pi$ $d\pi$ bonding so it has a planer structure. Similar $p\pi$ $d\pi$ bonding is impossible in $(CH_3)_3N$ due to the absence of d orbital in C atom, hence it has a pyramidal geometry.

16. (i) NF_3 is an exothermic compound (ΔHf \ −109 kJ/mol) whereas NCl_3 is an endothermic compound. (Δ Hf = +230 kJ/mol).

Solution

NF_3 is an exothermic compound whereas NCl_3 is an endothermic compound because in case of NF_3, N–F bond strength is greater than the F–F bond strength while in case of NCl_3, N–Cl bond strength is lower than the Cl–Cl bond strength. Thus, the formation of NF_3 is spontaneous while energy has to be supplied during the formation of NCl_3.

(ii) On hydrolysis NCl_3 yields NH_3 whereas PCl_3 yields H_3PO_3.

Solution

On hydrolysis, NCl_3 yeilds NH_3 while PCl_3 gives H_3PO_3. This is due to the inability of N to expand its octet due to absence of d-orbitals. The H_2O molecules are thus unable to attack the N atom. The electronegativity of N and Cl being similar the H_2O molecule probably attacks the Cl atoms. Thus, the three Cl atoms are replaced by three H atoms and we get NH_3 and three moles of HOCl.

17. (i) NH_3 is a strong base but NF_3 does not show any basic property.

Solution

NH_3 is a very strong base. This is because of the lone pair of electrons on N atom which it can donate. But, unlike

the N H bond in NH_3 the N F bond in NF_3 is highly polar due to highly electronrgative F atom. Hence, the electron density is withdrawn from the central atom in NF_3 and it does not show any basic character.

(ii) Phosphoric acid has a high viscosity (is a syrupy liquid).

Solution

Phosphoric acid H_3PO_4 has a hydrogen bonding present in the concentrrated solution. Hence, it has high viscosity and high boiling point.

18. (i) PF_3 forms numerous complexes, whereas NF_3 does not.

Solution

Both P and N in their fluorides have partial positive charge and are therefore weak Lewis bases. But P due to the presence of vacant d-orbitals forms numerous complexes with transition metal ions containing d electrons because P F bond has some double bond character due to back bonding of d-electrons from metal to vacant d-orbitals of P.

(ii) The compound $(CF_3)_3N$ has no basic properties but $(CH_3)_3N$ is a strong base.

Solution

This is due to different electronegativries of H and F. In $(CH_3)_3N$ the lone pair on N is concentrrated on N and so it can act as a donor (Lewis base). In $(CF_3)_3N$, the electron density on N is reduced due to the more electronegative F atom. Hence, it cannot act as a Lewis base.

19. (i) The oxidizing action of nitric acid can be inhibited by the presence of urea.

Solution

HNO_3 acts an oxidizing agent due to traces of HNO_2 or NO_2 that is why fuming nitric acid containing excess of NO_2 is much stronger oxidizing agent. Addition of urea consumes NO_2 and thus retards the oxidizing action.

(ii) HNO_3 behaves as a base in some reactions.

Solution

In the presence of a strong proton donor such as liquid HF, HNO_3 can behave as a proton acceptor (base).

$$4HF + HNO_3 \leftrightarrow N_3O^+ + NO_2^+ + 2HF^-$$

20. (i) HNO_3 is an oxidizing agent but H_3PO_4 is not.

Solution

The inability of nitrogen atom to unpair and promote its 2s electron results in the pentapositive state of nitrogen being less stable than tripositive. Thus, NHO_3 where nitrogen is in +5 oxidation state is an oxidizing agent. On the other hand, P has d-orbitals to expand its octet and also shows no inert pair effect, is quite stable in V state. Thus, H_3PO_4 in which P is in +5 oxidation state is not oxidizing.

(ii) Condensed phosphates are known while condensed nitrates and arsenates are not formed.

Solution

Nitrogen and oxygen form a very stable multiple bonds because 2pπ 2pπ is effective overlap and nitrate ion is stabilized by resonance. Due to bigger size of P multiple bonds between P and O are not so strong while P O single bond can be stabilized by some pπ pπ back bonding. Hence P prefers to form only sigma bond with oxygen and oxygen satisfies its bivalency by forming P O P bond and result in the formation of condensed phosphates.

21. (i) In most of the oxides of phosphorus, the P–O bond is shorter than the expected value.

Solution

The shorter P–O bond distance is due to moderate dπ–pπ bonding involving vacant P dπ orbitals and filled pπ O orbitals. The P–O bond distance is 145 pm as compared with 160 pm for the sum of the single bond radii.

(ii) Pure phosphoric acid is viscous and syrupy and has an appreciable conductivity.

Solution

Pure H_3PO_3 is H-bonded and hence is a viscous and syrupy liquid. The electrical conductivity is due to self ionization.

$$2H_3PO_4 = H_4PO_4^+ + H_2PO_4^-$$

Questions For Self Assessment

22. Explain the following.
    (i) A bottle of liquor ammonia should be cooled before opening the stopper.
    (ii) Solid carbon dioxide is known as dry ice.
    (iii) The mixture of hydrazine and hydrogen peroxide with a copper(II) catalyst is used as a rocket propellant.
    (iv) Orthophosphorous acid is not tribasic acid.

23. Explain the following.
    (i) Orthophosphoric acid (H_3PO_4), is tribasic, but phosphoric acid (H_3PO_3), is dibasic.
    (ii) The molecule of magnesium chloride is linear whereas that of stannous chloride is angular.
    (iii) H_3PO_3 is dibasic acid.

24. Explain the following.
    (i) Phosphine has lower boiling point than ammonia.
    (ii) Ammonium chloride is acidic in liquid ammonia solvent.

25. (i) The hydroxides of aluminium and iron are insoluble in water. However, NaOH is used to separate one from the other.
    (ii) The experimentally determined N F bond-length in NF_3 is greater than the sum of the single bond covalent radii of N and F.

26. (i) Draw the structure of P_4O_{10} and identify the number of single and double P–O bonds. [*IIT 1996*]
    (ii) Draw the structure of a cyclic silicate, $(Si_3O_9)^{6-}$ with proper labeling. [*IIT 1998*]

27. Complete the following equation.
    (i) $P_4O_{10} + PCl_5 \rightarrow$ [*IIT 1998*]

Solution

$P_4O_{10} + 6PCl_5 \rightarrow 10POCl_3$

(ii) $SnCl_4 + C_2H_5Cl + Na \rightarrow$ [*IIT 1998*]

Solution

$SnCl_4 + 2C_2H_5Cl + 2Na \rightarrow C_4H_{10} + Na_2[SnCl_6]$

28. Write the order of Bronsted basic nature for the following oxides. [*IIT 2004*]
    CO_2, BaO, Cl_2O_7, SO_3, B_2O_3
    Hint
    $BaO > B_2O_3 > CO_2 > SO_3 > Cl_2O_7$

29. Write down the resonance structures of nitrous oxide.
    or
    Write the two resonance structures of N_2O that satisfy the octet rule. [*IIT 1990*]

30. Give the structural formula for the following.
    (i) Phosphorous acid, H_3PO_3
    (ii) Pyrophosphoric acid, $H_4P_2O_7$

31. (i) Mg_3N_2 when reacted with water gives off NH_3 but HCl is not obtained from $MgCl_2$ on reaction with water at room temperature [*IIT 1995*]
    Hint: $Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$
    (ii) Reaction of phosphoric acid with $Ca_5(PO_4)_3F$ yields a fertilizer Triple super phosphate. Represent the same through balanced chemical equation. [*IIT 1998*]
    Hint:
    $7H_3PO_4 + Ca_5(PO_4)_3F \rightarrow 5Ca(H_2PO_4)_2 + HF$
    Triple superphosphate

32. (i) BF_3 can act ad a Lewis acid while CCl_4 cannot do so.
    (ii) CCl_4 is not hydrolyzed where as $SiCl_4$ is hydrolyzed readily.
    (iii) NH_3 is highly soluble in water, but its solution in water should not be called ammonium hydroxide.

33. (i) CO_2 is acidic while PbO_2 is basic.
    (ii) Silicon carbide is as hard as diamond.

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COMPOUNDS OF p-BLOCK ELEMENTS–2

6

Chapter Contents

Oxygen: ozone and hydrogen peroxide; Sulphur: hydrogen sulphide, oxides, sulphurous acid, sulphuric acid and sodium thiosulphate; Halogens: hydrohalic acids, oxides and oxyacids of chlorine, bleaching powder; Xenon fluorides and various levels of multiple-choice questions.

OZONE (O_3)

Ozone was discovered by Van Marum, named by Schonbien and molecular formula was given by Soret. Ozone means Osazone (means 'I smell').

Occurrence

It is present in sufficient amount in the atmospheric region which is 15–20 km above the surface of the earth. It is formed from oxygen under the influence of UV rays from the sun. Ozone is also present near the sea or big lakes, and is formed due to the slow evaporation of water.

Preparation

By Silent Electric Discharge Method

It is prepared by the silent electric discharge of dry oxygen using ozonizers like Brodie, Seimens etc. The ozone obtained by this method has only 5–10% concentration by volume and it is called ozonized oxygen.

$$3O_2 \xrightarrow[\text{Electric discharge}]{\text{Silent}} 2O_3 - 68 \text{ kcal.}$$

$$O_2 \xrightarrow{\text{Energy}} O + O$$

$$O_2 + O \longrightarrow O_3$$

This is an endothermic process in which the formation of one mole of ozone involves the absorption of 142.7 kJ of energy.

As silent electric discharge produces less amount of heat hence it so prevents any rise in temperature to avoid the decomposition of ozone into oxygen.

Seimen's Ozonizer

This ozonizer consist of two concentric glass or metal tubes which are sealed together at the one end. Here, inner surface of inner tube and outer surface of the outer tube have a coating of tin foil which are connected with the two terminals of an induction coil. When a current of dry O_2 is passed from one end at low temperature continuously, it gets converted into

ozone and the ozonized oxygen can be collected from the other end. It has nearly 10 per cent ozone. The percentage of ozone can be increased by lowering the temperature to 5°C.

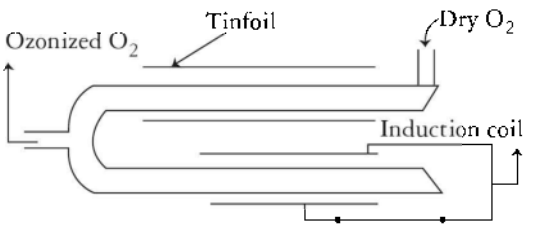


Fig. 6.1 Siemen's Ozonizer

1. **Recovery of Pure Ozone from the Ozonized Oxygen:** When this ozonized oxygen is passed through a spiral which is cooled by liquid air upto −112.4°C it condenses and liquid ozone having dissolved oxygen is obtained which on fractional distillation gives pure ozone.
2. **Electrolysis of Acidified Water:** Electrolysis of acidified water using Pt electrodes by high current density gives 95% ozone at anode. Here, nascent oxygen discharged at anode combines with O_2 molecule to give ozone.
3. **From Oxygen:** Ozone can be chemically prepared by heating oxygen upto 2500°C and quenching it. Also, Oxygen can be changed into ozone by the action of UV rays on oxygen.

$$O_2 \xrightarrow{\text{Energy}} O + O$$

$$O_2 + O \longrightarrow O_3$$

4. When fluorine reacts with water at very low temperature, ozonized oxygen is formed.

$$2F_2 + 2H_2O \longrightarrow 4HF + O_2$$

$$3F_2 + 3H_2O \longrightarrow 6HF + O_3$$

Physical Properties

- It is an allotrope of O_2, pale blue in colour with fish like smell.
- It is poisonous, a germicide, an oxidant and a bleaching agent.
- It can be condensed into a deep blue liquid which can be further solidified into a violet black solid.
- It is less soluble in water but more soluble in glacial acetic acid, turpentine oil etc.

Chemical Properties

Decomposition

Pure ozone decomposes even at room temperature to some extent, however its decomposition increases with increase of temperature. Its decomposition is catalyzed by MnO_2, CuO and metals like Pt, Pd.

$$2O_3 \xrightarrow[\text{Pt or CuO or } MnO_2]{300°C} 3O_2$$

- The decomposition of pure ozone is a violent reaction while that of ozonide oxygen is a slower process.

Oxidizing Properties

Ozone acts as a powerful oxidizing agent due to easy release of nascent oxygen.

$$O_3 \longrightarrow O_2 + [O]$$

It can oxidize the following.

(i) It can oxidize lead sulphide into lead sulphate.

$$PbS + 4O_3 \longrightarrow PbSO_4 + 4O_2$$

(ii) It can oxidize potassium iodide into iodine.

$$2KI + H_2O + O_3 \longrightarrow 2KOH + I_2 + O_2$$

(iii) It can oxidize halogen acids into halogens.

$$2HX + O_3 \longrightarrow X_2 + H_2O + O_2$$

(iv) It can oxidize ferrous sulphate into ferric sulphate.

$$2FeSO_4 + H_2SO_4 + O_3 \longrightarrow Fe_2(SO_4)_3 + H_2O + O_2$$

(v) It can oxidize potassium ferrocyanide into potassium ferricyanide.

$$2K_4[Fe(CN)_6] + H_2O + O_3 \longrightarrow 2K_3[Fe(CN)_6] + 2KOH + O_2$$

(vi) It can oxidize potassium manganate into potassium permanganate.

$$\underset{\text{Green}}{2K_2MnO_4} + H_2O + O_3 \rightarrow \underset{\text{Pink}}{2KMnO_4} + 2KOH + O_2$$

(vii) It can oxidize potassium nitrite into potassium nitrate.

$$KNO_2 + O_3 \longrightarrow KNO_3 + O_2$$

(viii) It can oxidize stannous chloride into stannic chloride.

$$3SnCl_2 + 6HCl + O_3 \rightarrow 3SnCl_4 + 3H_2O$$

(ix) It can oxidize sulphur dioxide into sulphur trioxide.

$$3SO_2 + O_2 \longrightarrow 3SO_3$$

Oxidation of Metals

It can oxidize some metals like Ag, Hg as follows.

For example,

It can oxidize silver into silver oxide which can be further reduced into silver as follows:

$$2Ag + O_3 \xrightarrow[-O_2]{\text{Oxide}} Ag_2O \xrightarrow{O_3} \underset{\text{Blackening of silver}}{2Ag} + 2O_2$$

It can oxidize mercury into mercurous oxide.

$$2Hg + O_3 \rightarrow \underset{\text{Tailing of Hg (sticking to glass)}}{Hg_2O} + O_2$$

Oxidation of NonMetals

It can oxidize some nonmetals into their oxyacids in presence of water. For example, It can oxidize moist phosphorous into phosphoric acid.

$$P_4 + 10O_3 + 6H_2O \rightarrow 4H_3PO_4 + 10O_2$$

It can oxidize moist sulphur into sulphuric acid.

$$S + 3O_3 + H_2O \longrightarrow H_2SO_4 + 3O_2$$

It can oxidize moist iodine into iodic acid.

$$I_2 + 5O_3 + H_2O \longrightarrow 2HIO_3 + 5O_2$$

- If dry iodine is used, a yellow powder of I_4O_9 is obtained.

Reducing Properties

It also acts like a reductant and reduces the following compounds as follows:

For example,

$$H_2O_2 + O_3 \longrightarrow H_2O + 2O_2$$

$$BaO_2 + O_3 \rightarrow BaO + 2O_2$$

$$Ag_2O + O_3 \rightarrow 2Ag + 2O_2$$

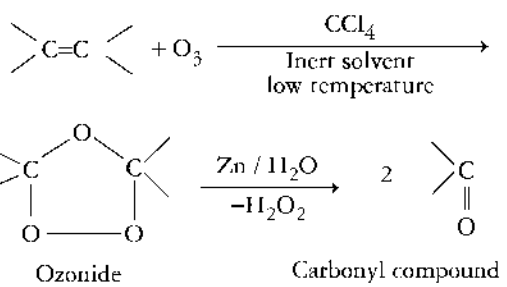
Bleaching Action

It has permanent bleaching action due to nascent oxygen. It can bleach starch, ivory, wood pulp and organic colouring matter etc.

$$\text{Coloured matter} + O_3 \rightarrow \text{Colourless matter} + O_2$$

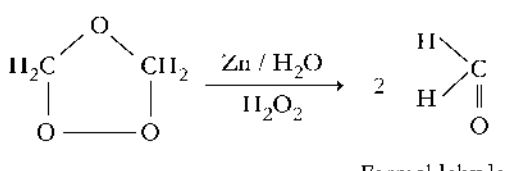
Ozonolysis

It is an important reaction to find the number of double bonds and their location in alkenes. Here, ozone reacts with alkene to give an adduct known as ozonide as follows:



eg $H_2C = CH_2 + O_3 \xrightarrow{CCl_4}$

Ethene



Tests of Ozone

Starch iodide paper + O_3 ➤ Blue colour

Benzidine solution + O_3 ⟶ Brown colour

Alcoholic solution of retramethyl base + O_3 ➤ Violet colour

Structure of O_3

It is an angular or bent molecule having a bond angle of 116 .8°. Ozone is considered to be a resonance hybrid of these two structures.

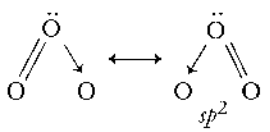


Fig. 6.2

Here O–O–O bond angle is 116°.49' while O–O bond length is 1.278 Å.

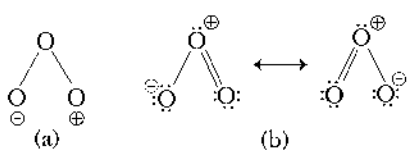


Fig. 6.3 (a) Geometry of ozone molecule
(b) Resonance in ozone molecule

Uses

- It is a germicide, oxidant and bleaching agent.
- In the manufacture of artificial silk, synthetic camphor.
- A mixture of ozone and cyanogen [$(CN)_2$] is used as a rocket fuel.
- It is also used to purify drinking water.
- To detect number and position of double bonds.

HYDROGEN PEROXIDE (AUXOCHROME) H_2O_2

Hydrogen peroxide was discovered by Thenard. It is also called oxygenated water and perhydrol (30 % H_2O_2)

Methods of Preparation

It is prepared by the following methods.

From True Peroxide

True peroxides are peroxides of IA and IIA groups like Na_2O_2, BaO_2.

$$3BaO_2 + 2H_3PO_4 \rightarrow Ba_3(PO_4)_2 + 3H_2O_2$$

HNO_3 cannot be taken as it oxidizes H_2O_2 into O_2.

BaO_2 is the best choice as $BaSO_4$ can be easily removed. H_3PO_4 is better than H_2SO_4 as it does not react with H_2O_2 and prevents reverse reaction. Here BaO_2 should be hydrated and not in excess.

Merck Method

Here, CO_2 is passed through ice cold BaO_2 solution to get H_2O_2.

$$BaO_2 + H_2O + CO_2 \longrightarrow BaCO_3\downarrow + H_2O_2$$

By the Electrolysis of (50%) or Moderate Concentrated H_2SO_4

Here, 30 % H_2O_2 is obtained at anode (Pt) by the electrolysis of 50 % H_2SO_4

$$2H_2SO_4 \longrightarrow 2H^+ + 2HSO_4^-$$

$$2HSO_4^- \longrightarrow H_2S_2O_8 + 2e \quad \text{(at anode)}$$

$$2H^+ + 2e^- \longrightarrow H_2 \text{ (at cathode)}$$

REMEMBER

- Here in place of 50% H_2SO_4, NH_4HSO_4 dissolved in excess of H_2SO_4 can also be used for electrolysis.

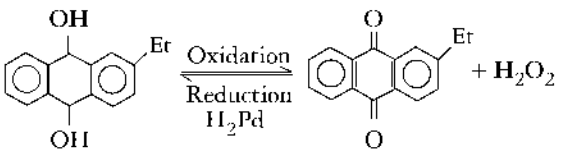
$$(NH_4)_2S_2O_8 + 2H_2O \rightarrow 2NH_4HSO_4 + H_2O_2$$

Distillation of Perdisulphuric acid also gives H_2O_2 as follows.

$$H_2S_2O_8 \xrightarrow[0°C,\ Pt]{2H_2O} H_2O_2 + 2H_2SO_4$$

By Redox reaction

2-Ethyl anthraquinol on oxidation by air gives H_2O_2. It is a cyclic process.



2-Ethyl-anthraquinol 2-Ethylanthra-quinone

Fig. 6.4

When a mixture of ammonium sulphate and sulphuric acid is taken in 1:1 ratio H_2O_2 is obtained.

Concentration of H_2O_2

- It is concentrated below 70°C by vacuum distillation or by distillation under reduced pressure.
- When dilute solution of H_2O_2 is evaporated on a water bath at 152°C and low pressure 20–30°C, H_2O_2 is obtained.
- When 20–30°C H_2O_2 undergo reduced pressure distillation, 90 % H_2O_2 is obtained.
- When 90% H_2O_2 is crystallized by freezing mixture of solid CO_2 and ether 100%, H_2O_2 is obtained.

Storage of H_2O_2

It is stored in wax-lined, wax amber colour bottles to avoid decomposition from alkali metal oxides present in glass.

- To avoid decomposition of, H_2O_2, negative catalyst like H_3PO_4, glycerol, acetanilide, sodium stannate, sodium pyrophosphate are added in it, in small amounts.
- Dilute solution of H_2O_2 is stable in presence of alcohol or ether or in acidic medium.
- The presence of sand, MnO_2, Fe, Al_2O_3 increase decomposition (positive catalysis).

Physical Properties

- It is a colourless, odourless, syrupy liquid and highly soluble in water.
- In H_2O_2 oxidation state of oxygen is −1.
- H_2O_2 is more hydrogen bonded than water and hence has a higher boiling point (425 K). It is more dense than water (density 1.4 gcm^{-3}).

Chemical Properties

Decomposition of H_2O_2

- Pure H_2O_2 is not very stable and undergo composition which is an example of disproportion reaction or auto oxidation reaction.

$$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2$$

- Here, positive catalysts are MnO_2, Pt, Fe etc., and negative catalysts are H_3PO_4, glycerol etc.

Oxidizing Properties

$$H_2O_2 \rightarrow H_2O + [O]$$

Due to nascent oxygen, it can act not only as an oxidant but also as a permanent bleaching agent.

Oxidation and reduction by H_2O_2 in acidic medium is generally slow while it is rapid in alkaline medium.

In acidic medium:

$$H_2O_2 + 2H^+ + 2e \rightarrow 2H_2O$$

In basic medium:

$$H_2O_2 + 2e^- \longrightarrow 2OH^-$$

For example,

$$\underset{\text{Ferrous sulphate}}{2FeSO_4} \xrightarrow[H_2O_2]{H_2SO_4} \underset{\text{Ferric sulphate}}{Fe_2(SO_4)_3} + 2H_2O$$

$$\underset{\substack{\text{Potassium}\\\text{ferro cyanide}}}{2K_4Fe(CN)_6} \xrightarrow{H_2O_2} \underset{\text{Potassium ferricyanide}}{K_3Fe(CN)_6} + 2KOH$$

It oxidizes sulphites to sulphates. For example:

$$H_2O_2 \longrightarrow H_2O + [O]$$

$$Na_2SO_3 + [O] \rightarrow Na_2SO_4$$

$$Na_2SO_3 + H_2O_2 \rightarrow Na_2SO_4 + H_2O$$

$$\underset{\text{Metal nitrite}}{MNO_2} + H_2O_2 \rightarrow \underset{\text{Metal nitrate}}{MNO_3} + H_2O$$

It oxidizes arsenites to arsenates. For example,

$$H_2O_2 \longrightarrow H_2O + [O]$$

$$Na_3AsO_3 + [O] \longrightarrow Na_3AsO_4$$

$$Na_3AsO_3 + H_2O_2 \longrightarrow Na_3AsO_4 + H_2O$$

$$2KI + H_2O_2 \longrightarrow 2KOH + I_2$$

$$HCHO + H_2O_2 \xrightarrow{\text{Pyrogallol}} HCOOH + H_2O$$

$$PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$$

$$Hg + H_2O_2 \longrightarrow HgO + H_2O$$

When a precipitate chromium hydroxide suspended in sodium hydroxide is treated with hydrogen peroxide, it gets oxidized into chromate and a yellow solution of sodium chromate is obtained.

For example,

$$2Cr(OH)_3 + 4NaOH + 3H_2O_2 \longrightarrow \underset{\text{Sodium chromate}}{2Na_2CrO_4} + 8H_2O$$

- It gives blue chromium peroxide (CrO_5) with acidic $K_2Cr_2O_7$.

$$H_2Cr_2O_7 + 4[O] \longrightarrow 2CrO_5 + H_2O$$

- CrO_5 has butterfly structure with two peroxy bonds and oxidation number of chromium is +6.
- CrO_5 is stable in ether layer forming a blue layer having $CrO(O_2)_2$.

Bleaching Action of H_2O_2

It is due to nascent oxygen. It has permanent bleaching action on human hair, silk, wool etc.

Coloured substance + [O] ➤ Colourless substance

Reducing Properties

H_2O_2 can act as a weak reducing agent in both acidic and basic mediums.

In acidic medium

$$H_2O_2 \longrightarrow 2H^+ + O_2 + 2e$$

In basic medium

$$H_2O_2 + 2OH \longrightarrow 2H_2O + O_2 + 2e$$

For example,

$$H_2O_2 \xrightarrow{\text{Acid}} 2H^- + O_2 + 2e^-$$

$E^\circ = -0.67$ V

$$H_2O_2 + 2OH^- \longrightarrow 2H_2O + O_2 + 2e^-$$

$E^\circ = +0.08$ V

For example,

$$X_2 + H_2O_2 \longrightarrow 2HX + O_2$$

$$2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$$

$$3K_3[Fe(CN)_6] + 2KOH + H_2O_2 \longrightarrow 2K_4[Fe(CN)_6] + 2H_2O + O_2$$

$$PbO_2 \xrightarrow{H_2O_2} PbO + H_2O + O_2$$

$$Ag_2O \xrightarrow{H_2O_2} 2Ag + H_2O + O_2$$

$$MXO \xrightarrow{H_2O_2} MX + H_2O + O_2$$

It dissolves manganese dioxide (black) suspended in dilute H_2SO_4 due to its reduction into pale ink manganese sulphate which is soluble.

$$MnO_2 + H_2SO_4 + H_2O_2 \longrightarrow MnSO_4 + 2H_2O + O_2$$

Acidic Properties

Pure H_2O_2 is a very weak acid ($K_a = 1.55 \times 10^{-12}$ at 25°C). It can also act as a dibasic acid.

$$H_2O_2 \rightleftharpoons H^+ + HO_2^-$$
(Hydroperoxide ion)

$$HO_2^- \rightleftharpoons H^+ + O_2^{2-}$$
(Peroxide ion)

For example,

$$2NaOH + H_2O_2 \longrightarrow Na_2O_2 + 2H_2O$$

$$Ba(OH)_2 + H_2O_2 \longrightarrow BaO_2 + 2H_2O$$

Addition Reaction

It is capable of showing addition reaction and forms addition compounds

For example,

$$\begin{array}{lll} CH_2 + HO & \longrightarrow & CH_2OH \\ \| \quad | & & | \\ CH_2 \quad HO & & CH_2OH \end{array}$$

Ethylene glycol

It can also form sodium perborate ($NaBO_3.H_2O_2 3H_2O$), $(NH_4)_2SO_4.H_2O_2$, $NH_2CONH_2.H_2O_2$.

Test of H_2O_2

- It gives I_2 with KI which turns starch paper blue.

$$2KI + H_2O_2 \longrightarrow 2KOH + I_2$$

$$\text{Starch} + I_2 \longrightarrow \text{Blue complex}$$

- With TiO_2

$$TiO_2 + H_2SO_4 + H_2O_2 \longrightarrow TiO_2.nH_2O$$

or

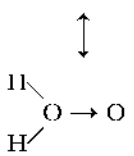
$$H_2[TiO_2(SO_4)_2]$$

Orange red
(Per titanic acid)

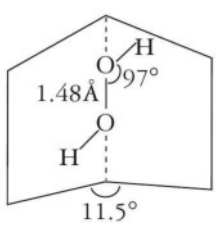
Structure of H_2O_2

The structure of H_2O_2 is given as

H – O – O – H
(Bacyer structure)



(Kingzett structure)



Open Book Structure

Fig. 6.5

The X-ray and dipole moment studies of H_2O_2 confirms that it has a nonplanar structure which is shown as in fig. 6.5. In this structure the dihedral angle is 111.5° in gaseous phase but in solid state due to hydrogen bonding it reduces to 90.2°. Here, the two oxygen atoms are linked by a peroxide linkage. Here, O – O bond length is 1.48 Å while O – H is 0.95 Å and the O – O – H bond is nearly 97°.

Uses of H_2O_2

1. H_2O_2 is used to bleach hair, teeth etc.
2. As a fuel in rockets.
3. As an antiseptic and germicides in wounds under the name Perhydrol (30% H_2O_2).
4. To clear paintings by converting PbS (black) into $PbSO_4$ (white).
5. The addition compound of H_2O_2 with urea is called hyperal.

Strength of H_2O_2

It can be given by

1. Volumetric strength (For example: 10 volume, 20 volume, 30 volume)

   X-volume H_2O_2 means that one ml of it gives X-ml of O_2 at STP.

   For example,

   15 volume of H_2O_2 solution means 1 ml of this solution on decomposition releases 15 ml of O_2 at NTP.

2. Percentage by Weight: The concentration of H_2O_2 in a solution can be expressed as percentage of H_2O_2 in solution (W/V).

   For example,

   20% solution of H_2O_2 means 20 gm of H_2O_2 are present in 1000 ml of water.

3. Concentration (g/litre), normality and molarity

REMEMBER

- Concentration or strength of 10 volume H_2O_2 solution = 30.35 g/lit.

  Volume strength = 5.5 × Normality
  Volume strength = 11.2 × Molarity

- 10 vol H_2O_2 = 3.036%H_2O_2 by wt. = 1.786 N = 0.893 M

  or

  11.2 vol H_2O_2 = 3.4 % H_2O_2 by wt. = 2N = 1M

COMPOUNDS OF SULPHUR

Hydrogen Sulphide (H_2S)

It is also known as sulphuretted hydrogen.

Occurrence

It is present in volcanic gases, sewage gases, coal gas and in many spring waters also. It is also present in small quantity in atmosphere, where it is due to the heating of coal and decay of animal and vegetable matter having sulphur compounds.

Preparation

Laboratory Method

In laboratory hydrogen sulphide is obtained by the action of dilute sulphuric acid on ferrous sulphide in Kipp's apparatus as follows:

$$FeS + H_2SO_4 \rightarrow FeSO_4 + H_2S$$

- Here, the obtained H_2S is always impure as it has hydrogen with it.

From Antimony Sulphide

Pure form of hydrogen sulphide can be obtained by the action of pure HCl on antimony sulphide as follows:

$$Sb_2S_3 + 6HCl \longrightarrow 2SbCl_3 + 3H_2S$$

Physical Properties

1. It is a colourless gas having an unpleasant odour of rotten eggs.
2. It is poisonous in nature and causes headache when inhaled in small amounts but may also cause death when inhaled for a long time.
3. It can be easily liquefied by applying pressure (b.p. 213K) and can be frozen into a transparent solid at 188K.
4. It is slightly heavier than air and quite soluble in cold water but less soluble in hot water.

Chemical Properties

Combustibility

It burns with a blue flame in oxygen or air giving sulphur dioxide and water.

$$2H_2S + 3O_2 \rightarrow 2H_2O + 2SO_2$$

However, in restricted supply of oxygen sulphur is formed due to incomplete combustion.

$$2H_2S + O_2 \rightarrow 2H_2O + 2S$$

Thermal Decomposition

It decomposes on heating at 1973K.

$$H_2S \longrightarrow H_2 + S$$

Acidic Nature

The aqueous solution acts as a weak dibasic acid and its ionization occurs as follows:

$$H_2S \rightleftharpoons H^+ + HS^- \rightleftharpoons 2H^+ + S^{2-}$$

It forms two type of salts, hydrosulphides and sulphides.

$$NaOH + H_2S \rightarrow \underset{\text{Sodium hydrosulphide}}{NaHS} + H_2O$$

$$2NaOH + H_2S \rightarrow \underset{\text{Sodium sulphide}}{Na_2S} + 2H_2O$$

Reducing Nature

It can act as a strong reductant as on decomposition it gives hydrogen.

For example,

It reduces sulphur dioxide into sulphur in presence of moisture.

$$SO_2 + 2H_2S \rightarrow 3S + 2H_2O$$

The occurrence of sulphur in volcanic regions is due to this reaction.

It reduces H_2O_2 into water.

$$H_2O_2 + H_2S \longrightarrow 2H_2O + S$$

It reduces O_3 into O_2.

$$H_2S + O_3 \longrightarrow H_2O + S + O_2$$

It reduces HNO_3 into NO_2.

$$2HNO_3 + H_2S \rightarrow 2NO_2 + S + 2H_2O$$

It reduces H_2SO_4 into SO_2.

$$H_2SO_4 + H_2S \longrightarrow 2H_2O + SO_2 + S$$

It reduces halogens into hydracids.

$$H_2S + X_2 \rightarrow 2HX + S$$

(X may be F_2, Cl_2, Br_2 or I_2)

It reduces ferric chloride to ferrous chloride.

$$2FeCl_3 + H_2S \rightarrow 2FeCl_2 + 2HCl + S$$

It reduces acidified $KMnO_4$ and decolourizes it.

$$2KMnO_4 + 3H_2SO_4 + 5H_2S \longrightarrow K_2SO_4$$

$$+ 2MnSO_4 + 8H_2O + 5S$$

It reduces acidified $K_2Cr_2O_7$ into green chromic sulphate.

$$K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3S$$

Reaction with Metals and Metal Oxides

It reacts with metals and their oxides to give corresponding sulphides as follows:

$$2Na + H_2S \longrightarrow Na_2S + H_2$$

$$CaO + H_2S \xrightarrow{\Delta} CaS + H_2O$$

$$Pb + H_2S \longrightarrow PbS + H_2$$

$$Cu + H_2S \longrightarrow CuS + H_2$$

$$2Ag + H_2S \longrightarrow Ag_2S + H_2$$

$$ZnO + H_2S \xrightarrow{\Delta} ZnS + H_2O$$

Reaction with Salts

It reacts with many metal salts to give corresponding sulphides.

These metal sulphides can be divided into three groups as follows:

(a) Sulphides Precipitated in Acidic Medium: These are the sulphides of Hg, Ag, Pb, Cu, Bi, Cd, As, Sb and Sn.

$$Pb(CH_3COO)_2 + H_2S \longrightarrow PbS + 2CH_3COOH$$

$$CuSO_4 + H_2S \longrightarrow CuS + H_2SO_4$$

$$CdSO_4 + H_2S \longrightarrow CdS + H_2SO_4$$

REMEMBER

SnS_2, As_2S_3	Sb_2S_3	SnS
Yellow	Orange	Chocolate

These are soluble in yellow ammonium sulphide.

HgS, CuS, PbS	Bi_2S_3	CdS
Black	Brown	Yellow

These are insoluble in yellow ammonium sulphide

(b) Sulphides Precipitated in Alkaline Medium: These are the sulphides of Mn, Fe, Co, Ni and Zn.

$$MnCl_2 + H_2S \longrightarrow MnS + 2HCl$$

$$Ni(NO_3)_2 + H_2S \longrightarrow NiS + 2HNO_3$$

$$ZnSO_4 + H_2S \longrightarrow ZnS + H_2SO_4$$

MnS	CoS, NiS	ZnS
Flesh coloured	Black	White

- Sulphides of Na, K, Mg, Ca, Sr, Ba, Al and Cr can neither be precipitated in acidic medium and nor in alkaline medium.

Formation of Polysulphides

Polysulphides of metals can be obtained by passing H_2S through their hydroxides.

For example,

$$Ca(OH)_2 + H_2S \longrightarrow CaS + 2H_2O$$

$$CaS + 4H_2S \longrightarrow CaS_5 + 4H_2$$

- When excess of H_2S is passed through ammonium sulphide, yellow ammonium sulphide is obtained.

$$2NH_4OH + H_2S \longrightarrow (NH_4)_2S + 2H_2O$$

$$(NH_4)_2S + xH_2S \longrightarrow \underset{\text{Ammonium Polysulphide}}{(NH_4)_2S_{x+1}} + xH_2$$

Uses

1. It is an important laboratory reagent used for the detection of basic radicals in qualitative analysis.
2. It can be used as a reducing agent.
3. It can be used for the preparation of many metal sulphides used in paint industry.

Tests of Hydrogen Sulphide

1. It has an unpleasant rotten eggs smell.
2. It turns lead acetate paper into black.
3. It gives a violet colouration with a solution of sodium nitroprusside due to the formation of sodium nitrothioprusside.

Structure of H_2S

The shape of the H_2S molecule is angular or V-shaped with bond length (H–S) 1.35 Å and bond angle (H–S–H) 92.5°.

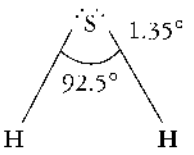


Fig. 6.5

Sulphur Dioxide (SO_2)

Preparation

It can be prepared by the following methods:

1. **Laboratory Method**

In laboratory it is obtained by the action of hot and concentrated sulphuric acid on copper turnings as follows:

$$Cu + 2H_2SO_4 \xrightarrow[\text{Hot \& conc.}]{} CuSO_4 + SO_2 + 2H_2O$$

2. **By the Action of conc. H_2SO_4 on C, S, Cu, Ag etc**

$$C + 2H_2SO_4 \longrightarrow CO_2 + 2SO_2 + 2H_2O$$

$$S + 2H_2SO_4 \longrightarrow 3SO_2 + 2H_2O$$

$$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2H_2O$$

$$2Ag + 2H_2SO_4 \rightarrow Ag_2SO_4 + SO_2 + 2H_2O$$

3. **Industrial Methods**

(i) By Burning Sulphur in Air

$$S + O_2 \longrightarrow SO_2$$

(ii) By Heating Iron Pyrite

$$4FeS_2 + 11\,O_2 \longrightarrow 2Fe_2O_3 + 8SO_2$$

(iii) By Zinc Blende

$$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_3$$

4. **By the Decomposition of Sulphides or Bisulphites**

$$Na_2SO_3 + H_2SO_4 \rightarrow Na_2SO_4 + SO_2 + H_2O$$

$$2NaHSO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2SO_2 + 2H_2O$$

- Sulphur dioxide can be dried by passing it through concentrated H_2SO_4. For this purpose, CaO cannot be used as it forms $CaSO_4$ with sulphur dioxide.

Physical Properties

1. SO_2 is a colourless, acidic gas with pungent and suffocating smell.
2. It can be easily liquefied.
3. It is highly soluble in water and its aqueous solution (H_2SO_3) is acidic in nature.
4. It is anhydrous product of H_2SO_3.
5. It freezes at −72.7° C to snow like mass.

Chemical Properties

1. Combustibility

It is neither combustible nor helps in combustion however K, Mg burns in its atmosphere.

$$4K + 3SO_2 \rightarrow \underset{\text{Pot. thiosulphate}}{K_2S_2O_3} + K_2SO_3$$

$$2Mg + SO_2 \rightarrow 2MgO + S$$

2. Thermal Dissociation

On strong heating at 1473K, it dissociates as follows:

$$3SO_2 \longrightarrow S + 2SO_3$$

3. Acidic Nature

It is an acidic oxide as it gives sulphurous acid in water and form salts with bases as follows:

$$SO_2 + H_2O \rightarrow H_2SO_3$$

$$NaOH + SO_2 \rightarrow \underset{\text{Sodium bisulphite}}{NaHSO_3}$$

$$2NaOH + SO_2 \longrightarrow \underset{\text{Sodium sulphite}}{Na_2SO_3} + H_2O$$

$$Ca(OH)_2 + SO_2 \longrightarrow \underset{\substack{\text{Calcium sulphite}\\ \text{(sparingly soluble)}}}{CaSO_3} + H_2O$$

$$CaSO_3 + H_2O + SO_2 \longrightarrow \underset{\substack{\text{Calcium bisulphite}\\ \text{(soluble)}}}{Ca(HSO_3)_2}$$

$$Na_2CO_3 + 2SO_2 + H_2O \rightarrow 2NaHSO_3 + CO_2$$

4. Oxidizing Properties

It acts as an oxidizing agent also.

It oxidizes H_2S into sulphur.

$$2H_2S + SO_2 \longrightarrow 2H_2O + 3S$$

It oxidizes iron into ferrous oxide.

$$3Fe + SO_2 \longrightarrow 2FeO + FeS$$

It oxidizes tin into stannous oxide.

$$3Sn + SO_2 \rightarrow 2SnO + SnS$$

It oxidizes stannous and mercurous chlorides as follows:

$$2SnCl_2 + SO_2 + 4HCl \rightarrow 2SnCl_4 + 2H_2O + S$$

$$2Hg_2Cl_2 + SO_2 + 4HCl \rightarrow 4HgCl_2 + 2H_2O + S$$

5. Reducing Properties

Its aqueous solution acts as a reducing agent as sulphurous acid gives nascent hydrogen by changing into sulphuric acid.

$$SO_2 + 2H_2O \longrightarrow H_2SO_4 + 2[H]$$

It reduces ferric salts into ferrous salts as follows:

$$Fe_2(SO_4)_3 + SO_2 + 2H_2O \rightarrow 2FeSO_4 + 2H_2SO_4$$

$$SO_2 + 2FeCl_3 + 2H_2O \rightarrow H_2SO_4 + 2FeCl_2 + 2HCl$$

It decolourizes the solution of $KMnO_4$ and $K_2Cr_2O_7$ by reducing them as follows:

$$\underset{\text{Purple}}{2KMnO_4} + 5SO_2 + 2H_2O \rightarrow K_2SO_4 + \underset{\text{Colourless}}{2MnSO_4} + 2H_2SO_4$$

$$\underset{\text{Orange}}{K_2Cr_2O_7} + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + \underset{\text{Green}}{Cr_2(SO_4)_3} + H_2O$$

It reduces halogens into halogen acids.

$$Cl_2 + SO_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$$

$$I_2 + SO_2 + 2H_2O \rightarrow H_2SO_4 + 2HI$$

It reduces acidified iodates to iodine.

$$2KIO_3 + 5SO_2 + 4H_2O \rightarrow K_2SO_4 + 4H_2SO_4 + I_2$$

6. Bleaching Action

In presence of water it has temporary bleaching action due to nascent hydrogen (i.e., due to reduction).

$$SO_2 + 2H_2O \longrightarrow H_2SO_4 + 2[H]$$

$$\text{Coloured substances} + 2[H] \rightarrow \text{Colourless substance}$$

- Here, the bleached matter regains its colour when exposed to atmosphere due to its oxidation.

7. Unsaturated Nature

It shows its unsaturated nature by its reaction with O_2, Cl_2 etc. to form adducts.

For example,

$$2SO_2 + O_2 \xrightarrow[573\ K]{V_2O_5} \underset{\text{Sulphur trioxide}}{2SO_3}$$

$$SO_2 + Cl_2 \xrightarrow{\text{Sun light}} SO_2Cl_2$$

$$PbO_2 + SO_2 \xrightarrow{\text{heat}} PbSO_4$$

Uses

1. It is used in the refining of cane juice in sugar industry.
2. For fumigation, it is used as a germicide and to preserve fruits.
3. Liquid SO_2 is used as a nonaqueous solvent and as a refrigerant and antichlor.
4. It is used in the manufacture of H_2SO_4.
5. SO_2 acts as an oxidising agent, reducing agent and bleaching agent (temporary bleaching property).

Structure: It has angular or V-shaped structure in which sulphur atom is sp^2 -hybridized. Here S–O bond length is 1.43 Å and the bond angle between O–S–O is 119.5°.

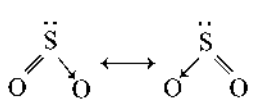


Fig. 6.6

Sulphur Trioxide (SO_3)

Preparation

1. By passing a mixture of SO_2 and O_2 over heated Pt or V_2O_5 as follows:

$$2SO_2 + O_2 \xrightarrow{V_2O_5} 2SO_3$$

- Here, SO_2 and O_2 must be completely dry.

2. By dehydration of H_2SO_4 with P_2O_5

$$H_2SO_4 + P_2O_5 \longrightarrow SO_3 + 2HPO_3$$

3. By heating Ferric Sulphate.

$$Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + 3SO_3$$

Physiochemical Properties

1. It is an acidic oxide and dissolves in H_2O to give H_2SO_4.
2. SO_3 is anhydrous product of H_2SO_4.
3. **Heating effect**

$$2SO_3 \rightleftharpoons 2SO_2 + O_2$$

4. **Reaction with water**

$$SO_3 + H_2O \rightleftharpoons H_2SO_4$$

5. **Acidic nature:** Being an acidic oxide it can react with basic oxides to form salts.

$$Na_2O + SO_3 \longrightarrow Na_2SO_4$$

$$MgO + SO_3 \longrightarrow MgSO_4$$

6. **Oxidizing Properties:** It can act as an oxidizing agent.

For example,

It can oxidize hydrogen bromide into bromine.

$$SO_3 + 2HBr \longrightarrow H_2O + Br_2 + SO_2$$

It can oxidize phosphorous into P_2O_5.

$$5SO_3 + 2P \longrightarrow 5SO_2 + P_2O_5$$

It can oxidize PCl_5 into phosphonyl chloride ($POCl_3$).

$$SO_3 + PCl_5 \longrightarrow POCl_3 + SO_2 + Cl_2$$

7. **With Sulphuric Acid:** It dissolves in concentrated sulphuric acid to give oleum.

$$H_2SO_4 + SO_3 \longrightarrow \underset{\text{Oleum}}{H_2S_2O_7}$$

Structure: It has a planar trigonal structure in which sulphur atom is sp^2 hybridized.

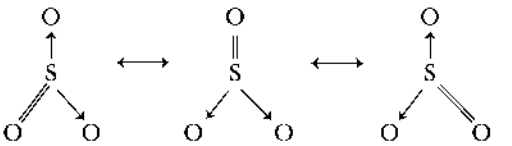
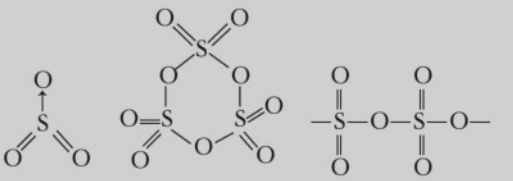


Fig. 6.7

REMEMBER

SO_3 has three allotropic forms α-SO_3, β-SO_3 and γ-SO_3.



Gaseous | Solid (α-SO_3) | (β-γ SO_3)

- α-SO_3 is most stable having ice like crystals and a melting point of 290K.
- β-SO_3 is needle like.
- γ-SO_3 is needle like.

Sulphurous Acid (H_2SO_3)

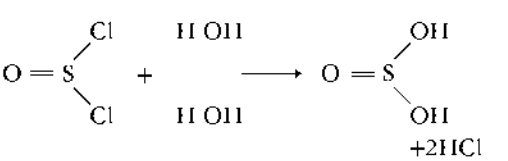
As this acid is unstable hence is unknown in the free state. It is known only in solution of SO_2 in water.

Preparation

1. It is prepared by dissolving SO_2 in water.

$$SO_2 + H_2O \longrightarrow H_2SO_3$$

2. It is also prepared by the reaction of thionyl chloride with water.



Physiochemical Properties

1. The acidic solution gives a smell of SO_2 and SO_2 is evolved on heating this solution. Hence, it is present in equilibrium with the free SO_2 gas.

$$H_2SO_3 \rightleftharpoons H_2O + SO_2$$

2. Dibasic Acidic Nature: It is a dibasic acid as it ionizes in two steps and forms two type of salts i.e. bisulphites and sulphites ($NaHSO_3$ and Na_2SO_3)

$$H_2SO_3 \rightleftharpoons H^- + HSO_3^- \quad K_1 = 1.3 \times 10^{-2}$$
$$HSO_3^- \rightleftharpoons H^+ + SO_3^{2-} \quad K_2 = 6.24 \times 10^{-8}$$

3. Decomposition: On heating it decomposes to give nascent sulphur.

$$3H_2SO_3 \xrightarrow{\text{Heat}} 2H_2SO_4 + H_2O + [S]$$

4. Reducing Properties: It can act as a reductant as it gives nascent hydrogen as follows:

$$H_2SO_3 + H_2O \longrightarrow H_2SO_4 + 2[H]$$

For example,

- It reduces halogens into haloacids.

$$H_2SO_3 + H_2O + X_2 \rightarrow H_2SO_4 + 2HX$$

- It reduces ferric sulphate into ferrous sulphate.

$$H_2SO_3 + H_2O + Fe_2(SO_4)_3 \rightarrow 2H_2SO_4 + 2FeSO_4$$

- It reduces and decolourizes $KMnO_4$ as follows:

$$2KMnO_4 + 5H_2SO_3 \longrightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4 + 3H_2O$$

- It reduces and decolourizes $K_2Cr_2O_7$ as follows:

$$K_2Cr_2O_7 + H_2SO_4 + 3H_2SO_3 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O$$

- It reduces potassium iodate into iodine.

$$5H_2SO_3 + 2KIO_3 \rightarrow 4H_2SO_4 + K_2SO_4 + H_2O + I_2$$

$$5SO_3^{2-} + 2IO_3^- + 2H^+ \rightarrow 5SO_4^{2-} + I_2 + H_2O$$

5. Oxidizing Properties: It can also act as an oxidizing agent due to release of nascent oxygen as follows:

$$H_2SO_3 \rightarrow S + H_2O + 2[O]$$

For example,

- It can oxidize H_2S into Sulphur.

$$H_2SO_3 + 2H_2S \rightarrow 3S + 3H_2O$$

- It can oxidize HI into I_2.

$$H_2SO_3 + 4HI \rightarrow S + 3H_2O + 2I_2$$

- It can oxidize CO into CO_2.

$$H_2SO_3 + 2CO \longrightarrow S + H_2O + 2CO_2$$

- It can oxidize stannous chloride into stannic chloride.

$$H_2SO_3 + 2SnCl_2 + 4HCl \rightarrow S + 3H_2O + 2SnCl_4$$

- It can oxidize mercurous chloride into mercuric chloride.

$$H_2SO_3 + 2Hg_2Cl_2 + 4HCl \rightarrow S + 3H_2O + 4HgCl_2$$

It can oxidize metals into their oxides and sulphides as follows:

$$2Mg + H_2SO_3 \longrightarrow 2MgO + MgS + H_2O$$

$$4K + 3H_2SO_3 \rightarrow K_2SO_3 + K_2S_2O_3 + 3H_2O$$

$$3Fe + H_2SO_3 \longrightarrow 2FeO + FeS + H_2O$$

6. Bleaching Action: It can show temporary bleaching action due to nascent hydrogen as the colourless substance can be further coloured by oxidation.

$$H_2SO_3 + H_2O \rightarrow H_2SO_4 + 2H$$

Coloured matter + [H] ➤ Colourless matter

Colourless matter + [O] ➤ Coloured matter

Uses

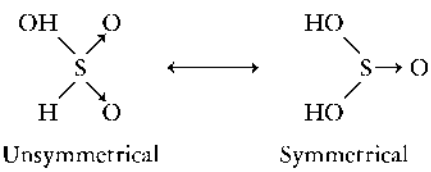
1. It is used as an oxidizing, bleaching and reducing agent.
2. It is used to prepare H_2SO_4 and $H_2S_2O_5$.

$$H_2SO_3 \xrightarrow{[O]} H_2SO_4$$

$$H_2SO_3 \xrightarrow{[S]} \underset{\text{Thiosulphuric acid}}{H_2S_2O_5}$$

Structure

It exists in the dynamic equilibrium between these two structures.



Unsymmetrical Symmetrical

Fig. 6.8

- In SO_3^{2-} ion, sulphur atom is sp^3 hybridized and the shape of the molecule is pyramidal.

Sulphuric Acid (H_2SO_4)

It has a very wide application in industries hence it is called king of chemicals.

It is also known as oil of vitriol since it was prepared by the distillation of green vitriol.

$$2FeSO_4.7H_2O \xrightarrow{\Delta} Fe_2O_3 + H_2SO_4 + SO_2 + 13H_2O$$

Preperation

It is manufactured by these two methods.

Lead–Chamber Process

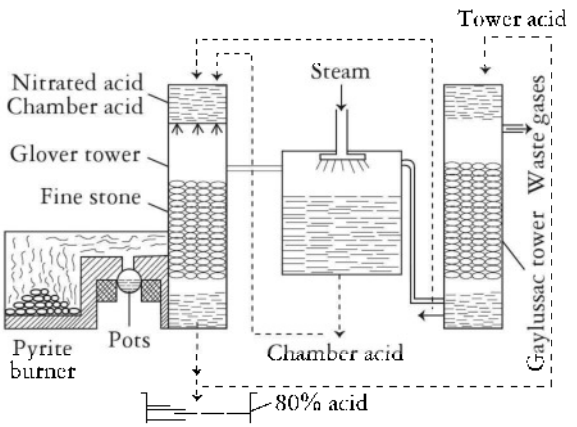


Fig. 6.9

Principle: By this method sulphuric acid is manufactured as follows:

(i) ***Formation of SO_2:*** It is formed by burning sulphur in air or by roasting iron pyrite in excess of air as follows:

$$S + O_2 \xrightarrow{\text{Burn}} SO_2$$

or

$$4FeS_2 + 11O_2 \xrightarrow{\text{Roasting}} 2Fe_2O_3 + 8SO_2$$

(ii) ***Oxidation of SO_2 into SO_3:*** SO_2 is oxidized into SO_3 in presence of nitrogen oxides (mainly NO_2) in large lead sheet chambers.

$$2SO_2 + O_2 \xrightarrow[NO]{\text{Oxides of N}_2} 2SO_3$$

Or

$$SO_2 + NO_2 \rightarrow SO_3 + NO$$

$$2NO + O_2 \longrightarrow 2NO_2$$

Here, NO and NO_2 act as an oxygen carriers and this phenomenon is known as Bergelius mechanism.

(iii) ***Conversion of SO_3 into Sulphuric Acid:*** SO_3 dissolves in steam to give H_2SO_4.

$$SO_3 + H_2O \rightarrow H_2SO_4$$

According to Davy and Lunge mechanism, when steam is in insufficient amount, first of all nitrososulphuric acid is obtained as an intermediate product which further reacts with water (steam) to give sulphuric acid as follows:

$$NO + NO_2 \rightarrow N_2O_3$$

$$2SO_2 + N_2O_3 + O_2 + H_2O \rightarrow 2HSO_4.NO$$
(Intermediate product)

$$2HSO_4.NO + H_2O \rightarrow 2H_2SO_4 + NO + NO_2$$

From here 78% H_2SO_4 is obtained. It has impurities of As_2O_3, $PbSO_4$, and oxides of nitrogen.

Contact process

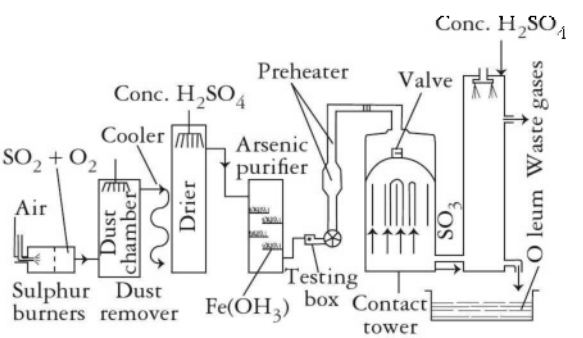


Fig. 6.10

Principle: Here H_2SO_4 is formed as follows:

(i) ***Formation of SO_2:*** It is formed by burning sulphur in air or by roasting iron pyrite in excess of air as follows:

$$S + O_2 \xrightarrow{\Delta} SO_2$$

Or

$$4FeS_2 + 11O_2 \xrightarrow{\Delta} 2Fe_2O_3 + 8SO_2$$
Iron pyrite

(ii) ***Oxidation of SO_2 into SO_3***: SO_2 is oxidized into SO_3 by air in presence of catalysts like vanadium pentaoxide as follows:

$$2SO_2 + O_2 \xrightarrow[450°C\ 1.5-1.7\ atm]{V_2O_5\ (Pt\ Asbastos)} 2SO_3 + 196.6\ kJ$$

Favourable Conditions for the Formation of SO_3: According to Le Chatelier's principle, the favourable conditions for the formation of SO_3 are as follows:

Low temperature: As the formation of SO_3 is an exothermic reaction so low temperature is favourable i.e., 397°– 447°C.

High pressure: Here high pressure is favourable as volume or number of molecules decreases ($\Delta_n = -1$).

Excess of O_2: O_2 in excess also favours the formation of SO_3 and the best proportion taken for SO_2 and O_2 is 2:3 to get the best yield of SO_3.

(iii) ***Conversion of SO_3 into H_2SO_4***: When SO_3 is dissolved in 98% sulphuric acid, oleum is formed which on dilution with water can be changed into sulphuric acid of any desired concentration as follows:

$$SO_3 + H_2SO_4 \rightarrow \underset{\text{Oleum or Pyrosulphate acid}}{H_2S_2O_7} \xrightarrow{H_2O} 2H_2SO_4$$

Physical Properties

1. In pure form colourless, dense oily liquid with a specific gravity of 1.84 at room temperature.
2. Due to intermolecular hydrogen bonding it has high boiling point (340°C) and viscous nature.
3. During hydrolysis of H_2SO_4 heat is evolved due to formation of hence hydrates hence to avoid any accident it is necessary that concentrated acid must be added to water and not water to acid.

   For example, $H_2SO_4.2H_2O$
4. It forms a constant boiling mixture (98.3% acid) which boils at 338°C hence it is not possible to concentrate aqueous solution beyond 98.3% by boiling.
5. It is the strongest dibasic acid, a powerful dehydrating agent hence used in industry to a large extent.

Chemical Properties

1. Decomposition or Dissociation: On boiling pure water-free sulphuric acid, it dissociates at 715K as follows:

   $$H_2SO_4 \rightleftharpoons H_2O + SO_3$$
2. Acidic Nature: It is a strong dibasic acid which turns blue litmus red and forms two type of salts with bases as follows:

   $$H_2SO_4 \rightleftharpoons H^+ + HSO_4^- \rightleftharpoons 2H^+ + SO_4^{2-}$$

   $$NaOH + H_2SO_4 \rightarrow \underset{\text{Sodium bisulphate}}{NaHSO_4} + H_2O$$

   $$2NaOH + H_2SO_4 \longrightarrow \underset{\text{Sodium sulphate}}{Na_2SO_4} + 2H_2O$$
3. Oxidizing Properties: It is a moderately strong oxidizing agent as it gives nascent oxygen easily as follows:

   $$\underset{\text{Hot and conc.}}{H_2SO_4} \rightarrow H_2O + SO_2 + [O]$$

   (i) *Oxidation of Non-Metals*: It can oxidize non metals like C, S, P into their oxides or oxyacids as follows:

   $$C + 2H_2SO_4 \longrightarrow CO_2 + 2SO_2 + 2H_2O$$

   $$2P + 5H_2SO_4 \longrightarrow 3H_3PO_4 + 5SO_2$$

   $$S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$$

   (ii) *Oxidation of Metals*: It can oxidize some metals like Cu, Pb, Hg, Zn etc., into their sulphates as follows:

   $$\underset{\text{dil.}}{H_2SO_4} + \underset{\text{(Above } H_2 \text{ in E.C.S)}}{\text{Metal}} \rightarrow M - SO_4 + H_2\uparrow$$

   For example, Na, Zn, Li etc.

   $$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$$

$$2H_2SO_4 + \text{Metal} \rightarrow M - SO_4 + SO_2 + H_2O$$
Conc. (Below H_2 in E.C.S)

For example, Cu, Pt, Ag, Au.

$$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2H_2O$$

It can oxidize metal halides (bromides and iodides) into halogens.

$$M - X + MnO_2 + H_2SO_4 \rightarrow X_2 + SO_2 + H_2O$$
(Above H_2 in E.C.S) dil.

For example,

$$2KI + 2H_2SO_4 \longrightarrow K_2SO_4 + SO_2 + I_2 + 2H_2O$$

$$2NaI + 3H_2SO_4 \longrightarrow 2NaHSO_4 + SO_2 + I_2 + 2H_2O$$

■ E.C.S. means Electro chemical series.

4. Dehydrating Properties: It is a powerful dehydrating agent due to a great affinity towards water and used for the dehydration of carbohydrates, acids etc. as follows:

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{conc. } H_2SO_4} 6C + 6H_2O$$

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} \xrightarrow{\text{conc. } H_2SO_4} 12C + 11H_2O$$

$$\underset{\text{Formic acid}}{HCOOH} \xrightarrow{\text{conc. } H_2SO_4} H_2O + CO$$

$$\underset{\text{Oxalic acid}}{COOH \,.\, COOH} \xrightarrow{\text{conc. } H_2SO_4} H_2O + CO + CO_2\uparrow$$

5. **Reaction with Salts:** Due to its low volatility it decomposes the salts of more volatile acids and liberates the corresponding acids.

$$\text{Any salt} + \underset{\text{Conc.}}{H_2SO_4} \xrightarrow{\Delta} SO_4 \text{ salt} + SO_2 + H_2O$$

For example,

$$2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$$

$$2NaNO_2 + H_2SO_4 \rightarrow Na_2SO_4 + 2HNO_2$$

$$2HNO_2 \longrightarrow NO + NO_2 + H_2O$$

$$CaC_2O_4 + H_2SO_4 \longrightarrow CaSO_4 + \underset{\text{Oxalic acid}}{H_2C_2O_4}$$

$$CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$$

$$FeS + H_2SO_4 \longrightarrow FeSO_4 + H_2S$$

6. Precipitation Reactions: Sulphuric acid on reaction with aqueous solutions of the salts of Ba, Pb etc., forms the precipitate of their insoluble sulphates.

$$BaCl_2 + H_2SO_4 \longrightarrow BaSO_4\downarrow + 2HCl$$

$$Pb(NO_3)_2 + H_2SO_4 \longrightarrow PbSO_4\downarrow + 2HNO_3$$

7. Sulphonation: Many organic compounds like aromatic compounds react with concentrated sulphuric acid to form their sulphonic acids.

$$C_6H_6 + \underset{\text{conc.}}{H_2SO_4} \longrightarrow \underset{\text{Benzene sulphonic acid}}{C_6H_5SO_3H} + H_2O$$

$$C_6H_{14} + \underset{\text{conc.}}{H_2SO_4} \longrightarrow \underset{\text{Hexane sulphonic acid}}{C_6H_{13}SO_3H} + H_2O$$

8. With P_2O_5: When it reacts with P_2O_5 it gives meta phosphoric acid and SO_3 as follows:

$$H_2SO_4 + P_2O_5 \rightarrow 2HPO_3 + SO_3$$

9. With Potassium Chlorate: When it is heated with potassium chlorate explosion occurs as follows:

$$3KClO_3 + \underset{\text{conc.}}{3H_2SO_4} \rightarrow 3KHSO_4 + HClO_4 + 2ClO_2 + H_2O$$

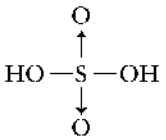
10. **With Potassium Ferrocyanide:** When it is heated with potassium ferrocyanide, CO is evolved as follows:

$$K_4Fe(CN)_6 + \underset{\text{conc.}}{6H_2SO_4} + 6H_2O \rightarrow 2K_2SO_4 + FeSO_4 + 3(NH_4)_2SO_4 + 6CO$$

Uses

1. It is used in lead storage battery (38% by mass).
2. It is used as a dehydrating agent (For example: alcohols) due to high affinity to H_2O.
3. It is used to prepare drugs, dyes, explosives like T, N, T, T, N, P, fertilizers etc.
4. It is used in petroleum refining and tanning of leather.
5. It is also used in cleansing of metals (pickling) before electroplating galvanizing etc.
6. It is also used as a laboratory reagent in petroleum industry etc.

Structure: In its structure the 2-OH groups are directly attached to sulphur as follows:



In crystals sulphate ion as a tetrahedral structure and the sulphur atom is sp^3 hybridized here.

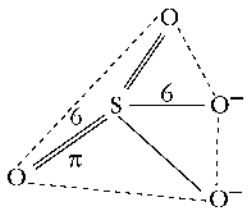


Fig. 6.11

Test of Sulphuric Acid

1. It gives white precipitate with $BaCl_2$ solution which is insoluble in concentrated HCl or HNO_3.
2. On heating it with copper, SO_2 gas is evolved which turns acidified $K_2Cr_2O_7$ solution green.

Sodium Thiosulphate or Hypo ($Na_2S_2O_3\ 5H_2O$)

Preparation

It is prepared by the following methods:

1. By boiling Sulphur with Caustic soda:

$$12S + 6NaOH \longrightarrow Na_2S_2O_3 + 2Na_2S_5 + 3H_2O$$

Sodium penta sulphide

$$2Na_2S_5 + 3O_2 \xrightarrow{air} 2Na_2S_2O_3 + 6S$$

2. By passing SO_2 gas into Sodium sulphite solution:

$$2Na_2S + 3SO_2 \longrightarrow 2Na_2S_2O_3 + S$$

Physical Properties

1. It is a white crystalline solid which is high soluble in water.
2. It is an efflorescent substance which can form a super saturated solution.
3. On heating upto 480 K it loses water of crystallization and on further heating it decomposes into SO_2, H_2S, S.

Chemical Properties

1. With Iodine: When it is treated with iodine, sodium tetra thionate is formed.

$$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$$

2. With Copper Sulphate: When hypo is treated with copper sulphate sodium cupro thiosulphate is formed.

$$CuSO_4 + Na_2S_2O_3 \longrightarrow Cu.S_2O_3 + Na_2SO_4$$

$$2CuS_2O_3 + Na_2S_2O_3 \longrightarrow Cu_2S_2O_3 + Na_2S_4O_6$$

$$3Cu_2S_2O_3 + 2Na_2S_2O_3 \rightarrow Na_4[Cu_6(S_2O_3)_5]$$

Sodium cupro thiosulphate

3. With Silver Nitrate: When silver nitrate is treated with dilute solution of hypo, silver thiosulphate is formed (white precipitate) which slowly turns black.

$$2AgNO_3 + Na_2S_2O_3 \rightarrow Ag_2S_2O_3 + 2NaNO_3$$

White ppt.

$$Ag_2S_2O_3 + H_2O \rightarrow Ag_2S + H_2SO_4$$

Black

When $AgNO_3$ is dissolved in excess of hypo a soluble complex sodium argentothiosulphate is formed.

$$2AgNO_3 + Na_2S_2O_3 \longrightarrow Ag_2S_2O_3 + 2NaNO_3$$

$$Ag_2S_2O_3 + 3Na_2S_2O_3 \longrightarrow 2Na_3[Ag(S_2O_3)_2]$$

Sodium argento thiosulphate

4. With AgBr: When AgBr is dissolved in sodium thiosulphate or hypo a complex compound sodium argentothiosulphate is formed. That is why hypo is used in photography as a fixing agent.

$$AgBr + 2Na_2S_2O_3 \rightarrow Na_3Ag(S_2O_3)_2 + NaBr$$

Sodium argento thiosulphate

Uses

1. It is used in the extraction of gold and silver.
2. It is used in photography as a fixing agent with AgBr.
3. It is used in iodometric titration for the estimation of copper.
4. It is used in textile industries as antichlor to remove excess of chlorine left in bleaching.
5. It is used as an antiseptic.

Hydra Acids (HX)

All halogen atoms can react with hydrogen to give HX type of halogen acids which are covalent in nature like HF, HCl, HBr, HI.

Methods of Preparation

1. **By the combination of H_2 and X_2**

   When chlorine is burnt in excess of hydrogen, HCl is formed as follows:

   $$H_2 + Cl_2 \rightarrow 2HCl$$

   When a mixture of H_2 and Br_2 is passed over red heated platinum spiral, HBr is formed as follows:

   $$H_2 + Br_2 \xrightarrow{Pt} 2HBr$$

   When vapours of H_2 and I_2 are passed over red heated fine platinum, HI is formed as follows:

   $$H_2 + I_2 \xrightarrow{Pt,\ 450°C} 2HI$$

2. **By passing H_2S or SO_2 through aqueous solution of halogens**

   $$X_2 + H_2S \rightarrow 2HX + S$$

   $$X_2 + SO_2 + 2H_2O \rightarrow 2HX + H_2SO_4$$

3. **By the Reaction of Phosphorous, X_2 and H_2O**

   $$P_4 + 6X_2 \longrightarrow 4PX_3$$

   $$PX_3 + 3H_2O \longrightarrow 3HX + H_3PO_3$$

   Here, X_2 may be Br_2 or I_2.

4. **From metal halides:** When metal halides are heated with conc. H_2SO_4 (only for HCl) or H_3PO_4 (for HBr, HI), HX are formed as follows:

   $$KCl + H_2SO_4 \rightarrow KHSO_4 + HCl$$

   $$KHSO_4 + KCl \rightarrow K_2SO_4 + HCl$$

   $$3KX + H_3PO_4 \longrightarrow K_3PO_4 + 3HX$$

Physical Properties

1. These are colourless gases with pungent odour, except H_2F_2 or HF which is a liquid due to hydrogen bonding.
2. These are soluble in water and form constant boiling mixtures also with water i.e., azeotropes.
3. Melting and Boiling Points:

	HCl	HBr	HI
M.P. (K)	162	187	222.2
B.P. (K)	188	206	237.5

■ Some Orders Related to HX:

Acidic strength: HF < HCl < HBr < HI

Boiling point: HF > HI > HBr > HCl

Due to

H-bonding

Stability or Bond strength: HF > HCl >HBr>HI

Reducing nature: HF < HCl < HBr < HI

Non reducing

Chemical Properties

1. **Acidic Properties:** Being acidic all HX can react with metals and their oxides, hydroxides carbonates etc., to form their salts.

   For example,

   $$Ca + 2HX \longrightarrow CaX_2 + H_2$$

   $$CaO + 2HX \longrightarrow CaX_2 + H_2O$$

   $$NaOH + HX \rightarrow NaX + H_2O$$

   $$MgCO_3 + 2HX \longrightarrow MgX_2 + H_2O + CO_2$$

2. **Reaction with Ammonia:** All HX can react with ammonia to give white fumes of ammonium halides.

   $$NH_3 + HX \rightarrow NH_4X$$

3. **Precipitation Reactions**

   (i) All HX can react with silver nitrate to give precipitate of AgX.

$$AgNO_3 + HX \rightarrow AgX + HNO_3$$

AgX may be AgCl (white ppt.), AgBr (pale yellow ppt.) and AgI (yellow ppt.).

(ii) All HX can react with lead acetate to give precipitate of PbX_2.

$$(CH_3COO)_2Pb + 2HBr \rightarrow PbX_2 + 2CH_3COOH$$

PbX_2 may be $PbCl_2$ (white), $PbBr_2$ (white) and PbI_2 (yellow).

4. **With Halogens:** An upper halogen can displace lower halogen from HX i.e., F_2 can displace Cl_2, Br_2 and I_2 from HCl, HBr and HI, respectively.

$$2HX + F_2 \rightarrow 2HF + X_2$$

$$X_2 = Cl_2, Br_2, I_2$$

$$2HX + Cl_2 \longrightarrow 2HCl + X_2$$

$$X_2 = Br, I$$

5. **Reducing Properties:** Among all the HX, HI is the strongest reducing agent. It is such a strong reducing agent that its aqueous solution get oxidized even by air.

$$4HI + O_2 \longrightarrow 2H_2O + 2I_2$$

Reducing nature α Size of X

For example,

$$H_2SO_4 + 8HI \rightarrow H_2S + 4I_2 + 4H_2O$$

$$2HNO_3 + 2HI \rightarrow 2NO_2 + 2H_2O + I_2$$

$$2FeCl_3 + 2HI \longrightarrow 2FeCl_2 + I_2 + 2HCl$$

$$2CuSO_4 + 4HI \longrightarrow Cu_2I_2 + 2H_2SO_4 + I_2$$

Uses

1. HF is used in the manufacture of F_2, etching of glass, making fluorides and for removing silica from artificial graphite.
2. HCl is used in the preparation of Cl_2, aqua regia, chlorides and in the cleaning of iron sheets during tin plating and galvanization.
3. HBr is used in making of AgBr (used in photography), NaBr, KBr (sedatives etc.).
4. HI is used as a reducing agent and for making KI which is used in a number of medicines.

REMEMBER

- It is an associated molecule which is shown as H_2F_2.
- It is prepared in pure anhydrous form by heating dry KHF_2, in a platinum retort.

  $$2KHF_2 \longrightarrow K_2F_2 + H_2F_2$$
- It is a poisonous liquid, with high boiling point, viscosity and solubility in water due to hydrogen bonding.
- It can form HF_2^{-2} due to hydrogen bonding however other HX_2^{-2} are not possible due to lack of hydrogen bonding.
- It attacks even glass and silica. With silica it forms SiF_4 and H_2SiF_6.

  $$SiO_2 + 2H_2F_2 \longrightarrow SiF_4 + 2H_2O$$

  $$SiF_4 + H_2F_2 \longrightarrow H_2SiF_6$$

OXIDES OF CHLORINE

Chlorine reacts with oxygen to form a number of oxides like Cl_2O, ClO_2, Cl_2O_6 and Cl_2O_7. All these oxides are highly reactive and very unstable.

Dichloro Oxide (Cl_2O)

Preparation

- It can be prepared by passing dry chlorine over a fresh precipitate of yellow mercuric oxide as follows:

$$HgO + 2Cl_2 \text{ (dry)} \rightarrow HgCl_2 + Cl_2O$$

Physiochemical Properties

1. It is a brownish yellow gas with a penetrating odour.
2. It can be condensed to an orange coloured liquid in a freezing mixture (boiling point = 275K).
3. Decomposition: It decomposes on heating or in sunlight as follows:

$$2Cl_2O \rightarrow 2Cl_2 + O_2$$

4. With Water: It dissolves in water forming a golden yellow solution of hypochlorous acid (HClO) i.e., it is an anhydride of hypochlorous acid.

$$Cl_2O + H_2O \rightleftharpoons 2HClO$$

5. Oxidizing Action: Being a strong oxidizing agent, it oxidizes HCl to Cl_2 as follows:

$$Cl_2O + 2HCl \longrightarrow 2Cl_2 + H_2O$$

6. With Ammonia: It reacts with ammonia with an explosion as follows:

$$3Cl_2O + 10NH_3 \longrightarrow 2N_2 + 6NH_4Cl + 3H_2O$$

Structure: It has angular or V-shaped structure. Here, oxygen atom is sp^3 hybridized. Cl – O – Cl bond angle is 110.9°C and O – Cl bond length is 1.71 Å

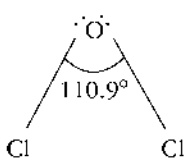


Fig. 6.12

Chlorine Dioxide (ClO_2)

Preparation

1. Pure ClO_2 can be obtained by passing dry Cl_2 over $AgClO_3$ heated to 363K as follows:

$$2AgClO_3 + Cl_2 \text{ (dry)} \longrightarrow 2AgCl + 2ClO_2 + O_2$$

2. It can also be prepared by the action of Cl_2 on sodium chlorite as follows:

$$2NaClO_2 + Cl_2 \longrightarrow 2NaCl + 2ClO_2$$

Physiochemical Properties

1. It can be condensed by cooling to a coloured liquid (boiling point 284K).
2. It is a powerful oxidizing and bleaching agent.
3. Decomposition: It explodes and decomposes to Cl_2 and O_2 by an electric spark.
4. With H_2O: It dissolves in water to give a mixture of chlorous acid and chloric acid.

$$2ClO_2 + H_2O \longrightarrow HClO_2 + HClO_3$$

5. With Alkalies: It gives a mixture of chlorite and chlorate with alkalies as follows:

$$2ClO_2 + 2KOH \longrightarrow KClO_2 + KClO_3 + H_2O$$

Structure: It is an odd electron molecule and paramagnetic in nature. It is an angular molecule with sp^3 hybridization of chlorine. Here, O – Cl – O bond angle is 118° and Cl – O bond length is 1.47 Å.

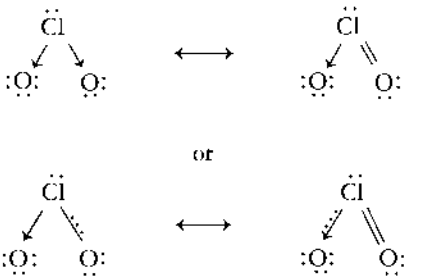


Fig. 6.13

Dichlorine Hexoxide (Cl_2O_6)

Preparation

1. It can be prepared by mixing chlorine dioxide with ozonized air at 273K.

$$2ClO_2 + 2O_3 \longrightarrow Cl_2O_6 + 2O_2$$

Physiochemical Properties

1. It is a dark red unstable liquid.
2. Decomposition: It is quite unstable and decomposes into ClO_2 and O_2.
3. It is a strong oxidizing agent.
4. With Water: It reacts with water to give chloric acid and perchloric acids.

$$Cl_2O_6 + H_2O \longrightarrow HClO_3 + HClO_4$$

5. With HF: It gives $HClO_4$ and chloryl fluoride with HF as follws:

$$Cl_2O_6 + HF \longrightarrow HClO_4 + ClO_2F$$

6. With Alkalies: It reacts with alkalies to give chlorates and perchlorates.

$$Cl_2O_6 + 2KOH \longrightarrow KClO_3 + KClO_4 + H_2O$$

Strcuture: In liquid state it is a diamagnetic molecule with uncertain structure which is not known so far.

$$2ClO_3 \rightleftharpoons Cl_2O_6$$

In the vapour state it exists as ClO_3 molecule which has odd number of electrons and hence paramagnetic

In solid state it has ClO_4^-, ClO_2^+ ions.

Although the exact structure is unknown for it yet the following structure having Cl – Cl linkage is proposed for it in which each chlorine atom is sp^3 hybridized.

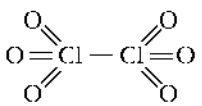


Fig. 6.14

Chlorine Heptoxide (Cl_2O_7)

Preparation

It can be formed by the dehydration of perchloric acid with P_2O_5 at 263K.

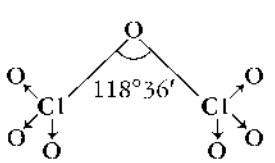
$$2HClO_4 \xrightarrow{P_2O_5} Cl_2O_7 + H_2O$$

Physiochemical Properties

1. It is a colourless oily explosive liquid.
2. With Water: It slowly dissolves in water to form perchloric acid i.e., it is an anhydride of perchloric acid.

$$Cl_2O_7 + H_2O \longrightarrow 2HClO_4$$

Structure: In Cl_2O_7 two ClO_3 units are joined through oxygen at an angle of 118.5° or 118°36′ giving tetrahedral structure. Here, Cl – O_b bond length is 1.72Å while Cl – O_t is 1.42Å.



Or

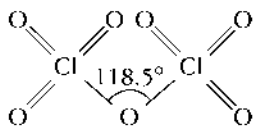


Fig. 6.15

Oxyacids of Chlorine

Chlorine forms all the four oxyacids, which are hypochlorous acid (HCl^+O), chlorous acid ($HCl^{3+}O_2$), chloric acid ($HCl^{5+}O_3$) and perchloric acid ($HCl^{7+}O_4$). It may be noted that Cl-atom is in +1, +3, +5 and +7 oxidation state, respectively in these acids.

$$\overset{+1}{HClO},\ \overset{+3}{HClO_2},\ \overset{+5}{HClO_3},\ \overset{+7}{HClO_4}$$

The acidic strength of these acids increases as follows:

$$\overset{+1}{HClO} < \overset{+3}{HClO_2} < \overset{+5}{HClO_3} < \overset{+7}{HClO_4}$$

Hypochlorous Acid (HClO)

Preparation

1. From Chlorine

(i) It can be prepared by passing Cl_2 into H_2O or into a suspension of $CaCO_3$ in water or into an aqueous solution of potassium hypochlorite (KClO) or bleaching powder ($CaOCl_2$)

$$Cl_2 + H_2O \longrightarrow HCl + HClO$$

$$Cl_2 + H_2O \longrightarrow Cl^- + ClO^- + 2H^+$$

(ii) It can be prepared by shaking Cl_2 water with freshly precipitated HgO.

$$2Cl_2 + 2HgO + H_2O \longrightarrow HgCl_2.HgO + 2HClO$$

The insoluble $HgCl_2.HgO$ is removed by filtration. The filtrate is distilled when dilute HClO is passed over it.

2. It can be prepared by the action of atmospheric CO_2 on sodium hypochlorite, NaOCl as follows:

$$NaOCl + CO_2 + H_2O \longrightarrow NaHCO_3 + HOCl$$

3. From Bleaching Powder

It can be prepared by distilling the aqueous solution of bleaching powder ($CaOCl_2$) with a calculated quantity of 5 % HNO_3 or by passing CO_2 into aqueous solution of $CaOCl_2$ and then distilling.

$$2CaOCl_2 + 2HNO_3 \rightarrow CaCl_2 + Ca(NO_3)_2 + 2HClO$$

$$CaOCl_2 + H_2O + CO_2 \longrightarrow CaCO_3 + 2HClO$$

Physiochemical Properties

1. Its dilute solution is colourless while the concentrated solution is yellow in colour.
2. It is a weak acid, even weaker than H_2CO_3 with a dissociation constant of 3×10^{-8} at 20°C.
3. Acidic Nature (Monobasic nature): It is a monobasic acid as its aqueous solution gives only one H ion on ionization.

$$HClO\ (aq) \rightleftharpoons H^-\ (aq) + ClO^-\ (aq)$$

Hypochlorite ion

Its monobasic nature shows that this molecule has one –OH group attached directly with the central Cl atom.

Being an acid, it reacts with alkalies to give salts which are called hypochlorites.

For example,

$$NaOH + HClO \longrightarrow NaOCl + H_2O$$

$$OH^- + HClO \rightarrow OCl^- + H_2O$$

4. Disproportion
   (i) The dilute solution of this acid is quite stable in the dark but when concentrated solution is exposed to light, it becomes unstable and undergoes disproportionation into Cl_2 and $HClO_3$ as follows:

   $$5HClO \rightarrow 2Cl_2 + HClO_3 + 2H_2O$$

   (ii) When aueous solution of HClO is heated, it undergoes disproportionation into HCl and $HClO_3$.

   $$3HClO \xrightarrow{\Delta} 2HCl + HClO_3$$

   $$3ClO^- \xrightarrow{\Delta} 2Cl^- + ClO_3^-$$

   (iii) On distillation it decomposes into H_2O and Cl_2O.

   $$2HClO \rightarrow Cl_2O + H_2O$$

5. Reaction with Metals: HClO reacts with Mg with the evolution of H_2.

   $$Mg + 2HClO \rightarrow Mg(OCl)_2 + H_2$$

   It reacts with Fe and Al with the evolution of H_2 and Cl_2 while with Co, Ni and Cu, Cl_2 and O_2 are evolved.

6. With Mercury: When it is shaken with Hg, a light brown precipitate of basic mercuric chloride, HgCl(OH) is formed which is soluble in HCl.

   $$2Hg + 2HClO \rightarrow 2HgCl(OH)$$

   Light brown ppt.

7. Oxidizing and bleaching properties: The aqueous solution of HClO and its salts (NaOCl) are oxidizing and bleaching agents. It is due to the fact that HClO or NaOCl decomposes to give nascent oxygen as follows:

   $$HClO \xrightarrow{\text{Reduction}} HCl + [O]$$

   $$NaOCl \xrightarrow{\text{Reduction}} NaCl + [O]$$

   $$ClO^- + 2H^+ + 2e^- \xrightarrow{\text{Reduction}} Cl^- + H_2O$$

   (Ion electron equation)

8. Action of $AgNO_3$: It reacts with $AgNO_3$ to give silver hypochlorite (AgClO) which is unstable and undergoes disproportionation into AgCl and $AgClO_3$.

   $$HOCl + AgNO_3 \rightarrow AgOCl + HNO_3$$

   $$3AgOCl \rightarrow AgCl + AgClO_3$$

Uses

It is used for bleaching paper pulp etc.

Structure: It is a linear molecule in which chlorine atoms is sp^3 hybridized.

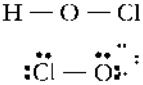


Fig. 6.16

Chlorous Acid ($HClO_2$)

Preparation

1. **From Barium hypochlorite**

   It is prepared in the aqueous solution by treating a suspension of barium hypochlorite, $Ba(ClO_2)_2$ with dil. H_2SO_4 and filtering off the precipitate of $BaSO_4$.

   $$Ba(ClO_2)_2 + H_2SO_4 \rightarrow 2HClO_2 + BaSO_4$$

2. **From ClO_2**

   It can also be obtained by the action of ClO_2 on H_2O_2.

   $$2ClO_2 + H_2O \rightarrow 2HClO_2 + O_2$$

Physiochemical Properties

1. The freshly prepared solution of chlorous acid is colourless but it soon decomposes to ClO_2 which colours the solution yellow.
2. It gives a violet colour with $FeSO_4$.
3. Decomposition: In acid solution, $HClO_2$ rapidly decomposes as follows:

   $$4HClO_2 \rightarrow 2ClO_2 + HClO_3 + HCl + H_2O$$

4. Disproportion: $HClO_2$ and its salts undergo disproportionation on heating in an alkaline solution as follows:

   $$3HClO_2 \rightarrow 2HClO_3 + HCl$$
   $$(Cl = +3) \quad (Cl = +5) \quad (Cl = -1)$$
   $$3ClO_2 \longrightarrow 2ClO_3 + Cl$$

   It undergoes autooxidation.

   $$2HClO_2 \rightarrow HClO + HClO_3$$

5. Oxidizing Properties: Both $HClO_2$ and its salts show oxidizing properties due to nascent oxygen.

   $$HClO_2 \longrightarrow HCl + 2[O]$$
   $$ClO_2 + 4H^+ + 4e \rightarrow Cl + 2H_2O$$

   For example, $HClO_2$ and its salts liberate I_2 from KI solution.

   $$HClO_2 + 2H_2O + 4KI \longrightarrow 4KOH + HCl + I_2$$
   $$4I^- + ClO_2^- + 4H^+ \longrightarrow Cl^- + 2H_2O + 2I_2$$

Structure: ClO_2 is angular in shape as chlorine atom is sp^2 hybridized.

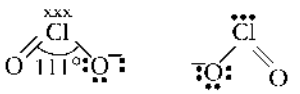


Fig. 6.17

Chloric Acid ($HClO_3$)

Preparation

1. From Barium Chlorate: It is prepared by the action of dil.H_2SO_4 on $Ba(ClO_3)_2$.

   $$Ba(ClO_3)_2 + H_2SO_4 \rightarrow \underset{\text{ppt.}}{BaSO_4} + 2HClO_3$$

   The precipitate of $BaSO_4$ is obtained by filtration. The unused H_2SO_4 is precipitated with baryta water. The filtrate is evaporated in vacuum desicator over conc. H_2SO_4 until a 40% solution of $HClO_3$ is obtained.

REMEMBER

If the solution having $HClO_3$ is evaporated further more, it gets decomposed into perchloric acid as follows:

$$3HClO_3 \longrightarrow HClO_4 + Cl_2 + 2O_2 + H_2O$$

2. From Potassium Chlorate: It is prepared by the action of hydrofluosilicic acid (H_2SiF_6) on $KClO_3$ as follows:

   $$2KClO_3 + H_2SiF_6 \longrightarrow \underset{\text{ppt.}}{K_2SiF_6} + 2HClO_3$$

Physiochemical Properties

1. The concentrated solution of the acid is a colourless and pungent smelling liquid.
2. Decomposition: It is quite stable in dark but in light it decomposes and becomes yellow. On heating it decomposes to give $HClO_4$.

   $$3HClO_3 \xrightarrow{\Delta} HClO_4 + Cl_2 + 2O_2 + H_2O$$

3. It is a powerful oxidizing and bleaching agent.
4. With Iodine: When iodine is evaporated with 25% $HClO_3$, iodic acid (HIO_3) is formed:

$$2HClO_2 + I_2 \rightarrow 2HIO_3 + Cl_2$$

Structure: Chlorate ion (ClO_3^-) is pyramidal in shape as chlorine atom is sp^3 hybridized. Here, Cl – O bond energy is 244 kJ per mol and O – Cl – O bond angle is 106°.

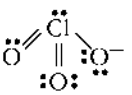


Fig. 6.18

- When organic substances like corron, wool, paper etc., come in contact with the acid, they catch fire.

Perchloric Acid ($HClO_4$)

$HClO_4$ is the strongest acid of all the acids and it is highly dangerous acid and produces severe wounds on the skin.

Preparation

1. From Chloric Acid: It is prepared by heating $HClO_3$.

$$3HClO_3 \xrightarrow{\Delta} HClO_4 + Cl_2 + 2O_2 + H_2O$$

2. From Potassium Chlorate: Anhydrous form of the acid can be obtained by distilling a mixture of potassium perchlorate ($KClO_4$) with conc. H_2SO_4 under reduced pressure.

$$KClO_4 + H_2SO_4 \longrightarrow HClO_4 + KHSO_4$$

3. From Barium Chlorate: An aqueous solution of this acid can be obtained by treating $Ba(ClO_4)_2$ with calculated quantity of dil. H_2SO_4 and then removing the insoluble $BaSO_4$ by filtration.

$$Ba(ClO_4)_2 + H_2SO_4 \longrightarrow 2HClO_4 + BaSO_4$$

4. From Ammonium Chlorate: An aqueous solution of this acid can be obtained by adding NH_4ClO_4 dissolved in conc. HCl to warm conc. HNO_3 and then evaporating as follows:

$$NH_4ClO_4 + 8HCl + 3HNO_3 \rightarrow HClO_4 + 2N_2O + 4Cl_2 + 7H_2O$$

Physiochemical Properties

1. Anhydrous $HClO_4$ is a colourless mobile hygroscopic and oily liquid.
2. It fumes strongly in moist air and dissolves in water with a hissing sound due to the liberation of much heat.
3. Formation of Hydrates: It can form hydrates with 1, 2, 2.5, 3 and 3.5 molecules of water of crystallization.
4. Decomposition: It is unstable and decomposes with explosion on heating and sometimes merely on standing for a few days even in the dark.

- Aqueous solution of the acid is quite and does not decompose and hence can be kept indefinitely.

5. Oxidizing Property: It is a powerful oxidizing agent and inflames paper and wood.
6. Dehydration: On dehydration with P_2O_5 at 263K, it gives Cl_2O_7.

$$2HClO_4 + P_2O_5 \rightarrow Cl_2O_7 + 2HPO_3$$

7. With Metals: Metals like Zn, Fe etc., dissolve in the aqueous solution of the acid to give the soluble perchlorates.

$$Zn + 2HClO_4\,(aq) \longrightarrow \underset{\text{Soluble}}{Zn(ClO_4)_2\,(aq)} + H_2$$

8. Reduction: The acid can not be reduced by nascent hydrogen but can be reduced to chloride by strong reducing agents like $SnCl_2$, $CrCl_2$ etc.
9. With Iodine: When a suspension of iodine is heated with $HClO_4$, paraperiodic acid (H_5IO_6) is obtained.

$$2HClO_4 + I_2 + 4H_2O \longrightarrow 2H_5IO_6 + Cl_2$$

Structure: Per-chlorate ion is tetrahedral in shape has Cl atom is sp^3-hybridized. Here, O – Cl – O bond angle is 109°28'.

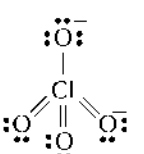


Fig. 6.19

Uses

- The aqueous solution of the acid is used for the estimation of potassium gravimetrically.

Bleaching Powder $CaOCl_2$ or $Ca(OCl)Cl$

[Calcium chlorohypochlorite or chloride of lime and calcium oxychloride]

It is a mixture of $CaOCl_2.4H_2O + CaCl_2.Ca(OH)_2.4H_2O$.

Preparation

It is prepared by passing a current of chlorine over dry slaked lime. This is possible by Bachmann method or Hasenclever method.

$$Ca(OH)_2 + Cl_2 \longrightarrow \underset{\text{Bleaching powder}}{CaOCl_2} + H_2O$$

- According to Odling's view, bleaching powder is a mixture of calcium chlorohypochlorite or chloride of lime and calcium oxychloride hence its formation can be written as

$$2Ca(OH)_2 + 2Cl_2 \rightarrow \underset{\text{Bleaching powder}}{Ca(ClO)_2} + CaCl_2 + 2H_2O$$

- According to Bunn, Clarke and Chifford's view it is a mixture of calcium hypochlorite and basic calcium chloride so its formation can be written as

$$2Cl_2 + 3Ca(OH)_2 \rightarrow \underset{\text{Bleaching powder}}{Ca(ClO)_2 + CaCl_2.Ca(OH)_2.H_2O} + H_2O$$

Physiochemical Properties

1. It is a white yellowish powder with smell of chlorine and soluble in cold water also.
2. A good quality bleaching powder contains 35 per cent – 38 per cent of chlorine out of total 55.9 per cent chlorine.
3. Cl_2 is generally obtained is less amount from bleaching powder by using moist CO_2

$$CaOCl_2 + CO_2 \longrightarrow CaCO_3 + Cl_2\uparrow$$

4. Decomposition: $CoCl_2$ acts as a catalyst to decompose bleaching powder to liberate oxygen.

$$2CaOCl_2 \xrightarrow{CoCl_2} 2CaCl_2 + O_2\uparrow$$

5. With Dilute Acids: When bleaching powder is treated with excess of dilute acids, chlorine is liberated.

$$CaOCl_2 + H_2SO_4 \rightarrow CaSO_4 + H_2O + Cl_2$$

6. Autooxidation: On standing for a long time it undergoes auto-oxidation as follows:

$$6CaOCl_2 \longrightarrow Ca(ClO_3)_2 + 5CaCl_2$$

7. Formation of Chloroform: Bleaching powder on reaction with ethyl alcohol or acetone gives chloroform (lab method).

$$CaOCl_2 + H_2O \rightarrow Ca(OH)_2 + Cl_2$$

$$H_2O + Cl_2 \longrightarrow HCl + [O]$$

$$CH_3CH_2OH \xrightarrow{[O]} CH_3CHO + H_2O$$

$$CH_3CHO \xrightarrow[-HCl]{3Cl_2} CCl_3CHO$$

$$Ca(OH)_2 + 2CCl_3CHO \xrightarrow{\Delta} 2CHCl_3 + \begin{matrix} HCOO \\ HCOO \end{matrix}\!\!>\!Ca$$

- $CH_3COCH_3 \xrightarrow[-HCl]{Cl_2} \underset{2\text{ moles}}{CCl_3COCH_3} \xrightarrow{Ca(OH)_2} 2CHCl_3 + (CH_3COO)_2Ca$

8. Oxidizing and Bleaching Action: In presence of dilute acids it looses nascent oxygen hence it can act as an oxidizing and a bleaching agent.

$$2CaOCl_2 + H_2SO_4 \longrightarrow CaCl_2 + CaSO_4 + 2HCl + 2O$$

$$\text{Coloured matter} + [O] \longrightarrow \text{Colourless matter}$$

- It oxidizes KI to I_2.

$$Ca(ClO)Cl + 2KI + 2CH_3COOH \rightarrow 2KCl + I_2 + H_2O + Ca(CH_3COO)_2$$

$$Ca(ClO)Cl + 2K + 2HCl \rightarrow 2KCl + I_2 + H_2O + CaCl_2$$

- It oxidizes PbO into PbO_2.

$$PbO + Ca(ClO)Cl \rightarrow PbO_2 + CaCl_2$$

$$PbO + ClO^- \longrightarrow PbO + Cl^-$$

- It oxidizes MnO into MnO_2.

$$MnO + Ca(ClO)Cl \rightarrow MnO_2 + CaCl_2$$

$$MnO + ClO^- \longrightarrow MnO + Cl^-$$

- It oxidizes H_2S into Sulphur.

$$Ca(ClO)Cl + H_2S \rightarrow CaCl_2 + H_2O + S$$

$$ClO^- + H_2S \rightarrow Cl^- + H_2O + S$$

- It oxidizes KNO_2 into KNO_3.

$$Ca(ClO)Cl + KNO_2 \rightarrow CaCl_2 + KNO_3$$

$$ClO^- + NO_2^- \rightarrow Cl^- + NO_3^-$$

Uses

1. It is used to purify water as a disinfectant and a germicide.
2. It is used to prepare $CHCl_3$.
3. It is used for bleaching cotton, wood pulp etc.

Estimation of Available Chlorine

The maximum percentage of available chlorine calculated by using Odling formula is 49 per cent as follows:

$$CaOCl_2.H_2O + H_2SO_4 \longrightarrow CaSO_4 + 2H_2O + Cl_2$$

145g 71g

As 145g of bleaching powder contains = 71g Cl_2

$$\text{So } 100g = \frac{71}{145} \rightarrow \times 100$$

$$= 49g\ Cl_2$$

The amount of available chlorine in a given sample of bleaching powder can be calculated volumetrically by using iodometric method or arsenite method.

Iodometric Method

Here, a weighted amount of bleaching powder is suspended in water and reacted with excess of KI and acetic acid. Here, iodine is liberated which can be estimated by treating it with a standard solution of sodium thiosulphate (hypo) using starch as an indicator.

$$CaOCl_2 + 2CH_3COOH \longrightarrow (CH_3COO)_2Ca + Cl_2 + H_2O$$

$$2KI + Cl_2 \longrightarrow 2KCl + I_2$$

$$2Na_2S_2O_3 + I_2 \rightarrow 2NaI + Na_2S_4O_6$$

Suppose V ml of N/x $Na_2S_2O_3$ be used for W g of a sample of bleaching powder.

V ml N/x $Na_2S_2O_3$ ≡ V ml N/x Iodine

= V ml N/x chlorine

$$= \frac{35.5}{\text{'x'}} \times \frac{V}{1000}\ \text{g chlorine}$$

Hence, percentage of available chlorine

$$= \frac{35.5 \times V \times 100}{\text{'x'} \times 1000 \times W} = \frac{35.5 \times V}{\text{'x'} \times W}$$

- Normally, percentage of chlorine is in between 33–38 per cent. The low availability is due to incomplete reaction between chlorine and slaked lime, impurities associated with slaked lime and decomposition of bleaching powder in air.

Example

0.5 g of bleaching powder was suspended in water and excess of KI added. On acidifying with dil. H_2SO_4, I_2 was liberated which required 50 ml of N/10 – $Na_2S_2O_3.5H_2O$ solution for exact oxidation. Calculate the percentage of available chlorine in bleaching powder.

Solution

Percentage of available chlorine in bleaching powder

$$= \frac{35.5 \times V \times 100}{\text{'x'} \times 1000 \times W}$$

$$= \frac{0.1 \times 50 \times 35.5 \times 100}{25 \times 4 \times 0.5} = 34.5\%$$

FLUORIDES OF XENON

Xenon Difluoride (XeF_2)

Preparation

1. XeF_2 is prepared when a mixture of xenon and fluorine (2:1) is passed through a nickel tube at 400°C and 1 bar.

$$Xe + F_2 \xrightarrow[\text{Ni or monel vessel}]{400^\circ C} XeF_2$$

XeF_2 is isolated by vacuum sublimation.

2. $Xe + O_2F_2 \xrightarrow{118^\circ C} XeF_2 + O_2 \uparrow$

3. $Xe + F_2 \xrightarrow[\text{Arc light}]{Hg} XeF_2$

Physiochemical Properties

1. It is a colourless, crystalline solid with a melting point of 140°C.

2. Decomposition: It is decomposed by H_2 or H_2O.

$XeF_2 + H_2 \rightarrow Xe + 2HF$ or H_2F_2

$2XeF_2 + 2H_2O \rightarrow 2Xe + 2H_2F_2 + O_2$

3. Substitution of F –arom

$$XeF_2 + HX \longrightarrow F\,Xe\,X + HF$$
$$\downarrow HX$$
$$XeX_2 + HF$$

It dissolves HF without any reaction. The solution does not have any change in conductivity also.

4. Oxidizing Agent: As in XeF_2, Xe – F bond has lowest bond energy so XeF_2 is a good oxidizing agent.

It oxidise I_2 into IF in preparation of BF_3.

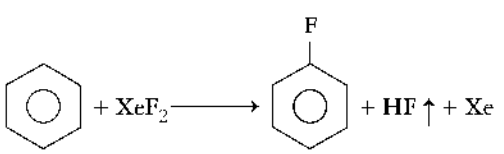
$I_2 + XeF_2 \xrightarrow{BF_3} 2IF + Xe\uparrow$

$IF + XeF_2 \xrightarrow{BF_3} IF_3 + Xe\uparrow$

5. As fluorinating Agent

$2NO + XeF_2 \rightarrow Xe + 2NOF$
Nitrosyl fluoride

$2NO_2 + XeF_2 \longrightarrow Xe + 2NO_2F$
Nitro fluoride



6. Reduction

$XeF_2 + H_2 \rightarrow Xe\uparrow + 2HF$

7. As Lewis Base: XeF_2 acts like a lewis base and form adduct with lewis acid.

For example,

$XeF_2 + 2SbF_5 \rightarrow XeF_2.2SbF_5$ or $[XeF]^+[Sb_2F_{11}]^-$ (F Donar)

$XeF_2 + IF_5 \longrightarrow [XeF]^+[IF_6]^-$

$2XeF_2 + AsF_5 \longrightarrow 2XeF_2.AsF_5 \longrightarrow [Xe_2F_3]^+[AsF_6]^-$

8. Hydrolysis in Aqueous Medium

$2XeF_2 + 4OH \rightarrow 2Xe + 4F + 2H_2O + O_2$

Structure of XeF_2: According to Infrared, Raman spectra it is a linear molecule in vapour state.

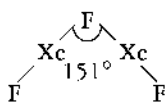
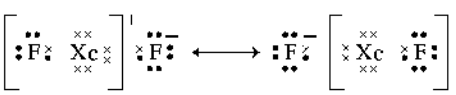


Fig. 6.20

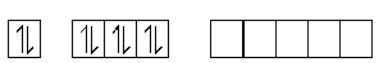
Crystal structure of solid XeF_2 shows that it has linear units. It is a resonance hybrid of these two structures.



VBT Structure: According to valence bond theory (VBT), XeF_2 is a linear molecule in which equatorial position is occupied by lone pair of electrons to minimize electron repulsion. It is explained as follows:

[Xe] In ground state:

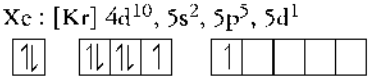
$[Kr], 4d^{10}, 5s^2, 5p^6, 5d^0$



In excited state, the configuration is as follows:

$_{54}Xe$: $[Kr]\ 4d^{10}, 5s^2, 5p^5, 5d^1$

$_{54}Xe$: $[Kr]\ 4d^{10}, 5s^2, 5p^5, 5d^1$



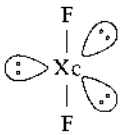


Fig. 6.21

Xenon Tetrafluoride (XeF_4)

Preparation

1. $Xe + 2F_2 \xrightarrow[5-6\ atm]{400^\circ C} XeF_4$ (1 : 5)

2. $Xe + 2F_2 \xrightarrow[2-15mm\ Hg]{-80^\circ C} XeF_4$ (1 : 2)

Physiochemical Properties

1. It is a colourless crystalline solid with a melting point of 117°C.

2. Reduction:

$XeF_4 + 2H_2 \xrightarrow{130^\circ C} Xe + 2H_2F_2$

$XeF_4 + 2Hg \longrightarrow Xe + 2HgF_2$

$3XeF_4 + 4BCl_3 \rightarrow 4BF_3 + 3Xe + 6Cl_2$

3. With H_2O:

$2XeF_4 + 3H_2O \rightarrow Xe + XeO_3 + 3H_2F_2 + F_2$

$XeF_4 + H_2O \xrightarrow{-80^\circ C} XeOF_2 + 2HF$

Xenon oxyfluoride

■ It is also soluble in CH_3COOH

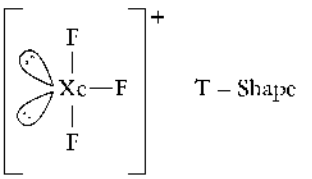
4. With Xe:

$XeF_4 + Xe \xrightarrow[Excess]{warm} 2XeF_2$

5. Formation of Adducts: It forms adduct with lewis acids like SbF_5, PF_5, AsF_5, NbF_5, RuF_5, OsF_5 etc.

$XeF_4 + SbF_5 \longrightarrow [XeF_3]^+ [SbF_6]^-$

Lewis base



T – Shape

Fig. 6.22

6. **Fluorinating Agent:** It is a better fluorinating agent than XeF_2.

$XeF_4 + 4NO \longrightarrow Xe + 4NOF$

$XeF_4 + 4NO_2 \longrightarrow Xe + 4NO_2F$

$XeF_4 + Pr \rightarrow Xe + PrF_4$

$2SF_4 + XeF_4 \rightarrow Xe + 2SF_6$

$2Hg + XeF_4 \rightarrow Xe + 2HgF_2$

7. It also dissolves TaF_5.

$XeF_4 + 2TaF_5 \longrightarrow Xe(TaF_6)$ or

$XeF_2\ 2TaF_5 + F_2$

Straw coloured

Structure: According to vibrational spectra, and electron diffraction, it has a planar structure.

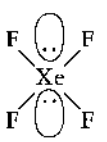
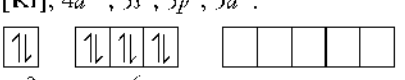


Fig. 6.23

VBT Structure: According to VBT, XeF_4 is a planar molecule in which axial position is occupied by lone pair of electrons to minimize electron repulsion. It is explained as follows:

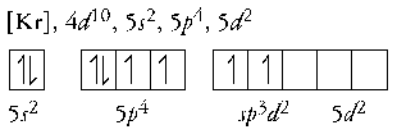
[Xe] In ground state

[Kr], $4d^{10}$, $5s^2$, $5p^6$, $5d^0$.



$5s^2$ $5p^6$ $5d$

In excited srate:



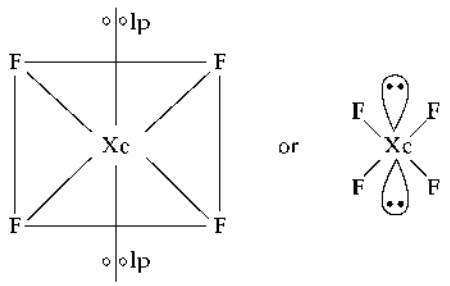


Fig. 6.24

Xenon hexafluoride (XeF_6)

Preparation

1. $Xe + 3F_2 \xrightarrow[60\ atm.]{300^\circ C} XeF_6$
   1 : 20

2. $XeF_4 + O_2F_2 \xrightarrow{-80^\circ C} XeF_6 + O_2$

3. $Xe + 3F_2 \xrightarrow[Low\ temperature]{Electrolysis} XeF_6$

   XeF_6 is most difficult to be prepared as maximum number of electron are required to be excited from 5p to 5d orbitals to prepare it i.e., large amount of energy is needed.

Physiochemical Properties

1. It is a colourless, Solid with a melting point of 49.5°C. It is very rapidly hydrolyzed and it is a strong oxidizing agent and fluorinating agent.

2. With H_2O:

   $XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$

3. Reduction:

   $XeF_6 + 3H_2 \rightarrow Xe + 6HF$

   $XeF_6 + 6HCl \longrightarrow Xe + 6HF + 3Cl_2$

   $XeF_6 + 8NH_3 \longrightarrow Xe + 6NH_4F + N_2$

4. It cannot be stored in glass or quartz as it reacts with silica.

   $2XeF_6 + SiO_2 \rightarrow 2XeOF_4 + SiF_4$

   $2XeOF_4 + SiO_2 \rightarrow 2XeO_2F_2 + SiF_4$

   $2XeO_2F_2 + SiO_2 \longrightarrow 2XeO_3 + SiF_4$

5. XeF_6 dissolve in HF.

   $XeF_6 + HF \longrightarrow [XeF_5]^- [HF_2]^-$

6. Thermal decomposition:

   $2XeF_6 \xrightarrow{\Delta} XeF_2 + XeF_4 + 3F_2$

7. Adduct formarion: It forms adduct with lewis acids as follows:

   $XeF_6 + AsF_5 \rightarrow [XeF_5]^+ [AsF_6]^-$

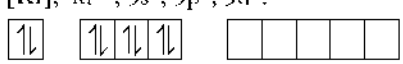
   $XeF_6 + PtF_5 \rightarrow [XeF_5]^+ [PtF_6]^-$

8. Hydrolysis: It undergoes hydrolysis in strong alkaline medium as follows:

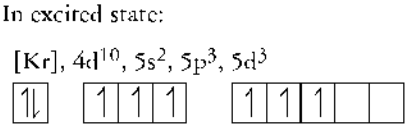
   $2XeF_6 + 16OH^- \rightarrow 8H_2O + 12F^- + XeO_6^{4-} + Xe + O_2$

Structure: According to VBT, XeF_6 has a distorted octahedral structure. It is explained as follows:

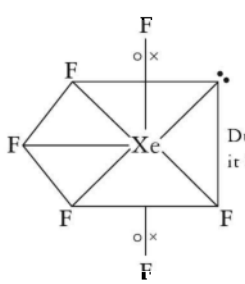
Xe In ground state:



In excited state:



sp^3d^3



Due to lone pair, (capped octahedron) it has distorted octahedron structure.

Fig. 6.25

Oxides of Xenon: Xenon forms two oxides: Xenon trioxide and Xenon tetroxide.

- **XeO_3:** It is prepared by the hydrolysis of XeF_4 or XeF_6.

$$6XeF_4 + 12H_2O \longrightarrow 2XeO_3 + 24HF + 3O_2 + 4Xe$$

$$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$$

- It is a colourless, highly explosive and a powerful oxidizing agent. It has pyramidal shape in which xenon atom is sp^3 hybridized and have one lone pair electron.

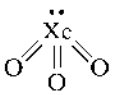


Fig. 6.26

- **XeO_4:** It is prepared by the action of anhydrous concentrated H_2SO_4 on sodium or barium per xenate at room temperature.

$$Na_4XeO_6 + 2H_2SO_4 \longrightarrow XeO_4 + 2Na_2SO_4 + 2H_2O$$

$$Ba_2XeO_6 + 2H_2SO_4 \longrightarrow XeO_4 + 2BaSO_4 + 2H_2O$$

- It is less stable than XeO_3 and decomposes into XeO_2.
- It has tetrahedral structure in which xenon atom is sp^3 hybridized.

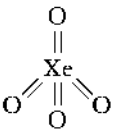
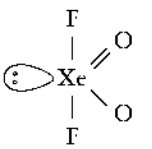


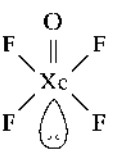
Fig. 6.27

Unforgettable Guidelines

⇒ XeO_2F_2 has Xe-atom in sp^3d hybridized state and the shape of this compound is sea saw.



⇒ $XeOF_4$ has Xe-atom in sp^3d^2 hybridized state and the shape of the molecule is square pyramidal.



Straight Objective Type Questions (Single Choice)

1. The oxidation states of the most electronegative element in the products of the reaction, BaO_2 with dil. H_2SO_4 are
   (a) 0 and −1 (b) −1 and −2
   (c) −2 and 0 (d) −2 and +1

2. The volume strength of 1.5N H_2O_2 solution is
   (a) 4.8 (b) 8.4
   (c) 4.0 (d) 8.0

3. In comparison to water, heavy water
   (a) has lower density
   (b) has lower boiling point
   (c) has lower dissociation constant
   (d) in general, reacts faster

4. Which among the following is paramagnetic?
   (a) ClO_2 (b) Cl_2O_6
   (c) Cl_2O (d) Cl_2O_7

5. Which one of the following on heating gives a mixture of SO_2 and SO_3?
   (a) $CuSO_4$ (b) $ZnSO_3$
   (c) $FeSO_4$ (d) Na_2SO_4

6. What are the products formed in the reaction of xenon hexafluoride with silicon dioxide?
   (a) $XeO_3 + SiF_2$ (b) $XeOF_4 + SiF_4$
   (c) $XeSiO_4 + HF$ (d) $XeF_2 + SiF_4$

7. Ozone reacts with dry iodine to give
   (a) I_4O_9 (b) I_2O_4
   (c) IO_2 (d) I_2O_3

8. A commercial sample of hydrogen peroxide is labeled as 10 volume, its percentage strength is
   (a) 1% (b) 3%
   (c) 10% (d) 90%

9. The catalyst used in lead chamber process is
   (a) Pt (b) V_2O_5
   (c) Cu-fillings (d) NO

10. When zeolite is treated with hard water, the hardness is reduced due to exchange of
    (a) Ca^{2-} ions of water with Na^+ ions
    (b) Ca^{2-} ions of water with H^+ ions
    (c) Cl^- ions of water with OH^- ions
    (d) SO_4^{2} ions of water with OH ions

11. The impurity "oxides of nitrogen" present in sulphuric acid obtained in lead chamber process is removed by
    (a) Passing H_2S gas
    (b) Diluting the acid with water
    (c) Treating impure sulphuric acid with $(NH_4)_2SO_4$
    (d) Both (a) and (c)

12. Excess of PCl_5 reacts with conc. H_2SO_4 giving
    (a) Sulphuryl chloride
    (b) Thionyl chloride
    (c) Sulphurous acid
    (d) Chlorosulphonic acid

13. In S_8 each sulphur atom is
    (a) sp^2 hybridized with a nonplanar ring
    (b) sp^3 hybridized with a nonplanar ring
    (c) sp^3d hybridized with a nonplanar ring
    (d) sp hybridized with a planar ring

14. Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. The explanation for it is that
    (a) strong affinity of HCl gas for moisture in air results in formation of droplets of liquid solution which appeas like a cloudy smoke.
    (b) oxygen in air reacts with the emitted HCl gas to form a cloud of chlorine gas.
    (c) due to strong affinity for water, concentrated hydrochloric acid pulls moisture of air towards itself. This moisture forms droplets of water and hence the cloud.
    (d) concentrated hydrochloric acid emits strongly smelling HCl gas all the time.

15. Phosphoric acid is preferred to sulphuric acid in the preparation of H_2O_2 from barium peroxide because
    (a) phosphoric acid is available at low cost than sulphuric acid.
    (b) phosphoric acid acts as a preservator by retarding decomposition of H_2O_2.
    (c) sulphuric acid reacts with explosion.
    (d) all of the above.

16. What products are expected from the disproportionation reaction of hypochlorous acid?
(a) $HClO_3$ and Cl_2O
(b) $HClO_2$ and $HClO_4$
(c) HCl and Cl_2O
(d) HCl and $HClO_3$

17. Low volatile nature of H_2SO_4 is due to
(a) Strong bonds
(b) Hydrogen bonding
(c) Van der Waal's forces
(d) None of these

18. Which one of the following statements is correct?
(a) H_2S is a weaker acid than H_2O.
(b) Hydrazine (N_2H_4) is a stronger base than NH_3.
(c) The bond dissociation energy of F_2 is less than that of Cl_2.
(d) Pure HBr can be prepared by the treatment of NaBr with concentrated HNO_3.

19. Which among these is an ozonide?
(a) Cr_2O_3 (b) KO_3
(c) NH_4O_3 (d) Both (b) and (c)

20. The H – O – O bond angle in H_2O_2 is
(a) 97° (b) 106°
(c) 104.5° (d) 109.28°

21. $K_4[Fe(CN)_6]$ reacts with ozone to give
(a) KNO_3 (b) $K_3[Fe(CN)_6]$
(c) Fe_2O_3 (d) $Fe(OH)_2$

22. The term thio cannot be used in the name of which of the following compound?
(a) NaSCN (b) $NaCS_3$
(c) Na_2SO_3 (d) $Na_2S_2O_3$

23. The ratio of the gases obtained on dehydration of HCOOH and $H_2C_2O_4$ by conc. H_2SO_4 is
(a) 2:1 (b) 1:2
(c) 3:1 (d) 1:3

24. The element evolving two different gases on reaction with conc. sulphuric acid is
(a) S (b) C
(c) Sn (d) P

25. Identify the incorrect statement with respect to ozone.
(a) Ozone is more reactive than oxygen.
(b) Ozone is formed in the upper atmosphere by a photochemical reaction involving dioxygen.
(c) Ozone protects the earth's inhabitants by absorbing γ-radiations.
(d) Ozone is diamagnetic whereas dioxygen is paramagnetic.

26. The reaction in the Kipp's apparatus stops on closing the outlet because
(a) Gas starts coming out from top.
(b) The acid becomes weak.
(c) The contact between sulphide and the acid is broken by the pressure of gas collected in the free surface of the middle chamber.
(d) A protective film is formed on iron sulphide.

27. The final acid obtained during the manufacture of H_2SO_4 by contact process is
(a) H_2SO_4 (dil.) (b) $H_2S_2O_7$
(c) H_2SO_4 (conc.) (d) H_2SO_4 (dil.)

28. When sulphur is boiled with Na_2SO_3 solution, the compound formed is
(a) sodium sulphide
(b) sodium sulphate
(c) sodium persulphate
(d) sodium thiosulphate

29. Sulphuric acid reacts with PCl_5 to give
(a) Sulphuryl chloride
(b) Sulphur tetrachloride
(c) Thionyl chloride
(d) Sulphur monochloride

30. Concentrated nitric acid reacts with iodine to give
(a) $HOIO_2$ (b) $HOIO_3$
(c) HOI (d) HI

31. The product obtained at anode when 50% H_2SO_4 aqueous solution is electrolyzed using platinum electrodes is
(a) H_2 (b) O_2
(c) H_2SO_3 (d) $H_2S_2O_8$

32. In the preparation of sulphuric acid, V_2O_5 is used in the reaction which is
(a) $N_2 + 3H_2 \rightarrow 2NH_3$
(b) $SO_2 + H_2O \rightarrow H_2SO_3$
(c) $2SO_2 + O_2 \longrightarrow 2SO_3$
(d) $S + O_2 \longrightarrow SO_2$

33. H_2SO_4 has a very corrosive action on skin because
(a) It acts as oxidizing agent.
(b) It acts as dehydrating agent and absorption of water is highly exothermic.
(c) It reacts with proteins.
(d) Both (a) and (b).

34. By the action of hot conc. H_2SO_4, phosphorous changes to
(a) phosphorous acid
(b) orthophosphoric acid
(c) metaphosphoric acid
(d) pyrophosphoric acid

35. Formula of iodine phosphate is
(a) $I_2(PO_4)_3$ (b) I_3PO_4
(c) I_2PO_4 (d) IPO_4

36. Chlorine water on cooling deposits greenish-yellow crystals of
(a) $Cl_2.8H_2O$ (b) $Cl_2.3H_2O$
(c) $Cl_2.2H_2O$ (d) $Cl_2.H_2O$

37. There is no S–S bond in
(a) $S_2O_4^{2-}$ (b) $S_2O_5^{2-}$
(c) $S_2O_3^{2-}$ (d) $S_2O_7^{2-}$

38. Which one is not a characteristic of ozone, O_3?
(a) The O–O distances are not equivalent.
(b) It is a nonlinear triatomic system.
(c) It is made by passing an electrical discharge through oxygen.
(d) It is an extremely powerful oxidizing agent.

39. In the manufacture of sulphuric acid by contact process, Tyndall box is used to
(a) remove impurities
(b) filter dust particles
(c) test the presence of dust particles
(d) convert SO_2 to SO_3

40. Which compound acts as an oxidizing as well as reducing agent?
(a) SO_2 (b) MnO_2
(c) Al_2O_3 (d) CrO_3

41. Conc. H_2SO_4 is diluted
(a) By adding glacial acetic acid in H_2SO_4
(b) By adding H_2SO_4 in water
(c) By adding water in H_2SO_4
(d) None of these

42. In which of the following reaction, conc. H_2SO_4 acts only as an acid?
(a) $P_4 + H_2SO_4$ ➤
(b) $Cu + H_2SO_4 \longrightarrow$
(c) $NaI + MnO_2 + H_2SO_4$ ➤
(d) $NaBr + H_2SO_4$ ➤

43. Which of the following pairs of acids are called super acids?
(a) FSO_3H and $ClSO_3H$
(b) FSO_3H and SbF_5
(c) $HClO_4$ and H_2SO_4
(d) SbF_6 and $HClO_4$

44. Which of the following is not an example of oxyacids of sulphur ?
(a) H_2SO_3 (b) SO_3
(c) H_2SO_4 (d) $H_2S_2O_3$

45. During bleaching of chlorine, an antichlor is used to
(a) Remove greases from the fibre.
(b) Liberate oxygen.
(c) Eliminate last traces of bleaching agent.
(d) Enhance bleaching action.

46. Hypo is used in photography for
(a) developing picture
(b) picture printing
(c) colouring of the picture
(d) fixation of the picture

47. Complete hydrolysis of XeF_6 forms a compound (a) which explodes violently in dry state. Here the compound (a) is
(a) XeO_2F_2 (b) XeO_3
(c) $XeOF_4$ (d) None of these

48. The structure of sulphuric acid is similar to
(a) Ascorbic acid (b) Telluric acid
(c) Selenic acid (d) Both (b) and (c)

49. SF_6 is unreactive towards water because
(a) Sulphur has very small size.
(b) Fluorine is the most electronegative element.
(c) 'S' shows the oxidation state of +6.
(d) Due to steric hindrance water molecule cannot attack on S atom in SF_6.

50. Which of the following is the strongest acid?
(a) $ClO_3(OH)$ (b) $ClO_2(OH)$
(c) $SO(OH)_2$ (d) $SO_2(OH)_2$

51. The products of the chemical reaction between $Na_2S_2O_3$, Cl_2 and H_2O are
(a) $S + HCl + Na_2SO_4$
(b) $S + NaClO_3 + H_2O$
(c) $S + HCl + Na_2S$
(d) $S + HCl + Na_2SO_3$

52. The catalysts used in the manufacture of H_2SO_4 by contact process and lead chamber process are, respectively
(a) V_2O_5 and platinum asbestos
(b) Oxides of nitrogen and V_2O_5
(c) V_2O_5 and oxides of nitrogen
(d) V_2O_5 and Cu_2Cl_2

53. How would one synthesise a perchlorate salt?
(a) Oxidation of a solution of chlorate salt by a perbromate salt.
(b) Electrolytic oxidation of a solution of hypochlorite salt.
(c) Electrolytic oxidation of a solution of chlorite salt.
(d) Electrolytic oxidation of a solution of chlorate salt.

54. In Deacon's process for the manufacture of Cl_2 the intermediate formed is
(a) $Cu_2Cl_2.Cu_2(OH)_2$
(b) Cu_2OCl_2
(c) $CuOCl_2$
(d) $Cu_2O_2Cl_2$

55. The compound which gives off oxygen on moderate heating is
(a) cupric oxide (b) mercuric oxide
(c) zinc oxide (d) aluminium oxide

56. In which of the following pairs, the two species are isostructural?
(a) SF_4 and XeF_4
(b) SO_3^{2-} and NO_3^-
(c) BF_3 and NF_3
(d) BrO_3 and XeO_3

57. Choose the correct statement regarding ozone.
(a) O_3 is a good oxidizing agent.
(b) O_3 is an unstable, dark blue diamagnetic gas.
(c) O_3 is toxic and have characteristic sharp smell.
(d) All of these.

58. Laughing gas cannot be used in which of the following?
(a) In the preparation of hydrazoic acid
(b) As a fuel for rockets
(c) As an anaesthetic
(d) As a propellant for whipped ice-cream

59. Which of the following metal nitrate does not give NO_2 on heating?
(a) $LiNO_3$ (b) $Mg(NO_3)_2$
(c) $NaNO_3$ (d) $AgNO_3$

60. Which of these statements is/are correct?
(a) Conc. HNO_3 reacts with Sn to give H_2SnO_3.
(b) Cold and highly conc. HNO_3 reacts with copper to give N_2.
(c) Cold and very dilute HNO_3 reacts with zinc to give ammonium nitrate.
(d) All are correct.

Brainteasers Objective Type Questions (Single Choice)

61. Given that hypohalous acids form by the following reaction

$$X_2(aq) + H_2O(l) \rightleftharpoons HOX(aq) + H^+(aq) + X^-(aq)$$

Which of the following changes will increase the yield of HOX?
(a) Add X^-
(b) Add water
(c) Increase the pH
(d) Decrease the pH

62. The percentage by weight of hydrogen in H_2O_2 is
(a) 6.50 (b) 5.88
(c) 25 (d) 50

63. 'S' on reaction with conc. HNO_3 gives (P) and on reaction with NaOH gives (Q). (P) and (Q) are, respectively
(a) H_2SO_4, Na_2SO_4
(b) $H_2S_2O_3$, $Na_2S_2O_3$
(c) NO_2, Na_2S
(d) H_2SO_3, $Na_2S_2O_3$

64. Consider the following statements
I. Atomic hydrogen is obtained by passing hydrogen through an electric arc.
II. Hydrogen gas will not reduce heated aluminium oxide.
III. Finely divided palladium absorbs large volume of hydrogen gas.
IV. Pure nascent hydrogen is best obtained by reacting Na with C_2H_5OH.
Which of the above statements is/are correct?
(a) I alone (b) II alone
(c) I, II, and III (d) II, III and IV

65. The oxidation states of sulphur in the anions SO_3^{2-}, $S_2O_4^{2-}$ and $S_2O_6^{2-}$ follow the order:
(a) $S_2O_6^{2-} < S_2O_4^{2-} < SO_3^{2-}$
(b) $S_2O_4^{2-} < S_2O_6^{2-} < SO_3^{2-}$
(c) $SO_3^{2-} < S_2O_4^{2-} < S_2O_6^{2-}$
(d) $S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$

66. Sulphur reacts with chlorine in 1:2 ratio and forms (X). (X) on hydrolysis gives a sulphur compound (Y). What is the hybridized state of central atom in the anion of (Y)?
(a) sp^3d (b) sp^2
(c) sp^3 (d) sp

67. Consider the following substances:
I. OF_2 II. Cl_2O III. Br_2O
The correct sequence of X–O–X bond angle is
(a) 3 > 2 > 1 (b) 1 > 2 > 3
(c) 2 > 1 > 3 (d) 1 > 3 > 2

68. When a white hygroscopic powder (X) is treated with cold water, a compound (Y) is formed which forms a white precipitate with $BaCl_2$ in acidic medium. (Y) also gives a white amorphous precipitate with neutral $AgNO_3$. Compound (Y) can also be obtained by heating H_3PO_4 here (X) and (Y), are respectively
(a) P_4O_{10} and H_3PO_2 (b) P_4O_{10} and HPO_3
(c) P_4O_6 and H_3PO_2 (d) N_2O_5 and HPO_3

69. A certain compound (X) when treated with copper sulphate solution yields a brown precipitate. On adding hypo solution, the precipitate turns white. The compound is
(a) KI (b) K_3PO_4
(c) KBr (d) K_2CO_3

70. Among the following molecules
I. XeO_3 II. $XeOF_4$ III. XeF_6
Those having same number of lone pairs on Xe are
(a) (II) and (III) only
(b) (I) and (II) only
(c) (I), (II) alnd (III) only
(d) (I) and (III) only

71. If 1000 cc of air is passed again and again over heated Cu and Mg till the reduction in volume stops the volume finally obtained would be approximately
(a) 800 cc (b) 200 cc
(c) 10 cc (d) zero cc

72. Boiling of dil. HCl acid does not increase its concentration beyond 22.2 per cent because hydrochloric acid
(a) forms a constant boiling mixture.
(b) forms a saturated solution at this concentration.
(c) is very volatile.
(d) is extremely soluble in water.

73. A compound (A) of S, Cl and O has vapour density of 67.5. It reacts with water to form two acids and reacts with KOH to form two salts (B) and (C). while (B) gives white precipitate with $AgNO_3$ solution, (C) gives white precipitate with $BaCl_2$ solution. Identify (A) here.
(a) $SOCl_2$ (b) SO_2Cl_2
(c) SO_2Cl (d) None of these

74. H_2SO_4 is not used for the preparation of CO_2 from marble chips because
(a) calcium sulphate is sparingly soluble and get deposited on marble chips and stops the reaction
(b) it does not react.
(c) the reaction is vigorous.
(d) huge amount of heat is evolved.

75. Oxalic acid is heated with concentrated H_2SO_4. When the resultant gases are passed over red hot carbon, X is obtained. X is
(a) CO (b) CO_2
(c) C_2H_4 (d) CH_4

76. In the following sequence of reaction the product formed (A) and the necessary conditions (B) are

$$PbS \xrightarrow[\text{in air}]{\text{Heat}} A + PbS \xrightarrow{B} Pb + SO_2$$

(a) $PbSO_4$ and high temperature, absence of air
(b) PbO and high temperature, absence of air
(c) PbO and low temperature
(d) both (a) and (b)

77. $KClO_3$ on reaction with SO_2 gives (X) and on reaction with conc. H_2SO_4 gives (Y). (X) and (Y) are
(a) KCl, $HClO_3$ (b) KCl, $HClO_2$
(c) KCl, $HClO_4$ (d) Cl_2, ClO_2

78. Which of the following pairs are correctly matched here?
I. Solvay process . . . . . Manufacture of sodium carbonate
II. Baeyer process . . . . . Manufacture of sulphuric acid
III. Haber process . . . . . Manufacture of ammonia
IV. Birkland-Eyde process . . . . . Manufacture of nitric acid
Select the correct answer:
(a) 1, 2 and 4 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) 2 and 3 only

79. The compound in which the number of $d\pi$-$p\pi$ bonds are equal to those present in ClO_4^-
(a) XeO_4 (b) XeF_6
(c) XeO_3 (d) XeF_4

80. Consider the following transformations:
1. $XeF_6 + NaF \rightarrow Na\,[XeF_7]$
2. $2PCl_5\,(s) \rightarrow [PCl_4]\,[PCl_6]$
3. $[Al(H_2O)_6]^{3} + H_2O \rightarrow [Al(H_2O)_5 OH]^{2} + H_3O$
Possible transformations are
(a) 1, 2 and 3 (b) 1 and 3 only
(c) 1 and 2 only (d) 2 only

81. Consider the two reactions:
I. $H_2O_2 + H_2S \rightarrow S + 2H_2O$
II. $H_2O_2 + O_3 \longrightarrow 2O_2 + H_2O$
Here, H_2O_2 acts as
(a) Reducing in both (I) and (II)
(b) Oxidizing in both (I) and (II)
(c) Oxidizing in (I) and reducing in (II)
(d) Reducing in (I) and oxidizing in (II)

82. In which of the following pairs both members show reducing property but not oxidizing property?
(a) H_3PO_2, H_2S (b) PH_3, H_2S
(c) H_2S, H_2O_2 (d) SO_2, $Na_2S_2O_3$

83. The oxidation state of xenon atom in XeF_4, $HXeO_4^-$, Na_4XeO_6 are, respectively
(a) +4, +6, +8 (b) +4, +6, +6
(c) +4, +6, +7 (d) +4, +5, +8

84. A reaction mixture was prepared at 25°C by filling a 1.0 litre nickel vessel with F_2 gas at 8 atm and Xe gas at 1.7 atm. The reaction mixture was maintained at 400°C for one hour. Then it was cooled to 25°C and the contents of the nickel container were analysed. All the xenon gas had reacted to form a solid Xe-F compound, but some of the fluorine gas had not reacted. The pressure of the F_2 gas is now 4.6 atm. What formula would you propose for the xenon–fluorine compound?
(a) XeF_2 (b) XeF_4
(c) XeF_6 (d) None of these

85. A certain compound when burnt gave three oxides. The first turned lime water milky, the second turned cobalt chloride paper pink and the third formed an aqueous solution of pH 3 nearly. The elements present in the compound are
(a) C, H, Ca (b) C, H, S
(c) C, H, Na (d) C, S, O

86. In the reaction $A + 2B + H_2O \longrightarrow C + 2D$, if A is HNO_2 and C is NH_2OH then B and D, are respectively
(a) H_2SO_4 and SO_3
(b) H_2SO_4 and H_2SO_3
(c) H_2SO_3 and H_2SO_4
(d) H_2SO_3 and SO_2

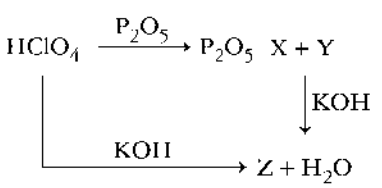
87. Arrange the acids (I) H_2SO_3, (II) H_3PO_3 and (III) $HClO_3$ in the decreasing order of acidity.
(a) I > III > II (b) I > II > III
(c) III > II > I (d) II > III > I

88. The number of sigma and pi bonds in peroxodisulphuric acid are, respectively
(a) 4 and 9 (b) 11 and 4
(c) 9 and 4 (d) 4 and 8

89. Which of the following oxyacids of chlorine is formed on shaking chlorine water with freshly precipitated yellow oxide of mercury?
(a) $HClO_4$ (b) HClO
(c) $HClO_3$ (d) $HClO_2$

90. Maximum percentage of available chlorine on the basis of $CaOCl_2.H_2O$ formula is
(a) 40 (b) 35
(c) 49 (d) 45

91. In the following sequence of reactions X, Y and Z are, respectively



(a) $KClO_3$, HPO_3 and Cl_2O_7
(b) $KClO_4$, HPO_3 and Cl_2O_7
(c) Cl_2O_7, HPO_3 and $KClO_4$
(d) $KClO_3$, H_3PO_4 and Cl_2O_7

92. Arrange the following acids in the decreasing order of acid strength.
1. H_2SO_4 2. H_3PO_4
3. $HClO_4$ 4. HNO_3
(a) 4 > 2 > 3 > 1 (b) 3 > 1 > 4 > 2
(c) 1 > 3 > 4 > 2 (d) 3 > 1 > 2 > 4

93. A yellow metallic powder is burnt in a stream of fluorine to obtain a colourless gas (A) which is thermally stable and chemically inert. Its molecule has octahedral geometry. Another colourless gas (B) with same constituent atoms as that of (A) is obtained when sulphur dichloride is heated with sodium fluoride. Its molecule has trigonal bipyramidal geometry. Gases (A) and (B) are, respectively
(a) SF_6 and SF_4 (b) SF_4 and SF_6
(c) SF_4 and S_2F_2 (d) KCl and KBr

94. CS_2 on reaction with KN_3 followed by reaction with H_2O_2 gives a pseudo halogen whose structure contains
(a) Two carbon – nitrogen bonds, the charge of each nitrogen is dispersed through resonance just on other nitrogen only.
(b) No carbon – nitrogen bonds and 6 lone pairs of electrons per molecule.
(c) No S – S bond and 12 lone pairs per molecule.
(d) One S – S bond and 14 lone pairs per molecule.

95. In analogy to $O_2^+[PtF_6]^-$ a compound $N_2^+[PtF_6]^-$ will not be formed because
(a) The I.E. of N_2 gas is higher than that of N atom
(b) The I.E. of N_2 gas is lower than that of O_2 gas
(c) The I.E. of N_2 gas is higher than that of O_2 gas
(d) none of these

96. $HClO_4 + P_2O_5 \rightarrow (P) + (Q)$
(P) and (Q) are respectively
(a) Cl_2O_6, HPO_3 (b) $HClO_3$, H_3PO_4
(c) Cl_2O_7, HPO_3 (d) ClO_2, H_3PO_4

97. When Cl_2 is passed through hot conc. NaOH, Salts (A) and (B) are formed which are separated by fractional crystallization. When (B) is treated with a calculated amount of KCl, a substance (C) is produced. When (C) is gently heated, it disproportionates to give (D) and (E). When (E) is heated with conc. H_2SO_4, the strongest bronsted acid (F) is formed which is isoelectronic with H_2SO_4. Identify F here.
(a) $HClO_2$ (b) $HClO_3$
(c) $HClO_4$ (d) HCl

98. In disulphurous acid or pyrosulphurous acid, the oxidation states of sulphur can be
(a) +6 and −2 (b) +5 and +3
(c) +6 and 0 (d) +5 and −2

99. Consider the given statements and select the set which represents only correct statements.
I. 50 per cent by mass alcoholic solution of iodine is known as tincture of iodine.
II. An oxohalide is always formed from larger halogen present during hydrolysis.

III. In lab, we use 38 per cent H_2SO_4 by mass (density 1.84 g ml^{-1}).

IV. Pure sulphuric acid on heating gives an azeotropic mixture of 98.3 per cent H_2SO_4 and 1.7 per cent water

(a) II and IV only (b) I, II and III

(c) I, II and IV (d) II, III and IV

100. Which of the following does not represent the correct order?

(a) Boiling point:

HF > HI > HBr > HCl

(b) Dipole moment:

HF > HCl > HBr > HI

(c) Gibb's free energy:

HI > HBr > HCl > HF

(d) Acidic strength:

HI > HBr > HCl > HF

Multiple Correct Answer Questions (More Than One Choice)

101. The species that undergoes disproportionation in an alkaline medium are

(a) Cl_2 (b) MnO_4^{2-}

(c) NO_2 (d) ClO_4^-

102. SO_2 will be obtained in

(a) PbS + air (b) FeS_2 + air

(c) ZnS + air (d) $H_2S + O_3$

103. Consider the following properties of the noble gases

(a) They readily form compounds which are colourless.

(b) They generally do not form ionic compounds.

(c) They have variable oxidation states in their compounds.

(d) Generally, do not form covalent compounds.

104. SO_2 behaves as

(a) Reducing agent

(b) Oxidizing agent

(c) Titrating agent

(d) Bleaching agent

105. In which of the following molecules xenon has same number of lone pair of electrons

(a) XeO_3 (b) $XeOF_4$

(c) XeF_2 (d) XeF_4

106. Pick out the correct statement/s about noble gases

(a) He cannot be used in preference to nitrogen (N_2) to dilute the oxygen in the gas cylinders used by divers.

(b) He is used in weather balloons and airships.

(c) He is used in cryoscopy to obtain the very low temperatures required for superconductivity and lasers.

(d) Ar is used in metallurgical processes.

107. HI can be prepared by which of the following methods

(a) $I_2 + H_2S$ (b) $PI_3 + H_2O$

(c) $KI + H_2SO_4$ (d) $H_2 + I_2$ Pt

108. Hydrogen peroxide.

(a) Give/s silver peroxide with moist silver oxide.

(b) Turn/s the mixture of aniline, $KClO_3$ and dil. H_2SO_4 violet.

(c) Liberate/s iodine from KI.

(d) Turn/s the titanium salt yellow.

109. In which of the following reactions, H_2O_2 acts as an oxidant?

(a) $PbO_2(s) + H_2O_2(aq) \rightarrow PbO(s) + H_2O(l) + O_2(g)$

(b) $KNO_2(aq) + H_2O_2(aq) \rightarrow KNO_3(aq) + H_2O(l)$

(c) $2KI(aq) + H_2O_2(aq) \rightarrow 2KOH(aq) + I_2(s)$

(d) $Na_2SO_3(aq) + H_2O_2(aq) \longrightarrow Na_2SO_4(aq) + H_2O(l)$

110. Which of the following is correct?
(a) H_2SO_4 forms various hydrates with H_2O.
(b) H_2SO_4 is colourless syrupy liquid.
(c) It is weakly acid and reductant also.
(d) Highly corrosive in nature k.

111. Which two of the following substances are used for preparing iodized salt?
(a) KI (b) I_2
(c) HI (d) KIO_3

112. Which one of the following reactions represents the reducing property of H_2O_2?
(a) $2NaI + H_2SO_4 + H_2O_2 \longrightarrow Na_2SO_4 + I_2 + 2H_2O$
(b) $PbO_2 + H_2O_2 \rightarrow PbO + H_2O + O_2$
(c) $2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \rightarrow K_2SO_4 + 8H_2O + 5O_2$
(d) $2K_3[Fe(CN)_6] + 2KOH + H_2O_2 \longrightarrow 2K_4[Fe(CN)_6] + 2H_2O + O_2$

113. Concentrated sulphuric acid is
(a) Oxidizing agent
(b) Efflorescent
(c) Sulphonating agent
(d) Hygroscopic

114. Which of the following statement is/are not correct?
(a) The hybridisation of Xe in XeF_4 is sp^3d^2.
(b) XeO_3 has four σ and four π bonds.
(c) In $SeCl_4$, Se atom is sp^3d^2 hybridized.
(d) In SF_4 sulphur atom has one lone pair of electron.

115. Which are correct statements for XeF_2?
(a) It oxidizes Cl^- and I^- to Cl_2 and I_2, respectively.
(b) It has linear structure.
(c) It cannot act as F^- donor.
(d) It is hydrolyzed rapidly in aqueous solution of a base.

116. XeO_3 can be prepared by
(a) $XeF_2 \xrightarrow{\text{hydrolysis}}$
(b) $XeF_6 + SiO_2 \longrightarrow$
(c) $XeF_4 \xrightarrow{\text{hydrolysis}}$
(d) $XeF_6 \xrightarrow{\text{hydrolysis}}$

117. Pick out the correct statements for XeF_6
(a) It reacts with SiO_2 to form $XeOF_4$.
(b) XeF_6 is hydrolyzed partially to form $XeOF_4$.
(c) It acts as F^- acceptor when treated with alkali metal fluoride, but cannot act as F^- donor to form complexes.
(d) On complete hydrolysis, it forms XeO_3.

118. Which of the following salts will evolve halogen on reaction with conc. H_2SO_4?
(a) NaBr (b) NaCl
(c) KI (d) $CaCl_2$

119. Which of the following is/are solid here?
(a) XeF_2 (b) XeF_4
(c) XeF_6 (d) HF

120. Which among the following are not peroxo acid of sulphur?
(a) H_2SO_4 (b) $H_2S_2O_8$
(c) H_2SO_3 (d) H_2SO_5

121. Which among the following statements is/are correct here?
(a) He and Ne do not form clathrate.
(b) XeF_4 and SbF_5 combine to form salt.
(c) He has highest boiling point in its group.
(d) He diffuses through rubber and polyvinyl chloride.

122. H_2S can be used as
(a) Oxidising agent
(b) Acid
(c) Reducing agent
(d) Amphoteric species

123. Select the correct statements about $Na_2S_2O_3.5H_2O$.
(a) It cannot used to remove stains of I_2.
(b) It is also called as hypo.
(c) It can be used as antichlor.
(d) It is used in photography to form complex with AgBr.

124. Which reagents give oxygen as one of the product during oxidation with ozone?
(a) H_2S (b) PbS
(c) SO_2 (d) $SnCl_2/HCl$

125. Which of the following pairs is/are correctly matched here
(a) The strongest oxidizing agent—Iodine.
(b) The most reactive halogen—Fluorine.

(c) A halogen which is liquid at room temperature—Bromine.

(d) The most electronegative element—Fluorine.

126. Which of these is/are not true about the oxoacids of halogens?

(a) All of them are good reducing agents.

(b) All of them are monobasic.

(c) They may be monobasic as well as polybasic.

(d) They have general formula HXO.

127. Which of the following is/are correct regarding the electrolytic preparation of H_2O_2?

(a) Sulphuric acid undergoes oxidation.

(b) Hydrogen is liberated at anode.

(c) 50 per cent H_2SO_4 is used.

(d) Lead is used as cathode.

128. Select the correct statement(s)

(a) Cl_2O and ClO_2 are used as bleaching agents and as germicides.

(b) ClO_2 is the anhydride of $HClO_2$ and $HClO_3$.

(c) I_2O_5 is used in the quantitative estimation of CO.

(d) Cl_2O_7 is anhydride of $HClO_3$.

129. XeF_6 on reaction with H_2 gives

(a) XeF_2 (b) XeF_3

(c) Xe (d) HF

130. Which of the following compounds can not be prepared by direct reaction between the constituent elements?

(a) XeO_2F_2 (b) XeO_3

(c) XeF_4 (d) XeF_6

131. Which of the following species involve M–M bond in addition to M–O bonds?

(a) Carbon suboxide

(b) Pyrosulphate ion

(c) Pyrophosphoric acid

(d) Dithionate ion

132. XeF_6 on hydrolysis gives

(a) XeO_2F_2 (b) $XeOF_4$

(c) XeO_4 (d) XeO_3

Linked-Comprehension Type Questions

Comprehension–1

An aqueous solution of a gas (P) gives the following reactions.

(i) It decolourizes an acidified $K_2Cr_2O_7$ solution.

(ii) On boiling with H_2O_2, cooling it and then adding an aqueous solution of $BaCl_2$, a precipitate insoluble in dil. HCl is obtained.

(iii) On passing H_2S in the solution, white turbidity is obtained.

133. Here the gas (P) is

(a) Cl_2 (b) N_2O

(c) SO_2 (d) H_2O_2

134. Which is correct about the gas (P)?

(a) It can act as an oxidant as well as a reductant.

(b) It has bleaching action.

(c) It has a linear shape.

(d) Both (a) and (b).

135. Here the colour of potassium dichromate becomes

(a) Orange to colourless

(b) Orange to green

(c) Orange to yellow

(d) Pink to green

Comprehension–2

Xenon atom has larger size and lower ionization energy in comparison to other inert gases (He, Ne, Ar, Kr). The outermost energy orbit has d-orbitals. The paired electrons of valence orbit can be unpaired and the electrons are shifted to d-orbitals under suitable conditions. The unpaired electrons are shared by fluorine or oxygen atoms and covalent bonds are formed.

136. Which of the following fluoride of xenon is not possible?

(a) XeF (b) XeF_3

(c) XeF_4 (d) Both (a) and (b)

137. Which is not correctly matched here?

(a) XeO_2F_2 See-saw

(b) XeF_6 — Distorted octahedral

(c) XeO_3 tetrahedral

(d) XeF_4 — Planar

138. In which of the following compound the number of lp and bp electrons are not given correct, respectively

(a) XeF_4 (2 and 4)

(b) XeO_3 (1 and 3)

(c) XeO_2F_2 (2 and 3)

(d) $XeOF_4$ (1 and 6)

Comprehension–3

When $Fe_2(SO_4)_3$ is heated, gas (X) is evolved along with a brown residue. Gas (X) has triangular structure. At room temperature, gas (X) is a solid which exists in three distinct forms. An aqueous solution of (X) gives a white precipitate, with $BaCl_2$ solution. (X), however does not decolourize acidified $KMnO_4$ or $K_2Cr_2O_7$ solution. When (X) is reacted with urea, a solid (Y) is formed which is used for cleaning the plant of sugar refineries and breweries.

139. Here the gas (X) and brown residue are, respectively of

(a) SO_2 and Fe_2O_3

(b) SO_3 and Fe_2O_3

(c) SO_3 and FeO

(d) SO_2 and Fe_3O_4

140. Which form of (X) is more stable?

(a) Alpha form (b) Beta form

(c) Gamma form (d) All are equally stable

141. Here, the solid compound (Y) is

(a) $(NH_2)_3SO_3$ (b) NH_2SO_3H

(c) $(NH_2)_2SO_4$ (d) $(NH_4)_2SO_3H$

142. (X) can act as a

(a) Lewis acid

(b) Oxidant

(c) Reductant

(d) Both (a) and (b)

Comprehension–4

A certain compound (M) shows the following reactions.

(I) When KI is added to an aqueous suspension of (M) containing acetic acid, iodine is liberated.

(II) When CO_2 is passed through an aqueous suspension of (M), the turbidity transforms to a precipitate.

(III) When the paste of (M) in water is heated with ethyl alcohol, a product of anaesthetic use is obtained.

143. Here the compound (M) is ..... and in it oxidation number of chlorine atom are

(a) $CaCl_2$ with oxidation number of chlorine as −1.

(b) $CaOCl_2$ with oxidation number of chlorine as −1 and +1.

(c) $Ca(OCl)_2$ with oxidation number of chlorine as +1.

(d) $Ca(ClO_3)_2$ with oxidation number of chlorine as +5.

144. The reaction of compound (M) with KI shows its

(a) Reducing nature

(b) Oxidising nature

(c) Basic nature

(d) Both (b) and (c)

145. Here, the compound formed that can be used as an anaesthetic will be

(a) Laughing gas (b) Chloroform

(c) Chloral (d) Calcium hydroxide

Comprehension–5

The noble gases have closed shell electronic configuration and are monoatomic gases under normal conditions. The low boiling points of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions.

The direct reaction of xenon with fluorine leads to a series of compounds with oxidation numbers +2, +4 and +6. XeF_4 reacts violently with water to give XeO_3. The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valence shell.

[*IIT 2007*]

146. Argon is used in arc welding because of its

(a) low reactivity with metal.

(b) ability to lower the melting point of metal.

(c) flammability.

(d) high calorific value.

147. The structure of XeO_3 is
(a) linear (b) planar
(c) pyramidal (d) T-shaped

148. XeF_4 and XeF_6 are expected to be
(a) reducing (b) oxidizing
(c) unreactive (d) strongly basic

Assertion and Reasoning Questions

In the following question two statements (Assertion) A and Reason (R) are given. Mark
(a) If A and R both are correct and R is the correct explanation of A.
(b) If A and R both are correct but R is not the correct explanation of A.
(c) A is true but R is false
(d) A is false but R is true

149. (A): SO_3 has a planar structure.
(R): S atom in SO_3 is sp^2 hybridized and O–S–O bond angle is 120°.

150. (A): Fresh stain of iodine on the cloth can be removed by washing it with sodium thiosulphate solution.
(R): Sodium thiosulphate reduces iodine to colourless iodide ion.

151. (A): OF_2 is named as oxygen difluoride.
(R): In OF_2, oxygen is less electronegative than fluorine.

152. (A): Reaction of SO_2 and H_2S in the presence of Fe_2O_3 catalyst gives elemental sulphur.
(R): SO_2 is a reducing agent.

153. (A): In SO_2, the bond angle is 119° whereas in SO_3, the bond angle is 120°.
(R): S atom in both SO_2 and SO_3 is sp^2 hybridized.

154. (A): $SeCl_4$ does not have a tetrahedral structure.
(R): Se in $SeCl_4$ has two lone pairs.

155. (A): Hydrofluoric acid is weaker acid than boric acid.
(R): Higher the ionization constant of acid, stronger is the acid.

156. (A): Ozone is a powerful oxidizing agent in comparison to oxygen.
(R): Ozone is diamagnetic but oxygen is paramagnetic.

157. (A): An acidified aqueous solution of $KClO_3$ when boiled with iodine produces KIO_3.
(R): $KClO_3$ is an oxidizing agent while KIO_3 is not.

158. (A): Salts of ClO_3^- and ClO_4^- are well known.
(R): F is more electronegative than O, while Cl is less electronegative than O.

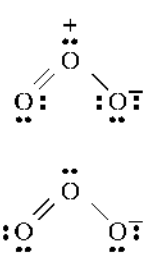
159. (A): H_2SO_4 is more viscous than water.
(R): In H_2SO_4 the S atom exhibit its highest oxidation state.

160. (A): Marshall acid has sulphur atom in +6 oxidation state.
(R): Marshall acid contains 1-peroxy group.

161. (A): Sulphur (IV) oxide can act as oxidizing as well reducing agent.
(R): In sulphur (IV) oxide sulphur assumes sp^3d hybrid state.

162. (A): In aqueous solution SO_2 reacts with H_2S liberating sulphur.
(R): SO_2 is an effective reducing agent.

163. (A): The electronic structure of O_3 is



(R): Structure is not allowed because octet around O cannot be expanded.

164. (A): Concentrated H_2SO_4 reacts with KCl to give Cl_2 gas.
(R): HCl cannot be oxidized by concentration H_2SO_4

165. (A): The O–O bond length in H_2O_2 is larger than that of O_2F_2.
(R): H_2O_2 is an ionic compound.

166. (A): In SO_2, the bond angle is 119° whereas in SO_3, the bond angle is 120°.
(R): S atom in both SO_2 and SO_3 is sp^2 hybridized.

167. (A): Ozone reacts with BaO_2 to give BaO.
(R): Ozone acts as an oxidizing agent in this reaction.

168. (A): Xenon forms fluorides.
(R): Because 5-d orbitals are available for valence shell expansion.

Matrix–Match Type Questions

	p	q	r	s
(A)	O	O	O	O
(B)	O	O	O	O
(C)	O	O	O	O
(D)	O	O	O	O

169. Match the following:

Column I	Column II
A. H_3PO_4	(p) Can react with $AgNO_3$
B. HO_3	(q) In water softener
C. $H_4P_2O_7$	(r) sp^3 hybridization
D. H_3PO_2	(s) Monobasic

170. Match the following:

Column I (Manufacturing process)	Column II (Catalyst used)
A. Deacon's process for chlorine	(p) Finely divided iron with molybdenum as promoter
B. Hydrogenation of vegetable oils	(q) Copper (II) chloride
C. Ostwald's process for nitric acid	(r) Finely divided nickel powder
D. Haber's process for ammonia	(s) Platinum gauze

171. Match the following:

Column I	Column II
A. H_2SO_5	(p) S–S linkage
B. $H_2S_2O_8$	(q) +6 oxidation state
C. $H_2S_2O_6$	(r) Peroxy linkage
D. $H_2S_2O_3$	(s) −2 oxidation state

172. Match the following:

Column I	Column II
A. XeF_2	(p) One lone pair electron on Xe-atom
B. XeF_4	(q) sp^3d
C. XeF_6	(r) sp^3d^2
D. $XeOF_4$	(s) sp^3d^3
	(t) +6 oxidation state of xenon

173. Match the following:

Column I	Column II
A. HF	(p) Liquid
B. HCl	(q) Gas
C. HBr	(r) Reducing agent
D. HI	(s) Chocking smell

174. Match the following:

Column I	Column II
A. P_4O_6	(p) White crystalline solid
B. P_2O_5	(q) Waxy solid with garlic smell
C. N_2O_3	(r) Blue coloured liquid
D. N_2O_5 (s)	(s) Coordinate bonds

175. March the following:

Column I	Column II
A. XeF_4	(p) Distorted octahedral
B. XeF_6	(q) Tetrahedral
C. XeO_3	(r) Square planar
D. XeO_4	(s) Pyramidal

176. Match the following:

Column I	Column II
A. Aqueous solution of NaOCl	(p) Anhydrone
B. $KClO_3$	(q) Javelle water
C. $Mg(ClO_4)_2$	(r) Euchlorine
D. Cl_2 and ClO_2 mixture	(s) Berrhelot's salt

177. March the following:

Column I	Column II
A. 10 volumes	(p) 5.358 N
B. 20 volumes	(q) 3.036 % (w/v)
C. 30 volumes	(r) 3.4 gm H_2O_2/100 ml solution
D. 11.2 volumes	(s) 1.785M

178. Match the following:

Column I	Column II
A. Peroxide	(p) C_3O_2
B. Superoxide	(q) PbO_2
C. Dioxide	(r) KO_2
D. Suboxide	(s) H_2O_2
	(r) RbO_2

179. March the following:

Column I	Column II (No. of lone pair electron)
A. XeF_2	(p) 0
B. XeF_4	(q) 1
C. XeF_6	(r) 2
D. XeO_3	(s) 3

The IIT–JEE Corner

180. Hydrolysis of one mole of peroxodisulphuric acid produces
(a) two moles of sulphuric acid.
(b) two moles of peroxomonosulphuric acid.
(c) one mole of sulphuric acid and one mole of peroxomonosulphuric acid.
(d) one mole of sulphuric acid, one mole of peroxomonosulphuric acid and one mole of hydrogen peroxide.
[*IIT 1996*]

181. The following acids have been arranged in the order of decreasing acid strength. Identify the correct order
(I) ClOH, (II) BrOH, (III) IOH
(a) I > II > III (b) II > I > III
(c) III > II > I (d) I > III > II
[*IIT 1996*]

182. Sodium thiosulphate is prepared by
(a) reducing Na_2SO_4 solution with H_2S.
(b) boiling Na_2SO_3 solution with S in alkaline medium.
(c) neutralizing $H_2S_2O_3$ solution with NaOH.
(d) boiling Na_2SO_3 solution with S in acidic medium.
[*IIT 1996*]

183. The critical temperature of water is higher than that of O_2 because the H_2O molecule has
(a) fewer electrons than oxygen.
(b) two covalent bonds.
(c) V-shape.
(d) dipole moment.
[*IIT 1997*]

184. The geometry of H_2S and its dipole moment are
(a) angular and non-zero
(b) angular and zero

(c) linear and nonzero
(d) linear and zero

[*IIT 1999*]

185. Which statement about H_2S is false?
(a) It is a covalent compound.
(b) It is a gas with bad smell.
(c) It is a weak base in water.
(d) It is a stronger reducing agent than H_2O.

[*IIT 2000*]

186. Which of the following ions does not have S–S linkage?
(a) $S_2O_6^{2-}$ (b) $S_2O_8^{2-}$
(c) $S_2O_5^{2}$ (d) $S_2O_3^{2}$

[*IIT 2000*]

187. Molecular shapes of SF_4, CF_4, and XeF_4 are
(a) The same, with 2, 0, and 1 lone pairs of electrons, respectively.
(b) The same, with 1, 1, and 1 lone pairs of electrons, respectively.
(c) Different, with 0, 1, and 2 lone pairs of electrons, respectively.
(d) Different, with 1, 0, and 2 lone pairs of electrons, respectively.

[*IIT 2000*]

188. The correct order of acidic strength is
(a) $Cl_2O_7 > SO_2 > P_4O_{10}$
(b) $CO_2 > N_2O_5 > SO_3$
(c) $Na_2O > MgO > Al_2O_3$
(d) $K_2O > CaO > MgO$

[*IIT 2000*]

189. The number of S–S bonds in sulphur trioxide trimer (S_3O_9) is
(a) Three (b) Two
(c) One (d) Zero

[*IIT 2000*]

190. The set with correct order of acidity is
(a) $HClO < HClO_2 < HClO_3 < HClO_4$
(b) $HClO_4 < HClO_3 < HClO_2 < HClO$
(c) $HClO < HClO_4 < HClO_3 < HClO_2$
(d) $HClO_4 < HClO_2 < HClO_3 < HClO$

[*IIT 2001*]

191. The acid having O–O bond is
(a) $H_2S_2O_3$ (b) $H_2S_2O_6$
(c) $H_2S_2O_8$ (d) $H_2S_4O_6$

[*IIT 2004*]

192. Total number of lone pair of electrons in $XeOF_4$ is
(a) 0 (b) 1
(c) 2 (d) 3

[*IIT 2004*]

193. The gas O_3 (ozone) can not oxidize
(a) KI (b) $FeSO_4$
(c) $KMnO_4$ (d) K_2MnO_4

[*IIT 2005*]

194. Which species has the maximum number of lone pair of electrons on the central atom?
(a) $[I_3]$ (b) XeF_4
(c) SF_4 (d) $[ClO_3]^-$

[*IIT 2005*]

195. The percentage of p-character in the orbitals forming P–P bonds in P_4 is
(a) 25 (b) 50
(c) 33 (d) 75

[*IIT 2007*]

196. Aqueous solution of $Na_2S_2O_3$ on reaction with Cl_2 gives
(a) NaOH (b) $NaFSO_4$
(c) $Na_2S_4O_6$ (d) NaCl

[*IIT 2008*]

ANSWERS

Straight Objective Type Questions

1. (b)	2. (b)	3. (c)	4. (a)
5. (c)	6. (b)	7. (a)	8. (c)
9. (d)	10. (a)	11. (c)	12. (a)
13. (b)	14. (a)	15. (b)	16. (d)
17. (b)	18. (b)	19. (d)	20. (a)
21. (b)	22. (c)	23. (b)	24. (b)
25. (c)	26. (c)	27. (b)	28. (d)
29. (a)	30. (a)	31. (d)	32. (c)
33. (b)	34. (b)	35. (d)	36. (a)
37. (d)	38. (a)	39. (c)	40. (a)
41. (b)	42. (c)	43. (b)	44. (b)
45. (c)	46. (a)	47. (b)	48. (c)
49. (d)	50. (a)	51. (a)	52. (c)
53. (d)	54. (b)	55. (b)	56. (d)
57. (d)	58. (b)	59. (c)	60. (d)

BRAINTEASERS OBJECTIVE TYPE QUESTIONS

61. (c)	62. (b)	64. (c)	65. (d)
66. (c)	67. (a)	69. (a)	70. (c)
71. (c)	72. (a)	74. (a)	75. (a)
76. (d)	77. (c)	79. (c)	80. (a)
81. (c)	82. (b)	84. (a)	85. (b)
86. (c)	87. (c)	89. (b)	90. (c)
91. (c)	92. (b)	94. (d)	95. (c)
96. (c)	97. (c)	99. (c)	100. (c)

Multiple Correct Answer Questions

101. (a), (c)	102. (a), (b), (c)
103. (b), (c)	104. (a), (b), (d)
105. (a), (b)	106. (b), (c), (d)
107. (a), (b), (d)	108. (b), (c), (d)
109. (b), (c), (d)	110. (a), (b), (d)
111. (a), (d)	112. (b), (c), (d)
113. (a), (c), (d)	114. (b), (c)
115. (a), (b), (d)	116. (b), (c), (d)
117. (a), (b), (d)	118. (a), (c)
119. (a), (b), (c)	120. (a), (c)
121. (a), (b), (d)	122. (b), (c)
123. (b), (c), (d)	124. (a), (b)
125. (b), (c), (d)	126. (a), (c), (d)
127. (a), (b), (d)	128. (a), (b), (c)
129. (c), (d)	130. (a), (b)
131. (a), (d)	132. (a), (b), (d)

Comprehension–1

133. (c) 134. (d) 135. (b)

Comprehension–2

136. (d) 137. (c) 138. (c)

Comprehension–3

139. (b) 140. (a) 141. (b) 142. (d)

Comprehension–4

143. (b) 144. (d) 145. (b)

Comprehension–5

146. (a) 147. (c) 148. (b)

Assertion and Reasoning Questions

149. (a) 150. (a) 151. (a) 152. (b)
153. (b) 154. (c) 155. (d) 156. (b)
157. (a) 158. (b) 159. (b) 160. (a)
161. (c) 162. (c) 163. (a) 164. (d)
165. (c) 166. (b) 167. (c) 168. (a)

Matrix–Match Type Questions

169. (a)-(p), (r), (b)-(p), (q), (c)-(p), (r), (d)-(p), (r), (s)

170. (a)-(q), (b)-(r), (c)-(s), (d)-(p)

171. (a)-(q), (r), (b)-(q), (r), (c)-(p), (s), (d)-(p), (s)

172. (a)-(q), (b)-(r), (c)-(p), (s), (r), (d)-(p), (r), (r)

173. (a)-(p), (s), (b)-(q), (r), (s), (c)-(q), (r), (d)-(q), (r)

174. (a)-(q), (b)-(p), (s), (c)-(r), (d)-(p), (s)

175. (a)-(r), (b)-(p), (c)-(s), (d)-(q)

176. (a)-(q), (b)-(s), (c)-(p), (d)-(r)

177. (a)-(q), (b)-(s), (c)-(p), (d)-(r)

178. (a)-(s), (b)-(r), (r), (c)-(q), (d)-(p)

179. (a)-(s), (b)-(r), (c)-(q), (d)-(q)

The IIT–JEE Corner

180. (c) 181. (a) 182. (b) 183. (d)
184. (a) 185. (c) 186. (b) 187. (d)
188. (a) 189. (d) 190. (a) 191. (c)
192. (b) 193. (d) 194. (a) 195. (d)
196. (b)

HINTS AND EXPLANATIONS

Straight Objective Type Questions

1. $BaO_2 + H_2SO_4 \rightarrow BaSO_4\downarrow + H_2O_2$

2. Volume strength of H_2O_2
= normality × 5.6

Volume strength of 1.5 N H_2O_2 = 1.5 × 5.6
= 8.4

3. Dissociation constant: Water = 1.0×10^{-14} and heavy water = 0.3×10^{-14}.

4. ClO_2 contains 7 + 12 i.e., 19 electrons (valence) which is an odd number, i.e., there is (are) free electron (s). Hence, it is paramagnetic in nature.

7. $9O_3 + 2I_2 \longrightarrow I_4O_9 + 9O_2$

10. When zeolite is treated with hard water the hardness is reduced due to exchange of Ca^{2+} or Mg^{2+} ions with Na ions.

$$Na_2Al_2Si_2O_8.\times H_2O + Ca^{2+} \rightarrow CaAl_2Si_2O_8.\times H_2O + 2Na^+$$

or

$$Na_2Al_2Si_2O_8.\times H_2O + Mg^{2+} \longrightarrow MgAl_2Si_2O_8\times H_2O + 2Na^+$$

12. Excess of PCl_5 reacts with conc. H_2SO_4 to give sulphuryl chloride.

$$SO_2(OH)_2 + 2PCl_5 \rightarrow SO_2Cl_2 + 2POCl_3 + 2HCl$$

Sulphuryl acid — Sulphuryl Chloride

14. Concentrated hydrochloric acid produces vapours of HCl. As HCl has strong affinity for moisture, it yields the formation of droplets of liquid solution which appears like a cloudy smoke.

15. It liberates H_2O_2 and also acts as a preservator by retarding decomposition.

17. H bonding in H_2SO_4 makes it a viscous liquid.

19. KO_3 and NH_4O_3 are ozonide.

23. $HCOOH \xrightarrow{H_2SO_4} H_2O + CO$

$H_2C_2O_4 \xrightarrow{H_2SO_4} H_2O + CO + CO_2$

The ratio is 1:2.

24. $C + 2H_2SO_4 \longrightarrow CO_2 + 2H_2O + 2SO_2$

Here, carbon is oxidized to CO_2 and H_2SO_4 is reduced to SO_2.

27. $H_2SO_4 + SO_3 \longrightarrow \underset{\text{Oleum}}{H_2S_2O_7}$

28. $Na_2SO_3 + S \longrightarrow \underset{\text{Sodium thiosulphate}}{Na_2S_2O_3}$

29. $HO - SO_2 - OH + PCl_5 \longrightarrow Cl - SO_2 - OH + POCl_3 + HCl$

$HO - SO_2 - OH + 2PCl_5 \longrightarrow \underset{\text{Sulphuryl chloride}}{Cl - SO_2 - Cl} + 2POCl_3 + 2HCl$

33. The corrosive action is due to dehydration of skin.

34. $P_4 + 10\,H_2SO_4 \longrightarrow \underset{\text{Orthophosphoric acid}}{4H_3PO_4} + 10SO_2 + 4H_2O$

37. In $S_2O_7^{2-}$, no S–S bond is present.

40. SO_2 has an oxidation state of +4 for S. Its oxidation state can increase as well as decrease. Therefore, it can act both as oxidizing agent as well as reducing agent.

41. Concentrated H_2SO_4 is diluted by adding the conc. H_2SO_4 in the water drop-by-drop with constant stirring because it is an exothermic reaction and by doing so the heat is generated slowly and dissipated in the atmosphere.

43. SbF_5 and $FeSO_3H$ are super acids.

44. SO_3 is an oxide of sulphur.

46. Hypo removes undecomposed AgBr as a soluble complex and thus image is fixed.

$AgBr + 2Na_2S_2O_3 \longrightarrow \underset{\text{Soluble complex}}{Na_3[Ag(S_2O_3)_2]} + NaBr$

47. XeF_6 on hydrolysis produces XeO_3 which explodes.

48. H_2SO_4 is similar to selenic acid (H_2SeO_4) in structure.

50. More is the electronegativity and oxidation number of the central atom (Cl), more is the acidic character.

55. Heavy metal oxides (like HgO, Pb_3O_4, Ag_2O, MnO_2) give O_2 gas on heating.

$2HgO \xrightarrow{\Delta} 2Hg + O_2$

Brainteasers Objective Type Questions

62. $H_2O_2 \longrightarrow 2H$;

34g 2g

34g of H_2O_2 gives 2g of hydrogen

100g of H_2O_2 will give hydrogen

65. Oxidation states of sulphur in $S_2O_4^{2-}$ = +3, SO_3^{2-} = +4, and $S_2O_6^{2-}$ = +5. Hence, the correct order is $S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$

66. $S + 2Cl_2 \longrightarrow SCl_4$ (X)

$SCl_4 + 4H_2O \longrightarrow S(OH)_4 + 4HCl$

$\downarrow$

$H_2SO_3 + H_2O$

(Y)

The hybridized state of S in (Y) is sp^3.

67. $\underset{103^\circ}{F-O-F}$ $\underset{111^\circ}{Cl-O-Cl}$ $\underset{>110^\circ}{Br-O-Br}$

68. $\underset{(X)}{P_4O_{10}} + \underset{\text{Cold}}{2H_2O} \longrightarrow \underset{(Y)}{4HPO_3}$

$\underset{(Y)}{2HPO_3} + BaCl_2 \longrightarrow \underset{\text{White ppt.}}{Ba(PO_3)_2} + 2HCl$

$\underset{(Y)}{HPO_3} + AgNO_3 \longrightarrow \underset{\text{White ppt.}}{AgPO_3} + HNO_3$

$2H_3PO_4 \xrightarrow[-H_2O]{250^\circ C} H_4P_2O_7 \xrightarrow[-H_2O]{600^\circ C} 2HPO_3$

69. KI reacts with $CuSO_4$ solution to produce cuprous iodide (white ppt.) and I_2 (which gives brown colour) iodine reactions with hypo solution, decolourization of solution shows the appearance of white ppt.

$$2CuSO_4 + 4KI \longrightarrow 2K_2SO_4 + 2CuI + I_2$$
Cuprous iodide white ppt. — (brown colour) in solution

$$2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$$
Sodium tetra thionate — colourless

70. XeO_3 has one lone pair on Xe atom.

$XeOF_4$ also have one lone pair on Xe atom.

XeF_6 also have one lone pair on Xe atom.

71. If 1000 cc of air is passed over heated Cu and Mg the reactive gases like N_2 and O_2 are consumed by reacting with these metals.

$$2Cu + O_2 \longrightarrow 2CuO$$

$$3Mg + N_2 \longrightarrow Mg_3N_2$$

The volume consumed will be approximately 99 per cent (nitrogen 78 per cent and oxygen 21 per cent) and remaining 1 per cent gases will be argon and other noble gases.

Therefore, the obtained volume

$$= 1000 \times \frac{1}{100} = 10 \text{ cc}$$

73. (X) is sulphuryl chloride of vapour density 67.5 and molecular weight 135.

$$\underset{(A)}{SO_2Cl_2} + 2H_2O \longrightarrow 2HCl + H_2SO_4$$

$$SO_2Cl_2 + 4KOH \longrightarrow \underset{(B)}{2KCl} + \underset{(C)}{K_2SO_4}$$

$$KCl + AgNO_3 \longrightarrow AgCl\downarrow + KNO_3$$

$$K_2SO_4 + BaCl_2 \longrightarrow BaSO_4\downarrow + 2KCl$$

74. Marble and H_2SO_4 is not used for preparation of CO_2 because $CaSO_4$ is sparingly soluble and get deposited on marble chips and stops the reaction.

75. $$H_2C_2O_4 \xrightarrow[-H_2O]{H_2SO_4} CO + CO_2 \longrightarrow CO$$

CO_2 is reduced by C to CO.

79. Both ClO_4^- and XeO_3 contain 3 $d\pi$-$p\pi$ bonds.

84. If RT/V is same, $p \propto n$

Moles Xe = 1.7 and moles F_2
$= 8.0 - 4.6 = 3.21$

Moles Xe : moles F_2

1.7 : 3.4

1 : 2

Hence, the formula of the compound is XeF_2.

85. The oxides are CO_2, H_2O and SO_2, respectively.

86. $$\underset{(A)}{HNO_2} + \underset{(B)}{2H_2SO_3} + H_2O \longrightarrow \underset{(C)}{NH_2OH} + \underset{(D)}{2H_2SO_4}$$

87. When the electronegativity and the oxidation state of the nonmetal is higher, the acid is stronger. Now the oxidation state of Cl in $HClO_3$ and P in H_3PO_3 both are +5, therefore $HClO_3$ is more acidic than H_3PO_3 as Cl is more electronegative than P. H_2SO_3 is weakest of all because S is in +4 oxidation state.

88. Peroxodisulphuric acid ($H_2S_2O_8$) contains 11 sigma and 4 pi-bonds.

89. $$HgO + 2Cl_2 + H_2O \longrightarrow HgCl_2 + 2HClO$$

90. $$\underset{145}{CaOCl_2.2H_2O} \longrightarrow \underset{71}{Cl_2}$$

$$\% \frac{71}{145} \times 100 = 49$$

95. The I.E. of N_2 gas (1503 kJmol^{-1}) is higher than that of O_2 gas (1175 kJmol^{-1}) and it cannot lose its electron, so easily as O_2 lose in forming $O_2[PtF_6]$ compounds.

96. $$2HClO_4 + P_2O_5 \longrightarrow \underset{(P)}{Cl_2O_7} + \underset{(Q)}{2HPO_3}$$

97. $$3Cl_2 + 6NaOH \longrightarrow \underset{(A)}{5NaCl} + \underset{(B)}{NaClO_3} + 3H_2O$$

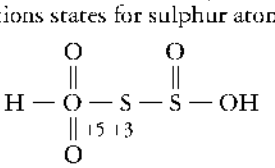
$$NaClO_3 + KCl \longrightarrow \underset{(C)}{KClO_3} + NaCl$$

+5 −1 +7

$$2KClO_3 \xrightarrow{\Delta} \underset{(D)}{2KCl} + \underset{(E)}{KClO_4}$$

$$\underset{(E)}{KClO_4} + H_2SO_4 \longrightarrow \underset{(F)}{HClO_4} + KHSO_4$$
(F): Perchloric acid

98. Disulphurous acid is $H_2S_2O_5$ having +5 and +3 oxidatations states for sulphur atom.



99. Here, only statement (III) is wrong as in lab we use 98 % by mass sulphuric acid (density 1.84 g ml^{-1}).

100. As the correct order of Gibb's free energy (ΔG°_{f}): HF > HCl > HBr > HI

Multiple Correct Answer Questions

101. Both Cl_2 and NO_2 undergo disproportionation in alkaline medium. The reactions are

$$Cl_2 + 2NaOH \longrightarrow NaCl + NaOCl + H_2O$$

$$2NO_2 + 2NaOH \longrightarrow NaNO_2 + NaNO_3 + H_2O$$

107. As H_2SO_4 is a strong oxidizing agent so it will oxidize HI to iodine.

$$2HI + H_2SO_4 \longrightarrow SO_2 + I_2 + 2H_2O$$

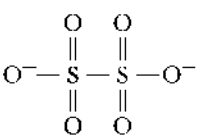
109. As in reaction (A) it act as a reductant in which PbO_2 is reduced to PbO.

112. As in reaction (A) it is an oxidant and oxidizes NaI into I_2.

131. The formula of carbon suboxide is as follows:

O – C – C – C – O (C_3O_2)

The formula of dithionate ion is as follows:



Linked–Comprehension Type Questions

Comprehension–1

133. Here, step (III) indicates that (P) will be SO_2.

$$SO_2 + 2H_2S \longrightarrow \underset{\text{Turbidity}}{3S} + 2H_2O$$

134. SO_2 can act both as an oxidant as well as a reductant and it has temporary bleaching action also.

135. $K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + \underset{\text{Colourless solution}}{Cr_2(SO_4)_3} + H_2O$

Comprehension–2

136. As by the unpairing of one paired orbital, two singly occupied orbitals are generated. Hence, either two or four or six singly occupied orbitals will be formed as a result only XeF_2, XeF_4, XeF_6 can be formed and not XeF, XeF_3 and XeF_5.

Comprehension–3

139. $Fe_2(SO_4)_3 \xrightarrow{BaCl_2 \text{ solution}} BaSO_4\downarrow$

$Fe_2(SO_4)_3 \xrightarrow{\text{Heat}} \underset{\text{Brown}}{Fe_2O_3} + \underset{(X)}{3SO_3}$

141. $2SO_3 + 2H_2O + NH_2{-}CO{-}NH_2 \longrightarrow NH_2{-}CO{-}NH_3 + 2HSO_4$

$NH_2 - CO - NH_2 + H_2S_2O_7 \longrightarrow \underset{\substack{(Y)\\ \text{Sulphanic acid}}}{2NH_2SO_3H} + CO_2$

Comprehension–4

144. The compound (M) is bleaching powder ($CaOCl_2$).

$$CaOCl_2 + 2CH_3COOH + 2KI \longrightarrow Ca(CH_3COO)_2 + 2KCl + H_2O + I_2$$

145. $Ca(OCl)Cl + H_2O \longrightarrow Ca(OH)_2 + Cl_2$

$C_2H_5OH + Cl_2 \longrightarrow CH_3CHO + HCl$

$CH_3CHO + 3Cl_2 \longrightarrow CCl_3CHO + 3HCl$

$2CCl_3CHO + Ca(OH)_2 \longrightarrow 2CHCl_3 + (HCOO)_2Ca$

Comprehension–5

146. Argon is used mainly to provide an inert atmosphere in high temperature metallurgical (arc welding of metals/alloys) extraction.

148. $6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$

$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$

Assertion and Reasoning Questions

152. Although SO_2 can act both like a reductant and an oxidant but this reaction is because of oxidizing nature of SO_2.

155. Hydrofluoric acid is stronger acid than boric acid.

156. Ozone is a powerful oxidizing agent as it can easily liberate nascent oxygen.

157. $KClO_3 + I_2 + H \rightarrow KIO_3 + Cl_2$

$+7 \qquad\qquad 0$

Here, $KClO_3$ works as an oxidizing agent.

158. Both assertion and reason are correct but does not give correct explanation. FO_3 and FO_4 are nonexistent due to absence of d-orbitals.

162. Assertion is correct, reason is wrong because SO_2 is an effective reducing as well as oxidizing agent.

164. Concentrated H_2SO_4 reacts with KCl to give HCl gas.

165. The O–O bond length is shorten in O_2F_2 than in H_2O_2 due to higher electronegativity. Moreover, H_2O_2 is a non-ionic compound.

166. S atom in both SO_2 and SO_3 is sp^2 hybridized but it contains a lone pair of electrons in SO_2.

167. Ozone acts as a reducing agent in this reaction.

The IIT–JEE Corner

180. $HO - SO_2 - O - O - SO_2 - OH + H_2O \rightarrow$

Peroxodisulphuric acid

$HOSO_2OH + HO - O - SO_2OH$

Sulphuric acid — permonosulphuric acid

181. Higher the electronegativity of the halogen, more easily it will withdraw shared O – H bond electron pair towards itself and hence stronger is the acid.

Thus, the order is

ClOH (I) > BrOH (II) > IOH (III).

182. $Na_2SO_3 + 1/8\ S_8 \xrightarrow[\text{Alkaline}]{373\ K} Na_2S_2O_3$

183. This is because water molecule has dipole moment.

184. The geometry is angular and it has nonzero dipole moment because the two individual dipoles add up.

185. H_2S acts as a weak acid in water and not as a weak base.

186. $S_2O_8^{2-}$ ion has no S–S linkage.

187. SF_4 has trigonal bipyramidal geometry (sp^3d hybridization) with 1 lone pair of electrons.

CF_4 is tetrahedral (sp^3 hybridization) with no lone pair of electrons.

XeF_4 is square planar (sp^3d^2) in shape with 2 lone pair of electrons (octahedral geometry).

188. Acidic strength increases as the electronegativity of the atom increases i.e.,

$Cl_2O_7 > SO_2 > P_4O_{10}$.

189. There is no S–S bond.

190. The order of acidic nature increases with the increase in the oxidation state of halogen.

195. P is sp^3 hybridized in P_4.

196. Sodium thiosulphate , $Na_2S_2O_3$ gets oxidized by chlorine water

$$Na_2S_2O_3 + 4Cl_2 + 5H_2O \longrightarrow 2NaHSO_4 + 8HCl$$

$Na_2S_2O_3$ gets oxidized by $FeCl_3$ into $Na_2S_4O_6$

$$2Na_2S_2O_3 + 2FeCl_3 \longrightarrow Na_2S_4O_6 + 2FeCl_2 + 2NaCl$$

Further, $Na_2S_2O_3$ decomposes on treatment with dilute acids

$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S + H_2O$

Solved Subjective Questions

1. State with balanced equations what happens when:

(i) Sodium chlorate reacts with sulphur dioxide in dilute sulphuric acid medium.

Solution

$NaClO_3 + SO_2 + 10H^+ \rightarrow NaCl + S + 5H_2O$

(ii) Sodium iodate is added to a solution of sodium bisulphate.

Solution

$2NaIO_3 + 5NaHSO_3 \rightarrow 2Na_2SO_4 + 3NaHSO_4 + I_2 + H_2O$

2. Complete and balance the following equations:

(i) $H_2SO_4 + HI \rightarrow ....+ .....+ .....$

(ii) $CaOCl_2 + NaI + HCl \rightarrow ...+ CaCl_2 + H_2O + NaCl$

(iii) $Ag_2S + 2CuCl_2 + 2Hg \rightarrow ... + ...+ S + 2Ag$

Solution

(i) $H_2SO_4 + 2HI \rightarrow I_2 + SO_2 + 2H_2O$

(ii) $CaOCl_2 + 2NaI + 2HCl \rightarrow I_2 + CaCl_2 + H_2O + 2NaCl$

(iii) $Ag_2S + 2CuCl_2 + Ag \rightarrow Cu_2Cl_2 + Hg_2Cl_2 + 2Ag + S$

3. What happens when:

(i) Hydrogen sulphide is bubbled through an aqueous solution of sulphur dioxide.

Solution

Here, H_2S oxidizes into S.

$SO_2 + 2H_2S \rightarrow 3S + 2H_2O$

(ii) Sulphur dioxide gas, water vapour and air are passed over heated sodium chloride.

Solution

$SO_2 + H_2O + ½\ O_2 \rightarrow H_2SO_4$

$H_2SO_4 + 2NaCl \rightarrow Na_2SO_4 + 2HCl$

(iii) Potassium permanganate is reacted with warm solution of oxalic acid in the presence of sulphuric acid.

Solution

$2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5[O]$

$5H_2C_2O_4 + 5[O] \rightarrow 10\ CO_2 \uparrow + 5H_2O$

(iv) Iodate ion reacts with bisulphite ion to liberate iodine.

Solution

$2\ IO_3^- + 5HSO_3^- \rightarrow I_2 + H_2O + 3HSO_4^- + 2SO_4^{2}$

(v) Dilute nitric acid is slowly reacted with metallic tin.

Solution

$4Sn + dil.\ 10HNO_3 \rightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$

4. March the following choosing one item from Column X and the appropriate item from Column Y:

X	Y
A. SO_2Cl_2	(i) Paramagnetic
B. Ice	(ii) Refrigeration
C. $CuSO_4$(anhydrous)	(iii) Testing NH_3
D. $K_2HgI_4 + NaOH$	(iv) Testing H_2O
E. Fluorocarbons	(v) Hydrogen bonding
F. NO	(v) Tetrahedral

Solution

X	Y
A. SO_2Cl_2	(i) Tetrahedral
B. Ice	(ii) Hydrogen bonding

C.	$CuSO_4$(anhydrous)	(iii) Testing H_2O
D.	$K_2HgI_4 + NaOH$	(iv) Testing NH_3
E.	Fluorocarbons	(v) Refrigeration
F.	NO	(v) Paramagnetic

5. Arrange the following in

(i) Increasing bond strength

HCl, HBr, HF, HI

Solution

HI < HBr < HCl < HF

(ii) HOCl, $HOClO_2$, $HOClO_3$, HOClO in increasing order of thermal stability.

Solution

HOCl < HOClO < $HOClO_2$ < $HOClO_3$

(iii) CO_2, N_2O_5, SiO_2, SO_3 in the order of increasing acidic character.

Solution

$SiO_2 < CO_2 < N_2O_5 < SO_3$

6. Complete the following chemical equations.

(a) $KI + Cl_2 \rightarrow$

(b) $KClO_3 + I_2 \rightarrow$

Justify the formation of the products in the above reactions.

[*IIT 1996*]

Solution

(a) $2KI + Cl_2 \rightarrow 2KCl + I_2$

As Cl_2 is more powerful oxidizing agent than I_2, therefore Cl_2 displaces I^- to form I_2.

(b) $2KClO_3 + I_2 \rightarrow 2KIO_3 + Cl_2$

Here, ClO_3^- is more powerful oxidizing agent than IO_3^-, so Cl is displaced by I.

7. Write down the balanced equations for the reactions when

(i) A mixture of potassium chlorate, oxalic acid and sulphuric acid is heated.

Solution

$KClO_3 + 2H_2C_2O_4 + H_2SO_4 \rightarrow$

$KHSO_4 + HCl + 6CO_2 + 3H_2O$

(ii) Ammonium sulphate is heated with a mixture of nitric oxide and nitrogen dioxide.

Solution

$(NH_4)_2SO_4 + NO + NO_2 \rightarrow 2N_2 + 3H_2O + H_2SO_4$

8. A certain compound (X) shows the following reactions.

(i) When KI is added to an aqueous suspension of (X) containing acetic acid, iodine is liberated.

(ii) When CO_2 is passed through an aqueous suspension of (X), the turbidity transforms to a precipitate.

(iii) When the paste of (X) in water is heated with ethyl alcohol, a product of anesthetic use is obtained.

Identify (X) and write down chemical equations for reactions at step (i), (ii) and (iii).

Solution

The compound (X) is bleaching powder ($CaOCl_2$).

(i) $CaOCl_2 + 2CH_3COOH + 2KI \rightarrow$

$Ca(CH_3COO)_2 + 2KCl + H_2O + I_2$

(ii) $CaOCl_2 + CO_2 \rightarrow CaCO_3 + Cl_2$

(iii) $Ca(OCl)Cl + H_2O \rightarrow Ca(OH)_2 + Cl_2$

$C_2H_5OH + Cl_2 \rightarrow CH_3CHO + HCl$

$CH_3CHO + 3Cl_2 \rightarrow CCl_3CHO + 3HCl$

$2CCl_3CHO + Ca(OH)_2 \rightarrow 2CHCl_3 + (HCOO)_2Ca$

9. An aqueous solution of a gas (X) gives the following reactions:

(i) It decolourizes an acidified $K_2Cr_2O_7$ solution.

(ii) On boiling with H_2O_2, cooling it and then adding an aqueous solution of $BaCl_2$, a precipitate insoluble in dil. HCl is obtained.

(iii) On passing H_2S in the solution, white turbidity is obtained. Identify (X) and give equations for steps (i), (ii) and (iii).

Solution

Step (iii) indicates that (X) will be SO_2.

$SO_2 + 2H_2S \rightarrow 3S + 2H_2O$

Turbidity

(i) $K_2Cr_2O_7 + H_2SO_4 + SO_2 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$

Colourless solution

(ii) $H_2O_2 + SO_2 \rightarrow H_2SO_4$

$H_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2HCl$

White ppt.

10. Write balanced equations for the reactions of the following compounds with water.
(i) Al_4C_3 (ii) CaNCN
(iii) BF_3 (iv) NCl_3
(v) XeF_4

[*IIT 2002*]

Solution

(i) $Al_4C_3 + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_4\uparrow$

(ii) $CaNCN + 3H_2O \rightarrow CaCO_3\downarrow + 2NH_3$

Here, ammonia formed dissolved in water to form NH_4OH

$CaNCN + 5H_2O \rightarrow 2NH_4OH + CaCO_3\downarrow$

(iii) $4BF_3 + 3H_2O \rightarrow 3HBF_4 + B(OH)_3$

(iv) $NCl_3 + 3H_2O \rightarrow NH_3 + 3HOCl$

(v) $3XeF_4 + 6H_2O \rightarrow$

$XeO_3 + 2Xe + 3/2\ O_2 + 12HF$

11. Identify the following:

$$Na_2CO_3 \xrightarrow[\text{excess}]{SO_2} A \xrightarrow{Na_2CO_3} B$$

$$\xrightarrow{\text{Elemental S, }\Delta} C \xrightarrow{I_2} D$$

Also, mention the oxidation state of S in all the compounds.

[*IIT 2003*]

Solution

$Na_2CO_3 + 2SO_2 + H_2O \rightarrow \underset{(A)}{2NaHSO_3} + CO_2$

$2NaHSO_3 + Na_2CO_3 \rightarrow \underset{(B)}{2Na_2SO_3} + H_2O + CO_2$

$Na_2SO_3 + S \rightarrow \underset{(C)}{2Na_2S_2O_3}$

$2Na_2S_2O_3 + I_2 \rightarrow \underset{(D)}{2Na_2S_4O_6} + 2NaI$

Oxidation states of S are: +4 in (A) and (B), +6 and −2 in (C) and +2.5 in (D)

12. (a) Give an example of oxidation of one halide by another halogen, Explain the feasibility of the reaction.
(b) Write the M.O., electron distribution of O_2. Specify its bond order and magnetic property.

[*IIT 2000*]

Solution

(a) $2KI(aq.) + Cl_2 \rightarrow 2KCl(aq) + I_2$

In the reaction Cl_2 oxidizes iodide ion (−1 oxidation state) to I_2 (0 oxidation state). Cl_2 has higher reduction potential than I_2 and thus oxidizes iodide to iodine getting itself reduced to chloride ion. Similarly,

$2KI + X_2$ (X=Cl, Br, F) $\rightarrow 2KX + I_2$

(b) M.O. configuration of O_2:

$\sigma 1S^2$, σ^*1s^2, $\sigma 2s^2$, σ^*2s^2, $\sigma 2p^2$, $\pi 2p^2x$, $\pi 2p^2y$, $\pi^*2p^1x\pi^*2p^1y$

Bond order = ½ [No. of bonding electrons − No. of antibonding electrons]

$$= \frac{1}{2}[10 - 6] = 2$$

Also, O_2 molecule is paramagnetic as it possesses two unpaired electrons.

13. Explosive substance toluene Intermediate

$$\xleftarrow{\text{Conc. } HNO_3} \quad (B)$$

Brown coloured gas of pungent odour

Identify (A), (B), (C) and (D). Give the reaction for (A) → (B) and (A) → (C)

[*IIT 2005*]

Solution

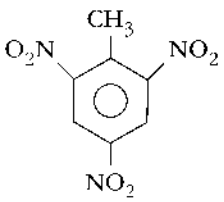
Here

(A) Conc. H_2SO_4

(B) Br_2

(C) NO_2^+

(D)



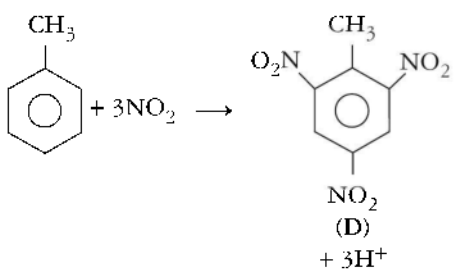
$$2NaBr + 2H_2SO_4 + MnO_2 \rightarrow Na_2SO_4 + MnSO_4 + Br_2\uparrow + 2H_2O$$

(B)

Brown fumes

$$H_2SO_4 + HNO_3 \rightarrow HSO_4^- + NO_2^+ + H_2O$$

(C)



14. Thionyl chloride can be synthesized by chlorinating SO_2 using PCl_5. Thionyl chloride is used to prepare anhydrous ferric chloride starting from its hexahydrated salt. Alternatively, the anhydrous ferric chloride can also be prepared from its hexahydrated salt by treating with 2,2-dimethoxypropane. Discuss all this using balanced chemical equation.

[*IIT 1998*]

Solution

$$SO_2 + PCl_5 \rightarrow SOCl_2 + POCl_3$$

$$FeCl_3.6H_2O + 6SOCl_2 \rightarrow FeCl_3 + 6SO_2 + 12HCl$$

$$FeCl_3.6H_2O + 6CH_3C(OCH_3)_2.CH_3 \rightarrow FeCl_3 + 12CH_3OH + 6CH_3COCH_3$$

15. In the contact process for industrial manufacture of sulphuric acid some amount of sulphuric acid is used as a starting material. Explain briefly. What is the catalyst used in the oxidation of SO_2?

[*IIT 1999*]

Solution

The SO_2 is obtained by burning sulphur in air as,

$$S + O_2 \rightarrow SO_2$$

The SO_2 so obtained is impure. Dust present in sulphur is removed by allowing the gas to expand, when some dust settles, down by passage through electrostatic precipitators and finally washing with water. The moistened gas is now treated with conc. H_2SO_4 to dry it which is kept in use until its concentration falls to 94 %.

The SO_2 so obtained then catalytically converted to SO_3.

$$2SO_2 + O_2 \xrightarrow{\text{Pr. asbestos or}} 2SO_3$$

Finely divided Pt. deposited on $MgSO_4$
or Fe_2O_3 and CuO
or Pt. –silica gel

Questions For Self-Assessment

16. Complete and balance the following equations.

(i) $H_2S + H_2SO_4$(conc.) $\rightarrow$.....+ .....+.....

(ii) NaOH (excess) + $I_2 \rightarrow$ ....+.....+ H_2O

(iii) NH_3 + NaOCl $\rightarrow$ ....+ NaCl + H_2O

Solution

(i) $H_2S + H_2SO_4$ (conc.) $\rightarrow 2H_2O + SO_2 + S$

(ii) NaOH (excess) + $I_2 \rightarrow$ 5NaI+$NaIO_3$ +3 H_2O

(iii) $2NH_3 + 3NaOCl \rightarrow N_2 + 3NaCl + 3H_2O$

17. (i) In the preparation of hydrogen iodide from alkali iodides, phosphoric acid is preferred to sulphuric acid.

(ii) Anhydrous HCl is a bad conductor of electricity but aqueous HCl is a good conductor.

18. (i) Sulphur dioxide is a more powerful reducing agent in an alkaline medium than in acidic medium.

[*IIT 1992*]

(ii) In the contact process for industrial manufacture of sulphuric acid, some amount of sulphuric acid is used as a starting material. Explain briefly. What is the catalyst used in the oxidation of SO_2?

[*IIT 1999*]

19. Draw the molecular structures of XeF_2, XeF_4 and XeO_2F_2 indicating the location of lone pairs of electrons.

[*IIT 2000*]

20. Compound (X) on reduction with $LiAlH_4$ gives a hydride (Y) containing 21.72 per cent hydrogen along with other products. The compound (Y) reacts with air explosively resulting in boron trioxide. Identify (X) and (Y). Give balanced reactions involved in the formation of (Y) and its reaction with air. Draw the structure of (Y).

[*IIT 2001*]

21. (i) Show with equations how the following compound is prepared (equations need not be balanced): Sodium thiosulphate from sodium sulphite

Hint

$Na_2SO_3 + S \rightarrow Na_2S_2O_3$

(ii) Give an example of oxidation of one halide by another halogen. Explain the feasibility of the reaction.

[*IIT 2000*]

Hint

$Cl_2 + 2KBr$ (or $2KI$) $\rightarrow 2KCl + Br_2$ (or I_2)

TRANSITION ELEMENTS AND COORDINATION CHEMISTRY

7

Chapter Contents

Definition, general characteristics, oxidation states and their stabilities; Colour (excluding the details of electronic transitions) and calculation of spin only magnetic moment; Coordination compounds nomenclature of mononuclear coordination compounds (linear, tetrahedral, square planar and octahedral) and various levels of multiple-choice questions.

Table 7.1

	IIIB	IVB	VB	VIB	VIIB		VIII		IB	IIB
3d Series	$_{21}Sc$	$_{22}Ti$	$_{23}V$	$_{24}Cr$	$_{25}Mn$	$_{26}Fe$	$_{27}Co$	$_{28}Ni$	$_{29}Cu$	$_{30}Zn$
4d Series	$_{39}Y$	$_{40}Zr$	$_{41}Nb$	$_{42}Mo$	$_{43}Tc$	$_{44}Ru$	$_{45}Rh$	$_{46}Pd$	$_{47}Ag$	$_{48}Cd$
5d Series	$_{57}La$	$_{72}Hf$	$_{73}Ta$	$_{74}W$	$_{75}Re$	$_{76}Os$	$_{77}Ir$	$_{78}Pt$	$_{79}Au$	$_{80}Hg$
6d Series	$_{89}Ac$	$_{104}Rf$								

TRANSITION ELEMENTS

- Those elements which have their last valence electron in d orbital or those elements which have two outermost shells incomplete are called d-block elements.
- They are called transition elements as their properties lie in between the properties of s and p block elements.
- Here, the first transition d-series is 3d series having ten elements from $_{21}Sc$ to $_{30}Zn$.
- Here, the second transition d-series is 4d series having ten elements from $_{39}Y$ to $_{48}Cd$.
- Here, the third transition d-series is 5d series having ten elements from $_{57}La$, $_{72}Hf$ to $_{80}Hg$.

- Here, the fourth transition d-series is 6dseries having elements from $_{89}Ac$ onwards.
- Here, IIIB and IIB group elements are nontypical transition elements.

Electronic Configuration

The general electronic configuration of transition elements is $(n-1)\ d^{1-10},\ ns^{1-2}$.

In case of 3d-Series

- $3d^{1-10},\ 4s^{1-2}$

For example,

- $_{21}Sc$: (Ar) $3d^1\ 4s^2$
- $_{24}Cr$: (Ar), $3d^5,\ 4s^1$
- $_{29}Cu$: (Ar), $3d^{10},\ 4s^1$
- $_{30}Zn$: (Ar) $3d^{10}\ 4s^2$

In Case of 4d-Series

- $4d^{1-10},\ 5s^{0-2}$

Some Exceptional Configurations

- $_{41}Nb$: [Kr], $4d^4,\ 5s^1$
- $_{42}Mo$: [Kr], $4d^5,\ 5s^1$
- $_{44}Ru$: [Kr], $4d^7,\ 5s^1$
- $_{45}Rn$: [Kr], $4d^8,\ 5s^1$
- $_{46}Pd$: [Kr], $4d^{10},\ 5s^0$
- $_{47}Ag$: [Kr], $4d^{10},\ 5s^1$

Reason of Exceptional Configuration: It is due to nuclear–electron and electron – electron forces present in these atoms.

In case of 5d Series

- $4f^{0-14}$, $5d^{1-10}$, $6s^{1-2}$
- $_{78}Pt$: $[_{54}Xe]$, $4f^{14}$, $5d^9$, $6s^1$
- $_{79}Au$: $[_{54}Xe]$, $4f^{14}$, $5d^{10}$, $6s^1$

Physiochemical Properties

Metallic Character and Related Properties

- As in transition elements the penultimate shell of electrons is expanding hence they have many common physiochemical properties.
- All transition elements are metals with metallic lustrure due to the low ionization energy values. All of these are generally hard, ductile, malleable, brittle and form alloys.
- These are good conductors of heat and electricity due to the presence of free or mobile electrons. For example, Cu, Ag, Au have high conductivity.
- They are hard due to strong metallic bonding.

  Hardness ∝ Metallic bond strength

  ∝ No. of unpaired electron

  For example, Cr, Mo W are extremely hard due to maximum number of unpaired electrons or stronger covalent bonding.

  Zn, Cd, Hg are not very hard as no unpaired electron is present in them.

  Hg is sofr and liquid due to weakest metallic bond.

Atomic Volume and Density

These elements have low atomic volume but high density than s- or p-block elements because of more nuclear pull as nuclear charge is weekly screened.

Volume decreases from IIIB to VIII and then increases from IB to IIB. Density increases from IIIB to VIII then decreases.

Reason: It is due to small radius and close packed structure of atoms of these elements.

Density increases from top to bottom. Density of IIIrd series is nearly double when compared to IInd series as the atomic weight is nearly double. Most of these elements have a density greater than $5g/cm^3$

Sc ~ $3g/cm^3$

y ~ $4.5g/cm^3$.

For example,

Ti	Zr	Hf
4.5	6.9	13.2 g/cm³

- Iridium has maximum density in the periodic table (22.61 g/ml) followed by Osmium (22.57 g/ml).

Melting Point and Boiling Point

These elements have high values of B.P., M.P. Here, tungsten has highest M.P. (3410°C) among these metals.

Melting point and boiling point ∝ Metallic bond strength.

∝ Number of unpaired electrons

- Melting point and boiling point increases from IIIB to VIII B then decrease due to decrease in number of unpaired electrons.
- Melting point and boiling point increase from top to bottom except in IB , IIB groups.

  W > Mo > Cr

  Cu > Ag > Au
- Zn, Cd and Hg have relatively lower melting points than the other d-block elements as they have completely filled (n 1) d-orbitals, their atoms are not expected to form covalent bonding amongst themselves.

Atomic Radii

In a Period: The value of atomic radii first decreases with increase of atomic number and then becomes almost constant and finally increases slightly.

IIIB ——— VII B ⟵ VIII B ⟶ IB IIB

Decrease in size ⟶ constant Increase slightly.

- Slight increase in radii in IA and IIB is due to pairing of electrons in d-orbitals i.e., more e – e repulsion occurs hence $Z_{eff.}$ decreases.

For example,

Sc	Ti	V	Cr	Mn	Fe
1.44	1.32	1.22	1.17	1.17	1.17
Co	Ni	Cu	Zn		
1.16	1.15	1.17	1.25Å^0		

In a Group: Atomic radii increases from top to bottom, however, sizes of 4d and 5d series elements are nearly same in groups due to lanthanide contraction.

Lanthanide Contraction: It is a steady or very little decrease in size due to poor screening effect of 4f electrons because of more diffused shape of f-orbitals.

Ionic Radii

Ionic size of cations with same oxidation number decreases in a period by the increase of atomic number.

Ti^{+2}	>	V^{-2}	>	Cr^{+2}	>	Mn^{+2}	>	Fe^{-2}	>
90		88		84		80		76	

Co^{-2}	>	Ni^{-2}	>	Cu^{+2}
74		72		69 pm

- When oxidation numbers are different, ionic size decreases with the increase in oxidation number. $Ti^{+2} > Ti^{+3} > Ti^{+4} > Ir^{+2} > Cr^{+3} > Cr^{+4} > Cr^{+5}$ ---------- $Cu^{-} > Cu^{+2}$ etc.

In a Group: Ionic radii increases from top to bottom.

$La^{+3} > Y^{+3} > Sc^{+3}$

- 4d, 5d same group cations will have nearly same ionic size due to lanthanide contraction.

Ionization Potential

The ionization energy of these elements is more than that of s-block elements but less than that of p-block elements.

- In a period on moving from left to right, there is a slow increase in ionization potential value as Zeff is reduced by shielding effect of (n – 1) d electrons.

For example,

V < Cr < Mn < Fe < Co < Ni < Cu < Zn

- Element of groupt IIB will have the highest value of ionization potential here.
- The value of ionization energy for Cr and Cu are comparatively higher because of $3d^5$ and $3d^{10}$ stable configurations.

  Ionization energy decrease down the group in Cu of group IIBond IB elements. In Groups, ionization energy decreases from top to bottom in case of only IIIB group and IB group (Cu > Ag).

  In rest of the groups there is an increase in ionization potential value as poor shielding occurs (un-effective) by 4f electrons hence greater effective nuclear charge (Zeff values) exists.

  The ionization energy of 5d series is more than that of 4d and 3d elements due to higher effective nuclear charge which is due to poor screening effect caused by 4f electrons.

For example,

Pt (IV) compounds are thermodynamically more stable than Ni (IV) compounds as formation of Pt^{+4} and Ni^{+4} requires 9.36 and 11.29 kJ of energy.

Table 7.2 Ionization Energy Values

First Transition Series										
3d	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
I.E ink.j	631	656	650	652	717	762	758	737	745	906
4d	Y	Zr	N_6	Mo	Tc	Ru	Rh	Pd	Ag	Cd
I.E ink.j	616	674	664	685	703	711	720	804	731	876
5d	La	Hj	Ta	W	Re	Os	Ir	Pt	Au	Hg
I.E ink.j	541	760	760	770	758	840	900	870	888	1008

Variable Oxidation State

Most of the transition metals show variable oxidation states except the first and the last member of each series.

- It is due to the fact that the energy of (n 1) d and ns electrons are nearly same hence the removal of electron is easy. When ns electron are removed the reminder is called core or kernel (unstable) hence one or more electrons can be removed further for stability hence more oxidation states are possible.

For example,

Sc: +2, +3 , Ti: +2, +3, +4, +5,

V: 0, +2, ro +5, Cr: 0, +1 ro +6

Zn: +2, Cd: +2 only

- Highest oxidation state is shown by Ru, Os.

Ru : + 8 in RuO_4

Os : +8 in OsO_4

Some specific example of low oxidation state are:

$(Fe^0(CO)_5)$, $(Cr^0(CO)_6)$,

$(Cr^0(C_6H_6)_2)$, $Na^1(V(CO)_6)$,

$(V^{-1}(di\ Py)_3)I$

Table 7.3 Oxidation States of Metals

First Transition Series										
Element:	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Oxidation states:	(+2)	(+2)	+2	+2	+2	(+2)			+1	
	+3	(+3)	+3	+3	(+3)	+3	+2	+2	+2	+2
		(+4)	+4	+4	+4	+4	+3	(+3)	(+3)	
			+5	(+5)	(+6)	+5	(+4)	(+4)		
				+6	+7	+6				

REMEMBER

In a general view the first four elements of each transition series have more stability in their higher oxidation state while for the next five elements more stability is in the lower oxidation state.

(i) Covalent nature α Oxidation number

For example,

VCl_4 > VCl_3 > VCl_2

More covalent

(ii) Acidic character α Oxidation number

For example,

VO < V_2O_3 < VO_2 < V_2O_5

Basic basic amphoteric acidic

CrO < Cr_2O_3 < CrO_2 < CrO_3

Basic amphoteic acidic

$MnO < Mn_3O_4 < Mn_2O_3 < MnO_2 < MnO_3 < Mn_2O_7$

Basic amphoteric strongly Acidic

Reactivity

These elements are less reactive or unreactive due to following reasons–high ionization energy, low heat of hydration and high heat of sublimation.

- The property to remain unreactive is more common in platinum and gold in the third transition series (e.g., Au, Pt, Ru, Rh, Os etc.)

Complex Formation

The cations of these elements have a great tendency to form complexes with several molecules or ions known as ligands (CO, CN, NH_3, X^- etc.).

It is due to the following factors:

- Cations have high effective nuclear charge and small size, that is, high positive charge density hence they can easily accept lone pair of electrons from ligands.
- Cations also have vacant inner d orbitals having enough energy to accept lone pair of electrons from the ligands by forming coordinate bond with ligands.

For example, $K_4[Fe(CN)_6]$, $[Ni(CN)_4]^{-2}$

- Transition metal in high oxidation state form stable complex with small, highly electronegative, basic ligands like X^- NH_3 etc.
- Transition metal in low oxidation state forms stable complex with acidic strong ligands like Co, CN etc.
- If two ligands have same stability then the tendency to form complex is directly proportion that of the central metal atom.

For example, $[Co(NH_3)_6]^{+3} > [Co(NH_3)_6]^{+2}$

Magnetic Properties

Most of these elements show magnetic properties.

- Most of the transition metals show paramagnetism due to the presence of unpaired electrons.
- Paramagnetic nature α Number of unpaired electrons.
- The magnetic nature of substances can be determined by weighing the substance in Gouy magnetic balance.

  $\mu = \sqrt{n(n+2)}$ B.M. Where,

  μ = magnetic moment

  B.M. = Bohr's magneton

  n = number of unpaired electrons

  It can also be calculated as follows:

  $\mu = \sqrt{4S(S+1) + L(L+1)}$

  Here, S = spin quantum number (n/2)

  L = orbital quantum number

- When n = 0 (d^0 or d^{10} state), μ is 0 and diamagnetic in nature.

Examples, Zn, Cd, Hg and ions like Cu^+, Ag^+, Zn^{+2}, Cd^{+2}, Hg^{+2} (all have d^{10} state)

- When n = 1, 2, 3, μ is +ve and paramagneric in nature.

Examples, Cu^{+2}, Ni^{2+}, Co^{2+} etc.

- When n = 4, 5, 6 then μ has highly positive value and ferromagneric in nature.

Examples, Mn^{-2}, Fe^{-3}, Fe^{+2} etc.

Coloured Ion or Coloured Complex Formation

- These elements and carions can easily form coloured ions or complexes due to the presence of incomplete d orbitals.
- Complex ion or compounds with transition metals can absorb light and then transmit the light. The colour exhibited is due to the transmited light.

For example,

(i) Cupric salts or compounds are blue coloured as Cu^{2-} absorbs blue yellow light but transmits blue colour.

(ii) Ti^{3} ion absorbs green radiarion but transits purple or violet colour.

(iii) Co^{21} absorbs blue green light and gives Red colour.

Reason: The main reason for the colour formation is d-d electron transition or crystal field splitting (electron transition from lower energy d orbital to higher energy d orbital).

Complex with transition metal with d^0 or d^{10} orbital will be colourless, as no d-d electron transirion takes place.

Examples, – Sc^{3-}, Ti^{+4},

$3d^0$

Cu^{+}, Ag^{-}, Zn^{+2}, Cd^{+2}, Hg^{+2}

$3d^{10}$

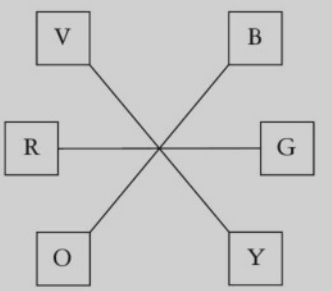
The colour of a transition metal complex is dependent on how big the energy difference is between the two d levels. This in turns depends upon the nature of ligand and complex formed.

For example,

$[Ni(H_2O)_6]^{-2}$ is Green while $[Ni(NH_3)_6]^{2+}$ is blue.

REMEMBER

- Complementary colours can be identified by using Munshel colour wheel.



Here,

V stands for Violet
R stands for Red
O stands for Orange
B stands for Blue
G stands for Green
Y stands for Yellow

Table 7.4

Magnetic Moments and Colour of Some Transition Metal Ions

Ion	Electronic configuration	Number of unpaired electrons	Magnetic moment (BM)		Colour of the ion
			Calculated	Observed	
Sc^{3-}	$3d^0$	0	0	0	Colourless
Ti^{3}	$3d^1$	1	1.73	1.75	Purple
Ti^{2}	$3d^2$	2	2.84	2.79	
V^{2-}	$3d^3$	3	3.87	3.86	Violet
Cr^{3-}	$3d^3$	5	5.92	5.96	Green
Mn^{21}	$3d^5$	5	5.92	5.96	Pink
Fe^{2-}	$3d^6$	4	4.90	5.0 5.5	Green
Co^{2}	$3d^7$	3	3.87	4.4–5.2	Pink
Ni^{2-}	$3d^8$	2	2.84	2.9–3.4	Green
Cu^{2-}	$3d^9$	1	1.73	1.8 2.2	Blue
Zn^{2+}	$3d^{10}$	0	0	0	Colourless

REMEMBER

Colour of $KMnO_4$, $K_2Cr_2O_7$, Cr_2O_3 Fe_4 $[Fe(CN)_6]_3$ is due to charge transfer. Charger transfer needs the energy levels on the two different atoms to be fairly close and thus the intersity of colour is more.

Catalytic Activity

Most of the transition metals [Fe, Ni, Pr], alloys [Fe – Mo] and compounds [V_2O_3, V_2O_5, MnO_2, Co^{2} salt] are used as catalysts in various processes.

Reason for Catalytic Property: It is due to the presence of vacant d orbital, tendency to show variable oxidation state, and their larger surface area. Due to these properties they can easily form various intermediates during a reaction.

For example,

$$2H_2O_2 \xrightarrow[\text{black}]{Pt} 2H_2O + O_2$$

$$4NH_3 + 5O_2 \xrightarrow[\text{black}]{\text{Pt-gauge}} 4NO + 6H_2O$$

$$2CH_3OH + O_2 \xrightarrow{Pt} 2HCHO + 2H_2O$$

- Fenton's Reagent [$FeSO_4 + H_2O_2$] is used to convert R CH_2OH into R CHO.

Alloys Formation

These elements can easily form alloys due to their almost equal atomic sizes, they can mutually substitute one another in the crystal lattice. Alloys having mercury as one of the constituent element are called amalgams. Alloy formation enriches properties also.

Formation of Non-Stoichiometric Compounds and Interstitial Compounds

These elements can trap some of the small size atoms such as H, C, N, B etc., in the vacant spaces between the crystal lattice forming interorbital or interstitial compounds like FeC, TiH_2, Fe_8N etc. Nonstoichiometry is shown specially among transition metal compounds of the group 16 elements (O, S, Te). This is due to their tendency to show variable valency.

- This tendency differs these metals from nontransition elements and products obtained are hard and rigid. For example, Cast iron and steel.

- Compounds like $Fe_{0.86}S$, $Fe_{0.98}O$, $VH_{0.56}$, $TiH_{1.7}$, $VSc_{0.98}$ are classified as interstitial or nonstoichiometric compounds.

Standard Reduction Potential and Reducing Property

The value of $E°_{R.P}$ of most of the transition elements except Cu and Hg is lower than that of hydrogen, that is, a negative value, hence they can easily displace hydrogen, from dilute acids.

For example,

$$M + 2H^+ \rightarrow M^{2+} + H_2 \uparrow$$

- The value of E° is governed mainly by these factors: heat of hydration, heat of sublimation and heat of ionization.
- The value of $E°_{R.P}$ in 3d-series (M2+) varies with some irregularities as follows:

For example,

Ti	V	Cr	Mn	Fe	Co	Ni	Cu
−1.6	−1.17	−0.91	−1.18	−0.44	−0.28	−0.25	0.35

COORDINATION CHEMISTRY

When two or more saturated solutions of neutral compounds are mixed in stoichiometric proportion and left undisturbed by standing, a solid adduct is formed.

These adducts or addition compound can be of two types:

Double Salt

It is formed by the combination of two salts.

- It retains its identity only in solid state but losses in solution.
- On dissolution in water it dissociates into original ions.

For example,

$$K_2SO_4.Al_2(SO_4)_3.24H_2O\ (s) + aq \rightarrow$$

Potash alum

$$2K^+(aq) + 2Al^{3-}(aq) + 4SO_4^{2-}(aq) + 24H_2O$$

$$FeSO_4.(NH_4)_2SO_4.6H_2O\ (s) + aq \longrightarrow$$

Mohr salt

$$Fe^{2+}(aq) + 2NH_4\ (aq) + 2SO_4^{2}\ (aq) + 6H_2O$$

Coordination Complex

It retains its identity not only in solid state but also in solution state.

- It gives complex or different ions on dissociation.

$$Fe(CN)_2\ (aq) + 4KCN(aq) \longrightarrow K_4[Fe(CN)_6]\ (aq)$$
$$\longrightarrow 4K^+\ (aq) + [Fe(CN)_6]^{4-}\ (aq)$$

REMEMBER

A complex ion is defined as an electrically charged radical which is formed with the combination of a simple cation with one or more simple ions or neutral molecules having two constituents an acceptor (central metal atom) and a donor (ligand).

Type of Coordination Complexes

According to the stability of complexes these are of two types:

Perfect or Penetrating Complex

In such a complex, complex ion is either completely stable i.e., undissociated or very feebly dissociated in solution also.

For example,

$$K_4[Fe(CN)_6] \longrightarrow 4K^+ + [Fe(CN)_6]^{4-}$$

Feebly dissociated

$$Fe^{2+} + 6CN^-$$

Imperfect or Normal Complexes

In such a complex, complex ion is either less stable or dissociates appreciably.

For example:

$$K_2[Cd(CN)_4] \longrightarrow 2K^+ + [Cd(CN)_4]^{2}$$

Appreciably dissociated

$$Cd^{2+} + 4CN^-$$

Representation of Coordinate Complex

A coordinate complex is represented as:

Ax [M (K)$_a$ (L)$_b$] B$_y$

- Here, A and B are cationic and anionic species which are ionizable and precipitable.
- M is central metal atom like Cr, Mn, Fe, Co, or Ni.
- K, L are ligands i.e., electron-pair donor species, nonionizable and nonprecipitable species.
- Sum of a and b is equal to coordination number.
- ( ) represents coordination sphere.

TERMS RELATED TO CO-ORDINATE COMPLEX

Coordination Sphere

The central metal atom and ligands directly attached to it are collectively known as coordination sphere.

The central metal atom or ion along with ligands surrounding it are written in a square bracket, ( ) called coordination sphere. The atoms, ions or molecules in this sphere are nonionizable.

Central Metal Atom or Ion or Centre of Coordination

The central metal atom or ion to which one or more neutral molecules or ions are attached is called the centre of coordination.

Ligand

The molecules and ions that surround the metal ion in a complex are known as ligands or complexing agents.

- An atom or group of atoms which binds to the central metal atom or ion through the lone pair of electrons present on its donor atom.
- Depending on number of pairs donated by different atoms. It is of following types:

Unidentate

These have one electron pair to donate.

For example,

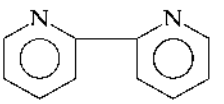
X$^-$, NH$_2$, CN$^-$, OH$^-$, SCN, etc.

Bidentate

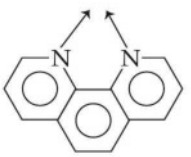
These have two electron pairs to donate.

For example,

1. $C_2O_4^{2-}$ (Oxalato)
2. $H_2NCH_2COO^-$ (Glycinato)
3. $H_2NCH_2CH_2NH_2$ or en (Ethylene diammine)
4. Bipyridyl (bipy)



5. 1, 10-phenanthroline (phen)



6. Acetyl acetonate $CH_3–CO–CH_2–CO–CH_3$

REMEMBER

- The attachment of a symmetrical bidentate ligand to the central metal atom is shown by a curve



where L is the abbreviation of the symmetrical bidentate ligand and two A's are two (similar) donor atoms.

Example, phen

- The attachment of an unsymmetrical bidentate ligand to the central metal atom is shown by a curve



where L is the abbreviation of the ligand and A and B are two (different) donor atoms.

Example, Glycinato (gly$^-$)

Tridentate

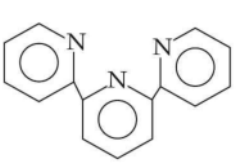
These have three electron pairs to donate.

For example,

$H_2N(CH_2)_2NH(CH_2)_2NH_2$

Di ethylene tri ammine (dien)

Ter pyridyl (terpy)



Tetradentate

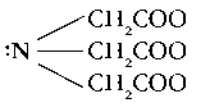
These have four electron pairs to donate.

For example,

$H_2N(CH_2)_2NH(CH_2)_2NH(CH_2)_2NH_2$

Tri ethylene tetra amine or trien

Tri amino tri ethyl amine or tren

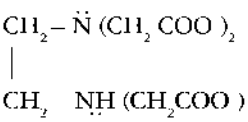


Nitriloacerato

Pentadentate

These have five electron pairs to donate.

For example,

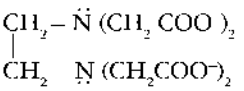


Erhylene di ammine tri acerare

Hexadentate

These have six electron pairs to donare.

For example,



Erhylene diammine terra acerare (E.D.T.A.)

Chelating Ligand

When a polydentate ligand binds to the central metal atom by more than one donor site, a cyclic ring like structure i.e., chelate is formed and this ligand is called chelating ligand.

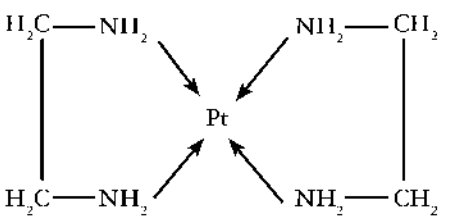


Fig. 7.1

- Such a cyclic or chelare complex is more srable than an open complex due ro chelaring effect or macro cyclic effect.

  Example, in case of $[Ni(en)_3]^{2+}$ stability is nearly 10^{10} more than $[Ni(NH_3)_6]^{2-}$.

REMEMBER

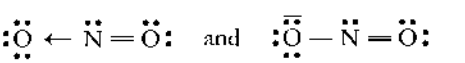
- Chelates are used in softening of hard water and in the separation of lanthanides and actinides.
- Chelating agents are used in medicine to remove metal ions like Hg^{2+}, Pb^{2+} and Cd^{2+}.

For example, EDTA is used to remove lead poisoning.

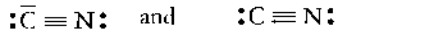
Ambidentate Ligand

Such ligands have rwo or more donor sires however during rhe complex formarion only one donor sire is used.

For example,



Nitro group (I) Nitrito group (II)



Cyano group (I) Isocyano group (II)

SCN and NCS

Thiocyanide Isothiocyanide

REMEMBER

- Thiosulphato ($S_2O_3^{2-}$) can also act as an ambidentate ligand i.e., ($O^- - S_2O_2^-$) ion and ($S^- - SO_3^-$) ion.

Flexidentate ligand

Here, rhe ligand has many donor sires, however nor necessarily all of rhem are used.

Example, EDTA.

It oftenly acts as a penta coordinate ligand leaving one of the acetate group dangling free.

π–acid Ligand

Such ligands are capable of accepting an appreciable amount of π electron density from the central metal atom into their empty π or π* orbitals. Example, CO.

Coordination Number or Ligancy

The total number of ligands surrounding the central metal atom in the coordination sphere is called coordination number.

Example, $[Ni(CN)_4]^{-2}$

Here, coordination number is 4.

$K_4[Fe(CN)_6]$

Here, coordination number is 6.

$[Fe(C_2O_4)_3]^{-3}$

Here, coordination number is 6 since $C_2O_4^{-2}$ is bidentate in nature.

Sidgwick Theory of Complexes

- Transition metal or ions are lewis acids.
- Ligands are electron donor species, that is, Lewis bases.
- Transition metal ions can form complex compounds by accepting the lone pair of electrons from ligands.
- The bond formed between the transition metal ion and the ligand is a coordinate covalent bond (L ⟶ M Here, L = ligand, M = central metal atom).
- Transition metal ion must contain vacant orbitals to accept the electron pairs from ligands.
- Ligands should contain at least one lone pair of electrons to donate.

Effective Atomic Number (EAN)

It was introduced by Sidgwick and it is defined as "Effective atomic number is the total number of electrons present around central metal ion in a complex".

EAN = [Atomic number of the metal] [Number of electrons lost in the formation of its ion] + [Number of electrons gained from ligands].

Example, EAN = Z – Oxidation number + 2 × Number of ligands

- Sidgwick proposed that metal ion is stable if E.A.N. is equal to the atomic number of the nearest inert gas. For example,

  $K_4[Fe(CN)_6]$

  EAN = 26 – 2 + 12 = 36

  In some complexes, EAN is not equal to the atomic number of the nearest inert gas.

  Example, in $K_3[Fe(CN)_6]$

  EAN = 26 – 3 + 12 = 35

  In $[Ni(CN)_4]^{-2}$

  EAN = 28 – 2 + 8 = 34

Table 7.5

Complex	Electrons lost in ion formation	Electrons gained from ligands	EAN
$[Ni(CO)_4]$	0	8	28 – 0 + 8 = 36
$[Cu(CN)_4]^{-3}$	1	8	29 – 1 + 8 = 36
$[Ag(NH_3)_4]^{+}$	1	8	47 1 + 8 = 54
$[Co(CN)_5]^{-4}$	2	12	27 2 + 12 = 37
$[Ni(NH_3)_6]^{+2}$	2	12	28 – 2 + 12 = 38
$[PdCl_4]^{-2}$	2	8	46 – 2 + 8 = 52
$[Pt(NH_3)_2Br_2]$	2	8	78 2 + 8 = 84
$[Ni(py)(en)(NH_3)_2]^{2+}$	2	12	28 2 + 12 = 38

PREPARATION OF COMPLEXES

Complexes are mainly prepared by the following methods:

By Substitution Reaction

$$[Cu(H_2O)_4]SO_4 + 4NH_3 \longrightarrow [Cu(NH_3)_4]SO_4 + 4H_2O$$

$$[Co(NO_2)_6]^{-3} + 2NH_2 - CH_2 - CH_2 - NH_2 \longrightarrow [Co(en)_2(NO_2)_2]^{+} + 4NO_2$$

By Direct Combination

$NiCl_2 + 6NH_3 \rightarrow [Ni(NH_3)_6]Cl_2$

$CoCl_3 + 6NH_3 \longrightarrow [Co(NH_3)_6]Cl_3$

$PrCl_4 + 2KCl \rightarrow K_2[PrCl_6]$

By Redox Reactions

$$CoCl_2 + 2NH_4Cl + 10\,NH_3 + H_2O_2 \longrightarrow 2[Co(NH_3)_6]Cl_3 + 2H_2O$$

$$2[Co(H_2O)_6](NO_3)_2 + 8NH_3 + 2NH_4NO_3 + H_2O_2 \longrightarrow 2[Co(NO_3)_2(NH_3)_5](NO_3)_2 + 14H_2O$$

Stability of Complexes

The stability of a complex depends upon these factors.

Stability α K (Stability Constant)

$M^{a+} + nL^{x-} \longrightarrow (MLn)^{b+}$

$$K = \frac{(MLn)^{b+}}{[M^{a+}][L^{x-}]^n}.$$

It means, that higher the stability of the complex higher is the value of K for it.

Example, $[Cu(CN)_4]^{2-}$ for it $K = 2 \times 10^{27}$

$[Fe(CN)_6]^{3-}$ for it $K = 7.7 \times 10^{43}$

- Stability α Charge on cation or oxidation number
- Stability α 1/ Size of cation

  Example, cation with higher charge and smaller size will form more stable complex.

  Example, $Fe^{3+} > Fe^{2+}$
- Stability α Basic nature of ligands or electron density

  Example, Cyano, amine complex > X complexes more basic ligands
- Stability α Number of chelate rings (macro cyclic effect)

REMEMBER

- Complex formation tendency of some divalent cations decreases as shown $Cu^{2+} > Ni^{2+} > Co^{2+} > Fe^{2+} > Mn^{2+}$

NOMENCLATURE OF COORDINATION COMPOUNDS

The basic rules are as follows:

- While naming the salts, the positive ion (cation) is named first followed by the negative ion (anion).

  Example, $[Co(NH)_5Br]Br_2$ here the name is first given for $[Co(NH_3)_5Br]^{2+}$ then Br .
- Name of the nonionic or neutral complex must be written in one word.
- With in a complex species the ligands are named before the meral arom.
- Names of the neutral ligands should be written as it is. Some exceptions are as follows:

  H_2O—aqua

  NH_3 ammine

  CO—carbonyl

  NO nitrosyl

  CS—rhiocarbonyl

  NS thionitrosyl
- Names of the negative ligands must be ended with 'O'. For example,

  Cl chloro

  CH_3COO^- acetato

  CO_3^{2-} carbanato

  $C_2O_4^{2-}$ oxalato

  NO_2 nirro

  NO_3^- nitrato

  S^{2-} sulphido

  SO_4^{2-} sulphato

  NH_2 amine or amino

  NH^{2-} imido

  OH hydroxo

  O_2^{2-} peroxo
- The names of the positive ligands should be ended with 'ium'. For example,

H_3O^+ hydronium

NO_2^+ nitronium

$N_2H_5^+$ hydrazinium

- If the same complex compound contain different ligands their names should be written in alphabetical order.
- Prefixes di, tri, tetra, penta, ... are used before the ligands to mention their numbers.
- The number of complex ligands like bidentate, tridentate is mentioned with bis, tris, tetrakis etc., if they already contain di, tri, in their names.
- While naming the ambidentate ligands, the ligands are named after point of attachment.

  For example,

  SCN– , S – thio cyanato (here S is donor)

  NCS–, N – isothio cyanato (here N is donor)
- If the complex contains two or more metal atoms, the bridging ligands are indicated by prefix μ before the names of such ligands.

  For example,

  $[(NH_3)_5 Cr - OH - Cr(NH_3)_5] Cl_5$

  Pentaamminechromium (III)-μ hydroxopentaamminechromium (III) chloride

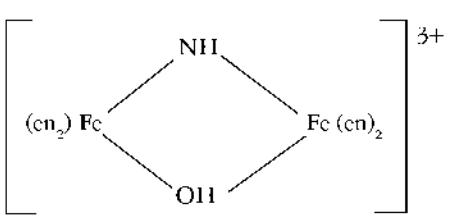


Fig. 7.2

Bis (ethylenediamine) cobalt (III)- μ- hydroxo- μ-imido bis (ethylenediamine) cobalt (III)

or

Tetrakis (ethylenediamine)- μ-hydroxo- μ-imido dicobalt (III) ion.

- The oxidation number of the central metal ion should be mentioned in the roman numerals in the parenthesis immediately after the name of the metal ion.
- If the complex ion is positive, then the name of metal ion is written as it is.

  Example, $[Cr (en)_3]Cl_3$ tris ethylene di amine cobalt(III) bromide
- If the complex ion is anion and symbol of the metal is taken from latin language, their names should be taken from latin language.

  For example,

  Ferrum—ferrate

  Argentums—argenate

  Stannum—stannate

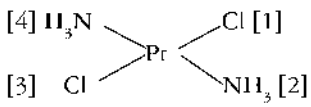
  Example, $K_4 [Fe (CN)_6]$ Potassium hexa cyano ferrate(II).
- Water molecule of crystallization are indicated after the name of the complex, arabic numerals are used to indicate the number of such molecules.

  For example,

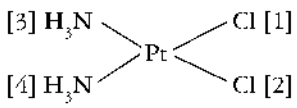
  $Al K(SO_4)_2.12H_2O$

  Aluminium potassium sulphate 12-water
- Geometrical isomers are named either by using the prefixes cis for adjacent (90° apart) positions and trans for opposite (180° apart) positions before the name of the ligands.
- In square planar complexes the groups at positions (1, 2) and (3, 4) are cis to each other while those at positions (1, 3) and (2, 4) are trans to each other.

  For example,



  Trans-diammine dichloro platinum (II)



  Cis-diammine dichloro platinum (II)
- In mononuclear octahedral complexes of Ma_4b_2 type (1, 2), (1, 3), (1, 4), (1, 5), (6, 2), (6, 3), (6, 4), (6, 5), (2, 3), (3, 4) and (5, 2) are cis positions while (1, 6), (2, 4) and (3, 5) are trans positions.
- In mononuclear complexes of Ma_3b_3 type (1, 2, 5) are cis while 1, 2, 6 are trans positions.
- Dextro and levo rotatory optically active compounds are designated by (+) and (–) or by d- and l-, respectively.

For example,

(+) or d $K_3[Ir(C_2O_4)_3]$ is potassium (+) or d-trioxalato iridate (III).

- The names of some coordination compounds on the basis of IUPAC rules are given below:

Formula	Name
$K_3[Cr(CN)_6]$	Porassium hexa cyano chromare (III)

(a) Complex Cations IUPAC Name

$[Ti(H_2O)_6]Cl_3$	hexa aqua titanium (III) chloride
$[Cr(NH)_6]Cl_3$	hexa amine chromium (III) chloride
$[CrSO_4(NH_3)_4]NO_3$	tetraamminesulphato chromium (III) nitrate
$[Co(en)_2F_2]ClO_4$	bis-(ethylenediamine)-difluoro cobalt (III) perchlorate
$[Cu(acac)_2]$	bis (acetylacetonato) copper (II)
$[CoCl.CN.NO_2.(NH_3)_3]$	triamminechloro cyano-nitro cobalt (III)
$[Pt(NH_3)_4(NO_2)Cl]SO_4$	tetraamminechloro nitro platinum (IV) sulphate
$[Cr(NO_2)_3(NH_3)_3]$	triammine nitrochromium (III)
$[Co(en)_2Cl(ONO)]^+$	chloro-bis-(ethylenediamine)-nitro cobalt-(III) ion

(b) Complex Anions

$[Pt(NH_3)_4(ONO)Cl]^{2-}$	tetraammine chloro nitrito palatinate (IV) ion
$K_2[Cr(CN)_2O_2(O_2)NH_3]$	potassium amminedicyano di oxo peroxo chromate(VI)
$Na_2[ZnCl_4]$	sodium tetrachloro zincate (II)
$(NH_4)_3[Co(C_2O_4)_3]$	ammonium tris-(oxalato)- cobaltate-(III)
cis $[PtBrCl(NO_2)_2]^{2-}$	cis-bromochloronitro palatinate (II) ion
$Na_3[Ag(S_2O_3)_2]$	sodiums (thiosulphato) argentate (I)
$K_2[OsCl_5N]$	potassium pentachloronitrido osmate (VI)
$Na_3[Ag(S_2O_3)_{12}]$	sodium bis-(thiosulphato) argenate (I)
$K_3[Fe(CN)_5NO]$	potassium pentacyanonitrosyl ferrate (II)

(c) Neutral Complexes

$[Cr(C_6H_6)_2]$	bis-(benzene) chromium–(0)
$[Ni(CO)_4]$	tetracarbonyl ickel-(0)
$[Cr(en)_3]Cl_3$	d or l tris (ethylenediamine) chromium (III) chloride
$[CoCl_2((NH_2)_2CO)_2]$	dichloro-bis-(urea)-cobalt-(II)
$Fe(C_5H_5)_2$	Bis(cyclopentadienyl) iron(II)
$[Fe(C_2H_5N)_2]$	dipyridineiron-(0)

(d) Cationic as well as Anionic Complexes [ ]$^{n+}$ [ ]$^{n-}$

$[Pt^{IV}(NH_3)_4Cl_2][Pt^{II}Cl_4]$	tetraamine dichloro platinum (IV) tetrachloro palatinate(II)
$[Pt(py)_4][PtCl_4]$	tetrapyridine platinum-(II) tetrachloro platinate-(II)
$[Pt(py)_4][PtCl_4]$	tetrapyridine platinum (II) tetrachloro platinate(II)

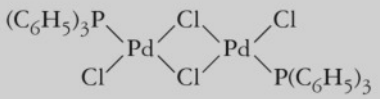
(e) Bridging Groups

$[Be_2O(CH_3COO)_6]$ Hexa-μ-acetato (O,O')- μ_4-oxo-tetraberylium(II)

$[(CO)_3Fe(CO)_3Fe(CO)_3]$ Tri-μ-carbonyl-bis[tricarbonyl iron(0)]

$[(NH_3)_5Co.NH_2.Co(NH_3)_5](NO_3)_5$

μ-amidobis[pentaammine cobalt (III)] nitrate



trans-bis-chloro, μ-chloro triphenyl phosphine palladium(II)

(f) Hydrates

$AlK(SO_4)_2.12H_2O$	Aluminium potassium sulphate 12-water
$[Cr(H_2O)_4Br_2]Br.2H_2O$	Tetraaqua dibromo chromium (III) bromide 2 water
$AlCl_3.4(C_2H_5OH)$	Aluminium trichloride-4-ethanol

ISOMERISM IN COORDINATION COMPOUNDS

Compounds that have same chemical formula but different structural arrangements are called isomers. Isomerism is of two types: structural and stereo isomerism.

Structural Isomerism

Ionization Isomerism

Compounds having same compositions but giving different ions in solution are called ionization isomers.

- Ionization isomers are formed by the interchange of the position of ligands inside and outside the coordination sphere.
- Ionization isomers can be detected by the conductance measurement or by chemical tests.

  Example

  $[Co(NH_3)_5SO_4]Br$ red colour

  $[Co(NH_3)_5Br]SO_4$ violet colour

  Here, the ions precipitable are Br^- and SO_4^{2-}, respectively in these isomers.

  Example

  $[Pt(NH_3)_4Br_2]Cl_2$

  $[Pt(NH_3)_4Cl_2]Br_2$

  Here, the ions precipitable are Cl^- and Br^- respectively in these isomers.

Hydrate Isomerism

It is due to the difference in the position of water molecules in a complex as ligand and hydrated molecules i.e., number of water molecules differ in the coordination sphere.

Example

$CrCl_3.6H_2O$ has 3 isomers: Example,

$[Cr(H_2O)_6]Cl_3$ violet

$[Cr(H_2O)_5]Cl_2.H_2O$ green

$[Cr(H_2O)_4]Cl.2H_2O$ dark green

Here, a fourth isomer $[CrCl_3(H_2O)_3].3H_2O$

Example

$CoCl_3(H_2O)_6$ has 3 isomers, i.e.,

$[Co(H_2O)_6]Cl_3$

$[CoCl_2(H_2O)_4]Cl.2H_2O$

$[CoCl_3(H_2O)_3].3H_2O$

Linkage Isomerism

This isomerism occurs when in a ligand more than a single atom can act as a donor site as, in case of an ambidentate ligand.

Example, CN, NC

SCN, NCS

NO_2, ONO

$[(NH_3)_5CoONO]Cl_2$ Red colour

$[(NH_3)_5CoNO_2]Cl_2$ Yellow colour

$[Cr(NH_3)_5(SCN)]Cl_2$; $[Cr(NH_3)_5(NCS)]Cl_2$

Coordination Isomerism

This isomerism occurs when the compounds contain both cationic and anionic complex and there is an exchange of ligands between these complexes.

Example

$[Cr(NH_3)_6]\ [Co(CN)_6]$ and

$[Cr(CN)_6]\ [Co(NH_3)_6]$

Example

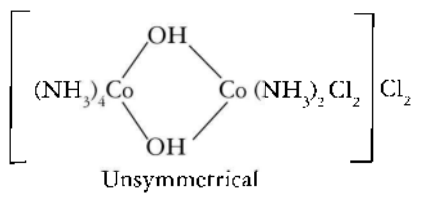
$[Cu\ (NH_3)_4]\ [PtCl_4]$ and

$[Cu(NH_3)_3\ Cl]\ [Pr(NH_3)\ Cl_3]$

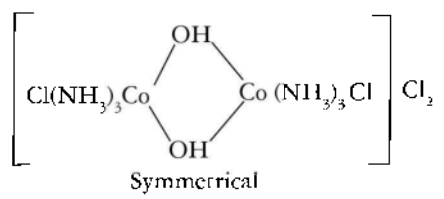
Coordination Position Isomerism

It arises in polynuclear complexes due to interchange of ligands between the metal atoms.

For example,



Unsymmetrical



Symmetrical

Fig. 7.3

Stereo Isomerism

It arises when the isomers differ in the arrangement of the atoms or groups in space. It is of two types: Geometrical isomerism and Optical isomerism.

Geometrical Isomerism

It arises when exchange of atoms or groups or ions within the coordination sphere of the complex changes its geometry.

- In geometrical isomers when two identical ligands are present in adjacent positions it is called cisisomerism and when such ligands are present diagonal (opposite) it is called transisomerism.
- Complexes in which the coordination number of central metal ion is 2, 3 or 4 (having tetrahedral structure) do not exhibit geometrical isomerism.
- Complexes in which the coordination number of central metal ion is 4 and have square planar structure will exhibit geometrical isomerism.
- Square planar complexes of the type Ma_4, Ma_3b, Mab_3, $M(aa)_2$ do not exhibit geometrical isomerism.
- Ma_2b_2, Ma_2bc, $M(aa)b_2$ or ab and Mabcd type square planar complexes exhibit geometrical isomerism.

  For example:

  $[Pr\ (NH_3)_2Br_2]$ and $[Pr(NH_3)_2\ (Cl)\ (Br)]$

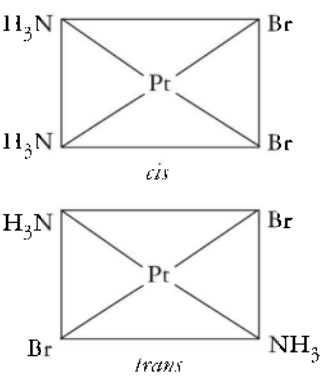


Fig. 7.4

Example, $[Pt\ (gly)_2]$

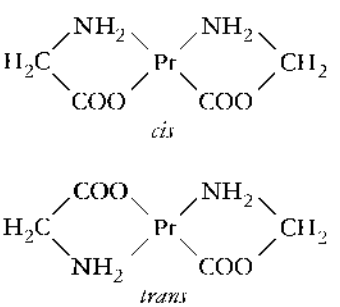


Fig. 7.5

- Mabcd type square planar complexes exists in three isomerism forms.

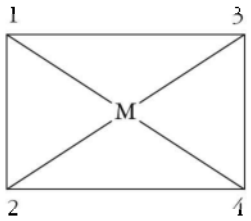


Fig. 7.6

- In square planar complexes positions 1, 4 and 2, 3 are cis while 1, 3 and 2, 4 are trans.

  For example:

  (Pt (NH_3) (Py) (Cl) (Br))

- Octahedral complexes of the type Ma_6, Ma_5b, or Mab_5, $M(aa)_3$ do not exhibit geometrical isomerism.
- Octahedral complexes of the type Ma_4b_2, Ma_4bc, Ma_3b_3, $Ma_2b_2c_2$ etc. type complexes exhibit geometrical isomerism.

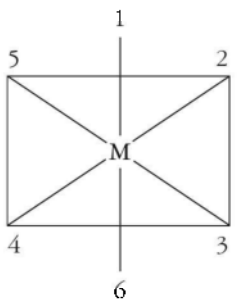


Fig. 7.7

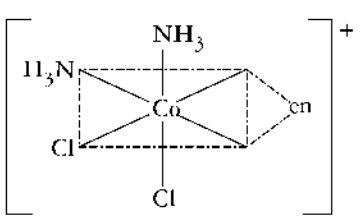
- In octahedral complexes, positions 1, 6 and 2, 4 and 3, 5 are trans.

  For example:

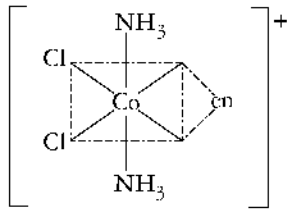
  $[Co(NH_3)_4Cl_2]^+$

- Complexes of the type [M (aa) x_2y_2] can show geometrical (cis and trans) isomerism.

  Example, [Co (en) $(NH_3)_2$ $Cl_2]^+$ exhibits cis and trans isomerism.



cis



trans

Fig. 7.8

- Octahedral complexes of the type $M(aa)_2b_2$ where (aa) is a symmetric bidentate ligand also exhibit geometrical isomerism.

  Example, [Co $(en)_2$ $Cl_2]^+$

- Octahedral complex of the type [M(abcdef)] has 15 different geometrical isomers with a pair of enantiomers.

  For example:

  [Pt (py) (NH_3) (Cl) (Br) (I) (NO_2)]

Optical Isomerism

Complexes which do not contain any centre of symmetry or plane of symmetry or axis of symmetry exhibit optical isomerism.

- Complexes with coordination number four having tetrahedral structure can exhibit optical isomerism when all the four ligands are of different types.
- Compound of type $Ma_2x_2y_2$, Ma_2x_2yz, Ma_2xyzl, Mabcxyz, $M(aa)_3$, $M(aa)_2x_2$ show optical isomerism.
- Octahedral complexes of the type $Ma_2b_2c_2$ type can exist in two optical isomers.

  Example, [Pr $(NH_3)_2$ $(Py)_2$ $Cl_2]^{+2}$ exhibits optical isomerism.

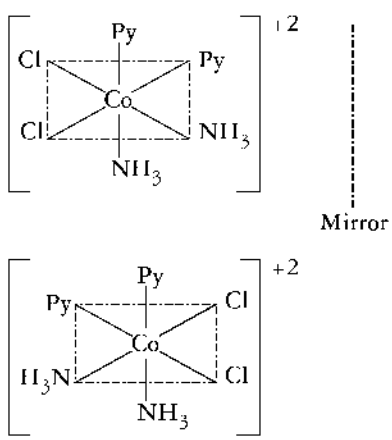


Fig. 7.9

In the octahedral complexes of the type $M(aa)_2b_2$ the cis isomer exists in two optical isomers. In case of trans isomer optical isomerism is not possible.

Example, $[Co\ (en)_2\ Cl_2]^+$

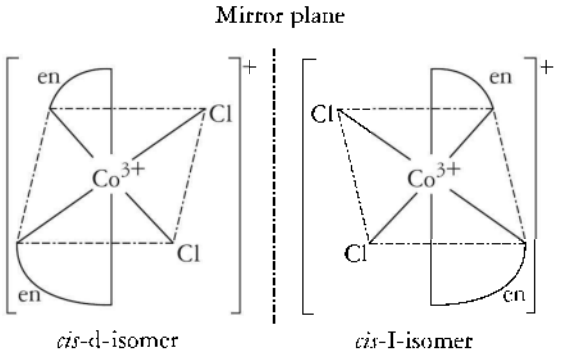


Fig. 7.10

$M(aa)_3$ type octahedral complexes also exists in two optical isomers.

Example, $[Co\ (en)_3]^{3+}$

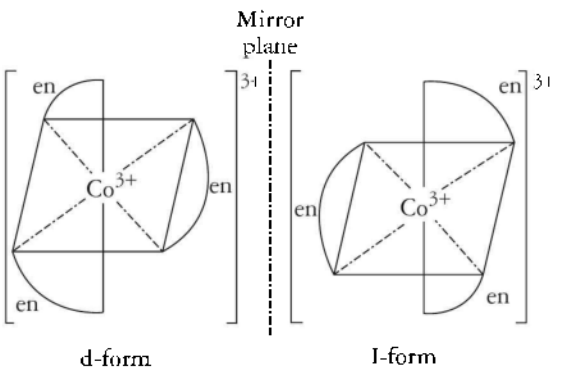


Fig. 7.11

[Mabcd] type Complexes

This type of complex ion/compound should exhibit optical activity. However, it has not been possible to isolate optically active d- and l-forms of such a complex due to its labile nature.

Example, $[As(CH_3)\ (C_2H_5)\ (S)(C_6H_5COO)]^{2-}$ ion. Since it does not posses a plane of symmetry, hence it is optically active.

BONDING IN COMPLEXES

Werner's Theory

Central metal atom shows two types valencies in coordination compounds, that is, primary (Principal), secondary (auxillary valency) and the metal atom tries to satisfy both of its valencies.

Primary Valency

- Primary valency represents oxidation number of the central metal atom.
- Primary valency is always satisfied by anions. Example, $[Co\ (NH_3)_4\ Cl_2]\ Cl$, $[Co\ (NH_3)_5\ Cl]\ Cl_2$, $[Co\ (NH_3)_6]\ Cl_3$

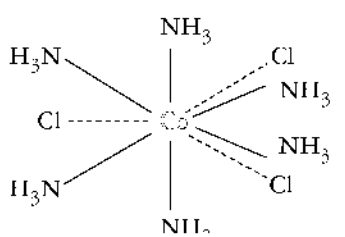
  Here Primary valency is 3 and it is satisfied by 3Cl ions.
- Primary valency is written outside the coordination sphere, but if species showing primary valency is also showing secondary valency it is also written inside the coordination sphere.
- Primary valency present outside the coordination sphere is nondirectional.
- Primary valency does not give the idea of geometry of the coordination compounds.
- Primary valency satisfying species can be obtained in free state by ionization of aqueous solution of coordination compounds.

  $[Co\ (NH_3)_5\ Cl]\ Cl_2 + AgNO_3 \longrightarrow$ --------

  $2Cl^- + 2Ag^+ \rightarrow 2AgCl$

  White ppt.
- Primary valency is shown by dotted (------) lines.

  For example,



$[Co\ (NH_3)_6]\ Cl_3$

Fig. 7.12

Secondary Valency

- Secondary valency represents coordination number of the central metal atom.
- Secondary valency is satisfied by ligands (anions and neutral molecules). For example – in $K_4[Fe(CN)_6]$, secondary valency of Fe is 6 and it is satisfied by 6 CN ligands.

  In $[Co(NH_3)_6]Cl_3$, secondary valency of Co is 6 and it is satisfied by 6 NH_3 molecules.

  In $[Co(NH_3)_5Cl]Cl_2$, Co has secondary valency 6 and it is satisfied by 5 NH_3 and Cl^- ion molecule.
- Secondary valencies are written inside the coordination sphere as in $[Co(NH_3)_5Cl]Cl_2$ and $[Co(NH_3)_4Cl_2]Cl$
- Secondary valency is directional in nature.
- Secondary valency gives a definite geometry to coordination compounds. Secondary valency is 2 for linear, 3 for trigonal planar, 4 for square planar or tetrahedral, and 6 for octahedral.

  Example, $[Co(NH_3)_6]Cl_3$ and $[Co(NH_3)_5Cl]Cl_2$, $[Co(NH_3)_4Cl_2]Cl$ will have octahedral shape.
- Secondary valency cannot be obtained in free state by the ionization of aqueous solution of coordination compounds. For example,

  $[Co(NH_3)_4Cl_2]Cl \rightarrow [Co(NH_3)_4Cl_2]^+ + Cl^-$
- Secondary valency is shown by thick (—) lines.

VALENCE BOND THEORY

To explain bonding in complex compounds, Linus Pauling proposed valence bond theory.

Its main postulates are as follows:

- The central atom loses a requisite number of electrons to form the cation. The number of electrons lost is equal to the oxidation number of the resulting cation.

Table 7.6

Coordination no.	Type of hybridization	Molecular geometry	Examples
2	sp	Linear	$[CuCl_2]^-$, $[Cu(NH_3)_2]^+$, $[Ag(NH_3)_2]^+$
3	sp^2	Trigonal planar	$[CuCl(CO)_2]$
4	sp^3	Tetrahedral	$[Ni(CO)_4]$, $[NiCl_4]^{2-}$, $[Cu(CN)_4]^{3-}$
4	dsp^2	Square planar	$[Ni(CN)_4]^{2-}$, $[PdCl_4]^{2-}$, $[Cu(NH_3)_4]^{2+}$, $[Pt(NH_3)_4]^{2+}$
5	sp^3d or dsp^3	Trigonal bipyramidal	$[Fe(CO)_5]$, $[CuCl_5]^{3-}$, $[Ni(CN)_5]^{3-}$
6	sp^3d^2 or d^2sp^3	Octahedral	$[Fe(CN)_6]^{3-}$, $[Ti(H_2O)_6]^{3+}$, $[Fe(CN)_6]^{4-}$, $[Co(NO_2)_6]^{3-}$, $[CoF_6]^{3-}$, $[FeF_6]^{3-}$, $[Ni(NH_3)_6]^{2+}$

- The central cation makes available a number of empty orbitals equal to its coordination number for the formation of dative bonds with the ligands.
- The cation orbitals hybridize to form new set of equivalent hybrid orbitals with definite directional characteristics.
- The nonbonding metal electrons occupy the inner d orbitals and do not participate in the hybridization.
- If the ligands are strong like CN^-, CO, NH_3 the delectrons are rearranged vacating some d orbitals (when the number of d electrons are more than 3 only) which can participate in hybridization.
- If the ligands are weak like F^-, Cl^- and H_2O, the delectrons are not rearranged.
- The d orbitals involved in the hybridization may be either (n 1) d orbitals or outer nd orbitals.
- The complexes formed by the involvement of (n 1) d orbitals in hybridization are called inner orbital complexes or low spin complexes.
- The complexes formed by the involvement of nd orbitals in hybridization are called outer orbital complexes or high spin complexes.
- Each ligand contains a lone pair of electrons. A dative bond is formed by the overlap of a vacant hybrid orbital of metal ion and a filled orbital of ligand.
- The complex will be paramagnetic, if any unpaired electrons are present, otherwise diamagnetic.

Table 7.7

Inner orbital octahedral complexes	Outer orbital octahedral complexes
These are formed by dsp^2, d^2sp^3 type of hybridization.	These are formed by sp^3, sp^3d^2 type of hybridization.
These complexes have less number of unpaired electrons therefore show low magnetic moment or no magnetic moment.	These complexes have greater number of unpaired electrons therefore show high magnetic moment.
These are less reactive, therefore substitution of ligands is fairly difficult.	These are more reactive, therefore substitution of ligands is easy.
These are formed by strong ligands.	These are formed by weak ligands.

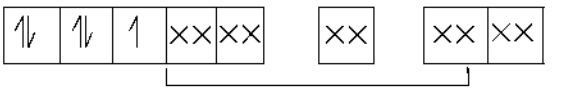
SOME COMPLEXS AND THEIR FORMATION ETC

(i) In case of $[Fe(CN)_6]^{3-}$:- In if Fe-arom is in +3 oxidarion state and rhe comples invoives d^2 and p^3 hybridizarion. It conrains unpaired electrons so ir is weakly paramagnatic in narure. The formation can be explained as follows:

$Fe^{3+} \rightarrow (Ar), 3d^5$



Now Fe^{3-} will rake $6e^-$ pair from 6CN legands which being strong pair up inner 3d electrons which means an inner orbital complex is formed.

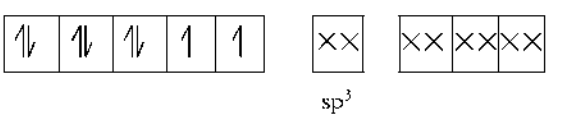


As n=1, **II**= $\sqrt{1(1+2)} = \sqrt{3}$ B.M

(ii) In Case of $(Ni(H_2O)_4)^{2+}$:- In it Ni atom is in +2 oxidation state and the complex has sp^3 hybridization with tetrahedral geometry. As it has an outer orbital complex with 2 unpaired electrons so it is paramagnetic in nature $Ni^{2+} \longrightarrow (Ar)\ 3d^8$.



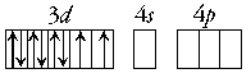
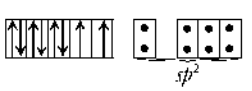
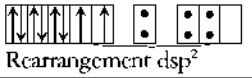
Now Ni^{2+} takes $4e^-$ pairs from $4H_2O$ legands which being weak one cannot pair up inner 3d-electrons resulting into an outer orbital complex.

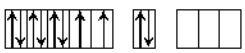
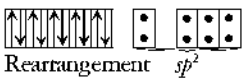
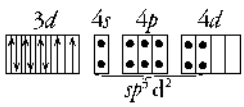
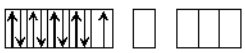
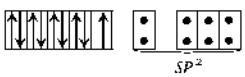
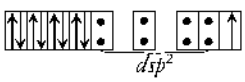
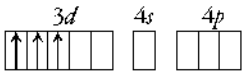
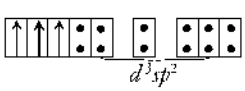
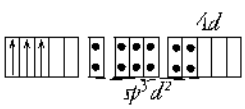
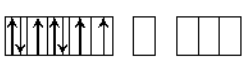
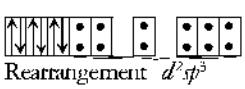
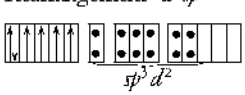
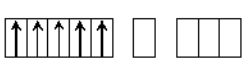
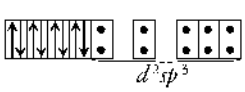
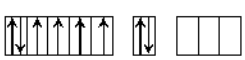
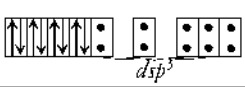


As n=2, π= $\sqrt{2(2+2)} = \sqrt{8}$ B.M.

Table 7.8

Geometry (shape) and magnetic nature of some of the complexes (On the Basic valence bond theory)

Atom/ion/ complex (1)	Configuration (2)	Oxidation state of metal (3)	Type of hybridization (4)	Geometry shape (5)	No. of unpaired electrons (6)	Magnetie nature (7)
1. Ni^{2+} (d^8)		+2			2	Paramagnetic
$[NiCl_4]^{2-}$		+2	sp^3	Tetrahedral	2	Paramagnetic
$[Ni(CN)_4]^{2-}$		+2	dsp^2	Square plannar	2	Diamagnetic

Atom/ion/ complex (1)	Configuration (2)	Oxidation state of metal (3)	Type of hybridization (4)	Geometry shape (5)	No. of unpaired electrons (6)	Magnetie nature (7)
2. Ni		0			2	Paramagnetic
$Ni(CO)_4$	 Rearrangement sp^2	0	sp^3	Tetrahedral	0	Diamagnetic
3. $[Ni(NH_3)_6]^{2+}$	 sp^3d^2	+2	sp^3d^2 (Outer)	Octahedral	2	Paramagnetic
4. $Cu^{2+}(d^9)$		+2			1	Paramagnetic
$[CuCl_4]^{2-}$	 sp^2	+2	sp^3	Tetrahedral	1	Paramagnetic
$[Cu(NH_3)_4]^{2+}$	 dsp^2	+2	dsp^2	Square plannar	1	Paramagnetic
$[Cu(CN)_4]^{3-}$	Here one electron is shifted from 3d-to 4p-orbital					
5. $Cr^{3+}(d^3)$		+3			3	Paramagnetic
$[Cr(NH_3)_6]^{3+}$	 d^3sp^2	+3	sp^3d^2 (Inner)	Octahedral	3	Paramagnetic
$[Cr(H_2O)_6]^{3+}$	 sp^3d^2	+3	sp^3d^2 (Outer)	Octahedral	3	Paramagnetic
6. $Fe^{2+}(d^6)$		+2			4	Paramagnetic
$[Fe(CN)_6]^{4-}$	 Rearrangement d^2sp^3	+2	d^2sp^3 (Inner)	Octahedral	4	Paramagnetic
$[Fe(H_2O)_6]^{2+}$	 sp^3d^2	+2	sp^3d^2 (Outer)	Octahedral	4	Paramagnetic
$[Fe(NH_3)_6]^{2+}$	Same	+2	sp^3d^2 (Outer)	Octahedral	4	Paramagnetic
7. $Fe^{3+}(d^5)$		+3			5	Paramagnetic
$[Fe(CN)_6]^{3-}$	 d^2sp^3	+3	d^2sp^3 (Inner)	Octahedral	1	Paramagnetic
8. Fe		0			4	Paramagnetic
$Fe(CO)_5$	 dsp^3	0	d^3sp^3 (Inner)	Trigonal bipyramidal	0	Diamagnetic

Applications

(i) In photography

$$AgBr + 2S_2O_3^{2-} \longrightarrow [Ag(S_2O_3)_2]^{3-} + Br^-$$

Excess — Soluble complex

(ii) In first group analysis [AgCl, Hg_2Cl_2, $PbCl_2$]

$$AgCl(s) + 2NH_3(aq) \longrightarrow [Ag(NH_3)_2]Cl$$

(iii) In second group, Cd^{+2} can be precipitated as CdS in presence of Cu^{+2} with excess of CN^-.

$$2Cu^{2+} + 10\,CN^- \longrightarrow 2[Cu(CN)_4]^{3-} + C_2N_2$$

Stable

$$Cd^{2+} + 4CN^- \longrightarrow [Cd(CN)_4]^{2-} \xrightarrow{H_2S} CdS$$

Unstable

(iv) Ag, Cu, Au are plated from solutions of their cyano complex ions by electroplating. (electroplating bath)

At Anode

$$Cu + 4\bar{C}N \longrightarrow [Cu(CN)_4]^{3-} + e$$

At Cathode

$$[Cu(CN)_4]^{3-} + e^- \longrightarrow Cu + 4CN^-$$

(v) Purification of H_2O by using EDTA to remove impurities of Ca^{+2}, Mg^{+2}.

(vi) EDTA is used in the treatment of metal (lead etc.) poisoning.

(vii) Extraction of Ag and Au is done by the formation of cyanide complexes.

(viii) Purification of Ni is done by Mond's method.

$$Ni + 4CO \longrightarrow Ni(CO)_4$$

$$[Ni(CO)_4] \xrightarrow{\Delta} Ni + 4CO$$

(ix) Test of Ni^{2+} from D.M.G. (Dimethyl glyoximate): A red blood colour chelate is formed here.

(x) Cis platin i.e., cis diammine dichloro platinum (II) [$Pt(NH_3)_2Cl_2$] is used in the treatment of cancer. It can damage kidney as now dinuclear pt-complex is used in place of it.

(xi) Animal and plant world, for example, chlorophyll is a complex of Mg^{2+} and haemoglobin is a complex of Fe^{2+}.

(xii) Vitamin B_{12} is a complex of Co^{2+}.

(xiii) Aryl arsenic compounds are used as chemotherapeutic agents.

(xiv) $(C_2H_5)HgCl$ is used as a fungicide for the protection of young plants.

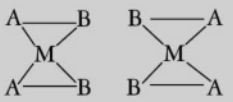
(xv) In heterogeneous catalysis eg., Zeigler Natta catalyst, $TiCl_4 + (C_2H_5)_3Al$ for polymerization of olefins.

(xvi) In homogeneous catalysis e.g., hydrogenation of alkenes by using the Wilkinson's catalyst, $(Ph_3P)_3RhCl$.

(xvii) Electroplating of metals involves the use of complex salts as electrolytes eg., $K[Ag(CN)_2]$ in silver plating.

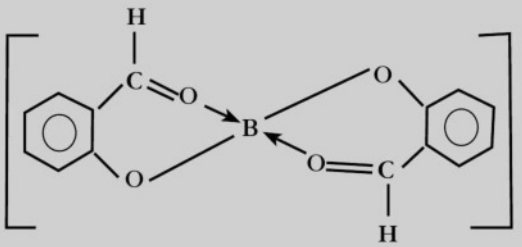
REMEMBER

- Optical isomerism in chelated tetrahedral and square planar complxes: In tetrahedral complexes optical isomerism is observed only in case of bis-chelates having unsymmetrical ligands. This has been observed in Be(II), B(III), Zn(II), Co(II) complexes which are generally represented as



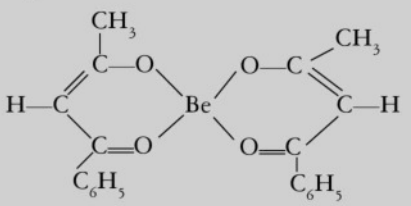
Example 1

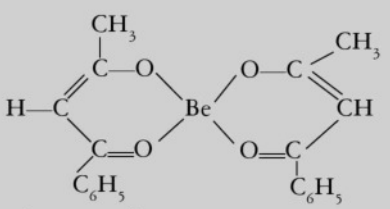
bis(salicylaldehyde) boron (III) cation has been found to be racemic and resolution has been accomplished.



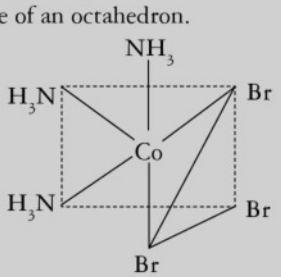
Example 2

The two enatiomers of bis(benzoylacetonato) beryllium (II) are shown as follows:



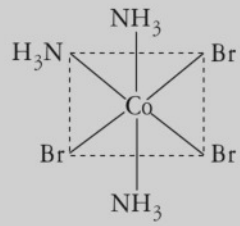


- $K_3[Co(NO_2)_6]$ i.e., potassium hexanitrocobaltate (III) is called Fischer's salt. Its common name is potassium cobaltinitrite.
- The geometric isomer is called facial (fac) when each trio of donor atoms of the similar ligands occupy adjacent positions at the corners of the same face of an octahedron.



Facial or fac-isomer

- The other geometric isomer of the compound is called meridional (mer) if the positions occupied are around the meridian of the octahedron.



Meridional or mer-isomer

Unforgettable guidelines

Complexes where the metal is (+3) oxidation state are more stable than those where te metal is in (+2) oxidation state

$$[Cr(H_2O)_6]^{+3} > [Cr(H_2O)_6]^{+2}$$

Irving William Order

In case of M^{2+} 3d-series the stability of complexes increases as follows:

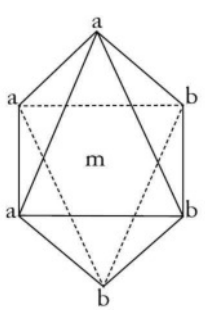
$$Mn^{2+} < Fe^{2+} < Co^{2+} < Ni^{2+} < Cu^{2+} < Zn^{2+}$$

Spectrochemical Series

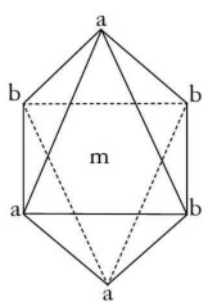
Ligands can be arranged in increasing order of their strength (ability to cause crystal field splitting) and this series is called spectrochemical series. It is as follows:

$$I^- < Br^- < S^{2-} < SCN^- < Cl^- < F^- < OH^- < Ox^{2-} < O^{2-} < H_2O < NCS^- < py < NH_3 < en < NO_2^- < CN^- < CO.$$

'Fac,' 'Mer' isomer of an ma_3b_3 octahedral complex can show optical isomerism.



1,2,3 or Fac



1,2,6 or Mer

Straight Objective Type Question (Single Choice)

1. The correct ground state electronic configuration atom (Z = 24) is
   (a) [Ar] 3d5 4s1 (b) [Ar] 3d4 4s2
   (c) [Ar] 3d6 4s0 (d) [Ar] 3d5 4s2

2. How many unpaired electrons are present in Ni^{2-}?
   (a) 8 (b) 4
   (c) 2 (d) 0

3. Which of rhe following has rhe maximum number of unpaired electrons?
   (a) Mg^{2+} (b) Ti^{3+}
   (c) V^{3+} (d) Fe^{2+}

4. While Ti^{3-}, V^{3+}, Fe^{3-} and Co^{2+} afford a large number of tetrahedral complexes, Cr^{3+} never does this, the reason being
   (a) Crystal field stabilization energy in octahedral vis-à-vis tetrahedral Cr^{3+} system plays the deciding role.
   (b) Cr^{3+} forces high crystal field splitting with a varieties of ligands.
   (c) Electronegativity of Cr^{3+} is the largest among these trivalent 3d-metals and so chromium prefers to be associated with as many ligands as its radius permits.
   (d) both (b) and (c).

5. The EAN of platinum in potassium hexachloroplatinate (IV) is
   (a) 46 (b) 86
   (c) 36 (d) 84

6. Which of the following ions has a magnetic moment of 5.93 BM? (At. no. V = 23, Cr = 24, Mn = 25, Fe = 26)
   (a) Cr^{2+} (b) V^{3-}
   (c) Mn^{2+} (d) Cr^{3+}
   (e) Fe^{2-}

7. Assign rhe hybridizarion, shape and magnetic moment of $K_2[Cu(CN)_4]$.
   (a) dsp^2, square planar, 1.73 BM
   (b) sp^3, tetrahedral, 1.73 BM
   (c) dsp^2, square planar, 2.44 BM
   (d) sp^3, rerrahedral, 2.44 BM

8. The oxidation states of Co and Cr respectively in the following complex are: $[Co(NH_3)_6][Cr(NH_3)_2Cl_4]_3$
   (a) +2, +3 (b) +3, +2
   (c) +3, +3 (d) +3, +4

9. IUPAC name of rhe compound $K_3[Fe(CN)_5CO]$ is
   (a) Potassium pentacyanocarbonylferrate (III)
   (b) Potassium carbonylpentacyanoferrate (III)
   (c) Potassium penracyanocarbonylferrare (II)
   (d) Potassium carbonylpentacyanoferrate (II)

10. Which one amongsr rhe following, exhibir geomerrical isomerism?
    (a) $[Pt^{II}(N11_3)_2Cl_2]$
    (b) $[Co^{III}(NH_3)_5Br]SO_4$
    (c) Co^{III} [EDTA] ^{1}
    (d) $[Cr^{III}(SCN)_6]^{3}$

11. The aqueous solution of the following salts will be coloured in the case of
    (a) $Zn(NO_3)_2$ (b) $LiNO_3$
    (c) $Co(NO_3)_2$ (d) porash alum

12. Which one of the following can show optical isomerism?
    (a) $K_3[Fe(CN)_6]$ (b) $Cr[(NH_3)_6]Cl_3$
    (c) $FeSO_4.7H_2O$ (d) $K_3[Cr(C_2O_4)_3]$

13. Which of rhe following will give four isomers?
    (a) $[Co(en)_2Cl_2]Br$
    (b) [Co (en) $(NH_3)_2Cl_2$] Cl
    (c) [Co $(PPh_3)_2$ $(NH_3)_2Cl_2$] Cl
    (d) [Co $(en)_3$] Cl_3

14. Which one of the following high spin complexes has the largest C.F.S.E. (crystal field stabilization energy)?
    (a) $[Cr(H_2O)_6]^{2+}$ (b) $[Cr(H_2O)_6]^{3+}$
    (c) $[Mn(11_2O)_6]^{2+}$ (d) $[Mn(II_2O)_6]^{3}$

15. Which one is the most likely structure of $CrCl_3.6H_2O$ if 1/3 of total chlorine of the compound is precipitated by adding $AgNO_3$ to its aqueous solution?

(a) $[Cr(H_2O)_5Cl_3].(H_2O)_3$

(b) $CrCl_3.6H_2O$

(c) $[CrCl(H_2O)_5]Cl_2.H_2O$

(d) $[CrCl_2(H_2O)_4].Cl.H_2O$

16. Tetrahedral complexes of the types of (Ma_4) and (Ma_3b) (here M = metal, a, b = achiral ligands) are not able to show optical isomerism because

(a) These molecules/ions have nonsuperimposable mirror images.

(b) These molecules possess a centre of symmetry.

(c) These molecules/ions possess a plane of symmetry and hence are achiral.

(d) These molecules/ions possess C_n axis of symmetry.

17. The complex salt having the molecular composition—$[Co(NO_2)(SCN)(en)_2]Br$ exhibits

(a) linkage isomerism only

(b) ionization isomerism only

(c) cis-trans isomerism only

(d) all of these

18. The equivalent weight of $MnSO_4$ is half of its molecular weight when it is converted to

(a) Mn_2O_3 (b) MnO_2

(c) MnO_4^- (d) MnO_4^{2-}

19. The number of chloride ion produced by complex tetraamminechloroplatinum(IV) chloride in an aqueous solution is

(a) 1 (b) 2

(c) 3 (d) 4

20. The compound which does not show paramagnetism is

(a) $[Cu(NH_3)_4]Cl_2$ (b) $[Ag(NH_3)_2]Cl$

(c) NO (d) NO_2

21. Amongst the following, the lowest degree of paramagnetism per mole of the compound at 298 K will be shown by

(a) $MnSO_4.4H_2O$ (b) $CuSO_4.5H_2O$

(c) $FeSO_4.6H_2O$ (d) $NiSO_4.6H_2O$

22. Amongst the following ions which one has the highest paramagnetism?

(a) $[Cr(H_2O)_6]^{3+}$ (b) $[Fe(H_2O)_6]^{2+}$

(c) $[Cu(H_2O)_6]^{2+}$ (d) $[Zn(H_2O)_6]^{2+}$

23. Which one of the following octahedral complexes will not show geometric isomerism? (A and B are monodentate ligands)

(a) (MA_4B_2) (b) (MA_5B)

(c) (MA_2B_4) (d) (MA_3B_3)

24. The formula of tetrachloro diamine platinum (IV) is

(a) $[Pt(NH_3)_2]Cl_4$ (b) $[Pt(NH_3)_2Cl_4]$

(c) $[Pt(NH_3)_2Cl_2]Cl_2$ (d) $K_4[Pt(NH_3)_2Cl_4]$

25. The possible numbers of isomers for the complex $(MCl_2Br_2)SO_4$ will be

(a) 5 (b) 4

(c) 3 (d) 2

26. Benzoyl acetonato beryllium exhibits the isomerism of

(a) Geometrical

(b) Optical

(c) Conformational

(d) Structural

27. Which of the following statements is not correct?

(a) The complexes $(NiCl_4)^{2-}$ and $[Ni(CN)_4]^{2-}$ differ in geometry.

(b) The complexes $(NiCl_4)^{2-}$ and $[Ni(CN)_4]^{2-}$ differ in primary valencies of nickel.

(c) Nickel ion has the same secondary valency in the complexes $(NiCl_4)^{2-}$ and $[Ni(CN)_4]^{2-}$.

(d) The complexes $(NiCl_4)^{2-}$ and $[Ni(CN)_4]^{2-}$ differ in the state of hybridization of nickel.

28. Both Co^{3+} and Pt^{4+} have a coordination number of six. Which of the following pairs of complexes will show approximately the same electrical conductance for their 0.001 M aqueous solutions?

(a) $CoCl_3.6NH_3$ and $PtCl_4.5NH_3$

(b) $CoCl_3.6NH_3$ and $PtCl_4.3NH_3$

(c) $CoCl_3.5NH_3$ and $PtCl_4.6NH_3$

(d) $CoCl_3.4NH_3$ and $PtCl_4.4NH_3$

29. In the complexes $[Fe(H_2O)_6]^{3+}$, $[Fe(CN)_6]^{3-}$, $[Fe(C_2O_4)_3]^{3-}$ and $[FeCl_6]^{3-}$ more stability is shown by
(a) $[Fe(H_2O)_6]^{3+}$ (b) $[Fe(C_2O_4)_3]^{3-}$
(c) $[FeCl_6]^{3-}$ (d) $[Fe(CN)_6]^{3-}$

30. A cation which does not form an amine complex ion with excess of NH_3 is
(a) Cu^{2+} (b) Al^{3+}
(c) Ag (d) Co^{2+}

31. From the stability constant (hypothetical values) given below, predict which is the strongest ligand.
(a) $Cu^{2+} + 4H_2O \rightleftharpoons [Cu(H_2O)_4]^{2+}$ $K = 9.5 \times 10^{8}$
(b) $Cu^{2+} + 2en \rightleftharpoons [Cu(en)_2]^{2+}$ $K = 3.0 \times 10^{15}$
(c) $Cu^{2+} + 4en \rightleftharpoons [Cu(CN)_4]^{2-}$ $K = 2.0 \times 10^{27}$
(d) $Cu^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+}$ $K = 4.5 \times 10^{11}$

32. In which of the following pair the EAN of central metal atom is not same?
(a) $(FeF_6)^{3+}$ and $[Fe(CN)_6]^{3-}$
(b) $[Fe(CN_6)]^{3-}$ and $[Fe(CN)_6]^{4-}$
(c) $[Cr(NH_3)_6]^{3+}$ and $[Cr(CN)_6]^{3-}$
(d) $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$

33. Which of the following complex ions will not show optical activity?
(a) $[Co(en)(NH_3)_2Cl_2]^{+}$
(b) $[Cr(NH_3)_4Cl_2]^{+}$
(c) [Pt (Br) (Cl) (I) (NO_2) (Py) NH_3]
(d) cis – $[Co(en)_2Cl_2]^{+}$

34. The number of d-electrons in $[Cr(H_2O)_6]^{3+}$ (at. no. of Cr = 24) is
(a) 2 (b) 3
(c) 4 (d) 5

35. The number of ions produced from one molecule of [Pr $(NH_3)_5$ Br] Br_2 in the aqueous solution will be
(a) 4 (b) 5
(c) 6 (d) 7

36. The stability constants of the complexes formed by a metal ions (M^{2+}) with NH_3, CN^-, H_2O and 'en' are of the order of 10^{11}, 10^{27}, 10^{15} and 10^{8} respectively. Then
(a) en is the strongest ligand.
(b) These values cannot predict the strength of the ligand.
(c) CN is the strongest ligand.
(d) All ligands are equally strong.

37. For a complex MA_3B_3 possessing a trigonal prismatic geometry, the number of possible isomers is
(a) 3 (b) 4
(c) 5 (d) 6

38. The coordination number and oxidation number of M in the compound $[M(SO_4)(NH_3)_5]$ will be
(a) 6 and 3 (b) 2 and 6
(c) 6 and 2 (d) 3 and 6

39. The complex with spin only magnetic moment of ~ 4.9 B.M. is
(a) $[Fe(CN)_6]^{3-}$ (b) $[Fe(H_2O)_6]^{3+}$
(c) $[Fe(CN)_6]^{4-}$ (d) $[Fe(H_2O)_6]^{2+}$

40. The number of isomers possible for complex K_2[Pd Cl Br $(SCN)_2$] is
(a) 1 (b) 2
(c) 3 (d) 4

41. The value of magnetic moment for a complex ion is 1.73 BM. The complex ion is
(a) $[Mn(CN)_6]^{4-}$ (b) $[Co(NH_3)_6]^{2+}$
(c) $(MnF_6)^{3-}$ (d) $[Fe(CN)_5NO]^{2-}$

42. The number of unpaired electrons in the complex ion $(CoF_6)^{3-}$ is
(Atomic no. of Co = 27)
(a) 4 (b) Zero
(c) 2 (d) 3

43. Why is $[Ni(en)_3]^{2+}$, nearly 10^{10} times more stable than $[Ni(NH_3)_6]^{2+}$?
(a) NH_3 is the weakest ligand.
(b) 'en' is a chelating ligand and forms thermodynamically more stable complexes.
(c) Six NH_3 ligands cause steric hindrance around the Ni^{2+} centre.
(d) NH_3 evaporates easily and causes instability to the $[Ni(NH_3)_6]^{2+}$ complex.

44. When concentrated HCl is added to a solution of $[Co(H_2O)_6]^{2}$ ion, an intense blue colour develops due to the formation of which one of the following?
(a) $[CoCl_4]^{2}$ (b) $[CoCl_6]^{4}$
(c) $[CoCl(H_2O)_5]^{1}$ (d) $[CoCl_2(H_2O)_4]$

45. Which of the following is not the correct systematic name of corresponding compound?
(a) $Ni(CO)_2(PPh_3)_2$: Dicarbonylbis (triphenyl phosphine) nickel(0)
(b) $Na_2[Fe(CN)_5NO]$: Sodium pentacyanonitrosyl ferrate(II)
(c) $NaMn(CO)_5$: Sodium pentacarbonyl manganate (I)
(d) $SnCl_4(Et_2NH)_2$: Tetrachlorobis (diethylamine) tin(IV)

46. Which of the following sequence is incorrect?
(a) Sc, V, Cr, Mn — increasing number of oxidation state
(b) MO, M_2O_3, MO_2, MO_5 — decreasing basic strength
(c) d^5, d^3, d^1, d^4 increasing magnetic moment
(d) Co^{2+}, Fe^{3+}, Cr^{3-}, Sc^{3+} increasing stability

47. Which of the following is not an organometallic compound?
(a) $(C_2H_5)_4Pb$ (b) C_2H_5–O–Na
(c) C_4H_9Li (d) $[(C_5H_5)_2Fe]$

48. The atomic number of V, Cr, Mn and Fe are respectively 23, 24, 25 and 26. Which one of these may be expected to have the highest second ionization enthalpy?
(a) Mn (b) Fe
(c) Cr (d) V

49. Find out the wrong statement for an octahedral complex.
(a) An ion with d^1 configuration has one unpaired electron both in weak and strong fields.
(b) A central metal ion with d^8 configuration has two unpaired electrons.
(c) An ion with d^6 configuration is diamagnetic in a strong field.
(d) In d^4, d^5, d^6, and d^7 configurations, weak and strong field complexes have different numbers of unpaired electrons.

50. Which of the following transition metal oxides is (normally) neither acidic nor amphoteric?
(a) CrO_3 (b) Fe_2O_3
(c) Mn_2O_7 (d) CuO

51. A certain complex ion has the formula, $[M(en)_2Br_2]^{+}$ where M is the metal ion and en = ethylene diamine. How many isomers are possible for this?
(a) 1 (b) 2
(c) 3 (d) 4

52. What is the effective atomic number, (EAN) of Cr and/or Cu in their complexes, $[Cr(CN)_6]^{3-}$ and $[Cu(CN)_4]^{3-}$?
(Atomic numbers: Cr = 24, Cu = 29)
(a) 36 for both
(b) 36 for neither
(c) 36 for Cr in $[Cr(CN)_6]^{3-}$
(d) 36 for Cu in $[Cu(CN)_4]^{3}$

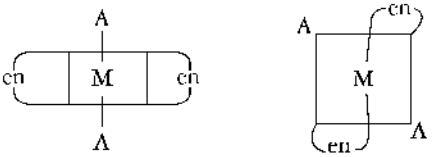
53. Which of the following is not a bidentate ligand?
(a) Acetyl acetonate (acac)
(b) Bis (dimethyl glyoximato) system (dmg H)
(c) Bis (diphenyl phosphino) ethane (dppe)
(d) Glyme, CH_3 O CH_2 CH_2 O CH_3

54. A certain complex ion, with octahedral geometry has six different ligands: $(ML_1L_2L_3L_4L_5L_6)^{n}$. How many isomeric structures are possible, counting optical isomers separately?
(a) 15 (b) 30
(c) 45 (d) 120

55. Ferrous ion change to X ion, on reacting with acidified hydrogen peroxide. The number of d-electrons present in X and its magnetic moment (in BM) are, respectively
(a) 5 and 4.9 (b) 4 and 5.92
(c) 6 and 6.95 (d) 5 and 5.92

Brainteasers Objective Type Questions (Single Choice)

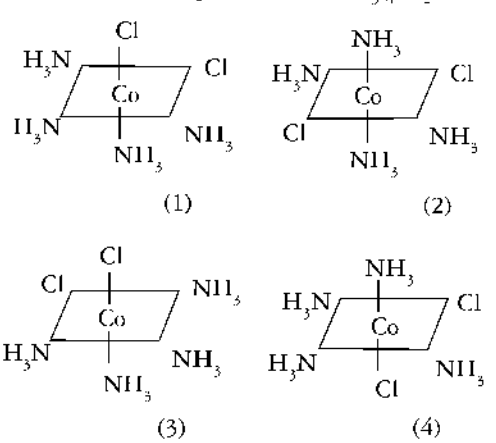
56. The two complexes given below are



(a) Geomerrical isomers
(b) Position isomers
(c) Optical isomers
(d) Identical

57. The correct IUPAC name of $KAl(SO_4)_2.12H_2O$ is
(a) Aluminium potassium sulphate-12-water
(b) Potassium aluminium (III) sulphate-12-water
(c) Potassium aluminium(III) sulphare hydrare
(d) Aluminium(III) potassium sulphate hydrate-12

58. Consider rhe following sparial arrangemenrs of rhe octahedral complex ion $[Co(NH_3)_4Br_2]$.



Which of rhe following sraremenrs is incorrecr regarding rhese srrucrures?
(a) 2 and 3 are cis and trans isomers, respectively
(b) 1 and 3 are enanriomers
(c) 3 and 4 have identical structures
(d) 2 and 4 are trans and cis isomers, respectively

59. Consider the following complex $[Co(NH_3)_5CO_3]ClO_4$ The coordination number, oxidation number, number of d electrons and number of unpaired d-electrons on the metal are, respectively
(a) 6, 3, 6, 0 (b) 6, 2, 7, 1
(c) 6, 3, 6, 4 (d) 6, 2, 7, 3

60. Each of the compounds $Pt(NH_3)_6Cl_4$, $Cr(NH_3)_6Cl_3$, $Co(NH_3)_4Cl_3$ and K_2PtCl_6 has been dissolved in water to make its 0.001 M solution. The order of their increasing conductivity in solution is
(a) $K_2PrCl_6 < Pr(NH_3)_6Cl_4 < Cr(NH_3)_6Cl_3 < Co(NH_3)_4Cl_3$
(b) $Cr(NH_3)_6Cl_3 < Co(NH_3)_4Cl_3 < Pr(NH_3)_6Cl_4 < K_2PrCl_6$
(c) $Co(NH_3)_4Cl_3 < K_2PtCl_6 < Cr(NH_3)_6Cl_3 < Pr(NH_3)_6Cl_4$
(d) $Pt(NH_3)_6Cl_4 < K_2PtCl_6 < Co(NH_3)_4Cl_3 < Cr(NH_3)_6Cl_3$

61. The magnetic moment of a transition metal of 3d series is 6.92 BM. Its electronic configuration would be
(a) $3d^{16}$ (b) $3d^5\,4s^0$
(c) $3d^4\,4s^2$ (d) $3d^5\,4s^1$

62. Pick our rhe correcr sraremenrs from here.
1. Both Fe(II) and Fe(III) salts react with NO to give brown compound.
2. Fe(III) forms octahedral complexes but Fe(II) form either tetrahedral or square planar complex.
3. Hexacyane ferrare (II) ion is diamagneric bur hexacyanoferrare (III) is paramagneric.
4. A pale yellow precipirare is formed when H_2S is passed rhrough acidic solution of Fe(III).

(a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) 3 and 4

63. Which of the following set represents only coloured ions?

(I) Fe^{3+} (II) Sc^{3}

(III) Cu^{2} (IV) Cr^{3+}

(a) I and III only (b) II, III and IV

(c) I, II and III (d) I, III and IV

64. Choose the correct statements.

1. Geometry of the coordination entity can be predicted if its magnetic behaviour is known.
2. $[Ni(CN)_4]^{2-}$ involves dsp^2 hybridization.
3. For analogous entities within a group, Δ_0 value follows 3d > 4d > 5d.
4. $[Pt(CN)_4]^{2}$ ion is square planar and diamagnetic.

(a) 1, 2 and 3 (b) 1, 2 and 4

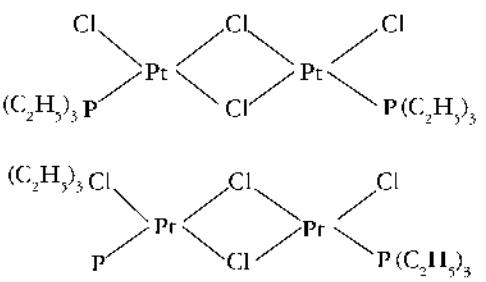
(c) 2, 3 and 4 (d) 1, 3 and 4

65. The number of donor sites in dimethyl glyoxime, diethylene triamine, glycinato and EDTA are, respectively

(a) 2, 2, 3, 6 (b) 2, 3, 2, 6

(c) 3, 2, 3, 6 (d) 3, 2, 2, 6

66. The complex given below shows.



(a) Optical isomerism

(b) Coordination isomerism

(c) Geometrical isomerism

(d) Bridged isomerism

67. On treatment of 100 ml of 0.1M solution of the complex $CoCl_3.6H_2O$ with excess of $AgNO_3$, 4.305 g of AgCl was obtained. The complex is

(a) $[Co(H_2O)_6]Cl_3$

(b) $[Co(H_2O)_5Cl]Cl_2.H_2O$

(c) $[Co(H_2O)_3Cl_3].3H_2O$

(d) $[Co(H_2O)_4Cl_2]Cl.2H_2O$

68. Which of the following statement is/are correct?

I. The ligand thiosulpharo, $S_2O_3^{2-}$ can give rise to linkage isomers.

II. In metallic carbonyls the ligand CO molecule acts both as donor and acceptor.

III. The complex $[Pt(py)(NH_3)(NO_2)ClBrI]$ exists in eight different geometrical isomeric forms.

IV. The complex ferricyanide does not follow effective atomic number (EAN) rule.

(a) 1 and 2 only (b) 2 and 4 only

(c) 1, 2 and 3 (d) 1, 3 and 4

69. The elements which exist in the liquid state at room temperature are

I. Na II. Br

III. Hg IV. Ga

(a) 1, 2, 3 (b) 2, 3

(c) 2, 4 (d) 1, 2, 3

70. Which of the following are diamagnetic?

1. $K_3[Fe(CN)_6]$ 2. $K_3[Cr(CN)_6]$

3. $K_3[Co(CN)_6]$ 4. $K_2[Ni(CN)_4]$

Select the correct answer using the codes given below.

(a) 1 and 2 (b) 2, 3 and 4

(c) 1, 3 and 4 (d) 2 and 4

71. In the complexes $[Cr(CN)_6]^{3-}$, $[CuCl_4]^{2-}$, $[Cu(NH_3)_2]$. The number of unpaired electrons are, respectively.

(a) 1, 3 and 0 (b) 3, 2 and 1

(c) 3, 2 and 0 (d) 3, 1 and 0

72. Which one of the following complex is diamagnetic in nature?

(I) $K_2[Ni(CN)_4]$ (II) $[Ni(H_2O)_6](NO_3)_2$

(III) $[Co(NH_3)_6]Cl_3$ (IV) $[Pt(NH_3)_4]Cl_2$

Select the correct answer:

(a) 1 and 2 (b) 2 and 3

(c) 1, 2 and 4 (d) 1, 3 and 4

73. The oxidation number of Fe in $[Fe(CN)_6]^{4-}$, Cr in $[Cr(NH_3)_3(NO_2)_3]$ and Ni in $[Ni(CO)_4]$ are, respectively.

(a) 0, +3, +2 (b) +3, +3, 0

(c) +3, 0, +3 (d) +2, +3, 0

74. Predict the correct statement about A and B in the previous question.

I. A is diamagnetic and B is paramagnetic with two unpaired electrons.

II. A is diamagnetic and B is paramagnetic with one unpaired electron.

III. The hybridization of A and B are dsp^2 and sp^3, respectively.

IV. The hybridization of A and B are sp^3 and dsp^2, respectively.

(a) 1 and 3 (b) 2 and 4
(c) 2 and 3 (d) 3 and 4

75. In the complex $[Cu(CN)_4]^{3-}$ the hybridization state, oxidation state and number of unpaired electron are, respectively.

(a) dsp^2, +1, 1 (b) sp^3, +1, zero
(c) sp^3, +2, 1 (d) dsp^2, +2, zero

76. Consider the following complex:

$[Cr(NH_3)_5CO_3]ClO_4$

The coordination number, oxidation number, number of d-electrons and number of unpaired d-electrons on the metal are, respectively

(a) 6, 3, 6, 0 (b) 6, 3, 6, 3
(c) 6, 0, 6, 3 (d) 6, 2, 6, 3

77. Among $[Ni(CO)_4]$, $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$

(a) $[Ni(CO)_4]$ and $[NiCl_4]^{2-}$ are diamagnetic and $[Ni(CN)_4]^{2-}$ is paramagnetic.

(b) $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $[Ni(CO)_4]$ is paramagnetic.

(c) $[Ni(CO)_4]$ and $[NiCN_4]^{2-}$ are diamagnetic and $[Ni(Cl)_4]^{2-}$ is paramagnetic.

(d) $[Ni(CO)_4]$ is diamagnetic and $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are paramagnetic.

78. Which of the following will have three stereoisomeric forms?

I. $[Cr(NO_3)_3(NH_3)_3]$

II. $K_3[Co(C_2O_4)_3]$

III. $K_3[Co(C_2O_4)_2Cl_2]$

IV. $[Co(en)_2ClBr]$

(Here en = ethylene diamine)

(a) 1 and 2 (b) 1 and 3
(c) 1 and 4 (d) 3 and 4

79. The correct number of unpaired electrons in Fe^{2+}, Mn^{2+}, Cr^{3+}, V^{5+} and Cu will be, respectively

(a) 5, 4, 3, 0, 1 (b) 4, 5, 3, 0, 1
(c) 5, 4, 2, 1, 1 (d) 4, 5, 3, 1, 1

80. Which of the following exhibit geometrical isomerism? (M stands for a metal, and a and b are achiral ligands).

1. Ma_2b_2 2. Ma_4b_2
3. Ma_5b 4. Ma_6

(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 2 and 4

81. Which of the following pairs of isomers and types of isomerism are correctly matched?

(I) $[CO(NH_3)_5(NO_2)]Cl_2$ and $[CO(NH_3)_5(ONO)]Cl_2$ .... Linkage

(II) $[Cu(NH_3)_4][PtCl_4]$ & $[Pt(NH_3)_4][CuCl_4]$ ..... Coordination

(III) $[Pt(NH_3)_4Cl_2]Br_2$ & $[Pt(NH_3)_4Br_2]Cl_2$ ..... Ionization

Select the correct answer using the codes given below:

(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

82. The correct order of magnetic moment (spin only values in B.M.) among is

(a) $(MnCl_4)^{2-} > (CoCl_4)^{2-} > Fe(CN_6)^{4-}$

(b) $[Fe(CN)_6]^{4-} > (MnCl_4)^{2-} > (CoCl_4)^{2-}$

(c) $[Fe(CN)_6]^{4-} > (CoCl_4)^{2-} > (MnCl_4)^{2-}$

(d) $(MnCl_4)^{2-} > [Fe(CN)_6]^{4-} > (CoCl_4)^{2-}$

(Atomic number Mn = 25, Fe = 26, Co = 27, Ni = 28)

83. Select the correct increasing order of 10 Dq value for chromium complexes using the given codes.

(I) $[Cr(en)_3]^{3+}$ (II) $[Cr(ox)_3]^{3-}$

(III) $(CrF_6)^{3-}$ (IV) $[Cr(dtc)]^{3+}$

(Here, dtc = dithiocarbamate)

(a) 1 < 2 < 3 < 4 (b) 3 < 4 < 2 < 1
(c) 4 < 1 < 2 < 3 (d) 3 < 1 < 4 < 2

84. The sums of first and second ionization energies and those of third and fourth energies (in MJ mol^{-1}) of Ni and Pt are

	$(IE_1 + IE_2)$	$(IE_3 + IE_4)$
Ni	2.49	8.80
Pr	2.66	6.70

Complexes formed by Ni and Pr have greater srability, respecrively in

(a) +2 states in both cases

(b) +4 states in both cases

(c) +2 and +4 states

(d) +4 and +2 states

85. Arrange the following in order of decreasing number of unpaired electrons.

I. $[Fe(H_2O)_6]^{2+}$ II. $[Fe(CN)_6]^{4-}$

III. $[Fe(CN)_6]^{4-}$ IV. $[Fe(H_2O)_6]^{3+}$

(a) IV, I, II, III (b) I, II, III, IV

(c) IV, II, I, III (d) II, III, I, IV

Multiple Correct Answer Questions (More Than One Choice)

86. Which of the following is/are characteristics of d-block elements?

(a) They are generally diamagnetic.

(b) They form coloured complexes.

(c) They show variable oxidation states.

(d) Their ionization energies are very high.

87. Which of the following metals have both valence shell and penultimate shell partially filled?

(a) Cu (b) Zn

(c) Cr (d) Mn

88. Which of the following d-block elements do not posses characteristic properties of transition elements?

(a) Cadmium (b) Manganese

(c) Zinc (d) Copper

89. Which of the following srarement(s) can be applied to Zieses salt?

(a) Its aqueous solution gives test for Cl^- ions.

(b) Its IUPAC name is potassium trichloro (η^2 ethylene) platinate (II).

(c) Its formula is $K[Pt(C_2H_4)Cl_3]$.

(d) It contains one π-acid ligand.

90. Which of the following is/are not bidentate ligands?

(a) 1,2-diamino propane

(b) Hydrazinium

(c) Cyano

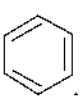
(d) Oxalato

91. Which of the following is/are incorrect statement(s)?

(a) Equivalent weight of $KMnO_4$ in acidic medium is M/5.

(b) In acidic medium MnO_4^{2-} disproportionates ro MnO_2 and MnO_4^-.

(c) $KMnO_4$ spor can be bleached by H_2O_2.

(d) Alkaline $KMnO_4$ can be used to rest unsaturation in  .

92. Which is correct about EDTA?

(a) The abbreviation represents ethylenediamine triacetate.

(b) It is flexidentate ligand.

(c) The ligand cannot form chelate complexes.

(d) The abbreviation stands for ethylenediamine tetraacetate.

93. Which is false about DMG?

(a) It is an ambidentate ligand.

(b) It is a bidentate ligand.

(c) It is called dimethyl glyoximato.

(d) It is a flexidentate ligand.

94. Which of the following srarement is/are correct?

(a) In $K_3[Fe(CN)_6]$, the ligand has satisfied both primary and secondary valencies of ferric ion.

(b) In $K_3[Fe(CN)_6]$ the ligand has satisfied only the secondary valency of ferric ion.

(c) In $K_4[Fe(CN)_6]$ the ligand has satisfied both primary and secondary valencies of ferrous ion.

(d) In $[Cu(NH_3)_4]SO_4$, the ligand has satisfied only the secondary valency of copper.

95. What is true about acetylacetonato ion?

(a) It is divalent ion and bidentate ligand.

(b) It is chelating ligand.

(c) It is divalent ion and ambidentate ligand.

(d) It is monovalent ion but bidentate ligand.

96. Which among the following complexes is/are not expected to be coloured?

(a) $[Cr(NH_3)_6]Cl_3$

(b) $K_3(VF_6)$

(c) $[Ti(NO_3)_4]$

(d) $[CrO(NCCH_3)_4]$ BF_4

97. Which statement about $[Cu(NH_3)_4]^{2+}$ is/are incorrect?

(a) Cu assumes dsp^2 hybrid state.

(b) Complex ion is deep blue in colour.

(c) Cu atom is in sp^3 hybrid state.

(d) The complex is white in colour.

98. Silver chloride dissolves in ammonium hydroxide forming P. What is not true about P?

(a) P is called Tollen's reagent.

(b) P is a double salt.

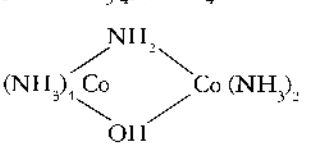
(c) P is paramagnetic in nature.

(d) P is carbonic complex.

99. Which of the following complex is/are polynuclear?

(a) $[Pt^{II}(NH_3)_4]\,(CuCl_4)$

(b)



(c) $[Co(en)_2Cl_2]_2SO_4$.

(d) $(NH_3)_5Cr \quad O \quad O \quad Cr(NH_3)_5$

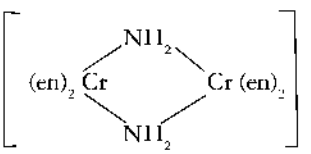
100. Which of the following compounds are coloured due to charge transfer spectra?

(a) $AgNO_3$ (b) $CuSO_4$

(c) $K_2Cr_2O_7$ (d) $KMnO_4$

101. Which of the following complexes is/are able to exhibit optical isomerism?

(a)



(b) $[Cr(Ox)_3]^{3-}$

(c) $[Co(EDTA)]^-$

(d) Trans-$[Co(en)_2Cl.NH_3]^{2+}$

102. What is correct about O $(CO)_2$ O ?

(a) It is symmetrical bidentate ligand.

(b) It is a tridentate ligand.

(c) It is called oxalato.

(d) It can produce chelation.

103. What is correct about the following compound? $K[Pt(\eta^2 - C_2H_4)Cl_3]$

(a) Four ligands surround the platinum atom.

(b) It is called Zeises salt.

(c) It is π-bonded complex.

(d) Oxidation number of Pt is +4.

104. Which of the following matching is/are correct here?

(a) Secondary valencies oxidation number

(b) Primary valencies hold ionizable species

(c) Primary valencies oxidation state

(d) Secondary valencies—coordination number

105. Which of the following can show geometrical isomerism?

(a) $[Cr(Ox)_3]^{3-}$ (b) $[Co(en)_2Cl_2]Cl$

(c) $[Cr(NH_3)_4Cl_2]Cl$ (d) $[NiCl_4]^{2-}$

106. What is correct about $Ni_3[Co(ONO)_6]$?

(a) Its aqueous solution is slightly basic in nature.

(b) Its IUPAC name is sodium hexanitrito cobaltate (III).

(c) It is called sodium cobaltinitrite.

(d) Its IUPAC name is sodium hexanitrocobaltate (III).

107. The mixture of which of the following can produce blue colouration?
   (a) Iron (III) chloride and $K_4[Fe(CN)_6]$ (aq)
   (b) NH_4OH (aq) and $CuSO_4$ (aq)
   (c) Adding anhydrous $CuSO_4$ to water
   (d) $ZnCl_2$ (aq) and $K[Fe(CN)_6]$(aq)

108. Which of the following ligand is/are negatively charged?
   (a) Isothiocyanate ion
   (b) Ammonium ion
   (c) Hydrazinium ion
   (d) Sodium ion

109. A magnetic moment of 1.73 BM, will be shown by
   (a) $[Zn(CN)_4]^{2-}$ (b) $TiCl_3$
   (c) $[Fe(CN)_6]^{3-}$ (d) $[Cu(NH_3)_4]^{2+}$

110. Which of the following conditions is/are suitable for the stability of the complex?
   (a) Chelation
   (b) Larger basic nature of the ligand
   (c) Larger charge on the central metal ion
   (d) Smaller charge on the central metal ion

Linked–Comprehension Type Question

Comprehension–1

The transition elements with some exceptions can show a large number of oxidation states. The various oxidation states are related to the electronic configuration of their atoms. The variable oxidation states of a transition metal is due to the involvement of (n 1)d and outer ns-electrons. For the first five elements of 3d-transition series. The minimum oxidation state is equal to the number of electrons in 4s shell and the maximum oxidation state is equal to the sum of 4s and 3d-electrons. The relative stability of various oxidation state of a given element can be explained on the basis of stability of d^0, d^5 and d^{10} configurations.

111. In 3d-series, the maximum oxidation state is shown by
   (a) Fe (b) Mn
   (c) Cr (d) V

112. In which of the following pair, the first species is more stable than second one?
   (a) Mn^{2+}, Mn^{3+} (b) Sc^{2+}, Sc^{3}
   (c) Ti^{3} , Ti^{4+} (d) Fe^{2+}, Fe^{3+}

113. Identify the correct statement.
   (a) Ti^{4-}, Mn^{2-} are stable oxidation states.
   (b) The most common oxidation state of 3d-series is +2.
   (c) The lowest oxidation state of Cr and Cu is +1 while for others it is +2.
   (d) All of these.

114. Which does not represent the correct order of stability here?
   (a) $Ti^{4+} > Ti^{3+}$
   (b) $Fe^{3} > Fe^{2}$
   (c) $Mn^{3+} > Mn^{2+}$
   (d) $V^{5+} > V^{3+}$

Comprehension–2

Coordination compounds may show two types of isomerism.

In constitutional isomerism, isomers differ in bond connectivities. It is further divided into ionization, hydrate isomerism etc.

In stereoisomerism, isomers differ in spatial arrangement of atoms. It is either geometrical or optical type.

115. Which of these pair does not represent any type of constitutional isomerism here?
   (a) $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5 SO_4]Br$
   (b) $[CrCl(H_2O)_5]Cl_2.H_2O$ and $[CrCl_2(H_2O)_4]Cl.2H_2O$
   (c) Cis-$[CrCl_2(Ox)_2]^{3}$ and trans-$[CrCl_2(Ox)_2]^{3}$
   (d) $[Pr(NH_3)_4](PtCl_6)$ and $[Pr(NH_3)_4Cl_2](PrCl_4)$

116. In which pair, both the compounds do not show geometrical isomerism?)

(a) $[Pr(NH_3)_2Cl_2]$ and $[Co(en)_2Cl_2]$
(b) $[Co(NH_3)_4Cl_2]^-$ and $[Co(NH_3)_3(NO_2)_3]$
(c) $[Fe(NH_3)_2(CN)_4]^{2-}$ and $[CoCl_2(Ox)_2]^{3-}$
(d) $[Co(NH_3)_5Cl]SO_4$ and $[Cr(en)_3]^{3+}$

117. Which of the following compounds can show geometrical isomerism here?

I. $[Pt(gly)_2]$
II. $[Pt(NH_3)_2ClNO_2]$
III. $[Ni(CN)_4]^{2-}$
IV. $[Co(NH_3)_3Cl_3]$

(a) I, II and III (b) I, II and IV
(c) II and III only (d) II, III and IV

Comprehension–3

Most of the transition metal ions and their compound are coloured and paramagnetic. The paramagnetism of the transition metal ion is calculated by using the relation

$$\mu = \sqrt{n(n+2)}\ \text{B.M.}$$

where n is number of unpaired electrons. The colour of the compound is due to d-d electron transition, charge transfer and polarization.

118. $K_2Cr_2O_7$ is coloured due to
(a) Charge transfer
(b) d-d electron transition
(c) Polarization
(d) Both (a) and (b)

119. A green coloured metal sulphate has $\sqrt{24}$ B.M. magnetic momentum. The metal ion in that compound is
(a) Mn^{2+} (b) Fe^{3+}
(c) Fe^{2+} (d) Cr^{2+}

120. $KFe^{III}[Fe(CN)_6]$ and $KFe^{II}[Fe(CN)_6]$
1. 2.
Which of the following is correct about these two?
(a) Both are blue coloured because colour arises due to d-d electron transition in Fe ion present outside the complex ion.
(b) Both are blue coloured because colour arises due to transfer of electron between Fe^{II} and Fe^{III}.
(c) Complex 1 has blue colour while complex 2 has brown colour.
(d) Both are blue coloured because in complex ion Fe^{II} shows same d-d transition bond.

121. Which of the following set of ions are paramagnetic are coloured?
(a) Mn^{2+}, Zn^{2+}
(b) Sc^{3+}, Cu^{2+}
(c) Fe^{3-}, Cu^{2-}
(d) V^{5+}, Ni^{2+}

Comprehension–4

The valence bond theory was extended to coordination compounds by Linus Pauling on the basis of orbital hybridization, bonding between ligand and the metal atom/ion and relation between the observed magnetic behaviour and the bond type.

It explains successfully the formation, geometrical shapes and magnetic nature of the complexes, however it fails in explaining quantitative interpretation of magnetic behaviour and optical properties of complexes.

122. Which of the following assumption cannot be explained with the help of VBT?
(a) Spectral properties of coordination compounds.
(b) Distinction between strong and weak ligands.
(c) Interpretation of the thermodynamic and kinetic stabilities of complexes.
(d) All of these.

123. Which is correct about $[Pt(CN)_4]^{2-}$?
(a) dsp^2, square planar and diamagnetic.
(b) sp^3, tetrahedral and paramagnetic.
(c) dsp^2, square planar and paramagnetic.
(d) sp^3, tetrahedral and diamagnetic.

124. $[Cu(NH_3)_4]^{2+}$ has hybridization and magnetic moment, respectively
(a) dsp^2, zero B.M.
(b) dsp^2, 1.73 B.M.
(c) sp^3, 1.73 B.M.
(d) sp^3d, 2.76 B.M.

125. The correct order of magnetic moments (spin only values in B.M.) among the following is (At. number of Mn = 25, Fe = 26, Co = 27)

(a) $(MnCl_4)^{2-} > (CoCl_4)^{2-} > [Fe(CN)_6]^{4-}$
(b) $(MnCl_4)^{2-} > [Fe(CN)_6]^{4-} > (CoCl_4)^{2-}$
(c) $[Fe(CN)_6]^{4-} > (MnCl_4)^{2-} > (CoCl_4)^{2-}$
(d) $[Fe(CN)_6]^{4-} > (CoCl_4)^{2-} > (MnCl_4)^{2-}$

Assertion and Reasoning Questions

In the following questions two statements (Assertion) A and Reason (R) are given. Mark

(a) If A and R both are correct and R is the correct explanation of A;
(b) If A and R both are correct but R is not the correct explanation of A;
(c) A is true but R is false;
(d) A is false but R is true.

126. (A): Transition metals form a large number of interstitials compounds.
(R): They have high melting point and boiling point.

127. (A): Transition metals form complexes.
(R): Transition metals have unpaired electrons.

128. (A): Zn, Cd, Hg are not regarded as transition metals.
(R): Zn, Cd, Hg have low melting and boiling points therefore they are regarded as volatile metals.

129. (A): $K_2Cr_2O_7$ is preferred to $Na_2Cr_2O_7$ for use in volumetric analysis as a primary standard.
(R): $Na_2Cr_2O_7$ is hydroscopic while $K_2Cr_2O_7$ is not.

130. (A): Tetrahedral complexes with chiral structure exhibit optical isomerism.
(R): They lack plane of symmetry.

131. (A): The $[Ni(en)_3]Cl_2$ has higher stability than $[Ni(NH_3)_6]Cl_2$.
(R): Ethylene diamine show chelation with Ni^{2+} ion.

132. (A): When $KMnO_4$ solution is added to hot oxalic acid solution, the decolourization is slow in the beginning but becomes spontaneous after sometime.
(R): Mn^{2+} acts as autocatalyst.

133. (A): $[Co(NH_3)_5Cl]Cl_2$ reacts with excess of $AgNO_3$ to form 2 moles of AgCl (white ppt.)
(R): $[Co(NH_3)_6]Cl_3$ gives 2 moles of Cl^- which react with $AgNO_3$ to forms 2 moles of AgCl.

134. (A): $KMnO_4$ is stored in dark bottles.
(R): On heating with alkalies, $KMnO_4$ is converted to manganate.

135. (A): $Ti(H_2O)_6^{3+}$ is a coloured ion.
(R): Ti shows +2, +3, +4 oxidation due to $4s^2$ and $3d^2$ electrons.

136. (A): $[Co(NO_2)_3(NH_3)_3]$ does not show optical isomerism.
(R): It has a plane of symmetry.

137. (A): CN^- and CO ligands usually form low spin complexes with metal ions.
(R): CN and CO cause the metal ions splitting of d-orbitals with greater Δ_0 values.

138. (A): There is an increase in oxidation potential of Fe^{2+} when it combined with a ligand CN^-.
(R): Ionic activity of Fe^{2+} increases during complex formation of $[Fe(CN)_6]^{3-}$.

139. (A): Thiocarbonyl is a neutral ligand.
(R): Thiocarbonyl has three donor atoms but behaves as a bidentate ligand.

140. (A): The ligands nitro and nitrito are called ambidentate ligands.
(R): These ligands give linkage isomers.

141. (A): Zeise's salt is a π-bonded organometallic compound.
(R): The oxidation number of platinum in Zeise's salt is +2.

142. (A): $[Fe(CN)_6]^{3-}$ is weakly paramagnetic while $[Fe(CN)_6]^{4-}$ is diamagnetic.

(R): $[Fe(CN)_6]^{3-}$ has +3 oxidation state while $[Fe(CN)_6]^{4-}$ has +2 oxidation state.

143. (A): Complex of calcium with EDTA is used to treat lead poisoning.

(R): EDTA is hexadentate ligand.

144. (A): $[Cu(NH_3)_4]^{2+}$ is coloured while $[Cu(CN)_4]^{3-}$ ion is colourless.

(R): $[Cu(NH_3)_4]^{2+}$ has dsp^2 hybridization.

145. (A): $[Cr(NH_3)_6]^{3+}$ is paramagnetic.

(R): $[Cr(NH_3)_6]^{3+}$ shows d^2sp^3 hybridization.

Matrix–Match Type Questions

	p	q	r	s
(A)	O	O	O	O
(B)	O	O	O	O
(C)	O	O	O	O
(D)	O	O	O	O

146. Match the following:

Column I	Column II
(a) $[Ag(CN)_2]^-$	(p) Square planar and 1.73 BM
(b) $[Cu(CN_4)]^{3-}$	(q) Linear and zero
(c) $[Cu(CN)_6]^{3-}$	(r) Octahedral and zero
(d) $[Cu(NH_3)_4]^{2+}$	(s) Tetrahedral and zero
(e) $[Fe(CN)_6]^{4-}$	(t) Octahedral and 1.73 BM

147. Match the following:

Column I	Column II
A. Highest density	(p) Os
B. Colourless salts	(q) Cr
C. Maximum magnetic moment	(r) Zn
D. Variable oxidation state	(s) Mn

148. Match the following:

Column I	Column II
A. Coloured ion	(p) Cu
B. μ = 1.73 B.M.	(q) Cu^{2+}
C. d^{10} configuration	(r) Fe^{2+}
D. More than 3 unpaired electrons	(s) Mn^{2+}

149. Match the following:

Column I	Column II
A. Mercury	(p) Liquid metal
B. Tungsten	(q) Nontransition metal
C. Astatine	(s) Transition metal
D. Iron	(t) Shows +2 and +3 oxidation states

150. Match the following:

Column I	Column II
A. Ti^{2+}	(p) Paramagnetic
B. Cu^{2+}	(q) Coloured
C. Co^{2+}	(r) One unpaired electron
D. Zn^{2+}	(s) Diamagnetic

151. Match the following:

Column I	Column II
A. $[Ni(CN)_4]^{2-}$	(p) +2 oxidation state
B. $[Cu(NH_3)_4]^{2+}$	(q) Paramagnetic
C. $[Fe(NH_3)_6]^{2+}$	(r) Diamagnetic
D. $[Fe(CN)_6]^{4-}$	(s) Outer orbital complex
	(t) Inner orbital complex

152. Match the following:

Column I	Column II
(a) $[Co(NH_3)_4Cl_2]$	(p) Optical isomerism
(b) cis-$[Co(en)_2Cl_2]$	(q) Ionization isomerism
(c) $[Co(en)_2(NO_2)Cl]SCN$	(r) Coordination isomerism
(d) $[Co(NH_3)_6][Cr(CN)_6]$	(s) Geometrical isomerism

153. Match the following:

Column I	Column II (Configuration)
A. $[Co(NH_3)_6]^{3+}$	(p) sp^3
B. $[Cu(NH_3)_4]^{2+}$	(q) dsp^2
C. $(MnCl_4)^{2-}$	(r) d^2sp^3
D. $(PtCl_4)^{2-}$	(s) Diamagnetic
	(t) Paramagnetic

154. Match the following:

Column I (Transition elements)	Column II (Properties)
(a) Cr	(p) Highest oxidation state
(b) Os	(q) Highest density
(c) Tc	(r) Maximum unpaired electrons
(d) Ru	(s) Radioactive nature

155. Match the following:

Column I (Transition metal ions)	Column II
(a) Cr^{3+}	(p) Violet in colour
(b) Mn^{3+}	(q) Green in colour
(c) V^{3+}	(r) 3 unpaired electrons
(d) Ni^{2+}	(s) 4 unpaired electrons
	(t) 2 unpaired electrons

The IIT–JEE Corner

156. Ammonium dichromate is used in some fireworks. The green coloured powder blown in the air is
(a) CrO_3 (b) Cr_2O_3
(c) Cr (d) $CrO(O_2)$
[*IIT 1997*]

157. Which of the following is an organometallic compound?
(a) lithium methoxide
(b) lithium acetate
(c) lithium dimethylamide
(d) methyl lithium
[*IIT 1997*]

158. Which of the following compounds is expected to be coloured?
(a) Ag_2SO_4 (b) CuF_2
(c) MgF_2 (d) CuCl
[*IIT 1997*]

159. Which of the following statement is correct with reference to the ferrous and ferric ions?
I. Fe^{3+} gives brown colour with potassium ferricyanide.
II. Fe^{2+} gives blue precipitate with potassium ferricyanide.
III. Fe^{3+} gives red colour with potassium thiocyanate.
IV. Fe^{2+} gives brown colour with ammonium thiocyanate.
(a) 1, 4 (b) 1, 2
(c) 2, 3 (d) all of these
[*IIT 1998*]

160. Which of the following statement is correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
I. A deep red vapour is evolved.
II. The vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4.
III. Chlorine gas is evolved.
IV. Chromyl chloride is formed.
(a) 1, 2, 4 (b) 1, 2 ,3
(c) 2, 3, 4 (d) All are correct
[*IIT 1998*]

161. In nitroprusside ion, the iron and NO exist as Fe^{II} and NO rather than Fe^{III} and NO. These forms can be differentiated by

(a) Estimating the concentration of iron.
(b) Measuring the concentration of CN^-.
(c) Measuring the solid state magnetic moment.
(d) Thermally decomposing the compound.

[*IIT 1998*]

162. The geometry of $Ni(CO)_4$ and $Ni(PPh_3)_2Cl_2$ are
(a) both square planar
(b) tetrahedral and square planar
(c) both tetrahedral
(d) square planar and tetrahedral

[*IIT 1999*]

163. In the standardization of $Na_2S_2O_3$ using $K_2Cr_2O_7$ by iodometry, the equivalent weight of $K_2Cr_2O_7$ is
(a) (molecular weight)/2
(b) (molecular weight)/6
(c) (molecular weight)/3
(d) same as molecular weight

[*IIT 2000*]

164. Amongst the following, identify the species with an atom in +6 oxidation state.
(a) MnO_4 (b) $Cr(CN)_6^{3-}$
(c) NiF_6^{2-} (d) CrO_2Cl_2

[*IIT 2000*]

165. The complex ion which has no 'd' electrons in the central metal atom is (At. no Cr = 24, Mn = 25, Fe = 26, Co = 27)
(a) (MnO_4) (b) $[Co(NH_3)_6]^{3+}$
(c) $[Fe(CN)_6]^{3-}$ (d) $[Cr(H_2O)_6]^{3+}$

[*IIT 2001*]

166. Anhydrous ferric chloride is prepared by
(a) Heating hydrated ferric chloride at a high temperature in a stream of air.
(b) Heating metallic iron in a stream of dry chlorine gas.
(c) Reaction of ferric oxide with hydrochloric acid.
(d) Reaction of metallic iron with hydrochloric acid.

[*IIT 2002*]

167. Identify the correct order of solubility of Na_2S, CuS and ZnS is aqueous medium.
(a) CuS > ZnS > Na_2S
(b) ZnS > Na_2S > CuS
(c) Na_2S > CuS > ZnS
(d) Na_2S > ZnS > CuS

[*IIT 2002*]

168. When MnO_2 is fused with KOH, coloured compound is formed, the product and its colour is
(a) K_2MnO_4, purple green
(b) $KMnO_4$, purple
(c) Mn_2O_3, brown
(d) Mn_3O_4, black

[*IIT 2003*]

169. In the process of extraction of gold, roasted gold
$Ore + CN^- + H_2O \xrightarrow{O_2}$
$(X) + Zn \rightarrow (Y) + Au$ (X) and (Y) are
(a) $(X) = [Au(CN)_2]^-$, $(Y) = [Zn(CN)_4]^{2-}$
(b) $(X) = [Au(CN)_4]^{3-}$, $(Y) = [Zn(CN)_4]^{2-}$
(c) $(X) = [Au(CN)_2]^-$, $(Y) = [Zn(CN)_6]^{-4}$
(d) $(X) = [Au(CN)_4]^-$, $(Y) = [Zn(CN)_4]^{-2}$

[*IIT 2003*]

170. Mixture of (X) = 0.02 moles of $[Co(NH_3)_5SO_4]Br$ and 0.02 mole of $[Co(NH_3)_5Br]SO_4$ was prepared in 2 litre of solution:
1 litre of mixture (X) + excess $AgNO_3 \rightarrow$ (Y)
1 litre of mixture (X) + excess $BaCl_2 \rightarrow$ (Z)
Number of moles of (Y) and (Z) are
(a) 0.01, 0.01 (b) 0.02, 0.01
(c) 0.01, 0.02 (d) 0.02, 0.02

[*IIT 2003*]

171. The pair of the compounds in which both the metals are in the highest possible oxidation state is
(a) $[Fe(N)_6]^{3-}$, $[Co(CN)_6]^{3-}$
(b) CrO_2Cl_2, MnO_4^-
(c) TiO_3, MnO_2
(d) $[Co(CN)_6]^{3-}$, MnO_2

[*IIT 2004*]

172. The product of oxidation of I^- with MnO_4^- in alkaline medium is
(a) IO_3^- (b) I_2
(c) IO (d) IO_4^-

[*IIT 2004*]

173. The species having tetrahedral shape is
(a) $(PdCl_4)^2$ (b) $[Ni(CN)_4]^{2-}$
(c) $[Pd(CN)_4]^2$ (d) $(NiCl_4)^{2-}$
[*IIT 2004*]

174. The spin magnetic moment of cobalt in the compound $Hg[Co(SCN)_4]$is
(a) $\sqrt{3}$ (b) $\sqrt{8}$
(c) $\sqrt{15}$ (d) $\sqrt{24}$
[*IIT 2004*]

175. The pair of which salts is expected to have same colour in their freshly prepared aqueous solutions?
(a) $VOCl_2$, $CuCl_2$ (b) $CuCl_2$, $FeCl_2$
(c) $FeCl_2$, $VOCl_2$ (d) $MnCl_2$, $FeCl_2$
[*IIT 2005*]

176. Isomerism shown by octahedral complex $Co(NH_3)_4(Br_2)Cl$ is
(a) geometrical and ionization
(b) optical and ionization
(c) geometrical and optical
(d) only geometrical
[*IIT 2005*]

177. A solution, when diluted with H_2O and boiled, gives a white precipitate. On addition of excess NH_4Cl/NH_4OH, the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in NH_4OH/NH_4Cl.
(a) $Zn(OH)_2$ (b) $Al(OH)_3$
(c) $Mg(OH)_2$ (d) $Ca(OH)_2$
[*IIT 2006*]

178. If the bond length of CO bond in carbon monoxide is 1.128 Å, then what is the value of CO bond length in $Fe(CO)_5$?
(a) 1.15Å (b) 1.128Å
(c) 1.72Å (d) 1.118Å
[*IIT 2006*]

179. $CuSO_4$ decolourise on addition of KCN, the product is.
(a) $[Cu(CN)_4]^2$
(b) Cu^{2+} gets reduced to form $[Cu(CN)_4]^3$
(c) $Cu(CN)_2$
(d) CuCN
[*IIT 2006*]

The coordination number of Ni^{2+} is 4.

$NiCl_2 + KCN$ (excess) ⟶ A (Cyano complex)

$NiCl_2$ + conc. HCl (excess) ➤ B (chloro complex)
[*IIT 2006*]

180. The IUPAC name of A and B are
(a) Potassium tetracyanonickelate (II), potassium tetrachloronickelate (II).
(b) Tetracyanopotassiumnickelate(II), tetrachloropotassiumnickelate(II).
(c) Tetracyanonickel(II), tetrachloronickel(II).
(d) Potassium tetracyanonickel(II), potassium tetrachloronickel(II).
[*IIT 2006*]

181. Predict the magnetic nature of A and B.
(a) Both are diamagnetic.
(b) A is diamagnetic and B is paramagnetic with one unpaired electron.
(c) A is diamagnetic and B is paramagnetic with two unpaired electrons.
(d) Both are paramagnetic.
[*IIT 2006*]

182. The hybridization of A and B are
(a) dsp^2, sp^3 (b) sp^3, sp^3
(c) dsp^2, dsp^2 (d) sp^3d^2, d^2sp^3
[*IIT 2006*]

183. A solution of a metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt(II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is
(a) Pb^2 (b) Hg^2
(c) Cu^{2-} (d) Co^{2+}
[*IIT 2007*]

184. Among the following metal carbonyls, the C–O bond order is lowest in
(a) $[Mn(CO)_6]^-$ (b) $[Fe(CO)_5]$
(c) $[Cr(CO)_6]$ (d) $[V(CO)_6]$
[*IIT 2007*]

185. Match the following:

List I (Complex)	List II (Properties)
(a) $[Co(NH_3)_4(H_2O)_2]Cl_2$	(p) Geometrical isomers
(b) $[Pt(NH_3)_2Cl_2]$	(q) Paramagnetic
(c) $[Co(H_2O)_5Cl]Cl$	(r) Diamagnetic
(d) $[Ni(H_2O)_6]Cl_2$	(s) Metal ion with +2 oxidation state

[*IIT 2007*]

186. (A): $[Fe(H_2O)_5NO]SO_4$ is paramagnetic.

(R): The Fe in $[Fe(H_2O)_5NO]SO_4$ has three unpaired electrons.

[*IIT 2008*]

187. (A): The geometrical isomer of the complex $[M(NH_3)_4Cl_2]$ are optically inactive.

(R): Both geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ posses axis of symmetry.

[*IIT 2008*]

188. The IUPAC name of $[Ni(NH_3)_4][NiCl_4]$ is

(a) Tetrachlornickel (II) tetraamminenickel (II)

(b) Tetrachlornickel (II) –tetraamminenickel (II)

(c) Tetrachlornickel (II) tetraamminenickel (II)

(d) Terrachlornickelare (II) –tetraamminenickelare (II)

[*IIT 2008*]

189. Both $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$ are diamagnetic. The hybridization of nickel in these complexes, respectively, are

(a) sp^3, sp^3 (b) sp^3, dsp^3

(c) dsp^3, sp^3 (d) dsp^3,dsp^3

[*IIT 2008*]

190. The spin only magnetic moment value (In Bohr-magnetic units of $Cr(CO)_6$ is

(a) 0 (b) 2.84

(c) 4.9 (d) 5.92

[*IIT 2009*]

191. The compounds that exhibit (s) geomarrical isomerism is

(a) $[Pt(en)Cl_2]$ (b) $[Pt(en)_2]Cl_2$

(c) $[Pt(en)Cl_2]Cl_2$ (d) $[Pt(NH_3)_2Cl_2]$

[*IIT 2009*]

0	1	2	3	4	5	6	7	8	9

192. The number of water molecules bounded to the metal centre in $CuSO_4 . 5H_2O$ is?

[*IIT 2009*]

193. The coordination number of AQ in the Crystalline state of $AlCl_3$ is?

[*IIT 2009*]

Note: In MCQ. 192, 193 one has to give answer in one integer number only by blackening the circle.

ANSWERS

Straight Objective Type Questions

1. (a)	2. (c)	3. (d)	4. (a)
5. (b)	6. (c)	7. (a)	8. (c)
9. (d)	10. (a)	11. (c)	12. (d)
13. (a)	14. (b)	15. (d)	16. (c)
17. (d)	18. (b)	19. (c)	20. (b)
21. (b)	22. (b)	23. (b)	24. (b)
25. (d)	26. (b)	27. (b)	28. (a)
29. (b)	30. (b)	31. (c)	32. (b)
33. (b)	34. (b)	35. (a)	36. (c)
37. (a)	38. (c)	39. (d)	40. (b)
41. (a)	42. (a)	43. (b)	44. (a)
45. (c)	46. (c)	47. (b)	48. (c)
49. (a)	50. (d)	51. (c)	52. (d)
53. (b)	54. (b)	55. (d)	

Brainteasers Objective Type Questions

56. (a) 57. (b) 58. (b) 59. (a)
60. (c) 61. (d) 62. (d) 63. (d)
64. (b) 65. (b) 66. (c) 67. (a)
68. (d) 69. (b) 70. (c) 71. (d)
72. (d) 73. (d) 74. (a) 75. (b)
76. (a) 77. (c) 78. (c) 79. (b)
80. (a) 81. (d) 82. (a) 83. (b)
84 (c) 85. (a)

Multiple Correct Answer Questions

86. (b), (c) 87. (c), (d)
88. (a), (c) 89. (b), (c), (d)
90. (b), (c) 91. (b), (d)
92. (a), (b), (d) 93. (a), (d)
94. (a), (c), (d) 95. (a), (c)
96. (c), (d) 97. (c), (d)
98. (b), (c) 99. (b), (d)
100. (c), (d) 101. (a), (b), (c)
102. (a), (c), (d) 103. (a), (b), (c)
104. (b), (c), (d) 105. (b), (c)
106. (a), (b), (c) 107. (b), (c), (d)
108. (a), (b), (d) 109. (b), (d)
110. (a), (b), (c)

Linked–Comprehension Type Questions

Comprehension–1

111. (b) 112. (a) 113. (d) 114. (c)

Comprehension–2

115. (c) 116. (d) 117. (b)

Comprehension–3

118. (a) 119. (c) 120. (b) 121. (c)

Comprehension–4

122. (d) 123. (a) 124. (b) 125. (a)

Assertion and Reasoning Questions

126. (b) 127. (b) 128. (b) 129. (a)
130. (a) 131. (a) 132. (a) 133. (a)
134. (b) 135. (b) 136. (a) 137. (a)
138. (c) 139. (c) 140. (a) 141. (b)
142. (b) 143. (b) 144. (b) 145. (b)

Matrix–Match Type Questions

146. (a)(-(q), (b)-(s), (c)-(r), (d)-(p), (e)-(r)
147. (a)-(p), (b)-(r), (c)-(q), (d)-(p), (q), (s)
148. (a)- (q), (r), (s), (b)- (q), (c)- (p), (d)- (r), (s)
149. (a)-(p), (s), (b)-(r), s), (c)-(q), (d)-(s), (r)
150. (a)-(p), (q), (r), (b)-(p), (q), (r), (c)-(p), (q), (d)-(s)
151. (a)-(p), (r), (r), (b)-(p), (q), (r), (c)-(p), (q), (s), (d)-(p), (r), (r)
152. (a)- (s), (b)- (p), (s), (c)-(q), (d)- (r)
153. (a)-(r), (r), (b)-(q), (r), (c)-(q), (r), (d)-(q), (s)
154. (a)-(r), (b)-(p), (q), (c)-(s), (d)-(p)
155. (a)-(p), (r), (b)-(p), (s), (c)-(q), (r), (d)-(q), (r)

The IIT–JEE Corner

156. (b) 157. (d) 158. (b) 159. (c)
160. (a) 161. (c) 162. (c) 163. (b)
164. (d) 165. (a) 166. (b) 167. (d)
168. (a) 169. (a) 170. (a) 171. (b)
172. (a) 173. (d) 174. (c) 175. (a)
176. (a) 177. (a) 178. (a) 179. (d)
180. (a) 181. (c) 182. (a) 183. (b)
184. (b)
185. (a)-(p), (q), (s), (b)-(p), (r), (s), (c)-(q), (s), (d)-(q), (s)
186. (a) 187. (a) 188. (c) 189. (d)
190. (a) 191. (c, d) 192. (4) 193. (6)

HINTS AND EXPLANATIONS

Straight Objective Type Questions

1. It is Cr. It has ground state electronic configuration: (Ar) $3d^5\ 4s^1$.

2. 3d orbital of Ni^{2+} ion. Atomic number of Ni = 28

3. Mg^{2+} has electronic configuration $1s^2\ 2s^2\ 2p^6$; no unpaired electrons.

   $_{22}Ti^{3}$ has electronic configuration (Ar) $3d^1$; one unpaired electron.

   V^{3+} has electronic configuration (Ar) $3d^2$; two unpaired electrons.

   Fe^{2} has electronic configuration (Ar) $3d^6$; four unpaired electrons.

5. EAN = At. no. of central atom − Oxidation state + 2 x (No. of ligands)

   EAN = 78 − 4 + 2 x 6 = 86.

7. $Cu[(CN)_4]^{2-}$ is a square planar complex. Cu in this complex is in +2 (d^9) oxidation state. There is only one unpaired electron (n = 1).

   Magnetic moment = $\sqrt{n(n+2)}$

   = $\sqrt{3}$ = 1.73 BM.

9. As there are two ligands, carbonyl and cyano hence these must be arranged alphabetically and iron has oxidation state +2.

11. Co^{2+} has configuration (Ar) $3d^7$ and has unpaired d-electrons. Hence, it is coloured.

12. The anion of the compound $K_3[Cr(C_2O_4)_3]$ is $[Cr(C_2O_4)_3]^{3-}$ or $[Cr(ox)_3]^{3-}$, here ox = $C_2O_4^{2-}$. Since such octahedral complexes have non-superimposable mirror images, so they show optical isomerism.

13. Here, complex compound $[Co(en)_2Cl_2]Br$ shows ionization isomerism, cis-isomerism, trans-isomerism and optical isomerism i.e., have four isomers.

14. Cation with higher oxidation state has a larger value of C.F.S.E and C.F.S.E decreases with the increase of the number of d-electrons.

17. $[Co(NO_2)(SCN)(en)_2]Br$ is an octahedral complex of the type $(Mxy(aa)_2)$. This type of octahedral complexes show geometrical isomerism.

18. $\overset{+2}{MnSO_4} \longrightarrow \overset{+4}{MnO_2}$

    There is transfer of two electrons

    $$\text{Eq. mass} = \frac{\text{Molecular mass}}{\text{Number of electrons transferred}}$$

    $$= \frac{\text{Molecular Mass}}{2}$$

19. $[Pt(NH_3)_4Cl]Cl_3 \leftrightarrow [Pt(NH_3)_4Cl]^{3+} + 3Cl^-$

20. Ag^+ has d^{10} configuration with no unpaired electron.

21. $CuSO_4.5H_2O$ has lowest number of unpaired d-electrons and lowest degree of paramagnetism.

22. Fe^{2+} has d^6 configuration and is expected to be most paramagnetic with four unpaired electrons.

23. (MA_5B) will not show geometrical isomerism.

24. The coordination entity has four chloride ions, each having a charge of −1, two neutral NH_3 groups, and platinum metal with an oxidation state of +4.

    Let X = Charge on coordination entity.

    4 (−1) x 2 (0) + 4 = X

    X = 0

    This shows that the coordination entity is not an ion and hence, the formula of the tetrachloro diamine platinum(IV) is $[Pt(NH_3)_2Cl_4]$.

25. The complex compound $(MCl_2Br_2)SO_4$ gives $(MCl_2BrSO_4)Br$ as ionization isomer and also cis-isomers, trans-isomers as geometrical isomers.

29. Complex having more rings in the structure will be more stable.

30. Al^{3+} cannot form an amine complex ion with excess of NH_3.

31. The complex having higher value of stability constant, is most stable one.

32. E.A.N. of central metal atom of $[Fe(CN_6)]^{3-}$ and $[Fe(CN)_6]^{4-}$ is not same.

33. As for optical activity at least one bidentate ligand should be there.

34. Cr^{3+}: (Ar) $3d^3$

Here d^2sp^3 hybridization is present and three unpaired d-electrons are there in it.

35. $[Pt(NH_3)_5Br]Br_3$ ionizes to $[Pt(NH_3)_5Br]^{3+}$ and 3Cl ions giving total four ions.

36. CN^- is the strongest ligand which gives highest value 10^{27} of the stability constant here.

42. Cobalt assumes sp^3d^2 hybrid state and number of unpaired electron in Co^{3+} is 4.

45. The correct IUPAC name of $NaMn(CO)_5$ is Sodium pentacarbonyl manganate (−I).

48. It is due to stable $3d^5$ configuration in Cr.

49. A d^5-ion has 5 unpaired electrons in weak field.

54. $5 \times 3 \times 2$ ways of accommodating the ligands.

Brainteasers Objective Type Questions

60. The conductivity of solution increases as the number of ions increases. Number of free ions in $Co(NH_3)_4Cl_3$, K_2PrCl_6, $Cr(NH_3)_6Cl_3$ and $Pr(NH_3)_6Cl_4$ is 1, 2, 3 and 4, respectively. Increasing conductivity order 1 < 2 < 3 < 4.

61. As $\mu = \sqrt{n(n+2)}$

Here, n = number of unpaired electrons

$6.92 = \sqrt{n(n+2)}$

On solving, we get

$n = 6$

Hence, the electronic configuration is $3d^5\ 4s^1$.

67. As mol of $AgCl = \frac{4.305}{143.5} = 0.03$

= mol of Cl given by the complex

Mol of the complex = $100 \times 10^{-3} \times 1 = 0.01$

$[Co(H_2O)_6]Cl_3 \rightarrow [Co(H_2O)_6]^{3+} + 3Cl^-$

0.01 mol 0.01 mol 0.03 mol

68. As $[Pt(py)(NH_3)(NO_2)ClBrI]$ exists in 15 different geometrical isomers, each of which would also have an optical isomer so it is incorrect.

69. Both bromine and mercury are liquids at room temperature. Ga is low melting solid. It is not liquid at room temperature.

72. As $[Ni(H_2O)_6](NO_3)_2$ has two unpaired electrons hence it will be paramagnetic while rest complexes are diamagnetic in nature as they do not have any unpaired electron.

73. $[Fe(CN)_6]^{4-}$

$x + (6) = -4$

$x = +2$

$[Co(NH_3)_3(NO_2)_3]$

$x + (3 \times 0) + (-3) = 0$

$x - 3 = 0$

$x = +3$

$[Ni(CO)_4]$

$x + (4 \times 0) = 0$

$x = 0$

77. $[Ni(CO)_4]$ and $[NiCN_4]^{2-}$ are diamagnetic (number unpaired electrons) while $[Ni(Cl)_4]^{2-}$ is paramagnetic with two unpaired electrons.

82. $[Fe(CN)_6]^{4-}$ is a inner orbital complex, i.e., has d^2sp^3 hybridization with no unpaired electron.

$(MnCl_4)^{4-}$ is a tetrahedral complex (sp^3) with 5 unpaired electrons.

$(CoCl_4)^{2-}$ is a tetrahedral hybridization (sp^3) with two unpaired electrons.

85. IV, I, II and III have 5, 4, 1 and 0 number of unpaired electrons, respectively.

Multiple Correct Answer Questions

89. Zieses does not give Cl ions in solution as these are the part of coordination sphere.

92. As EDTA form chelate complexes.

96. These complexes are coloured as they can allow d-d transitions.

105. As $[Cr(Ox)_3]^{3-}$ and $(NiCl_4)^{2-}$ cannot show geometrical isomerism.

Comprehension–1

114. As Mn^{2+} $(3d^5)$ is more stable than Mn^{3+} $(3d^4)$.

Comprehension–2

115. As pairs in option (a), (b), (d) show ionization, hydrate and coordination isomerism while option (b) shows geometrical isomerism.

Comprehension–4

124. As in it Cu^{2} ion is dsp^2 hybridized and contain one unpaired electron only hence its magnetic moment is 1.73.

125. As Mn^{2-}, Co^{2+} and Fe^{2+} have 5, 3 and zero unpaired electrons, respectively.

Assertion and Reasoning Questions

126. Some nonmetallic atoms (e.g., H, B, C, N etc.) are able to fit in the interstitial sites of transition metal lattice to form interstitial compounds.

127. Complex ion formation is a typical property of transition elements because they possess small size, high nuclear charge, and vacant d orbitals of equivalent energy.

134. $KMnO_4$ is stored in dark bottles because it is decomposed in light.

135. Ti^{3+} has $(Ar)3d^1$ configuration. Thus, d-d transition is possible and thereby it shows colour.

136. The essential requirement for a substance to be optically active is that the substance should not have a plane of symmetry in its structure.

139. Thiocarbonyl (CS) has two donor atoms but behaves as a monodentate ligand.

140. When a monodentate ligand has two possible donor atoms and attached in two ways to the central metal atoms are called ambidentate ligands.

141. In these complexes, the metal and ligand form a bond that involves the π-electrons of the ligand and so it is a π-bonded organometallic compound.

143. Ca-EDTA complex replaces Ca by lead making a soluble complex (Pb-EDTA) which is excreted through urine.

144. $[Cu(NH_3)_4]^{2+}$ is coloured due to the presence of unpaired electron whereas $[Cu(CN)_4]^{3-}$ has no unpaired electron .

The IIT–JEE Corner

156. $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

(green)

157. Methyl lithium (Li-CH_3) is organometallic compound due to the presence of M C bond.

158. CuF_2 is coloured due to the presence of one unpaired d-electron in Cu^{2} . It can undergo d–d transition.

159. $Fe^{2+} + [Fe(CN)_6]^{3-} \longrightarrow Fe^{3-} + [Fe(CN)_6]^{4-}$

$Fe^{3+} + [Fe(CN)_6]^{4-} + K^{+} \longrightarrow K[Fe(Fe(CN)_6)]$

Turnbull's blue

$Fe^{3+} + 3KCNS \longrightarrow 3K^{+} + Fe(SCN)_3$

Red ppt.

160. $4NaCl + K_2Cr_2O_7 + 6H_2SO_4$ (conc.) $\xrightarrow{\Delta}$

$2KHSO_4 + 4NaHSO_4 + 2CrO_2Cl_2 + 3H_2O$

(orange red)

Chromyl chloride

Chromyl chloride vapours when passed through NaOH solution gives a yellow solution of Na_2CrO_4.

161. This is clear from magnetic moment studies.

162. Both are tetrahedral with sp^3 hybridization.

163. In iodometry KI reacts with $K_2Cr_2O_7/H_2SO_4$ to give I_2 which is titrated against $Na_2S_2O_3$.

$K_2Cr_2O_7 + 7H_2SO_4 + 6KI \rightarrow$ 1 mole

$4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3I_2$

3 mole

$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$

2 mole 1 mole

1 mole $K_2Cr_2O_7 = 3$ mole I_2

1 mole $I_2 = 2$ mole $Na_2S_2O_3$

1 mole of $Na_2S_2O_3 = ½$ mole I_2

$= 1/6$ mole $K_2Cr_2O_7$

Thus, Eq. mass of $K_2Cr_2O_7 = \frac{\text{Mol. Mass}}{6}$

164. X 2 1

CrO_2Cl_2

$x - 4 - 2 = 0$

$x = +6$

165. MnO_4^- has no d electron

166. Anhydrous ferric chloride is obtained by passing dry chlorine gas over heated metallic iron.

$2Fe + 3Cl_2 \longrightarrow 2FeCl_3$

Choice (a) is not correct because it gives

Fe_2O_3.

$2[FeCl_3.6H_2O] \xrightarrow{\Delta} Fe_2O_3 + 6HCl + 9H_2O$

167. Solubility depends upon the lattice energy and hydration energy.

168. $MnO_2 + 2KOH + ½\ O_2 \rightarrow K_2MnO_4 + H_2O$

Purple green

169. $2Au + 4CN^- + H_2O + ½\ O_2 \rightarrow$

$2[Au(CN)_4]^- + 2OH^-$

(X)

$2[Au(CN)_2]^- + Zn \rightarrow [Zn(CN)_4]^{-2} + 2Au$

(Y)

170. Ag^+ (excess) + Br (0.01 mole) $\rightarrow$

AgBr (0.01 mole)

Ba^{+2} (excess) + SO_4^{-2} (0.01 mole) $\longrightarrow$

$BaSO_4$ (0.01 mole)

171. In (MnO_4) and CrO_2Cl_2 oxidation states of Mn and Cr are +7 and +6, respectively.

172. When I^- is oxidized by MnO_4^- in alkaline medium I^- converts into IO_3^-.

$2KMnO_4 + 2KOH \rightarrow 2K_2MnO_4 + H_2O + (O)$

$2K_2MnO_4 + 2H_2O \longrightarrow 2MnO_2 + 3KOH + 2(O)$

$2KMnO_4 + H_2O \xrightarrow{\text{alkaline}} 2MnO_2 + 2KOH + 3(O)$

$KI + 3\ (O) \rightarrow KIO_3$

$2KMnO_4 + KI + H_2O \longrightarrow 2KOH + 2MnO_2 + KIO_3$

174. Magnetic moment $(\mu) = \sqrt{n(n+2)}$

$= \sqrt{3(3+2)} = \sqrt{15}$ BM.

177. Due to formation of tetrammine zinc(II) complex; $Zn^{2+} + 4NH_4OH \rightarrow [Zn(NH_3)_4]^{2+}$

178. Due to synergic bond formation between metal and CO, the bond order of CO decreases.

179. $Cu^{+2} + 2CN^- \longrightarrow Cu(CN)_2$

$2Cu(CN)_2 \rightarrow 2CuCN + (CN)_2$

183. $Hg^{2+} + 2I^- \longrightarrow HgI_2$

$HgI_2 + 2I^- \rightarrow (HgI_4)^{2-}$

(excess) Soluble

$Hg^{2+} + Co(SCN)_2 \rightarrow Hg(SCN)_2$

Blue crystalline ppt.

184. $Fe^0 = 3d^6\ 4s^2$ in presence of CO effective configuration = $3d^8$.

Four lone pair for back bonding with CO.

186. The oxidation of Fe in the complex, $[Fe(H_2O)_5NO]SO_4$ is +1 (No has + 1 charge)

$Fe^1 = (Ar)\ 3d^6\ 4s^1$

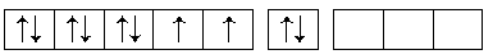
NO causes pairing of 4s electron inside.

Thus, the configuration is $3d^7$ and number of unpaired electrons = 3

187. The *cis* and *trans* both form of complex $[M(NH_3)_4Cl_2]$ are optically inactive due to plane of symmetry.

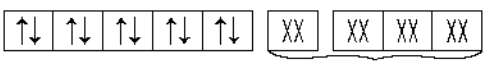
188. IUPAC name is tetrachlornickel (II) tetrachlornickelare (II)

189. In $[Ni(CO)_4]$, the oxidation state of Ni is zero (0). Ni (28) Ar $4s^2\ 4s^2\ 3d^8$



3d 4s 4p

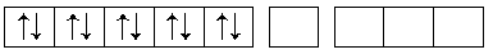
CO is a strong ligands, causes coupling, thus



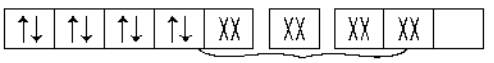
Hybridization sp^3

In $[Ni(CN)_4]^{2}$, the oxidation state is + 2

Ni^{2+} Ar $3d^8\ 4s^0$



CN is strong ligands causes coupling



Hybridization dsp^2

190. As in $Cr(CO)_6$ n = 0 so $\mu = 0$.

191. Here, compounds $[pt(en)_2Cl_2]Cl_2$ and $[pt(NH_3)_2Cl_2]$ can show geometrical isomerism.

192. The number of H_2O molecules directly bonded to the meral centre in $CuSO_4.5H_2O$ is 4 as it is represented as $[Cu(H_2O)_4SO_4]H_2O$.

193. The coordination number of Al in the crystalline state of $AlCl_3$ is (6) as if exists in C.C.P. lattice with 6 coordinate layer structures.

Solved Subjective Questions

1. Give reasons for the following:

(i) Silver is used in phorography.

Solution

AgBr is used in phorography as it converts into metallic silver when light is incident on it.

(ii) Most transition metal compounds are coloured.

Solution

Most of the transition meral compounds are coloured due to the presence of vacant d-orbirals or d-d electron transition mainly. Some compounds like $KMnO_4$, $K_2Cr_2O_7$, CrO_3 are coloured due to charge transfer.

(iii) The species $[CuCl_4]^{2}$ exist but $[CuI_4]^{2}$ does not.

[IIT 1992]

Solution

As I ion is a stronger reducing agent than Cl ion. It reduces Cu^{2+} ion Cu^{+2} ion. Hence, cupric iodide is converted into cuprous iodide. Hence, the species $[CuI_4]^{2}$ does not exist.

(iv) CrO_3 is an acid anhydride.

[IIT 1999]

Solution

It is an acid anhydride of chromic acid (H_2CrO_4)

$$CrO_3 + H_2O \rightarrow H_2CrO_4$$

2. Identify the complexes which are expected to be coloured. Explain?

(i) $[Ti(NO_3)_4]$

(ii) $[Cu(NCCH_3)_4]^{+}\ BF_4^{-}$

(iii) $[Cr(NH_3)_6]^{3+}\ 3Cl$

(iv) $K_3[VF_6]$

[IIT 1994]

Solution

Here, $K_3[VF_6]$ is coloured as in it V^{3+} ion has two unpaired electrons due to d-d electron transition it

is coloured, similarly $[Cr(NH_3)_6]^{3-}$ $3Cl^-$, in it Cr^{3+} has three unpaired electrons. In other two compounds d-d electron transition is not possible so they are colourless.

3. Write down the IUPAC names of the following compounds:

(i) $[Co(NH_3)_5 ONO] Cl_2$

[*IIT 1995*]

Solution

Penta ammine nitritocobalt (III) chloride

(ii) $K_3 [Cr(CN)_6]$

[*IIT 1995*]

Solution

Potassium hexa cyanochromate (III)

(iii) $[Cr(NH_3)_5 CO_3] Cl$.

[*IIT 1996*]

Solution

Penta ammine carbonato chromium (III) chloride

4. Write the IUPAC name of the compound $[Cr(NH_3)_5 (NCS)] [ZnCl_4]$. Is this compound coloured?

[*IIT 1997*]

Solution

Penta ammine isothiocyanato chromium (III) tetra chlorozincate.

The compound is coloured as Cr^3 has $3d^3$ configuration so it has 3 unpaired electrons in d-orbitals and it shows colour, through d-d e-transition.

5. (a) Write the equation for the reaction of Silver bromide with hypo in photographic process.

[*IIT 1997*]

Solution

$AgBr + 2Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaBr$

(Hypo) Soluble complex

(b) Compare qualitatively the first and second ionization potentials of Cu and Zn.

Solution

[*IIT 1996*]

Cu:$3d^{10}, 4s^1$

Zn:$3d^{10}, s^2$

IP_2 values of Cu shows a jump whereas no such jump is noticed is IP_2 values of Zn.

6. Write the formula of the following complexes:

(i) Penta amine chloro cobalt (III)

(ii) Lithium tetra hydro aluminate(III)

[*IIT 1997*]

Solution

(i) $[CoCl (NH_3)_5]^{2+}$

(ii) $LiAlH_4$

7. A, B and C are three complexes of chromium (III) with the empirical formula $H_{12}O_6Cl_3Cr$. All the three complexes have water and chloride ion as ligands. Complex A does not react with concentrated H_2SO_4, whereas complexes B and C lose 6.75% and 13.5% of their original mass, respectively on treatment with concentrated H_2SO_4. Identify A, B and C.

[*IIT 1999*]

Solution

Here A = $[Cr(H_2O)_6] Cl_3$

$[Cr(H_2O)_6]Cl_3 + H_2SO_4 \rightarrow$ No reaction

This reaction is not possible as in it all H_2O molecules are present in co-ordination sphere.

$B = [Cr(H_2O)_5Cl]Cl_2.H_2O$

$[Cr(H_2O)_5Cl]Cl_2.H_2O + H_2SO_4 \rightarrow$

One molecule of H_2O is removed since it is present outside the coordination sphere.

As Molecular weight of complex = 266.5

$$\text{So loss } \% = \frac{18}{266.5} \times 100 = 6.75\ \%$$

$C = [Cr(H_2O)_4Cl_2]Cl.2H_2O$

$[Cr(H_2O)_4Cl_2]Cl.2H_2O + H_2SO_4 \rightarrow$ Two molecules of H_2O are removed since these are present outside the coordination sphere.

So loss % = $2 \times \frac{18}{266.5} \times 100 = 13.50\ \%$

So compounds,

A = $[Cr(H_2O)_6]\ Cl_3$

B = $[Cr(H_2O)_5Cl]Cl_2.H_2O$

C = $[Cr(H_2O)_4Cl_2]Cl.2H_2O$

8. A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms (A) and (B). The form (A) reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, whereas (B) gives a pale yellow precipitate soluble in concentrated ammonia. Write the formula of (A) and (B) and state the hybridization of chromium in each. Calculate their magnetic moments (spin-only value).

[IIT 2001]

Solution

A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has two forms (A) and (B).

The form (A) reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia as the white precipitate is of AgCl and the reactions are as follows:

$[Cr(NH_3)_4\ ClBr]Cl + AgNO_3 \rightarrow$

$AgCl \downarrow + [Cr(NH_3)_4ClBr] + NO_3$

White ppt.

$AgCl + 2NH_4OH \rightarrow [Ag(NH_3)_2\ Cl] + 2H_2O$

Complex salt

Hence, the compound (A) must be $[Cr(NH_3)_4\ ClBr]Cl$

(B) gives a pale yellow precipitate soluble in concentrated ammonia.

$AgNO_3 + [Cr(NH_3)_4Cl_2]Br \rightarrow$

$AgBr \downarrow + [Cr(NH_3)_4Cl_2]^+ + NO_3^-$

Pale yellow ppt.

$AgBr + 2NH_4OH \rightarrow [Ag(NH_3)_2Br] + 2H_2O$

Hence, the compound (B) must be $[Cr(NH_3)_4Cl_2]Br$.

In both these complexes, chromium is in Cr^{3+} state hence the number of unpaired electrons = 3

Magnetic moment (μ) = $\sqrt{[n(n+2)]}$

= $\sqrt{[3(3+2)]}$

= $\sqrt{15}$ = 3.872 BM.

9. Write the IUPAC nomenclature of the given complex along with its hybridization and structure.

$K_2\ [Cr(NO)(NH_3)(CN)_4]$, μ = 1.73 BM.

[IIT 2003]

Solution

The IUPAC name of $K_2\ [Cr(NO)(NH_3)(CN)_4]$ is potassium amine tetracyano nitroso chromate (I).

Here, chromium atom is in Cr^+ state so its configuration is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$.

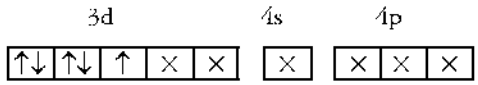
As here (μ) = $\sqrt{[n(n+2)]}$

1.73 = $\sqrt{[n(n+2)]}$

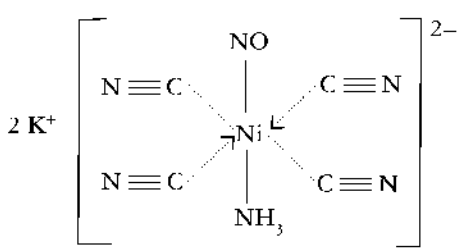
1.73 x 1.73 = $n^2 + 2n$

Here, n = 1

On the basis of value of n = 1, an unpaired electron present in chromium (I) of this complex ion, hence in excited state of Cr (I)



d^2sp^3-hybridization



10. $Fe^{3+} \xrightarrow[\text{excess}]{SCN^-}$ (A) $\xrightarrow[\text{excess}]{F^-}$ (B) colourless

blood red colouration

What are (A) and (B)? Give IUPAC name of (A). Find the spin only magnetic moment of (B).

[IIT 2003]

Solution

$Fe^{3+} \xrightarrow[\text{excess}]{SCN^-} \underset{(A)}{Fe(SCN)_3} \xrightarrow[\text{excess}]{F} \underset{(B)}{[FeF_6]^{3-}}$

IUPAC name of $Fe(SCN)_3$ is tri thiocyano iron(III)

Spin only magnetic moment of B = $\sqrt{[n(n+2)]}$

$\sqrt{[5(5+2)]}$ = 5.92 B.M.

Here, n = number of unpaired electrons which is 5 in Fe^3 .

11. Explain the following:

(i) Copper is regarded as transition metal though it has completely filled d orbital.

Solution

Although copper has $3d^{10}$ configuration it can lose one electron from this arrangement. Hence, Cu^{2-} ion has $3d^9$ configuration. So, according to the definition that transition metal cations have partially filled (n-1)d-subshell, copper can be regarded as a transition metal.

(iii) Why are Sm^{21}, Eu^{21} and Yb^2 good reducing agent?

Solution

The most stable oxidation state of lanthanides is +3. Hence, ions in +2 state act end to change to +3 oxidation state by loss of electron and act as reducing agents.

12. Explain the following:

(i) Out of cobalt and zinc salt, which is arracted in a magnetic field? [Roorkee 1995]

Solution

Out of cobalt and zinc salts, the cobalt salts are attracted in a magnetic field, as cobalt ion having unpaired electrons is characterized by a permanent magnetic moment. Zn^{2+} ion contained $3d^{10}$ configuration, i.e. has no unpaired electrons, so zinc salts are not attracted in magnetic field.

(ii) Vanadium salts/oxides (II, III, IV) are coloured and even V_2O_5 (d^{10}) is coloured.

Solution

V has configuration $3d^34s^2$. The oxidation state II upto IV correspond to d^3, d^2 and d^1 configuration. The d electrons permit d-d transition and hence give d-d spectra. Hence V-compound (II-IV) are all coloured. The V state means d^0 configuration so that d-dspectra are not feasible. Nevertheless V_2O_5 is coloured because of defects in the solid.

13. Write the balanced chemical equation for developing photographic film.

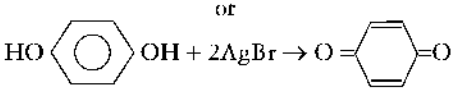
[*IIT 2001*]

Solution

AgBr is reduced to Ag by

$$\overset{II}{K_2Fe(C_2O_4)_2} + AgBr \rightarrow \overset{III}{KFe(C_2O_4)_2} + Ag + KBr$$

or



$$+2HBr + 2Ag$$

14. Explain why

(i) $[Ni(H_2O)_6]^{2+}$ is green but $Ni(en)_3^{2+}$ is blue.

Solution

Enthylenediamine (en) is a stronger ligand than H_2O so that it provides greater CFSE (crystal field stabilization energy) than the latter. Hence, absorption bands of $Ni(en)_3^{2+}$ are shifted toward shorter wavelength. Thus, while $Ni(H_2O)_6^{2-}$ is green, $Ni(en)_3^{2+}$ is blue.

(ii) Cu^+ is a d^{10} ion and colourless but Cu_2O yellow red depending in its mode of preparation and Cu_2S is black.

Solution

Cu is a d^{10} ion so that there is no scope for d-d transition and the ion is colourless. But Cu_2O and Cu_2S are coloured because of charge-transfer spectra arising from transfer of electrons from O^2 and S^2 to the vacant d-orbital of Cu .

15. Explain the following:

(i) Cu(I) is more stable than Cu (II) in non-aqueous solvent like CH_3CN etc.

Solution

Cu(I) halides are insoluble in H_2O but highly soluble in CH_3CN, Cu(I) is more stable than Cu (II) in CH_3CN. The tetrahedral $[Cu(CH_3CN)_4]$ can be isolated in salts such as $[Cu(CH_3CN)_4]^1ClO_4$

(ii) Why do Zr and Hf exhibit similar properties?

Solution

Due to lanthanide contraction, the atomic radii of both Zr and Hf are same. Hence, their properties are similar.

Questions For Self Assessment

16. Draw the structures of $[Co(NH_3)_6]^3$, $[Ni(CN)_4]^2$, and $[Ni(CO)_4]$. Write the hybridization of atomic orbitals of the transition metal in each case.

[IIT 2000]

17. Deduce the structure of $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ considering the hybridization of the metal ion. Calculate the magnetic moment (spin only) of the species.

[IIT 2002]

18. Explain the following :

(i) Ce^{+3} can be easily oxidized into Ce^{+4}

(ii) Cu^{+} undergoes disproportion into Cu^{+2} and Cu

[IIT 1991]

(iii) $CuSO_4$ is paramagnetic while $ZnSO_4$ is diamagnetic.

(iv) Mn^{+3} is less stable then Mn^{+2} and Mn^{+4} ions.

19. Explain the following:

(i) All the octahedral complexes of Ni^{+2} are outer orbital complexes.

(ii) NH_4^+ form can not act like a ligand.

(iii) Hexa synocomplexes of metals in their +2 oxidation state are mostly yellow. While corresponding hexa aqua compounds are blue or green.

20. Explain the following:

(i) $Co(NH_3)_6^{3+}$ is diamagnetic while CoF_6^{3-} is strongly paramagnetic.

(ii) Co(III) forms more complexes than any other element.

21. Explain the following:

(i) CrO_4^{-2} is a d^9–complex and yet it is coloured.

(ii) Mn^{2-} is more stable than other divalent ions and more difficult to oxidize than Cr^{2+} or Fe^{2+}.

(iii) $[\pi (C_5H_5)_2Fe]$ is a very stable compound whereas the corresponding Co (II) compound is unstable and prone to oxidation.

22. Explain the following

(i) While Co (III) forms a host of tetrahedral complex, Ni (II) forms only a limited number of them.

(ii) $Co(NH_3)_6Cl_2$ is strongly paramagnetic but $Co(NH_3)_6Cl_3$ is diamagnetic.

(iii) Both Ni (II) and Co (III) are d^6 systems but $K_2[NiF_6]$ is diamagnetic while $K_3[CoF_6]$ is paramagnetic.

23. Explan why a tetrahedral complex ion in the type $[Co\ cLR]^2$ shows intense blue colour while the octahedral Complex of the type $[Co\ (H_2O)_6]^{3+}$ is palepink.

24. Most of the trarsiotion mutal complexes are fuintly coloured how ever Mno_4^- is intense violet explain why is it so?

25. The base hydrolysis of $[Co\ (NH_3)_5\ Cl]^{2+}$ is quite fast and is dependent on OH^- ion concentration. Explain it?

METALLURGY

8

Chapter Contents

Ores and Minerals: Commonly occurring ores and minerals of Fe, Cu, Sn, Pb, Mg, Al, Zn and Ag Extractive Metallurgy: Chemical principles and reactions, carbon reduction method–Fe and Sn; Self reduction method–Cu and Pb, Electrolytic reduction method–Mg and Al, Cyanide process–Ag and Au and various levels of multiple-choice questions.

OCCURRENCE OF THE ELEMENTS

Elements in Atmosphere

The atmosphere mainly contains nitrogen (78.09%), oxygen (20.95%), and other gases (abour 1%).

Elements in Sea

Sea is the major source of elements like Br, I, Ni, Cu, Zn, Sn, Au etc.

Elements in Earth Crust (Lithosphere)

Elements occur in two states in the earth crust:

1. Free or Native State

Less reactive metals or noble metals with least electropositive nature are present in free or native state like copper, silver, gold, platinum etc.

Most Abundant Elements in the Lithosphere

	O	Si	Al	Fe	Ca	Na
%	48.60	26.30	7.73	4.75	3.45	2.74

2. Combined State (Minerals)

Reactive metals occur in combined state known as minerals. Those minerals from which metals can be profitably extracted are called ores. These metals are generally associated with rocky materials, sand, clay etc., known as *Gangue* or *Matrix* or *Nuggets*.

Classification of Ores of Elements

1. Free or Native Ores

Copper, silver, gold and platinum exist in free state.

2. Oxide Ores

Bauxite ($Al_2O_3.2H_2O$) of aluminium, haematite (Fe_2O_3) of iron, zincite (ZnO) of zinc, pyrolussite

(MnO_2) of manganese, tin stone (SnO_2) of tin are main oxide ores.

3. Carbonate Ores

Calcite ($CaCO_3$) of calcium, dolomite ($MgCO_3$. $CaCO_3$) of magnesium, malachite [$CuCO_3$. $Cu(OH)_2$] of copper, azurite [$2CuCO_3$. $Cu(OH)_2$] are main carbonate ores.

4. Sulphide Ores

Iron pyrites (FeS_2) of iron, galena (PbS) of lead, copper pyrites (CuS.FeS) of copper, cinnabar (HgS) of mercury, zinc blende (ZnS) of zinc are the main sulphide ores.

5. Sulphate Ores

Barytes ($BaSO_4$) of barium, anglesite ($PbSO_4$) of lead, zypsum ($CaSO_4$. $2H_2O$) of calcium are the main sulphate ores.

6. Halide Ores

Carnallite ($KCl.MgCl_2.6H_2O$) of porassium, rock salt (NaCl) of sodium, cryolite (Na_3AlF_6) of aluminium, horn silver (AgCl) of silver, fluorspar (CaF_2) of calcium are the main halide ores.

7. Silicate Ores

Silicon does not occur in free state but it is commonly found combined with oxygen known as silicates.

Many elements like, Fe, Mg, K, Na, Ca, Al are found combined with silicates.

Important Ores of Fe, Cu, Sn, Pb, Mg, Al, Zn and Ag

Ores of Fe

Magnetite: Fe_3O_4
Limonite: $3\ Fe_2O_3.3H_2O$
Iron pyrite: FeS_2
Haematite: Fe_2O_3
Copper pyrite: $CuFeS_2$
Spathic iron: $FeCO_3$

Ores of Cu

Cuprite or Ruby copper: Cu_2O
Copper glance: Cu_2S
Malachite: $Cu(OH)_2.CuCO_3$
Azurite: $Cu(OH)_2.2CuCO_3$

Ores of Sn

Cassiterite: SnO_2
Tin pyrite: $SnS_2.Cu_2S.FeS$

Ores of Pb

Galena: PbS
Anglesite: $PbSO_4$
Stolzite: $PbWO_4$
Cerrusite: $PbCO_3$
Wulfenite: $PbMnO_4$

Ores of Mg

Magnesite ($MgCO_3$)
Dolomite ($MgCO_3$. $CaCO_3$)
Kieserite ($MgSO_4$. H_2O)
Asbestos [$CaMg_3(SiO_3)_4$] etc.

Ores of Al

Alunite or alum stone
$K_2SO_4.Al_2(SO_4)_3.4Al(OH)_3$
Bauxite $Al_2O_3.2H_2O$
Corundum, Al_2O_3
Cyrolite, Na_3AlF_6
Diaspore, $Al_2O_3.H_2O$
Feldspar, $KAlSi_3O_8$
Kaolinite, $Al_2O_3.2SiO_2.2H_2O$
Mica, $K_2O.3Al_2O_3.6SiO_2.2H_2O$

Ores of Zn

Zinc blende or Sphalerite: ZnS
Zincite or Red zinc: ZnO
Calamine or Zinc spar: $ZnCO_3$
Franklinite: $ZnO.Fe_2O_3$
Willemite: Zn_2SiO_4

Ores of Ag

Argentite or silver glance: Ag_2S
Pyrargurite: $3\ Ag_2S.Sb_2S_3$

Proustite: $3Ag_2S.As_2O_3$
Horn silver: AgCl

Ores of Au

Sylvanite: $AgAuTeO_2$
Bismithaurite: $BiAu_2$
Calverite : $AuTe_2$

Extraction of Metals (Metallurgy)

The process of extraction of pure metal from its ore is called metallurgy.

It involves following processes:

1. Crushing of Ore

Big lumps of ore obtained from earth crust are crushed into smaller pieces with the help of jaw crushers and grinders. This process is known as crushing of ore.

2. Removal of Impurities from the crushed ore

It is known as concentration or dressing of ore.

(i) ***Hand Picking***: Selected pieces of ores are picked up.

(ii) ***Levigation—Washing***: The crushed ore is washed in a stream of water. The lighter impurities are swept away while heavier ore particles settle down. Iron ores and tin ores are concentrated by this method (i.e., mainly oxides and carbonates ores). It is also known as gravity separation or hydraulic washing.

(iii) ***Magnetic Separation***: The process is meant for the separation of magnetic ore from impurities. In this method the powdered ore is placed over leather belt which moves over two rollers one of which is magnetic.

When the crushed ore is passed over magnetic roller, magnetic ore particles are attracted by it and fall below it while impurities fall away from the magnetic roller.

For example:

Chromite $[Fe(CrO_2)]$ from siliceous gangue, rutile (TiO_2) from chloraparite and wolframite $(FeWO_4)$ from cassiterite are separated by this method.

(iv) ***Froth Floatation Process***: This process is commonly used for sulphide ores and is based upon different wetting characteristics of ore and gangue particles.

Here, finely powdered ore is mixed with water, pine oil (frother) and ethylxanthate or potassium ethyl xanthate (collector) in a big tank. The whole mixture is agitated with air. The ore particles wetted with oil come in froth, are taken off while impurities wetted with water settle at the bottom.

Here, the foaming agent is pine oil and froth stabilizers are cresol and anisole etc. Collectors are ethylxanthate and potassium ethylxanthate etc. The activator is $CuSO_4$ while the depressant is KCN.

(v) ***Chemical Method (Leaching)***: In this method the powdered ore is treated with a suitable chemical reagent which dissolves the ore while impurities remain insoluble in that reagent.

For example:

Bauxite is separated from Fe_2O_3, SiO_2, TiO_2 with the help of NaOH in which Al_2O_3 gets dissolved while rest are insoluble.

For example:

$$Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$$

$$NaSiO_2 + 2H_2O \longrightarrow Al(OH)_3 \downarrow + 2NaOH$$

$$2Al(OH)_3 \longrightarrow Al_2O_3 + 3H_2O$$

For example:

$$Ag_2S + 4NaCN \longrightarrow 2Na[Ag(CN)_2] + Na_2S$$

Electrostatic separation is used for the separation of PbS from zinc sulphide.

3. Calcination

Calcination is the process of heating the ores below their melting points in absence of air to remove volatile impurities like water, CO_2 and organic matter etc.

For example:

$$\underset{\text{Lime stone}}{CaCO_3} \longrightarrow \underset{\text{Calcium oxide}}{CaO} + CO_2 \uparrow$$

$$\underset{\text{Bauxite}}{Al_2O_3.2H_2O} \longrightarrow \underset{\text{Alumina}}{Al_2O_3} + 2H_2O \uparrow$$

During calcination the ore becomes porous, volatile impurities are removed and carbonate ore decomposes into oxides.

4. Roasting

Roasting is the process of heating the ore in excess of air in order to convert metals into their oxides and water insoluble sulphides into water soluble sulphates. It is also called de-electronation of ores.

For example:

$$\underset{\text{Pyrite}}{2FeS} + 3O_2 \xrightarrow{\Delta} 2FeO + 2SO_2$$

$$\underset{\text{Cinnabar}}{2HgS} + 3O_2 \xrightarrow{\Delta} 2HgO + 2SO_2$$

$$\underset{\text{Zinc sulphide}}{ZnS} + 2O_2 \xrightarrow{\Delta} ZnSO_4$$

$$\underset{\text{Lead sulphide}}{PbS} + 2O_2 \xrightarrow{\Delta} PbSO_4$$

During roasting volatile impurities are removed, S, As, Sb are removed as SO_2, As_2O_3, Sb_2O_3 respectively and sulphides ores are converted into their oxides.

- Both calcination and roasting occur in mainly reverberatory furnace.

5. Reduction of Metal Oxides into Free Metal

It is possible as follows:

(i) ***Reduction With Carbon (Smelting)***: Reduction of oxides of less electropositive metals like Pb, Fe, Zn, Sn and Cu is carried out by heating them with coal or coke in a blast furnace in presence of limited air at a temperature range nearly 200°C 1500°C.

For example:

$$\underset{\text{Zincoxide}}{ZnO} + \underset{\text{coke}}{C} \xrightarrow{\Delta} \underset{\text{zinc}}{Zn} + CO$$

$$\underset{\substack{\text{Lead mono}\\\text{oxide}}}{PbO} + \underset{\text{coke}}{C} \xrightarrow{\Delta} \underset{\text{lead}}{Pb} + CO$$

$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$$

Flux: It is the substance added to remove non-fusible impurities from roasted or calcined ore as fusible substance known as slag.

Flux + non-fusible impurity ⟶ Fusible slag

Acidic flux is used to remove basic impurities, example silica (SiO_2), Borax etc.

$$\underset{\substack{\text{Ferricoxide}\\\text{(basic impurity)}}}{Fe_2O_3} + \underset{\substack{\text{Silica}\\\text{(acid flux)}}}{3SiO_2} \longrightarrow \underset{\substack{\text{Ferric silicate}\\\text{(slag)}}}{Fe_2(SiO_3)_3}$$

Basic flux is used to remove acidic impurities e.g CaO, MgO, Fe_2O_3 etc.

$$\underset{\substack{\text{Phosphorus}\\\text{pentoxide}\\\text{(acidic impurity)}}}{P_2O_5} + \underset{\substack{\text{Calcium oxide}\\\text{(basic flux)}}}{3CaO} \longrightarrow \underset{\substack{\text{Calcium phosphate}\\\text{(slag)}}}{Ca_3(PO_4)_2}$$

In case of tin, the concentrated cassiterite ore is mixed with 1/5th of its weight of powdered anthracite (carbon) and some limestone for heating in a reverberatory furnace at 1473–1573K. Here the ore gets reduced to metallic tin and the impurity of silica can be removed as calcium silicate (slag).

$$SnO_2 + 2C \longrightarrow Sn + 2CO$$

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

- Tin obtained from here is 99.5% pure and known as black tin. Here, use of excess of lime is avoided otherwise calcium stannate will also be formed.

(ii) ***Reduction with Carbon Monoxide***: Carbon monoxide produced by heating coke in limited supply of oxygen, is also used as a reducing agent.

$$\underset{\text{Ferric oxide}}{Fe_2O_3} + 3CO \longrightarrow \underset{\text{Iron}}{2Fe} + 3CO_2$$

$$\underset{\text{Lead monoxide}}{PbO} + CO \longrightarrow \underset{\text{Lead}}{Pb} + CO_2$$

(iii) ***Reduction With Highly Electropositive Metal***: Some metal oxides which are not reduced by carbon, like chromium trioxide (Cr_2O_3), tiranium chloride ($TiCl_4$), manganese oxide (Mn_3O_4) are reduced by using highly electropositive metals like Na, K, Al, Mg etc. It is also called electrometallurgy.

Example 8.1

The anhydrous $MgCl_2$ is fused with NaCl and anhydrous calcium chloride in the ratio of 35 percent, 50 percent and 15 percent, respectively. This mixture is subjected to electrolysis at 973–1023 K in presence of an inert gas in an electrolytic cell. Here, NaCl and anhydrous $CaCl_2$ are used to lower the fusion temperature and to increase the conductivity of the fused mass.

After electrolysis magnesium gets discharged at cathode in molten state. It being lighter than the electrolyte floats over the surface of the fused mass so can be easily removed with perforatory ladles it is 99.9 percent pure. It can be further purified by remelting it by the flux of anhydrous $MgCl_2$ and NaCl.

$$MgCl_2 \rightleftharpoons Mg^{2+} + 2Cl^-$$

At cathode:

$$Mg^{2+} + 2e \longrightarrow Mg$$

At anode:

$$2Cl \longrightarrow Cl_2 + 2e$$

Example 8.2

Aluminium is obtained by the electrolysis of alumina (Al_2O_3) using cryolite.

The mechanism of electrolysis is given by two methods:

(i) $Na_3AlF_6 \rightleftharpoons 3NaF + AlF_3$

$AlF_3 \rightleftharpoons Al^{3+} + 3F$

At cathode: $Al^{3-} + 3e^- \longrightarrow Al$

At anode : $2F^- \longrightarrow F_2 + 2e^-$

The liberated fluorine reacts with alumina to give AlF_3 and O_2 as follows:

$$2Al_2O_3 + 6F_2 \longrightarrow 4AlF_3 + 3O_2$$

$$2C + O_2 \longrightarrow 2CO$$

$$2CO + O_2 \longrightarrow 2CO_2$$

(ii) $Al_2O_3 \rightleftharpoons \underset{\text{Cathode}}{Al^{3+}} + \underset{\text{Anode}}{AlO_3^{3-}}$

At cathode: $Al^{3} + 3e \longrightarrow Al$

At anode:

$$4AlO_3^{3-} \longrightarrow 2Al_2O_3 + 3O_2 + 12e^-$$

The net chemical reaction taking place during electrolysis is

$$2Al_2O_3 \longrightarrow 4Al + 3O_2$$

From this process, 99.8% pure aluminium is formed.

(v) ***Self Reduction***: Sulphide ores of less electro-positive metals like Hg, Cu, Pb and Sb undergo self reduction.

For example:

$$\underset{\text{Cinnabar}}{2HgS} + 3O_2 \longrightarrow \underset{\text{Mercury (II) oxide}}{2HgO} + 2SO_2$$

$$2HgO + \underset{\text{Mercury (II) sulphide}}{HgS} \longrightarrow 3Hg + SO_2$$

$$\underset{\text{Lead sulphide}}{2PbS} + 3O_2 \longrightarrow \underset{\text{Lead oxide}}{2PbO} + 2SO_2$$

$$2PbO + \underset{\text{Lead sulphide}}{PbS} \longrightarrow \underset{\text{Lead}}{3Pb} + SO_2$$

(vi) ***Reduction by Electrolysis***: Alkali and alkaline metals are extracted by this method. For example sodium metal is obtained by the electrolysis of fused sodium chloride.

(vii) ***Reduction by Water Gas***:

$$2NiO + CO + H_2 \longrightarrow 2Ni + CO_2 + H_2O$$

(viii) ***Alumino–Thermic Reduction***: It involves reduction of Fe_2O_3, Cr_2O_3, Mn_3O_4 with the help of thermite mixture (Fe_2O_3 + Al in 3:1 ratio).

$$Cr_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Cr + Heat$$

$$3Mn_3O_4 + 8Al \longrightarrow 4Al_2O_3 + 9Mn + Heat$$

$$Fe_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Fe$$

(ix) ***Hydrometallurgy***: It is based on the fact that more electropositive metals displace less electropositive metals from their salts. This process is also called wet process.

For example:

(i) **Extraction of Silver**: Silver can be extracted by Mac Arthur Forest Cyanide method. Here silver compounds are treated with a dilute solution of NaCN (0.03–0.8 percent) to get sodium silvercyanide in which zinc is added to precipitate silver.

$$4Ag + 8NaCN + O_2 + 2H_2O \longrightarrow 4Na[Ag(CN)_2] + 4NaOH$$

$$AgCl + 2NaCN \longrightarrow Na[Ag(CN)_2] + NaCl$$

$$2Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Ag \downarrow$$

For example:

(ii) Extraction of Gold: Gold can be extracted by Mac Arther Forest Cyanide method like silver. Here, gold slurry is treated with a dilute solution of NaCN (0.03–0.8 percent) to get sodium aurocyanide in which zinc is added to precipitate gold.

$$4Au + 8NaCN + O_2 + 2H_2O \longrightarrow 4Na[Au(CN)_2] + 4NaOH$$

$$AuCl + 2NaCN \rightarrow Na[Au(CN)_2] + NaCl$$

$$2Na[Au(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2Au \downarrow$$

6. Refining or Purification of Metals

The metals obtained after reduction may still contain some objectionable impurities which are removed by refining using the following methods:

(a) Physical Methods

(i) *Liquation*: This method is meant for refining of readily fusible metals like Sn, Pb and Bi having less fusible impurities.

For example:

Zn–Pb mixture is separated by placing crude Zn in the upper part of the sloping hearth maintained at the temperature just above the melting point of Zinc. The nonfusible mass of Pb remains behind the hearth while Zn melts and flows down.

(ii) *Distillation*: Volatile metals like zinc and mercury are refined by distillation. Here, crude metal is heated in retort as a result, pure metal gets distilled and impurities are left in the retort.

(iii) *Zone Refining (Fractional Crystallization)*: Highly pure metal is obtained by this method.

For example: Ge, Si, Ga etc.

It is based on the fact that metal and impurities have difference in the solubilities.

A circular heater fitted around a rod of impure metal is slowly moved along the length of the rod. At the heated zone, the rod melts and as the heater moves on, impurities pass into the molten zone while the pure metal crystallizes.

(iv) *Park's Distribution Process*: This process is used for extraction of Ag and Au from Pb.

The principle underlying this process is that out of two phase system of molten zinc and molten lead, silver and gold are more soluble in molten zinc. Zn–Ag alloy thus formed freezes out first and is removed and zinc is distilled off.

(b) Chemical Methods

These include cupellation, poling, Van-Arkel, electrorefining etc.

(i) *Cupellation*: By this method, the impurity of lead is removed from silver.

(ii) *Poling*: It is used for the metals having impurities of their oxides. For example: Cu_2O is removed from blister copper; SnO_2 from tin.

(iii) *Van-Arkel Method*: It is used mainly for the purification of Ti and Zr (space technology metals) using iodine. It is called vapour phase refining.

$$\underset{\text{Impure metal}}{Ti} + 2I_2 \xrightarrow{500\ K} TiI_4 \xrightarrow{1700\ K} \underset{\text{Pure metal}}{Ti} + 2I_2$$

(iv) *Mond Process*: It is used to purify Ni using carbon monoxide.

$$Ni + 4CO \xrightarrow{330-350\ K} Ni(CO)_4$$

$$Ni(CO)_4 \xrightarrow{450-470\ K} Ni + 4CO \uparrow$$

(v) *Electrorefining*: High electropositive metals like Cu, Ag, Au, Al, Zn, Sn, Pb are purified by this method. Here, pure metal gets deposited at cathode while impurities get deposited at anode after electrolysis.

Extracition of Fe, Cu, Sn, Pb, Mg, Al, Zn and Ag

METALLURGY OF IRON

Haematite Ore (Fe_2O_3)

↓

Concentration of ore by gravity process followed by electromagnetic separation

↓

Calcination and Roasting

Moisture, CO_2,SO_2, As_2O_3 are removed and
FeO is oxidized into ferric oxide (Fe_2O_3)

$$S + O_2 \longrightarrow SO_2$$

$$4As + 3O_2 \rightarrow 2As_2O_3$$

$$2Fe_2O_3.3H_2O \longrightarrow 2Fe_2O_3 + 3H_2O$$

$$FeCO_3 \longrightarrow FeO + CO_2$$

↓

Smelting

Smelting is made in a blast furnace (Ore + Coke + Limestone).
Here, the following reactions occur.

Zone of Reduction

$$Fe_2O_3 + 3CO \xrightarrow{400\ 700°C} 2Fe + 3CO_2$$

Spongy Iron

Zone of Slag Formation

$$CaCO_3 \xrightarrow{1000°C} CaO + CO_2$$

$$CaO + SiO_2 \rightarrow CaSiO_3 \text{ (slag)}$$

Zone of Combustion

$$2CO \xrightarrow{\text{Hot Iron}} CO_2 + C$$

$$SiO_2 + 2C \xrightarrow{1200°C} Si + 2CO$$

$$MnO_2 + 2C \rightarrow Mn + 2CO$$

$$P_4O_{10} + 10C \longrightarrow 4P + 10CO$$

Zone of Fusion

$$C + O_2 \xrightarrow{1600°C} CO_2$$

$$CO_2 + C \rightarrow 2CO$$

Spongy Iron + C, Mn, Si, etc.
(Impure iron)

↓

Pig Iron

↓ Remelted and cooled

Cast Iron (Fe = 93%, C = 5%, impurity = 2%)

- Cast iron on further purification changes into wrought iron (Purest form of iron having only 0.25 2% C) from which steel is obtained by adding a little amount of spiegeleisen.

METALLURGY OF COPPER

Copper pyrites ($CuFeS_2$)

↓

Crushed and sieve the ore

↓

Concentration by Froth Floatation Method

Powdered Ore + water + pine oil + air ➤ Sulphide ore in the froth

↓

Roasting in reverberatory furnace in presence of air

$S + O_2 \rightarrow SO_2$, $4As + 3O_2 \rightarrow 2As_2O_2$

$2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$

$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$

↓ Smelting

Silica, coke, roasted ore are smelted in a blast furnace in the presence of air to get matte.

$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$

$FeO + SiO_2 \longrightarrow FeSiO_2$ (Slag)

↓

Matte (Cu_2S FeS)

↓ Bessemerization

Bessemerization in bassemeter converter in presence of air

$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$

$FeO + SiO_2 \longrightarrow FeSiO_3$ (Slag)

$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$

$2Cu_2O + Cu_2S \xrightarrow{\text{Autoreduction}} 6Cu + SO_2$

Blister Copper (98% Cu + 2% Impurities)

↓

Electrolytic refining

Anode: Impure copper plates,**Cathode:** Pure copper plates

Electrolyte: $CuSO_4$ solution + H_2SO_4

Pure copper deposits at cathode (99.6–99.9% pure)

METALLURGY OF SILVER

Argentite (Ag_2S)

↓

Concentration by Froth Floatation Process

Powdered ore + water + pine oil + air ⟶ Froth carrying sulphide ore particles

↓

Cyanidation

Concentrated ore + aq. NaCN solution (0.4–0.6 %) + Air

$$Ag_2S + 4NaCN \longrightarrow 2NaAg(CN)_2 + Na_2S$$

Sodium argento cyanide

$$4Na_2S + SO_2 + 2H_2O \longrightarrow 2Na_2SO_4 + 4NaOH + 2S$$

↓ Filtration

Precipitation of Silver with Zinc

$$2NaAg(CN)_2 + Zn \longrightarrow 2Ag + Na_2Zn(CN)_4$$

Black ppt. of Ag + KNO_3

Compact mass (silver metal)

↓ Filtration

2.5 Electrolytic Refining

Anode: Impure silver, Cathode: Pure silver plate, **Electrolyte:** $AgNO_3$ solution + HNO_3

Pure silver deposits on cathode

METALLURGY OF TIN (SN)

Cassiterite (SnO_2)

↓

Concentration by levigation and Electromagnetic separation

Crushed powdered ore is washed with water to remove lighter siliceous impurities and the impurities of $FeWO_4$ and $MnWO_4$ are removed by electromagnetic separation

↓

Roasting in Reverberatory Furnace

Here, the impurities of S, As are removed as SO_2 and As_2O_3

$$S + O_2 \longrightarrow SO_2$$

$$4As + 3O_2 \longrightarrow 2As_2O_3$$

↓

Smelting

Roasted ore + carbon + lime stone are heated in reverberatory furnace at 1200–1300°C

$$CaCO_3 + SiO_2 \longrightarrow CaSiO_3 + CO_2$$

$$SnO_2 + 2C \longrightarrow Sn + 2CO$$

Black tin (99.5%)

Purification by Liquation and Poling or Electrolytic Method

↓

Pure Sn

METALLURGY OF LEAD

Galena (PbS)

↓

Concentration by Froth Floatation Process, Reduction

↓

Air Reduction Process

It is in reverberatory furnace and in presence of air.

$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$

$$PbS + 2O_2 \longrightarrow PbSO_4$$

↓

Mix it with more galena smelted in blast furnace ro in furnace.

↓

PbO and $PbSO_4$ are reduced by PbS into lead meral (crude lead).

$$PbS + 2PbO_2 \longrightarrow 3Pb + SO_2$$

$$PbS + PbSO_4 \longrightarrow 2Pb + 2SO_2$$

Purification: By Perk method and Cupellation

Carbon Reduction Process

The ore is mixed with lime and heated in sinterer

$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$

$$PbS \longrightarrow PbO$$

↓

Mixed it with C and CaO and and heat at higher temperature get crude lead.

METALLURGY OF ZINC

Zinc Blend (ZnS)

↓

Concentration by Froth Floatation Process

Powdered ore + water + pine oil ⟶ Froth carrying sulphide ore particles

↓

Roasting in Reverberatory Furnace

$$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$$

$$ZnS + 2O_2 \longrightarrow ZnSO_4,$$

$$2ZnSO_4 \longrightarrow 2ZnO + 2SO_2 + O_2$$

↓

Reduction ⟶ $ZnO + C \longrightarrow Zn + CO$

Purification by Distillation 950°C–1000°C or Electrolytic Refining

↓

Anode: Impure metal, **Cathode:** Pure Al sheet, **Electrolyte:** solution of Zn sulphate
Pure Zn deposits at cathode

METALLURGY OF ALUMINIUM

$Al_2O_3.2H_2O$ (Bauxite)

↓

Refining of Bauxite

(a) **Baeyer's Method,** Bauxite ore $\xrightarrow{\text{Roasted}}$ to convert FeO into Fe_2O_3

Roasted ore + NaOH $\xrightarrow[80\text{ atm.}]{150^\circ C}$ $NaAlO_2$ Sod. meta aluminate
Solution

$\xrightarrow[\text{In the presence of little } Al(OH)_3]{\text{Hydrolysis}}$ $Al(OH)_3 + NaOH$
ppt.

(b) **Hall's Method**

Bauxite ore + Na_2CO_3 $\xrightarrow{\text{Fused}}$ $NaAlO_2$

↓ Extract with water

Solution

↓ heat 50–60°C

CO_2 is circulated

↓

$Al(OH)_3 + Na_2CO_3$

(c) **Serpeck's Method**

Bauxite ore + Coke + N_2 $\xrightarrow{1800^\circ C}$ AlN $\xrightarrow{3H_2O}$ $Al(OH)_3 + NH_3$
ppt.

↓

Calcination

$2Al(OH)_3$ $\xrightarrow{1500^\circ C}$ $Al_2O_3 + 3H_2O$
Anhydrous alumina

↓

Electrolytic Reduction

Electrolyte Al_2O_3 dissolved in Na_3AlF_6 and CaF_2

Cathode ⟶ Carbon lining

$2Al_2O_3$ $\xrightarrow[950^\circ C]{\text{Electrolysis}}$ $2Al + 3O_2$

Anode ⟶ Graphite rods

$4Al + 3O_2$

99.8% pure

↓

Electrolytic Refining

(By Hoope's process)

↓

Al (99.98% Pure)

Unforgettable guidelines

1. NaCN acts as a depressant in preventing ZnS from forming the froth by formation a layer of zinc complex $Na_2[Zn(N)_4]$.
2. Carbon in blast furnase acts like a reducing agent and a spacer during smelting.
3. Metals are not common as nitrates as they are un-stable. [Except Nitrates of Na^+, K^-.)
4. Cu^1, Cu^2 , Ag^1 being soft acids can easily combine with soft base S^{-2} that is why sulphides of Cu, Ag are stable.

Straight Objective Type Questions (Single Choice)

1. Matte contains
   (a) Cu_2S, FeO and Silica
   (b) Cu_2S, FeS and Silica
   (c) Cu_2S, Cu_2O and Silica
   (d) Cu_2S, CuO and Silica

2. By which of the following reactions is blister copper obtained?
   (a) $Cu_2S + FeS \longrightarrow 2Cu + FeS_2$
   (b) $Cu^2 + Fe \rightarrow Fe^2 + Cu$
   (c) $Cu_2S + FeO \rightarrow 2Cu + FeO$
   (d) $Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$

3. Copper is extracted from its sulphide ore by
   (a) Self-reduction of the oxide and sulphide of copper
   (b) Electrolytic reduction
   (c) The carbon reduction process
   (d) Displacement reaction

4. During the extraction of copper from chalcopyrites, iron is removed as
   (a) $Fe_2(SiO_3)_3$ (b) $FeSiO_3$
   (c) $Fe_3(PO_4)_2$ (d) Fe_2O_3

5. Blister copper is
   (a) A mixture of impure copper and silver.
   (b) Electrolytically refined copper.
   (c) Present in the anode mud in an electrolytic process.
   (d) Copper containing 2 per cent impurity.

6. The reactivity of copper is low because of its
   (a) Low enthalpy of sublimation and low ionization energy.
   (b) Low enthalpy of sublimation and high ionization energy.
   (c) High enthalpy of sublimation and high ionization energy.
   (d) High enthalpy of sublimation and low ionization energy.

7. The melting point of copper is higher than that of zinc because
   (a) The d-electrons of copper are involved in metallic bonding.
   (b) The s- as well as d-electrons of copper are involved in metallic bonding.
   (c) Copper has a bcc structure.
   (d) The atomic volume of copper is higher.

8. The melting points of Cu, Ag and Au follow the order
   (a) Au > Ag > Cu (b) Cu > Ag > Au
   (c) Cu > Au > Ag (d) Ag > Au > Cu

9. The extraction of silver from its ore involving KCN, air and an active metal is known as
   (a) The Mc Arthur Forrest process
   (b) Parke's process
   (c) Parrinson's process
   (d) The amalgamation process

10. The metallic radius of gold is almost identical with that of silver because of
    (a) The high electropositive character of gold in comparison to silver.
    (b) The effect of lanthanide contraction in gold.
    (c) Transition metal contraction.
    (d) The same crystal structure of silver and gold.

11. Silver is extracted from Ag_2S by
    (a) Reducing it with zinc.
    (b) Fusing it with KCl, and electrolyzing the melt.
    (c) Roasting it and reducing the resultant product by smelting.
    (d) Treating it with sodium cyanide followed by zinc.

12. Silver is refined by
    (a) Poling
    (b) Cupellation
    (c) Liquation
    (d) Van Arkel method

13. The ores of Ag and Cu are concentrated using their solubility in
(a) KCN (b) HNO_3
(c) H_2SO_4 (d) HCl

14. Zinc is used to extract silver by
(a) Carbon monoxide reduction in Mond's process.
(b) Solvent extraction from molten iron in the LD process.
(c) Solvent extraction from molten lead in Parke's process.
(d) Solvent extraction from molten gold in the cyanide process.

15. In the electrolytic refining of silver, the anode mud obtained contains
(a) Au (b) Zn, Ag and Au
(c) Cu, Ag and Au (d) Zn, Cu, Ag and Au

16. During the extraction of Ag and Au, using an excess of KCN, soluble complexes are formed. These complexes, which have 10 d-electrons in Ag and Au, are
(a) $K_2[Ag(CN)_3]$ and $K_2[Au(CN)_3]$
(b) $K_4[Ag_6(CN)_{10}]$ and $K_4[Au_6(CN)_{10}]$
(c) $K_2[Ag(CN)_4]$ and $K_3[Ag(CN)_4]$
(d) $K[Ag(CN)_2]$ and $K[Au(CN)_2]$

17. Zinc is extracted from ZnS by the
(a) Roasting of ZnS followed by aluminium reduction at 1200°C in a muffle furnace.
(b) Roasting of ZnS followed by carbon monoxide reduction at 1200°C in a smelter.
(c) Calcination of ZnS followed by hydrogen reduction at 400°C.
(d) Calcination of ZnS followed by carbon dioxide reduction.

18. The purest zinc is made by
(a) Van Arkel method
(b) Poling
(c) Mond's process
(d) Zone refining

19. Granulated zinc is made by
(a) Pouring molten zinc into water
(b) Pouring molten zinc into molten nickel
(c) Displacing Zn from a $ZnSO_4$ solution
(d) Zone refining

20. Galvanized iron pipes are made by 'hot dipping'
(a) Iron in molten magnesium
(b) Iron in molten zinc
(c) Zinc in molten iron
(d) Iron in molten nickel

21. Zinc reacts with very dilute nitric acid acid to produce
(a) $ZnO_2^{2-} + Zn^{2+}$
(b) $NH_4NO_2 + ZnO_2^{2-}$
(c) $NH_4NO_3 + Zn^2$
(d) $N_2O + N_2 + Zn^{2+}$

22. The chemical processes in the production of iron from haematite ore involve
(a) oxidation
(b) reduction
(c) oxidation followed by reduction
(d) reduction followed by oxidation

23. In the manufacture of iron, the principal reaction in the zone of heat absorption of the blast furnace is
(a) $CO_2(g) + C(s) \rightarrow 2CO(g)$
(b) $2C(s) + O_2(g) \longrightarrow 2CO_2(g)$
(c) $3C(s) + O_2(g) \longrightarrow C_3O_2(g)$
(d) $2C(s) + O_2(g) \rightarrow 2CO(g)$

24. In the extraction of iron, Fe_2O_3 is reduced by
(a) Carbon dioxide
(b) Carbon monoxide
(c) Calcium carbonate
(d) Carbon

25. Among the following, the maximum amount of carbon is present in
(a) steel (b) stainless steel
(c) pig iron (d) wrought iron

26. When hard steel is heated to bright redness and then allowed to cool slowly, it gets softened. The process is called
(a) Hardening (b) Tempering
(c) Quenching (d) Annealing

27. In the manufacture of iron from haematite, limestone acts as a

(a) Flux (b) Slag
(c) Reducing agent (d) Matrix

28. Which of the following statement is correct?
(a) Pig iron is soft and brittle.
(b) Slag floats on molten iron, thus protecting iron from reduction.
(c) Molten slag and molten iron are drawn off through separate openings.
(d) Molten slag and molten iron are drawn off through the same openings.

29. Iron reacts with P_4O_{10} in the presence of O_2 at a high temperature to produce
(a) Fe_3P (b) $Fe_3(PO_3)_2$
(c) $FePO_4$ (d) $Fe_3(PO_4)_2$

30. Iron is not attacked by
(a) steam
(b) dilute H_2SO_4
(c) dilute NaOH
(d) concentrated NaOH

31. The alloy of nichrome contains
(a) Ni, Cr, Fe and C
(b) Ni, Cr, Fe and Zn
(c) Ni, Cr, Fe and Mn
(d) Ni, Cr, Fe and Zn

32. The molten metal, obtained after the treatment of copper pyrites in the blast furnace, has the composition of
(a) $CuS + Fe_2S_3$ (b) $Cu_2S + Fe_2S_3$
(c) Cu_2S (d) $Cu_2S + FeS$

33. The main function of roasting is
(a) reduction
(b) to remove volatile matter
(c) oxidation
(d) to make slag

34. Which of the following reactions is used to estimate copper volumetrically?
(a) $2Cu^{2+} + 4I^- \longrightarrow Cu_2I_2 + I_2$
(b) $2Cu^{2+} + 2CNS^- + SO_2 + 2H_2O \longrightarrow Cu_2(CNS)_2 + H_2SO_4 + 2H^+$
(c) $2Cu^{2+} + 4CN^- \longrightarrow Cu_2(CN)_2 + (CN)_2$
(d) $Cu^{2+} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{2+}$

35. Oxygen is absorbed by molten Ag, which is evolved on cooling and the silver particles are scattered; the phenomenon is known as
(a) Spitting of silver
(b) Silvering of mirror
(c) Hairing of silver
(d) Frosting of silver

36. From gold aurocyanide $Na[Au(CN)_2]$, gold can be precipitated by adding powder of
(a) Hg (b) Ag
(c) Zn (d) None of these

37. Which method is based on distribution law?
(a) Poling process
(b) Mond's process
(c) Cupellation process
(d) Parke's process

38. Wrought iron is
(a) Pig iron
(b) Pure iron with 0.1 to 0.2% C
(c) Impure sulphide ore of iron
(d) An alloy of steel

39. The axles are made by heating rods of iron embedded in charcoal powder. This process is known as
(a) Annealing (b) Tempering
(c) Case hardening (d) Nitriding

40. Which of the following metals is extracted on smelting of its ore in blast furnace?
(a) Sodium (b) Iron
(c) Magnesium (d) Potassium

41. Monel metal is an alloy of
(a) Cu, Zn (b) Cu, Sn, Zn
(c) Cu, Ni, Fe, Mn (d) Cu, Sn, P

42. Composition of azurite mineral is
(a) $CuCO_3.2Cu(OH)_2$
(b) $CuCO_3.CuO$
(c) $2CuCO_3.Cu(OH)_2$
(d) $Cu(HCO_3)_2.Cu(OH)_2$

43. Which one of the following reactions will occur on heating $AgNO_3$ above its melting point?

(a) $2AgNO_3 \rightarrow 2Ag + N_2 + 3O_2$
(b) $2AgNO_3 \longrightarrow Ag_2O + N_2O_3 + O_2$
(c) $2AgNO^3 \longrightarrow 2AgNO_2 + O_2$
(d) $2AgNO_3 \longrightarrow 2Ag + 2NO_2 + O_2$

44. Identify the reaction that does not take place in a blast furnace.
(a) $CaCO_3 \rightarrow CaO + CO_2$
(b) $2Fe_2O_3 \longrightarrow 4Fe + 3CO_2$
(c) $CO_2 + C \rightarrow 2CO$
(d) $CaO + SiO_2 \rightarrow CaSiO_3$

45. In the aluminothermite process, aluminium acts as:
(a) reducing agent (b) a solder
(c) an oxidizing agent (d) a flux

Brainteasers Objective Type Questions (Single Choice)

46. Leaching can be used for which metal/s?
(I) Pb (II) Al
(III) Ag (IV) Au
(a) (I), (II) and (III) (b) (II) and (III)
(c) (II), (III) and (IV) (d) (II) and (IV)

47. The oxide of a metal (R) can be reduced by the metal (P) while metal (R) can reduce the oxide of metal (Q). Here the decreasing order of the reactivity of metal (P), (Q) and (R) with oxygen is
(a) P > R > Q (b) Q > P > R
(c) P > Q > R (d) R > P > Q

48. Carbon cannot be used in the reduction of Al_2O_3 because
(I) It is nonmetal.
(II) The heat of formation of CO_2 is more than that of Al_2O_3.
(III) Pure carbon is not easily available.
(IV) The heat of formation of Al_2O_3 is too high.
(a) (I) and (II) (b) (I) and (III)
(c) (II) and (III) (d) (I), (III) and (IV)

49. The anode mud obtained during electro refining of copper contain metals is/are
(I) Ag (II) Fe
(III) Mg (IV) Au
(a) (I) and (II) (b) (II) and (III)
(c) (III) and (IV) (d) (I) and (IV)

50. $ACl_2 + BCl_2$ (excess) $\longrightarrow ACl_4 + B \downarrow$
$BO \xrightarrow[>400°C]{\text{Heat}} \frac{1}{2} O_2 + B$,
Ore of B would be
(a) Galena (b) Azurite
(c) Cinnabar (d) Siderite

51. In zone refining
(I) Metal rod is moved.
(II) The heater ring is moved.
(III) The impurities get accumulated at one end of the rod.
(IV) The impurities get accumulated at the centre of the rod.
(a) (I) and (II) (b) (I) and (III)
(c) (II) and (III) (d) (I), (III) and (IV)

52. MgO can be used as a refractory material because
(a) It is a good electrical insulator.
(b) It has high melting point.
(c) It is a good conductor of heat.
(d) All of these.

53. Which of the following reaction(s) occur during calcination?
(I) $CaCO_3 \longrightarrow CaO + CO_2$
(II) $4FeS_2 + 11O_2 \longrightarrow 2Fe_2O_3 + 8SO_2$
(III) $2Al(OH)_3 \longrightarrow Al_2O_3 + 3H_2O$
(IV) $CuS + CuSO_4 \longrightarrow 2Cu + 2SO_2$

(a) (I) and (II) (b) (I) and (III)
(c) (III) and (IV) (d) (I), (II) and (IV)

54. When ZnS and PbS minerals are present together, then NaCN is added to separate them in the froth floatation process as a depressant because
(a) $Pb(CN)_2$ is precipitated while there is no effect on ZnS.
(b) ZnS forms soluble complex $Na_2[Zn(CN)_4]$
(c) PbS forms soluble complex $Na_2[Pb(CN)_4]$
(d) Both (a) and (b)

55. In the commercial electrochemical process for aluminium extraction, the electrolyte used is
(a) A molten mixture of Al_2O_3 and Na_3AlF_6.
(b) A molten mixture of Al_2O_3 and $Al(OH)_3$.
(c) $Al(OH)_3$ in NaOH solution.
(d) An aqueous solution of $Al_2(SO_4)_3$.

56. Which of the following is not true for the calcinations of a metal ore?
(I) It makes the ore more porous.
(II) The ore is heated to a temperature when fusion just begins.
(III) Hydrated salts lose their water of crystallization.
(IV) Impurities of S, As and Sb are removed in the form of their volatile oxides.
(a) (I) and (II) (b) (II) and (III)
(c) (II) and (IV) (d) (II), (III) and (IV)

57. $Ag_2S + NaCN \rightarrow A$,
$A + Zn \rightarrow B + Ag$
A and B are, respectively
(a) $Na_3[Ag(CN)_3]$, $Na_2[Zn(CN)_4]$
(b) $Na[Ag(CN)_2]$, $Na[Zn(CN)_3]$
(c) $Na_2[Zn(CN)_4]$, $Na_2[Ag(CN)_4]$
(d) $Na[Ag(CN)_2]$, $Na_2[Zn(CN)_4]$

58. Which of the following reaction in the blast furnace are not endothermic?
(I). $C(s) + O_2(g) \rightleftharpoons CO_2(g)$
(II). $CO_2(g) + C(s) \rightleftharpoons 2CO_2(g)$
(III). $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
(IV). $Fe_2O_3(s) + 3CO(g) \rightleftharpoons 2Fe(l) + 3CO_2(g)$
(a) (I) and (II) (b) (I) and (III)
(c) (I) and (IV) (d) (II) and (IV)

59. Which of the following is/are correct statement/s for the Hoope's process for the refining of aluminium?
(I) It is an electrolysis process.
(II) It consists of three layers in which the bottom layer is molten, impure aluminium.
(III) By using fractional distillation to remove zinc as an impurity.
(IV) It involves the electronation of aluminium ion at cathode.
(a) (I) and (II) (b) (I), (II) and (IV)
(c) (II) and (IV) (d) (I), (III) and (IV)

60. Sulphide ores of metals are usually concentration by froth flotation process. Which of the following sulphide ores offers an exception and is concentration by chemical leaching?
(a) Argentite (b) Galena
(c) Copper pyrite (d) Sphalerite

61. Which of the following pair consists of ores of the same metal?
(a) bauxite, limonite
(b) haematite, siderite
(c) galena, cerussite
(d) both (b) and (c)

62. Select the correct statement/s regarding roasting process:
(I) It is the process of heating ore in air to obtain the oxide.
(II) It is an exothermic process.
(III) It is used for hydrated oxide and oxysalt ore.
(IV) It is used after the concentration of ore.
(a) (I) and (II) (b) (I) and (III)
(c) (I), (II) and (IV) (d) (I), (III) and (IV)

63. In the extraction of gold from gold bearing rocks and minerals by the cyanide leaching process, the gold is brought into solution as an anionic complex of which composition?
(a) $Au(CN)_6^{3-}$ (b) $Au(CN)_4^-$
(c) $Au(CN)_2^-$ (d) $Au(CN)_4^{3-}$

64. The furnace lining in steel manufacture consists of
(I) CaO (II) SiO_2
(III) MgO (IV) $CaCO_3$

(a) (II) and (IV) (b) (I) and (III)
(c) (III) and (IV) (d) (II), (III) and (IV)

65. Why is CaF_2 added to the molten mixture of alumina and cryolite in the extraction of aluminium?
(a) It forms slag with gangue.
(b) To remove impurities.
(c) To increase the temperature of mixture.
(d) To increase conductivity.

66. Which of the following best explains why the free energy of formation of Fe_2O_3 become less negative as the temperature increases?
(a) Entropy drops as a result of the consumption of oxygen.
(b) The free energy of formation is independent of the absolute temperature.
(c) As entropy increases, the free energy of formation increases.
(d) At low temperature, the free energy of formation become less dependent on the enthalpy of formation.

67. Elemental carbon is most likely to reduce Fe_2O_3 in which of the following temperature ranges?
(a) Below 1000K
(b) Above 1000K
(c) 500–1000K
(d) 500 1500K

68. Zn gives H_2 gas with H_2SO_4 and HCl but not with HNO_3 because
(a) in electrochemical series Zn is above hydrogen .
(b) NO_3 ion is reduced in preference to hydronium ion.
(c) Zn acts as an oxidizing agent when reacts with HNO_3.
(d) HNO_3 is weaker acid than H_2SO_4 and HCl .

69. Consider the following reactions at 1000°C.
(I) $Zn(s) + \frac{1}{2} O_2(g) \rightarrow ZnO(s)$; $\Delta G^\circ = -360$ kJ mol^{-1}
(II) $C(s) + \frac{1}{2} O_2(g) \longrightarrow CO(g)$; $\Delta G^\circ = -460$ kJ mol^{-1}
Choose the correct statement at 1000°C.
(a) Both statements (I) and (II) are true
(b) Zinc can be reduced by C
(c) Both statements (a) and (b) are false.
(d) Zinc can be oxidized by CO.

70. Froth floatation process may be used to increase the concentration of the mineral in
(a) chalcopyrites (b) bauxite
(c) haematite (d) calamine

71. Copper is extracted from copper pyrites ore by heating in a blast furnace. The method is based on the principle that
(a) copper has less affinity for oxygen that sulphur at high temperature.
(b) sulphur has less affinity for oxygen at high temperature.
(c) iron has less affinity for oxygen than sulphur at high temperature.
(d) copper has more affinity for oxygen than sulphur at high temperature.

72. For which of the following metals, hydrometallurgical extractive method may be employed?
(a) copper (b) iron
(c) chromium (d) tin

73. $FeCr_2O_4$ (Chromite) is converted to Cr by following steps:

$$\text{Chromite} \xrightarrow{\text{I}} NaCrO_4 \xrightarrow{\text{II}} Cr_2O_3 \xrightarrow{\text{III}} Cr$$

Reagents in I, II and III step might be

	Step I	Step II	Step III
(a)	NaOH/air, Δ	C	C
(b)	Na_2CO_3/air, Δ	C, Δ	Al, Δ
(c)	Conc.H_2SO_4, Δ	NH_4Cl, Δ	C, Δ
(d)	NaOH/air, Δ	C, Δ	C, Δ

74. Iron is rendered passive by treatment with concentrated
(a) HCl (b) HNO_3
(c) H_3PO_4 (d) H_2SO_4

75. In the metallurgy of iron, when limestone is added to the blast furnace, the calcium ion ends up in
(a) gangue (b) calcium carbonate
(c) slag (d) metallic calcium

76. Zinc copper couple that can be used as a reducing agent is obtained by
(a) zinc coated with copper.
(b) zinc and copper wires welded together.

(c) mixing zinc dust and copper gauze.
(d) copper coated with zinc.

77. Name the metal M which is extracted on the basis of following reactions:

$$4M + 8CN^- + 2H_2O + O_2 \longrightarrow 4[M(CN)_2]^- + 4OH^-$$

$$2[M(CN)_2]^- + Zn \longrightarrow [Zn(CN)_4]^{2-} + 2M$$

(a) Ag (b) Cu
(c) Hg (d) Ni

Multiple Correct Answer Questions (More Than One Choice)

78. Which of the following statement is/are not correct?
(a) All minerals are ores.
(b) Mercury is transported in containers made of iron.
(c) Calcination is the process of heating the ore strongly in the presence of air.
(d) Cassiterite is an ore of iron.

79. Which of the following are correctly matched?
(a) Malachite → $CuCO_3.Cu(OH)_2$
(b) Chalcopyrites → $CuFeS_2$
(c) Turquoise → $(CuAl_6PO_4)_4(OH)_8.4H_2O$
(d) Peacock ore → Cu_4FeS_2

80. Which of the following are incorrectly matched?
(a) Nickel silver → Cu, Ni, Zn
(b) Brass → Cu, Sn
(c) Fool's fold → CuS_2
(d) Phosphor bronze → Cu, Sn, P

81. Bessemerization is carried out for
(a) Magnesium (b) Iron
(c) Silver (d) Copper

82. Which of the following statement is/are correct with the extraction of silver?
(a) Zinc is used to extract silver by solvent extraction from molten lead in Parke's process.
(b) Silver is obtained as a by-product in the extraction of copper, lead and zinc.
(c) Gold can also be used to extract silver from Sodium argentocyanide in cyanidation.
(d) Silver is obtained from the anode slime formed in the electrolytic refining of copper and zinc.

83. In the extraction of Ag and Au using a KCN solution, cyanide ions react with metal ions as
(a) An oxidizing agent
(b) A Lewis acid
(c) A reducing agent
(d) A complexing agent

84. Which of the following alloys contain Cu and Zn?
(a) Bronze (b) Brass
(c) Gun metal (d) Type metal

85. Which of the following are correctly matched?
(a) Zone refining: Ultra pure Ge
(b) Leaching: Extraction of Au, Ag
(c) Cyanide process: Extraction of Pb
(d) Electrolyte reduction: Extraction of Al

86. Which of the following statements are correct?
(a) Quenched steel is produced by heating steel to redness and allowing it to cool slowly.
(b) Stainless steel is produced by heating wrought iron in molten chromium.
(c) The process of producing a hard coating of iron nitride on the surface of steel is called nitriding.
(d) The process of producing a thin coating of hardened steel on the surface of mild steel is called hardening.

87. Which of the following are correct processes?

(a) $Fe + Al_2O_3 \rightarrow 2Al + Fe_2O_3$

(b) $ZnO + C \longrightarrow Zn + CO$

(c) $Cr_2O_3 + 2Al \longrightarrow 2Cr + Al_2O_3$

(d) $2Ca_3(PO_4)_2 + 6SiO_2 + 10C \longrightarrow 6CaSiO_3 + P_4 + 10CO$

88. Which of the following statement is/are correct?

(a) Steel cannot be permanently magnetized.

(b) Cast iron cannot be permanently magnetized.

(c) Spiegeleisen is an alloy of iron, zinc and antimony.

(d) Steel can be permanently magnetized.

89. Which of the following metals have both valence shell and penultimate shell partially filled?

(a) Cu (b) Zn

(c) Cr (d) Mn

90. Find out the incorrect statement among the following.

(a) Ferric alum is commonly known as Mohr's salt.

(b) Cast iron cannot be welded.

(c) Heating steel to a high temperature followed by quick quenching in water is called annealing.

(d) Alnico is used for making utensils and automobile parts.

91. Which of the following are the advantages of using oxygen in place of air in steel industry?

(a) It gives more pure product.

(b) The surface is free from nitrides.

(c) It makes the procedure faster i.e., more production.

(d) Larger quantities can also be handled.

92. Which of the following statements is/are correct for electrolytic refining?

(a) The electric current used is AC.

(b) On account of electrolysis, the metal is indirectly transferred from anode to cathode.

(c) The impurity metal is made the anode while a thin sheet of pure metal is used as the cathode.

(d) The electrolyte used is a covalent compound of metal.

93. Which of the following is correctly matched? Reaction taking place in blast furnace/temperature range.

(a) $C + H_2O \rightarrow CO + H_2O > 1150°C$

(b) $Fe_2O_3 + CO \longrightarrow CO_2 + Fe > 900°C$

(a) $CaO + SiO_2 \longrightarrow CaSiO_3 > 1200°C$

(d) $C + O_2 \rightarrow CO_2 > 650°C$

Linked–Comprehension Type Questions

Comprehension–I

(I) An ore (a) on roasting with sodium carbonate and lime in the presence of air gives two compounds, (b) and (c).

(II) The solution of (b) in conc. HCl on reaction with potassium ferrocyanide gives a blue colour or precipitate of compound (d).

(III) The aqueous solution of (c) on treatment with conc. H_2SO_4 gives a yellow coloured compound (E).

(IV) Compound (E) when treated with KCl gives an orange-red compound (F) which is used as an oxidizing reagent.

94. Here the ore (a) and compound (c) are, respectively:

(a) $CuFeS_2$ and Cu_2O

(b) $FeO.Cr_2O_3$ and Na_2CrO_4

(c) $Al_2O_3.2H_2O$ and Al_2O_3

(d) PbS and $PbSO_4$

95. In this sequence of reactions the compound (b) and (d) are, respectively:

(a) Fe_2O_3 and Ferrithiocyanate

(b) FeO and Prussian blue

(c) Fe_2O_3 and Prussian blue

(d) FeO and Ferrithiocyanate

96. Here the yellow and orange precipitates are, respectively

(a) $Na_2Cr_2O_7$, $K_2Cr_2O_7$

(b) $K_2Cr_2O_7$, $Na_2Cr_2O_7$

(c) Na_2CrO_4, K_2CrO_4

(d) $Na_2Cr_2O_7$, K_2CrO_4

Comprehension–2

Some reactions of two ores, A_1 and A_2 of the metal M are given below:

$$[A_1] \xrightarrow{\text{Calcination}} [C] \downarrow + CO_2 + H_2O$$
Black

$$KI / HCl \qquad [D] \downarrow + I_2$$

$$[A_2] \xrightarrow{\text{Roasting}} [G] \downarrow + M$$

$$[G] + K_2Cr_2O_7 \xrightarrow{H^+} \text{Green solution}$$

97. Identify (A_1) and (A_2).

(a) $(A_1) = CuCO_3.Cu(OH)_2$, $(A_2) = Cu_2S$

(b) $(A_1) = Cu_2S$, $(A_2) = CuCO_3.Cu(OH)_2$

(c) $(A_1) = CuFeS_2$, $(A_2) = Cu_2S$

(d) None of these

98. In this sequence of reactions the black residue (c) and precipitate (d) are, respectively

(a) HgS and HgO (b) CuI and CuO

(c) CuO and CuI (d) PbS and PbO

99. Here the gas (G) and green solution are of

(a) SO_3, $Cr_2(SO_4)_3$

(b) SO_2, $Cr_2(SO_4)_3$

(c) NH_3, $(NH_4)_2CrO_4$

(d) Cl_2, $CrCl_3$

100. Here metal (M) is ... and it is obtained by ... respectively

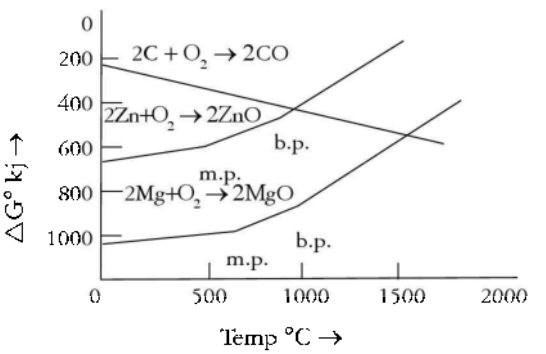
(a) Cu and auto reduction

(b) Lead and air reduction

(c) Iron and carbon reduction

(d) Silver and cyanidation

Comprehension–3



Ellingham diagram reveals that the curves involving the formation of MgO, ZnO and CO can be given as above. On the basis of these curves answer the following questions.

101. ΔG° vs T plot in the Ellingham diagram slopes downward for the reaction

(a) $2Ag + \frac{1}{2} O_2 \rightarrow Ag_2O$

(b) $Mg + \frac{1}{2} O_2 \longrightarrow MgO$

(c) $CO + \frac{1}{2} O_2 \longrightarrow CO_2$

(d) $C + \frac{1}{2} O_2 \rightarrow CO$

102. At what temperature, zinc and carbon have equal affinity for oxygen?

(a) 500°C (b) 1000°C

(c) 1200°C (d) 1500°C

103. At this temperature ΔG° of this reaction is

$ZnO + C \longrightarrow Zn + CO$

(a) positive (b) zero

(c) negative (d) cannot be predicted

Assertion and Reasoning Questions

104. (A): Pine oil act as frothing agent in froth floatation.
(R): Sulphide ores are concentrated by froth floaration method.

105. (A): Wolframite impurity is separated from SnO_2 by magnetic separation
(R): Tin stone is ferromagnetic, therefore attracted by magnet.

106. (A): Nitriding is process of heating steel in presence of N_2 to form iron nitrides.
(R): The surface of steel becomes hard after nitriding process.

107. (A): Zinc and not copper is used in the recovery of silver from the complex $[Ag(CN)_2]^-$.
(R): Zinc is powerful oxidizing agent than copper.

108. (A): Chalcosite is roasted and not calcined during recovery of copper.
(R): Copper pyrite is not ore of copper.

109. (A): Zinc is used in the galvanization of iron.
(R): Its coating on iron articles increases their life by protecting them from rusting.

110. (A): Hydrometallurgy is used for extraction of Ag and Au.
(R): Pyrometallurgy is another name of hydrometallurgy.

111. (A): NaCN acts as a depressant in preventing ZnS from forming the froth.
(R): NaCN combines with ZnS to form a complex $Na_2[Zn(CN)_4]$ on the surface of ZnS and thus prevents it from formation of froth.

112. (A): Levigation is used for the separation of oxides ores from impurities.
(R): Ore particles are removed by washing in a current of water.

113. (A): Titanium is purified by Van Arkel method.
(R): Ti reacts with I_2 to form TiI_4 which decomposes at 1700 K to give pure Ti.

114. (A): Leaching is a process of concentration.
(R): Leaching involves treatment of the ore with a suitable reagent so as to make it soluble while impurities remains insoluble.

115. (A): CuO can be reduced by C, H_2 as well as CO.
(R): CuO is basic oxide.

116. (A): In froth floaration process sodium ethyl xanthate can be used as collector.
(R): Sulphide ores are soluble in water.

117. (A): Lead, tin and bismuth can be purified by liquation method.
(R): Lead, tin and bismuth have low melting point than impurities.

118. (A): Desilverization of lead is made by Parke's method.
(R): When lead–silver alloy is poor in silver, zinc is added to the molten ore.

Matrix–Match Type Questions

	p	q	r	s
(A)	O	O	O	O
(B)	O	O	O	O
(C)	O	O	O	O
(D)	O	O	O	O

119. March the following:

Column I	Column II
A. Smelting	(p) Copper glance
B. Self reduction	(q) Malachite
C. Electrolytic reduction	(r) Haematite
D. Hydrometallurgy	(s) Bauxite

120. Match the following:

Column I	Column II
A. Bell metal	(p) Cu : 60–80%, Zn : 20–40 %
B. Gun metal	(q) Cu : 75 90%, Sn : 10 25%
C. Bronze	(r) Cu : 88 %, Sn : 10%, Zn : 2%
D. Brass	(s) Cu : 80%, Sn : 20%

121. March the following:

Column I	Column II
A. Nichrome	(p) Pb, Bi, Sn, Cd
B. Type metal	(q) Pb, Sb, Sn
C. Wood's metal	(r) Cu, Al, Zn
D. Devarda's alloy	(s) Fe, Ni, Cr

122. Match the following:

Column I (Ore)	Column II (Method)
A. Silver glance	(p) Self reduction
B. Bauxite	(q) Leaching
C. Copper pyrite	(r) Froth floatation
D. Calamine	(s) Calcination

123. March the following:

Column I	Column II
A. Mond's process	(p) Nickel
B. Poling	(q) Tin
C. Cupellation	(r) Copper
D. Electro-refining	(s) Silver

124. Match the following:

Column I
A. Van Arkel process
B. Mond's process
C. Thermite
D. Cyanidation process

Column II

(p) $Cr_2O_3 + 2Al \xrightarrow{\Delta} 2Cr + Al_2O_3$

(q) $ZrI_4 \xrightarrow{\Delta} Zn + 2I_2$

(r) $2[Au(CN)_2] + Zn \rightarrow [Zn(CN)_4]^{-2} + 2Au$

(s) $Ni + 4CO \xrightarrow{\Delta} Ni(CO)_4 \xrightarrow{\Delta} Ni + 4CO$

125. March the following:

Column I	Column II
A. Chalcopyrites or copper pyrites	(p) Cu_2S
B. Chalcocite or Copper glance	(q) Cu_2O
C. Bronite	(r) $CuFeS_2$
D. Cuprite (red)	(s) Cu_3FeS_3

The IIT–JEE Corner

126. Among the following statements, the incorrect one is
(a) calamine and siderite and carbonates
(b) argentite and cuprite are oxides
(c) zinc blende and pyrites are sulphides
(d) malachite and azurite are ores of copper
[*IIT 1997*]

127. Addition of high proportions of manganese makes steel useful in making rails of railroads, because manganese
(I) gives hardness to steel.
(II) helps the formation of oxides of iron.
(III) can remove oxygen and sulphur.
(IV) can show highest oxidation state of +7.
(a) (I), (II), (III) (b) (I), (III)
(c) (II) , (IV) (d) (I), (III), (IV)
[*IIT 1998*]

128. In the commercial electrochemical process for aluminium extraction, the electrolyte used is
(a) $Al(OH)_3$ in NaOH solution
(b) an aqueous solution of $Al_2(SO_4)_3$
(c) a molten mixture of Al_2O_3 and Na_3AlF_6
(d) a molten mixture of AlO(OH) and $Al(OH)_3$
[*IIT 1999*]

129. Electrolytic reduction of alumina to aluminium by Hall-Heroult process is carried out
(a) in the presence of NaCl.
(b) in the presence of fluorite.
(c) in the presence of cryolite which forms a melt with lower melting temperature.
(d) in the presence of cryolite which forms a melt with higher melting temperature.
[*IIT 2000*]

130. The process of converting hydrated alumina into anhydrous alumina is called
(a) Roasting (b) Smelting
(c) Dressing (d) Calcination
[*IIT Roorkee 2000*]

131. The chemical processes in the production of steel from haematite ore involve
(a) reduction
(b) oxidation
(c) reduction followed by oxidation
(d) oxidation followed by reduction
[*IIT 2000*]

132. The chemical composition of 'slag' formed during the smelting process in the extraction of copper is
(a) $Cu_2O + FeS$ (b) $FeSiO_3$
(c) $CuFeS_2$ (d) $Cu_2S + FeO$
[*IIT 2001*]

133. Which of the following processes is used in the extractive metallurgy of magnesium?
(a) Fused salt electrolysis
(b) Self reduction
(c) Aqueous solution electrolysis
(d) Thermite reduction
[*IIT 2002*]

134. In the process of extraction of gold,
roasted gold ore + CN $\xrightarrow{O_2 / |X| + OH^-}$
[X] + Zn $\rightarrow$ [Y] + Au
[X] and [Y] are
(a) $[X] = [Au(CN)_2]^-$, $[Y] = [Zn(CN)_4]^{-2}$
(b) $[X] = [Au(CN)_4]^{-3}$, $[Y] = [Zn(CN)_4]^{-2}$
(c) $[X] = [Au(CN)_2]^{-}$, $[Y] = [Zn(CN)_6]^{-4}$
(d) $[X] = [Au(CN)_4]^{-}$, $[Y] = [Zn(CN)_4]^{-2}$
[*IIT 2003*]

135. The ore which contains copper and iron both is
(a) cuprite (b) chalcocite
(c) chalcopyrite (d) malachite
[*IIT 2005*]

136. Extraction of zinc from zinc blende is achieved by
(a) roasting followed by reduction with another metal.
(b) electrolytic reduction.

(c) roasting followed by reduction with carbon.
(d) roasting followed by self reduction.

[*IIT 2007*]

137. Match the following:

Column I		Column II
A. PbS	➤ PbO	(p) Roasting
B. $CaCO_3$	➤ CaO	(q) Calcination
C. ZnS	➤ Zn	(r) Carbon reduction
D. Cu_2S	⟶ Cu	(s) Self reduction

[*IIT 2008*]

138. Native Ag forms a water soluble complex with a dil. aq. solution of NaCN in the presnece of
(a) Nitrogen (b) Oxygen
(c) CO_2 (d) Argon

[*IIT 2008*]

ANSWERS

Straight Objective Type Questions

1. (b)	2. (d)	3. (a)	4. (b)
5. (d)	6. (c)	7. (a)	8. (c)
9. (a)	10. (b)	11. (d)	12. (b)
13. (a)	14. (c)	15. (a)	16. (d)
17. (b)	18. (d)	19. (a)	20. (b)
21. (c)	22. (d)	23. (a)	24. (b)
25. (c)	26. (d)	27. (a)	28. (c)
29. (d)	30. (c)	31. (a)	32. (d)
33. (b)	34. (a)	35. (a)	36. (c)
37. (d)	38. (b)	39. (c)	40. (b)
41. (c)	42. (c)	43. (c)	44. (a)
45. (a)			

Brainteasers Objective Type Questions

46. (c)	47. (a)	48. (c)	49. (d)
50. (c)	51. (b)	52. (d)	53. (b)
54. (b)	55. (a)	56. (c)	57. (d)
58. (c)	59. (b)	60. (a)	61. (d)
62. (c)	63. (c)	64. (b)	65. (d)
66. (c)	67. (c)	68. (b)	69. (b)
70. (a)	71. (a)	72. (a)	73. (a)
74. (b)	75. (c)	76. (a)	77. (a)

Multiple Correct Answer Questions

78. (a), (c), (d)	79. (a), (b), (c)
80. (b), (c)	81. (b), (d)
82. (a), (b), (d)	83. (c), (d)
84. (a), (b)	85. (a), (b), (d)
86. (c), (d)	87. (a), (c), (d)
88. (b), (d)	89. (c), (d)
90. (a), (c), (d)	91. (a), (b), (c), (d)
92. (b), (c)	93. (b), (d)

Comprehension–1

94. (b) 95. (c) 96. (a)

Comprehension–2

97. (a) 98. (c) 99. (b) 100. (a)

Comprehension–3

101. (d) 102. (b) 103. (b)

Assertion and Reasoning Questions

104. (b)	105. (c)	106. (d)	107. (c)
108. (b)	109. (a)	110. (c)	111. (a)
112. (c)	113. (a)	114. (a)	115. (b)
116. (c)	117. (a)	118. (a)	

Matrix–Match Type Questions

119. a-(r), b-(p), c-(s), d-(q)

120. a-(s), b-(r), c-(q), d-(p)

121. a-(s), b-(q), c-(p), d-(r)

122. a-(q, r), b-(q, s), c-(p, r), d-(s)

123. a-(p), b-(q, r), c-(s), d-(q, r, s)

124. a-(q), b-(s), c-(p), d-(r)

125. a-(r), b-(p), c-(s), d-(q)

The IIT–JEE Corner

126. (b) 127. (b) 128. (c) 129. (c)

130. (d) 131. (c) 132. (b) 133. (a)

134. (a) 135. (c) 136. (c)

137. a-(p), b-(q) c-(p, r), d-(p, r, s) 138. (b)

HINTS AND EXPLANATIONS

Straight Objective Type Questions

36. $2Na[Au(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Au$

38. Wrought iron is the purest form of iron containing only 0.1 to 0.5% carbon.

39. It is the process of giving a thin coating of hardened steel to a strong, flexible and mild steel by heating it in charcoal and quenching in oil.

41. Monel metal contain 67% Ni, 29% Cu and 2–4% Fe and Mn.

42. Azurite is a basic carbonate ore of copper.

$2CuCO_3Cu(OH)_2$

43. Melting point of $AgNO_3$ is 212°C.

$2AgNO_3 \xrightarrow{\Delta,\ T > 212^\circ C} 2AgNO_2 + O_2$

44. Blast furnace is frequently used for the extraction of ion and copper from their ores. Carbon and carbon monoxide reduce the metallic oxides to the free metals. Slag formation plays an important role in blast furnace as it covers the melted metal and thus protects the metal from being reoxidized.

45. Al reduces Fe_2O_3 or Cr_2O_3 to respective metals and acts as a reducing agent.

$Fe_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Fe$

50. $SnCl_2 + HgCl_2 \longrightarrow SnCl_4 + Hg$

$(ACl_2)\ (BCl_2)\ (ACl_4)\ (B)$

$HgO \xrightarrow[>400^\circ C]{\text{heat}} \frac{1}{2}O_2 + Hg,$

So, ore of Hg (b) will be Cinnabar.

54. $ZnS + 4NaCN \rightleftharpoons$

$4Na + \underset{\text{Water soluble}}{[Zn(CN)_4]^{2}} + S^{2}$

60. Silver is extracted from argentite by the McArthur and Forest process (leaching process).

$Ag_2S + 4NaCN \rightleftharpoons 2NaAg(CN)_2 + Na_2S$

$2NaAg(CN)_2 + Zn \rightarrow Na_2Zn(CN)_4 + 2Ag$

63. Gold and silver are extracted from their native ores by leaching process (McArthur Forrest cyanide process). Both silver and gold particles dissolve in dilute solution of sodium cyanide in presence of oxygen of the air forming complex cyanides.

$Ag + NaCN + H_2O + O_2 \longrightarrow \underset{\text{Sod. argentocyanide}}{NaAg(CN)_2} + NaOH$

$Au + NaCN + H_2O + O_2 \rightarrow \underset{\text{Sod. aurocyanide}}{NaAu(CN)_2} + NaOH$

65. The addition of CaF_2 decreases the melting point of the mixture and increases the conductivity of the molten mixture.

66. Gibb's Helholtz equation is

$\Delta G = \Delta H \quad T\Delta S$

As the formation of Fe_2O_3 from Fe and O_2 occurs with a negative entropy change, so the term $T\Delta S$ becomes more negative as temperature increases i.e., free energy of formation of Fe_2O_3 increases with temperature.

67. $Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$

According to this reaction, the values of $\Delta G°$ for Fe_2O_3 and CO for per mole of oxygen consumed. For this reaction $\Delta G°$ has to be negative.

$\Delta G° = 3\Delta G°CO - \Delta G° Fe_2O_3$ has ro be negative. This seems to be possible for the value of temperatures above 1000 K.

68. $4Zn + 10HNO_3 \longrightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$

70. Froth floaration process is for sulphide ores. Of these, only chalcopyrite is a sulphide ore. ($CuFeS_2$)

71. The reaction at high temperature in the blast furnace is

$2CuFeS_2 + O_2 \longrightarrow Cu_2S + 2FeS + SO_2$

72. Poor pyrite ores of Cu are crushed, exposed to air and sprayed with water to yield $CuSO_4$ solution. This is treated with scrap iron to get Cu.

74. Conc. HNO_3 renders iron passive by forming a thin protective film of Fe_3O_4 on its surface.

75. $CaCO_3 \xrightarrow{Heat} CaO + CO_2$;

$CaO + SiO_2 \longrightarrow CaSiO_3$ (slag)

76. Zinc–copper couple is obrained by contraining zinc with copper.

77. Ag and Au is extracted by this process.

82. As gold is below silver in the electrochemical series so it cannot displace silver due to its lower reactivity and oxidation potential.

84. Brass gas Cu = 80% and Zn = 20%.
Bronze has Cu = 80%, Zn = 10% and Sn = 10%.
Gun metal has Cu = 90% and Sn = 10%
Type metal has Pb = 70%, Sb = 20% and Sn = 10%.
So both brass and bronze contains Cu and Zn.

94. The ore is chromate, $FeOCr_2O_3$.

$4FeO.Cr_2O_3 + 8Na_2CO_3 + 7O_2 \xrightarrow{Lime}$
(A)

$2Fe_2O_3 + 8Na_2CrO_4 + 8CO_2$
(B) (C)

95. $Fe_2O_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O$
(B)

$4FeCl_3 + 3K_4Fe(CN)_6 \longrightarrow Fe_4[Fe(CN)_6]_3 + 12KCl$
(D)
Prussian blue

96. $2Na_2CrO_4 + H_2SO_4 \longrightarrow$
(C)

$Na_2Cr_2O_7 + Na_2SO_4 + H_2O$
(E)
Yellow coloured

$Na_2Cr_2O_7 + 2KCl \longrightarrow K_2Cr_2O_7 + 2NaCl$
(F)
Orange red

Hints for Question 97, 98, 99, 100

$CuCO_3.Cu(OH)_2 \xrightarrow{Calcination} 2CuO\downarrow + CO_2\uparrow + H_2O$
(A_1) Malachite — Black residue (C)

$CuCO_3.Cu(OH)_2 + 4HCl \longrightarrow 2CuCl_2 + CO_2\uparrow + H_2O$

$2CuCl_2 + 4KI \longrightarrow 2CuI + 4KCl + I_2$
ppt. (D)

$2Cu_2S + 3O_2 \xrightarrow{Roasting} 2Cu_2O + 2SO_2\uparrow$
(A_2) (G)
Copper glance

$2Cu_2O + Cu_2S \xrightarrow[during\ roasting]{Auto\ reduction} 6Cu + SO_2$
(M)

$3SO_2 + K_2Cr_2O_7 + H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O$

103. $ZnO \longrightarrow Zn + \frac{1}{2} O_2$; $\Delta G°_1 = x$

$C + \frac{1}{2} O_2 \longrightarrow CO$; $\Delta G°_2 = x$

$ZnO + C \rightarrow Zn + CO \; ; \Delta G^\circ_5 = 0$

i.e., system is in equilibrium.

Here, $\Delta G^\circ_1 = \Delta G^\circ_2$ as zinc and carbon have equal affinity for oxygen.

109. Zinc has no action of air or water on it and thus its coating on iron articles increase their life by protecting them from rusting.

112. Oxide ores being heavier than the earthy or rocky gangue particles settle down while lighter impurities are washed away.

126. Argentite is sulphide ore, Ag_2S. Cuprite is oxide ore Cu_2O

128. A molten mixture of alumina (Al_2O_3) and cryolite (Na_3AlF_6) is used as an electrolyte in aluminium extraction.

129. Cryolite with Al_2O_3 and CaF_2 forms a melt which has lower melting point and is good conductor of electricity.

131. Reduction followed by oxidation.

132. During smelting process (in the extraction of copper), the slag formed is

$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$

$FeO + SiO_2 \rightarrow FeSiO_3$

(From ferrous sulphide) (Slag)

133. Magnesium is obtained by electrolysis of fused mixture of $MgCl_2$ with NaCl and $CaCl_2$.

$MgCl_2 \rightarrow Mg^{-2} + 2Cl^-$

At cathode: $Mg^{2} + 2e \longrightarrow Mg$

At anode: $Cl^- \rightarrow Cl + e^-$

$Cl + Cl \longrightarrow Cl_2 \uparrow$

134. $2Au + 4CN + 11_2O + \frac{1}{2} O_2 \rightarrow 2[Au(CN)_4] + 2OH^-$ (X)

$2[Au(CN)_2]^- + Zn \longrightarrow [Zn(CN)_4]^{-2} + 2Au$ (Y)

136. Zinc blende is roasted and then treated with coke for the reduction.

$2ZnS + 3O_2 \xrightarrow{\Delta} 2ZnO + 2SO_2 \uparrow$

$ZnO + C \xrightarrow{\Delta} Zn + CO$

138. This reaction takes place in presence of O_2 as follows.

$4Ag + 8NacN + 211_2O + O_2 \longrightarrow 4Na[Ag(CN)_2] + 4NaOH$

Sodium argentocyanide.

Solved Subjective Questions

1. Match the following choosing one item from column (X) and the appropriate item from column (Y). Write down the matched pair on the answer script.

(X)	(Y)
Al	Calamine
Cu	Cryolite
Mg	Malachite
Zn	Carnalite

Solution

(X)	(Y)
Al	Cryolite
Cu	Malachite
Mg	Carnalite
Zn	Calamine

2. State the conditions under which the preparation of alumina from aluminium is carried out.

Give the necessary equations which need not be balanced.

Solution

Alumina can be prepared from aluminium under the following conditions.

$$Al + NaOH \xrightarrow{\text{Aqueous}} NaAlO_2 \xrightarrow{CO_2}$$

$$Al(OH)_3 \xrightarrow{\text{Heat}} \underset{\text{Alumina}}{Al_2O_3}$$

3. Give reasons for the following.

(1) Metals can be recovered from their ores by chemical methods.

Solution

Metals can be recovered from their ores by chemical methods since they occur as oxides, carbonates, sulphides hence they can be calcined or roasted.

4. Write balanced chemical equation for the following.

"Gold is dissolved in aquaregia".

Solution

When gold is dissolved in aqua regia following reaction occurs.

$$Au + 3HNO_3 + 4HCl \rightarrow HAuCl + 3H_2O + 3NO_2$$

5. Answer the following questions.

(1) What is the actual reducing agent of haematite in blast furnace?

Solution

CO is the actual reducing agent of haematite in blast furnace

(2) Give the equations for the recovery of lead from galena by air reduction.

Solution

Here, galena is changed into a mixture of PbO and $PbSO_4$ by roasting as follows:

$$2PbS + 3O_2 \rightarrow 2PbO + SO_2 \uparrow$$

$$PbS + 2O_2 \rightarrow PbSO_4$$

Galena

When air supply is stopped, more galena is added and temperature is increased, lead is formed by auto-reduction as follows:

$$2PbO + PbS \rightarrow 3Pb + SO_2$$

$$\underset{\text{Galena}}{PbSO_4} + PbS \rightarrow \underset{\text{Lead}}{2Pb} + 2SO_2$$

(3) Why is sodium chloride added during electrolysis of fused anhydrous magnesium chloride?

Solution

Sodium chloride is added during electrolysis of fused anhydrous magnesium chloride as it not only prevents hydrolysis of $MgCl_2$ but also increases its conductivity.

(4) Why is chalcocite roasted and not calcinated during recovery of copper?

Solution

Chalcocite roasted and not calcinated during recovery of copper as it is a sulphide ore hence roasting is better here.

(5) Zinc and not copper is used for the recovery of metallic silver from complex $[Ag(CN)_2]^-$. Explain.

Solution

Zinc and not copper is used for the recovery of metallic silver from complex $[Ag(CN)_2]$ as being more electropositive and reactive zinc can easily substitute silver from $[Ag(CN)_2]$ while copper can not do that.

6. Each entry in column X is in some way related to the entries in columns Y and X. Match the appropriate entries.

X	Y	Z
Invar	Co, Ni	Cutlery
Nichrome	Fe, Ni	Heating element
Stainless-steel	Fe, Cr, Ni	Watch spring

Solution

X	Y	Z
Invar	Fe, Ni	Watch spring
Nichrome	Co, Ni	Hearing element
Stainless-steel	Fe, Cr, Ni	Cutlery

7. Write balanced equation for "the extraction of copper from copper pyrites by self reduction."

Solution

Roasting:

$$2CuFeS_2 + O_2 \xrightarrow{\Delta} Cu_2S + 2FeS + SO_2$$

Copper pyrites

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 3SO_2$$

$$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$$

Smelting with coke and sand:

$$FeO + SiO_2 \rightarrow FeSiO_3 \text{ (slag)}$$

$$Cu_2O + FeS \rightarrow Cu_2S + FeO$$

Bessemerization:

$$Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$$

8. (A_1) and (A_2) are two ores of metal M. (A_1) on calcinations gives black precipitate, CO_2 and water.

$$A_1 \xrightarrow{\text{Calcination}} \text{Black solid} + CO + H_2O$$

$$A_1 \xrightarrow{\text{dil. HCl, KI}} I_2 + \text{ppt.}$$

$$A_2 \xrightarrow{\text{Roasting}} \text{Metal} + \text{Gas} + K_2Cr_2O_7 + H_2SO_4$$

Green colour

[IIT 2004]

Solution

$$CuCO_3.Cu(OH)_2 \xrightarrow{\text{Calcination}}$$

A_1

$$2CuO + CO_2 + H_2O$$

Black solid

$$CuCO_3.Cu(OH)_2 \xrightarrow{HCl}$$

A_1

$$CuCl_2 + CO_2 + H_2O$$

$$CuCl_2 \xrightarrow{KI} \underset{\text{Ppt.}}{Cu_2I_2}\downarrow + KCl + I_2$$

$$Cu_2S \xrightarrow{\text{Roasting}} Cu_2O + SO_2\uparrow$$

B

$$Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2\uparrow$$

SO_2 is the gas which forms green colour with acidified $K_2Cr_2O_7$ as follows:

$$3SO_2 + K_2Cr_2O_7 + H_2SO_4 \rightarrow K_2SO_4 + \underset{\text{Green}}{Cr_2(SO_4)_3} + 4H_2O$$

9. Metal sulphides occur mainly in rocks and metal halides occur mostly in seas and lakes. Why?

Solution

Metal sulphides are usually insoluble compounds whereas most of the metal halides being water soluble pass into solution in a stream of rain water and are carried to sea or lakes.

10. Copper and silver are below hydrogen in electrochemical series and yet they are found in the combined state as sulphides in nature. Why?

Solution

Both copper and silver combines directly with sulphur at very high temperature to form sulphides under earth crust.

11. The choice of a reducing agent in a particular case depends on thermodynamic factor. How far do you agree with this statement? Explain with example.

Solution

As thermodynamic considerations are very essential is deciding the temperature and suitable reducing agent during extraction of metals, for example, In extraction of Fe from Fe_2O_3 by reducing it at 823 K by CO; SnO_2 is reduced by C at 1473K.

12. Sulphide ores are usually dressed by froth floaration process. Why?

Solution

Sulphide ores being lighter are easily wetted by oil to come on the surface of solution with froths during froth floatation process, leaving behind heavy matter of gangue wetted by water.

13. Why reduction of metal oxides by aluminium becomes faster just after ignition of mixture?

Solution

The reaction possesses exothermic nature and huge amount of heat released during the course of reaction makes it fast.

$$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$$

$$\Delta H = -Ve$$

14. Coke and flux are used in smelting.

Solution

Infusible mass present in ore on mixing with suitable flux are fused which are then reduced by coke to give free metal.

15. In the metallurgy of iron, limestone is added to the ore.

Solution

In the metallurgy of iron, limestone ($CaCO_3$) acts as a flux and forms calcium silicate $CaSiO_3$ (slag).

$$CaCO_3 \rightarrow CaO + CO_2$$

$$CaO + SiO_2 \rightarrow CaSiO_3$$

Flux Slag

16. Why graphite is used as anode but not diamond?

Solution

Graphite is good conductor of current due to sp^2-hybridisation having mobile pi-electrons. Diamond has sp^3-hybridisation and is insulator.

17. How can NaCN act as an depressant in preventing ZnS from forming froth in froth flotation process?

Solution

As NaCN forms a layer of zinc complex $Na_2[Zn(CN)_4]$ on the surface of ZnS and so prevents it from the formation of froth.

$$ZnS + 4NaCN \rightarrow Na_2[Zn(CN)_4] + Na_2S$$

Sodium tetracyanozincate (II)

18. When an inert atmosphere is needed for a metallurgical process, nitrogen is oftenly used. However during the reduction of $TiCl_4$ by magnesium, helium is preferred. Why?

Solution

$$TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$$

As here magnesium is used as a reducing agent, if N_2 is used to provide, an inert atmosphere, it may combine with Mg to form magnesium nitride as follows:

$$3Mg + N_2 \rightarrow Mg_3N_2$$

Hence, magnesium can not be able to reduce $TiCl_4$ to Ti that is why helium is preferred here.

Questions For Self Assessment

19. Copper pyrite is roasted instead of being directly reduced by carbon or in extraction of copper the sulphide ore is partially oxidized.
20. Polling process is used for the removal of Cu_2O from copper. Why?
21. Zinc is used in the galvanization of iron. Why?
22. Limestone is used in the manufacture of pig iron from haematite. Why?
23. Magnesium oxide is used for the lining of furnace for making steel. Why?
24. Cast iron is hard while pure iron is soft in nature. Why?
25. Why is pure iron not used for the manufacture of tools and machines?
26. Why is Fe_2O_3 amphoteric in nature?
27. Tin vessels are not used for packing in cold countries. Why?
28. A piece of tin foil is added to $SnCl_2$ solution for preserving it. Why?
29. What flux is used during smelting of iron ore? How does it remove the impurities?
30. Partial roasting of sulphide ore is done in the metallurgy of copper.
31. Zinc becomes dull in moist air.
32. In the metallurgy of iron, lime stone is added to the ore.
33. Magnesium oxide is used for lining of steel making furnace.
34. Cast iron is hard but pure iron is soft in nature.
35. Excess of carbon is added in the zinc metallurgy.
36. Anode mud in copper refining contains silver and gold.
37. Copper is largely used in electrical wiring.
38. Recovery of silver from silver coin.
39. Extraction of zinc from zinc blende.
40. Reactions occurring in blast furnace in the extraction of iron.

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COMPOUNDS OF HEAVY METALS

9

Chapter Contents

Oxides and Chlorides of tin and lead; Oxides, chlorides and sulphates of Fe^{2+}, Cu^{2+} and Zn^{2+}; Potassium permanganate, potassium dichromate, silver oxide, silver nitrate, silver thiosulphate and various levels of multiple-choice questions.

OXIDES AND CHLORIDES OF TIN

Stannous Oxide (SnO)

Preparation

1. **From $SnCl_2$ and NaOH:** When NaOH is added to $SnCl_2$, stannous hydroxide is obtained, which on heating in an atmosphere of CO_2 gives SnO.

$$SnCl_2 + 2NaOH \longrightarrow Sn(OH)_2 + 2NaCl$$

$$Sn(OH)_2 \xrightarrow{\Delta,\ CO_2} SnO + H_2O$$

2. **From $SnCl_2$ and Na_2CO_3:** When a mixture of $SnCl_2$ and Na_2CO_3 is heated in an atmosphere of CO_2, SnO is formed as follows:

$$SnCl_2 + Na_2CO_3 \xrightarrow{\Delta,\ CO_2} SnO + CO_2 + 2NaCl$$

3. **From Stannous Oxalate:** When stannous oxalate is heated in absence of air, SnO is formed.

$$SnC_2O_4 \xrightarrow{\Delta} SnO + CO + CO_2$$

Physiochemical Properties

1. It is a dark grey or blackish powder which is insoluble in water.
2. It is an amphoteric oxide and dissolves both in acids and bases.

$$SnO + 2HCl \longrightarrow \underset{\text{Stannous chloride}}{SnCl_2} + H_2O$$

$$SnO + 2NaOH \longrightarrow \underset{\text{Sodium stannite}}{Na_2SnO_2} + H_2O$$

Sodium stannite is stable only in aqueous solution and absorbs O_2 from air to form sodium stannate.

$$2Na_2SnO_2 + O_2 \longrightarrow 2Na_2SnO_3$$

3. **Burning in Air:** It burns in air with incandescence giving stannic oxide.

$$2SnO + O_2 \longrightarrow 2SnO_2$$

Stannic Oxide (SnO_2)

In nature it is found as tin stone or cassiterite.

Preparation

1. **By Heating Tin in Air:**

$$Sn + O_2 \rightarrow SnO_2$$

2. **From Meta Stannic Acid:**

$$Sn + 4HNO_3 \rightarrow H_2SnO_3 + 4NO_2 + 2H_2O$$

$$\underset{\text{Meta stannic acid}}{H_2SnO_3} \longrightarrow SnO_2 + H_2O$$

or

$$2SnO + O_2 \xrightarrow{\text{Heat}} 2SnO_2$$

$$H_2Sn_5O_{11}.4H_2O \xrightarrow{\text{Heat}} 5SnO_2 + 5H_2O$$

Properties

1. It is a white powder which is insoluble in water but soluble in concentrated H_2SO_4 and alkalies (KOH).

$$SnO_2 + 2H_2SO_4 \rightarrow Sn(SO_4)_2 + 2H_2O$$

$$SnO_2 + 2KOH \longrightarrow \underset{\text{Potassium stannates}}{K_2SnO_3} + H_2O$$

Uses

1. It is used in making pottery and glass white glazes.
2. It is used for making milky glass and as a polishing powder (putty powder).

Stannous Chloride ($SnCl_2$)

Preparation

1. **From Stannous Oxide and Tin:**

$$SnO + 2HCl \longrightarrow SnCl_2 + H_2O$$

$$Sn + 2HCl \rightarrow SnCl_2 + H_2$$

The solution of $SnCl_2$ on concentration and cooling gives the crystals of hydrated stannous chloride.

2. Anhydrous $SnCl_2$ can be prepared as follows:

$$\underset{\text{Dry}}{Sn + Cl_2} \rightarrow SnCl_2$$

$$Sn + HgCl_2 \longrightarrow SnCl_2 + Hg$$

Physiochemical Properties

1. It is a white crystalline hygroscopic solid which is soluble in water, alcohol and ether.

It undergoes hydrolysis as follows:

$$SnCl_2 + H_2O \rightleftharpoons \underset{\text{Stannous hydroxyl chloride}}{Sn(OH)Cl} + HCl$$

2. **With Sodium Hydroxide:** When NaOH is added in it sodium stannite is formed as follows:

$$SnCl_2 + 2NaOH \rightarrow \underset{\text{White ppt.}}{Sn(OH)_2} + 2NaCl$$

$$Sn(OH)_2 + 2NaOH \rightarrow \underset{\text{Sodium stannite}}{Na_2SnO_2} + 2H_2O$$

3. **With H_2S:** It gives a dark brown precipitate with H_2S as follows:

$$SnCl_2 + H_2S \rightarrow \underset{\text{Brown ppt.}}{SnS} + 2HCl$$

The brown precipitate dissolves in ammonium sulphide.

$$SnS + (NH_4)_2S_2 \rightarrow \underset{\text{Ammonium thiostannate}}{(NH_4)_2\,SnS_3}$$

4. **Reducing Properties:** It is a strong reducing agent

For example,

(i) It reduces mercuric chloride into mercury as follows:

$$2HgCl_2 + SnCl_2 \rightarrow \underset{\substack{\text{(White ppt.)}\\ \text{Mercurous chloride}}}{Hg_2Cl_2} + SnCl_4$$

$$Hg_2Cl_2 + SnCl_2 \rightarrow \underset{\text{(Grey black)}}{2Hg} + SnCl_4$$

For example,

(ii) It reduces ferric chloride into ferrous chloride as follows:

$$2FeCl_3 + SnCl_2 \longrightarrow 2FeCl_2 + SnCl_4$$

For example,

(iii) It reduces auric chloride into metallic gold as follows:

$$2AuCl_3 + 3SnCl_2 \rightarrow \underset{\text{Colloidal gold}}{2Au} + 2SnCl_4$$

For example,

(iv) It reduces nitro compounds into amines as follows:

$$RNO_2 + 6HCl + 3SnCl_2 \rightarrow RNH_2 + 3SnCl_4 + 2H_2O$$

Uses

It is used as a reducing agent, as a mordent in dyeing and for making purple of casius.

Stannic Chloride ($SnCl_4$)

Preparation

1. By passing dry chlorine over fused tin or stannous chloride:

   $$Sn + 2Cl_2 \longrightarrow SnCl_4$$

   $$SnCl_2 + Cl_2 \rightarrow SnCl_4$$

2. By distilling tin with excess of mercuric chloride:

   $$2HgCl_2 + Sn \rightarrow SnCl_4 + 2Hg$$

Physiochemical Properties

1. It is a colourless fuming liquid with unpleasant smell.
2. It is hygroscopic in nature and can form many crysralline hydrares having 3, 5, 6, 8 water molecules as water of crystallizarion.

   For example,

   $SnCl_4.5H_2O$ is called butter of tin or oxymuriate of tin.
3. It is soluble in water and undergoes hydrolysis as follows:

   $$SnCl_4 + 4H_2O \longrightarrow Sn(OH)_4 + 4HCl$$
4. It dissolves in conc. HCl giving chloro srannic acid.

   $$SnCl_4 + 2HCl \longrightarrow H_2SnCl_6$$
5. It combines with ammonia ro form crysralline adducr, $SnCl_4.4NH_3$ which can be sublimed wirhour any decomposirion.
6. With NH_4Cl:

   $$SnCl_4 + 2NH_4Cl \rightarrow \underset{\text{Pink salt}}{(NH_4)_2SnCl_6}$$

Uses

1. It is used for fire proofing corron and for increasing the weighr of silk.
2. Butter of tin acts as a mordent in dyeing (also pink salt).

OXIDES AND CHLORIDES OF LEAD

Lead Mono Oxide or Litharge (PbO)

It is also known as plumbous oxide or lead (II) oxide.

Preparation

1. **By Heating Lead in Air:** It can be prepared by hearing lead in air or oxygen ar 300°C. It changes to a yellow powder known as Massicor which on further hearing ar 880°C becomes reddish yellow known as litharge. Chemically, both are PbO.

   $$2Pb \xrightarrow{\Delta\ O_2} 2PbO$$
2. **By Heating Lead Nitrate:**

   $$2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$$
3. **By Heating Lead Carbonate:**

   $$PbCO_3 \xrightarrow{\Delta} PbO + CO_2\uparrow$$

Physiochemical Properties

1. When it is a yellow powder it is called massicor and when on heating it is in buff coloured crystalline form it is called litharge.

2. **Amphoteric Nature:** It is an amphoteric oxide which is insoluble in water but soluble in acids and alkalies.

$$PbO \xrightarrow{2HCl} 2PbCl_2 + H_2O$$

$$PbO \xrightarrow{2NaOH} \underset{\text{Sodium plumbate}}{Na_2PbO_2} + H_2O$$

3. **Heating Effect:**

$$\underset{\text{Litharge}}{6PbO} + O_2 \xleftarrow{\Delta} \underset{\text{Red lead}}{2Pb_3O_4}$$

4. **Reduction:** It can be reduced into metallic lead by heating it with H_2 or CO or with carbon.

$$PbO + H_2 \longrightarrow Pb + H_2O$$

$$PbO + CO \longrightarrow Pb + CO_2$$

$$PbO + C \longrightarrow Pb + CO$$

Uses

1. PbO is used in paints, utensils and flint glass.
2. Glycerol + Massicot: A cement for glass and stone.
3. It is used for making flint glass.
4. It is used for glazing pottery.

Plumbic Oxide or Lead Dioxide or Lead (IV) Oxide (PbO_2)

Preparation

From Litharge

Litharge on heating (fusion) with KNO_3 or $KClO_3$ gives lead dioxide.

$$PbO + KNO_3 \longrightarrow PbO_2 + KNO_2$$

$$3PbO + KClO_3 \longrightarrow 3PbO_2 + KCl$$

By Heating Red Lead with Dilute HNO_3

$$\underset{\text{Red lead}}{Pb_3O_4} + 4HNO_3 \xrightarrow{\Delta} 2Pb(NO_3)_2 + 2H_2O + \underset{\text{Brown}}{PbO_2}\downarrow$$

Laboratory Method

In laboratory it is prepared by treating lead acetate with bleaching powder having slaked lime as follows:

$$(CH_3COO)_2Pb + Ca(OH)_2 \longrightarrow Pb(OH)_2 + (CH_3COO)_2Ca$$

$$Pb(OH)_2 + \underset{\substack{\text{Bleaching}\\\text{powder}}}{CaOCl_2} \longrightarrow PbO_2 + CaCl_2 + H_2O$$

From Lead Hydroxide

$$Pb(OH)_2 + NaOCl \longrightarrow PbO_2 + H_2O + NaCl$$

Physiochemical Properties

1. It is a chocolate brown coloured powder which does not dissolve in water.
2. **Heating Effect:** On heating at 300°C it changes into litharge.

$$2PbO_2 \xrightarrow{300°C} 2PbO + O_2$$

As oxygen is liberated here hence it can act as a strong oxidant.

3. **Amphoteric Nature:** Being an amphoteric oxide it can react with both acids and bases

For example,

(i) It forms plumbic chloride on reaction with cold and concentrated HCl.

$$PbO_2 + \underset{\substack{\text{Cold and}\\\text{conc.}}}{4HCl} \longrightarrow PbCl_4 + 2H_2O$$

For example,

(ii) It dissolves in concentrated HCl and H_2SO_4 to give $PbCl_2$ and $PbSO_4$, respectively.

$$PbO_2 + \underset{\text{Conc.}}{4HCl} \longrightarrow PbCl_2 + Cl_2 + 2H_2O$$

$$2PbO_2 + 2H_2SO_4 \longrightarrow 2PbSO_4 + 2H_2O + O_2$$

For example,

(iii) It also dissolves in hot and concentrated solution of sodium hydroxide to give sodium plumbate.

$$PbO_2 + 2NaOH \longrightarrow \underset{\text{Sodium plumbate}}{Na_2PbO_2} + H_2O$$

4. **Oxidizing Action:** It can act as an oxidant also.

   For example,

   $PbO_2 + SO_2 \longrightarrow PbSO_4$

   $PbO_2 + 4HCl \rightarrow PbCl_2 + Cl_2 + 2H_2O$

Uses

It is used in match industry, lead storage battery and as an oxidizing agent.

Red Lead or Tri Lead Tetra Oxide (Pb_3O_4)

Pb_3O_4 is triplumbic tetra-oxide or minium or sindur. It is a mixed oxide of PbO_2 and PbO in 1 : 2 ratio.

Preparation

When massicor (litharge) is heated with air at 400°C in a reverberatory furnace red lead is formed.

$$\underset{\text{Litharge}}{6PbO} + O_2 \xrightarrow{470^\circ C} \underset{\text{Red lead}}{2Pb_3O_4}$$

Physiochemical Properties

1. It is a red powder which is insoluble in water.
2. **Heating Effect (above 470°C):** On heating, it turns violet or blackish and at temperature above 470°C, it decomposes into PbO and O_2.

   $2Pb_3O_4 \xrightarrow{\Delta} 6PbO + O_2$

3. **Oxidizing Properties:** Being an oxidant it can oxidize HCl into chlorine and evolve oxygen with hot and concentrated sulphuric acid as follows:

   $Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + 4H_2O + Cl_2$

   $2Pb_3O_4 + 6H_2SO_4 \rightarrow 6PbSO_4 + 6H_2O + O_2$

   It can oxidized CO into CO_2.

   $Pb_3O_4 + 4CO \rightarrow 3Pb + 4CO_2$

4. **With Nitric Acid:** On reaction with nitric acid it gives lead nitrate and a brown precipitate of PbO_2 which suggesrs that it is a mixture of PbO and PbO_2.

   $Pb_3O_4 + 4HNO_3 \rightarrow 2Pb(NO_3)_2 + PbO_2\downarrow + 2H_2O$

Uses

1. It is used in making flint glass, match industry and as a red pigment in making protective paints for coating iron and steel.
2. Pb_3O_4 and linseed oil is a protective paint for iron, silver mirror etc.

Lead (II) Halides or Plumbous Halides (PbX_2)

Lead can form stable dihalides of PbX_2 type. For examples, PbF_2, $PbCl_2$, $PbBr_2$, PbI_2 which are ionic in nature.

$$\underset{\text{White solids}}{PbF_2 > PbCl_2 > PbBr_2 >} \underset{\text{Yellow solid}}{PbI_2}$$

Decreasing order of ionic nature

Lead Chloride or Plumbous Chloride ($PbCl_2$)

Preparation

It is prepared by the reaction of hydrochloric acid or some soluble chloride with a lead salt solution as follows:

$Pb(NO_3)_2 + 2HCl \longrightarrow PbCl_2 + 2HNO_3$

$Pb(NO_3)_2 + 2NaCl \rightarrow PbCl_2 + 2NaNO_3$

$PbO + 2HCl \longrightarrow PbCl_2 + H_2O$

Physiochemical Properties

1. It is a white crystalline solid, soluble in hot water. It is soluble in excess of hydrochloric acid giving chloro plumbous acid.

   $PbCl_2 + 2HCl \longrightarrow H_2PbCl_4$

2. With Hot Lime Water: It reacts with hot lime water to give Pb(OH)Cl (a white pigment).

   $PbCl_2 + Ca(OH)_2 \rightarrow \underset{\text{Pattuson's white lead}}{Pb(OH)Cl} + CaO + HCl$

3. **Heating Effect:** On heating in air it forms lead oxychloride as follows:

$$2PbCl_2 + \tfrac{1}{2}O_2 \longrightarrow Pb_2OCl_2 + Cl_2$$

Lead (IV) Halides or Plumbic Halides (PbX_4)

Among PbX_4 only PbF_4 and $PbCl_4$ are known, however PbF_4 has not been obtained in the pure state.

Lead Tetrachloride or Plumbic Chloride ($PbCl_4$)

Preparation

It is obtained by the reaction of lead dioxide and well cooled hydrochloric acid as follows:

$$PbO_2 + 4HCl \xrightarrow{273\ K} PbCl_4 + 2H_2O$$

Physiochemical Properties

1. It is a yellow liquid compound, covalent in nature and soluble in organic solvents.
2. It is an unstable compound and decomposes easily as follows:

$$PbCl_4 \longrightarrow PbCl_2 + Cl_2$$

3. It is hydrolyzed by water as follows:

$$PbCl_4 + 2H_2O \longrightarrow PbO_2 + 4HCl$$

4. It forms stable double salt with ammonium chloride, which decomposes when reacts with sulphuric acid.

$$PbCl_4 + 2NH_4Cl \longrightarrow (NH_4)_2PbCl_6$$

Ammonium hexachloro plumbate

$$(NH_4)_2PbCl_6 + H_2SO_4 \longrightarrow (NH_4)_2SO_4 + PbCl_4 + 2HCl$$

5. With Concentrated HCl

$$PbCl_4 + 2HCl \longrightarrow H_2[PbCl_6]$$

OXIDES

Ferric Oxide (Fe_2O_3)

Occurrence: Haematite and Limonite

Methods of Preparation

1. **By Heating Ferrous Sulphate**

$$2FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_2 + SO_3$$

Bright Red
(Venetian Red Pigment)

2. **By Heating Iron Pyrite in Air**

$$4FeS_2 + 11O_2 \xrightarrow{\Delta} 2Fe_2O_3 + 8SO_2$$

3. **By Heating Ferric Carbonate**

$$Fe_2(CO_3)_3 \xrightarrow{\Delta} Fe_2O_3 + 3CO_2$$

Physiochemical Properties

1. It is a deep red powder and amphoteric in nature.
2. With HCl

$$Fe_2O_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O$$

3. Sodium Carbonate

$$Fe_2O_3 + Na_2CO_3 \longrightarrow Na_2Fe_2O_4 + CO_2$$

Sodium Ferrite
Or
$2NaFeO_2$

4. Hydrolysis

$$2NaFeO_2 + H_2O \longrightarrow Fe_2O_3 + 2NaOH$$

It is lowing process to prepare NaOH.

5. With NaOH

$$Fe_2O_3 + 2NaOH \longrightarrow 2NaFeO_2 + H_2O$$

$$2NaFeO_2 + Cl_2 + 4NaOH \longrightarrow 2Na_2FeO_4 + 2NaCl + 2H_2$$

6. Heating Effect

$$6Fe_2O_3 \xrightarrow[1400°C]{\Delta} 4Fe_3O_4 + O_2$$

or
$4(FeO.Fe_2O_3)$
Magnetic in nature

7. Some Other Reactions

$$Fe_2O_3 + 2LiOH \xrightarrow{\Delta} 2LiFeO_2 + H_2O$$

$$Fe_2O_3 + 3CO \xrightarrow{\Delta} 2Fe + 3CO_2\uparrow$$

$$Fe_2O_3 + \underset{\underset{6\ COOH}{|}}{COOH} \rightarrow 2[Fe(C_2O_4)_3]^{3-} + 3H_2O + 6H^+$$

Water soluble complex

$$Fe_2O_3.H_2O + 4NaOCl \rightarrow 2Na_2FeO_4 + 2HCl + Cl_2$$

$$Fe_2O_3 + 3H_2 \xrightarrow[900°C]{\Delta} 2Fe + 3H_2O$$

Uses

It is used as a catalyst in Bosch's process, as a polishing powder (jewellers rouge) and as a red pigment.

Cupric Oxide (CuO)

It is known as black oxide of copper and it is present as tenorite in nature.

Preparation

1. **By Heating Cupric Hydroxide or Copper Nitrate:**

$$Cu(OH)_2 \xrightarrow{\Delta} CuO + H_2O$$

$$2Cu(NO_3)_2 \longrightarrow 2CuO + 4NO_2 + O_2$$

2. **From Cuprous Oxide or Copper:** When cuprous oxide or copper is heated in air at nearly 1273–1373K, cupric oxide is formed as follows:

$$Cu_2O + \tfrac{1}{2}O_2 \xrightarrow{\Delta} 2CuO$$

$$2Cu + 2O_2 \xrightarrow{\Delta} 2CuO$$

3. **Industrial Method:** Industrially, it is prepared by heating malachite ore.

$$CuCO_3.Cu(OH)_2 \xrightarrow{\Delta} 2CuO + CO_2 + H_2O$$

Malachite

Physiochemical Properties

1. It is black powder which is insoluble in water but dissolves in acids.
2. With Acids: It dissolves in acids to form salts as follows:

$$CuO + H_2SO_4 \xrightarrow{\Delta} CuSO_4 + H_2O$$

$$CuO + 2HCl \xrightarrow{\Delta} CuCl_2 + H_2O$$

3. Heating Effect: On heating above 1373K it decomposes into cuprous oxide.

$$4CuO \xrightarrow{1100-1200°C} 2Cu_2O + O_2$$

4. Reduction: It can be reduced by H_2, Cu, C etc. as follows:

$$CuO + H_2 \xrightarrow{\Delta} Cu + H_2O$$

$$CuO + CO \xrightarrow{\Delta} Cu + CO_2$$

Uses

1. It is used to give greenish blue colour to glass.
2. It is used to remove sulphur from petroleum.
3. It is used in the estimation and detection of carbon during organic analysis.

Cuprous Oxide (Cu_2O)

Preparation

On heating above 1373K it decomposes into cuprous oxide.

$$4CuO \xrightarrow{1373-1473K} 2Cu_2O + O_2$$

Physiochemical Properties

1. It is a red brown powder which is insoluble in water but soluble in ammonia.

$$Cu + 2NH_3 \rightarrow [Cu(NH)_2]^+$$

2. It imparts red colour to glass.

Zinc Oxide (ZnO)

It is called zinc white or philisopher wool or Chinese white. In nature it is present as zincite or red zinc.

Preparation

It is prepared as follows:

1. $2Zn + O_2 \xrightarrow{\Delta} 2ZnO$
2. $ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2$

3. $Zn(OH)_2 \xrightarrow{\Delta} ZnO + H_2O$
4. $2Zn(NO_3)_2 \xrightarrow{\Delta} 2ZnO + 4NO_2 + O_2$
5. Pure ZnO is also prepared as follows:

$$4ZnSO_4 + 4Na_2CO_3 + 3H_2O \longrightarrow \underset{\text{ppt.}}{ZnCO_3.3Zn(OH)_2} + 4Na_2SO_4 + 3CO_2$$

$$ZnCO_3.3Zn(OH)_2 \xrightarrow{\Delta} 4ZnO + 3H_2O + CO_2$$

Physiochemical Properties

1. It is a white powder which becomes yellow on hearing and becomes white again on cooling.

$$\text{White Powder} \xrightarrow{\Delta} \text{Yellow Powder} \xrightarrow{\text{cool}}$$

2. It is insoluble in water and sublimates at 673K.
3. Amphoteric Nature: It is an amphoteric oxide in nature and can react with acids as well as bases.

(i) With Acids

$$ZnO + H_2SO_4 \rightarrow ZnSO_4 + H_2O$$

$$ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$$

(ii) With NaOH

$$ZnO + 2NaOH \longrightarrow \underset{\text{Sodium Zincate}}{Na_2ZnO_2} + H_2O$$

4. Reduction: It can be reduced into zinc as follows:

$$ZnO + H_2 \xrightarrow{673K} Zn + H_2O$$

$$ZnO + C \rightarrow Zn + CO\uparrow$$

7. With $CO(NO_3)_2$: On heating with cobalt nitrate it gives a green mass of cobalt zincate i.e., Rinmann's green.

$$2Co(NO_3)_2 \xrightarrow{\Delta} 2CoO + 2NO_2 + O_2$$

$$ZnO + CoO \rightarrow \underset{\substack{\text{Cobalt zincate}\\ \text{(Green ppt.)}}}{CoZnO_2 \text{ or } CoO.ZnO}$$

Uses

1. ZnO is used as a white pigment in paint. It is better than white lead as it does not turn black in atmosphere due to H_2S.
2. It is used as a catalyst in following reaction.

$$CO + H_2 + H_2 \xrightarrow{ZnO+Cr_2O_3} CH_3OH$$

3. It is used in making zinc ointment, creams, cosmetic powders etc.
4. It is used as an absorbent in surgical dressing.

HALIDES

Ferric Chloride ($FeCl_3$)

$FeCl_3$	$FeCl_3.6H_2O$
Anhydrous	Hydrated
Black	Yellow

Preparation

1. **Formation of Anhydrous $FeCl_3$:** Anhydrous $FeCl_3$ can be obtained by passing dry chlorine gas over heated iron filing as follows:

$$\underset{\text{Dry}}{2Fe + 3Cl_2} \longrightarrow \underset{\text{Anhydrous}}{2FeCl_3}$$

2. **Formation of Hydrated $FeCl_3$**

$$Fe_2(CO_3)_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O + 3CO_2$$

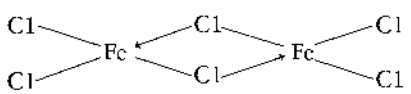
$$Fe(OH)_3 + 3HCl \rightarrow FeCl_3 + 3H_2O$$

$$Fe_2O_3 + 6HCl \rightarrow 2FeCl_3 + 3H_2O$$

The solution of $FeCl_3$ on evaporation and cooling gives yellow crystals of hydrated ferric chloride.

Physiochemical Properties

1. Anhydrous $FeCl_3$ is a dark reddish black deliquescent solid.
2. It is possible as a dimmer in gaseous state i.e Fe_2Cl_6



3. Decomposition: On decomposition it gives $FeCl_2$ and Cl_2.

$$2FeCl_3 \xrightarrow{\Delta} 2FeCl_2 + Cl_2$$

4. Dehydration of Hydrated Ferric Chloride: It can be dehydrated on heating or treating with thionyl chloride as follows:

$$2[FeCl_3.6H_2O] \xrightarrow{\Delta} Fe_2O_3 + 6HCl + 9H_2O$$

$$FeCl_3.6H_2O + 6SOCl_2 \xrightarrow{\Delta} FeCl_3 + 12HCl + 6SO_2$$

5. With H_2O: On hydrolysis, it gives ferric hydroxide as follows:

$$FeCl_3 + 3HOH \rightarrow Fe(OH)_3 + 3HCl$$

(Acidic solution)

6. With NH_4OH: It reacts with NH_4OH to give a reddish brown precipitate of ferric hydroxide as follows:

$$FeCl_3 + 3NH_4OH \rightarrow Fe(OH)_3 + 3NH_4Cl$$

Reddish Brown ppt.

7. Oxidizing Properties:
It can oxidize $SnCl_2$ into $SnCl_4$.

$$2FeCl_3 + SnCl_2 \rightarrow 2FeCl_2 + SnCl_4$$

It can oxidize KI into I_2.

$$2FeCl_3 + 2KI \rightarrow 2FeCl_2 + 2KCl + I_2$$

8. With $K_4Fe(CN)_6$: It reacts with porassium ferrocyanide to give Prussian blue as follows:

$$4FeCl_3 + 3K_4Fe(CN)_6 \longrightarrow Fe_4[Fe(CN)_6]_3 + 12\,KCl$$

Prussian blue or
Ferri ferrocyanide

9. With NH_4CNS: It reacts with NH_4CNS to give ferric thio cyanide.

$$FeCl_3 + 3NH_4CNS \longrightarrow Fe(SCN)_3 + 3NH_4Cl$$

Ferric thio cyanide
(Blood red colour)

Uses

1. It is used to prepare prussian blue.
2. Its alcoholic solution is used as medicine (tincher ferri per chloride).
3. In laboratory, it is used to detect acetates and phenols.

Cupric Chloride ($CuCl_2.2H_2O$)

Preparation

It is prepared as follows:

1. **Formation of Hydrated Cupric Chloride:** When Cu, CuO or Copper carbonate ($CuCO_3$) is dissolved in concentrated HCl and the solution is crystallized, green crystals of hydrated cupric chloride are formed as follows:

$$CuO + 2HCl \longrightarrow CuCl_2 + H_2O$$

$$2Cu + 4HCl + O_2 \rightarrow 2CuCl_2 + 2H_2O$$

$$Cu(OH)_2\,CuCO_3 + 4HCl \longrightarrow 2CuCl_2 + 3H_2O + CO_2$$

2. **Formation of Anhydrous Cupric Chloride**

(i) It can be obtained by heating Cu with excess of chlorine gas.

$$Cu + Cl_2 \xrightarrow{\Delta} CuCl_2$$

(ii) When hydrated cupric chloride is heated with HCl gas at 423K, anhydrous cupric chloride is formed.

$$CuCl_2.2H_2O \xrightarrow{HCl,\ \Delta} CuCl_2 + 2H_2O$$

Physiochemical Properties

1. It is a deliquescent compound and readily soluble in water.
2. Its dilute solution is blue and concentration solution is green which becomes yellow on adding concentrated HCl in it. Here the blue colour and yellow colour are due to $[Cu(H_2O)_4]^{2+}$ and $[CuCl_4]^{2-}$ ions, respectively.
3. Hydrolysis: On hydrolysis it gives acidic aqueous solution due to the formation of HCl.

$$CuCl_2 + 2H_2O \rightleftharpoons Cu(OH)_2 + 2HCl$$

4. Heating Effect:

(i) When anhydrous $CuCl_2$ is heated, Cu_2Cl_2 is formed as follows:

$$2CuCl_2 \xrightarrow{\Delta} Cu_2Cl_2 + Cl_2$$

(ii) When hydrated cupric chloride is strongly heated it decomposes as follows:

$$3CuCl_2.2H_2O \xrightarrow{\Delta} Cu_2Cl_2 + CuO + 2Cl_2 + 2HCl + 5H_2O$$

5. With NaOH: On adding NaOH in it, it gives a pale blue precipitate of basic cupric chloride.

$$CuCl_2 + 2NaOH \longrightarrow Cu(OH)_2 + 2NaCl$$

$$CuCl_2 + 3Cu(OH)_2 \longrightarrow \underset{\text{Basic cupric chloride}}{CuCl_2.3Cu(OH)_2}$$
Pale blue ppt.

6. With NH_4OH: It dissolves in it to give deep blue solution of tetraamine cupric chloride which can be crystallized on evaporation into deep blue crystals.

$$CuCl_2 + 4NH_4OH \longrightarrow \underset{\text{Tetraamine cupric chloride}}{Cu(NH_3)_4Cl.H_2O} + 3H_2O$$

7. Reduction: It can be easily reduced into Cu_2Cl_2 by reductants like $SnCl_2$, H_2, Cu etc. as follows

$$2CuCl_2 + SnCl_2 \longrightarrow Cu_2Cl_2 + SnCl_4$$

$$CuCl_2 + Cu \longrightarrow Cu_2Cl_2$$

Uses

It is used as a catalyst in Deacon's process to prepare chlorine and an oxygen carrier etc.

Zinc Chloride ($ZnCl_2.2H_2O$)

Preparation

1. From Zinc Compounds: It can be prepared by the action of HCl on these zinc compounds as follows:

$$ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$$

$$ZnCO_3 + 2HCl \longrightarrow ZnCl_2 + CO_2 + H_2O$$

$$Zn(OH)_2 + 2HCl \longrightarrow ZnCl_2 + 2H_2O$$

The solution of $ZnCl_2$ on concentration and cooling gives the crystals of hydrated zinc chloride.

2. Anhydrous $ZnCl_2$ can be prepared as follows:

$$Zn + Cl_2 \longrightarrow \underset{\text{Dry}}{ZnCl_2}$$

$$Zn + 2HCl \xrightarrow{\text{Dry}} ZnCl_2 + H_2$$

$$Zn + HgCl_2 \xrightarrow{\text{Distillation}} ZnCl_2 + Hg$$

Physiochemical Properties

1. It is a white deliquescent solid which is soluble in water.
2. It has a melting point of 933K and a boiling point of 1003K.
3. Heating Effect: Hydrated form on heating gives zinc oxychloride as follows:

$$ZnCl_2.2H_2O \xrightarrow{\Delta} Zn(OH)Cl + HCl + H_2O$$

$$2ZnCl_2.2H_2O \xrightarrow{\Delta} \underset{\substack{\text{Zinc}\\\text{oxychloride}}}{Zn_2OCl_2} + 2HCl + 3H_2O$$

4. With Sodium Hydroxide: On dissolving excess of NaOH in it, a white precipitate of sodium zincate is obtained.

$$ZnCl_2 + 2NaOH \longrightarrow \underset{\text{White ppt.}}{Zn(OH)_2} + 2NaCl$$

$$Zn(OH)_2 + 2NaOH \longrightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2} + 2H_2O$$

5. With ammonium hydroxide or aqueous ammonia: On dissolving it, in excess of NH_4OH a white complex tetraamine zinc chloride is formed.

$$ZnCl_2 + 2NH_4OH \longrightarrow Zn(OH)_2 + 2NH_4Cl$$

$$Zn(OH)_2 + 2NH_4OH + 2NH_4Cl \longrightarrow \underset{\substack{\text{Tetraamine zinc}\\\text{chloride}}}{[Zn(NH_3)_4]Cl_2} + 4H_2O$$

6. With Sodium Carbonate: On reaction with sodium carbonate it gives a white precipitate of basic of basic zinc carbonate.

$$4ZnCl_2 + 4Na_2CO_3 + 3H_2O \longrightarrow \underset{\text{Basic zinc carbonate}}{ZnCO_3.3Zn(OH)_2} + 8NaCl + 3CO_2$$

If we use sodium bicarbonate solution, zinc carbonate is formed.

$$ZnCl_2 + 2NaHCO_3 \longrightarrow ZnCO_3 + 2NaCl + H_2O + CO_2$$

7. With H_2S: On passing H_2S through its solution a white precipitate of ZnS is obtained as follows:

$$ZnCl_2 + H_2S \longrightarrow ZnS + 2HCl$$

Uses

1. Anhydrous $ZnCl_2$ is used as a dehydrating agent and in making dry cells, parchment paper and adhesives.
2. It is used to prevent timber against the action of microorganisms.
3. A mixture of syrupy $ZnCl_2$ and ZnO is used for dental filling.

SULPHATES

Ferrous Sulphate or Green Vitriol ($FeSO_4\ 7H_2O$)

It is also called Harakasis.

Preparation

1. **By the oxidation of Iron Pyrite:** Iron pyrite on oxidation in atmospheric air gives ferrous sulphate as follows:

$$\underset{\text{Iron pyrite}}{2FeS_2} + 7O_2 + 2H_2O \longrightarrow 2FeSO_4 + 2H_2SO_4$$

2. When scraps of iron are dissolved in dilute H_2SO_4 its solution is formed and it is crystallized by alcohol as $FeSO_4$ is partially soluble in it.

$$\underset{\text{Scrap}}{Fe} + H_2SO_4 \longrightarrow FeSO_4 + H_2$$
$$\text{dil.}$$

Physiochemical Properties

1. Atmospheric Oxidation: It is a light green powder which turns brown in air due to oxidation.

$$4FeSO_4 + 2H_2O + O_2 \longrightarrow \underset{\text{Basic ferric sulphate (Brown)}}{4Fe(OH).SO_4}$$

2. Heating Effect: On heating it decomposes as follows:

$$FeSO_4.7H_2O \xrightarrow[300^\circ C \quad 7H_2O]{\Delta} FeSO_4$$

$$2FeSO_4 \xrightarrow[\text{Temp.}]{\text{High}} Fe_2O_3 + SO_2 + SO_3$$

3. With KCN: With KCN, it forms potassium ferrocyanide.

$$FeSO_4 + 2KCN \longrightarrow Fe(CN)_2 + K_2SO_4$$

$$Fe(CN)_2 + 4KCN \longrightarrow K_4Fe(CN)_6$$

4. With Nitric Oxide: With nitric oxide it gives nitroso ferrous sulphate by absorbing NO.

$$FeSO_4 + NO \longrightarrow \underset{\substack{\text{Nitroso ferrous sulphate}\\ \text{(Brown coloured)}}}{FeSO_4.NO}$$

It is used in brown ring test for Nitrate ion.

5. Reducing Properties: It reduces others compounds and get oxidized into ferric sulphate.

For example,

With $KMnO_4$: It reduces and decolourizes acidified $KMnO_4$ solution into $MnSO_4$ as follows:

$$\underset{\text{Purple}}{10FeSO_4 + 2KMnO_4 + 8H_2SO_4} \longrightarrow$$

$$5Fe_2(SO_4)_3 + K_2SO_4 + \underset{\text{Colourless}}{2MnSO_4} + 8H_2O$$

With $K_2Cr_2O_7$: It reduces and decolourizes acidified $K_2Cr_2O_7$ solution into $Cr_2(SO_4)_3$ as follows:

$$\underset{\text{Orange}}{6FeSO_4 + K_2Cr_2O_7 + 7H_2SO_4} \longrightarrow$$

$$3Fe_2(SO_4)_3 + K_2SO_4 + \underset{\text{Green}}{Cr_2(SO_4)_3} + 7H_2O$$

With $HgCl_2$: It reduces mercuric chloride into mercurous chloride.

$$6HgCl_2 + 6FeSO_4 \longrightarrow 3Hg_2Cl_2 + 2Fe_2(SO_4)_3 + 2FeCl_3$$

With $AuCl_3$: It reduces auric chloride into gold as follows:

$$AuCl_3 + 3FeSO_4 \longrightarrow Au + FeCl_3 + Fe_2(SO_4)_3$$

6. With NH_4OH: When saturated solutions of pure ferrous sulphate and ammonium hydroxide are mixed and cooled in presence of sulphuric acid, the crystals of Mohr salt are formed.

$$FeSO_4 + (NH_4)_2SO_4 + 6H_2O \xrightarrow[40°C]{\Delta} FeSO_4.(NH_4)_2SO_4.6H_2O$$

Mohr's salt
[Pale green crystals]

Uses

1. It is used to prepare Mohr's Salt.
2. It is used to prepare Blue–Black ink (Tannin + $FeSO_4$).

Copper Sulphate or Blue Vitriol ($CuSO_4 \cdot 5H_2O$)

Preparation

1. It is prepared by dissolving cupric oxide or hydroxide or carbonate in dil. H_2SO_4 as follows.

$$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$$

$$Cu(OH)_2 + H_2SO_4 \longrightarrow CuSO_4 + 2H_2O$$

$$CuCO_3 + H_2SO_4 \longrightarrow CuSO_4 + CO_2 + H_2O$$

The solution of $CuSO_4$ on evaporation and crystallization gives blue crystals of Blue vitriol.

2. **Industrial Method:** When scraps of copper are taken in a perforated lead bucket in presence of dilute H_2SO_4 and air is blown, the crystals of copper sulphate is formed.

$$\underset{\text{dil.}}{Cu + H_2SO_4} + \tfrac{1}{2}O_2 \xrightarrow{\Delta} CuSO_4 + H_2O$$

Physiochemical Properties

1. It is a blue crystalline solid, quite soluble in H_2O.
2. Heating Effect: On heating it decomposes as follows:

$$CuSO_4.5H_2O \underset{2H_2O}{\overset{\text{Exposure}}{\rightleftharpoons}} \underset{\text{pale blue}}{CuSO_4.3H_2O}$$

$$CuSO_4.3H_2O \underset{2H_2O}{\overset{373K}{\rightleftharpoons}} \underset{\text{bluish white}}{CuSO_4.H_2O}$$

$$\underset{\text{White}}{CuSO_4\ (\text{anhydrous})} \underset{303\ K}{\rightleftharpoons}$$

$$CuSO_4 \xrightarrow{720°C} CuO + SO_3$$

$$SO_2 + \tfrac{1}{2}O_2$$

3. With KI: On reaction with KI, it gives cuprous iodide as follows:

$$2CuSO_4 + 4KI \longrightarrow \underset{\substack{\text{Cuprous iodide}\\ \text{White}}}{Cu_2I_2} + 2K_2SO_4 + I_2$$

4. With NH_4OH: On dissolving it in aqueous ammonia solution or NH_4OH, a blue complex tetraammine cupric sulphate is formed as follows:

$$CuSO_4 + 2NH_4OH \rightarrow Cu(OH)_2 + (NH_4)_2SO_4$$

$$Cu(OH)_2 + 2NH_4OH + (NH_4)_2SO_4 \longrightarrow \underset{\substack{\text{Tetraamine cupric}\\ \text{sulphate}}}{Cu(NH_3)_4SO_4} + 4H_2O$$

It is called Schwitzer's reagent which is used to dissolve cellulose during manufacture of artificial silk.

5. With KCN: With it, it forms a complex known as potassium cupro cyanide.

$$2CuSO_4 + 10KCN \longrightarrow \underset{\text{Potassium cupro cyanide}}{2K_3Cu(CN)_4} + 2K_2SO_4 + (CN)_2$$

6. With $K_4Fe(CN)_6$: With it, it forms a reddish brown precipitate of cupric ferrocyanide.

$$2CuSO_4 + K_4Fe(CN)_6 \rightarrow \underset{\substack{\text{Cupric ferrocyanide}\\ \text{[Red Brown ppt.]}}}{Cu_2Fe(CN)_6} + 2K_2SO_4$$

It is a test of Cu^{2} ions.

7. With KCNS: With it, it gives cupric sulphocyanide and when SO_2 is passed through the solution a white precipitate of cuprous sulphocyanide is obtained.

$$CuSO_4 + 2KCNS \longrightarrow \underset{\text{Cuptic sulphocyanide}}{Cu(CNS)_2} + K_2SO_4$$

$$2CuSO_4 + 2KCNS + SO_2 + 2H_2O \rightarrow \underset{\text{White ppt.}}{Cu_2(CNS)_2} + K_2SO_4 + 2H_2SO_4$$

$$2CuSO_4 + 2KCNS + SO_2 + 2H_2O \rightarrow Cu_2(CNS)_2 + K_2SO_4 + 2H_2SO_4$$

8. With $Na_2S_2O_3$ (Hypo): With excess hypo it gives sodium cupro thiosulphate as follows:

$$CuSO_4 + Na_2S_2O_3 \rightarrow Cu.S_2O_3 + Na_2SO_4$$

$$2CuS_2O_3 + Na_2S_2O_3 \rightarrow Cu_2S_2O_3 + Na_2S_4O_6$$

$$3Cu_2S_2O_3 + 2Na_2S_2O_3 \longrightarrow \underset{\substack{\text{Sodium cupro thio}\\\text{sulphate}}}{Na_4[Cu_6(S_2O_3)_5]}$$

Uses

1. It is used in electroplating, calicoprinting electrotyping and dyeing.
2. It is used as fungicide or germicide in agriculture.
3. Bordeaux Mixture ($CuSO_4$ + lime) is used to kill moulds and fungus on trees, potatoes etc.

Zinc Sulphate or White Vitriol ($ZnSO_4 \cdot 7H_2O$)

Preparation

It is prepared as follows:

$$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$$

$$ZnO + H_2SO_4 \rightarrow ZnSO_4 + H_2O$$

$$ZnCO_3 + \underset{\text{dil.}}{H_2SO_4} \longrightarrow ZnSO_4 + H_2O + CO_2$$

The solution of $ZnSO_4$ on conc. followed by crystallization gives colourless crystals of $ZnSO_4.7H_2O$.

Physiochemical Properties

1. It is a colourless, crystalline efflorescent solid which is soluble in H_2O.
2. Heating Effect: On heating it decomposes as follows:

$$ZnSO_4.7H_2O \xrightarrow[\text{Below } 70^\circ\ H_2O]{\text{Above } 40^\circ C} ZnSO_4.6H_2O$$

$$ZnSO_4.6H_2O \xrightarrow[-5H_2O]{\text{Above } 70^\circ C} ZnSO_4.H_2O$$

$$ZnSO_4.H_2O \xrightarrow[280^\circ C\ \ H_2O]{\Delta} \underset{\text{Anhydrous}}{ZnSO_4}$$

$$ZnSO_4 \xrightarrow{800^\circ C} ZnO + SO_3 \text{ or } SO_2 + \tfrac{1}{2} O_2$$

3. With NaOH: On adding excess of NaOH in its solution, sodium zincate is formed.

$$ZnSO_4 + 2NaOH \rightarrow Zn(OH)_2 + Na_2SO_4$$

$$Zn(OH)_2 + 2NaOH \longrightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2} + 2H_2O$$

4. With Na_2CO_3: On adding sodium carbonate solution in its solution a white precipitate of basic zinc carbonate is formed.

$$4ZnSO_4 + 4Na_2CO_3 + 3H_2O \rightarrow \underset{\substack{\text{Basic zinc carbonate}\\\text{(White ppt)}}}{ZnCO_3\ 3\,Zn(OH)_2} + 4Na_2SO_4 + 3CO_2$$

If we use sodium bicarbonate solution, zinc carbonate is formed.

$$ZnSO_4 + 2NaHCO_3 \rightarrow ZnCO_3 + Na_2SO_4 + H_2O + CO_2$$

Uses

1. It is used to prepare lithopone [$ZnO + BaSO_4$], a white famous pigment.
2. It is used in eye lotion.
3. It is used as a mordent in dyeing and calicoprinting.

Potassium Permanganate ($KMnO_4$)

Preparation

1. **From Pyrolusite (MnO_2):** When finely powdered ore is fused with KOH or K_2CO_3 in air, green coloured K_2MnO_4 is obtained.

$$2\,MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4 + 2H_2O$$

2. **Conversion of K_2MnO_4 into $KMnO_4$:** The green mass of K_2MnO_4 is extracted with H_2O and

the solution is treated with Cl_2 or O_3 or O_2 to oxidize K_2MnO_4 into $KMnO_4$ (purple solution).

$$2K_2MnO_4 + O_3 + H_2O \longrightarrow 2KMnO_4 + 2KOH + O_2$$

$$2K_2MnO_4 + Cl_2 \longrightarrow 2KMnO_4 + 2KCl$$

$$3K_2MnO_4 + 2CO_2 \longrightarrow 2KMnO_4 + 2K_2CO_3 + MnO_2$$

The purple solution of $KMnO_4$ is concentrated to get the purple crystals of $KMnO_4$.

3. Basic solution of K_2MnO_4 on electrolytic oxidation gives $KMnO_4$.

$$\underset{\text{Magnate}}{MnO_4^{-2}} \quad e^- \longrightarrow \underset{\text{Permagnetic}}{MnO_4^-}$$

At anode, O_2 is liberated.

At cathode, Hydrogen is liberated.

$$2H^+ + 2e^- \longrightarrow 2H \longrightarrow H_2$$

Physiochemical Properties

1. It has a dark purple needle like crystals with greenish metallic lustre.
2. Its melting point is 523K.
3. It is less soluble in cold water but readily in hot water (aqueous solution is pink).
4. Heating Effect: Here different products are obtained at different temperatures.

$$2KMnO_4 \xrightarrow{200°C} K_2MnO_4 + MnO_2 + O_2$$

$$3K_2MnO_4 \xrightarrow{500-700°C} \underset{\text{Pot. sub-magnate}}{2K_3MnO_4} + MnO_2 + O_2$$

$$2K_2MnO_4 \xrightarrow[500\ 700°C]{\Delta} \underset{\text{Pot. Maganite}}{2K_2MnO_3} + O_2$$

5. Reaction with Conc. H_2SO_4

$$2KMnO_4 + H_2SO_4 \longrightarrow Mn_2O_7 + K_2SO_4 + H_2O$$

$$2Mn_2O_7 \xrightarrow{\text{Explodes}} 4MnO_2 + 3O_2$$

Mn_2O_7 is a dark brown highly explosive liquid.

6. Oxidizing Action: It is a powerful oxidizing agent in neutral, acidic or alkaline medium.

(i) In Neutral Medium

$$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3[O]$$

It can oxidize H_2S into sulphur.

$$2KMnO_4 + 3H_2S \longrightarrow 2MnO_2 + 2KOH + 3S + 2H_2O$$

It can oxidize hypo into Na_2SO_4.

$$8KMnO_4 + \underset{\text{Hypo}}{3Na_2S_2O_3} + H_2O \longrightarrow 3Na_2SO_4 + 3K_2SO_4 + 8MnO_2 + 2KOH$$

(ii) In Alkaline Medium

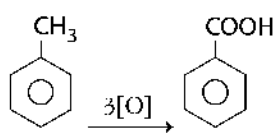
$$2KMnO_4 + 2KOH \longrightarrow \underset{\text{Colourless}}{2K_2MnO_4} + H_2O + [O]$$

For example,

It can oxide alkene into diol.

$$\underset{\text{Ethene}}{\begin{matrix}CH & H\\ \| & \\ CH & OH\end{matrix}} + [O] \longrightarrow \underset{\text{Glycol}}{\begin{matrix}CH_2OH\\ |\\ CH_2OH\end{matrix}}$$

It can oxidize toluene into benzoic acid.



It can oxidize KI into iodine.

$$KI + 3[O] \longrightarrow KIO_3$$

Alkaline $KMnO_4$ is called Baeyer's reagent and it is used to test unsaturation in organic compounds.

(iii) In Acidic Medium

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MSO_4 + 3H_2O + 5[O]$$

Or

$$2MnO_4^- + 6H^+ \longrightarrow 2M^{+2} + 3H_2O + 5[O]$$

For example,

It can oxidize $FeSO_4$ into $Fe_2(SO_4)_3$.

$$10FeSO_4 + 2KMnO_4 + 8H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O$$

It can oxidize HX into X_2.

$$2KMnO_4 + 3H_2SO_4 + 10HX \rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5X_2$$

It can oxidize oxalic acid into CO_2 and H_2O.

$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2$$

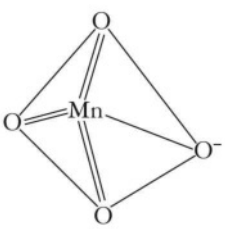
It can oxidize SO_2 into H_2SO_4.

$$2KMnO_4 + 5SO_2 + 2H_2O \rightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4$$

It can oxidize nitrites into nitrates.

$$2KMnO_4 + 5KNO_2 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5KNO_3 + 3H_2O$$

Structure



Tetrahedron structure

Mn–O is 1.63 Å
O–Mn–O is 109°

Uses

1. It used as an oxidizing agent.
2. It is used as a disinfectant and germicide.
3. It used in the manufacture of saccharin, boric acid, acetaldehyde etc.
4. It is used in volumetric estimation of Fe^{+2} salts, oxalate, H_2O_2 etc.

Potassium Dichromate ($K_2Cr_2O_7$)

Preparation

1. **From Chromite Ore ($FeCr_2O_4$ or $FeO\cdot Cr_2O_3$)** It is prepared by the following steps:

   (i) ***Formation of Na_2CrO_4***: Here, the ore is fused with molten alkali in presence of air in reverberatory furnace and the obtained fused mass is extracted with water followed by filtration gives Na_2CrO_4.

$$4FeCr_2O_4 + O_2 \rightarrow 2Fe_2O_3 + 4Cr_2O_3$$

$$4Cr_2O_3 + 8Na_2CO_3 + 6O_2 \longrightarrow 8Na_2CrO_4 + 8CO_2$$

$$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 2Fe_2O_3 + 8CO_2 + 8Na_2CrO_4$$

or

$$4FeCr_2O_4 + 16NaOH + 7O_2 \longrightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8H_2O$$

   (ii) ***Conversion of Na_2CrO_4 into $Na_2Cr_2O_7$:***

$$2Na_2CrO_4 + \underset{\text{dil.}}{H_2SO_4} \rightarrow \underset{\text{Separate first}}{Na_2Cr_2O_7} + Na_2SO_4 + H_2O$$

   (iii) ***Conversion of $Na_2Cr_2O_7$ into $K_2Cr_2O_7$:***

$$Na_2Cr_2O_4 + 2KCl \xrightarrow{\Delta} \underset{\text{Less soluble}}{K_2Cr_2O_7} + 2NaCl$$

   $K_2Cr_2O_7$ being less soluble can be easily obtained by fractional crystallization.

> **REMEMBER**
>
> $K_2Cr_2O_7$ is preferred over $Na_2Cr_2O_7$ in volumetric analysis as it is not hygroscopic like $Na_2Cr_2O_7$.

Physiochemical Properties

1. It has orange red coloured crystals with melting point 669K.
2. It is moderately soluble in cold water but readily soluble in hot water.
3. Heating Effect

$$4K_2Cr_2O_7 \xrightarrow{\Delta} \underset{\text{Potassium chromate}}{4K_2CrO_4} + \underset{\text{Chromic oxide}}{2Cr_2O_3} + 3O_2$$

4. Reaction with H_2SO_4

(i) With cold H_2SO_4, red crystals of chromic anhydride is formed.

$$K_2Cr_2O_7 + H_2SO_4 \longrightarrow \underset{\text{Conc.}}{2KHSO_4} + \underset{\text{Red crystals}}{2CrO_3} + H_2O$$

(ii) With hot Conc. H_2SO_4, chromium sulphate is formed.

$$2K_2Cr_2O_7 + 8H_2SO_4 \longrightarrow 2K_2SO_4 + 2Cr_2(SO_4)_3 + H_2O + 3O_2$$

5. Reaction with HCl: On reaction with HCl, Chromium chloride is formed.

$$K_2Cr_2O_7 + 14HCl \longrightarrow 2CrCl_3 + 2KCl + 7H_2O + 3Cl_2$$

6. Reaction with KOH: Here, a yellow solution of potassium chromate is formed.

$$2K_2Cr_2O_7 + 2KOH \longrightarrow \underset{\text{Yellow solution}}{4K_2CrO_4} + H_2O$$

On acidification, yellow solution again changes into orange-red potassium dichromate solution.

7. **Oxidizing Nature:** It is a very powerful oxidizing agent in acidic medium.

$$Cr_2O_7^{2-} + 14H^+ + 6e \rightarrow 2Cr^{3+} + 7H_2O$$

It can oxidize $FeSO_4$ into $Fe_2(SO_4)_3$.

$$\underset{\text{Orange}}{K_2Cr_2O_7} + 6FeSO_4 + 7H_2SO_4 \rightarrow K_2SO_4 + \underset{\text{Green}}{Cr_2(SO_4)_3} + 3Fe_2(SO_4)_3 + 7H_2O$$

It can oxidize KI into I_2.

$$K_2Cr_2O_7 + 7H_2SO_4 + 6KI \longrightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3I_2$$

It can oxidize SO_3^{2-} (Sulphite) into SO_4^{2-} (Sulphate).

$$Cr_2O_7^{-2} + 8H^+ + 3SO_3^{-2} \longrightarrow 2Cr^{+3} + 4H_2O + 3SO_4^{2-}$$

It can oxidize AsO_3^{3-} (arsenite) into AsO_4^{3-} (arsanate).

$$Cr_2O_7^{-2} + 8H^+ + 3AsO_3^{-3} \longrightarrow 2Cr^{+3} + 4H_2O + 3AsO_4^{-3}$$

It can oxidize HX into X_2.

$$K_2Cr_2O_7 + 4H_2SO_4 + 6HX \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3X_2$$

It can oxidize H_2S into sulphur.

$$K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3S$$

It can oxidize ethyl alcohol into acetaldehyde and acetaldehyde into acetic acid.

$$\underset{\text{Ethyl alcohol}}{C_2H_5OH} \xrightarrow{[O]} \underset{\text{Acetaldehyde}}{CH_3CHO} \xrightarrow{[O]} \underset{\text{Acetic acid}}{CH_3COOH}$$

8. Reaction with Hydrogen Peroxide: Acidified $K_2Cr_2O_7$ gives a deep blue colour with H_2O_2 due to formation of CrO_5.

$$K_2Cr_2O_7 + H_2SO_4 + 4H_2O \longrightarrow 2CrO_5 + K_2SO_4 + 5H_2O$$

9. Formation of Insoluble Chromates: It forms insoluble chromates with soluble salts of Pb, Ba, etc.

For example,

$$2Pb(NO_3)_2 + K_2Cr_2O_7 + H_2O \rightarrow \underset{\text{Lead chromate}}{PbCrO_4} + 2KNO_3 + 2HNO_3$$

10. Chromyl Chloride Test: When it is treated with NaCl and conc. H_2SO_4, orange-red vapours of chromyl chloride are formed which dissolve in NaOH to give yellow solution of sodium chromate.

$$K_2Cr_2O_7 + 4NaCl + \underset{\text{Conc.}}{6H_2SO_4} \xrightarrow{\Delta} 2KHSO_4 + 4NaHSO_4 + \underset{\text{Orange red vapours}}{2CrO_2Cl_2} + 3H_2O$$

$$CrO_2Cl_2 + 4NaOH \rightarrow 2NaCl + 2H_2O + \underset{\text{Yellow solution}}{Na_2CrO_4}$$

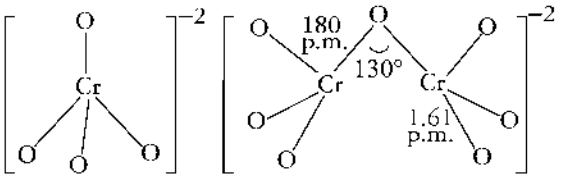
11. Sodium chromate gives yellow precipitate with lead acetate.

$$Na_2CrO_4 + Pb(CH_3COO)_2 \longrightarrow 2Na(CH_3COO)_2 + PbCrO_4$$

$$CrO_2Cl_2 + 2H_2O \rightarrow 2HCl + H_2CrO_4$$

Chromyl Chloride test is used for detection of Cl^- in any mixture.

Structure of Chromate and Dichromate



Chromate Dichromate

Uses It is used in photography for hardening of gelatin and in dyeing as mordant [$Cr(OH)_3$].

COMPOUNDS OF SILVER

Silver Oxide (Ag_2O)

Preparation

It is prepared by the reaction of silver nitrate with NaOH.

$$2AgNO_3 + 2NaOH \rightarrow Ag_2O + H_2O + 2NaNO_3$$

Silver nitrate

Physiochemical Properties

1. It is a brown precipitate which looks black when completely dried.
2. On slow heating it decomposes into Ag and O_2.

$$2Ag_2O \xrightarrow[330°C]{\Delta} 4Ag + O_2$$

Silver Bromide (AgBr)

Preparation

It is prepared by the reaction of silver nitrate with NaBr.

$$AgNO_3 + NaBr \rightarrow AgBr + NaNO_3$$

Physiochemical Properties

1. It is a pale yellow solid, insoluble in H_2O and concentrated acid.
2. It is slightly soluble in NH_4OH (strong solution) due to complex formation.

$$AgBr + 2NH_4OH \rightarrow Ag(NH_3)_2Br + 2H_2O$$

Diammine silver bromide

3. It is light sensitive and undergoes photoreduction.

$$2AgBr \xrightarrow{h\nu} 2Ag + Br_2$$

4. Reaction with Hypo: It dissolves in sodium thio sulphate or hypo to give soluble sodium argento thiosulphate.

$$AgBr + 2Na_2S_2O_3 \rightarrow Na_3Ag(S_2O_3)_2 + NaBr$$

Sod. argento thiosulphate

5. Reaction with KCN: It dissolves in KCN and forms soluble potassium argento cyanide.

$$AgBr + 2KCN \rightarrow K[Ag(CN)_2] + KBr$$

6. Reduction by dil. H_2SO_4 or Zn

$$Zn + H_2SO_4 \longrightarrow ZnSO_4 + 2H$$

$$2AgBr + 2(H) \longrightarrow 2Ag + 2HOH$$

Uses

AgBr is used in photographic plates as it is light sensitive. It gives metallic Ag strains when light is incident on it.

> **REMEMBER**
>
> AgF is fairly soluble in H_2O as the hydration energy of AgF is more than the lattice energy. However, other AgX are less soluble or insoluble as their hydration energy is less than lattice energy.

Silver Nitrate or Lunar Caustic $AgNO_3$

Preparation

It is prepared as follows:

$$3Ag + 4HNO_3 \text{ (dil.)} \xrightarrow{\Delta} 3AgNO_3 + NO + 2H_2O$$

Physiochemical Properties

1. It is a colourless, crystalline solid with a melting point of 485K and soluble in H_2O and alcohol.

2. It leaves black stains on skin or clothes due to metallic silver so it is called lunar caustic.

$$AgNO_3 \longrightarrow Ag + NO + O_2$$

3. Heating Effect:

$$2AgNO_3 \xrightarrow[\text{Above m.p.}]{\Delta} 2AgNO_2 + O_2$$

$$2AgNO_3 \xrightarrow[\text{Red Hot}]{\Delta} 2Ag + 2NO_2 + O_2$$

4. Precipitation Reaction: Solutions of many halides, sulphides, chromates etc., give a precipitate of silver salt with it.

For example:

$$AgNO_3 + NaCl \longrightarrow \underset{\text{white}}{AgCl\downarrow} + NaCl$$

$$AgNO_3 + NaBr \rightarrow \underset{\text{Pale yellow}}{AgBr\downarrow} + NaCl$$

$$AgNO_3 + NaI \rightarrow \underset{\text{Yellow}}{AgI\downarrow} + NaCl$$

$$3AgNO_3 + Na_3PO_4 \longrightarrow \underset{\substack{\text{Silver phosphate}\\ \text{[Yellow ppt.]}}}{Ag_3PO_4} + 3NaNO_3$$

$$2AgNO_3 + Na_2S \rightarrow \underset{\text{Black ppt.}}{Ag_2S} + 2NaNO_3$$

$$AgNO_3 + NaCNS \rightarrow \underset{\text{White ppt.}}{AgCNS} + NaNO_3$$

$$2AgNO_3 + K_2CrO_4 \longrightarrow \underset{\substack{\text{Silver chromate}\\ \text{Brick red ppt.}}}{Ag_2CrO_4} + 2KNO_3$$

5. Reaction with NaOH: With NaOH solution it gives silver oxide precipitate as follows:

$$2AgNO_3 + 2NaOH \rightarrow \underset{\substack{\text{Brown ppt but}\\ \text{turns black on standing}}}{Ag_2O} + 2NaNO_3 + H_2O$$

6. Reaction with KCN: It gives a white precipitate of potassium argento cyanide as follows:

$$AgNO_3 + KCN \longrightarrow AgCN + KNO_3$$

$$AgCN + KCN \rightarrow \underset{\substack{\text{Potassium argento}\\ \text{cyanide}}}{KAg(CN)_2}$$

7. Reaction with Hypo: On dissolving in excess of hypo it gives a precipitate of sodium argento thiosulphate.

$$2AgNO_3 + Na_2S_2O_3 \longrightarrow \underset{\substack{\text{White ppt.}\\ \text{which slowly turns black}}}{Ag_2S_2O_3} + 2NaNO_3$$

$$Ag_2S_2O_3 + H_2O \rightarrow \underset{\text{Black ppt.}}{Ag_2S} + H_2SO_4$$

$$2AgNO_3 + Na_2S_2O_3 \rightarrow Ag_2S_2O_3 + 2NaNO_3$$

$$Ag_2S_2O_3 + 3Na_2S_2O_3 \rightarrow \underset{\text{Sodium argento thiosulphate}}{2Na_3[Ag(S_2O_3)_2]}$$

8. Reaction with HX (except HF):

$$AgNO_3 + HX \rightarrow \underset{\text{ppt.}}{AgX} + HNO_3$$

9. Reaction with NH_4OH: When excess of NH_4OH is added in it diammine silver nitrate is formed.

$$2AgNO_3 + 2NH_4OH \rightarrow Ag_2O + 2NH_4NO_3 + H_2O$$

$$Ag_2O + 2NH_4NO_3 + 2NH_4OH \rightarrow \underset{\text{Di ammine silver nitrate}}{2[Ag(NH_3)_2]NO_3} + 3H_2O$$

10. Reactions of Ammonical Silver Nitrate Solution

$$2AgNO_3 + 2NH_4OH + C_2H_2 \rightarrow \underset{\text{Silver acetylide}}{Ag_2C_2} + 2NH_4NO_3 + 2H_2O$$

$$Ag_2O + C_6H_{12}O_6 \rightarrow 2Ag + \underset{\text{Gluconic acid}}{C_6H_{12}O_7}$$

$$Ag_2O + HCHO \rightarrow 2Ag + \underset{\text{Formic acid}}{HCOOH}$$

Uses

1. This ammonium solution of Ag_2O is called Tollen's reagent and it is used in testing –CHO group in aldehydes, glucose etc.
2. $AgNO_3$ is used in the preparation of AgX (used in photography) and making hair dye, ink etc.
3. It is also used in silvering of mirrors.

Straight Objective Type Questions (Single Choice)

1. An $FeCl_3$ solution reacts with sodium hydroxide to produce
   (a) Fe_3O_4
   (b) $Fe_2O_3.nH_2O$
   (c) Fe_2O_3 and FeO
   (d) FeO and $FeCl_3$

2. Which of the following is formed when ferrous oxalate is heated in the absence of air?
   (a) Fe_2O_3 (b) Fe_3O_4
   (c) Fe_2O_4 (d) FeO

3. An extremely hot copper wire reacts with steam to give
   (a) Cu_2O (b) CuO_2
   (c) Cu_2O_2 (d) CuO

4. Yellow coloured solution of $FeCl_3$ changes to light green when
   (a) Zn is added
   (b) $SnCl_2$ is added
   (c) H_2S gas is passed
   (d) All of these

5. $Fe(OH)_2$ dissolves in a concentrated NaOH solution giving a blue-green complex with the formula
   (a) $Na_2[Fe(OH)_6]$
   (b) $Na_4[Fe(OH)_4]$
   (c) $Na_4[Fe_2(OH)_6]$
   (d) $Na_4[Fe(OH)_6]$

6. In concentrated HCl, $FeCl_3.6H_2O$ forms an
   (a) $[FeCl_4]$ ion
   (b) $[Fe(H_2O)_6]^{3+}$ ion
   (c) $[FeCl_6]^{2-}$ ion
   (d) $[FeCl_4]^-$ ion

7. Which of the following methods is used to form a neutral ferric chloride solution?
   (a) Adding one to two drops of NH_3 to an $FeCl_3$ solution.
   (b) Addition dilute HCl to an $FeCl_3$ solution followed by the addition of an NaOH solution.
   (c) Adding an excess of NH_3 to an $FeCl_3$ solution.
   (d) Adding an excess of NaOH to an $FeCl_3$ solution.

8. On strong heating, $Fe_2(SO_4)_3$ gives
   (a) FeO and SO_3
   (b) Fe_2O_3 and SO_2
   (c) FeO and SO_2
   (d) Fe_2O_3 and SO_3

9. $FeCl_3$ reacts with $K_4[Fe(CN)_6]$ to give a blue precipitate known as prussian blue. prussian blue is
   (a) $K\,Fe^{III}[Fe^{II}(CN)_6]$
   (b) $Fe_4^{III}[Fe^{III}(CN)_6]$
   (c) $K\,Fe^{II}[Fe^{IV}(CN)_6]$
   (d) $Fe_4^{II}[Fe^{II}(CN)_6]$

10. Which of the following is calomel?
    (a) $HgBr_2$ (b) Hg_2Cl_2
    (c) $HgCl_2$ (d) $HgSO_4$

11. Which of the following statements about the oxidizing property of $KMnO_4$ in acidic medium is not correct?
    (a) H_2S is oxidized to S.
    (b) H_2S is oxidized to SO_4^{2-}
    (c) $C_2O_4^{2-}$ gets oxidized to CO_2
    (d) SO_3^{2} get oxidized to SO_4^{2}

12. Which of the following compounds can be used to standardize $KMnO_4$ solution?
    (a) Fe_3O_4 (b) $FeCl_3$
    (c) $FeSO_4.7H_2O$ (d) Mohr's salt

13. Sodium thiosulphate is used in photography because of its
    (a) reaction with light.
    (b) oxidizing behaviour.
    (c) reducing behaviour.
    (d) complex forming behaviour.

14. $FeSO_4$ reacts with $K_3[Fe(CN)_6]$ to give a blue precipitate known as Turnbull's blue. This precipitate is of

(a) $K\,Fe^{III}[Fe^{III}(CN)_6]$
(b) $K\,Fe^{II}[Fe^{III}(CN)_6]$
(c) $K\,Fe^{II}[Fe^{II}(CN)_6]$
(d) $K\,Fe^{III}[Fe^{II}(CN)_6]$

15. $KMnO_4$ can be decolourized by acidified
(a) $FeCl_2$
(b) $FeSO_4.7H_2O$
(c) $FeSO_4.(NH_4)_2SO_4.6H_2O$
(d) All of these

16. Amongst the following salts of iron, which is most unstable in aqueous solutions?
(a) FeI_3 (b) $FeSO_4.7H_2O$
(c) $K_3[Fe(CN)_6]$ (d) $Fe_2(SO_4).9H_2O$

17. Which of the following gives chocolate red precipitate with $K_4[Fe(CN)_6]$ in aqueous solution?
(a) $ZnSO_4$ (b) $Fe_2(SO_4)_3$
(c) $CuSO_4$ (d) $FeSO_4$

18. A $CuSO_4$ solution reacts with an Na_2CO_3 solution to give
(a) CuO
(b) $CuCO_3$
(c) $CuCO_3.Cu(HCO_3)_2$
(d) $CuCO_3.Cu(OH)_2$

19. Which of the following is formed when cupric oxide reacts with glucose?
(a) CuCHO (b) CuO
(c) Cu_2O (d) $C_5H_{11}O_6$

20. When excess of dilute NH_4OH is add to an aqueous solution of copper sulphate, an intense blue colour is obtained. This is due to formation of
(a) $Cu(OH)_2$ (b) $[Cu(NH_3)_4]^{2+}$
(c) $CuSO_4$ (d) $(NH_4)_2SO_4$

21. Which of the following compounds is known as bornite or peacock's ore?
(a) Cu_5FeS_4
(b) $CuAl_6(PO_4)_4(OH)_8.4H_2O$
(c) $YBa_2Cu_3O_7$
(d) $Cu_2S_2O_3$

22. When a $CuSO_4$ solution reacts with $K_4[Fe(CN)_6]$, the brown precipitate formed is of
(a) $Cu[Fe(CN)_6]$ (b) $Cu_2[Fe(CN)_6]_2$
(c) $Cu_2[Cu(CN)_6]$ (d) $Cu_2[Fe(CN)_6]$

23. When a $CuSO_4$ solution is treated with an excess of KCN, a colourless complex salt in which copper has a d^{10} configuration is obtained. This complex is
(a) $K_3[Cu(CN)_4]$ (b) $K_2[Cu_2(CN)_4]$
(c) $K_4[Cu(CN)_5]$ (d) $K_2[Cu(CN)_6]$

24. $CuSO_4$ reacts with an excess of hypo solution to give a Cu complex of
(a) $Cu_4[Na_6(S_2O_3)_5]$
(b) $Na_4[Cu_6(S_2O_3)_5]$
(c) $Cu_2S_2O_3$
(d) $Na_3[Cu(S_2O_3)_2]$

25. Silver bromide dissolves in hypo solution to give
(a) $[Ag(S_2O_3)_2]^5$
(b) $Ag_2S_2O_3$
(c) $[Ag_6(S_2O_3)_5]^{4+}$
(d) $[Ag(S_2O_3)_2]^-$

26. Which of the following reaction occurs on heating $AgNO_3$ to its melting point?
(a) $2AgNO_3 \rightarrow Ag + NO + O_2$
(b) $2AgNO_3 \rightarrow 2AgNO_2 + O_2$
(c) $2AgNO_3 \rightarrow 2Ag + 2NO_2 + O_2$
(d) $2AgNO_3 \longrightarrow 2Ag + N_2 + 3O_2$

27. A $ZnSO_4$ solution is boiled with an $NaHCO_3$ solution to produce
(a) $ZnHCO_3.ZnO$
(b) $Zn(OH)_2$
(c) $ZnCO_3$
(d) $ZnCO_3.Na_2SO_4$

28. $ZnSO_4$ is boiled with Na_2CO_3 to produce
(a) $ZnCO_3$
(b) $ZnCO_3.Zn(OH)_2$
(c) $ZnCO_3.Na_2SO_4$
(d) $ZnCO_3.ZnSO_4$

29. ZnO, a white solid, is a covalent molecule and adopts a
(a) Rock salt structure
(b) Spinel structure
(c) Diamond structure
(d) BCC structure

30. Zinc oxide is normally white but turns yellow on heating and becomes white again on cooling, because of

(a) Various types of lattice defects because of which the oxygen ion is lost during heating.
(b) Its high transition temperature (T_c).
(c) d-d transition spectra as well as crystal defect.
(d) The two dimensional network structure of ZnO.

31. Hg_2Cl_2 (calomel) and $HgCl_2$ (corrosive sublimate) react separately with liquor ammonia to produce respectively,
(a) $Hg(NH_2)Cl$ and $Hg(NH_2)Cl + Hg$
(b) $Hg(NH_2)Cl + Hg$ and $Hg(NH_2)Cl$
(c) $Hg(NH_2)Cl + Hg$ and $Hg(NH_2)Cl + Hg$
(d) $Hg(NH_2)Cl + HgCl_2$ and $Hg(NH_2)Cl + Hg_2Cl_2$

32. When an excess of $SnCl_2$ is added to an $HgCl_2$ solution, we get
(a) Hg (b) Sn
(c) Hg_2Cl_2 (d) $[SnCl_6]^{4}$

33. Calomel is made by treating
(a) $HgSO_4$ with NaCl
(b) $Hg_2(NO_3)_2$ and Hg with aqua regia
(c) $Hg_2(NO_3)_2$ with HCl
(d) $Hg(NO_3)_2$ with NaCl

34. Iron is rendered passive by treatment with concentrated
(a) HCl (b) HNO_3
(c) H_3PO_4 (d) H_2SO_4

35. Identify the reaction that does not take place during smelting process of copper extraction.
(a) $FeO + SiO_2 \rightarrow FeSiO_3$
(b) $Cu_2O + FeS \rightarrow Cu_2S + FeO$
(c) $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$
(d) $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$

36. $HgCl_2$ is a solid containing
(a) A pyramidal Cl–Hg–Cl molecule
(b) A T-shaped Cl–Hg–Cl molecule
(c) An angular Cl–Hg–Cl molecule
(d) A linear Cl–Hg–Cl molecule

37. $ZnSO_4$ reacts with the excess of a KCN solution to produce the complex ion
(a) $[Zn(CN)_4]^{2-}$ with a tetrahedral structure
(b) $[Zn(CN)_2]^{-}$ with a linear structure
(c) $[Zn(CN)_4]^{4-}$ with an octahedral structure
(d) $[Zn(CN)_4]^{2}$ with a square planar structure

38. In $[Ag(CN)_2]^{2-}$, the number of π bonds is
(a) 2 (b) 3
(c) 4 (d) 6

39. ZnO shows yellow colour on heating due to
(a) C-T spectra
(b) F-centres
(c) d-d transition
(d) Higher polarization caused by Zn^{2+} ion

40. Mercury on heating with aqua regia gives
(a) $Hg(NO_2)_2$ (b) Hg_2Cl_2
(c) $Hg(NO_3)_2$ (d) $HgCl_2$

41. Silver chloride dissolves in excess of NH_4OH. The cation present in this solution is
(a) $[Ag(NH_3)_6]^{+}$ (b) $[Ag(NH_3)_4]^{+}$
(c) $[Ag(NH_3)_2]^{+}$ (d) Ag^{-}

42. When excess of CN is added to $CuSO_4$ solution, the complex ion formed is
(a) $[Cu(CN_4]^{3}$ (b) $[Cu(CN)_4]^{2}$
(c) $[Cu(CN)_6]^{4}$ (d) $[Cu(CN)_5]^{4}$

43. Which of the following is highly corrosive salt?
(a) $FeCl_2$ (b) $HgCl_2$
(c) $PbCl_2$ (d) Hg_2Cl_2

44. Calomel (Hg_2Cl_2) on reaction with ammonium hydroxide gives
(a) HgO (b) Hg_2O
(c) $HgNH_2Cl$ (d) NH_2–Hg–Hg–Cl

45. With a very dilute solution of sodium thiosulphate, silver nitrate gives a white precipitate with quickly changes colour to yellow, brown and black due to formation of
(a) Ag_2S (b) $Ag_3S_4O_6$
(c) $Ag_2S_2O_3$ (d) Ag_2SO_4

46. When excess of KI is added to aqueous $CuSO_4$, the solution acquires dark brown colouration. This is due to the formation of
(a) Cu_2I_2 (s) (b) CuI_2 (s)
(c) I_2 (s) (d) I_3^{-} (aq)

47. On heating $AgNO_3$ above its melting point, the gas evolved is
(a) O_2 only (b) N_2 and O_2
(c) NO_2 only (d) NO_2 and O_2

48. $HgCl_2$ and $SnCl_2$ cannot coexist in a solution due to

(a) Solubility product
(b) Redox change
(c) Common ion effect
(d) All of these

49. Which of the following nitrates will leave behind a metal on strong heating?
(a) ferric nitrate
(b) copper nitrate
(c) manganese nitrate
(d) silver nitrate

50. Anhydrous $FeCl_2$ is made by
(a) Reacting Fe with conc. HCl
(b) Heating Fe with dilute HCl
(c) Heating Fe with gaseous HCl
(d) Heating Fe with Cl_2 gas

51. FeI_3 does not exist because
(a) of low lattice energy
(b) of large size
(c) iodine is not highly electronegative enough to oxidize Fe to Fe^{3+}
(d) Fe^{3+} oxidizes I^- to I_2

52. Silver nitrate produces a black stain on the skin due to
(a) Its reduction to metallic silver.
(b) Its corrosive action.
(c) The formation of a complex compound.
(d) Its strong reducing action.

53. Heating mixture of Cu_2O and Cu_2S will give
(a) $Cu + SO_2$ (b) $Cu + SO_3$
(c) $CuO + CuS$ (d) Cu_2SO_3

54. Cuprous chloride is obtained from cupric chloride
(a) By the electrolysis of cupric chloride containing HCl.
(b) By heating cupric chloride with chlorine.
(c) By passing H_2 over $CuCl_2$.
(d) By heating cupric chloride with conc. HCl and copper turnings.

55. The fraction of chlorine precipitated by $AgNO_3$ solution from $[Co(NH_3)_5Cl]Cl_2$ is
(a) 1/2 (b) 1/3
(c) 2/3 (d) 1/4

56. From an aqueous solution of zinc sulphate, normal zinc carbonate may be precipitated by
(a) Warming with $NaHCO_3$
(b) Passing CO_2
(c) Boiling with $CaCO_3$
(d) Adding Na_2CO_3

57. Identify the correct statement when $CuSO_4$ (aq) is mixed with KI
(a) Cu_2I_2 formed is white
(b) Cu^{2+} is reduced and I^- is oxidized
(c) The solution becomes brown due to liberated I_2
(d) All of these

58. Addition of $K_4[Fe(CN)_6]$ solution to $FeCl_3$ solution gives:
(a) Ferriferrocyanide
(b) Ferriferricyanide
(c) Ferroferricyanide
(d) None of these

59. For making good quality mirrors, plates of float glass are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be
(a) Mercury (b) Tin
(c) Iron (d) Zinc

60. When $AgNO_3$ is heated strongly the products formed are
(a) NO and NO_2 (b) NO_2 and O_2
(c) NO_2 and N_2O (d) NO and O_2

61. When mercury (II) chloride is treated with excess of stannous chloride, the products obtained are
(a) Liquid Hg and $[SnCl_4]^2$
(b) Hg_2Cl_2 and $[SnCl_4]^2$
(c) Liquid Hg and $SnCl_4$
(d) Hg_2Cl_2 and $SnCl_4$

62. In the silver plating of copper, $K[Ag(CN)_2]$ is used instead of $AgNO_3$. The reason is
(a) less availability of Ag^+ ions, as Cu cannot displace Ag from $[Ag(CN)_2]^-$ ion.
(b) more voltage is required.
(c) a thin layer of Ag is formed on Cu.
(d) Ag^+ ions are completely removed from the solution.

63. $HgCl_2$ is a covalent compound, sparingly soluble in water, the solubility increases by the addition of chloride ions due to
(a) Formation of complex $[HgCl_4]^{2-}$
(b) Strong ion dipole forces
(c) Common ion effect
(d) Weakening of Hg–Cl bonds

64. When Zn metal is added to $CuSO_4$ solution, Cu is precipitated, it is due to
(a) ionization of $CuSO_4$
(b) hydrolysis of $CuSO_4$
(c) reduction of Cu^2
(d) oxidation of Cu^{2+}

65. Ag_2S on reaction with NaCN, O_2 and water gives:
(a) $Na[Ag_2CN]$ (b) $Na[Ag(CN)_2]$
(c) $Na[Ag(CN)]$ (d) $Na_2[Ag(CN)_3]$

66. $CuSO_4.5H_2O$ on strong heating at 573K gives:
(a) $CuSO_4$ (b) $CuO + SO_2$
(c) $CuSO_4.3H_2O$ (d) $CuSO_4.H_2O$

67. Zinc carbonate is a precipitate from zinc sulphate solution by the addition of
(a) $MgCO_3$ (b) $NaHCO_3$
(c) Na_2CO_3 (d) $CaCO_3$

68. $Fe(OH)_3$ can be separated from $Al(OH)_3$ by the addition of
(a) NaOH solution
(b) NaCl solution
(c) Dil. HCl solution
(d) NH_4Cl and NH_4OH

69. Which of the following pairs of compounds can be precipitated when their aqueous solutions are mixed together?
(a) $Cu(NO_3)_2$ and $ZnCl_2$
(b) $AlCl_3$ and $ZnSO_4$
(c) KF and $AgNO_3$
(d) $FeSO_4$ and $BaCl_2$

70. Predict the product of the following reaction
$Cl_2 + HgO \longrightarrow$
(a) $Hg + Cl_2O$
(b) $HgCl_2 + Cl_2O$
(c) $Hg + HgCl_2 + O_2$
(d) $HgCl_2 + O_2$

Brainteasers Objective Type Questions (Single Choice)

71. In the brown ring test , the brown colour of the ring is due to
(a) Ferric nitrate
(b) Nitrosoferrous sulphate
(c) A mixture of NO and NO_2
(d) Ferrous nitrate

72. Compare the thermal stability of ZnO, CdO, HgO
(a) HgO < ZnO < CdO
(b) ZnO < CdO < HgO
(c) HgO < CdO < ZnO
(d) CdO < HgO < ZnO

73. An insoluble salt (X) is readily dissolves in concentrated ammonium acetate solution and the resulting solution on reaction with K_2CrO_4 gives a yellow precipitate which is insoluble in mineral acid. Here, (X) is
(a) Ag_2SO_4 (b) $PbSO_4$
(c) $BaSO_4$ (d) $HgSO_4$

74. Which compound is formed when excess of KCN is added to an aqueous solution of copper sulphate?
(a) $Cu(CN)_2$ (b) $K_2[Cu(CN)_6]$
(c) $K[Cu(CN)_2]$ (d) $K_3[Cu(CN)_4]$

75. Complete the following by identifying (A) to (D).

$CuSO_4.5H_2O \xrightarrow{100^oC} (A) \xrightarrow{230^oC}$

$(B) \xrightarrow{800^oC} (C) + (D)$

(a) A – $CuSO_4.2H_2O$, B – $CuSO_4$, C – Cu_2O, D = SO_3

(b) A = $CuSO_4.H_2O$, B = $CuSO_4$, C = CuO, D – SO_3

(c) A – $CuSO_4.H_2O$, B – $CuSO_4$, C – CuO, D = SO_2

(d) A = $CuSO_4.2H_2O$, B = $CuSO_4$, C = Cu, D – SO_3

76. $Fe(OH)_2$ is precipitated from Fe(II) solutions as a white solid but turns dark green and then brown due to the formation of

(a) $Fe(OH)_3$ (b) $Fe_2O_3.(H_2O)_n$

(c) $Fe_2O_3.2H_2O$ (d) $Fe(OH)_2$ and $Fe(OH)_3$

77. $FeSO_4$ can be estimated volumetricaly by titration with $K_2Cr_2O_7$ using $K_3[Fe(CN)_6]$ as an external indicator. How do you detect the end point?

(a) green to colourless

(b) blue to colourless

(c) colourless to green

(d) colourless to blue

78. A white precipitate of AgCl dissolves in excess of

(I) NH_3 (aq) (II) $Na_2S_2O_3$

(III) NaCN

(a) III only (b) I and II

(c) I, II and III (d) II and III

79. Copper sulphate is prepared by blowing a current of air through copper scrap and dilute H_2SO_4. Dilute HNO_3 is also added

(a) To oxidize Fe^{2+} to iron (III) sulphate, which remains in solution after crystallization of $CuSO_4$.

(b) To oxidize copper to Cu^{2+} which then form $CuSO_4$ with dilute H_2SO_4.

(c) Which combines with H_2SO_4 to give a very strong oxidizing mixture and oxidizes Cu to Cu^{2+}.

(d) To speed up the ionization of H_2SO_4 to give SO_4^{2-} ions.

80. A white precipitate of AgCl dissolves in excess of

(I) NH_3 (aq) (II) $Na_2S_2O_3$

(III) NaCN

(a) I only (b) III only

(c) I, II and III (d) I and II

81. $FeSO_4$ is used in brown ring test for nitrates and nitrites. In this test, a freshly prepared $FeSO_4$ solution is mixed with solution containing NO_2^- or NO_3^- and the conc. H_2SO_4 is run down the side of the test tube. If the mixture gets hot or is shaken,

(I) The brown colour disappear

(II) NO is evolved

(III) A yellow solution of $Fe_2(SO_4)_3$ is formed

(a) I, III correct (b) I, II, III correct

(c) Only I correct (d) II, II correct

82. $FeCl_3$ solution added to $K_4[Fe(CN)_6]$ gives (a) while with KSCN gives (b). (a) and (b) respectively are

(a) $Fe_4[Fe(CN)_6]_3$, $K_3[Fe(SCN)_6]$

(b) $Fe_4[Fe(CN)_6]_3$, $K_3[Fe(CNS)_6]$

(c) $Fe_3[Fe(CN)_6]_2$, $Fe(CNS)_3$

(d) $Fe_4[Fe(CN)_6]_3$, $KFe(CNS)_3$

83. $AgNO_3 \xrightarrow{\Delta} (A) + (B) + O_2$

$(B) + H_2O \longrightarrow HNO_2 + HNO_3$

$(A) + HNO_3 \longrightarrow (C) + NO + H_2O$

$(C) + Na_2S_2O_3$ (excess) $\rightarrow (D) + NaNO_3$

Identify (A) to (D).

(a) A – Ag_2O, B – N_2, C – $AgNO_3$, D = $Na[Ag(S_2O_3)_2]$

(b) A – Ag, B – NO_2, C – $AgNO_3$, D – $Na_3[Ag(S_2O_3)_2]$

(c) A = Ag, B = N_2O, C = $AgNO_3$, D – $Na_2[Ag(S_2O_3)_2]$

(d) A = Ag_2O, B = NO, C = $AgNO_3$, D = $Na_3[Ag(S_2O_3)_2]$

84. Aqueous solution of (P) can dissolve AgBr forming a soluble complex (Q). (P) also reacts with aqueous $AgNO_3$ solution giving white precipitate (R) changing to black precipitate (S). (P), (Q), (R) and (S) are

(a) P – $Na_2S_2O_3$, Q – $Ag_2S_2O_3$, R = Ag_2S, S = $Na_3[Ag(S_2O_3)_2]$

(b) P – $Na_2S_2O_3$, Q – $Na_3[Ag(S_2O_3)_2]$, R = $Ag_2S_2O_3$, S = Ag_2S

(c) P = NH_3, Q = $Na[Ag(OH)_2]$, R = Ag_2O, S = $[Ag(NH_3)_2]Br$
(d) P = NH_3, Q = $[Ag(NH_3)_2]Br$, R = $Ag(OH)$, S = Ag_2O

85. The black compound formed during the reaction between sodium thiosulphate and silver nitrate is
(a) silver sulphite (Ag_2SO_3)
(b) silver sulphate (Ag_2SO_4)
(c) silver sulphide (Ag_2S)
(d) silver thiosulphate ($Ag_2S_2O_3$)

86. The brown ring complex compound is formulated as $[Fe(H_2O)_5NO^+]SO_4$. The oxidation state of iron is
(a) 1 (b) 2
(c) 3 (d) 4

87. (A) is a coloured crystalline solid which is easily soluble in water. Addition of aqueous KCN gives a precipitate which dissolves in excess of reagent whereas addition of KI aqueous solution gives a white grey precipitate with the liberation of iodine. On gently heating, (A) loses 80 % of its water of crystallization. Identify (A) here?
(a) $ZnSO_4.7H_2O$ (b) $FeSO_4.7H_2O$
(c) $CuSO_4.5H_2O$ (d) Hg_2Cl_2

88. The fixing process of photographic film involves removal of unchanged silver bromide as
(a) $Na_2[AgBr(SO_3)]$
(b) $Na_3[AgBr(S_2O_3)]$
(c) $Na_3[Ag(SO_3)_2]$
(d) $Na_3[Ag(S_2O_3)_2]$

89. Amongst the following, the lowest degree of paramagnetism per mole of the compound at 298K will be shown by
(a) $MnSO_4.4H_2O$ (b) $CuSO_4.5H_2O$
(c) $FeSO_4.6H_2O$ (d) $NiSO_4.6H_2O$

90. A light blue coloured compound (A) on heating gives a black compound (B) which reacts with glucose to gives a red compound (C). (A), (B) and (C) are, respectively
(a) $[Cu(NH_3)_4]SO_4$, CuO, Cu_2O
(b) $Cu(OH)_2$, CuO, Cu_2O
(c) $CuSO_4.5H_2O$
(d) $Cu(OH)_2$, Cu_2O, CuO

91. A metal gives two chlorides (X) and (Y). (X) gives black precipitate with NH_4OH and (Y) gives white. With KI, (Y) gives a red precipitate soluble in excess of KI. (X) and (Y) are, respectively
(a) $ZnCl_2$ and $HgCl_2$
(b) $HgCl_2$ and $ZnCl_2$
(c) Hg_2Cl_2 and $HgCl_2$
(d) $HgCl_2$ and Hg_2Cl_2

92. Identify the compound (X) and the condition (B) here.

$$PbS \xrightarrow[\text{in air}]{\text{Heat}} (X) + PbS \xrightarrow{(Y)} Pb + SO_2$$

(a) X = $PbSO_3$, Y = low temperature
(b) X = PbO or $PbSO_4$, Y = high temperature (air supply is cut off)
(c) X = Pb_3O_4, Y = high temperature
(d) X = $PbSO_4$, Y = N_2

93. Fe^{3+} can be detected by the reaction

1. $Fe^{3+} + [Fe^{II}(CN)_6]^{4-} \longrightarrow \underset{\text{Deep blue}}{Fe^{III}[Fe^{II}(CN)_6]}$
2. $Fe^{3+} + [Fe^{III}(CN)_6]^{3-} \longrightarrow \underset{\text{Brown}}{Fe^{III}[Fe^{III}(CN)_6]}$
3. $Fe^{3+} + SCN^- \xrightarrow{H_2O} \underset{\text{Blood red}}{[Fe(SCN)(H_2O)_5]^{2+}}$

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1, 2 and 3
(d) 3 only

94. Sodium sulphide react with sodium nitroprusside to form a purple coloured compound. During the reaction, the oxidation state of iron
(a) Changes from +2 to +4
(b) Remains unchanged
(c) Changes from +2 to +3
(d) Changes from +3 to +2

95. The blue colour obtained in the Lassaigne test is due to the formation of the compound
(a) $Fe_4[Fe(CN)_6]_3$ (b) $Fe_4[Fe(CN)_6]$
(c) $Na_4[Fe(CN)_6]$ (d) $Fe_3[Fe(CN)_6]_4$

96. In the following sequence of reactions, identify (C) here?

$$FeCO_3 \xrightarrow{\text{Heat in air}} (A) \xrightarrow{\text{Boil with HCl}} (B) \xrightarrow{K_4Fe(CN)_6} (C)$$

(a) $K_3Fe(CN)_6$ (b) $Fe(CNS)_3$
(c) $Fe_4[Fe(CN)_6]_3$ (d) Both (b) and (c)

97. When dilute HCl is added to a solution of substance (X) which is water soluble, a white precipitate (Y) is formed. Treatment of (Y) with NH_4OH turns it black (Z). Treatment of (Y) with H_2S also turns it black. Identify (X) here?
(a) HgO (b) $Hg(NO_3)_2$
(c) $HgSO_4$ (d) $Hg_2(NO_3)_2$

98. $Zn(s) + Cr^{3+}(aq) + H_3O^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g) + H_2O(l) + A$.
A is
(a) $Cr^+(aq)$ (b) $CrO_4^{2-}(aq)$
(c) $Cr^{2+}(aq)$ (d) $Cr_2O_7^{2-}(aq)$

99. Arrange the following hydroxy compounds in order of increasing acid strength.
$CrO_2(OH)_2$ (I); $Cr(OH)_2$ (II); $Cr(OH)_3$ (III)
(a) II < III < I (b) III < I < II
(c) I < II < III (d) III < II < I

100. Compound (X)
1. On strongly heating it gives two oxides of sulphur.
2. On adding aqueous NaOH solution to its aqueous solution, a dirty green precipitate is obtained which starts turning brown on exposure to air.

Identify (X) here.
(a) $CuSO_4.5H_2O$ (b) $FeSO_4.7H_2O$
(c) $ZnSO_4.7H_2O$ (d) $Ag_2S_2O_3$

101. A white water insoluble solid (A) turns yellow on heating and becomes white on cooling (A) gives a clear solution (B) when treated with dilute HCl or NaOH. When H_2S is passed through a solution (B), after it is made neutral a white precipitate (C) is formed. Identify (A)?
(a) ZnO (b) CuO
(c) Fe_2O_3 (d) Ag_2O

102. A certain metal (a) is boiled in dilute nitric acid to give a salt (b) and an oxide of nitrogen (c). An aqueous solution of (b) with brine gives a precipitate (d) which is soluble in NH_4OH. On adding aqueous solution of (b) to hypo solution, a white precipitate (E) is obtained. (E) turns black on standing. Identify (a) and (E) here.
(a) Zn, Zn_2SO_4
(b) Ag, $Ag_2S_2O_3$
(c) Ag, Ag_2SO_4
(d) Fe, $Fe_2(SO_4)_3$

103. A colourless water soluble salt (X) on heating gives brown gas and leaves a metallic residue. Solution of (X) gives brick red precipitate with solution of K_2CrO_4. An ammonical solution of (X) gives a silver mirror with formic acid. Salt (X) is used in making indelible election ink. Identify the salt (X) here.
(a) AgCl (b) $HgNO_3$
(c) $AgNO_3$ (d) $Cu(NO_3)_2$

104. The number of ions formed on dissolving one molecule of $FeSO_4.(NH_4)_2SO_4.6H_2O$ is
(a) 4 (b) 5
(c) 6 (d) 3

105. An inorganic compound on heating gives a mixture of NO_2 and O_2 along with a residue and the aqueous solution of this compound gives a white precipitate with NaCl solution which dissolves in NH_4OH. Here, compound is
(a) $Pb(NO_3)_2$ (b) $Hg(NO_3)_2$
(c) $AgNO_3$ (d) $Cu(NO_3)_2$

Multiple Correct Answer Questions (More Than One Choice)

106. Select the correct statement(s).
   (a) $AgNO_3$ can be reduced into metallic silver by phosphine.
   (b) Organic tissues turn $AgNO_3$ black.
   (c) Minium is Hg_2Cl_2.
   (d) Vermilion is HgS.

107. Which statement is incorrect about $K_2Cr_2O_7$?
   (a) It gives coloured solution in water due to d-d electron transition.
   (b) It can act as an oxidant.
   (c) In acidic medium, its oxidation state becomes +6 to zero.
   (d) It is an orange-red crystalline solid.

108. Which of the following statement is correct when a mixture of $CaCl_2$ and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4 acid?
   (a) The vapours when passed into NaOH solution gives a yellow solution of Na_2CrO_4.
   (b) Chlorine gas is evolved.
   (c) Chromyl chloride is formed.
   (d) Deep red vapours are evolved.

109. Which is/are amphoteric oxide(s)?
   (a) CaO (b) Cs_2O
   (c) SnO (d) ZnO

110. SnO can be prepared by
   (a) boiling a stannous chloride solution with Na_2CO_3.
   (b) heating tin hydroxide in air.
   (c) heating tin oxalate (SnC_2O_4) in absence of air.
   (d) by heating Cassiterite with carbon monoxide.

111. Which of the following gives nitrogen dioxide gas on heating?
   (a) $Zn(NO_3)_2$ (b) $Pb(NO_3)_2$
   (c) KNO_3 (d) $AgNO_3$

112. Which of the following statement is/are correct here?
   (a) Mercury (II) oxide is thermally unstable and readily decomposes into mercury and oxygen on being heated above 400°C.
   (b) Mercury (II) oxide is thermally stable even at high temperatures.
   (c) Mercury (II) sulphide is precipitated from Hg^{2+} solution by passing H_2S in the presence of high concentration of hydrogen ions.
   (d) Mercury forms two types of oxides: HgO and Hg_2O.

113. The incorrect statement is/are
   (a) ZnO and Cu_2Cl_2 are colourless.
   (b) $KMnO_4$ and $K_2Cr_2O_7$ are coloured due to charge transfer spectra.
   (c) Green vitriol and blue vitriol are isomorphous.
   (d) Upon strong heating paramagnetic gases are evolved by $CuSO_4$ and $AgNO_3$.

114. Which statement(s) is/are correct regarding copper sulphate?
   (a) It gives CuO on strong heating in air.
   (b) It reacts with KI to give brown colouration.
   (c) It reacts with KCl to give Cu_2O.
   (d) It reacts with NaOH and glucose to give Cu_2O.

115. Which of the following does not give blue solution?
   (a) Addition of $CuSO_4$ in water.
   (b) Addition of $K_4[Fe(CN)_6]$ solution into $FeCl_3$ solution.
   (c) Addition of aq. NH_3 into $Cu(NO_3)_2$ solution.
   (d) A solution containing Fe^{2+} ion with $K_3[Fe(CN)_6]$ solution.

116. Select the correct statement(s).
   (a) In $K_2Cr_2O_7$, every chromium atom is linked with four oxygen atoms.
   (b) Blue colour of copper sulphate solution fades in excess of ammonia.

(c) A mixrure of alkaline $CuSO_4$ and sodium potassium rarrarare is Fehling's solurion.

(d) Mixrure of $CuSO_4$ + $Ca(OH)_2$ is called Bordeaux mixrure.

117. $CuSO_4 \xrightarrow{(X)} Cu(OH)_2$

$\rightarrow [Cu(NH_3)_4]\,SO_4$

(a) Borh $Cu(OH)_2$ and $[Cu(NH_3)_4]SO_4$ are pale blue precipirare.

(b) $Cu(OH)_2$ is paramagneric and $[Cu(NH_3)_4]$ is diamagneric.

(c) (X) is NaOH and (Y) is NH_4OH.

(d) Blue colour of solution is due to d-d transition.

118. The correct statement/s about FeO is/are

(a) Ir gives red colour with KCNS.

(b) Irs aqueous solurion changes ro $Fe(OH)_3$ and rhen ro $FeO_3.(H_2O)_n$ by atmospheric oxygen.

(c) Ir is non-stoichiomerric and metal deficienr.

(d) It is basic oxide.

119. Which one of the following statements are correct?

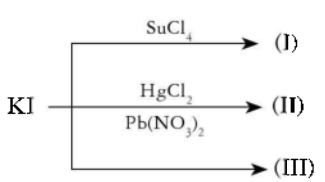
(a) $HgCl_2$ gives yellow precipirare with NaOH.

(b) $HgCl_2$ give HCl when rreared with sulphuric acid.

(c) $HgCl_2$ dissolve HCl is hot warer.

(d) Hg_2Cl_2 gives white precipitate with ammonium hydroxide.

120. The correct statement is/are



(a) A yellow precipirare of PbI_2 in (III).

(b) A whire precipirare of CuI, HgI_2 and PbI_2 in each case.

(c) A white precipitate of CuI in (I).

(d) An orange precipitate dissolving to HgI_4^{2-} in (II).

121. Find out the incorrect statement/s out of the following:

(a) Whire precipirare of $Zn(OH)_2$ is obrained on adding excess of NaOH ro aqueous $ZnSO_4$.

(b) Copper (I) salts are nor known in aqueous solurion.

(c) The srabiliry of eirher of $HgCl_2$ and $SnCl_2$ is nor affecred when presenr simulraneously in aqueous solurion.

(d) Borh $Cu(OH)_2$ and $Fe(OH)_2$ are soluble in aqueous NH_3.

122. Which of the following reagents is/are not used as primary standard?

(a) $KMnO_4$

(b) NaOH

(c) $K_2Cr_2O_7$

(d) $FeSO_4.(NH_4)_2\,SO_4.6H_2O$

123. NH_3 and its salts are identified by

(a) $HgCl_2$ + KI (excess) + NH_4OH

(b) Hg_2Cl_2 + excess of NH_4OH

(c) K_2HgI_4 and KOH

(d) $HgCl_2$ + KI (excess) + KOH

124. Which of the following acids, attack(s) on copper and silver?

(a) Dilute HCl (b) Aqua regia

(c) Dilute HNO_3 (d) Conc.H_2SO_4

125. Which of rhese are soluble in NH_3 solurion?

(a) AgI (b) Ag_2S

(c) AgCl (d) AgBr

Linked–Comprehension Type Questions

Comprehension–I

A pale yellow inorganic compound (K) is insoluble in mineral acid but dissolves in aqueous ammonia to give (L). It also dissolves in hypo solution to give (M). When an aqueous solution of (M) is boiled, a black precipitate of (N) is formed which on dissolved in HNO_3 and on adding HCl gives a white precipitate. When the compound (K) is heated with concentrated H_2SO_4 and MnO_2 brown fumes are observed.

126. The compound (K) is ... and the brown fumes are of ... respectively here
   (a) AgI, I^-
   (b) AgBr, Br
   (c) ZnS, S^2
   (d) $PbCl_2$, Cl

127. Here, the compound (M) and the black precipitate of (N) are, respectively
   (a) Ag_2S, $AgNO_3$
   (b) $AgNO_3$, AgCl
   (c) $Ag_2S_2O_3$, Ag_2S
   (d) AgCl, Ag_2S

128. Here, compound (K) on heating with H_2SO_4 and MnO_2 gives
   (a) Cl_2 (b) Br_2
   (c) I_2 (d) O_3

Comprehension–2

An aqueous solution of a white coloured compound (A) on reaction with HCl gives a white precipitate of compound (B). (B) becomes soluble in chlorine water with formation of (C). (C) reacts with KI to give a precipitate which becomes soluble in excess of it forming a compound (D) which is used for detecting a basic salt. When concentrated H_2SO_4 is added slowly into a mixture of cold solution of (A) and a metal sulphate, a brown ring compound (E) is formed.

129. Here, compounds (A) and (B) are, respectively
   (a) $Zn(NO_3)_2$ and $ZnCl_2$
   (b) $Hg_2(NO_3)_2$ and Hg_2Cl_2
   (c) $Hg(NO_3)_2$ and $HgCl_2$
   (d) $Ag(NO_3)_2$ and AgCl

130. Here, the compound (C) is
   (a) HgS (b) $ZnCl_2$
   (c) $HgCl_2$ (d) Hg_2Cl_2

131. Here, the compound (D) and the basic salt detected by it are, respectively
   (a) K_2HgI_4 and NH_4^+
   (b) Baeyer's reagent and NH_4^+
   (c) Tollen's reagent and NH_4^+
   (d) K_2HgI_4 and Ni^{2+}

132. Here, the brown ring compound is
   (a) Millon's reagent
   (b) $FeSO_4.NO$
   (c) $[Fe(H_2O)_5\ NO]SO_4$
   (d) $Fe_4[Fe(CN)_6]_3$

Comprehension–3

A scarlet compound (A) is treated with conc. HNO_3 to give a chocolate brown precipitate (B). The precipitate is filtered and the filtrate is neutralized with NaOH. Addition of KI to the resulting solution gives a yellow precipitate C. The precipitate B on warming with conc. HNO_3 in the presence of $Mn(NO_3)_2$ produces a pink coloured solution.

133. Here, scarlet coloured compound is
   (a) PbO_2 (b) Pb_3O_4
   (c) Fe_3O_4 (d) Fe_2O_3

134. Here, residue in the form of chocolate brown precipitate is of
   (a) PbO (b) Fe_2O_3
   (c) PbO_2 (d) FeO

135. Here, yellow precipitate (c) is of
   (a) $PbBr_2$ (b) PbI_2
   (c) I_2 (d) NaI

136. Here, pink coloured solution is of
   (a) $KMnO_4$ (b) $PbMnO_4$
   (c) $Pb(MnO_2)_2$ (d) $Pb(MnO_4)_2$

Comprehension–4

A bluish-green coloured compound (P) on heating gives two products (Q) and (R). A metal (S) is deposited on passing hydrogen through heated (Q). The compound (P) and (Q) are insoluble in water. (Q) is black in colour and when dissolved in HCl and treated with $K_4Fe(CN)_6$ gives a chocolate brown precipitate of complex (T). Here, (R) is a colourless, odourless gas which can turn lime water milky.

137. The compound (P) is
(a) $FeSO_4$ (b) $CuSO_4$
(c) $CuCO_3$ (d) $ZnCO_3$

138. The compounds (Q) and (R) are, respectively
(a) CuO, CO_2 (b) ZnO, CO_2
(c) CuS, SO_2 (d) FeO, H_2S

139. The compound (S) and (T) are, respectively
(a) Zn, $ZnCO_3$
(b) Cu, $Cu_2[Fe(CN)_6]$
(c) Zn, ZnO
(d) Fe, $Cu_2[Fe(CN)_6]$

Comprehension–5

A certain compound (X) is used in the laboratory for analysis, its aqueous solution gives the following reactions:

1. On addition to copper sulphate, a brown precipitate is obtained which turns white on addition of excess of $Na_2S_2O_3$ solution.
2. On addition of Ag^+ ion solution, a yellow curdy precipitate is obtained which is insoluble in ammonium hydroxide.

140. Here, (X) is
(a) PbI_2 (b) KI
(c) KBr (d) $PbCrO_4$

141. The white precipitate is of
(a) CuI_2 (b) PbI_2
(c) Cu_2Br_2 (d) Cu_2I_2

142. The yellow precipitate insoluble in NH_4OH is
(a) AgI (b) AgBr
(c) $AgCrO_4$ (d) Both (a) and (b)

Assertion and Reasoning Questions

In the following question two statements (Assertion) A and Reason (R) are given. Mark
(a) If A and R both are correct and R is the correct explanation of A.
(b) If A and R both are correct but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

143. (A): Concentrated aqueous solution of $CuCl_2$ is green in colour.
(R): The solution contains two complex ions i.e., $[Cu(H_2O)_4]^{2+}$ and $[CuCl_4]^{2-}$ in equilibrium.

144. (A): Magnetic moment value for copper (II) acetate is less than that for one unpaired electron.
(R): Interaction between unpaired electrons belonging to different copper atoms occurs via bridging acetate group.

145. (A): $KMnO_4$ is stored in dark bottles.
(R): On heating with alkalies, $KMnO_4$ is converted to manganate.

146. (A): All the iron–carbon bond distances in ferrocene are equal.
(R): The pi-electrons in the cyclopentadienyl group of ferrocene are delocalized.

147. (A): $KMnO_4$ cannot be used as primary standard.
(R): Its standard solution cannot be prepared as it is moderately soluble in water.

148. (A): $K_4[Fe(CN)_6]$ and $K_3[Fe(CN)_6]$ have same magnetic moment.
(R): Magnetic moment is controlled by the number of unpaired electrons.

149. (A): $AgNO_3$ is also called lunar caustic.
(R): $AgNO_3$ is photosensitive, therefore stored in dark coloured bottles.

150. (A): $HgCl_2$ and $SnCl_2$ exist together in aqueous solution.
(R): On heating, $HgCl_2$ sublimes.

151. (A): Silver fluoride is insoluble in water,
(R): Hydration energy of AgF is higher than its lattice energy.

152. (A): $K_2Cr_2O_7$ is preferred to $Na_2Cr_2O_7$ for use in volumetric analysis as a primary standard.
(R): $Na_2Cr_2O_7$ is hydroscopic while $K_2Cr_2O_7$ is not.

153. (A): A solution of ferric chloride on standing gives a brown precipitate.
(R): $FeCl_3$ possesses covalent bonds and chlorine bridge structure.

154. (A): The purple colour of $KMnO_4$ is due to the charge transfer transition.
(R): The intense colour, in most of the transition metal complexes, is due to d-d transition.

155. (A): When $KMnO_4$ solution is added to hot oxalic acid solution, the decolourization is slow in the beginning but becomes spontaneous after sometime.
(R): Mn^{2+} acts as autocatalyst.

156. (A): Mohr salt is used as a primary standard in volumetric analysis.
(R): Mohr salt contains both Fe^{2+} and Fe^{3+} ions in the crystalline salt.

157. (A): CuO can be reduced by C, H_2 as well as CO.
(R): CuO is basic oxide.

158. (A): Green vitriol always has traces of ferric ions.
(R): As air oxidizes ferrous ion into ferric ions.

159. (A): $KMnO_4$ and $K_2Cr_2O_7$ are intensely coloured compounds.
(R): Transition metal compounds having electrons in d-orbitals are coloured due to d-d transition.

160. (A): Pb^{4+} compounds are stronger oxidizing agents than Sn^{4+} compounds.
(R): The higher oxidation states for the group 14 elements are more stable for the heavier members of the group due to insert pair effect.

[IIT 2008]

Matrix–Match Type Questions

	p	q	r	s
(A)	O	O	O	O
(B)	O	O	O	O
(C)	O	O	O	O
(D)	O	O	O	O

161. Match the following:

Column I	Column II
a. $Fe(OH)_3$	(p) Green
b. $Cr(OH)_3$	(q) Red brown
c. $Ni(OH)_2$	(r) Blue
d. $Cu(OH)_2$	(s) +3 oxidation state of metal

162. Match the following:

Column I	Column II
a. Kipp's apparatus waste	(p) $CuCl_2.2H_2O$
b. Green coloured compound	(q) $FeSO_4$
c. Leave(s) brown residue on heating	(r) $Cu(OH)_2.CuCO_3$
d. Leave(s) black residue on heating	(s) $(NH_4)_2SO_4.FeSO_4.6H_2O$

163. Match the following:

Column I	Column II
a. Ag_2S	(p) Black precipitate
b. Hg_2O	(q) Blood red precipitate
c. Ag_2CrO_4	(r) White precipitate
d. $PbCl_2$	(s) Yellow precipitate

164. March the following:

Column I	Column II
a. Cu^{2+}	(p) Purple
b. Ni^{2+}	(q) Green
c. Fe^{2+}	(r) Yellow
d. Ti^{3+}	(s) Blue

165. Match the following:

Column I (Compound)	Column II (Can react with)
a. AgCl	(p) NH_4OH (aq)
b. $CuSO_4$	(q) $SnCl_2$
c. $FeCl_3$	(r) KI
d. $HgCl_2$	(s) NIO

166. March the following:

Column I (Metals)	Column II (Chlorides formed by them)
a. Fe	(p) MCl
b. Cu	(q) MCl_2
c. Th	(r) MCl_3
d. La	(s) MCl_4

167. March the following:

Column I (Compound)	Column II (Uses)
a. AgBr	(p) As purgative in medicine
b. Hg_2Cl_2	(q) In photographic films
c. $Na_2S_2O_3$	(r) As mordant in dyeing
d. $FeSO_4$	(s) In preparation of Mohr's salt

168. Match the following:

Column I	Column II
a. Baeyer's reagent	(p) $CuSO_4$ + $Ca(OH)_2$
b. Equivalent mass = 158	(q) Alkaline $KMnO_4$
c. Bordeaux mixture	(r) Detection of unsaturation in organic compounds
d. Nessler's reagent	(s) Complex of mercury

169. March the following:

Column I	Column II
a. $2CuSO_4 + 2NaCl + SO_2 + H_2O \rightarrow$	(p) Basic carbonate of metal
b. $Cu + H_2SO_4 \rightarrow$	(q) Oxidation of Cu to Cu^{2+}
c. $CuSO_4 + Na_2S_2O_3 \xrightarrow{\text{excess}}$	(r) Sodium cupro thiosulphate
d. $ZnSO_4 + Na_2CO_3 + 3H_2O \rightarrow$	(s) Reduction

170. March the following:

Column I	Column II
A. $FeO.Fe_2O_3$	(p) NaCl
B. FeC_2O_4	(q) Iodometrically with $Na_2S_2O_3$
C. $CuSO_4$	(r) $K_2Cr_2O_7$ in acidic medium
D. $AgNO_3$	(s) $KMnO_4$ in acidic medium

171. Match the following:

Column I	Column II
A. Baeyer's reagent	(p) Detection of unsaturation in organic compounds
B. Equivalent mass = 158	(q) 1 % alkaline $KMnO_4$
C. Bordeaux mixture	(r) Complex of mercury
D. Nessler's reagent	(s) $CuSO_4 + Ca(OH)_2$

The IIT–JEE Corner

172. Which of the following compounds is expected to be coloured?

(a) Ag_2SO_4 (b) CuF_2

(c) MgF_2 (d) CuCl

[*IIT 1997*]

173. Which of the following statement is correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?

1. a deep red vapour is evolved.
2. the vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4.
3. chlorine gas is evolved.
4. chromyl chloride is formed.

(a) 1, 2, 4 (b) 1, 2 ,3

(c) 2, 3, 4 (d) all are correct

[*IIT 1997*]

174. In nitroprusside ion, the iron and NO exist as Fe^{II} and NO^+ rather than Fe^{III} and NO. These forms can be differentiated by:

(a) Estimating the concentration of iron.

(b) Measuring the concentration of CN^-.

(c) Measuring the solid state magnetic moment.

(d) Thermally decomposing the compound.

[*IIT 1997*]

175. In the standardization of $Na_2S_2O_3$ using $K_2Cr_2O_7$ by iodometry, the equivalent weight of $K_2Cr_2O_7$ is

(a) (molecular weight)/2

(b) (molecular weight)/6

(c) (molecular weight)/3

(d) same as molecular weight.

[*IIT 1997*]

176. Anhydrous ferric chloride is prepared by:

(a) heating hydrated ferric chloride at a high temperature in a stream of air.

(b) heating metallic iron in a stream of dry chlorine gas.

(c) reaction of ferric oxide with hydrochloric acid.

(d) reaction of metallic iron with hydrochloric acid.

[*IIT 1997*]

177. When MnO_2 is fused with KOH, coloured compound is formed, the product and its colour is

(a) K_2MnO_4, purple green

(b) $KMnO_4$, purple

(c) Mn_2O_3, brown

(d) Mn_3O_4, black

[*IIT 2003*]

178. The product of oxidation of I with MnO_4^- in alkaline medium is

(a) IO_3^- (b) I_2

(c) IO^- (d) IO_4^-

[*IIT 2004*]

179. $CuSO_4$ decolourize on addition of KCN, the product is:

(a) $[Cu(CN)_4]^{2-}$

(b) Cu^{2+} gets reduced to form $[Cu(CN)_4]^{3-}$

(c) $Cu(CN)_2$

(d) CuCN

[*IIT 2006*]

180. A solution, when diluted with H_2O and boiled, gives a white precipitate. On addition of excess NH_4Cl/NH_4OH, the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in NH_4OH/NH_4Cl:

(a) $Zn(OH)_2$ (b) $Al(OH)_3$

(c) $Mg(OH)_2$ (d) $Ca(OH)_2$

[*IIT 2006*]

181. Sodium fusion extract, obtained from aniline, on treatment with iron (II) sulphate and H_2SO_4 in presence of air gives a Prussian blue precipitate. The blue colour is due to the formation of

(a) $Fe_4[Fe(CN)_6]_3$ (b) $Fe_3[Fe(CN)_6]_2$

(c) $Fe_4[Fe(CN)_6]_2$ (d) $Fe_3[Fe(CN)_6]_3$

[*IIT 2007*]

182. Among the following the coloured compound is

(a) CuCl (b) $K_3[Cu(CN)_4]$

(c) CuF_2 (d) $[Cu(CH_3CN)_4]BF_3$

[*IIT 2008*]

ANSWERS

Straight Objective Type Questions

1. (b)	2. (d)	3. (d)	4. (b)
5. (d)	6. (a)	7. (a)	8. (d)
9. (a)	10. (b)	11. (b)	12. (d)
13. (d)	14. (b)	15. (d)	16. (a)
17. (c)	18. (d)	19. (c)	20. (b)
21. (a)	22. (d)	23. (a)	24. (b)
25. (a)	26. (b)	27. (c)	28. (b)
29. (c)	30. (a)	31. (b)	32. (a)
33. (c)	34. (b)	35. (b)	36. (d)
37. (a)	38. (c)	39. (b)	40. (d)
41. (c)	42. (a)	43. (b)	44. (c)
45. (a)	46. (d)	47. (a)	48. (b)
49. (d)	50. (c)	51. (d)	52. (a)
53. (a)	54. (d)	55. (c)	56. (a)
57. (d)	58. (a)	59. (a)	60. (b)
61. (c)	62. (a)	63. (a)	64. (c)
65. (b)	66. (a)	67. (b)	68. (d)
69. (d)	70. (b)		

Brainteasers Objective Type Questions

71. (b)	72. (b)	73. (b)	74. (d)
75. (b)	76. (b)	77. (b)	78. (c)
79. (a)	80. (c)	81. (b)	82. (a)
83. (b)	84. (b)	85. (c)	86. (a)
87. (c)	88. (d)	89. (b)	90. (b)
91. (c)	92. (b)	93. (c)	94. (b)
95. (a)	96. (c)	97. (d)	98. (c)
99. (a)	100. (b)	101. (a)	102. (b)
103. (c)	104. (b)	105. (c)	

Multiple Correct Answer Questions

106. (a), (b), (d)	107. (a), (c)
108. (a), (c), (d)	109. (c), (d)
110. (a), (b), (c)	111. (a), (b), (d)
112. (a), (c), (d)	113. (b), (c), (d)
114. (a), (b), (d)	115. (a), (b), (c)
116. (a), (c), (d)	117. (a), (c), (d)
118. (b), (c), (d)	119. (a), (b), (c)
120. (a), (c), (d)	121. (a), (c), (d)
122. (a), (b), (c)	123. (c), (d)
124. (b), (c), (d)	125. (c), (d)

Comprehension–1

126. (b) 127. (c) 128. (b)

Comprehension–2

129. (b) 130. (c) 131. (a) 132. (b)

Comprehension–3

133. (b) 134. (c) 135. (b) 136. (d)

Comprehension–4

137. (c) 138. (a) 139. (b)

Comprehension–5

140. (b) 141. (d) 142. (a)

Assertion and Reasoning Questions

143. (a)	144. (a)	145. (b)	146. (a)
147. (a)	148. (c)	149. (b)	150. (d)
151. (a)	152. (a)	153. (b)	154. (c)
155. (a)	156. (c)	157. (b)	158. (a)
159. (b)	160. (c)		

Matrix–Match Type

161. (a)-(q), (s), (b)-(p), (s), (c)-(p), (d)-(r)

162. (a)-(q), (b)-(p), (q), (r), (s), (c)-(q), (s), (d)-(r)

163. (a)-(p), (b)-(p), (c)-(q), (d)-(r)

164. (a)-(s), (b)-(q), (c)-(r), (d)-(p)

165. (a)-(p), (b)-(p), (r), (c)-(p), (q), (s), (d)-(p), (q), (r)

166. (a)-(q), (b)-(p), (c)-(s), (d)-(r)

167. (a)-(q), (b)-(p), (c)-(q), (d)-(r), (s)

168. (a)-(r), (b)-(q), (c)-(p), (d)-(s)

169. (a)-(s), (b)-(q), (c)-(r), (d)-(p)

170. (a)-(p), (b)-(r), (c)-(q), (d)-(s)

171. (a)-(p), (q), (b)-(q), (c)-(s), (d)-(r)

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172. (b) 173. (a) 174. (d) 175. (b)

176. (b) 177. (a) 178. (a) 179. (d)

180. (a) 181. (a) 182. (c)

HINTS AND EXPLANATIONS

Straight Objective Type Questions

12. Only the Mohr's salt, containing Fe(II), is a substance of primary standard.

13. Hypo solution ($Na_2S_2O_3$) is used in photography to remove the unaffected AgBr in the form of soluble complex.

$$AgBr + Na_2S_2O_3 \longrightarrow \underset{\text{Sod. argentothiosulphate}}{Na_3[Ag(S_2O_3)_2]} + 2NaBr$$

16. Here I^- ion reduces Fe (III) to Fe (II).

$$FeI_3 \longrightarrow FeI_2 + \tfrac{1}{2} I_2$$

17. $2CuSO_4 + K_4[Fe(CN)_6] \longrightarrow Cu_2[Fe(CN)_6] + 2K_2SO_4$

34. Conc. HNO_3 renders iron passive by forming a thin protective film of Fe_3O_4 on its surface.

35. $Cu_2O + FeS \longrightarrow Cu_2S + FeO$

does not take place during the smelting of copper ore ($CuFeS_2$)

38. In $[Ag(CN)_2]^-$, CN has π bonds C ≡ N so two π-bonds are present in one CN so, the number of π bonds are $2 \times 2 - 4$.

41. Ag forms a complex ion, diammine silver(I) chloride, in which cation is represented as $[Ag(NH_3)_2]^+$.

42. $2Cu^{2+} + 4CN^- \longrightarrow Cu_2(CN)_2 + (CN)_2$

$Cu_2(CN)_2 + 6CN^- \longrightarrow 2[Cu(CN)_4]^{3-}$

43. Corrosive sublimate, $HgCl_2$ is highly corrosive salt.

44. $Hg_2Cl_2 + 2NH_4OH \longrightarrow$

$Hg + Hg(NH_2)Cl + 2H_2O + NH_4Cl$

46. $2CuSO_4 + 4KI \longrightarrow Cu_2I_2 + 2K_2SO_4 + I_2$

$I_2(s) + I^-(aq) \longrightarrow I_3^-(aq)$

47. $2AgNO_3 \xrightarrow{\text{heat}} 2AgNO_2 + O_2$

48. $2HgCl_2 + SnCl_2 \longrightarrow SnCl_4 + \underset{\text{White}}{Hg_2Cl_2}$

$SnCl_2 + Hg_2Cl_2 \longrightarrow SnCl_4 + \underset{\text{Grey}}{2Hg}$

52. Silver nitrate produces a black stain on skin due to its reduction into metallic silver.

$Ag^+ + e^- \longrightarrow Ag$,

53. $Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$

This is an example of auto reduction.

54. $CuCl_2 + Cu \xrightarrow{HCl} Cu_2Cl_2$

55. $[Co(NH_3)_5Cl]Cl_2$ ionises into $[Co(NH_3)_5Cl]^{2+}$ and Cl^- ion. These $2Cl^-$ react with Ag^+ to form white ppt. of AgCl.

56. $ZnSO_4 + 2NaHCO_3 \rightarrow ZnCO_3 + CO_2 + H_2O$

57. $2CuSO_4 + 4KI \longrightarrow \underset{\text{White}}{Cu_2I_2} + 2K_2SO_4 + I_2$

58. $FeCl_3 + K_4[Fe(CN)_6] \rightarrow$

$\underset{\substack{\text{Ferri-ferrocyanide}\\ \text{(Prussian blue)}}}{Fe_4[Fe(CN)_6]_3} + 12KCl.$

60. $2AgNO_3 \xrightarrow{\text{red heat}} 2Ag + 2NO_2 + O_2$

62. Due to less availability of Ag^+ ions, as Cu cannot displace Ag from $[Ag(CN)_2]^-$ ion.

64. Zinc being more electropositive displaces copper or reduces it.

68. Due to common ion effect of NH_4^+, concentration of OH^- decreases, as $K_{sp}Fe(OH)_3 < K_{sp}Al(OH)_3$ thus $Fe(OH)_3$ get precipitated first.

70. As chlorine reduces mercuric to mercurous and itself gets oxidized.

Brainteasers Objective Type Questions

71. NO forms dark brown nitrosoferrous sulphate.

$\underset{\text{Impure gas}}{FeSO_4} + NO \longrightarrow \underset{\substack{\text{dark brown}\\ \text{sulphate}}}{FeSO_4 \cdot NO\ (\text{Nitrosoferrous})}$

When this solution is heated, pure nitric oxide is liberated.

$FeSO_4 \cdot NO \xrightarrow{\text{heat}} \underset{\text{Pure gas}}{FeSO_4} + NO$

74. Firstly, Cu^{2+} ion is reduced to Cu^+ by CN which then forms complex with it.

$2Cu^{2+} + 2CN^- \rightarrow 2Cu^+ + (CN)_2$

$Cu^+ + CN^- \rightarrow CuCN \xrightarrow{3KCN}$

$K_3[Cu(CN)_4]$

75. $CuSO_4.5H_2O \xrightarrow{100^\circ C} \underset{\text{(A)}}{CuSO_4.H_2O} \xrightarrow{230^\circ C}$

$\underset{\text{(B)}}{CuSO_4} \xrightarrow{800^\circ C} \underset{\text{(C)}}{CuO} + \underset{\text{(D)}}{SO_3}$

77. Fe^{2+} salts give blue colour with $K_3[Fe(CN)_6]$ and it is used as an external indicator. After all the Fe^{2+} has been completely oxidized there will be no colour when tested with $K_3[Fe(CN)_6]$.

83. $AgNO_3 \xrightarrow{\Delta} \underset{\text{(A)}}{Ag} + \underset{\text{(B)}}{NO_2} + \frac{1}{2}O_2$

$\underset{\text{(B)}}{NO_2} + H_2O \rightarrow HNO_2 + HNO_3$

$\underset{\text{(A)}}{3Ag} + 4HNO_3 \rightarrow \underset{\text{(C)}}{3AgNO_3} + NO + 2H_2O$

$AgNO_3 + 2Na_2S_2O_3 \longrightarrow$

$\underset{\text{(D)}}{Na_3[Ag(S_2O_3)_2]} + NaNO_3$

85. $2AgNO_3 + Na_2S_2O_3 \rightarrow Ag_2S_2O_3 + 2NaNO_3$

$Ag_2S_2O_3 + H_2O \longrightarrow H_2SO_4 + \underset{\text{Black}}{Ag_2S}$

86. Let O. N. of Fe be x then $\{Fe(H_2O)_5NO\}^{2+}SO_4^{2-}$

$x + 1 = +2$

$x = 2 - 1 = +1$

87. $CuSO_4.5H_2O \xrightarrow{110^\circ C} \underset{\text{Slightly blue}}{CuSO_4.H_2O}$

$\%$ loss of water $= \frac{4}{5} \times 100 = 80\ \%$

$2CuSO_4 + 4KI \longrightarrow 8K_2SO_4 + \underset{\text{White}}{Cu_2I_2}\downarrow + I_2\uparrow$

$CuSO_4 + 2KCN \longrightarrow Cu(CN)_2 + K_2SO_4$

$2Cu(CN)_2 \rightarrow Cu_2(CN)_2 + \underset{\text{Cyanogen}}{(CN)_2}$

$Cu_2(CN)_2 + 6KCN \rightarrow \underset{\substack{\text{Potassium tetracyano}\\ \text{Copper(I) (colourless)}}}{2K_3Cu(CN)_4}$

88. $AgBr + 2Na_2S_2O_3 \longrightarrow$

$Na_3[Ag(S_2O_3)_2] + NaBr$

89. $CuSO_4.5H_2O$ has lowest number of unpaired d-electrons and lowest degree of paramagnetism.

90. $Cu(OH)_2 \xrightarrow{heat} CuO + H_2O$
Light blue Black

$2CuO + glucose \longrightarrow Cu_2O$
Red

91. Hg_2Cl_2 forms black precipitate of $Hg(Cl)NH_2 + Hg$ and $HgCl_2$ forms white precipitate of $Hg(Cl)NH_2$.

94. The reaction does not involve the change in oxidation number of iron.

$Na_2[IIIFe(CN)_5NO] + Na_2S \longrightarrow$

$Na_4[IIIFe(CN)_5NOS]$

95. The blue colour obtained in the Lassaigne's test is due to the formation of $Fe_4[Fe(CN)_6]_3$.

96. $Fe_2O_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O$
(A) (B)

$3K_4Fe(CN)_6 + 4FeCl_3 \longrightarrow$

$Fe_4[Fe(CN)_6]_3 + 12KCl$
(C) Prussian blue

97. $Hg_2(NO_3)_2 + 2HCl \longrightarrow Hg_2Cl_2 + 2HNO_3$
(X) White ppt.

$Hg_2Cl_2 \xrightarrow{2NH_3} Hg + Hg(NH_2)Cl + NH_4Cl$
Black

$Hg_2Cl_2 \xrightarrow{H_2S} Hg + HgS + 2HCl$
Black

98. Zinc reduces Cr^{3+} to Cr^{2+}.
$2Zn\ (s) + 2Cr^{3+}\ (aq) + 2H_3O^-\ (aq) \rightarrow$
$2Zn^{2+}\ (aq) + H_2\ (g) + 2H_2O\ (l) + 2Cr^{2+}\ (aq)$

99. Acid strength increases with the polarity of the bond which increases with the increase in oxidation state.

100. (1) (X) gives on heating two oxides of sulphur and so (X) is $FeSO_4.7H_2O$.

$FeSO_4.7H_2O \xrightarrow{\Delta} FeSO_4 + 7H_2O$
(X)

$2FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_2 + SO_3$

2. $FeSO_4 + 2NaOH \rightarrow Fe(OH)_2 + Na_2SO_4$

$Fe^{2} \xrightarrow[Alkaline\ medium]{Oxidation} Fe^{3+} + e$

101. $ZnO + 2HCl \longrightarrow ZnCl_2 + H_2O$
(A) (B)

$ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$

$ZnCl_2 + H_2S \xrightarrow{pH > 8} ZnS + 2HCl$
(C)

102. $3Ag + 4HNO_3 \rightarrow 3AgNO_3 + NO + 2H_2O$
(A) (B) (C)

$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
(B) (D)

$AgCl + 2NH_4OH \rightarrow Ag(NH_3)_2Cl + 2H_2O$
(D) Soluble

$2AgNO_3 + Na_2S_2O_3 \rightarrow Ag_2S_2O_3 + 2NaNO_3$
(B) (E) White

$Ag_2S_2O_3 \longrightarrow Ag_2S + SO_3$
(E) Black

103. $2AgNO_3 \xrightarrow{\Delta} 2Ag + 2NO_2 + O_2$
(X)

$2AgNO_3 + K_2CrO_4 \rightarrow Ag_2CrO_4 + 2KNO_3$
Red ppt.

$AgNO_3 + NH_4OH \rightarrow AgOH + NH_4NO_3$

$AgOH + 2NH_3 \longrightarrow Ag(NH_3)_2 + OH$

$HCOOH + Ag_2O \longrightarrow 2Ag + CO_2 + H_2O$
Silver mirror

104. Here the compound is silver nitrate.

$2AgNO_3 \longrightarrow 2Ag + 2NO_2 + O_2$

$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

$AgCl + 2NH_4OH \rightarrow Ag(NH_3)_2Cl + 2H_2O$

Multiple Correct Answers Type Questions

105. Since minium is Pb_3O_4.

106. Since blue copper sulphate solution becomes more blue in excess of ammonia.

107. Here option (b) is correct. As Cu^{+} ion in aqueous solution disproportionates to Cu (s) and Cu^{2+} (aq).

Linked–Comprehension Type Questions

Comprehension–I

108. As (K) is insoluble in mineral acid but dissolves in aqueous NH_3 so it is AgBr and the fumes are of Br .

$$\underset{(K)}{AgBr} + 2NH_3 \longrightarrow \underset{(L)}{Ag(NH_3)_2Br}$$

109. $$\underset{(K)}{2AgBr} + Na_2S_2O_3 \longrightarrow \underset{(M)}{Ag_2S_2O_3} + 2NaBr$$

$$Ag_2S_2O_3 \xrightarrow{\Delta,\ \text{boiling}} \underset{\substack{(N)\\ \text{Black ppt.}}}{Ag_2S} + SO_3$$

Comprehension–2

110. $$\underset{(A)}{Hg_2(NO_3)_2} + 2HCl \longrightarrow \underset{(B)}{Hg_2Cl_2} + 2HNO_3$$

111. $$\underset{\text{Chlorine water}}{Hg_2Cl_2} + 2Cl \longrightarrow \underset{(C)}{2HgCl_2}$$

112. $$HgCl_2 + 4KI \longrightarrow \underset{(D)}{K_2HgI_4} + 2KCl$$

113. $$Hg_2(NO_3)_2 + H_2SO_4 \longrightarrow Hg_2SO_4 + 2HNO_3$$

$$2HNO_3 \rightarrow H_2O + 2NO + 3[O]$$

$$2FeSO_4 + [O] + H_2O \longrightarrow Fe_2(SO_4)_3 + H_2O$$

$$FeSO_4 + NO \rightarrow \underset{\substack{\text{Brown ring}\\ (E)}}{FeSO_4.NO}$$

Comprehension–3

114. $$\underset{\substack{Pb_3O_4\\ \text{(scarlet)}}}{\text{Compound(A)}} \xrightarrow{\text{conc. } HNO_3} \underset{\substack{PbO_2\\ \text{(chocolate)}}}{\text{Compound(BS)}}$$

$$\xrightarrow{\text{filtered}} \underset{[Pb(NO_3)_2]}{\text{Filterate}}$$

115. $$Pb_3O_4 + \text{conc. } 4HNO_3 \rightarrow \underset{\substack{\text{Residue in form}\\ \text{of brown ppt.}}}{PbO_2\downarrow} + 2Pb(NO_3)_2 + 2H_2O \quad \underset{\text{Filtrate}}{}$$

116. Filtrate $Pb(NO_3)_2$ is neutralized with NaOH and on reaction with KI to give yellow precipitate of PbI_2.

$$Pb(NO_3)_2 + 2KI \longrightarrow \underset{\substack{\text{Yellow ppt.}\\ (C)}}{PbI_2\downarrow} + 2KNO_3$$

117. PbO_2 on warming with conc. HNO_3 in presence of $Mn(NO_3)_2$ produced pink coloured solution due to the formation of $Pb(MnO_4)_2$.

$$5PbO_2 + 2Mn(NO_3)_2 + 4HNO_3 \longrightarrow \underset{(D)}{Pb(MnO_4)_2} + 4Pb(NO_3)_2 + 2H_2O$$

Comprehension–5

118. $$\underset{(X)}{2CuSO_4} + 4KI \rightarrow 2CuI_2 + 2K_2SO_4$$

119. The solution becomes white as I_2 changes into NaI as follows:

$$2CuI_2 \longrightarrow \underset{\text{(White)}}{Cu_2I_2} + I_2$$

$$I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$$

120. $$\underset{(X)}{Ag} + KI \rightarrow \underset{\substack{\text{Yellow ppt.}\\ \text{Insoluble in } NH_4OH}}{AgI} + K^{+}$$

Assertion and Reasoning Questions

121. $KMnO_4$ is stored in dark bottles because it is decomposed in light.

122. $SnCl_2$ is strong reducing agent and reduces $HgCl_2$ first to Hg_2Cl_2 (white) and then to Hg (black)

$$SnCl_2 + 2HgCl_2 \longrightarrow SnCl_4 + Hg_2Cl_2$$

$$SnCl_2 + Hg_2Cl_2 \rightarrow SnCl_4 + 2Hg$$

So $HgCl_2$ and $SnCl_2$ cannot exist together in an aqueous solution.

123. Hydration energy of AgF is appreciably higher than its lattice energy because of smaller F^{-} ion and thus AgF is soluble in water.

124. Hydrated solution of $FeCl_3$ gets converted into brown $Fe_2O_3.XH_2O$ due to hydrolysis.

$$FeCl_3 + 3H_2O \rightarrow Fe(OH)_3 + 3HCl$$

$$Fe(OH)_3 \longrightarrow \underset{\text{Brown}}{Fe_2O_3} + 3H_2O$$

125. Assertion is correct but reason is wrong. Correct reason is that the colour is most of the transition metal complexes is due to d-d transition and charge transfer.

126. Mohr salt contains only Fe^{2+} ions without any trace of Fe^{3+} ions. Thus a standard solution of Fe^{2+} ions can be obtained directly by weighing a known amount of the Mohr salt.

127. $KMnO_4$ is coloured due to charge transfer and $K_2Cr_2O_7$ is coloured due to crystal defects.

128. In early p-block on moving top to bottom, the stability of lower oxidation state increases due to insert pair effect. Pb^{4+} is less stable than Sn^{4+}, making it a better oxidizing agent.

The lower oxidation state for the group 14 elements are more stable for the heavier members.

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129. CuF_2 is coloured due to the presence of one unpaired d-electron in Cu^{2+}. It can undergo d – d transition.

130. $4NaCl + K_2Cr_2O_7 + 6H_2SO_4\ (conc.) \xrightarrow{\Delta}$

$$2KHSO_4 + 4NaHSO_4 + \underset{\substack{\text{(orange red)}\\ \text{Chromyl chloride}}}{2CrO_2Cl_2} + 3H_2O$$

Chromyl chloride vapours when passed through NaOH solution gives a yellow solution of Na_2CrO_4.

131. This is clear from magnetic moment studies

132. In iodometry KI reacts with $K_2Cr_2O_7/H_2SO_4$ to give I_2 which is titrated against $Na_2S_2O_3$.

$$\underset{1\text{ mole}}{K_2Cr_2O_7} + 7H_2SO_4 + 6KI \rightarrow$$

$$4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + \underset{3\text{ mole}}{3I_2}$$

$$\underset{2\text{ mole}}{2Na_2S_2O_3} + \underset{1\text{ mole}}{I_2} \rightarrow Na_2S_4O_6 + 2NaI$$

1 mole $K_2Cr_2O_7$ = 3 mole I_2

1 mole I_2 = 2 mole $Na_2S_2O_3$

1 mole of $Na_2S_2O_3$ = ½ mole I_2

= 1/6 mole $K_2Cr_2O_7$

Thus, Eq. mass of $K_2Cr_2O_7$ = $\frac{\text{Mol. mass}}{6}$

133. Anhydrous ferric chloride is obtained by passing dry chlorine gas over heated metallic iron.

$$2Fe + 3Cl_2 \longrightarrow 2FeCl_3$$

Choice (A) is not correct because it gives Fe_2O_3.

$$2[FeCl_3.6H_2O] \xrightarrow{\Delta} Fe_2O_3 + 6HCl + 9H_2O$$

134. $MnO_2 + 2KOH + \frac{1}{2}O_2 \longrightarrow \underset{\text{Purple green}}{K_2MnO_4} + H_2O$

135. When I is oxidized by MnO_4^- in alkaline medium I converts into IO_3^-.

$$2KMnO_4 + 2KOH \rightarrow 2K_2MnO_4 + H_2O + [O]$$

$$2K_2MnO_4 + 2H_2O \longrightarrow 2MnO_2 + 3KOH + 2[O]$$

$$2KMnO_4 + H_2O \xrightarrow{\text{alkaline}} 2MnO_2 + 2KOH + 3[O]$$

$$KI + 3[O] \longrightarrow KIO_3$$

$$2KMnO_4 + KI + H_2O \longrightarrow 2KOH + 2MnO_2 + KIO_3$$

136. $Cu^{+2} + 2CN^- \longrightarrow Cu(CN)_2$

$$2Cu(CN)_2 \rightarrow 2CuCN + (CN)_2$$

137. Due to formation of tetrammine zinc(II) complex; $Zn^{2+} + NH_4OH \rightarrow [Zn(NH_3)_4]^{2+}$

138. In CuF_2, Cu^{2+} ions exits, having d^9 configuration unpaired electrode causes colour (d-d transition)

Solved Subjective Questions

1. Complete the following equation (no balancing is needed):
   (i) $SO_2 + MnO_4^- + \ldots\ldots \uparrow SO_4^{2-} + Mn^{2+} + \ldots$

Solution

$SO_2 + MnO_4^- + H^+ \rightarrow SO_4^{2-} + Mn^{2+} + H_2O$

(ii) $[MnO_4]^{2-} + H^+ \rightarrow \ldots\ldots + [MnO_4]^- + H_2O$

$3[MnO_4]^{2-} + 4H^+ \rightarrow MnO_2 + 2[MnO_4]^- + 2H_2O$

(iii) $SO_2(aq) + Cr_2O_7^{2-} + 2H^+ \rightarrow$ ..... + ... + ...

Solution

$3SO_2 + Cr_2O_7^{2-} + 2H^+ \rightarrow 2Cr^{3+} + 3SO_4^{2-} + H_2O$

(iv) $AgBr + Na_2S_2O_3 \rightarrow \ldots\ldots + \ldots\ldots$

Solution

$AgBr + 2Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaBr$

Sodium argento thiosulphate

(v) $(NH_4)_2S_2O_8 + H_2O + MnSO_4 \rightarrow$ ..... + ... + ...

Solution

$(NH_4)_2S_2O_8 + H_2O + MnSO_4 \rightarrow MnO_2 + 2H_2SO_4 + (NH_4)_2SO_4$

2. State with balanced equations what happens when:
   (i) Sulphur dioxide gas is bubbled through an aqueous solution of copper sulphate in presence of potassium thiocyanate.

Solution

$2CuSO_4 + SO_2 + 2H_2O + 2KCNS \rightarrow 2CuCNS\downarrow + K_2SO_4 + 2H_2SO_4$

Cuprous thiocyanate (white ppt.)

(ii) Aqueous solution of ferric sulphate and potassium iodide.

Solution

$Fe_2(SO_4)_3 + 2KI \rightarrow 2FeSO_4 + K_2SO_4 + I_2$

(iii) Aqueous solution of potassium manganate and acid are mixed.

Solution

$2MnO_4^{2-} + 4H^+ \rightarrow MnO_2 + MnO_4^- + 2H_2O$

(iv) Aqueous solution of potassium chromate and acid are mixed.

Solution

$2K_2CrO_4 + H_2SO_4 \rightarrow K_2Cr_2O_7 + K_2SO_4 + H_2O$

(Yellow) (Orange red)

(v) Potassium permanganate interacts with manganese dioxide in presence of potassium hydroxide:

Solution

$2KMnO_4 + 4KOH + MnO_2 \rightarrow 3K_2MnO_4 + 2H_2O$

or

$2KMnO_4 + H_2O \rightarrow 2MnO_2 + 2KOH + 3[O]$

(vi) Potassium ferrocyanide is heated with concentrated sulphuric acid.

Solution

$K_4Fe(CN)_6 + 6H_2SO_4 + 6H_2O \rightarrow K_2SO_4 + FeSO_4 + 3(NH_4)_2SO_4 + 6CO$

(vii) Gold is dissolved in aqua regia.

Solution

$3HCl + HNO_3 \rightarrow NOCl + 2H_2O + 2[Cl]$

(aqua regia)

$Au + 2[Cl] \xrightarrow{HCl} AuCl_3 \xrightarrow{HCl} HAuCl_4$

Aurochloric acid

(viii) Write balanced equations for the extraction of silver from silver glance by cyanide process.

Solution

$$Ag_2S + 4NaCN \rightleftharpoons 2NaAg(CN)_2 + Na_2S$$

$$4Na_2S + SO_2 + 2H_2O \rightarrow 2Na_2SO_4 + 4NaOH + 2S$$

(Here Na_2S is converted into Na_2SO_4 to avoid reversibility of first reaction)

$$2NaAg(CN)_2 + Zn \rightarrow Na_2Zn(CN)_4 + 2Ag$$

Sodium zincocyanide

(ix) Silver chloride is treated with aqueous sodium cyanide and the product thus formed is allowed to react with zinc in alkaline medium.

Solution

$$AgCl + 2NaCN \rightleftharpoons NaCl + Na[Ag(CN)_2]$$

$$2Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Ag\downarrow$$

Soluble

(x) Cobalt (II) solution reacts with KNO_2 in acetic acid medium.

Solution

$$CoCl_2 + 2KNO_2 \rightarrow Co(NO_2)_2 + 2KCl$$

$$KNO_2 + CH_3COOH \rightarrow CH_3COOK + HNO_2$$

$$Co(NO_2)_2 + 3KNO_2 + 2HNO_2 \rightarrow K_3[Co(NO_2)_6]\downarrow + NO + H_2O$$

Potassium cobaltinitrite

(Yellow ppt.)

(xi) A mixture of potassium dichromate and sodium chloride is heated with concentration H_2SO_4.

Solution

$$K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \rightarrow 2CrO_2Cl_2\uparrow + 4NaSO_4 + 2KHSO_4$$

Chromyl chloride

(Orange)

(xii) Iron reacts with cold dilute nitric acid.

Solution

$$4Fe + 10HNO_3 \rightarrow 4Fe(NO_3)_2 + NH_4NO_3 + 3H_2O$$

(xiii) Potassium permanganate is added to a hot solution of manganous sulphate.

Solution

$$2KMnO_4 + 3MnSO_4 + 2H_2O \rightarrow 5MnO_2 + K_2SO_4 + 2H_2SO_4$$

(xiv) Copper reacts with HNO_3 to give NO and NO_2 in molar ratio of 2 : 1.

Solution

The individual reactions are

$$3Cu + 8HNO_3 \text{ (dil.)} \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

$$Cu + 4HNO_3 \text{ (dil.)} \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

For the molar ratio of 2 : 1 of NO and NO_2,

$$7Cu + 20HNO_3 \rightarrow 7Cu(NO_3)_2 + 4NO + 2NO_2 + 10H_2O$$

(xv) Na_2CO_3 is added to a solution of copper sulphate.

Solution

$$2CuSO_4 + 2Na_2CO_3 + H_2O \rightarrow CuCO_3.Cu(OH)_2 + 2Na_2SO_4 + CO_2$$

(xvi) Potassium dichromate and concentrated hydrochloric acid are heated together.

Solution

$$K_2Cr_2O_7 + 14HCl \rightarrow 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$$

(xvii) Write balanced equations for the oxidation of cuprous oxide to cupric hydroxide by alkaline $KMnO_4$.

[*IIT 1997*]

Solution

$$KMnO_4 + Cu_2O + KOH \rightarrow 2Cu(OH)_2 + K_2MnO_4$$

(xviii) Write balanced equations for the reaction of alkaline perbromate with zinc giving tetra hydroxozincate anion.

[*IIT 1997*]

Solution

$$Zn + 2OH^- + BrO_4^- + H_2O \rightarrow Zn(OH)_4^{2-} + BrO_3^-$$

(xix) Write balanced equations for the reaction of zinc with dilute nitric acid.

[*IIT 1997*]

Solution

$$[Zn + 2HNO_3 \text{ (dil.)} \rightarrow Zn(NO_3)_2 + 2[H]] \times 4$$

$$2HNO_3 + 8[H] \rightarrow N_2O + 5H_2O$$

$$4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + N_2O + 5H_2O$$

3. What happens when:

(i) Aqueous ammonia is added dropwise to a solution of copper sulphate till it is in excess.

Solution

$$CuSO_4 + 4NH_4OH \rightarrow Cu(NH_3)_4SO_4 + 4H_2O$$

or

$$CuSO_4 + 2NH_4OH \rightarrow Cu(OH)_2 + (NH_4)_2SO_4$$

$$Cu(OH)_2 + 2(NH_4)_2SO_4 \rightarrow$$

$$Cu[(NH_3)_4]SO_4 + 2H_2O + H_2SO_4$$

Deep blue complex

(ii) $CrCl_3$ solution is treated with sodium hydroxide and then with hydrogen peroxide.

Solution

$$H_2O_2 \rightarrow H_2O + [O]$$

$$2CrCl_3 + 10NaOH + 3[O] \rightarrow$$

(Green)

$$2Na_2CrO_4 + 6NaCl + 3H_2O$$

(Yellow)

(iii) Cobaltous chloride with excess KNO_2 in aqueous acidic solution.

[*IIT 1997*]

Solution

$$CoCl_2 + 7KNO_2 + 2CH_3COOH \rightarrow$$

$$K_3Co(NO_2)_6 + 2CH_3COOK + 2KCl + NO$$

Yellow ppt. $+ H_2O$

(Potassium hexanitrocobaltate III)

4. Work out the following using chemical equations:

(a) In moist air copper corrodes to produce a green layer on the surface.

[*IIT 1998*]

Solution

$$2Cu + H_2O + CO_2 + O_2 \rightarrow CuCO_3.Cu(OH)_2$$

Green basic carbonate

(b) Give reasons for, "The colour of mercurous chloride, Hg_2Cl_2, changes from white to black when treated with ammonia."

Solution

When mercurous chloride reacts with aqueous ammonia it gives black colour due formation of mercury mercuric amino chloride as follows :

$$Hg_2Cl_2 + 2NH_4OH \rightarrow Hg + Hg(NH_2)Cl + NH_4Cl$$

Black ppt.

5. A Compound (A) impart a golden yellow flame and shows the following reactions:

(I) Zinc powder when boiled with a concentrated aqueous solution of (A), dissolve and hydrogen is evolved.

(II) When an aqueous solution of (A) is added to an aqueous solution of stannous chloride, a white precipitate is obtained first which dissolves in excess of solution of (A).

Identify (A) and write equation at steps (1) and (2).

Solution

(I) (A) imparts golden yellow flame so it contains Na^+

(II) Reaction suggests that (A) is NaOH as it reacts with Zn to give H_2 as follows

$$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$$

(A)

(III) (A) is also justified by these reactions as follows

$$2NaOH + SnCl_2 \rightarrow Sn(OH)_2 + 2NaCl$$

(A) white ppt.

$Sn(OH)_2 + 2NaOH \rightarrow Na_2SnO_2 + 2H_2O$
Excess(A) Soluble

6. A certain compound (A) is used in laboratory for analysis. Its aqueous solution gave the following reaction :
   (a) On addition to copper sulphate solution, a brown precipitate is obtained which turns white on addition of excess of $Na_2S_2O_3$ solution.
   (b) On addition to Ag^+ ion solution, a yellow curdy precipitate is obtained which is insoluble in NH_4OH.

   Identify (A),by giving reactions.

Solution

(1) (A) gives yellow ppt. insoluble in NH_4OH with Ag^+ and so contains I^- ions.

(2) Step (a) suggesrs ir ro be **KI**.
Reacrions are as follows:

(i) $2CuSO_4 + 2KI \rightarrow 2CuI_2 + K_2SO_4$
(A)

$2CuI_2 \rightarrow Cu_2I_2 + I_2$
yellow

$I_2 + 2Na_2S_2O_3 \rightarrow Na_2S_4O_6 + 2NaI$

(ii) $Ag^+ + KI \rightarrow AgI + K$
(A) Yellow ppt.
insoluble in NH_4OH

7. A scarler compound A is rreated with conc. HNO_3 to give a chocolare brown precipitate B. The precipirare is filtered and rhe filtrate is neutralized with NaOH. Addirion of KI ro rhe resulring solution gives a yellow precipirate C. The precipirare B on warming wirh conc. HNO_3 in rhe presence of $Mn(NO_3)_2$ produces a pink-coloured solution due ro rhe formarion of D. Idenrify A, B, C and D. Write rhe reacrion sequence. [*IIT 1995*]

Solution

Compound (A) $\xrightarrow{\text{conc. } HNO_3}$ Compound (B)
Pb_3O_4 PbO_2
(scarlet) (chocolare)

$\xrightarrow{\text{filtered}}$ Filterate
$[Pb(NO_3)_2]$

$Pb_3O_4 + \text{conc. } 4HNO_3 \rightarrow PbO_2\downarrow + 2Pb(NO_3)_2 + 2H_2O$
Residue in form Filtrate of brown ppt.

Filtrate $Pb(NO_3)_2$ is neutralized with NaOH and on reacrion with KI ir gives yellow precipirate of PbI_2.

$Pb(NO_3)_2 + 2KI \rightarrow PbI_2\downarrow + 2KNO_3$
Yellow ppt.
(C)

PbO_2 on warming with conc. HNO_3 in presence of $Mn(NO_3)_2$ gives pink coloured solution due to the formation of $Pb(MnO_4)_2$.

$5PbO_2 + 2Mn(NO_3)_2 + 4HNO_3 \rightarrow Pb(MnO_4)_2 + 4Pb(NO_3)_2 + 2H_2O$
(D)

So compounds (A) = Pb_3O_4 , (B) = PbO_2

(C) = PbI_2, (D) = $Pb(MnO_4)_2$

8. Give complere and balanced chemical equation for rhe following.
   (i) Mercurous nitrare reacrs with excess of **KI** solurion.
   (ii) Sodium chromite reacts with H_2O_2 in presence of NaOH.
   (iii) Nickel sulphare reacts with dimethyl glyoxime reagent in ammonical solurion.

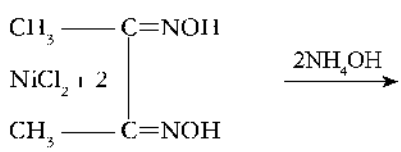
Solution

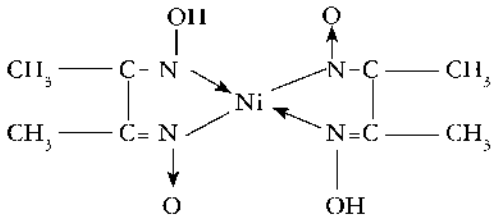
(i) $Hg_2(NO_3)_2 + 2KI \rightarrow Hg_2I_2 + 2KNO_3$

$Hg_2I_2 + 2KI \rightarrow K2HgI_4 + Hg$

(ii) $Na_2CrO_3 + H_2O_2 \rightarrow Na_2CrO_4 + H_2O$

(iii)





Red ppt. + $2NH_4Cl$

Nickel dimethyl glyoxime

9. An inorganic compound (P), transparent like glass is a strong reducing agent. Its hydrolysis in water gives a white turbidity (Q). Aqueous solution of (P) gives white precipitate (R) with NaOH (aq) which is soluble in excess NaOH. (P) reduces auric chloride to produce purple of cassius. (P) also reduces I_2 and gives chromyl chloride test.

Solution

Since (P) gives chromyl chloride test so, it has Cl^-. Since (P) is a strong reducing agent So (P) is $SnCl_2$ and the reactions are as follows:

(i) $SnCl_2 + H_2O \rightarrow Sn(OH)Cl + HCl$

(ii) $SnCl_2 + 2NaOH \rightarrow Sn(OH)_2 + 2NaCl$
(A) (C)

$SnCl_2 + 2HCl + I_2 \rightarrow SnCl_4 + 2HI$

(iii) $SnCl2 + 2HCl + I_2 \rightarrow SnCl_4 + 2HI$

(iv) $3SnCl_2 + 2AuCl_3 \xrightarrow{H_2O} SnCl_4 + 2Au$

$3SnCl_4 \rightarrow Sn(OH)_4 + HCl$

Stannic acid absorbs colloidal particles of gold, this beautiful purple colour composed is known as Purple of cassius. it is used for colouring glass & pottery.

10. A unknown inorganic compound (A) gave the following reactions: (i) on hearing 'X' gave a residue, oxygen and oxide of nitrogen. (ii) Addition of acetic acid and $K_2Cr_2O_7$ to its aqueous solution give a yellow precipitate. (iii) Addition of NaOH to its aqueous solution first forms a white precipitate, dissolve in the excess of the reagent. Identify the compound (A) and write balanced equation for step (i), (ii) and(iii).

Solution

(i) $Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
CH_3COOH

(ii) $Pb(NO_3)_2 + K2Cr_2O_2 \rightarrow PbCrO_4\downarrow$

(iii) $Pb(NO_3)_2 + 2NaOH \rightarrow Pb(OH)_2 + 2NaNO_3$

$Pb(OH)_2 + 2NaOH \rightarrow Na_2[Pb(OH)_4]$

11. An aqueous solution containing one mole of HgI_2 and two mole of NaI is orange in colour. On addition of excess NaI the solution becomes colourless. The orange colour reappears on subsequent addition of NaOCl. Explain with equations.

[*IIT 1999*]

Solution

A solution having one mole of HgI_2 and two moles of NaI is orange in colour due to the partial solubility of HgI_2. On addition of excess of NaI, the colourless complex Na_2HgI_4 is formed.

$2NaI + HgI_2 \rightarrow Na_2HgI_4$
Excess

The Na_2HgI_4 on addition of NaOCl, oxidizes as:

$3Na_2HgI_4 + 2NaOCl + 2H_2O \rightarrow$

$3HgI_2 + 2NaCl + 4NaOH + 2NaI_3$

So, colour of partially soluble HgI_2 is restored.

12. An aqueous blue coloured solution of a transition metal sulphate reacts with H_2S in acidic medium to give a black precipitate A, which is insoluble in warm aqueous solution of KOH. The blue solution on treatment with KI in weakly acidic medium, turns yellow and produces a white precipitate B. Identify the transition metal ion. Write the chemical reactions involved in the formation of A and B.

[*IIT 2000*]

Solution

The transition metal is Cu^{2+} and the compound is $CuSO_4.5H_2O$ which dissolves in water to give blue coloured solution due to presence of Cu^{2+} (d^9-configuration). On passing H_2S in acidic medium in such a solution, the black precipitate of CuS is obtained which is not soluble in aqueous KOH (warm) solution.

$$CuSO_4 + H_2S \xrightarrow{\text{Acidic medium}} \underset{\text{Black ppt.}}{CuS\downarrow} + H_2SO_4$$

(Insoluble in aq. KOH)

On addition of KI solution in aqueous solution of $CuSO_4$, it produces yellow coloured solution of CuI_2, which is decomposed into white ppt. of Cu_2I_2 and I_2.

$$CuSO_4 + 2KI \rightarrow CuI_2 + K_2SO_4$$

$$2CuI_2 \rightarrow \underset{\text{White ppt.}}{Cu_2I_2} + I_2$$

13. $MCl_4 \xrightarrow{Zn} (A)$

Colourless liquid — Purple colour compound

Here M = Transition metal

$$MCl_4 \xrightarrow{\text{moist air}} \underset{\text{White fumes}}{(B)}$$

Identify (A), (B) and MCl_4. Also explain colour difference between MCl_4 and (A).

[*IIT 2005*]

Solution

$$TiCl_4 \xrightarrow[\text{Heat}]{Zn} TiCl_3 \xrightarrow{H_2O} \underset{(A)}{[Ti(H_2O)_6]Cl_3}$$

$$TiCl_4 + (n+2)H_2O \rightarrow TiO_2(H_2O)_n + 4HCl\uparrow$$

(B)

White fumes

$Ti^{4+} = [Ar]\ 3d^0$

$Ti^{3+} = [Ar]\ 3d^1$

Ti(+IV) ion contains no d-electron, while d-d-transition of single electron in Ti (+III) causes colour change.

14. Some reactions of two ores, A_1 and A_2 of the metal M are given below:

$$[A_1] \xrightarrow{\text{calcination}} \underset{\text{Black}}{[C]\downarrow} + CO_2 + H_2O$$

$$[A_1] \xrightarrow{KI/HCl} [D]\downarrow + I_2$$

$$[A_2] \xrightarrow{\text{Roasting}} [G]\downarrow + M$$

$$[G] + K_2Cr_2O_7 \xrightarrow{H^+} \text{Green solution}$$

Identify A_1, A_2, M, C, D, G and explain using the required chemical reactions.

[*IIT 2004*]

Solution

$$CuCO_3.Cu(OH)_2 \xrightarrow{\text{Calcination}} \underset{(A_1)}{}$$

$$2CuO\downarrow + CO_2 + H_2O\ (C)$$

Black ppt.

$$CuCO_3.Cu(OH)_2 + 4HCl \rightarrow 2CuCl_2 + CO_2 + 3H_2O$$

$$2CuCl_2 + 4KI \rightarrow \underset{(D)}{Cu_2I_2}\downarrow + 4KCl + I_2$$

$$\underset{(A_2)}{2Cu_2S} + 3O_2 \xrightarrow{\text{roasting}} \underset{(G)}{2Cu_2O} + 2SO_2$$

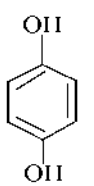
$$Cu_2S + 2Cu_2O \xrightarrow{\text{Self reduction}} \underset{(M)}{6Cu} + SO_2$$

$$3SO_2\ (G) + K_2Cr_2O_7 + H_2SO_4 \rightarrow K_2SO_4 + \underset{\text{(Green)}}{Cr_2(SO_4)_3} + 4H_2O$$

15. Write the chemical reaction involved in developing of a black and white photographic film. An aqueous $Na_2S_2O_3$ solution is acidified to give a milky white turbidity. Identify the product and write the balanced half chemical reaction for it.

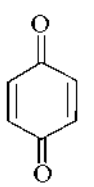
[*IIT 2005*]

Solution



Hydroquinone

+ 2AgBr → (Activated Br, exposed portion)



+ 2Ag + 2HBr

$AgBr + 2Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaBr$

Unexposed portion of photographic film

$Na_2S_2O_3 + 2H^+ \rightarrow 2Na^+ + H_2SO_3 + S\downarrow$
Colloidal sulphur

16. Write the balanced chemical equation for the following:
(a) Pb_3O_4 is treated with nitric acid.
(b) Ozone is passed through potassium ferrocyanide solution.
(c) Heating of potassium permanganate.

Solution

(a) Pb_3O_4 ia a mixed oxide ($2PbO, PbO_2$), with nitric acid PbO is converted to $Pb(NO_3)_2$ while PbO_2 remains as such.

(b) $2K_4Fe(CN)_6 + O_3 + H_2O \rightarrow 2K_3Fe(CN)_6 + 2KOH + O_2$

(c) $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

Questions For Self-Assessment

17. (i) A blue coloured compounds (A) on heating gives two of the products (B) and (C)
(ii) A metal (D) is deposited on passing hydrogen through heated(B)
(iii) The solurion of (B) in HCl on rrearmenr with $K_4[Fe(CN)_6]$ gives a chocolare brown coloured precipitare of compound (E).
(iv) (C) turns lime water milky which disappears on continuous passage of (C) forming a compound,(F) . Identrify (A) ro (F) and give chemical equarion for the reactions at srep (i) to (iv).

Ans (A)= $CuCO_3$
(B) = CuO
(C)- CO_2
(D)–Cu
(E) = $Cu_2[Fe(CN)_6]$
(F) = $Ca(HCO_3)_2$

18. A compound (P) is soluble in warer. When a few drops of sodium chloride solurion is added ro rhe solurion of (P), a whire precipirare of (Q) appears. The whire ppr of (Q) becomes soluble when heared but reappears on cooling. To this ppr of (Q) HCl is added and rhe ppr (Q) dissolves and yellow solurion (R). If ro yellow solurion more (P) is added again (Q) appears and yellow colour of the solurion fades. Here P, Q, R are respecrively?

Ans P= $PbNO_3$, Q= $PbCl_2$

19. Complete the following by identifying (A) to (F).
100°C 230°C 800°C
(i) $CuSO_4.5H_2O \rightarrow (A) \rightarrow (B) \rightarrow (C) + (D)$
(ii) $AgNO_3 \rightarrow (E) + (F) + O_2$

Ans (A) = $CuSO_4.H_2O$
(B) – $CuSO_4$
(C) – CuO (D) – SO_3
(E) Ag (F) – NO_2

20. A substance (A) is soluble in conc. HCl. When to this solution NaOH solution is added, a white precipitate is produced. This precipitate dissolves in excess of NaOH solution giving a strongly reducing solution. Heating of (A) with sulphur gives a brown powder (B) which is soluble in warm yellow ammonium sulphide solution. When HCl is added to the latter, a grey precipitate is produced. Heating of (A) in air gives compound (C) which is soluble in conc. H_2SO_4. It also gives awater soluble compound when fused with NaOH. The addition of mineral acid to the water soluble compound gives white gelatinous precipitate. Identify A, B, C here?

Ans A – Sn B – SnC_2 C – SnO_2

21. Write the balanced chemical equation for the following:
(a) Silver chloride is rreared with aqueous sodium cyanide and rhe product rhus formed is allowed to react with zinc in alkaline medium
(b) Cobalr (II) solurion reacrs with KNO_2 in aceric acid medium.

22. Explain and identify M, A, B, C the following reaction .

$$\text{Metal (M)} \xrightarrow[\text{in dil } H_2SO_4]{\text{Dissolves}} A \xrightarrow{\text{add alkali}} B$$

A = pale green solution

B – White precipitate quickly turning brown

$$B \xrightarrow{HCl} C$$

Brown residue (as above) — Dissolves giving yellow solution

Ans M – Fe, A – $FeSO_4$ B – $Fe(OH)_2$ C – $FeCl_3$

23. A salt of tin (P) gives a basic chloride when dissolved in excess of water. The salt (P) gives grey mass with corrosive sublimate and another compound (Q) which a fuming liquid and fumes more in moist air. Salt of (P) gives blue coloured precipitate (R) with acidified $(NH_4)_2MoO_4$. Identify (P), (Q) and (R) and give the reactions.

Ans P – $SnCl_2$, Q – $SnCl_4$ R – MO_3O_3

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PRINCIPLES OF QUALITATIVE ANALYSIS

10

Chapter Contents

Acidic radicals like nitrates, halides (excluding fluoride), Sulphate and Sulphide. Basic radicals from Group I to V (only Ag^{+}, Hg^{2+}, Cu^{2+}, Pb^{2+}, Bi^{3+}, Fe^{3+}, Cr^{3+}, Al^{3+}, Ca^{2+}, Ba^{2+}, Zn^{2+}, Mn^{2+} and Mg^{2+}) and various levels of multiple-choice questions.

It deals with qualitative and quantitative analysis of a substance or a compound. In qualitative analysis, the acidic and basic radicals are found.

PRELIMINARY TESTS

Physical State Examination

Here colour, smell, density etc., are noticed.

Colourless Gases

H_2S, O_2, CO, CO_2, SO_2, NH_3, CH_3COOH, HCl vapours etc.

Coloured Gases

NO_2 (brown), Cl_2 (yellow greenish), Br_2 (brown), I_2 (violet) etc.

Colour of Some Cations

Light green (Fe^{2+}), deep green (Cr^{3+}), greenish (Ni^{2+}), blue or bluish green (Cu^{2+} or Ni^{2+}), Pink (Co^{2+}), light pink, flesh colour or earthy colour (Mn^{2+}).

Smell

A pinch of mixture is rubbed between the fingers with a drop of water and smelled.

Ammonical smell (NH_4^{+}),
Rotten eggs smell (S^{2-}),
Vinegar like smell (CH_3COO^{-})
Burning sulphur smell (sulphates).
Pungent smell (Br^{-})

Density

Salts of Hg^{2+}, Pb^{2+} or Ba^{2+} are heavy and light fluffy powder (carbonates).

Some substances like $CaCl_2$, $MgCl_2$, $ZnCl_2$, and nitrite, absorb moisture and get wet, that is, become deliquescent.

Effect of Heating of Mixture

It undergoes sublimation and gives white sublimates. Example, NH_4X, $HgCl_2$, Hg_2Cl_2, $AlCl_3$, As_2O_3 and Sb_2O_3

Yellow Sublimate

As_2S_3, HgI_2 gives oxides of characteristic colours.

For example,

ZnO–yellow when hot and white when cold.

PbO reddish brown when hot and yellow when cold.

HgO and PbO black when hot and red when cold.

Alkalinitrates give O_2.

Carbonates and oxalates give CO_2 which turns lime water milky.

NH_4NO_2 gives N_2 gas.

Ammonium salt evolves NH_3 which turns litmus blue and mercurous nitrate paper black.

Sulphites and thiosulphates evolve SO_2 (smell of burning sulphur) which turns acidified $K_2Cr_2O_7$ paper green and lime water milky. (Similarity with CO_2)

Nitrites and nitrates of heavy metals evolve NO_2 (brown) which turns starch iodide paper blue.

CHARACTERISTIC TEST OF ANIONS (ACIDIC RADICALS)

Table 10.1 Analysis of Acidic Radicals

	Radical	Observation	Confirmatory Test
Observation With Dil. H_2SO_4			
1	CO_3^{2-}	Brisk effervences with evolution of colourless, odourless gas.	gas forms lime water milky and milkness disappears in excess of gas.
2	CH_3COO^-	Solution with Vinegar smell	Aqueous Solution · feels neutral. Blood Red colour Brownish ppt.
3	NO_2^-	Brown fumres	On adding KI and starch solution becomes blue.
4	S^{-2}	Rotten eggs Smell	Gas turns lead acetate paper into black colour. ppt. of P_bS.
5	SO_3^{2-}	Colourless gas with a pungent smell of burning sulphur.	Gas turns acidified k_2Cr_2O7 solution into green colour.
Observation With Conc H_2SO_4			
I	Cl	Colourless gas giving white fumes with NH_4OH.	On adding MnO_2 pale green colour of Cl_2. (Also by chromyl chloride Test)
II	Br^-	Reddish-Brown fumes	On adding MnO_2 yellowish brown Br_2 evolved.
III	I	Violet pungent vapours that turns starch paper blue.	Sodium extract $+HNO_3+AgNO_3$. Yellow ppt of AgI (Insoluble in NH_4OH)
IV	NO_3^-	Brown Pungent fumes.	By Brown Ring Test.
V	CO_3^{2-}	Colourless, odourless gas, burns with blue flame and turns line water millky	Decolourize acidified solution.

Chromyl Chloride Test for Chloride

Here, chloride salt is mixed with $K_2Cr_2O_7$ and heared with conc. H_2SO_4. Under this condition, orange-red vapours of chromyl chloride are evolved.

These vapours are passed through dil. NaOH solution to give yellow solution of Na_2CrO_4.

The yellow solution is acidified with CH_3COOH and lead acetate solution is added, formation of yellow precipitate of lead chromate shows the presence of chloride.

$$4NaCl + K_2Cr_2O_7 + 6H_2SO_4 \rightarrow 4NaHSO_4 + 2KHSO_4 + 3H_2O + 2CrO_2Cl_2$$
(Chromyl chloride)

$$CrO_2Cl_2 + 4NaOH \longrightarrow Na_2CrO_4 + 2NaCl + 2H_2O$$

$$Na_2CrO_4 + (CH_3COOH)_2Pb \longrightarrow PbCrO_4 + 2CH_3COONa$$
(Yellow ppt.)

CH_3COOH is used to neutralize excess of NaOH because $PbCrO_4$ is soluble in NaOH.

Chlorides of Hg, Pb, Ag, Sb and Sn do not give this test.

Test of Bromide (Br^-)

When mixture on treatment with conc. H_2SO_4, gives reddish brown fumes, Br may be present.

On adding a little MnO_2 and heating, yellow brown gas that is, Br_2 is obtained.

$$KBr + H_2SO_4 \xrightarrow{\Delta} KHSO_4 + HBr$$

$$4HBr + MnO_2 \xrightarrow{\Delta} Br_2 + 2H_2O + MnBr_2$$

When in water extract or sodium carbonate extract acidified by dilute HNO_3, a little $AgNO_3$ is added, a pale yellow precipitate (partially soluble in NH_4OH) is obtained.

$$NaBr + AgNO_3 \rightarrow AgBr + NaNO_3$$

$$AgBr + 2NH_3\,(aq) \longrightarrow [Ag(NH_3)_2]Br$$

Test of Iodide (I^-)

When mixture on treatment with conc. H_2SO_4, gives pungent violet vapours, which turns starch paper blue I may be present.

$$KI + H_2SO_4 \xrightarrow{\Delta} KHSO_4 + HI$$

$$2HI + H_2SO_4 \xrightarrow{\Delta} I_2 + 2H_2O + SO_2$$

When, in water extract or sodium carbonate extract acidified by dilute HNO_3, a little $AgNO_3$ is added, a yellow precipitate (insoluble in NH_4OH) is obtained.

$$NaI + AgNO_3 \longrightarrow AgI + NaNO_3$$

Test of Nitrate (NO_3^-) Ion

Any nitrate salt on decomposition by conc. H_2SO_4 produces reddish-brown fumes of NO_2.

$$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$$

$$4HNO_3 \longrightarrow 4NO_2 \uparrow + O_2 \uparrow + 2H_2O$$

Ring Test

Aqueous extract is acidified with dil. H_2SO_4, then freshly prepared $FeSO_4$ solution and few drops of conc. H_2SO_4 is added, a brown ring appears, confirming the presence of NO_3 radical.

$$2NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$$

$$2HNO_3 + 6FeSO_4 + 3H_2SO_4 \rightarrow 3Fe(SO_4)_3 + 2NO + 4H_2O$$

$$FeSO_4 + NO \longrightarrow \underset{\text{(Brown ring)}}{[Fe(NO)]SO_4}$$

Test of Sulphate (SO_4^{2-})

When water extract or sodium extract acidified with acetic acid is treated with lead acetate solution, a white precipitate is formed which is soluble in excess of ammonium acetate on warming.

$$Na_2SO_4 + (CH_3COO)_2Pb \rightarrow PbSO_4 + 2CH_3COONa$$

Test of Borate (BO_3^{3-})

When any borate salt is heated with con. H_2SO_4 and ethyl alcohol, volatile ethyl borate is formed, which burns at the mouth of a test tube with a green edged flame.

$$2Na_3BO_3 + 3H_2SO_4 \rightarrow 3Na_2SO_4 + 2H_3BO_3$$

$$H_3BO_3 + 3C_2H_5OH \rightarrow \underset{\text{(volatile)}}{(C_2H_5O)_3B} + 3H_2O$$

$(C_2H_5O)_3B$ burns with green flame.

Test of Phosphate (PO_4^{3-})

When phosphate salt is heated with conc. HNO_3 and excess of ammonium molybdate solution is added, a canary yellow precipitate of ammonium phosphomolybdate is formed.

$$Na_3PO_4 + 3HNO_3 \xrightarrow{\Delta} 3NaNO_3 + H_3PO_4$$

$$H_3PO_4 + 12(NH_4).12MoO_4 + 21HNO_3 \rightarrow \underset{\text{(yellow ppt.)}}{(NH_4)_3PO_4.12MoO_3\downarrow} + NH_4NO_3 + 12H_2O$$

When water extract or sodium extract acidified with acetic acid is treated with 3–4 ml of magnesia mixture, after five minutes, a white ppt. of magnesium ammonium phosphate is formed.

$$Na_2HPO_4 + Mg(NO_3)_2 + NH_3 \rightarrow \underset{\text{White ppt.}}{Mg(NH_4)PO_4} + 2NaNO_3$$

Test of Basic Radicals

Table 10.2 Test of Cations (Basic Radicals)

Group	Group Reagent	Basic Radical	Composition and Colour of the ppt.	
Zero	NaOH	NH_4^+	NH_3 gas is evolved.	
I	Dil. HCl	Ag^+	AgCl	White
		Pb^{2+}	$PbCl_2$	White
		Hg_2^{2+}	Hg_2Cl_2	White
II	H_2S in the presence of dil HCl	Hg^{2+}	HgS	Black
		Pb^{2+}	PbS	Black
		Bi^{3+}	Bi_2S_3	Black
		Cu^{2+}	CuS	Black
		Cd^{2+}	CdS	Yellow
		As^{3+}	As_2S_3	Yellow
		Sb^{3+}	Sb_2S_3	Orange
		Sn^{2+}	SnS	Brown
		Sn^{4+}	SnS_2	Yellow
III	NH_4OH in the presence of excess of NH_4Cl	Fe^{3+}	$Fe(OH)_3$	Reddish
		Cr^{3+}	$Cr(OH)_3$	Dirty green
		Al^{3+}	$Al(OH)_3$	White gelatinous ppt.
IV	H_2S in the presence of NH_4OH	Co^{2+}	CoS	Black
		Ni^{2+}	NiS	Black
		Zn^{2+}	ZnS white	Bluish
		Mn^{2+}	MnS	Buff (flesh) coloured
V	$(NH_4)_2CO_3$ in the presence of NH_4OH	Ba^{2+}	$BaCO_3$	White
		Sr^{2+}	$SrCO_3$	White
		Ca^{2+}	$CaCO_3$	White
VI	Na_2HPO_4 No group reagent	Mg^{2+}	$Mg(NH_4)PO_4$	White
	No group reagent	Na^+		
		K^+		

REMEMBER

- I group halides and all II group sulphides are insoluble in dil. HCl.
- All III group hydroxides, IV group sulphides and V group carbonates are insoluble in NH_4OH.

Test of NH_4^+ Ions

It gives a yellow precipitate with sodium cobaltrinitrite solution.

$$3NH_4Cl + Na_3[Co(NO_2)_6] \rightarrow (NH_4)_3[Co(NO_2)_6]\downarrow + 3NaCl$$

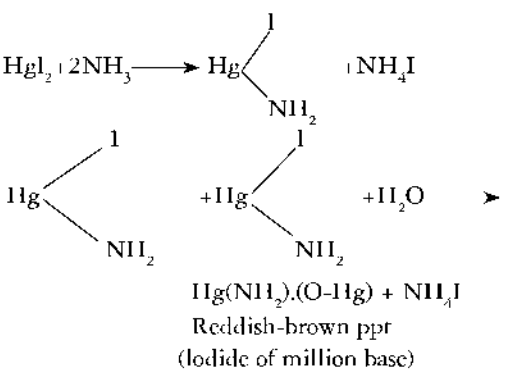
(yellow ppt.)

With NaOH, ammonium salts liberates NH_3 gas which can be tested with the help of Nessler's reagent.

$$NH_4Cl + NaOH \longrightarrow NH_3 + NaCl + H_2O$$

$$K_2HgI_4 \rightarrow 2KI + HgI_2$$

Nessler's Reagent



$Hg(NH_2).(O\text{-}Hg) + NH_4I$

Reddish-brown ppt

(Iodide of million base)

Ist GROUP BASIC CATIONS

Test of Ag^+ Ions

AgCl dissolves in NH_4OH solution to give a complex salt.

$$AgCl + 2NH_4OH \rightarrow [Ag(NH_3)_2]Cl + 2H_2O$$

On acidification, the complex gives AgCl.

$$Ag(NH_3)_2Cl + 2HNO_3 \longrightarrow AgCl\downarrow + 2NH_4NO_3$$

On addition of KI solution in the complex, a pale yellow ppt. of AgI is produced.

$$[Ag(NH_3)_2]Cl + KI \rightarrow AgI\downarrow + KCl + 2NH_3$$
(yellow ppt.)

On addition of K_2CrO_4 solution in the complex, a brick red ppt. of Ag_2CrO_4 is formed.

$$2[Ag(NH_3)_2]Cl + K_2CrO_4 \longrightarrow \underset{\text{Red ppt.}}{Ag_2CrO_4} + 2KCl + 4NH_3$$

Test of Pb^{2+}

The Pb^{2+} ions give yellow precipitate both with potassium chromate solution and KI solution.

$$PbCl_2 + K_2CrO_4 \rightarrow \underset{\text{Yellow ppt.}}{PbCrO_4} + KCl$$

$$PbCl_2 + 2KI \rightarrow \underset{\text{Yellow ppt.}}{PbI_2} + 2KCl$$

Test of Mercurous Ion (Hg_2^{2+})

NH_4OH converts Hg_2Cl_2 to a black residue which consists of white amino mercuric chloride and black finely divided Hg.

$$Hg_2Cl_2 + 2NH_4OH \rightarrow \underset{\text{White}}{\overset{NH_2}{Hg}\!\!\underset{Cl}{\diagup\diagdown}} + Hg + NH_4Cl + 2H_2O$$

Aquaregia dissolves the black residue forming soluble mercuric chloride.

$$\underset{\text{White}}{\overset{NH_2}{Hg}\!\!\underset{Cl}{\diagup\diagdown}} + 2HCl \rightarrow HgCl_2 + NOCl + 2H_2O$$

Addition of $SnCl_2$ reduces $HgCl_2$ to white Hg_2Cl_2.

$$2HgCl_2 + SnCl_2 \longrightarrow Hg_2Cl_2\downarrow + SnCl_4$$

$$Hg_2Cl_2 + SnCl_2 \rightarrow 2Hg + SnCl_4$$

IIND GROUP CATIONS

Test of Cu^{2+} Ions

Cupric ion reacts with excess of ammonia solution to give a deep blue colour of tetra ammine cupric ion.

$$3CuS + 8HNO_3 \xrightarrow{\Delta} 3Cu(NO_3)_2 + 2NO + 3S + 4H_2O$$

$$Cu(NO_3)_2 + 4NH_4OH \rightarrow \underset{\text{Deep blue colour}}{[Cu(NH_3)_4](NO_3)_2} + 4H_2O$$

Addition of potassium ferrocyanide solution to a cupric salt solution gives a reddish brown (chocolate) precipitate of cupric ferro cyanide,which is insoluble in CH_3COOH.

$$[Cu(NH_3)_4]^{2+} \rightleftharpoons Cu^{2+} + 4NH_3$$

$$2Cu(NO_3)_2 + K_4[Fe(CN)_6] \longrightarrow \underset{\text{Reddish brown ppt.}}{Cu_2[Fe(CN)_6]\downarrow} + 4KNO_3$$

Test of Bi^{3+} Ions

Bi_2S_3 dissolves in hot dil. HNO_3 to give soluble nitrate salt.

$$Bi_2S_3 + 8HNO_3 \longrightarrow 2Bi(NO_3)_3 + 2NO + 3S + 4H_2O$$

Addition of NH_4OH precipitates out $Bi(OH)_3$.

$$Bi(NO_3)_3 + 3NH_4OH \rightarrow Bi(OH)_3\downarrow + 3NH_4NO_3$$

$Bi(OH)_3$ dissolves in HCl.

$$Bi(OH)_3 + 3HCl \longrightarrow BiCl_3 + 3H_2O$$

Here, addition of water gives a white turbidity of bismuth oxychloride.

$$BiCl_3 + H_2O \rightarrow \underset{\text{White turbidity}}{BiOCl\downarrow} + 2HCl$$

Here, addition of alkaline sodium stannite gives a black ppt. of Bi.

$$3BiCl_3 + 6NaOH + 3Na_2(SnO_2) \rightarrow \underset{\text{Black ppt.}}{3Bi\downarrow} + 3NaSnO_3 + 3H_2O + 9NaCl$$

IIIRD GROUP CATIONS

Test of Al^{3+} Ions

White ppt. of $Al(OH)_3$ dissolves in NaOH solution forming sodium meta aluminate.

$$Al(OH)_3 + NaOH \longrightarrow \underset{\text{Soluble}}{NaAlO_2} + 2H_2O$$

On boiling $NaAlO_2$ with NH_4Cl again, $Al(OH)_3$ gets precipitated.

$$NaAlO_2 + NH_4Cl + H_2O \xrightarrow{\text{Boil}} Al(OH)_3\downarrow + NaCl + NH_3$$

An addition of dil. HCl, followed by heating with NH_4OH, gives precipitate of $Al(OH)_3$.

$$NaAlO_2 + 4HCl \longrightarrow AlCl_3 + NaCl + 2H_2O$$

$$AlCl_3 + 3NH_4OH \xrightarrow{\text{Heat}} Al(OH)_3\downarrow + 3NH_4Cl$$

Test of Cr^{3+}

$$CrCl_3 + 3NH_4OH \longrightarrow \underset{\text{Green ppt.}}{Cr(OH)_3} + 3NH_4Cl$$

Test of Fe^{3+} Ions

$Fe(OH)_3$ dissolves in conc. HCl to give $FeCl_3$.

$$Fe(OH)_3 + 3HCl \longrightarrow FeCl_3 + 3H_2O$$

Fe^{3+} ion gives a blood red colouration with thiocyanate ions.

$$FeCl_3 + 3NH_4SCN \longrightarrow \underset{\text{Ferric thiocyanate (red colour)}}{Fe(SCN)_3} + 3NH_4Cl$$

Fe^{2+} ions do not respond to thiocyanate test.

Fe^{2+} ion gives a deep blue precipitate with ferricyanide ion.

$$3FeCl_2 + 2K_3[Fe(CN)_6] \longrightarrow \underset{\substack{\text{Ferrous ferricyanide}\\ \text{(Turnbull's blue)}}}{Fe_3[Fe(CN)_6]_2} + 6KCl$$

Fe^{3+} ions produce brown colour with ferricyanide ion.

$$FeCl_3 + K_3[Fe(CN)_6] \longrightarrow \underset{\substack{\text{Ferric ferricyanide}\\ \text{(Brown colour)}}}{Fe[Fe(CN)_6]} + 3KCl$$

Fe^{3+} ions give deep blue colour with potassium ferrocyanide.

$$3K_4[Fe(CN)_6] + 4FeCl_3 \longrightarrow \underset{\substack{\text{Ferri ferrocyanide}\\ \text{(Prussian's blue)}}}{Fe_4[Fe(CN)_6]_3} + 12KCl$$

Fe^{2+} ions produce a white precipitate with $K_4[Fe(CN)_6]$ in the absence of air, which is oxidized into ferric ferrocyanide by air.

$$2FeCl_2 + K_4[Fe(CN)_6] \longrightarrow \underset{\substack{\text{Ferrous ferricyanide}\\ \text{(white ppt.)}}}{Fe_2[Fe(CN)_6]} + 4KCl$$

$$3Fe_2[Fe(CN)_6] \xrightarrow{\text{Oxidation}} \underset{\text{Ferri ferrocyanide}}{Fe_4[Fe(CN)_6]_3} + 2Fe$$

IVTH GROUP CATIONS

Test of Zn^{2+} Ions

Its sulphide dissolves in HCl.

$$ZnS + 2HCl \longrightarrow ZnCl_2 + H_2S$$

$ZnCl_2$ reacts with NaOH to form a white precipitate of zinc hydroxide which dissolves in excess of NaOH.

$$ZnCl_2 + 2NaOH \longrightarrow \underset{\text{White ppt.}}{Zn(OH)_2} + 2NaCl$$

$$Zn(OH)_2 + 2NaOH \longrightarrow \underset{\substack{\text{Sodium zincate}\\ \text{(Soluble)}}}{Na_2ZnO_2} + 2H_2O$$

$$Na_2ZnO_2 + H_2S \longrightarrow \underset{\text{White ppt.}}{ZnS} + 2NaOH$$

Test of Mn^{2+} Ions

MnS is dissolved in dil. HCl. The solution on treatment with NaOH gives a brown ppt. of MnO_2 which is dissolved in conc. HNO_3 to give magnese nitrate.

$$MnS + 2HCl \longrightarrow MnCl_2 + H_2S\uparrow$$

$$MnCl_2 + 2NaOH \longrightarrow Mn(OH)_2\downarrow + 2NaCl$$

$$Mn(OH)_2 + [O] \longrightarrow \underset{\text{Brown ppt.}}{MnO_2\downarrow} + H_2O$$

$$2MnO_2 + 4HNO_3 \xrightarrow{\Delta} 2Mn(NO_3) + 2H_2O + O_2\uparrow$$

When the solution is diluted and add sodium bismuthate, a purple colour is produced.

$$5NaBiO_3 + 2Mn(NO_3)_2 + 16HNO_3 \xrightarrow{\Delta} 2HMnO_4 + 5Bi(NO_3)_3 + 5NaNO_3 + 7H_2O$$

(Permanganic acid)

Purple colour

Take the solution in conc. HNO_3, add PbO_2 and then boil, the appearance of purple colour indicates the Mn^{2+} ion.

$$2Mn(NO_3)_2 + 5PbO_2 + 6HNO_3 \xrightarrow[\text{cool}]{\text{Boil}} 2HMnO_4 + 5Pb(NO_3)_3 + 2H_2O$$

Purple colour

Test of Ni^{2+} Ions

Black ppt. of NiS is boiled with aquaregia.

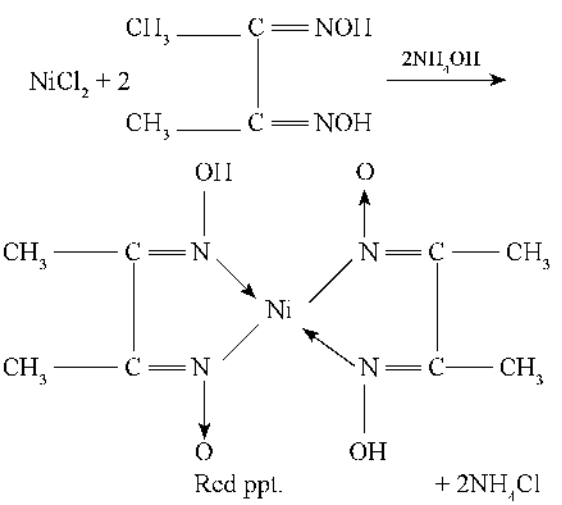
$$2HNO_3 \longrightarrow H_2O + 2NO_2 + [O]$$

$$2HCl + [O] \longrightarrow H_2O + 2[Cl]$$

$$NiS + 2[Cl] \longrightarrow NiCl_2 + S$$

Black ppt. soluble

Add dimethyl glyoxime to ammonical solution of $NiCl_2$. Formation of rosy red ppt. shows the presence of Ni^{2+} ions.



(Nickel dimenthyl glyoxime)

On hearing $NiCl_2$ with sodium bicarbonate, it gives nickel carbonate, which is converted into a black ppt. of nickel oxide followed by addition of NaOH and Br_2 water.

$$NiCl_2 + 2NaHCO_3 \longrightarrow NiCO_3 + 2NaCl + H_2O + CO_2$$

$$2NiCO_3 + 4NaOH + [O] \xrightarrow{\text{Boil}} Ni_2O_3\downarrow + 2Na_2CO_3 + 2H_2O$$

Black ppt.

Test of Co^{2+} Ions

Cobalt sulphide is dissolved in boiling aquaregia.

$$2HNO_3 \longrightarrow H_2O + 2NO_2 + [O]$$

$$CoS + 2HCl + 3[O] \longrightarrow CoCl_2 + H_2O + SO_2$$

Soluble

Cobalt salt solution reacts with KCN to give a reddish brown ppt. which dissolves in excess of KCN forming a yellowish brown solution of potassium cobalto cyanide. On boiling it is oxidized into bright yellow solution of potassium cobalticyanide.

$$CoCl_2 + 2KCN \longrightarrow Co(CN)_2\downarrow + 2KCl$$

(Reddish brown ppt.)

$$Co(CN)_2 + 4KCN \longrightarrow K_4[Co(CN)_6]$$

(Yellowish brown solution)

$$2K_4[Co(CN)_6] + H_2O + [O] \longrightarrow 2K_3[Co(CN)_6] + 2KOH$$

Bright yellow solution potassium cabalto cyanide

VTH GROUP CATION

Test of Ba^{2+} Ions

White ppt. of $BaCO_3$ is dissolved in hot dilute acid to form soluble barium acetate.

$$BaCO_3 + 2CH_3COOH \xrightarrow{\Delta} (CH_3COO)_2Ba + CO_2\uparrow + H_2O$$

Addition of K_2CrO_4 gives yellow ppt. of barium chromate.

$$(CH_3COO)_2Ba + K_2CrO_4 \longrightarrow BaCrO_4\downarrow + 2CH_3COOK$$

$BaCrO_4$ is soluble in conc. HCl.

$$2BaCrO_4 + 16HCl \longrightarrow 2BaCl_2 + 2CrCl_3 + 3Cl_2 + 8H_2O$$

With $(NH_4)_2SO_4$, Ba^{2+} ion gives a ppt. of $BaSO_4$, which is insoluble even in conc. HNO_3.

$$(CH_3COO)_2Ba + (NH_4)_2SO_4 \rightarrow BaSO_4\downarrow + CH_3COONH_4$$

With ammonium oxalate, it gives a white ppt. of barium oxalate.

$$(CH_3COO)_2Ba + (NH_4)_2C_2O_4 \rightarrow BaC_2O_4\downarrow + 2CH_3COONH_4$$

VIth GROUP CATION

Test of Mg^{2+} Ions

Mg^{2+} ions give a white precipitate with NH_4OH and $(NH_4)_2HPO_4$.

$$MgCl_2 + (NH_4)_2HPO_4 + NH_4OH \longrightarrow Mg(NH_4)PO_4\downarrow + 2NH_4Cl + H_2O$$

This test can also be done by using disodium hydrogen phosphate

$$MgCl_2 + Na_2HPO_4 + NH_4OH \rightarrow Mg(NH_4)PO_4\downarrow + 2NaCl + H_2O$$

Fusion Mixture Test

It is a mixture of Na_2CO_3 and KNO_3. When the salt is fused with the above mixture.

(i) Green colour _____ Mn is indicated

(ii) Yellow colour _____ Cr is indicated

$$MnSO_4 + 2KNO_3 + 2Na_2CO_3 \xrightarrow{\text{Fuse}} Na_2MnO_4 + 2KNO_2 + Na_2SO_4 + 2CO_2$$
[Green]

$$Cr_2(SO_4)_3 + 5Na_2CO_3 + 3KNO_3 \xrightarrow{\text{Fuse}} 2Na_2CrO_4 + 3KNO_2 + 5CO_2 + 3Na_2SO_4$$
[Yellow]

SOME DRY TESTS

1. Flame Test

(a) The uppermost part of the flame is called the nonluminous or oxidizing flame, while the middle and lower are called the luminous or reducing flame.

(b) Luminous character of the flame is due to the presence of carbon particles in the flame and it is used for performing charcoal cavity test.

(c) Nonluminous flame is the hottest part of the flame and it is used for performing flame test.

(d) Highest temperature attained in a Bunsen flame is about 1550°C.

(e) $Pb(NO_3)_2$ decrepitates (giving cracking sound) on heating.

Characteristic Flame Colour

- Pb imparts pale greenish colour to the flame.
- Cu and its salts impart blue or green colour to the flame.
- Borates also impart green colour to the flame.
- Ba and its salts imparts apple green colour to the flame.
- Sr imparts crimson red colour to the flame.
- Ca imparts brick red colour to the flame.
- Na imparts yellow colour to the flame.
- K imparts pink-violet(lilac) colour to the flame.
- Livid-blue flame is given by As, Sb and Bi.

2. Borax Bead Test

The transparent glassy mass of $(NaBO_2 + B_2O_3)$ is known as borax bead. It is obtained by heating borax on platinum wire as follows:

$$Na_2B_4O_7.10H_2O \xrightarrow{\Delta} Na_2B_4O_7 + 10H_2O$$

$$Na_2B_4O_7 \xrightarrow{\Delta} \underbrace{2NaBO_2 + B_2O_3}_{\text{Glassy bead}}$$

Borax bead is used to detect the cations like Cu, Cr, Co, Fe, Mn, Ni

Reactions

Copper

$$CuO + B_2O_3 \rightarrow Cu(BO_2)_2$$
$$CuO + NaBO_2 \rightarrow NaCuBO_3$$

$$2Cu(BO_2)_2 + 4NaBO_2 + 2C \rightarrow 2Cu + 2Na_2B_4O_7 + 2CO\uparrow$$
(Red) (Reducing flame)

Chromium

$$Cr_2(SO_4)_3 + 3B_2O_3 \longrightarrow 2Cr(BO_2)_3 + 3SO_3\uparrow$$
(Green) (Oxidizing flame)

Cobalt

$$2NaBO_2 + CoCO_3 \longrightarrow Co(BO_2)_2 + 3NaCl$$
(Blue)

Iron

$$FeCl_3 + 3NaBO_2 \longrightarrow Fe(BO_2)_3 + 3NaCl$$
(Yellow)

Manganese

$$MnO + B_2O_3 \longrightarrow Mn(BO_2)_2$$
(Colourless)-Mangnous metaborate

Nickel

$$NiO + B_2O_3 \rightarrow Ni(BO_2)_2$$
(Brown)

$$Ni(BO_2)_2 + C \rightarrow Ni + B_2O_3 + CO$$
(Grey)

REMEMBER

There are different colours in oxidizing flame and reducing flame.

Table 10.3 Borax Bead Test

Metal	Metaborate	Oxidizing flame		Reducing flame	
		Hot	Cold	Hot	Cold
Chromium	$Cr(BO_2)_3$	Green	Green	Green	Green
Cobalt	$Co(BO_2)_2$	Blue	Blue	Blue	Blue
Copper	$Cu(BO_2)_2$	Green	Blue	Colourless	Brown red
Iron	$Fe(BO_2)_3$	Brown yellow	Pale yellow	Bottle green	Bottle green
Manganese	$Mn(BO_2)_2$	Violet	Amethyst	Grey	Grey
Nickel	$Ni(BO_2)_2$	Violet	Brown	Grey	Grey

3. Microcosmic Salt Bead Test

It is used to identify cations like in borax bead test. Here, microcosmic salt is heated on platinum wire loop to get sodium meta phosphate as follows.

$$Na(NH_4)HPO_4.4H_2O \longrightarrow Na(NH_4)HPO_4 + 4H_2O$$

$$Na(NH_4)HPO_4 \longrightarrow NaPO_3 + NH_3 + H_2O$$
(Glassy mass)

Now, $NaPO_3$ reacts with metallic oxides to give coloured orthophosphates.

$$NaPO_3 + CoO \rightarrow NaCoPO_4 \text{(Blue)}$$

$$NaPO_3 + Cr_2O_3 \rightarrow NaPO_3.Cr_2O_3 \text{ (Green)}$$

$$NaPO_3 + CuO \longrightarrow NaCuPO_4 \text{ (Blue)}$$

Table 10.4 Microcosmic Salt Bead Test

Metal	Colour in Bead Oxidizing flame	
	Hot	Cold
1. Cu	Green	Blue
2. Fe	Yellow	Yellow
3. Cr	Green	Green
4. Mn	Violet	Violet
5. CO	Blue	Blue
6. Ni	Brown	Brown

4. Charcoal Cavity Test

Here, the mixture is heated with fusion mixture ($Na_2CO_3 + K_2CO_3$) in a charcoal cavity in reducing flame of the bunsen burner. Now the colour of the flame or residue suggests the possible cation.

For example:

If the residue or incrustation is yellow in hot and white in cold, it is Zn^{2+}.

If the residue or incrustation is brown in hot and cold, it is Cd^{2+}.

If the residue or incrustation is brown in hot and yellow in cold, it is Pb^{2+}.

Table 10.5 Charcoal Cavity Test

Incrustation or Residue	Metallic Bead	Cation Indicated
Brown when hot, brown when cold	None	Cd^{2+}
Brown when hot, yellow when cold	Grey bead which marks paper	Pb^{2+}
Characteristic residue	Red bead or scales	Cu^{2+}
No characteristic residue	Shining white	Ag^{+}
White residue which glows on heating	None	Ba^{2+}, Ca^{2-}, Mg^{2-}
Yellow when hot, white when cold	None	Zn^{2+}

5. Cobalt Nitrate Test

Here, a drop of cobalt nitrate solution is added in the white residue left in charcoal cavity. It is heated in an oxidizing flame and the colour of residue suggests cation.

Table 10.6 Cobalt Nitrate Test

Experiment	Observation (Residue colour)		Inference
Add a drop of cobalt nitrate solution to a white residue in the charcoal cavity and heat it in an oxidizing flame.	1	Blue infusible	Due to Al^{3}
	2	Blue fusible	Due to PO_4^{3-}, BO_3^{3-}
	3	Greenish	Due to Zn^{2}
	4	Pinkish	Due to Mg^{2+}
	5	Black residue	Absence of Al^{3-}, Sn^{2-}, Zn^{2+}, Mg^{2-}

Unforgettable Guidelines

- Nitrate gives a brown ring when it reacts with conc. H_2SO_4 in presence of $FeSO_4$ due to the formation of $FeSO_4.NO$ or $[Fe(H_2O)_5NO]SO_4$ complex compound.
- The black precipitate of mercury is dissolved in aqua regia to give $HgCl_2$, which forms white precipitate turning grey with $SnCl_2$ firstly due to the formation of Hg_2Cl_2 and then Hg.
- HgS is insoluble in 50 percent HNO_3 while sulphides of Pb^{2}, Bi^{3+}, Cu^{2} and Cd^{2} are soluble forming their nitrates.
- Cu^{2+} and Cd^{2+} are separated with the help of KCN solution where $K_3[Cu(CN)_4]$ and $K_2[Cd(CN)_4]$ complexes are formed. On passing H_2S gas, only Cd^{2+} complex is decomposed to give yellow precipitate of CdS.
- Fe^{3} ions give blood red colour with potassium thiocyanate due to the formation of $Fe(CNS)_3$ or $[Fe(SCN)_6]^{3}$ ion and Prussian blue colour with potassium ferrocyanide due to the formation of $Fe_4[Fe(CN)_6]_3$ (ferri ferrocyanide).
- **Interference of Acid Radicals in the Analysis of Basic Radicals:** Some acid radicals like $C_2O_4^{2-}$, F^{-}, BO_3^{3-} and PO_4^{3-} interfere in the systematic analysis of basic radicals after group II. This is because upto group II, the medium is highly acidic due to the presence of strong HCl, so the oxalate, fluorides, borates and phosphates of the basic radicals of group III, IV, V and Mg^{2+} are soluble. However, in group III, the solution is made alkaline by adding NH_4OH, the oxalates, fluorides, borates and phosphates of the basic radicals of group III, IV, V and Mg^{2+} form precipitates in the alkaline medium.

  Hence, interfering radicals cause precipitation of the cations of groups IV, V and magnesium in group III instead in their own groups.
- An asbestos fibre can be safely used in place of Pt wire for performing flame test. Glass rod should never be used as it gives a golden yellow persistent colour due to sodium present in it.
- Salts like sodium sulphide, potassium nitrite, sodium nitrite develop yellow colour.

- The flame test should be avoided in case of Sn, Pb, As, Sb and Bi salts since they corrode the Pt wire.
- Chlorides of mercury owing to little ionization do not respond to chromyl chloride test.
- In CS_2 layer test for I^- and Br^-, use of excess of Cl_2 water is to be avoided as it reacts with I^- and Br^- to give colourless HIO_3 and HBrO, respectively.
- Crystals of certain substances like KI, NaCl, $Pb(NO_3)_2$, etc. have minute quantities of mother liquor within their structure. On heating, such crystals burst into pieces with a crackling sound due to vapourization of enclosed water.
- Carbonates of bismuth and barium are not easily decomposed by dil. H_2SO_4. Use dil. HCl. This is because $BaSO_4$ and $Bi_2(SO_4)_3$ are insoluble in water.
- Reducing agents like S^{2-}, SO_3^{2-} etc. interfere in ammonium molybdate test as they reduce the reagent to molybdenum blue ($Mo_3O_8.xH_2O$). In such a case boil the salt with HNO_3 to oxidize these ions (S^{2-}, SO_3^{2-} etc.) before testing for phosphate.
- Before testing acetate in the aqueous solution by $FeCl_3$, it must be made clear that the solution does not contain CO_3^{2-}, PO_4^{3-}, SO_3^{2-} and I^- since these also combine with Fe^{3+}. Therefore, test of acetate should be performed by neutral ferric chloride only after the removal of these ions with $AgNO_3$.
- Ring test is not reliable in presence of nitrite, bromide and iodide.

Straight Objective Type Questions (Single Choice)

1. A metal hydrzoxide is precipitated as a white gelarinous substance when NH_4OH is added to the solution (after previously removing acid insoluble sulphides with H_2S). The metal in meral hydroxide is
   (a) Cr (b) Fe
   (c) Al (d) Zn

2. Which compound does not dissolve in hot dilute HNO_3?
   (a) HgS (b) PbS
   (c) CuS (d) CdS

3. A pale green crysralline metal salr of (X) dissolves freely in water. It gives a brown precipitate on the addition of aq. NaOH. This metal salt solurion also gives a black precipirare on bubbling H_2S in aqueous medium. An aqueous solution of the metal salt decolourizes the pink colour of rhe permanganare solurion. The meral in rhe meral salr solurion is
   (a) Fe (b) Pb
   (c) Cu (d) Al

4. $[Ni(DMG)_2]$, a cherry red coloured complex associate with extra stability due to
   (a) Small size of Ni^2
   (b) $3d^8$ electronic configuration of Ni^{2+}
   (c) Covalenr bonding
   (d) Hydrogen bonding and chelarion

5. When $K_2Cr_2O_7$ is hearcd with conc. H_2SO_4 and soluble chloride such as KCl
   (a) Red vapours of CrO_2Cl_2 are evolved.
   (b) Cl ion is oxidized ro Cl_2 gas.
   (c) $CrCl_3$ is formed.
   (d) $Cr_2O_7^{2-}$ ion is reduced to green Cr^{3+} ion.

6. An aqueous solurion of $FeSO_4$, $Al_2(SO_4)_3$ and chrome alum is heated with excess of Na_2O_2 and filtered. The material obtained are
   (a) a colourless filtrare and a green residue
   (b) a yellow filtrate and a green residue
   (c) a yellow filtrare and a brown residue
   (d) a green filtrare and a brown residue

7. A salt on hearing with dilure H_2SO_4 and subsequenrly rrearmenr with a few drops of dilure $K_2Cr_2O_7$, turns into green solution. The salt may be a
   (a) Sulphare (b) Bromide
   (c) Sulphide (d) Nitrire

8. The reagenrs, NH_4Cl and aqueous NH_3 will precipitate
   (a) Ca^{2+} (b) Al^{3-}
   (c) Mg^{2+} (d) Zn^{2+}

9. Cu^{2-} and Cd^{2-} are detected in a mixture of their solurions by using
   (a) Concentrated HNO_3 and H_2S
   (b) $K_4[Fe(CN)_6]$ and H_2S
   (c) KCN and H_2S
   (d) HCl and H_2S

10. When a reagent (X) reacts with Fe^{3-} the solution rurns red due ro rhe formarion of a compound (Y). This reagent causes no change in colour with Fe^{2-} in the pure state. Here (X) and (Y) are, respectively
    (a) NH_4CNS and $[Fe(SCN)]^{2-}$
    (b) $K_4[Fe(CN)_6]$ and $Fe_4[Fe(CN)_6]_3$
    (c) Na_2HPO_4 and $FeSO_4$
    (d) $K_3[Fe(CN)_6]$ and $K_2Fe[Fe(CN)_6]$

11. An aqueous solurion of $FeSO_4.Al_2(SO_4)_3.24H_2O$ and chrome alum on heating with an excess of Na_2O_2 and filtration gives
    (a) A brown filtrare and a yellow residue.
    (b) A yellow filtrare and a brown residue.
    (c) A green filtrare and a brown residue.
    (d) A yellow filtrate and a green residue.

12. The ion that cannot be precipitated by both HCl and H_2S is
    (a) Ag^- (b) Cu^+
    (c) Sn^{2+} (d) Pb^{2+}

13. In a mixrure of PbS, ZnS and FeS_2, each component is separated from rhe orher by using rhe reagents in which of the following sequence in froth floatation process.

(a) Porassium ethylxanthate, KCN, NaOH, copper sulphate, acid
(b) KCN, $CuSO_4$, acid
(c) Potassium ethylxanthate, KCN
(d) None of these

14. Which of the following reagents can be used to distinguish between a sulphite and a sulphate in solution?
(a) $Na_2[Fe(CN)_5NO]$
(b) $BaCl_2$
(c) $Na_3[Co(NO_2)_6]$
(d) $FeSO_4$

15. Which one among the following pairs of ions cannot be separated by H_2S in dilute hydrochloric acid?
(a) Bi^{3+}, Sn^{4+} (b) Al^{3+}, Hg^{2+}
(c) Zn^{2+}, Cu^{2+} (d) Ni^{2+}, Cu^{2+}

16. A doctor by mistake administers a $Ba(NO_3)_2$ solution to a patient for radiography investigations. Which of the following should be given as the best to prevent the absorption of soluble barium?
(a) Na_2SO_4 (b) NaCl
(c) NH_4Cl (d) Na_2CO_3

17. The colour of the iodine solution is discharged by shaking with
(a) Sodium sulphide
(b) Sodium sulphate
(c) Sodium bromide
(d) Aqueous sulphur dioxide

18. Which of the following salt gives green colour mass in cobalt nitrate charcoal cavity test?
(a) Al salts (b) Zn salts
(c) Copper salts (d) Alums

19. When H_2S is passed through Hg_2^{2+}, we get
(a) Hg_2S (b) HgS
(c) HgS + Hg_2S (d) HgS + Hg

20. Before adding the reagents of group III, the solution is heated with some concentrated HNO_3 in order to
(a) Increase the NO_3^-
(b) Lower than pH
(c) Oxidize Fe^{2+} to Fe^{3+}
(d) Oxidize Cr^{3+} to $Cr_2O_7^{2-}$

21. Which of the following, on treatment with KCN, will give cyanogens gas?
(a) $[Cu(NH_3)_4]^{2+}$ (b) $[Zn(NH_3)_4]^{2+}$
(c) $[Ag(NH_3)_2]^+$ (d) $[Cd(NH_3)_4]^{2+}$

22. Which of the following pairs of cations can be separated by using or adding NaOH solution?
(a) Sn^{2+}, Pb^{2+} (b) Zn^{2+}, Pb^{2+}
(c) Cu^{2+}, Zn^{2+} (d) Pb^{2+}, Al^{3+}

23. Which of these is the correct group reagent for group cations?
(a) Mn^{2+} Co^{2+} Zn^{2+} Ni^{2+} ; dil. HCl
(b) Mn^{2+} Co^{2+} Zn^{2+} Ni^{2+}; NH_4Cl + NH_4OH + H_2S
(c) Mn^{2+} Co^{2+} Zn^{2+} Ni^{2+}; NH_4Cl + NH_4OH
(d) Mn^{2+} Co^{2+} Zn^{2+} Ni^{2+}; HCl + H_2S

24. A white solid imparts a violet colour to a bunsen flame. On being heated with concentrated H_2SO_4, the solid gives violet vapours that turns starch paper blue. The salt may be
(a) KI (b) NaI
(c) MgI_2 (d) $CaBr_2$

25. Thenard blue is
(a) $Cu(NH_3)_4SO_4$
(b) $CoAl_2O_4$
(c) $K_2Fe[Fe(CN)_6]$
(d) $Fe_4[Fe(CN)_6]_3$

26. If NH_4OH in presence of NH_4Cl is added to a solution containing $Al_2(SO_4)_3$ and $MgSO_4$, which of the following will precipitate?
(a) $Mg(OH)_2$ only
(b) $Al(OH)_3$ only
(c) $Al(OH)_3$ and $Mg(OH)_2$
(d) None of these

27. The aqueous solution of mixture gives white precipitate with dil. HCl which dissolves in excess of dil. HCl. It confirms
(a) $ZnSO_4$ + NaOH
(b) $BaCl_2$ + NaOH
(c) $AgNO_3$ + NaOH
(d) Na_2SO_4 + NaOH

28. The chocolate coloured precipitate is
(a) $[Fe(H_2O)_5(NO)]SO_4$
(b) $Fe_2[Fe(CN)_6]$
(c) $Cu_2[Fe(CN)_6]$
(d) $HgSO_4$

29. How do we differentiate between Fe^{3+} and Cr^{3+} in group III?

(a) by increasing NH_4^+ ion concentration
(b) by decreasing OH^- ion concentration
(c) by adding excess of NH_4OH solution
(d) both (a) and (b)

30. Freshly prepared chloride water is added to the aqueous solution of some halide salt containing some CS_2. After shaking the contents, a violet colour appeared in CS_2 layer. The halide ion in solution is
(a) Bromide
(b) Iodide
(c) Chloride
(d) Iodide as well as bromide

31. Sulphuric acid is not used for the preparation of original solution in the analysis of basic radicals because:
(a) It is a strong reducing agent.
(b) It decomposes many of the anions.
(c) If forms insoluble sulphates with some of the basic radicals.
(d) It is a strong oxidizing agent.

32. Nitric acid is generally not used for preparation of original solution in analysis of basic radicals, because
(a) It is a reducing agent.
(b) It is an oxidizing agent.
(c) Nitrate ions cause interference with the scheme of analysis.
(d) It forms insoluble nitrates.

33. An aqueous solution contains Hg^{2+}, Hg_2^{2+}, Pb^{2+} and Cd^{2+}. The addition of dil. HCl (approx. 6M) will precipitate
(a) $PbCl_2$ only
(b) Hg_2Cl_2 only
(c) $PbCl_2$ and Hg_2Cl_2
(d) $PbCl_2$ and $HgCl_2$

34. Which one of the following statement is correct?
(a) Ferric ions give a deep green precipitate on adding potassium ferrocyanide solution.
(b) From a mixed precipitate of AgCl and AgI, ammonia solution dissolves only AgCl.
(c) Manganese salts give a violet borax bead test in the reducing flame.
(d) On boiling a solution having K^+, Ca^{2+} and HCO_3^- ions we get a precipitate of $K_2Ca(CO_3)_2$.

35. Two different salts (A) (zinc nitrate) and (B) (potassium bromide) were separately warmed with conc. H_2SO_4. Which of them will produce reddish brown fumes that dissolve in CS_2 giving yellow solution
(a) Both (A) and (B)
(b) Only (A)
(c) Only (B)
(d) Neither (A) nor (B)

36. Which of the following salt gives a white precipitate with a solution of $AgNO_3$; a white precipitate with dil. H_2SO_4 and a green flame test
(a) Lead nitrate (b) Barium chloride
(c) Copper chloride (d) Copper nitrate

37. Two colourless solutions are mixed. A white precipitate results which turns black on addition of ammonia. The two solutions are, respectively
(a) Washing soda and barium nitrate.
(b) Sodium bicarbonate and calcium nitrate.
(c) Mercurous nitrate and sodium chloride.
(d) Lunar caustic and hypo.

38. A colourless salt gives white precipitate with $CaCl_2$ solution and can also decolourise MnO_4^- / H^+. Salt is decomposed by conc. H_2SO_4 forming gases
(a) CO, SO_2 (b) N_2, CO, CO_2
(c) CO, CO_2 (d) CO_2, SO_2

39. Water soluble mixture $\xrightarrow[\text{(ii) Filter}]{\text{(i) } BaCl_2}$ white ppt.
Filtrate + Br_2 water + $BaCl_2$ ⟶ White ppt.
The mixture contains
(a) SO_4^{2-} only (b) SO_3^{2-} only
(c) Both of these (d) None of these

40. Which of these gives precipitate with K_2CrO_4?
(a) Pb^{2+}, Ag^+, Ba^{2+}
(b) Hg_2^{2+}, Pb^{2+}, Ag^+, Ba^{2+}
(c) Pb^{2+}, Ba^{2+}
(d) Ag^+, Ba^{2+}

41. Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. The explanation for it is that

(a) Concentrated hydrochloric acid emits strongly smelling HCl gas all the time.
(b) Oxygen in air reacts with the emitted HCl gas to form a cloud of chlorine gas.
(c) Strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke.
(d) Due to strong affinity for water, concentrated hydrochloric acid pulls moisture of air towards itself. This moisture forms droplets of water and hence the cloud.

42. Ecofriendly reagent that can be used instead of H_2S is
(a) $(NH_4)_2S_2$ (b) S_8
(c) Na_2S (d) $(NH_4)_2CS_3$

43. On adding KNO_2 and CH_3COOH solution to the neutral solution of $CoCl_2$, there is formation of yellowish orange precipitate of
(a) $K_3[Co(NO_2)_4(CH_3COO)_2]$
(b) $K_2[Co(NO_2)_4]$
(c) $K_3[Co(NO_2)_6]$
(d) $K_4[Co(NO_2)_6]$

44. In sodium carbonate bead test, chromium salts (green) change to coloured bead
(a) Yellow, Na_2CrO_4 (b) Orange, $Na_2Cr_2O_7$
(c) Yellow, $PbCrO_4$ (d) Yellow, $BaCrO_4$

45. The reagents, NH_4Cl and aqueous NH_3 will precipitate
(a) Ca^{2+} (b) Al^{3}
(c) Mg^{2+} (d) Zn^{2+}

46. Microcosmic salt and borax are used in the identification of cations by dry tests. They are, respectively
(a) $Na_2B_4O_7$ and $NaNH_4HPO_4$
(b) $NaPO_3$ and $NaBO_2$
(c) $NaNH_4HPO_4.4H_2O$ and $Na_2B_4O_7.10H_2O$
(d) $NaBO_2$ and $NaPO_3$

47. Aq. solution contains $Zn(CH_3COO)_2$, $Cd(CH_3COO)_2$ and $Cu(CH_3COO)_2$. On passing H_2S gas, there is a precipitation of ... as sulphide
(a) Zn^{2}, Cu^{2} (b) Zn^{2+}, Cu^{2+}, Cd^{2}
(c) Zn^{2}, Cd^{2} (d) Cu^{2+}, Cd^{2+}

48. H_2S would separate the following at pH < 7.
(a) Cu^{2+}, Cr^{3-} (b) Zn^{2-}, Co^{2-}
(c) Cu^{2+}, As^{5+} (d) Cu^{2}, Cd^{2}

49. The ion that cannot be precipitated by both HCl and H_2S is
(a) Ag^{+} (b) Cu
(c) Sn^{2+} (d) Pb^{2-}

50. Potassium ferrocyanide yields a chocolate brown precipitate with
(a) copper salts (b) ferric salts
(c) ferrous salts (d) silver salts

51. Which of the following is a reaction test for a metal ion?
(a) PbO_2 + conc. HNO_3
(b) K_2HgI_4 + NaOH
(c) use of anhydrous $CuSO_4$
(d) Lucas's reagent

52. Which of the following reagents can be used to distinguish between SO_2 and CO_2?
(a) H_2O_2 + $BaCl_2$
(b) Acidified dichromate paper
(c) Lime water
(d) Both (a) and (b)

53. Which of the following reactions is relevant to the microcosmic salt bead test?
(a) $CoO + ZnO \longrightarrow CoZnO_2$
(b) $CoO + NaPO_3 \longrightarrow NaCoPO_4$
(c) $Al_2(SO_4)_3 + 3Na_2CO_3 \longrightarrow Al_2O_3 + 3Na_2SO_4 + 3CO_2$
(d) $Cr_2O_3 + 3B_2O_3 \longrightarrow 2Cr(BO_2)_3$

54. Which of the following ions cannot be detected by the borax bead or microcosmic bead test?
(a) Zn^{2+} (b) Cr^{3+}
(c) Cu^{2+} (d) Fe^{3+}

55. Which of the following pairs of cations cannot be separated by using H_2S in the presence of 0.2 M HCl?
(a) Pb^{2+}, Fe^{3} (b) Pb^{2}, Cr^{3+}
(c) Hg^{2-}, Bi^{3+} (d) Bi^{3+}, Al^{3-}

Brainteasers Objective Type Questions (Single Choice)

56. A red solid is insoluble in water, however, it becomes soluble if some KI is added to water. Heating the red solid in a test tube results in liberation of some violet coloured fumes and droplets. Metal appears on the cooler parts of the test tube. The red solid is
(a) Pb_3O_4 (b) HgO
(c) $(NH_4)_2Cr_2O_7$ (d) HgI_2

57. Which of the following statement is/are correct?
I. In $S_2O_3^{2-}$ both sulphur are different in nature.
II. Sodium acetate and lead acetate on heating giving same type of product, whereas Mn, Sn, Fe oxalate salt giving different type of products.
III. Aqueous solution OCl^-, S^{2-} and CO_3^{2-} basic in nature.
IV. NO_2 oxidizes I whereas Br_2 and Cl_2 oxidizes NO_2
(a) II only (b) II and IV
(c) I, III and IV (d) I, II and IV

58. When a mixture containing phosphate is heated with conc. HNO_3 and ammonium molybdate solution, a cannery yellow precipitate is formed. The formula of the yellow precipitate is
(a) $(NH_4)_3PO_4\,12MoO_4$
(b) $(NH_4)_3PO_4\,12MoO_3$
(c) $(NH_4)_3PO_4(NH_4)_2MoO_4$
(d) $(NH_4)_3PO_4$

59. If Fe^{3+} and Cr^{3+} both are present in group III of qualitative analysis, then distinction can be made by
(a) Addition of NH_4OH in presence of NH_4Cl, when $Cr(OH)_3$ and $Fe(OH)_3$ both are precipitated and on adding Br_2 water and NaOH, $Cr(OH)_3$ dissolves.
(b) Precipitates of $Cr(OH)_3$ and $Fe(OH)_3$ as obtained in (A) are treated with conc. HCl, when only $Fe(OH)_3$ dissolves.
(c) Addition of NH_4OH in presence of NH_4Cl, when only $Fe(OH)_3$ is precipitated.
(d) Both (A) and (B)

60. Concentrated sulphuric acid is put into two test tubes, (A) containing nitrate salt and (B) containing bromide salts and the contents are heated to evolve reddish brown gases which were passed through water. Water will
(a) Turn yellow by gas coming from (B).
(b) Turn yellow by gas coming from (A).
(c) Turn blood red by gas coming from (B).
(d) Turn brown by gas coming from (A).

61. From the following information,
(A) + $H_2SO_4 \longrightarrow$ (B)
(a colourless and irritating gas)
(B) + $K_2Cr_2O_7$ + $H_2SO_4 \longrightarrow$ Green coloured solution
Identify the pair (A) and (B) from the list given below:
(a) Cl^-, HCl
(b) S^{2-}, H_2S
(c) SO_3^{2-}, SO_2
(d) CO_3^{2-}, CO_2

62. Among the species A (Sr_2S_3), B (CuS), C ($AlCl_3$), D ($ZnCl_2$), which will be soluble in excess of NaOH?
(a) A, C and D (b) C and D only
(c) B and C only (d) A and D only

63. Reddish brown (chocolate) precipitate are formed by mixing solutions containing, respectively:
(a) Ba^{2+} and SO_4^{2-} ions
(b) Cu^{2+} and $[Fe(CN)_6]^{4-}$ ions
(c) Pb^{2+} and SO_4^{2-} ions
(d) Pb^{2+} and I^- ions.

64. A green substance, sparingly soluble in water, dissolves in HCl to yield a green solution.
(1) The sodium carbonate extract of the substance, neutralized with dilute HNO_3 gives a canary-yellow precipitate with excess of ammonium molybdate in presence of concentrated HNO_3.
(2) The residue of the sodium carbonate extract, dissolved in HCl yields a beautiful rose-red precipitate when treated with a solution of dimethyl glyoxime in ethanol followed by excess of

NH_4OH. What may be inferred about the substance? The substance is

(a) Nickel fluoride

(b) Nickel phosphate

(c) Copper phosphate

(d) Ferrous oxalate

65. Before adding the reagents of group IIIA, the solution is heated with some concentrated HNO_3 in order to

(a) Oxidize Fe^{2+} to Fe^{3+}

(b) Lower the pH

(c) Increase the $[NO_3^-]$

(d) Oxidize Cr^{3+} to $Cr_2O_7^{2-}$

66. The gradual addition of ammonium hydroxide to the aqueous solution of a salt (A) results in a brownish precipitate which does not dissolve in excess of NH_4OH. However, when HCl is added to the original solution a white precipitate is formed. The salt (A) is

(a) Silver salt (b) Lead salt

(c) Copper salt (d) Mercurous salt

67. Pick out the wrong statement.

(a) Golden yellow PbI_2 dissolves in hot water to give a colourless solution.

(b) Ba^{2+} and Ca^{2+} ions can be separated by adding CrO_4^{2-} ion in acetic acid medium.

(c) Salts of calcium, copper and nickel give a green flame colour.

(d) The sulphide ion gives with alkaline sodium nitroprusside, a violet colour.

68. A white coloured salt forms white sublimate in dry heating test and also gives ammonia on heating with caustic soda solution. Which of the following test will be shown positive by the salt?

(a) It will give greenish yellow gas on reaction with conc. H_2SO_4.

(b) It will give red gas by heating with $K_2Cr_2O_7$ (s) and conc. H_2SO_4.

(c) It will give colourless gas with dil. H_2SO_4.

(d) It will give ring test.

69. Suggest the names of the products X and Y.

$$(1)\ 2Ag^+ \text{ (excess)} + S_2O_3^{2-} \longrightarrow Ag_2S_2O_3 \xrightarrow{H_2O} H_2OX + H_2SO_4$$

$$(2)\ 2NO_2 + 2I^- \xrightarrow{\text{Acid medium}} \text{acid medium Y} + I_2 + 2H_2O$$

(a) X = Ag_2O, Y = N_2

(b) X = Ag_2S, Y = N_2

(c) X = Ag_2O, Y = $2NO_2$

(d) X = Ag_2S, Y = $2NO$

70. Certain yellow coloured solid gives reddish brown precipitate in group III of qualitative analysis. The solid does not react with dilute H_2SO_4 but when few drops of $KMnO_4$ solution is added to the hot suspension of salt in dilute H_2SO_4, its pink colour is discharged with effervescence and evolution of CO_2. The salt is likely to be

(a) Ferric carbonate (b) Ferric chloride

(c) Iron (II) oxalate (d) Ferrous sulphate

71. $$\text{A (Colourless salt)} \xrightarrow{\Delta} \underset{\text{Gas}}{\text{B} + \text{C}} + \text{D}$$

$$\text{D} \xrightarrow{H_2O} \text{E}$$

Gas (C) turns solution (E) milky. (B) burns with blue flame. (A) also decolourises MnO_4^-/H^+. Thus (A), (B), (C), (D) and (E) are

(a) A = CaC_2O_4, B = CO_2, C = CO, D = CaO, E = $Ca(OH)_2$

(b) A = $CaCO_3$, B = CaO, C = CO, D = CO_2, E = $Ca(OH)_2$

(c) A = $CaCl_2$, B = Cl_2, C = O_2, D = CaO, E = $Ca(OH)_2$

(d) A = CaC_2O_4, B = CO, C = CO_2, D = CaO, E = $Ca(OH)_2$

72. A green coloured water soluble substance, A, forms a greenish precipitate, B with NH_4OH in presence of NH_4Cl (excess). When B is oxidized with a little sodium bismuthate in presence of H_2SO_4, a red-orange solution is obtained. If now a little solid Na_2O_2 is added, an intensely blue coloured compound is formed, extractable with ether. (A) yields a bulky white precipitate with $BaCl_2$ solution. Suggest the nature of the compound (A).

(a) Nickel (II) sulphate

(b) Copper (II) sulphate

(c) Chromium (III) sulphate

(d) Ferrous sulphate

73. $KCl + K_2Cr_2O_7 + H_2SO_4 \xrightarrow{\Delta}$ Red gas

dil. NaOH

Yellow solution

$Pb(Ac)_2$

(P)

The formula and colour of (P) are, respectively

(a) $PbCrO_4$, yellow (b) $Cr_2(SO_4)$, green

(c) CrO_2Cl_2, red (d) $BaCrO_4$, Green

74. $NaNO_2$ + (X) ➤ (Y) $\xrightarrow{\Delta}$ ➤ N_2

(X) + $AgNO_3$ ➤ White ppt. soluble in (Z)

(X) NaOH, $\xrightarrow{NaOH, \Delta}$ ➤ (Z)

Hence, (X) is

(a) NH_4Cl (b) NH_4NO_3

(c) NH_4NO_2 (d) NaCl

75. A certain pale green substance, (A) becomes dark brown on adding $NaNO_2$ in presence of dil. H_2SO_4 its aqueous solution gives precipitates with

(I) $BaCl_2$ and

(II) NaOH

The latter NaOH in separate tests.

The latter precipitate (B), gradually changes colour from green to brown, on exposure to air. What is A?

(a) $NiSO_4$ (b) $FeSO_4$

(c) $CuSO_4$ (d) $Cr_2(SO_4)_3$

76. A white crystalline salt gave colourless pungent smelling vapours with hot conc. H_2SO_4. On adding the piece of paper to the contents, the colourless vapours become reddish brown. The paper act as

(a) Oxidising agent (b) Reducing agent

(c) Dehydrating agent (d) Catalyst

77. A substance on treatment with dilute H_2SO_4 liberates a colourless gas which produces (i) turbidity with baryta water and (ii) turns acidified dichromate solution green. These reactions indicate the presence of

(a) CO_3^{2-} (b) S^{2-}

(c) SO_3^{2-} (d) NO_2^-

78. The ratio of the amounts of H_2S needed to precipitate all the metal ions separately from 100 ml. 1M-$AgNO_3$ and from 100 ml 1M-$CuSO_4$ respectively will be

(a) 1 : 1 (B) 2 : 1

(c) 1 : 2 (d) 2 : 3

79. An alloy which is golden yellow in colour can be dissolved in hot nitric acid to yield a blue solution. Adding excess of alkali to this solution in the cold yields a pale blue precipitate which may be filtered off. The colourless filtrate gives, on adding excess of solid NH_4Cl, a white gelatinous precipitate. The pale blue residue on the filter paper dissolves in HCl to yield a green solution. What is the name of the alloy?

(a) Bell metal

(b) German silver

(c) Aluminium bronze

(d) Monel metal

80. From the following information

(A) + H_2SO_4 ➤ (B)

(a colourless and irritating gas)

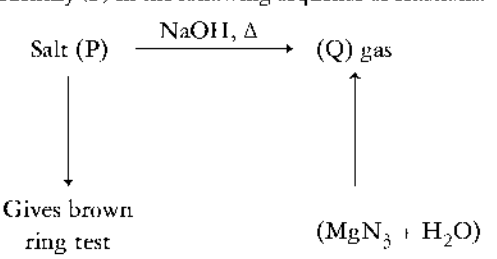
(B) + $K_2Cr_2O_7$ + H_2SO_4 ⟶ Green solution

Identify the pair (A) and (B) from the list given below:

(a) S^{2-}, H_2S (b) CO_3^{2-}, CO_2

(c) SO_3^{2-}, SO_2 (d) Cl , HCl

81. Identify (P) in the following sequence of reactions.



(a) NH_4Cl (b) KNO_3

(c) $Ba(NO_3)_2$ (d) NH_4NO_3

82. A compound (A) on heating gives a colourless gas. The residue is dissolved in water to obtain (B). Excess CO_2 is passed through aqueous solution of (B) when (C) is formed. (C) on gentle heating gives back (A). The compound (A) is

(a) Na_2CO_3 (b) $NaHCO_3$

(c) $CaCO_3$ (d) $Ca(HCO_3)_2$

83. A light green metal chloride gives a yellow solution when treated with NaOH and H_2O_2. The solution changes its colour to orange on

acidification with dilute H_2SO_4. The orange colour of the solution is due to

(a) $Cr_2O_7^{2-}$ (b) CrO_5

(c) $Cl_2 8H_2O$ (d) CrO_4^{2-}

84. Which of the following reaction(s) are not relevant to the microcosmic salt bead test?

I. $Cr_2O_3 + 3B_2O_3 \longrightarrow 2Cr(BO_2)_3$

II. $CoO + ZnO \longrightarrow CoZnO_2$

III. $CoO + NaPO_3 \longrightarrow NaCoPO_4$

IV. $Al_2(SO_4)_3 + 3Na_2CO_3 \longrightarrow Al_2O_3 + 3Na_2SO_4 + 3CO_2$

(a) I, III and IV (b) I, II and IV

(c) II, III and IV (d) II and IV only

85. Choose the correct code by identifying (A), (B) and (C) in each case for the changes indicated.

(i) $CrO_2Cl_2 \xrightarrow{KOH} (A) \xrightarrow{\text{conc. } H_2SO_4} (B) \xrightarrow{AgNO_3} (C)$

(ii) $CrCl_3$ (aq) $\xrightarrow[NaOH]{\text{excess}} (A) \xrightarrow[H_2O,\ \text{boil}]{Na_2O_2} (B) \xrightarrow{\text{Lead acetate}} (C)$

(iii) $ZnSO_4$ (aq) $\xrightarrow{Na_2CO_3} (A) \xrightarrow{\text{Cobalt nitrate, }\Delta} (B) \xrightarrow{\text{Excess KCN}} (C)$

(iv) $CuCl_2$ (aq) $\xrightarrow[H_2S]{NH_4OH} (A) \xrightarrow[\Delta]{HNO_3} (B) \xrightarrow{\text{Excess KCN}} (C)$

(a) A = CuS, B = $Cu(NO_3)_2$, C = $K_3[Cu(CN)_4]$

(b) A = $ZnCO_3$, B = ZnO, C = $CoZnO_2$

(c) A = K_2CrO_4, B = $K_2Cr_2O_7$, C = Ag_2CrO_4

(d) A = $Cr(OH)_3$, B = Na_2CrO_4, C = $PbCrO_4$

86. Three test tubes P, Q, R contain Pb^{2+}, Hg_2^{2+} and Ag^+ (but unknown). To each aqueous solution NaOH is added in excess. The following changes occur

P: Black precipitate

Q: Brown precipitate

R: White precipitate but dissolves in excess of NaOH

(P), (Q) and (R) contain respectively.

(a) Ag^+, Pb^{2+}, Hg_2^{2+} (b) Ag^+, Hg_2^{2+}, Pb^{2+}

(c) Hg_2^{2+}, Ag^+, Pb^{2+} (d) Pb^{2+}, Hg_2^{2+}, Ag^+

87. A sulphate of a metal (A) on heating evolves two gases (B) and (C) and an oxide (D). Gas (B) turns $K_2Cr_2O_7$ paper green while gas (C) forms a trimer in which there is no S-S bond. Compound (D) with conc. HCl forms a Lewis acid (E) which exists in a dimer. Compounds (A), (B), (C), (D) and (E) are, respectively

(a) FeS, SO_2, SO_3, $FeSO_4$, $FeCl_3$

(b) FeS, SO_2, SO_3, $Fe_2(PO_4)_3$, $FeCl_2$

(c) $FeSO_4$, SO_2, SO_3, Fe_2O_3, $FeCl_3$

(d) $Al_2(SO_4)_3$, SO_2, SO_3, Al_2O_3, $FeCl_3$

88. Identify (A) in the following sequence of reaction

A (Black) $\xrightarrow{\text{dil. } H_2SO_4}$ B (gas) $\xrightarrow{\text{dil. } HNO_3}$ Colloidal sulphur

(a) PbS (b) NiS

(c) CuS (d) FeS

89. Which of the following compound/s is/are partially soluble or insoluble in NH_4OH solution?

(I) $Fe(OH)_3$ (II) Ag_2CrO_4

(III) $Al(OH)_3$ (IV) Ag_2CO_3

(a) II and III (b) I and III

(c) I, III and IV (d) II, III and IV

90. Hg_2^{2+} when reacts with H_2S, black precipitate (X) formed which when reacts with Na_2S followed by filtration leaving behind black precipitate (Y). The filtrate with H^+ gives black precipitate (Z). (X), (Y) and (Z) are

(a) Hg + HgS, Hg, HgS

(b) Hg_2S, Hg, HgS

(c) Hg_2S, HgS, Hg

(d) Hg + HgS, HgS, Hg

91. Identify (P) to (R):

(P) $\xrightarrow{KOH}$ (Q) + (R) (gas turns red litmus blue)

$\xrightarrow{Zn + KOH}$ (Q) gas

(P) $\xrightarrow{\Delta}$ Gas (does not support combustion)

(a) P = $(NH_4)_2SO_4$, Q = NH_3, R = K_2SO_4

(b) P = NH_4NO_3, Q = NH_3, R = KNO_3

(c) P = NH_4NO_2, Q = NH_3, R = KNO_2

(d) P = $(NH_4)_2Cr_2O_7$, Q = NH_3, R = Cr_2O_3

92. In the following reactions, compound P is

(a) Red lead

(b) Barium carbonate

(c) Calcium carbonate

(d) Lead carbonate

(Clear Solution) S $\xleftarrow{\text{dil.HCl}}$ (P)

$\downarrow$ K_2CrO_4 (in acetic acid)

(Q) yellow ppt.

(P) $\xrightarrow{\text{dil. } H_2SO_4}$ (R) white ppt.

Multiple Correct Answer Questions (More Than One Choice)

93. On reaction with dilute H_2SO_4, an inorganic salt (A) gives a gas which gives a milky precipitate on passing through lime water. The salt is

(a) $NaNO_3$ (b) Na_2CO_3

(c) HCOONa (d) Na_2SO_3

94. Which of the following cation is/are detected by the flame test?

(a) Be^{2} (b) Mg^{2+}

(c) K^{+} (d) Ba^{2+}

95. Which statements is/are incorrect with reference to the ferrous and ferric ions?

(a) Fe^{2+} gives brown colour with potassium thicyanate.

(b) Fe^{3} gives red colour with potassium thiocyanate.

(c) Fe^{2+} gives blue colour with potassium ferricyanide.

(d) Fe^{2-} gives brown colour with potassium ferricyanide.

96. Heating with concentrated HNO_3 and an ammonium molybdate solution, a salt solution gives a yellow precipitate. It may be

(a) As_2O_3 (b) $CuSO_4$

(c) Na_2HPO_4 (d) $CaCl_2$

97. The presence of which of these radicals can be detected by using dilute H_2SO_4?

(a) Sulphide (b) Sulphite

(c) Chloride (d) Nitrate

98. Which of the following reagents will be useful in separating a mixture of Zn^{2} and Cu^{2} ?

(a) H_2S in an alkaline medium

(b) NH_3 solution

(c) H_2S in an acid medium

(d) Excess NaOH solution

99. On reaction with H_2S gas, the solution of a salt in dilute HCl gives a black precipitate. The salt is/are

(a) Nickel salt (b) Copper salt

(c) Calcium salt (d) Lead salt

100. Which of the following compounds can be used as a primary standard?

(a) $KBrO_3$

(b) $K_2Cr_2O_7$

(c) $Na_2B_4O_7.10H_2O$

(d) $Na_2S_2O_3.5H_2O$

101. Which of the following substance(s) is/are blue in colour?

(a) $CoAl_2O_4$ (b) $Co(BO_2)_2$

(c) $NaCoP_4$ (d) $Fe(BO_2)_2$

102. The sulphides of metals that is soluble in a yellow ammonium sulphide solution is/are

(a) Hg (b) Sn

(c) Sb (d) As

103. K_2CrO_4 gives a yellow precipitate on reaction with:
(a) Cu^{2+} (b) Fe^{3+}
(c) Ba^{2+} (d) Pb^{2+}

104. Which of the following ions can be separated by using NH_4Cl and NH_4OH?
(a) Cr^{3+} and Co^{2+} (b) Fe^{3+} and Cr^{3+}
(c) Al^{3+} and Ba^{2+} (d) Cr^{3+} and Al^{3+}

105. In an ammonical solution, a salt gives a black precipitate on passing H_2S through it. The salt is
(a) Lead salt (b) Cobalt salt
(c) Nickel salt (d) Mercury salt

106. Which of the following ions can be separated by using dilute HCl?
(a) Hg_2^{2+} and Cd^{2+} (b) Ag^+ and Al^{3+}
(c) Ag and Hg_2^{2+} (d) Ag^+ and Cu^{2+}

107. An inorganic salt (A) is mixed with an equal quantity of MnO_2, and heated with concentrated H_2SO_4. A gas is evolved which reacts with potassium iodide, and one of the products turns starch paper blue. Here, salt may be
(a) NaCl (b) $KHCO_3$
(c) KCl (d) $NaNO_2$

108. Potassium iodide is added to the freshly precipitated mercury (II) iodide. Which is/are the correct statement?
(a) The precipitate will dissolve.
(b) The anion present in the solution is $[HgI_4]^{2-}$.
(c) Nessler's reagent will be formed in the solution.
(d) The precipitate remain undissolved.

109. An aqueous solution containing S^{2-} ions will give
(a) White precipitate with $BaCl_2$ solution.
(b) Purple colour with sodium thiosulphate solution.
(c) Yellow precipitate with the suspension of $CdCO_3$ in water.
(d) Black precipitate with lead acetate solution.

110. Which of the following pair/s contain species, which react with each other on mixing their aqueous solutions to give a yellow precipitate?
(a) KI and KBr
(b) KI and I_2
(c) KI and silver nitrate
(d) KI and lead (II) nitrate

111. Which of the following salt can form coloured beads in borax bead test?
(a) $CoCl_2$
(b) $Ba(NO_3)_2$
(c) Potassium oxalate
(d) Cobalt oxalate

112. What is/are incorrect about carbonate salts?
(a) They produce CO_2 on treatment with dil. H_2SO_4.
(b) All carbonates are white or colourless.
(c) Heavy metal carbonates are generally insoluble in water.
(d) All carbonate decompose to give CO_2 and respective oxide.

113. Select the incorrect statement(s).
(a) $BaCO_3$ (s) + K_2CrO_4 ⟶ Yellow ppt.
(b) $SrCO_3$ (s) + K_2CrO_4 + AcOH ⟶ No ppt.
(c) $BaCl_2$ + AcOH + K_2CrO_4 ⟶ Yellow ppt
(d) $BaCO_3$ (s) + K_2CrO_4 + AcOH ⟶ No ppt.

114. Which of the following sulphate is/are soluble in water?
(a) Lead (II) sulphate
(b) Copper (II) sulphate
(c) Barium sulphate
(d) Silver (I) sulphate

115. A colourless salt on treatment with dilute HCl gave a colouress gas which turned lime water milky. The salt could be
(a) Sodium carbonate
(b) Sodium maleate
(c) Sodium tartarate
(d) Sodium bicarbonate

116. Which of the following metal sulphides are soluble in hot 50 % HNO_3?
(a) Bi_2S_3 (b) CuS
(c) HgS (d) Na_2S

117. Which of the following acid radical is/are not decomposed by dilute HCl?
(a) $C_2O_4^{2-}$ (b) I^-
(c) NO_2^- (d) SO_3^{2-}

118. Brown ring test for nitrate fails if the mixture of salts, contain along with nitrate,
(a) CO_3^{2-} ions (b) NO_2^- ions
(c) SO_4^{2-} ions (d) Br^- ions

119. Which statement is/are correct with reference to the ferrous and ferric ions?
(a) Fe^{2+} gives blue precipitate with potassium ferricyanide.
(b) Fe^{3+} gives brown colour with potassium ferricyanide.
(c) Fe^{2+} gives brown colour with ammonium thiocyanate.
(d) Fe^{3+} gives red colour with potassium thiocyanate.

120. Aq. solution of (A) $\xrightarrow{Na_3[Co(NO_2)_6]}$ Yellow ppt.

Aq. solution of (A) $\xrightarrow{H_2[PtCl_6]}$ Yellow ppt.

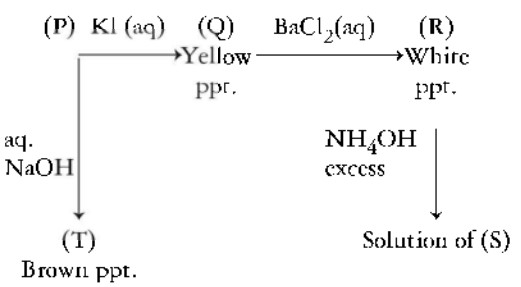
The cation(s) present in (A) is/are
(a) Na (b) NH_4^+
(c) K^+ (d) Mg^{2+}

121. $KMnO_4$ + Gas (X)
H_2O_2 + Gas (X) → Aq. Suspension
Br_2 Water + Gas (X)
Aq. Suspension —[Y] Reagent→ H_2SO_4

Which of the following option(s) is/are correct regarding (Y) among the following?
(a) HCl (b) conc. HNO_3
(c) O_3 (d) Excess Cl_2 water

Linked–Comprehension Type Questions

Comprehension–1



122. Here, compound (P) can be
(a) $AgNO_3$ (b) $Ca(NO_3)_2$
(C) $CuSO_4$ (d) $Pb(NO_3)_2$

123. Here, yellow precipitate (Q) is of
(a) CHI_3 (b) CaI_2
(c) AgI (d) PbI_2

124. Here, white precipitate (R) obtained on treatment with aqueous solution of $BaCl_2$ is of
(a) AgCl (b) $CaCl_2$
(c) $BaSO_4$ (d) $PbCl_2$

125. The compound (S) obtained when (R) dissolves in excess of NH_4OH is
(a) Ag_2O (b) $AgNO_3$
(c) $[Ag(NH_3)_2Cl]$ (d) AgOH

126. Molecular formula of the compound (T) given as
(a) $Ca(OH)_2$ (b) AgOH
(c) Ag_2O (d) $Pb(OH)_2$

Comprehension–2

A teacher gave a student two reagents (A) and (B) and told him to identify these reagents. The student heated reagent (A) strongly and observed two oxides of sulphur. He added NaOH solution to the aqueous solution of (A) and observed a dirty green precipitate, which turned brown on exposure to air.

When he took reagent (B) to flame test, a green colour was observed. On heating reagent (B) with a solid compound (X) and concentrated sulphur acid, orange red vapours are evolved. When this gas is passed through an aqueous solution of a base, the solution turns yellow.

127. The reagent (A) indicates the presence of which ion
(a) Fe^{2+} (b) Fe^{3+}
(c) Cr^{3+} (d) Cu^{2+}

128. The reagent (A) can be given as

(a) $Fe_2(SO_4)_3$ (b) $FeSO_4$
(c) $FeSO_4.7H_2O$ (d) Both (B) and (C)

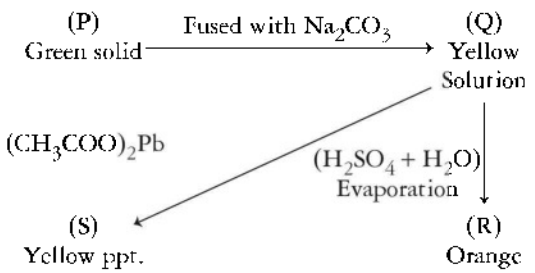
129. Here, reagent (B) suggest that the cation and anion in it, are respectively

(a) Ba^{2+} and SO_4^{2-} (b) Ba^{2} and Cl
(c) Ba^{2+} and S^{2-} (d) Ba^{2} and CO_3^{2-}

130. Here, the gas which turns aqueous solution of NaOH to yellow is

(a) Br_2 (b) SO_2
(c) Cr_2O_3 (d) Cl_2

Comprehension–3



Answer the following questions:

131. Here, green solid (P) is

(a) $FeSO_4$ (b) Cr_2O_3
(c) $FeCl_2$ (d) $FeSO_4$

132. Here, yellow solution (Q) can be

(a) $FeSO_4$ (b) $FeCl_3$
(c) $FeCO_3$ (d) Na_2CrO_4

133. Here, (R) can be

(a) $Fe(OH)_3$ (b) $PbSO_4$
(c) $Na_2Cr_2O_7$ (d) $Fe(OH)_3$

134. Here, yellow precipitate (S) is of

(a) $PbCrO_4$ (b) $PbCO_3$
(c) $PbCl_2$ (d) $Fe(OH)_3$

Comprehension–4

A black mineral (A) on heating in presence of air gives a gas (B). The mineral (A) on reaction with dilute H_2SO_4 gives a gas (C) and the solution of a compound (D). On passing gas (C) into an aqueous solution of (B), a white turbidity is obtained. The aqueous solution of compound (D) on reaction with potassium ferricyanide gives a blue compound (E).

135. Here the mineral (A) and gas (B) are, respectively

(a) PbS and SO_3

(b) FeS and SO_2

(c) FeS and H_2S

(d) PbS and SO_2

136. Here, the gas (C) and turbidity are

(a) SO_3 and sulphur

(b) SO_2 and sulphur

(c) H_2S and sulphur

(d) Oxygen and sulphur

137. Here, the oxidation number change for gas (C) is from

(a) +4 to Zero (b) −2 to Zero
(c) −2 to +4 (d) Zero to +2

138. Here, the blue coloured complex (E) is

(a) $NaFe[Fe(CN)_6]$ (b) $Fe_4[Fe(CN)_6]_3$
(c) $CoAl_2O_4$ (d) $KFe[Fe(CN)_6]$

Assertion and Reasoning Questions

In the following questions is two statements (Assertion) A and Reason (R) are given. Mark,

(a) If A and R both are correct and R is the correct explanation of A.

(b) If A and R both are correct but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

139. (A): Moistened ammonium salts give the smell of NH_3.

(R): Ammonium salts give NH_3 on heating with concentrated NaOH.

140. (A): If yellow precipitate is obtained on adding ammonium molybdate solution on boiling then phosphate radical is identified.

(R): Ammonium phosphomolybdate is a yellow compound.

141. (A): Green edge flame test confirms presence of borate ion.

(R): Green colour of the flame is due to burning of tri ethyl borate.

142. (A): The blue precipitate formed by the action of $K_4[Fe(CN)_6]$ on Fe^{3+} and by that of $K_3[Fe(CN)_6]$ on Fe^{2+} have the same composition.

(R): $[Fe(CN)_6]^{3-}$ oxidizes Fe^{2+} to Fe^{3+} and itself gets reduced to $[Fe(CN)_6]^{4-}$.

143. (A): Br^- ions do not interfere in the chromyl chloride test for chlorides.

(R): A bromide, on oxidation with $K_2Cr_2O_7$/ conc. H_2SO_4, liberates Br_2, which dissolves in NaOH to give a colourless solution.

144. (A): SO_4^{2-} ions in the salt are not decomposed by dil. H_2SO_4.

(R): Conc. H_2SO_4 decompose sulphates to give SO_2 gas.

145. (A): In charcoal cavity test an intimate mixture of salt and Na_2CO_3 is heated on a charcoal block.

(R): Charcoal cavity test is meant only for coloured salts.

146. (A): CdS and As_2S_3 both have yellow colour.

(R): CdS and As_2S_3 can be separated by yellow ammonium sulphide.

147. (A): Ring test fails if NO_3^- and Br^- ions are present together.

(R): Br ions are not decomposed by dil. H_2SO_4.

148. (A): Borax bead test is applicable only to coloured salts.

(R): In borax bead test coloured salts are decomposed to give metal meta borates, which are coloured.

149. (A): All soluble sulphides give white precipitate with $BaCl_2$ solution.

(R): BaS is soluble in water.

150. (A): A very dilute acidic solution of Cd^{2+} and Ni^{2+} gives yellow precipitate of CdS on passing hydrogen sulphide.

(R): Solubility product of CdS is more than that of NiS.

151. (A): All sulphates give white ppt. with $BaCl_2$ solution.

(R): $BaSO_4$ is insoluble in water.

152. (A): Both ZnS and MnS are insoluble in dil. HCl.

(R): These sulphides are precipitated in presence of NH_4OH.

153. (A): A brown gas which intensifies on adding copper turnings in conc. H_2SO_4 test for NO_3^-.

(R): Cu reacts with conc. HNO_3 to give brown coloured NO_2 gas.

154. (A): CO_2 containing H_2O changes CrO_4^{2-} (yellow) into $Cr_2O_7^{2-}$ (Orange).

(R): Aqueous CO_2 is acidic ($H_2CO_3 \leftrightarrow H^+ + HCO_3^-$) which converts CrO_4^{2-} to $Cr_2O_7^{2-}$

Matrix–Match Type Questions

	p	q	r	s
(A)	O	O	O	O
(B)	O	O	O	O
(C)	O	O	O	O
(D)	O	O	O	O

155. Match the following:

Column I	Column II
A. $ZnCl_2 + H_2S$	(p) Brown precipitate
B. $CuSO_4$ + Excess KI	(q) Pale green colouration
C. b_3O_4 + Conc. HNO_3	(r) No change is observed
D. $FeCl_3$ H_2S	(s) White turbidity

156. Match the following:

Column I	Column II
A. Fe^{2+}	(p) With $K_3[Fe(CN)_6]$, blue ppt.
B. Fe^{3}	(q) With $K_4[Fe(CN)_6]$, brown ppt.
C. Zn^{2+}	(r) With $K_4[Fe(CN)_6]$, bluish white ppt.
D. Cu^{2+}	(s) With $K_4[Fe(CN)_6]$, blue ppt.

157. Match the following:

Column I	Column II
A. Canary-yellow precipitate with ammonium molybdate	(p) NO_3^-
B. Ring test (brown)	(q) NO_2^-
C. Acid radical decomposed by dil. H_2SO_4	(r) PO_4^{3-}
D. Acid radical decomposed by conc. H_2SO_4	(s) As^3

158. Match the following:

Column I	Column II
A. $(NH_4)_2CO_3$ in presence of NH_4OH	(p) NH_4^+
B. NaOH	(q) V Group
C. Nessler's reagent	(r) Salts of Ca, Sr, Ba
D. Precipitate of carbonates	(s) Iodide of million's base

159. Match the following:

Column I	Column II
A. Pb^{2+}	(p) H_2S in presence of HCl
B. Hg_2^{2+}	(q) Soluble in ammonium sulphide
C. Hg^{2+}	(r) ppt. with $SnCl_2$
D. As^{3+}	(s) Dil. HCl

160. Match the following:

Column I (Sulphates)	Column II (Their properties)
A. $CuSO_4.5H_2O$	(p) Water soluble
B. $PbSO_4$	(q) Water insoluble
C. $BaSO_4$	(r) White residue (in solid state)
D. Ag_2SO_4	(s) Blue coloured

161. Match the following:

Column I (Sulphides)	Column II (Colour/group)
A. HgS	(p) Black
B. NiS	(q) Dirty white
C. MnS	(r) Buff colour
D. ZnS	(s) IInd group of basic radical

162. Match the following:

Column I (Radical)	Column II (Tested as)
A. Fe^{2+}	(p) Prussian blue
B. Fe^{3}	(q) Turnbull's blue
C. NH_4^+	(r) Iodide of Millons base
D. S^{2-}	(s) $Na_4[Fe(CN)_5NOS]$

163. Match the following:

Column I	Column II
A. Red vapours	(p) CrO_2Cl_2
B. NaOH solution is turned yellow by the vapours	(q) $MnSO_4$ + $NaBiO_3$ + conc. HNO_3
C. Purple solution	(r) $[Fe(CN)_5NOS]^{4-}$
D. A colourless solution results when the evolved gas is absorbed in an NaOH solution	(s) KBr heated with MnO_2 and conc. H_2SO_4

164. Match the following:

Column I (Base radical)	Column II (Flame colouration)
A. Na^+	(p) Violet
B. K^-	(q) Golden yellow
C. Ca^{2+}	(r) Crimson red
D. Sr^{2+}	(s) Brick red

The IIT–JEE Corner

165. Which compound does not dissolve in hot dilute HNO_3?
(a) HgS (b) PbS
(c) CuS (d) CdS
[*IIT 1996*]

166. An aqueous solution of $FeSO_4$, $Al_2(SO_4)_3$ and chrome alum is heated with excess of Na_2O_2 and filtered. The material obtained are
(a) a colourless filtrate and a green residue
(b) a yellow filtrate and a green residue
(c) a yellow filtrate and a brown residue
(d) a green filtrate and a brown residue
[*IIT 1996*]

167. Ammonium dichromate is used in some fire works. The green coloured powder blown in the air is
(a) CrO_3 (b) Cr_2O_3
(c) Cr (d) $CrO(O_2)$
[*IIT 1997*]

168. In nitroprusside ion, the iron and NO exist as Fe^{II} and NO^+ rather than Fe^{III} and NO. These forms can be differentiated by
(a) estimating the concentration of iron
(b) measuring the concentration of CN^-
(c) measuring the solid state magnetic moment
(d) thermally decomposing the compound
[*IIT 1998*]

169. Which of the following statement is correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
1. A deep red vapour is evolved
2. The vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4
3. Chlorine gas is evolved
4. Chromyl chloride is formed
(a) 1, 2, 4 (b) 1, 2 ,3
(c) 2, 3, 4 (d) All are correct
[*IIT 1998*]

170. An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a
(a) Hg_2^{2+} salt (b) Cu^{2+} salt
(c) Ag salt (d) Pb^{2+} salt
[*IIT 2002*]

171. Identify the correct order of solubility of Na_2S, CuS and ZnS in aqueous medium.

(a) $CuS > ZnS > Na_2S$

(b) $ZnS > Na_2S > CuS$

(c) $Na_2S > CuS > ZnS$

(d) $Na_2S > ZnS > CuS$

[*IIT 2002*]

172. A gas X is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white ppt. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas Y. Identify X and Y.

(a) X = CO_2, Y = Cl_2 (b) X = Cl_2, Y = CO_2

(c) X = Cl_2, Y = H_2 (d) X = H_2, Y = Cl_2

[*IIT 2002*]

173. [X] + H_2SO_4 ➤ [Y], a colourless gas with irritating smell.

[Y] + $K_2Cr_2O_7$ + H_2SO_4 ⟶ green solution.

[X] and [Y] are

(a) SO_3^{-2}, SO_2 (b) Cl^-, HCl

(c) S^{-2}, H_2S (d) CO_3^{-2}, CO_2

[*IIT 2003*]

174. A metal nitrate on reaction with KI gives black precipitate and with excess of KI gives orange solution. The metal is

(a) Hg^{2+} (b) Pb^{2}

(c) Cu^{2} (d) Bi^{3+}

[*IIT 2005*]

175. A solution, when diluted with H_2O and boiled, gives a white precipitate. On addition of excess NH_4Cl/NH_4OH, the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in NH_4OH/NH_4Cl.

(a) $Zn(OH)_2$ (b) $Al(OH)_3$

(c) $Mg(OH)_2$ (d) $Ca(OH)_2$

[*IIT 2006*]

176. A solution of a metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt(II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is

(a) Pb^{2+} (b) Hg^{2+}

(c) Cu^{2+} (d) Co^{2-}

[*IIT 2007*]

p-Amino-N, N-dimethylaniline is added to a strongly acidic solution of X. The resulting solution is treated with a few drops of aqueous solution of Y to yield blue coloration due to the formation of methylene blue. Treatment of the aqueous solution of Y with the reagent potassium hexacyanoferrate(II) leads to the formation of an intense blue precipitate. The precipitate dissovles on excess addition of the reagent. Similarly, treatment of the solution of Y with the solution of potassium hexacyanoferrate(III) leads to a brown coloration due to the formation of Z.

[*IIT 2009*]

13. The compound X is

(a) $NaNO_3$ (b) NaCl

(c) Na_2SO_4 (d) Na_2S

14. The compound Y is

(a) $MgCl_2$ (b) $FeCl_2$

(c) $FeCl_3$ (d) $ZnCl_2$

15. The compound Z is

(a) $Mg_2[Fe(CN)_6]$ (b) Fe $[Fe(CN)_6]$

(c) $Fe_4[Fe(CN)_6]_3$ (d) $K_2Zn_3[Fe(CN)_6]_2$

ANSWERS

Straight Objective Type Questions

1. (c)	2. (a)	3. (a)	4. (d)
5. (a)	6. (c)	7. (c)	8. (b)
9. (c)	10. (a)	11. (b)	12. (c)
13. (a)	14. (b)	15. a)	16. (a)
17. (d)	18. (b)	19. (d)	20. (c)
21. (a)	22. (c)	23. (b)	24. (a)
25. (b)	26. (b)	27. (a)	28. (c)
29. (d)	30. (b)	31. (c)	32. (b)
33. (d)	34. (b)	35. (c)	36. (b)
37. (c)	38. (c)	39. (c)	40. (b)
41. (b)	42. (d)	43. (c)	44. (a)
45. (b)	46. (c)	47. (b)	48. (a)
49. (c)	50. (a)	51. (a)	52. (d)
53. (b)	54. (a)	55. (c)	

Brainteasers Objective Type Questions

56. (d)	57. (c)	58. (b)	59. (a)
60. (a)	61. (c)	62. (a)	63. (b)
64. (b)	65. (a)	66. (d)	67. (c)
68. (b)	69. (d)	70. (c)	71. (d)
72. (c)	73. (a)	74. (a)	75. (b)
76. (b)	77. (c)	78. (c)	79. (c)
80. (c)	81. (d)	82. (c)	83. (a)
84. (b)	85. (a)	86. (c)	87. (c)
88. (d)	89. (b)	90. (a)	91. (c)
92. (b)			

Multiple Correct Answers Questions

93. (b), (d)	94. (c), (d)
95. (a), (d)	96. (a), (c)
97. (a), (b)	98. (a), (c), (d)
99. (b), (d)	100. a, b, c
101. (a), (b), (c)	102. (b), (c), (d)
103. (c), (d)	104. (a), (c)
105. (b), (c)	106. a, b, d
107. (a), (c)	108. (a), (b), (c)
109. (c), (d)	110. (c), (d)
111. (a), (d)	112. (b), (d)
113. (a), (d)	114. (b), (d)
115. (a), (d)	116. (a), (b)
117. (a), (b)	118. (b), (d)
119. (a), (d)	120. (b), (c)
121. (b), (c), (d)	

Linked–Comprehension Type Questions

Comprehension–1

122. (a)	123. (d)	124. (a)	125. (c)
126. (b)			

Comprehension–2

127. (a)	128. (d)	129. (d)	130. (d)

Comprehension–3

131. (b)	132. (d)	133. (c)	134. (a)

Comprehension–4

135. (b)	136. (c)	137. (b)	138. (d)

Assertion and Reasoning Questions

139. (b) 140. (d) 141. (a) 142. (a)

143. (a) 144. (c) 145. (c) 146. (b)

147. (b) 148. (a) 149. (d) 150. (b)

151. (a) 152. (d) 153. (a) 154. (a)

Matrix–Match Type Questions

155. (a)-(r), (b)-(p), (c)-(p), (d)-(q, s)

156. (a)-(p), (b)-(s), (c)-(r), (d)-(q)

157. (a)-(r, s), (b)-(p, q), (c)-(q), (d)-(p, q)

158. (a)-(q, r), (b)-(p), (c)-(p, s), (d)-(q, r)

159. (a)-(p, s), (b)-(r, s), (c)-(p, r), (d)-(p, q)

160. (a)-(p, s), (b)-(q, r), (c)-(q, r), (d)-(p, r)

161. (a)-(p, s), (b)-(p), (c)-(r), (d)-(q)

162. (a)-(q), (b)-(p), (c)-(r), (d)-(s)

163. (a)-(p, s), (b)-(p), (c)-(q, r), (d)-(s)

164. (a)-(q), (b)-(p), (c)-(s), (d)-(r)

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165. (a) 166. (c) 167. (b) 168. (c)

169. (a) 170. (d) 171. (d) 172. (c)

173. (a) 174. (d) 175. (a) 176. (b)

HINTS AND EXPLANATIONS

Straight Objective Type Questions

2. PbS, CuS and CdS (Gp II) dissolve in hot dil HNO_3 while HgS does not dissolve in dil HNO_3.

6. $FeSO_4$ first gets oxidized to $Fe_2(SO_4)_3$ and then get precipitated as brown $Fe(OH)_3$ precipitate. On the other hand, $Al(OH)_3$ and $Cr(OH)_3$ dissolve forming colourless and yellow solution of $NaAlO_2$ and Na_2CrO_4, respectively.

8. This is a group reagent for group III cations and hence only Al^{3+} will be precipitated.

12. Sn^{2+} can be precipitated by H_2S but not by HCl.

15. Cations belonging to group II of qualitative analysis (Bi^{3+}, Sn^{4+}) can not be separated by H_2S in dil HCl because both of these get precipitated.

16. $Ba(NO_3)_2 + Na_2SO_4 \longrightarrow \underset{\text{White ppt.}}{BaSO_4\downarrow} + 2NaNO_3$

30. Cl_2 oxidizes I^- ions to I_2 which dissolve in CS_2 to give violet colour.

31. H_2SO_4 forms insoluble sulphates with Pb^{2+}, Ba^{2+} and Sr^{2+}.

32. HNO_3 oxidizes H_2S to sulphur which may appear as yellowish ppt. or turbidity in IInd group.

34. Out of AgCl and AgI precipitate, only AgCl is soluble in NH_3 solution due to the formation of $[Ag(NH_3)_2]Cl$ complex.

35. KBr will give Br_2 vapour which dissolved in CS_2 readily to give yellow coloured solution.

36. Barium gives grassy green flame. Ba^{2+} ions also give white precipitate of $BaSO_4$ with H_2SO_4. At the same time Cl^- ions give white precipitate of AgCl with $AgNO_3$ solution.

37. $2HgNO_3 + 2NaCl \longrightarrow \underset{\text{White}}{Hg_2Cl} + 2NaNO_3$

$$Hg_2Cl_2 + 2NH_4OH \longrightarrow \underset{\text{Black}}{[Hg + Hg(NH_2)Cl]} + 2H_2O$$

45. This is a group reagent for group III cations and hence only Al^{3+} will be precipitated.

49. Sn^{2+} can be precipitated by H_2S but not by HCl.

Brainteasers Objective Type Questions

60. Br_2 gas will be evolved from bromide salt and will give brownish yellow solution on dissolving in water.

61. $SO_3^{2-} + H_2SO_4 \longrightarrow SO_4^{2-} + SO_2 + H_2O$

$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$ (Green)

63. $Cu^{2+} + [Fe(CN)_6]^{4-} \longrightarrow Cu_2[Fe(CN)_6]$ Chocolate ppt.

66. The observations show the presence of mercurous salt, which give white precipitate with dil. HCl.

68. White sublimate is obtained by NH_4Cl which support other observations also.

70. Observations support the presence of $Fe(C_2O_4)$ i.e., Ferrous oxalate salt.

76. The paper acts as reducing agent and reduces HNO_3 vapours to NO_2 which is reddish brown.

77. The substance must contain SO_3^{2-} ion because on reaction with dil H_2SO_4, SO_2 gas is released (colourless) which give turbidity with baryta water [$Ba(OH)_2$ solution] and also turns acidified $K_2Cr_2O_7$ solution green.

$SO_3^{2-} + 2H^+ \longrightarrow H_2O + SO_2$ (from dil H_2SO_4)

$Ba(OH)_2 + SO_2 \longrightarrow BaSO_3 + H_2O$ (insoluble turbidity)

$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$ (Green)

78. Ag^+ ions in 100 ml = 0.1 mol

Cu^{2+} ions in 100 ml = 0.1 mol

$2Ag^+ + H_2S \longrightarrow Ag_2S + 2H^+$

$Cu^{2+} + H_2S \longrightarrow CuS + 2H^+$

0.1 mol Ag^+ require H_2S = 0.05 mol;

0.1 mol Cu^{2+} require H_2S = 0.1 mol

Hence, ratio of H_2S required = 0.05 : 0.1 = 1 : 2

82. $CaCO_3 \xrightarrow{\Delta} CaO + CO_2\uparrow$
(A) Colourless gas

$CaO + H_2O \longrightarrow Ca(OH)_2$
Residue (B)

$Ca(OH)_2 + 2CO_2 \longrightarrow Ca(HCO_3)_2$
(B) excess (C)

$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + CO_2 + H_2O$
(C) (A)

83. Yellow solution is obtained due to oxidation of Cr^{3+} to CrO_4^{2-} ion. On acidification CrO_4^{2-} ions change to orange coloured $Cr_2O_7^{2-}$ ions.

85. (i) $CrO_2Cl_2 \xrightarrow{KOH} K_2CrO_4 \xrightarrow{conc.\ H_2SO_4}$
(A)

$K_2Cr_2O_7 \xrightarrow{AgNO_3} Ag_2Cr_2O_7$
(B) (C)

(ii) $CrCl_3$ aq $\xrightarrow[NaOH]{Excess} Na[Cr(OH)_4] \xrightarrow[H_2O,\ boil]{Na_2O_2}$
(A)

$Na_2CrO_4 \xrightarrow{Lead\ acetate} PbCrO_4$
(B) (C)

(iii) $ZnSO_4$ aq $\xrightarrow{Na_2CO_3} ZnCO_3.3Zn(OH)_2$
(A)

$\xrightarrow{\Delta} ZnO \xrightarrow{Cobalt\ nitrate,\ \Delta} CoZnO_2$
(B) (C)

(iv) $CuCl_2$ (aq) $\xrightarrow[H_2S]{NH_4OH} CuS \xrightarrow[\Delta]{HNO_3}$
(A)

$Cu(NO_3)_2 \xrightarrow{Excess\ KCN} K_2[Cu(CN)_4]$
(B) (C)

87. $2FeSO_4 \longrightarrow SO_2 + SO_3 + Fe_2O_3$
(A) (B) (C) (D)

$Fe_2O_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O$
(E)

88. FeS (Black) $\xrightarrow{dil.\ H_2SO_4} H_2S$ (gas)

$\xrightarrow{dil.\ HNO_3} S + NO_2 + 2H_2O$
Colloidal

89. Precipitates of Ag_2CrO_4, Ag_2CO_3 are soluble in NH_4OH due to formation of $[Ag(NH_3)_2]^+$.

$Ag_2CrO_4\downarrow + 4NH_4OH \longrightarrow 2[Ag(NH_3)_2] + CrO_4^{2-} + 4H_2O$

$Ag_2CO_3\downarrow + 4NH_4OH \longrightarrow 2[Ag(NH_3)_2] + CO_3^{2-} + 4H_2O$

$Fe(OH)_3$ is insoluble in NH_4OH.

$Al(OH)_3$ is insoluble in NH_4OH.

91. $NH_4NO_2 + KOH \longrightarrow NH_3\uparrow + KNO_2 + H_2O$
(P) (Q) (R)

$NO_2^- + 3Zn + 5OH^- + 5H_2O \longrightarrow$
(R)

$3[Zn(OH)_4]^{2-} + NH_3\uparrow$
Soluble (Q)

$NH_4NO_2 \xrightarrow{\Delta} N_2\uparrow + 2H_2O$
(P) (does not support combustion)

92. $BaCO_3 + K_2CrO_4 + H^+ \longrightarrow$
(P)

$BaCrO_4 + 2K^+ + CO_3^{2-}$
(Q) yellow ppt.

$BaCO_3 + H_2SO_4 \longrightarrow BaSO_4 + CO_2 + H_2O$
(R) white ppt.

$BaCO_3 + 2HCl \longrightarrow BaCl_2 + CO_2 + H_2O$
(S)
Clear solution

Multiple Correct Answers Type Questions

110. $KI + AgNO_3 \longrightarrow KNO_3 + AgI$
ppt.

$2KI + Pb(NO_3)_2 \longrightarrow 2KNO_3 + PbI_2$
ppt.

111. Both the cobalt salts can form bluish bead in borax bead test.

115. As HCO_3^- and CO_3^{2-} decompose by reaction with HCl to give CO_2 gas.

Comprehension–2

127. Formation of dirty green and brown precipitate confirms that it is ferrous ion as follows.

$Fe^{2+} + 2OH^- \longrightarrow Fe(OH)_2\downarrow$
Dirty green colour

$2Fe(OH)_2 + \frac{1}{2}O_2 + H_2O \longrightarrow 2Fe(OH)_3$
Brown colour

128. As both $FeSO_4$ and $FeSO_4.7H_2O$ on heating gives a mixture of two oxides SO_2 and SO_3.

$2FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_2 + SO_3$

Comprehension–4

135. $2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$
(A) (B)

136. $FeS + H_2SO_4 \longrightarrow FeSO_4 + H_2S$
(D) (C)

$2H_2S + SO_2 \longrightarrow 2H_2O + 3S$
(C) (B) Turbidity

138. $FeSO_4 + K_3Fe(CN)_6 \longrightarrow$

$KFe[Fe(CN)_6] + K_2SO_4$
(E) blue.

The IIT–JEE Corner

165. PbS, CuS and CdS (Gp II) dissolve in hot dil HNO_3 while HgS does not dissolve in dil HNO_3.

166. $FeSO_4$ first gets oxidized to $Fe_2(SO_4)_3$ and then get precipitated as brown $Fe(OH)_3$ precipitate. On the other hand, $Al(OH)_3$ and $Cr(OH)_3$ dissolve forming colourless and yellow solution of $NaAlO_2$ and Na_2CrO_4, respectively.

167. $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

168. This is clear from magnetic moment studies.

169. $4NaCl + K_2Cr_2O_7 + 6H_2SO_4$ (conc.) $\xrightarrow{\Delta}$
$2KHSO_4 + 4NaHSO_4 + 2CrO_2Cl_2 + 3H_2O$
(orange red)
Chromyl chloride

Chromyl chloride vapours when passed through NaOH solution gives a yellow solution of Na_2CrO_4.

170. The substance is a Pb^{2+} salt which give white precipitate with dil. HCl. The ppt. of $PbCl_2$ dissolves in hot solution. When H_2S gas is passed through the hot acidic solution, a black ppt. of PbS ($Pb^{2+} + S^{2-} \longrightarrow PbS$) is formed.

171. Solubility depends upon the lattice energy and hydration energy.

172. Since the saturated solution gives white precipitate with $AgNO_3$, so the solution must contain Cl^- ions. Thus, the gas X is Cl_2

Saturated solution + Mg $\longrightarrow MgCl_2 + H_2$
(colourless gas)

Thus, Y is I_2.

173. $\underset{(X)}{SO_3^{2-}} + H_2SO_4 \rightarrow \underset{(Y)}{SO_2} + H_2O + SO_4^{2-}$

SO_2 is a colourless gas with irritating smell.

$3SO_2 + K_2Cr_2O_7 + H_2SO_4 \rightarrow$
(Y)

$K_2SO_4 + Cr_2(SO_4)_3 + H_2O$
Green coloured solution
(it is due to Cr^{3+})

Hence, (X) is SO_3^{2-} and (Y) is SO_2.

174. Bismuth nitrate, $Bi(NO_3)_2$, reacts with KI to give a black precipitate of BiI_3 which on addition of excess of KI, dissolved to give orange colour solution of $KBiI_4$ complex salt.

$Bi(NO_3)_2$ (aq) + 3KI (aq) $\rightarrow$
BiI_3 (s) + $3KNO_3$ (aq)
Black ppt.

BiI_3 (s) + KI (aq) $\rightarrow K[BiI_4]$
Orange colour

175. Due to formation of tetrammine zinc(II) complex; $Zn^{2+} + NH_4OH \rightarrow [Zn(NH_3)_4]^{2}$

176. $Hg^{2+} + KI \longrightarrow \underset{\text{Red ppt.}}{HgI_2}$

$HgI_2 + KI$ (excess) $\rightarrow \underset{\text{Soluble}}{K_2HgI_4}$

$HgI^{2+} + Co(SCN)_2 \longrightarrow \underset{\text{Blue crystalline precipitate}}{Hg(SCN)_2}$

Solved Subjective Questions

1. A mixture of two salts was treated as follows:
   (i) The mixture was heated with manganese dioxide and concentrated sulphuric acid when yellowish-green gas was liberated.
   (ii) The mixture on heating with sodium hydroxide solution gave a gas which turned red litmus blue.
   (iii) Its solution in water gave blue precipitate with potassium ferricyanide and red colouration with ammonium thiocyanate.
   (iv) The mixture was boiled with potassium hydroxide and the liberated gas was bubbled through an alkaline solution of K_2HgI_4 to give brown precipitate.

Solution

(i) Mixture + MnO_2 + conc. $H_2SO_4 \rightarrow$
Yellowish green gas

The reaction indicates that the mixture contains Cl ions.

$2Cl^- + MnO_2 + \text{conc. } H_2SO_4 + 2H^- \rightarrow$
$MnSO_4 + Cl_2\uparrow + 2H_2O$
Yellowish green

(ii) Mixture + NaOH $\rightarrow$ Gas turning red litmus blue.

This reaction indicates that the gas is ammonia (i.e., basic in nature which turns red litmus blue).

$NH_4^+ + NaOH \rightarrow NH_3 + H_2O + Na^+$

(iii) Solution of mixture + $K_3[Fe(CN)_6] \rightarrow$
Blue ppt.

This reaction indicates that the mixture contains a Fe (II) salt. Fe (II) reacts with $K_4[Fe(CN)_6]$ to give a blue precipitate of prussian blue.

$3Fe^{2+} + 2[Fe(CN)_6]^{3-} \rightarrow \underset{\text{Prussian blue (Blue ppt.)}}{Fe_3[Fe(CN)_6]_2}$

Red colouration with ammonium thiocyanate indicates that some Fe(III) is also present. It is likely that a part of Fe(II) is oxidized to Fe(III) by air:

$Fe^{3+} + 3NH_4SCN \rightarrow \underset{\text{Red colour}}{Fe(SCN)_3} + 3NH_4^+$

$2Fe^{2+} + [O] + 2H^+ \rightarrow 2Fe^{3+} + H_2O$

(iv) Mixture + KOH $\longrightarrow$ Gas $\uparrow$

$$K_2HgI_4 \xrightarrow{K_2HgI_4} \text{Brown ppt.}$$

This reaction can be explained as follows: Ammonia is evolved on boiling the mixture with potassium hydroxide. The evolved ammonia reacts with an alkaline solution of K_2HgI_4 to give a brown precipitate.

So the mixture has Fe^{2+}, NH_4^+ and Cl^- ions with an impurity of Fe^{3+} ions hence the two salts are $FeCl_2$ and NH_4Cl.

2. The gas liberated on heating a mixture of two salts with NaOH, gives a reddish brown precipitate with an alkaline of K_2HgI_4. The aqueous solution of the mixture on treatment with $BaCl_2$ gives a white precipitate which is sparingly soluble in concentrated HCl. On heating the mixture with $K_2Cr_2O_7$ and conc. H_2SO_4, red vapours A are produced. The aqueous solution of the mixture gives a deep blue colouration B with potassium ferricyanide solution. Identify the radicals in the given mixture and write the balanced equations for the formation of A and B

[*IIT 1991*]

Solution

The Formation of reddish brown precipitate with alkaline K_2HgI_4 solution shows the presence of radical ammonium NH_4^+ and the gas evolved is ammonia and the reactions here are as follows:

$$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$$
$$K_2HgI_4 \rightarrow 2KI + HgI_2$$
$$HgI_2 + 2NH_3 \rightarrow Hg(NH_2)I + NH_4I$$
$$Hg(NH_2)I + HgI \rightarrow NH_2Hg_2I_2$$

On reaction with $BaCl_2$, a white precipitate is formed which confirms the presence of sulphate anion (SO_4^{2-}) sparingly soluble in conc. HCl

With $K_2Cr_2O_7$ and conc. H_2SO_4 red vapours are liberated which confirms the presence of chloride ion (Cl^-). The reaction is as follows:

$$K_2Cr_2O_7 + 4Cl^- + \text{conc. } 6H_2SO_4 \rightarrow 2CrO_2Cl_2 + 2KHSO_4 + 4HSO_4^- + 3H_2O$$
(A)

The reaction with potassium ferricyanide shows the presence of ferrous ions.

$$K^+ + Fe^{2+} + [Fe(CN)_6]^{3-} \rightarrow KFe[Fe(CN)_6]$$
(B)

3. A light bluish green crystalline compound responds to the following tests.
   (1) Its aqueous solution gives a brown precipitate or colour with alkaline $K_2[HgI_4]$ solution.
   (2) Its aqueous solution gives a blue colour with $K_3[Fe(CN)_6]$ solution.
   (3) Its solution in hydrochloric acid gives a white precipitate with $BaCl_2$ solution.

   Identify the ions present and suggest the formula of the compound.

[*IIT 1992*]

Solution

(1) Brown precipitate or colour with alkaline $K_2[HgI_4]$ (Nessler's reagent), confirms the presence of NH_4^+ ion.

(2) Aqueous Fe^{2+} ion solution gives a blue colour with $K_3[Fe(CN)_6]$ due to formation of complex $[Fe(Fe(CN)_6)]^-$ ion.

$$Fe^{2+} + [Fe(CN)_6]^{3-} \rightarrow \underset{\text{Blue colour}}{[Fe(Fe(CN)_6)]^-}$$

(3) With $BaCl_2$ solution, aqueous solution of sulphate ion in hypochloric acid gives white precipitate of $BaSO_4$.

$$SO_4^{2-} + BaCl_2 \rightarrow \underset{\text{White ppt.}}{BaSO_4\downarrow} + 2Cl^-$$

4 An orange solid (A) on heating gave a green residue (B), a colourless gas (C) and water vapour. The dry gas (C) on passing over heated Mg gave a white solid (D). (D) on reaction with water gave a gas (E) which formed dense white fumes with HCl. Identify (A) to (E) and give reaction involved.

[*IIT 1993*]

Solution

$$\underset{\text{Orange Solid}}{(A)} \xrightarrow{\Delta} \underset{\text{Green residue}}{(B)} + \underset{\text{Colourless gas}}{(C)} + H_2O$$

$$\underset{\text{(Dry gas)}}{(C)} + Mg \rightarrow \underset{\text{White solid}}{(D)}$$

$$(D) + H_2O \rightarrow \underset{\text{(Gas)}}{(E)} \xrightarrow{HCl} \text{White fumes}$$

$$\underset{\text{Orange solid}}{(NH_4)_2Cr_2O_7} \rightarrow \underset{\text{Green residue}}{Cr_2O_3} + \underset{\text{(C)}}{N_2} + 4H_2O$$

$$\underset{\text{(C)}}{N_2} + 3Mg \rightarrow \underset{\text{(D)}}{Mg_3N_2}$$

$$\underset{\text{(D)}}{Mg_3N_2} + 6H_2O \rightarrow 3Mg(OH)_2 + \underset{\text{(E)}}{2NH_3}$$

$$\underset{\text{(E)}}{NH_3} + HCl \rightarrow NH_4Cl \text{ (White fumes)}$$

Therefore, (E) is NH_3 gas, so (D) is Mg_3N_2 and (C) is N_2. This N_2 is obtained with $(NH_4)_2Cr_2O_7$ on strongly heating as $(NH_4)_2Cr_2O_7$ have orange colour due to $Cr_2O_7^{2-}$ and produces Cr_2O_3 which have green colour due to Cr^{3+} ion.

5. During the qualitative analysis of a mixture containing Cu^{2+} and Zn^{2+} ions, H_2S gas is passed through an acidified solution having these ions in order to test Cu^{2+} alone. Explain briefly.

[*IIT 1998*]

Solution

Ksp (solubility product) of CuS is less than Ksp of ZnS. On passing H_2S in acidic medium, the dissociation of H_2S is suppressed due to common ion effect(H^+) and it provides lower $[S^{2-}]$ but it is just sufficient to exceed ionic product to Ksp of CuS and not Ksp of ZnS. Hence, only CuS gets precipitated.

6. An aqueous solution having one mole of HgI_2 and two moles of NaI is orange in colour. On addition of excess NaI the solution becomes colourless. The orange colour reappears on subsequent addition of NaOCl. Explain with equations.

[*IIT 1999*]

A solution having one mole of HgI_2 and two moles of NaI is orange in colour due to the partial solubility of HgI_2. On addition of excess of NaI, the colourless complex Na_2HgI_4 is formed as follows

$$2NaI + HgI_2 \rightarrow Na_2HgI_4$$

(Excess)

Na_2HgI_4 on addition of NaOCl, oxidizes as:

$$3Na_2HgI_4 + 2NaOCl + 2H_2O \rightarrow$$

$$3HgI_2 + 2NaCl + 4NaOH + 2NaI_3$$

Hence, colour of partially soluble HgI_2 can be restored.

7. An aqueous blue coloured solution of a transition metal sulphate reacts with H_2S in acidic medium to give a black precipitate A, which is insoluble in warm aqueous solution of KOH. The blue solution on treatment with KI in weakly acidic medium, turns yellow and produces a white precipitate B. Identify the transition metal ion. Write the chemical reactions involved in the formation of A and B.

[*IIT 2000*]

Solution

The transition metal is Cu^{2+} and the compound is $CuSO_4.5H_2O$ which dissolves in water to give blue coloured solution due to presence of Cu^{2+} (d^9-configuration). On passing H_2S in acidic medium in the solution, the black precipitate of CuS is obtained which is not soluble in aqueous KOH (warm) solution.

$$CuSO_4 + H_2S \xrightarrow{\text{Acidic medium}} \underset{\text{Black ppt.}}{CuS\downarrow} + H_2SO_4$$

(Insoluble in aq. KOH)

On addition of KI solution in aqueous solution of $CuSO_4$, it generates yellow coloured solution of CuI_2, which is decomposed in white ppt. of Cu_2I_2 and I_2.

$$CuSO_4 + 2KI \downarrow CuI_2 + K_2SO_4$$

$$2CuI_2 \rightarrow Cu_2I_2 + I_2$$

White ppt.

8. Write the chemical reactions associated with the 'borax bead test' of cobalt(II) oxide.

[*IIT 2000*]

Solution

$$Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$$

$$\text{Salt of Co} \xrightarrow{\Delta} CoO + \text{Gas}$$

$$CoO + B_2O_3 \xrightarrow{\Delta} \underset{\substack{\text{Cobalt metaborate}\\ \text{(Blue colour)}}}{Co(BO_2)_2}$$

9. A white substance (A) reacts with dilute H_2SO_4 to produce a colourless gas (B) and a colourless solution (C). The reaction between (B) and acidified $K_2Cr_2O_7$ solution produces a green solution and a slightly coloured precipitate (D). The substance (D) burns in air to produce a gas (E) which reacts with (B) to yield (D) and a colourless liquid. Anhydrous copper sulphate is turned blue on addition of this colourless liquid. Addition of aqueous NH_3 or NaOH to (C) produces first a precipitate, which dissolves in the excess of the respective reagent to produce a clear solution in each case. Identify (A), (B), (C), (D) and (E). Write the equations of the reactions involved.

[*IIT 2001*]

Solution

As the white substance (A) gives a colourless gas (B) with dil. H_2SO_4. Such a gas may be H_2S gas so substance (A) may be a metal sulphide (Na K Zn etc.)

$$\underset{\text{(A)}}{MS} + \text{dil. } H_2SO_4 \rightarrow \underset{\text{Colourless solution (C)}}{MSO_4} + \underset{\text{(B)}}{H_2S}$$

or

$$\underset{\text{(A)}}{M_2S} + \text{dil. } H_2SO_4 \rightarrow \underset{\text{Colourless solution (C)}}{M_2SO_4} + \underset{\text{(B)}}{H_2S}$$

When H_2S gas reacts with acidified $K_2Cr_2O_7$ solution, it gives green coloured solution of $Cr_2(SO_4)_3$ along with slightly yellow coloured precipitate of (D) as sulphur.

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3[O]$$

$$[H_2S + [O] \rightarrow H_2O + S] \times 3$$

$$K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \rightarrow \underset{\text{Green solution}}{K_2SO_4 + Cr_2(SO_4)_3} + 7H_2S + \underset{\text{(D)}}{S}\downarrow$$

Sulphur on burning in oxygen gives SO_2 (E).

$$\underset{\text{(D)}}{S} + O_2 \rightarrow \underset{\text{(E)}}{SO_2}$$

Substance (E) reacts with B (H_2S) to give (D).

$$\underset{\text{(B)}}{2H_2S} + \underset{\text{(E)}}{SO_2} \rightarrow 2H_2O + \underset{\text{colourless (D)}}{3S}\downarrow$$

Anhydrous $CuSO_4$ gives blue colour with water.

$$\underset{\text{(Anhydrous)}}{CuSO_4} \xrightarrow{H_2O\ (l)} \underset{\text{Blue colour}}{Cu^{2+}} + SO_4^{2-}$$

Hence due to formation of Cu^{2+} ion, such a solution generates blue colour.

As Solution (C) produces precipitate first with NH_3/NaOH which dissolves in excess NH_3/NaOH. So substance (C) may be $ZnSO_4$ or $Al_2(SO_4)_3$ (nor Na_2SO_4 or K_2SO_4 or $MgSO_4$ etc.)

$$\underset{\text{(A)}}{ZnS} + \text{dil. } H_2SO_4 \rightarrow \underset{\text{(C)}}{ZnSO_4} + \underset{\text{(B)}}{H_2S}$$

$$\underset{\text{(C)}}{ZnSO_4} + 2NaOH \rightarrow \underset{\text{White ppt.}}{Zn(OH)_2}\downarrow + Na_2SO_4$$

$$Zn(OH)_2 + \underset{\text{Excess}}{2NaOH} \rightarrow \underset{\text{soluble}}{Na_2ZnO_2} + 2H_2O$$

Therefore,

(A) – ZnS, (B) – H_2S

(C) – $ZnSO_4$, (D) – S

(E) = SO_2

10. When a white crystalline compound X is heated with $K_2Cr_2O_7$ and concentrated H_2SO_4, a reddish brown gas A is evolved. On passing A into caustic soda solution, a yellow coloured solution of B is obtained. Neutralizing the solution B with acetic acid and on subsequent addition of lead acetate, a yellow precipitate C is obtained. When X is heated with NaOH solution, a colourless gas is evolved and on passing this gas into K_2HgI_4 solution, a reddish brown precipitate D is formed. Identify A, B, C, D and X. Write the equations of reactions involved.

[*IIT 2002*]

Solution

Compound (X) – NH_4Cl, (A) – CrO_2Cl_2

(B) – Na_2CrO_4, (C) – $PbCrO_4$,

(D) = $NH_2(HgO)HgI$

The reactions involved can be as follows:

(i) $\underset{\text{(X)}}{4NH_4Cl} + K_2Cr_2O_7 + 6H_2SO_4 \rightarrow \underset{\text{(A)}}{2CrO_2Cl_2} + 4NH_4HSO_4 + 2KHSO_4 + 3H_2O$

(ii) $\underset{\text{(A)}}{CrO_2Cl_2} + 2NaOH \rightarrow \underset{\text{(B)}}{Na_2CrO_4} + 2HCl$

(iii) $\underset{\text{(B)}}{Na_2CrO_4} + (CH_3COO)_2Pb \rightarrow$

$$PbCrO_4\downarrow + 2CH_3COONa$$
(C)

(iv) $NH_4Cl + NaOH \rightarrow NaCl + H_2O + NH_3$
(X)

(v) $2K_2HgI_4 + NH_3 + 3KOH \rightarrow NH_2(HgO)HgI + 7KI + 2H_2O$
(D)

11. A mixture consisrs of A (yellow solid) and B (colourless solid) which gives lilac colour in flame.

(a) Mixture gives black precipitate C on passing H_2S gas through its aqueous solution.

(b) C is soluble in aqua regia and on evaporation of aqua-regia and adding $SnCl_2$ gives grayish black precipitate D. The salt solution with NH_4OH gives a brown precipitate.

(i) The sodium extract of the salt with CCl_4/ $FeCl_3$ gives a violet layer.

(ii) The sodium extract gives yellow precipitate with $AgNO_3$ solution which is insoluble in NH_3. Identify A and B, and the precipitates C and D. [*IIT 2003*]

Solution

A + B → Lilac colour (light purple colour) (mixture)

(a) $A + B \xrightarrow{H_2S\,(g)}$ Black ppt,
Mixture (C)

(b) (C) $\xrightarrow[\text{residue}]{\text{Soluble in}}$ Soluble $\xrightarrow{\text{Evaporation}}$

Residue $\xrightarrow{SnCl_2}$ Greyish black ppt. (D)

Thus, (C) is HgS.

(c) $A + B \xrightarrow{NH_4OH}$ Brown ppt.
solution of mixture

(i) Sodium extract of salt $\xrightarrow{CCl_4 / FeCl_3}$
Violet layer

(ii) Sodium extract $\xrightarrow{AgNO_3}$ yellow ppt.

$\xrightarrow{NH_3}$ Insoluble

Hence, (A) and (B) are KI and HgI_2 respectively.
KI + HgI_2 → Lilac colour of flame

(a) $HgI_2 + H_2S \xrightarrow{\text{Aqua regia}} HgS\downarrow + 2HI$
(B) Black ppt.

(b) $HgI_2 \xrightarrow[(B)]{\text{Aqua regia}} \underset{\text{Soluble}}{HgCl_2} \xrightarrow{\text{Evaporation}}$

$HgCl_2 \xrightarrow{SnCl_2} SnCl_4 + Hg$ Greyish black

$2KI + HgI_2 \rightarrow K_2(HgI_4)$
(A) (B) Orange

$2K_2HgI_4 + NH_3 + 3KOH \rightarrow [HgO\,Hg(NH_2)I]$

$Na_2CO_3 + HgI_2 \rightarrow Hg\downarrow + NaI + CO_2 + O_2\uparrow$

$2NaI + 2Fe^{3+} \xrightarrow{CCl_4} I_2 + 2Na^{-}\,2Fe^{2+}$

$AgNO_3 + NaI \rightarrow AgI\downarrow + NaNO_3$
Yellow ppt.
(Insoluble in NH_3)

Hence, (A) = KI, (B) = HgI_2, (C) = HgS,
(D) Hg

12. Nickel chloride is mixed with dimethyl glyoxime. When ammonium hydroxide is slowly added, a shining red precipitate is formed.

(a) Give the structure of complex showing hydrogen bonds.

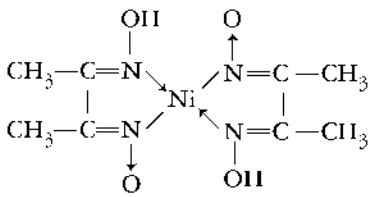
(b) Give the charge and the state of hybridisation of the central metal ion.

(c) Predict the magneric behaviour of complex. [*IIT 2004*]

Solution

(a) When nickel chloride is mixed with dimethyl glyoxime and ammonium hydroxide is slowly added, a shining red precipitate is formed as follows:

$$NiCl_2 + 2\begin{array}{l}CH_3-C=NOH\\ \quad\quad|\\ CH_3-C=NOH\end{array} \xrightarrow{2NH_4OH}$$



Red ppt. $+ 2NH_4Cl$

(Nickel dimethyl glyoxime)

(b) In this complex the charge on Nickel is +2 and the hybridization is dsp^2 (square planer shape)

(c) This complex is diamagnetic as it has no unpaired electron.

13. (i) The yellow coloured precipitate of compound (A) is formed on passing H_2S through a neutral solution of a salt (B).

(ii) (A) is soluble in hot dilute HNO_3, but insoluble in yellow ammonium sulphide.

(iii) The solution of (B) on treatment with small quantity of NH_3 gives white precipitate which becomes soluble in excess of it forming a compound (C).

(iv) The solution of (B) gives white precipitate with small concentration of KCN which becomes soluble in excess of this reagent forming a compound (D).

(v) The solution of (D) on reaction with H_2S gives (A).

(vi) The solution of (B) in dilute HCl on reaction with a solution of $BaCl_2$ gives white precipitate of compound (E) which is insoluble in conc. HNO_3.

Identify (A) to (E) and give chemical equations for the reactions at steps (i) and (iii) to (vi).

Solution

(B) is $CdSO_4$ which gives a yellow compound by passing H_2S through its solution. It gives a white precipitate with $BaCl_2$ solution. This precipitate is insoluble in conc. HNO_3. It indicates that SO_4^{2-} ion is present in the salt.

(i) $CdSO_4 + H_2S \rightarrow CdS + H_2SO_4$

(B) (A)

(ii) $CdSO_4 + BaCl_2 \rightarrow BaSO_4 + CdCl_2$

(B) (E)

Insoluble in water

$CdS + 2HNO_3 \rightarrow Cd(NO_3)_2 + H_2S$

Soluble

$CdS + (NH_4)_2Sx \rightarrow$ Insoluble

Yellow ammonium sulphide

(iii) $CdSO_4 + 2NH_4OH \rightarrow Cd(OH)_2 + (NH_4)_2SO_4$

$Cd(OH)_2 + 4NH_4OH \rightarrow Cd(NH_3)_4(OH)_2 + 4H_2O$

(C) soluble

(iv) $CdSO_4 + 2KCN \rightarrow Cd(CN)_2 + K_2SO_4$

White ppt.

$Cd(CN)_2 + 2KCN \rightarrow K_2Cd(CN)_4$

(D) soluble

(v) $K_2Cd(CN)_4 + H_2S \rightarrow CdS + 2KCN + 2HCN$

(A)

14. A hydrated metallic salt (A), light green in colour on careful heating give a white anhydrous (B). (B) is soluble in water and its aqueous solution reacts with NO to give a dark brown compound (C). (B) on heating gives a brown residue (D) and a mixture of two gases (E) and (F). The gaseous mixture when passed through acidified permanganate, discharges the pink colour and when passed through $BaCl_2$ solution gives a white precipitate. Identify (A) to (F).

Solution

(i) $FeSO_4.7H_2O \rightarrow FeSO_4 + 7H_2O$

(A) (B)

$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$

(B) (D) (E) (F)

Brown

(ii) (B) is soluble in water and reacts with NO to give brown compound.

$FeSO_4(aq) + NO \rightarrow FeSO_4.NO$

Brown ring(C)

(iii) Gaseous mixture decolourises acidified $KMnO_4$

$5SO_2 + 2KMnO_4 + 2H_2O \rightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4$

(iv) $SO_3 + H_2O \rightarrow H_2SO_4$

$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$

white ppt.

15. An aqueous solution of a compound (A) when treated with $BaCl_2$ solution gives a white

precipitate insoluble in concentrated HCl. Another sample of (A) gives first white precipitate with NaOH which is soluble in excess of NaOH solution. The resulting solution of (A) gives white precipitate on passing H_2S gas. Identify the compound (A) and gives necessary reaction.

Solution

(A) Gives white precipitate with $BaCl_2$ which is insoluble in concentrated HCl hence the anion of (A) must be SO^2_4 ion. Because the cation gives white precipitate with NaOH which is soluble in excess of NaOH, therefore cation of compound (A) may be Zn^{2+} or Al^3 ion. But Zn^2 cannot be cation of the compound (A).

16. (A) gives a blue solution in H_2O. On passing H_2S, a black precipitate (B) is formed which is soluble in HNO_3. On addition of $NaOH_2$, the solution gives blue precipitate (C) which becomes black on boiling in NaOH. On passing ammonia into solution of (A) in water, a drop blue precipitate is formed, which dissolves in excess of NH_3 giving deep blue colouration (D). Treatment of KCN with aqueous solution of (A) gives a yellow ppt. (E) which dissolves in excess of KCN giving a colourless solution.

Solution

$$\underset{(A)}{CuSO_4} \xrightarrow{H_2S} \underset{(B)\ \text{Black}}{CuS} \xrightarrow{HNO_3} \underset{(C)\ \text{Blue ppt.}}{Cu(NO_3)_2} \xrightarrow{NaOH} Cu(OH)_2$$

$$CuSO_4 \xrightarrow{NH_3} [Cu(NH_3)_4]SO_4$$

(D) Deep Blue sol.

$$CuSO_4 \rightarrow \underset{\text{Yellow ppt.(E)}}{Cu(CN)_2} \xrightarrow{KCN} \underset{\text{Colourless}}{K_3[Cu(CN)_4]}$$

Questions For Self Assessment

17. A mixture of two salts was treated as follows:

    (i) The mixture was heated with manganese dioxide and concentrated H_2SO_4 when yellowish green gas was liberated.

    (ii) The mixture on heating with NaOH solution gave a gas which turned red litmus blue.

    (iii) Its solution in water gave blue precipitate with potassium ferricyanide and red colouration with NH_4CNS.

    (iv) The mixture was boiled with potassium hydroxide and the liberated gas was bubble through an alkaline solution of K_2HgI_4 to give brown precipitate.

    Identify the two salts. Give ionic reactions involved in the tests (i), (ii) and (iii)

    (i) Cl_2 gas

    (ii) Ammonia

    (iii) $Fe[Fe(CN)_6]_2$
    $Fe(CNS)_3$)

18. A hydrated metallic salt (A), light green in colour, on careful heating gives a white anhydrous residue (B). (B) is soluble in water and its aqueous solution reacts with NO to give a dark brown compound (C). (B) gives a brown residue (D) and a mixture of two gases (E) and (F). The gases when passed through an acidified $KMnO_4$ solution discharges the pink colour and when passed through acidified $BaCl_2$ solution gave a white precipitate. Identify (A), (B), (C), (D), (E) and (F).

    (A) $FeSO_4.7H_2O$ (B) $FeSO_4$

    (C) $FeSO_4NO$ (D) Fe_2O_3

    (E) SO_2 (F) SO_3

19. The acidic, aqueous solution of ferrous ion forms a brown complex in the presence of NO_3^-, by the following two steps. Complete and balance the equations:

$$[Fe(H_2O)_6]^2 + NO_3 + H^+ \rightarrow$$
$$\ldots\ldots + [Fe(H_2O)_6]^{3+} + H_2O$$

$[Fe(H_2O)_6]^{2+}$ + …… → …… + H_2O

$3[Fe(H_2O)_6]^{2+} + NO_3^- + 4H^+ \rightarrow$

$NO + 3[Fe(H_2O)_6]^{3+} + 2H_2O$

$[Fe(H_2O)_6]^{2+} + NO \rightarrow [Fe(H_2O)_5NO]^{2+} + H_2O$

20. A hydrated metallic salt A, light green in colour, on careful heating gives a white anhydrous residue B. B is soluble in water and its aqueous solution reacts with NO to give a dark brown compound C. B on strong heating gives a brown residue D and a mixture of two gases E and F. The gaseous mixture when passed through acidified permanganate, discharges the pink colour and when passed through acidified $BaCl_2$ solution gave a white precipitate. Identify A, B, C, D, E and F.

(A) = $FeSO_4.7H_2O$,

(B) = $FeSO_4$,

(C) = $FeSO_4.NO$

(D) = Fe_2O_3

(E) = SO_2 and (F) = SO_3

21. When 20.02 g of a white solid X is heated 4.4 g of an acid gas A and 1.8 g of a neutral gas B are evolved, leaving behind a solid residue Y of weight 13.8 g. A turns lime water milky and B condenses into a liquid which changes anhydrous copper sulphate blue. The aqueous solution of Y is alkaline to litmus and gives 19.7 g of white precipitate Z with barium chloride solution. Z gives carbon dioxide with an acid. Identify A, B, X, Y and Z.

(A) = CO_2,

(B) = H_2O (X) = $KHCO_3$

(Y) = K_2CO_3, (Z) $BaCO_3$

22. A light bluish green compound (P) responds to the following tests.

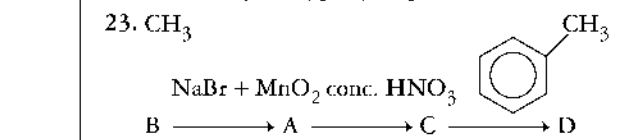
(a) Its aqueous solution gives a brown ppt. with alkaline $K_2[HgI]_4$ solution.

(b) Its aqueous solution gives a blue colour with $K_3[Fe(CN)_6]$ solution

(c) Its solution in HCl gives a white ppt. with.

Suggest the formula of (P).

P = $FeSO_4.(NH_4)_2SO_4.6H_2O$

23. 

Here B has brown fumes with pungent smell C is intermediate and D is an Explosive product

Identify A, B, C and D also write the balanced equation.

[*IIT 2005*]

A = Con.H_2SO_4, B = Br_2

C = NO_2^-, D = Trinitrotoluene (T.N.T.)

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Appendix 11

FACTS TO REMEMBER

Important Facts To Memorize

Most abundant element in human body	Oxygen
Most abundant gas in sun	H_2
Most abundant gases in universe	H_2, He
Element having maximum tendency for catenation	Li
Most abundant metal of earth	Al
Rarest element of earth	Astatine (At)
Most abundant element of earth	Oxygen (O)
Element containing no neutron	$_1H^1$
Amphoteric metals	Zn, Al, Sn, Pb
Noble metals	Au, Pt
Elements having highest tensile strength	Boron
Metals showing highest oxidation number	Ru, Os
Non-metal having highest m.p., b.p.	Diamond
Lightest element	H
Heaviest naturally occurring element	^{238}U
Best electricity conductor among metals	Ag
Best conductor among non metals	Graphite
Most reactive solid element	Li
Most reactive liquid element	Cs
Most reactive gaseous element	F
Amphoteric non metal	Si
Elements showing diagonal relationship	Li-Mg, Be-Al, B-Si
Highest electronegativity	F
Highest ionization potential	He
Lowest ionization potential	Cs
Lowest electron affinity	Noble gases (zero)
Highest electron affinity	Cl
Non-metals having metallic lusture	iodine, graphite
Element sublime on heating	I
Coolant in nuclear reactors	D_2O
Most poisonous element	Pu
Liquid non-metal	Br_2
Total number of radioactive elements in periodic table	25
Volatile d-block elements	Zn, Cd, Hg
Liquid metals s	Hg, Ga, Cs, Fr, Eka
Element kept in water	P
Elements kept in kerosene oil	Na, K, I_2, Cs
Metal with highest m.p.	W
Metal with lowest m.p.	Hg
Non metal with highest refractive index	Diamond
Lowest refractive index	Air
Lowest b.p.	H_2
Heaviest solid metal	Os
Lightest solid metal	Li
Lightest solid non metal	B

Hardest naturally occurring non metal	Diamond
Hardest artificial substance good conductor of heat	B_4C (norbide)
Strongest basic hydroxide	CsOH
Strongest basic oxide	CsO_2 (caesium peroxide)
Most stable metal carbonate	Cs_2CO_3
Element with highest radioactivity	Ra
Strongest reducing agent	Azide (N_3^-)
Strongest oxidising agent	OF_2
Smallest anion	H (hydrides)
Smallest atomic size	H
Largest atomic size	Cs
Element with maximum number of isotopes	Ag (46)
Element with minimum number of isotopes	H (3)
Element with maximum number of allotropes	Sn
Liquid element of radioactive nature	Francium (Fr)
Poorest conductor of current	Pb (metal), S (non-metal)
Metalloids elements	B, Si, Ge, As, Sb, Te
Dry ice	CO_2
Most recently elements name by IUPAC	Ds (atomic number 110)
'All purpose' grease	Lithium stearate
Old name of astatine	Albamine
Most abundant gas in atmosphere	N_2
Rarest gas in atmosphere	Rn
Lightest gas in atmosphere	H_2
Groups containing higher no. of gaseous elements	Group 18 (Noble gases)
Most electrovalent compound	CsF
Bad conductor of electricity	Mica
Lightest radio isotope	Tritium ($_1H^3$)
Compound with maximum covalent nature	H_2, N_2, O_2, Cl_2 (diatomic gases)
First noble prize in chemistry	Van't Hoff
Softest form of carbon	Lamp black
Latest allotrope of carbon	fullerene or bucky ball
Strongest acid	HSO_3F + 90 % SbF_5 called magic acid
Most reactive form of P	White
Least reactive form of P	Red
Purest form of silica	Quartz
Most ductile metal	Gold

IMPORTANT ORES

Fe

- Magnetite → Fe_3O_4
- Limonite → $3\,Fe_2O_3.3H_2O$
- Iron Pyrite → FeS_2
- Haematite → Fe_2O_3
- Copper Pyrite → $CuFeS_2$
- Spathic Iron → $FeCO_3$

Cu

- Cuprite or Ruby copper → Cu_2O
- Copper Glance → Cu_2S
- Malachite → $Cu(OH)_2.CuCO_3$
- Azurite → $Cu(OH)_2.2CuCO_3$

Pb

- Galena → PbS
- Anglesite → $PbSO_4$
- Stolzite → $PbWO_4$
- Cerrusite → $PbCO_3$
- Wulfenite → $PbMnO_4$

Ag

- Argentite or Silver Glance → Ag_2S
- Pyragurite → $3\,Ag_2S.\,Sb_2S_3$
- Proustite → $3Ag_2S.\,As_2O_3$
- Horn Silver → AgCl

Mg

- Magnesite → $MgCO_3$
- Carnalite → $MgCl_2.KCl.6H_2O$
- Kiesserite → $MgSO_4.H_2O$
- Schonite → $MgSO_4.K_2SO_4.6H_2O$
- Dolomite → $MgCO_3.CaCO_3$
- Epsomite → $MgCO_3.7H_2O$
- Kainite → $MgSO_4.KCl.3H_2O$
- Olivine → Mg_2SiO_4
- Spinel → $Mg_2Al_2O_4$

Al

- Corundum → Al_2O_3
- Diaspore → $Al_2O_3.H_2O$
- Bauxite → $Al_2O_3.2H_2O$
- Cryolite → Na_3AlF_6
- Felspar → $KAlSi_3O_8$
- Talc → $Mg_3H_2(SiO_3)_4$
- Asbestos → $CaMg_3(SiO_3)_4$
- Hg → HgS (Cinabar)

Mica ⟶ $K Al Si_3 O_{10} (OH)_2$
Kaolinite ⟶ $Al(OH)_4 Si_2O_5$
Turquoise ⟶ $AlPO_4.Al(OH)_3.H_2O$
Alum Stone or Alunite ⟶ $K_2SO_4.Al_2(SO_4)_3.4Al(OH)_3$
Cassetirite or Tin Stone ⟶ SnO_2
Rutile ⟶ TiO_2
Ilmenite ⟶ $FeOTiO_2$
Beryl ⟶ $3BeO.Al_2O_3.6SiO_2$
Florapatite ⟶ $3Ca_3 (PO_4)_2 .CaF_2$

Be

Beryl ⟶ $3BeO Al_2O_3.6SiO_2$ or $Be_3Al_2Si_6O_{18}$

Phenacite ⟶ $2BeO SiO_2$ or Be_2SiO_4
Cryso beryl ⟶ $BeO Al_2O_3$

Ca

Anhydrite ⟶ $CaSO_4$
Dolomite ⟶ $CaCO_3.MgCO_3$
Fluorapatite ⟶ $3Ca_3 (PO_4)_2. CaF_2$
Gypsum ⟶ $CaSO_4.2H_2O$

Sr

Strontionite ⟶ $SrCO_3$
Celestite ⟶ $SrSO_4$

Ores of Ba

Barytes ⟶ $BaSO_4$
Whitherite ⟶ $BaCO_3$

Ores of Ra

Carnotite ⟶ $K_2O, U_2O_3, (VO_4)_2, 3H_2O$
Pitch blende oxide ⟶ Uranium

Ores of Zn

Zinc blends or Sphalerite ⟶ ZnS
Zincite or Red zinc ⟶ ZnO
Calamine or Zinc spar ⟶ $ZnCO_3$
Franklinite ⟶ $ZnO.Fe_2O_3$
Willemite ⟶ Zn_2SiO_4

Some Important Alloys

1. *Alclad* It is an alloy of aluminium and used in making sea planes.
2. *Alnico* It is an alloy of steel (77%), nickel (2%), aluminium (20%) and cobalt (1%). It is used in making permanent magnets.
3. *Aluminium bronze* It is an alloy of 90% copper and 10% aluminium. It is used in making coins, trays and picture frames.
4. *Aluminium bronze* contains 90% Cu and 10% Al phosphor bronze consisting of 94% Cu, 5% Sn and 1% P is hard, elastic and used for pump rods, valves, axle bearing and certain other equipments. These are malleable, corrosion resistant and suitable for cold working.
5. *Babbit metal* It is an alloy of Sn (88-90%), Sb (7-4%) and Cu (3-7%). Hard babbit composition is (a) Sn = 91%, Pb = 4.5% and Sb = 4.5% and (b) Sn = 83%, Cu = 8.5% and Sb = 8.5%. It is used as bearing metal.
6. *Bell metal* It is an alloy of Cu and Sn having 80% Cu and 20% Sn. It is hard, brittle and sonorous. It is used for fabricating machine parts and bells, gongs etc.
7. *Brass* It is an alloy of 70% copper and 30% zinc. It is used in making utensils etc.
8. *Britania metal or pewter* It is an alloy of Sn (85-95%), Sb (6-10%), and Cu (1-3%). It is used for making cups, mugs and other utensils.
9. *Bronzes* These are mostly the alloys of copper and tin and contain 75 90% Cu and 25 10% Sn. These bronzes are mainly used for making coins, statues an special type of utensils.
10. *Constantan* It is an alloy of nickel (40%) and copper (60%). It is used in electrical work such as for making resistance boxes and thermo couples etc.
11. *Delta metal* It is an alloy of Cu (55%), Zinc (41%) and Fe(4%). It is used in making ships, bearing, and propellers.
12. *Duralumin* It is an alloy of Al(95.5%), Copper(4%), Mg(0.5%) and Mn(0.5%). It is used in making aeroplanes parts.
13. *Dutch metal* It is an alloy of copper and Zinc and is used in gold coverings.
14. *Electron* It is an alloy of Mg and Zn with small amounts of Al, Cu and Mn. It is a hard metal alloy used for making propellers of engines and air-crafts.
15. *Ferro alloys – Ferro molubdenum* is an alloy of Mo. Usually the percentage of Mo is less than 1%, but about 1.5-2% Mo has been used for high speed steel and 5% Mo in resisting steels. 6-10% Mo is used for preparing special steels. **Ferro silicon** has a composition of Si = 90-95%, C = 0.15%, S = 0.01% and P = 0.05%.
16. *Ferro manganese or Spiegeleisen* has average composition Mn = 78-82%, C = 7.5%, P = 0.35%, S = 0.5% and Si = 1.25%.
17. *Ferro nikel* contains Ni = 2.5-5%. It is hard, tough and rustless. It is used in the manufacture of cables. Propeller shaft, armor plates etc.
18. *Ferro titanium* has a composition of Ti = 38-45%, C = 0.1-6%, Si = 15-25% and Al = 9-10%, **Ferro tungsten** is hard and strong and contains W = 14-20%. It is used in the manufacture of high speed tools.
19. *Ferro vanadium* has composition V = 30-40%, C = 3.5%, P = 0.25%, S = 0.4%, Si = 13% and Al = 1.5%. It has high tensils strength and is used for making springs, axles, shafts etc. Both tungsten and vanadium make steel hard. Such hard steel is used for making high speed tools.

20. *German silver or Nickel silver* These are Cu –Zn – Ni alloys containing about 50% Cu, 25% Zn and 25% Ni. Nickel is used for fancy articles, forks, spoons, cigarette cases etc.
21. *Gun metal* It is an alloy of Cu, Sn and Zn. It contains 88%Cu, 10%Sn and 2%Zn. It is used in making guns. Gears and bearings.
22. *Invar* It has 64% Fe, 35% Ni and some traces of Mn, C. It is used in making pendulum rods.
23. *Magnalium* It is an alloy of 90% Al and 10% Mg. It is used in making balance beams.
24. *Monel metal* It is an alloy of 30% Cu, 67%Ni and 3%Mn or Fe. It is used for the construction of household sinks and containers and alkali-resisting equipments.
25. *Nichrome* It is an alloy of Ni(60%), Cr(15%) and Fe(25%) and is used in making electrical resistance.
26. *Nickel coinage alloy* for coinage purpose, an alloy of Ni(25%) and Cu (75%) is used. Another coinage alloy is silver coinage which contains 5% nickel.
27. *Pewter* It is an alloy of 24% Pb and 76% Sn and used for making utensils.
28. *Rosemetal* It is an alloy of Bi(50%), Pb(25%) and Sn(25%). It is used in making stereo metal in printing and safety Plugs in boilers.
29. *Silicon Bronze* contains upto 4% Si and upto 1% Fe, Mn, Zn and Al, but does not contain Sn. They have strength like mild steel, excellent corrosion resistance and also have welding prop.
30. *Solder* It is an alloy of 67% Sn and 33% Pb. It is used in soldering. Soft solder contains 3-80% Pb and 97-20% Sn. This tin-lead alloy is used for joining metal parts because of its low melting point.
31. *Stainless steel* It contains Cr (about 11%) and Ni (about 7%). It is used in making utensils and surgical instruments.
32. *Sterite* It is an alloy of chromium, tungsten and nickel and is used for the manufacture of high speed tools and cutlery. This alloy is also used for making surgical instruments.
33. *Tiscor* It contains maximum 0.1% carbon, Mn = 0.1 - 0.4% (maximum), Cr = 0.7 1.1%, Cu = 0.3 - 0.5%, Si = 0.5 1.0% , P = 0.1 0.2 % and S = 0.05% .These two alloys (Tiscorn and Tiscor) are high strength engineering steels prepared by **Tata Iron and Steel company** in India.
34. *Tiscorn* It contains maximum 0.3% carbon, Mn = 0.5-1.3%, Cr = 1.00%, Ca = 0.25-0.6%, Si = 0.3% (maximum), P = 0.05% (maximum) and S = 0.05% (maximum).
35. *Type metal* It is an alloy of Pb(82%), Sb(15%) and Sn(3%) and is used for making type for printing.
36. *Wood metal* It is alloy of Pb(25%), Sn (12.5%), Cd (12.5%) and Bi (50%). It is used as automatic sprinkles. It melts in hot water as its melting point is 68°C.

Important Compounds

1. Agate is silicon dioxide, SiO_2.
2. Ammonal is a mixture of ammonium nitrate and Al powder (NH_4NO_3 + Al). It is used as an explosive.
3. Alum is $(NH_4)_2SO_4.Al_2(SO_4)_3.24H_2O$. It is used as mordant by dyers of clothes. Potash alum is $K_2SO_4.Al_2(SO_4)_3.24H_2O$.
4. Aqua fortis is nitric acid, HNO_3.
5. Antichlor is sodium thiosulphate, $Na_2S_2O_3.5H_2O$. It is also called Hypo.
6. Aqua-Regia is a mixture of conc. HNO_3 and conc. HCl in the ratio of 1 : 3. It is also known as kingly water.
7. Baking soda is sodium bicarbonate, $NaHCO_3$.
8. Baryres is barium sulphate, $BaSO_4$.
9. Brine is sodium chloride (NaCl) solution.
10. Blue vitriol is copper sulphate, $CuSO_4.5H_2O$.
11. Bone ash is mainly calcium phosphate, $Ca_3(PO_4)_2$.
12. Borax is the name of sodium tetraborate hydrate $Na_2B_4O_7.10H_2O$. Borax ($Na_2B_4O_7$) is also called tincal.
13. $B(OH)_3$ is an acid.
14. Brown Ring is of $FeSO_4.NO$
15. Butter of tin is $SnCl_4.5H_2O$.
16. B.O.D. is biological oxygen demand.
17. Cuprite is Cu_2O.
18. Calomel is Hg_2Cl_2.
19. Caustic potash is KOH.
20. Caustic soda is sodium hydroxide, NaOH.
21. Chile saltpeter is sodium nitrate, $NaNO_3$.
22. Cinnabar is HgS.
23. Carbonic acid is hydrogen carbonate, H_2CO_3.
24. Carbolic acid is phenol, C_6H_5OH.
25. Carborundum is silicon carbide, SiC.
26. Copper glance is Cu_2S.
27. Carbogen is a mixture of 1% CO_2 and O_2. It is used as antidote for for CO poisoning.
28. Corrosive sublimate is mercuric chloride, $HgCl_2$.
29. Corundum is aluminium oxide, Al_2O_3.
30. Chromyl chloride is CrO_2Cl_2.
31. Cream of tartar is $KHC_4H_4O_6$.
32. Cyanogen is C_2N_2.
33. Dead burnt plaster is anhydrous $CaSO_4$.
34. Dry ice is solid carbon dioxide (CO_2).

35. Epsom salt is the name of magnesium sulphate, $MgSO_4.7H_2O$.
36. Energy is a mixture of Al_2O_3 and Fe_2O_3.
37. Eka aluminium is gallium.
38. Fluorspar is CaF_2.
39. Freon is CCl_2F_2.
40. Formalin is 40% formaldehyde (HCHO).
41. Fremy's salt is potassium hydrogen fluoride KHF_2.
42. Foul air is nitrogen, N_2. It is also called azote.
43. Fischer's salt is potassium cobalt nitric $K_3[Co(NO_2)_6]$.
44. Fowler's solution is $NaAsO_2$ solution.
45. Fusion mixture is $Na_2CO_3 + K_2CO_3$.
46. Fluorine is called super halogen.
47. Fulminating gold is $Au(NH_2) = NH$.
48. Grain alcohol is ethyl alcohol, C_2H_5OH.
49. Grape sugar is dextrose, $C_6H_{12}O_6$.
50. Glauber's salt is the name of sodium sulphate, $Na_2SO_4.10H_2O$.
51. Gypsum is calcium sulphate, $CaSO_4.2H_2O$
52. Green vitriol is ferrous sulphate, $FeSO_4.7H_2O$
53. Gammexane is benzene hexachloride (BHC), $C_6H_6Cl_6$.
54. Gun powder is a mixture of sulphur, charcoal and nitre.
55. Graham's salt is $(NaPO_3)_3$.
56. Hydrolith is calcium hydride, CaH_2.
57. Halite is common rock salt (NaCl).
58. Horn silver is AgCl.
59. Hair salt is $Al_2(SO_4)_3.18H_2O$.
60. Hypo is $Na_2S_2O_3.5H_2O$.
61. King of chemicals is H_2SO_4.
62. Kali : Germans used the word 'Kali' for Potash.
63. Limestone is calcium carbonate, $CaCO_3$.
64. Lunar caustic is silver nitrate, $AgNO_3$.
65. Laughing gas is nitrous oxide, N_2O. It is also known as laughing grites.
66. Lithia water is aqueous solution of lithium bicarbonate ($LiHCO_3$).
67. Lapis Lazuli is blue coloured mineral used as semi-precious stone. It is sodium alumino silicate.
68. Milk of magnesia is magnesium hydroxide, $Mg(OH)_2$.
69. Marshall's acid is persulphuric acid, $H_2S_2O_8$.
70. Milk of lime is calcium hydroxides, $Ca(OH)_2$. It is also called slaked lime.
71. Magnesite is $MgCO_3$.
72. Microscopic salt is $Na(NH_4)HPO_4.4H_2O$.
73. Mica is $KH_2Al_2(SiO_4)_3$.
74. Magnesia is MgO.
75. Mosaic gold is SnS_2.
76. Marsh gas or fire damp is CH_4.
77. Mohr's salt is $FeSO_4.(NH_4)_2SO_4.6H_2O$.
78. Muriatic acid is hydrochloric acid, HCl
79. Nessler's reagent is K_2HgI_4. It contains $HgCl_2$, KI and NaOH. The ions present in it is HgI_4^{2-}.
80. Nitre Cake is $NaHSO_4$.
81. Nitrolim is $CaCN_2 + C$ (graphite)
82. Norweigian saltpetre is basic calcium nitrate, $Ca(NO_3)_2$.
83. Oxone is sodium peroxide, Na_2O_2.
84. Oil of vitriol is sulphuric acid, H_2SO_4.
85. Oleum is fuming sulphuric acid, concentrated $H_2SO_4 + SO_3$.
86. Oil of mirbane is $C_6H_5NO_2$.
87. Oil of winter green is methyl salicylate.
88. Plaster of paris is calcium sulphate hemihydrate, $CaSO_4.\ ^1/_2\ H_2O$.
89. Philosopher's wool is zinc oxide, ZnO.
90. Phosgene is carbonyl chloride, $COCl_2$.
91. Picric acid is 2,4,6-trinitrophenol.
92. Paris green is double salt of copper acetate and copper arsenate.
93. Pearl white is BiOCl and is used as a pungent.
94. Prussian blue is $Fe_4[Fe(CN)_6]_3$.
95. Pearl ash is K_2CO_3.
96. Perhydrol is 30% H_2O_2.
97. Prussic acid HCN.
98. Quick lime is calcium oxide, CaO.
99. Quartz is silicon dioxide, SiO_2.
100. Quick silver is mercury, Hg.
101. Realgar is As_2S_2.
102. Rectified spirit is 95% ethyl alcohol, C_2H_5OH.
103. Reagar is As_4S_4.
104. Red lead is lead tetroxide, Pb_3O_4. It is also called Minium.
105. Red liquor is aluminium acetate, $(CH_3COO)_3$ Al.
106. Rochelle salt is sodium potassium tartrate, $NaKC_4H_2O_6$.
107. Rock salt is NaCl.
108. Ruby or sapphire or Emery is Al_2O_3.
109. Salt cake is sodium sulphate, Na_2SO_4.
110. Sand is silicon dioxide, SiO_2.
111. Scheeb's green is $CuHAsO_3$.
112. Selidlitz powder is $NaHCO_3$.
113. Smelting salt is $(NH_4)_2CO_3$.
114. Soda lime is a mixture of NaOH and CaO.
115. Soda ash is sodium carbonate, Na_2CO_3.
116. Sodamide is $NaNH_2$.
117. Spirit of wine is C_2H_5OH.

118. Spirit of salt is HCl.
119. Sterling silver is a solution of Cu and Ag.
120. Stranger gas is xenon (Xe).
121. Sugar of lead is lead acetate, $(CH_3COO)_2Pb$.
122. Super phosphate of lime contains $Ca(H_2PO_4)_2$. H_2O and $2CaSO_4.2H_2O$
123. Syvine is KCl.
124. Tartar emetic is potassium antimony tartrate, $K(SbO)C_4H_4O_6$.
125. TNT is trinitrotoluene, an explosive.
126. Tear gas is chloropicrin, CCl_3NO_2.
127. TEL is terra ethyl lead, $Pb(C_2H_5)_4$.
128. Thomas slag is calcium phosphate, $Ca_3(PO_4)_2$.
129. Tincture of iodine is I_2 and KI solution in alcohol.
130. Thermite is a mixture of iron oxide (Fe_3O_4) and Al powder.
131. Tinstone or Cassiterite is SnO_2.
132. Vinegar is dilute acetic acid, CH_3COOH.
133. Washing soda is Na_2CO_3.
134. Water glass is sodium silicate, Na_2SiO_3.
135. White lead is $Pb(OH)_2.2PbCO_3$.
136. Wackenroder's liquid is $H_2SO_4 + H_2S$.
137. White vitriol is Zinc sulphate, $ZnSO_4.7H_2O$
138. Wood spirit is CH_3OH.
139. Verdigris is the name of basic copper acetate, $(CH_3COO)_2$ Cu $Cu(OH)_2$.
140. Yellow ammonium sulphide is $(NH_4)_2Sx$.
141. Zincite is ZnO.

Water Solubility of Some Common Inorganic Compounds

1. Allv Na^+, K^- and NH_4^- compounds are soluble.
2. All nitrates, nitrites and acetates are soluble.
3. All chlorides, except AgCl, $PbCl_2$ and Hg_2Cl_2 are soluble. $PbCl_2$ is soluble in hot water.
4. All bromides, except AgBr, $PbBr_2$, Hg_2Br_2 and $HgBr_2$ are soluble.
5. All iodides, except AgI, PbI_2, Hg_2I_2 and HgI_2 are soluble.
6. All sulphates except $BaSO_4$, $CaSO_4$, $SrSO_4$, $PbSO_4$, Hg_2SO_4 and Ag_2SO_4 are soluble.
7. All carbonates except those of group 1 elements and $(NH_4)_2CO_3$ are insoluble.
8. All hydroxides except those of group 1 elements, $Ba(OH)_2$, $Ca(OH)_2$ and $Sr(OH)_2$ are insoluble.
9. All sulphides except those of group 1 & 2 elements and $(NH_4)_2S$ are insoluble.
10. All phosphates except those of group 1 elements and $(NH_4)_3PO_4$ are insoluble.
11. All sulphites, except those of group 1 elements and $(NH_4)_2SO_3$ are insoluble.

Action of Heat on Some Salts

$ZnCO_3$ (white) $\rightarrow ZnO + CO_2$ (yellow-hot; white-cold)

$CuCO_3$ (green) $\longrightarrow CuO + CO_2$ (black)

$CaCO_3 \longrightarrow CaO + CO_2$

$2Ag_2CO_3 \longrightarrow 4Ag + 2CO_2 + O_2$

$2NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$

$NH_4HCO_3 \longrightarrow NH_3 + CO_2 + H_2O$

$2FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$ $2CaSO_4.2H_2O$ 393K $2CaSO_4.\ ^1/_2\ H_2O + 3H_2O$ (Plaster of paris)

$CuSO_4.5H_2O$ (blue) $\longrightarrow CuSO_4 + 5H_2O$ (white)

$CuSO_4 \longrightarrow CuO + SO_3$

$(NH_4)_2Cr_2O_7$ (orange) $\longrightarrow N_2 + Cr_2O_3 + 4H_2O$ (green)

$2KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$

$2KClO_3 \longrightarrow 2KCl + 3O_2$

$4K_2Cr_2O_7 \longrightarrow 4K_2CrO_4 + 2Cr_2O_3 + 3O_2$

$(COO)_2Fe \longrightarrow FeO + CO + CO_2$ (black)

$2Ag_2O \longrightarrow 4Ag + O_2$

$2HgO$ (red) $\longrightarrow 2Hg + O_2$ (silver deposit)

$2Pb_3O_4$ (red) $\longrightarrow 6PbO + O_2$ (yellow)

$2PbO_2$ (brown) $\longrightarrow 2PbO + O_2$ (black)

$2NaNO_3 \longrightarrow 2NaNO_2 + O_2$

$NH_4NO_2 \longrightarrow N_2 + 2H_2O$

$2AgNO_3 \longrightarrow 2Ag + 2NO_2 + O_2$

$2Cu(NO_3)_2$ (brown) $\longrightarrow 2CuO + 4NO_2 + O_2$

$2Zn(NO_3)_2$ (white) $\rightarrow 2ZnO + 4NO_2 + O_2$ yellow-hot white-cold (brown)

$2Ca(NO_3)_2 \longrightarrow 2CaO + 4NO_2 + O_2$

$2Pb(NO_3)_2$ (white) $\longrightarrow 2PbO + 4NO_2 + O_2$ (yellow) (brown)

$2Mg(NO_3)_2 \longrightarrow 2MgO + 4NO_2 + O_2$

$NH_4NO_3 \longrightarrow N_2O + 2H_2O$

Important Processes

1. Bosch process $\longrightarrow H_2$
2. Down, Castner $\longrightarrow$ Na
3. Nelson, Castner – Kellner, Solvey Droney, Lowing $\longrightarrow$ NaOH
4. Ammonia soda process (Solvay process) $\longrightarrow$ Na_2CO_3, $NaHCO_3$
5. Leblanc, Pretch process $\longrightarrow K_2CO_3$

6. MacArthur forest or Cyanidarion ⟶ Ag, Au
7. Perk, Parrinson ⟶ Ag
8. Cupellarion ⟶ Ag (Purification)
9. Mund's process ⟶ Ni (Purification)
10. Baeyer's or Serpeck's process ⟶ Al
11. Hoope's process ⟶ Al (Purification)
12. Hall, Heroult Process ⟶ Al
13. Gold schmidt Process ⟶ Thermite Welding
14. Carrer's process ⟶ White lead
15. Haeber's process ⟶ NH_3
16. Deacon's process ⟶ Cl_2
17. Contact, Lead Chamber process ⟶ H_2SO_4
18. Berkland-Eyde,Ostwald ⟶ HNO_3, NO
19. Kaldo, L.D. ⟶ steel
20. Corey – House ⟶ Alkane
21. Oxo ⟶ R – OH
22. Dow's process ⟶ Mg
23. Pidgeon ⟶ Mg
24. Cyanamide ⟶ NH_3
25. IMI ⟶ Ti
26. frash ⟶ 'S'
27. Siemens, Basemer Thomass ⟶ Steel.
28. Lane's process ⟶ H_2
29. Gossage process ⟶ NaOH

MINERALS WITH METALLIC LUSTURE

Mineral (formula)	*Colour*	*Crystal system*	*Uses and their properties*
Augite [(Ca, Na) (Mg, Fe, Al) (Al, Si)$_2$O$_6$]	Black	Monoclinic	Square or 8-sided cross section
Bornite [Cu_5FeS]	Bronze, tarnishes to dark blue, purple	Tetragonal	Source of copper; called "Peakcock ore" due to its purple shine when it tarnishes
Chalcopyrite ($CuFeS_2$)	Brassy to golden yellow	Tetragonal	Main ore of copper
Chromite ($FeCr_2O_4$)	Black or brown	Cubic	Ore of chromium, stainless steel, metallurgical bricks
Copper (Cu)	Copper red	Cubic	Pipes, coins, gutters, wire, cooking utensils, jewellery, decorative plaques; malleable and ductile
Corundum (Al_2O_3)	Colourless, blue, brown, green, white, pink, red	Hexagonal	Gemstones; ruby is red, sapphire is blue, industrial abrasive
Feldspar (orthoclase) ($KAlSi_3O_8$)	Colourless, white to gray, green and yellow	Monoclinic	Insoluble in acids, used in the manufacture of porcelain
Feldspar (plagioclase) ($NaAlSi_3O_8$) ($CaAl_2Si_2O_8$)	Gray, green white	Triclinic	Used in ceramics; striations present on some faces
Fluorite (CaF_2)	Colourless, white, blue, green, red, yellow, purple	Cubic	Used in the manufacture of optical equipment, glows under ultraviolet light
Galena (PbS)	Gray	Cubic	Source of lead, shields for X-rays, fishing equipment sinkers, used in pipes
Garnet (Mg, Fe, Ca)$_3$ ($Al_2Si_3O_{12}$)	Deep yellow-red, green, black	Cubic	Used in jewellery, also used as an abrasive
Gold (Au)	Pale to golden	Cubic	Medicines, jewellery, money, gold leaf, fillings for teeth, does not tarnish
Graphite (C)	Black to grey	Hexagonal	Pencil lead, rods to control some small nuclear reactions, lubricants for locks, battery poles
Haematite (Fe_2O_3)	Black or reddish brown	Hexagonal	Source of iron; roasted in a blast furnace, converted to "pig" iron, made into steel

Mineral (formula)	*Colour*	*Crystal system*	*Uses and their properties*
Hornblende [Ca Na (Mg, Al, Fe)$_5$ (Al, Si)$_2$ Si$_6$ O$_{22}$ (OH)$_2$]	Green to black	Monoclinic	Will transmit light on thin edges; 6-sided cross section
Limonite (hydrous iron oxides)	Yellow, brown, black	-	Source of iron, weathers easily colouring matter of soils
Magnetite (Fe_3O_4)	Black	Cubic	Source of iron, naturally magnetic, called lodestone
Olivine [(Mg, Fe)$_2$ SiO_4]	Olive green	Ortho-rhombic	Gemstones, refractory sand
Pyrite (FeS_2)	Light, brassy, yellow	Cubic	Source of iron, "fools's gold" alters to limonite
Pyrrhotite (FeS)	Bronze	Hexagonal	Often found with pentlandite, an ore of nickel; may be magnetic
Quartz (SiO_2)	Colourless, various colorus	Hexagonal	Used in glass manufacture, electronic equipment, radios, computers, watches, gemstones
Silver (Ag)	Silvery white, tarnishes to black	Cubic	Coins, jewellery, silverplate, fillings for teeth, wires, malleable and ductile
Topaz [(Al_2SiO_4) (F, OH)$_2$]	White, pink yellow, pale blue, colourless	Ortho-rhombic	Valuable gemstone
Bauxite (hydrous aluminium compound)	Gray, red, white, brown	-	Source of aluminum; used in paints, aluminium foil, and airplane parts
Biotite [K(Mg,Fe)$_3$ $AlSi_3O_{10}$ (OH)$_2$]	Black to dark brown	Monoclinic	Occurs in large flexible plates
Calcite ($CaCO_3$)	Colourless, white pale, blue	Hexagonal	Fizzes when HCl is added; used in cement and other building materials
Dolomite [$CaMg(CO_3)_2$]	Colourless, white, pink, green, gray, black	Hexagonal	Concrete and cement, used as an ornamental building stone
Gypsum ($CaSO_4.2H_2O$)	Colourless, gray, white, brown	Monoclinic	Used extensively in the preparation of plaster of paris, alabaster, and dry wall for building construction
Halite (NaCl)	Colourless, red, white, blue	Cubic	Salt; very soluble in water; a preservative
Kaolinite [$Al_4Si_2O_5(OH)_4$]	White, red, reddish brown, black	Triclinic	Clays; used in ceramics and in china dishes; common in most soils; often microscopic sized particles
Muscovite [$KAl_3Si_3O_{10}(OH)_2$]	White, light gray, yellow, rose, green	Monoclinic	Occurs in large flexible plates, used as an insulator in electrical equipment, lubricant
Sphalerite (ZnS)	Brown	Cubic	Main ore of zinc; used in paints, dyes, and medicine
Sulphur (S)	Yellow	Ortho-rhombic	Used in medicine, fungicides for plants, vulcanisatoin of rubber, production of sulphuric acid
Talc [$Mg_3(OH)_2$ Si_4O_{10}]	White, greenish	Monoclinic	Easily cut with fingernail; used for talcum powder; soapstone is used in paper and for table tops

ELEMENT SCIENTISTS AND THEIR CONTRIBUTIONS

Name	*Contributions*
Avogadro	Avogadro hypothesis
Aston	mass spectrograph
Arrhenius	Theory of electrolytic dissociation
Bronsted & Lowry	Acid-base concept

Name	*Contributions*
Brown	Brownian movement
Boyle	Gas laws
Bohr	Atomic structure
Becquerel	Radioactivity

Barlett	Compounds of inert gases
Charles	Gas laws
Chadwick	Discovery of neutron
de Broglie	Wave nature of electron
Dalton	Atomic theory, law of partial pressures, law of multiple proportions
Einstein	Theory of relatively, mass energy relation-ship
Faraday	Law of electrolysis
Fajan	Polarisation of ions
Fajan, Russel & Soddy	Radioactive group dis-placement law
Graham	Law of diffusion
Goldstein	Discovery of anode rays
Guldberg & Waage	Law of mass action
Gay-Lussac	Law of combining volumes
Hund	Multiplicity rule
Hund & Mulliken	Molecular orbital theory
Heitlor & London	Valence bond theory
Heisenberg	Uncertainty principle
Helmholtz	First law of thermo-dynamics
Hahn & Strassman	Nuclear fission
Irene Curie & F.Joliot	Artificial radioactivity
Kossel & Lewis	Electrovalency
Keyser	Adsorption
Kekule	Structure of benzene
Lockyer & Frankland	Discovery of helium
Libby	Radioactive dating
Lewis	Covalency, acid-base concept
Le Chatelier	Effect of P, T and C on equilibrium
Lawrence	Cyclotron
Lavoisier	Laws of combination of mass
Langmuir	Adsorption isotherm
Moseley	Modern periodic law
Milliken	Charge of electron
Mendeleev	Periodic classification of elements
Maxwell	Kinetic theory of gases
Marie Curie	Discovery of radium and polonium
Nernst	Electrode potential
Ostwald	Dilution law
Proust	Law of reciprocal proportions
Planck	Wave nature of light
Pauling and Slatter	Valence bond theory
Pauling	Chemical bonding
Pauli	Exclusion principle
Raoult	Relation between vapour pressure and mole fraction
Raman	Scattering of light by liquid molecules
Rutherford	Nuclear model of atom
Rutherford & Soddy	Radioactive disintegration
Roentgen	Discovery of X-ray
Ritcher	Law of definite proportions
Rayleigh & Ramsay	Discovery of argon
Ramsay & Travers	Discovery of neon, krypton and xenon
Sorel	Composition of ozone
Sorensen	pH
Soddy & Aston	Isotopes
Sidgwick & Powell	Hybridisation
Seaborg	Transuranium elements
Schulze & Hardy	Coagulation of colloids by electrolytes
Scheele	Discovery of nitrogen
Schrodinger	Wave equation
Tyndal	Scattering of light by colloidal particles
Thomson	Discovery of electron, photoelectric effect
Thenard	Discovery of hydrogen peroxide
Urey	Discovery of D_2 and D_2O
Van't Hoff	Relation between rate constant, equilibrium constant & temperature
Wohler	Synthesis of first organic compound (urea)
Werner	Theory of coordinate linkage
Yukawa	Discovery of mesons

IMPORTANT REAGENTS AND MIXTURES

Ammonal	Al powder + NH_4NO_3 [used as an explosive]
Amatol	NH_4NO_3 (80%) + trinitrotoluene (20%) [used as an explosive]
Aqua regia	Conc. HNO_3 + conc. HCl (1 : 3) [used as a laboratory reagent]
Bordeaux mixture	A solution of $CuSO_4$ + lime [used to kill moulds and fungi on plants]

Black ash	Na_2CO_3 + CaS [impure Na_2CO_3 obtained in Leblanc process]
Benedict solution	A solution of $CuSO_4.5H_2O$ + NaOH + sodium citrate [used for detecting aldehydes]
Baeyer reagent	Alkaline $KMnO_4$ solution [used for detecting ethylene and acetylene linkages]
Baking powder	$NaHCO_3$ + sodium potassium tartarate
Carbon oil	Vegetable oil + lime water [used for treatment of burns]
Carbogen	O_2 (90-95%) + CO_2 (5-10%) [used for artificial respiration]
Coal gas	H_2 (47%) + CH_4 (32%) + CO (7%) + N_2 (4%) + C_nH_n (3%) + C_2H_2 (2%) + CO_2 (1%) + other gases (1%) [used to produce reducing atmosphere in metallurgical operations]
Euchlorine	Cl_2 + ClO_2
Fusion mixture	Na_2CO_3 + K_2CO_3 [used as a laboratory reagent]
Freezing mixture	NaCl + ice [used for lowering temperature]
Fehling solution	$CuSO_4.5H_2O$ + NaOH + sodium potassium tartarate [used for detecting aldehydes]
Fenton reagent	H_2O_2 + few drops of $FeCl_3$
Gun powder	KNO_3 (75%) + S (12%) + charcoal (13%) [used as an explosive]
Granite	Mica + rock + clay + sand
Gobar gas	CH_4 + CO + H_2 [used as a domestic fuel]
Ignition mixture	BaO_2 + Mg + Al [used in aluminothermit process]
Lucas reagent	Conc. HCl + anhy. $ZnCl_2$ [used for distinguishing three types of alcohols]
Lithopone	ZnS + $BaSO_4$ [used as a white paint]
Lindlar catalyst	Pd/$BaSO_4$, S [used for hydrogenation of alkyne to alkene]
Mortar	(slaked lime + sand) (1 : 3 in water)
Molish reagent	An ethanolic solution of α-naphthol [used for detecting carbohydrates]
Milk of magnesia	An aqueous suspension of $Mg(OH)_2$ [used as a antacid]
Methylated spirit	Rectified spirit (85-90%) + CH_3OH (10-15%) + pyridine + acetone [used as a solvent]
Matte	Cu_2S + FeS
Nitrophos	$Ca(H_2PO_4)_2$ + $Ca(NO_3)_2$ [used as a fertilizer]
Nitrolim	$CaCN_2$ + graphite [used as a fertilizer]
Nitro chalk	NH_4NO_3 + $CaCO_3$ [used as a fertilizer]
Nessler reagent	A solution of K_2HgI_4 + KOH [used for detecting NH_4^+ ion]
Oil gas	H_2 (50-55%) + CH_4 (25-30%) + CO (10-12%) + CO_2 (3%) [used in laboratory]
Purple of cassius	Colloidal sol of Au + $Sn(OH)_4$ [used for colouring of glass and pottery red]
Producer gas	N_2 (52-55%) + CO (22-30%) + H_2 (8-12%) + CO_2 (3%) [used as a fuel]
Power alcohol	(petrol + C_2H_5OH) (1 : 1) + little benzene [used as a motor fuel]
Rectified spirit	C_2H_5OH (95.87%) + H_2O (4.13%) [used as a solvent]
Super phosphate of lime	$Ca(H_2PO_4)_2$ + $CaSO_4$ [used as a fertilizer]
Sublimated white lead	PbO + $PbSO_4$ + ZnO [used as a white paint]
Sorel's cement	MgO + $MgCl_2$ [used as a substitute for tiles]
Sodalime	NaOH + $Ca(OH)_2$ [used in decarboxylation of carboxylic acids]
Soda bleach	Na_2O_2 + HCl [used for bleaching of fabrics]
Schweitzer reagent	$[Cu(NH_3)_4]SO_4$ [used in the manufacture of artificial silk]
Schiff's reagent	An aqueous solution of rosaniline hydrochloride whose red colour has been discharged by SO_2 [used for detecting aldehydes]
Thomas slag	$Ca_3(PO_4)_2$ + $CaSiO_3$ [used as a fertilizer]
Tincture of iodine	I_2 + KI + C_2H_5OH + water [used as an antiseptic]
Thermite mixture	Al powder + metal oxide [used in metallurgy]
Water gas	H_2 (51%) + CO (41%) + N_2 (4%) + CO_2 (4%) [used for manufacture of CH_3OH]

Solved Question Papers 2011

CHEMISTRY

PAPER–I

SECTION–I

(Single Correct Answer Type)

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

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1. Extra pure N_2 can be obtained by heating
   (A) NH_3 with CuO
   (B) NH_4NO_3
   (C) $(NH_4)_2Cr_2O_7$
   (D) $Ba(N_3)_2$

Sol. **(D)**

$$Ba(N_3)_2 \xrightarrow{\Delta} Ba + 3N_2$$

2. Dissolving 120 g of urea (mol. wt. 60) in 1000 g of water gave a solution of density 1.15 g/mL. The molarity of the solution is
   (A) 1.78 M
   (B) 2.00 M
   (C) 2.05 M
   (D) 2.22 M

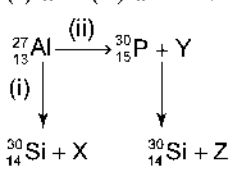
Sol. **(C)**

$$\text{Total mass of solution} = 1000 + 120 = 1120 \text{ g}$$

$$\text{Total volume of solution in (L)} = \frac{1120}{1.15} \times 10^3$$

$$M = \frac{W}{M} \times \frac{1}{V(\text{in L})} = \frac{120}{60} \times \frac{1.15 \times 10^3}{1120} = 2.05 \text{ M}$$

3. Bombardment of aluminium by α-particle leads to its artificial disintegration in two ways, (i) and (ii) as shown. Products X, Y and Z respectively are,



   (A) proton, neutron, positron
   (B) neutron, positron, proton
   (C) proton, positron, neutron
   (D) positron, proton, neutron

Sol. **(A)**

$$^{27}_{13}Al + {}_2\alpha^4 \rightarrow {}^{30}_{14}Si + {}_1P^1(X)$$

$$^{27}_{13}Al + {}_2\alpha^4 \rightarrow {}^{30}_{15}P + {}_0n^1(Y)$$

$$^{30}_{15}P \rightarrow {}^{30}_{14}Si + {}_{+1}\beta^0(Z)$$

4. Geometrical shapes of the complexes formed by the reaction of Ni^{2+} with Cl , CN and H_2O, respectively, are

(A) octahedral, tetrahedral and square planar

(B) tetrahedral, square planar and octahedral

(C) square planar, tetrahedral and octahedral

(D) octahedral, square planar and octahedral

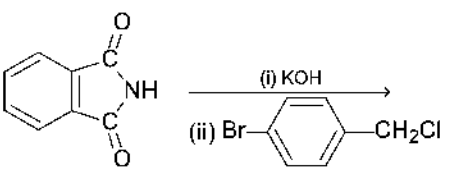
Sol. **(B)**

$[NiCl_4]^{2-} \rightarrow$ Tetrahedral

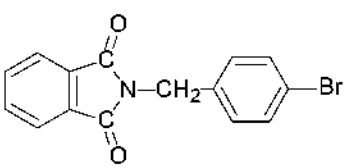
$[Ni(CN)_4]^{2} \rightarrow$ Square Planar

$[Ni(H_2O)_6]^{2+} \rightarrow$ Octahedral

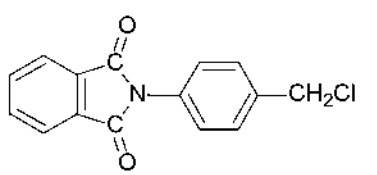
5. The major product of the following reaction is



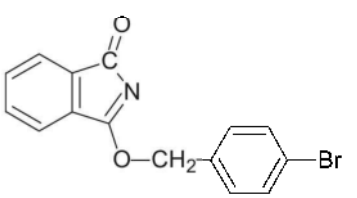
(A)



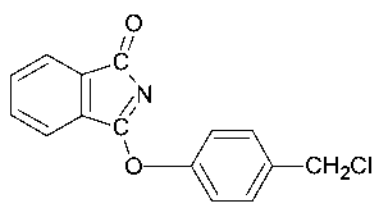
(B)



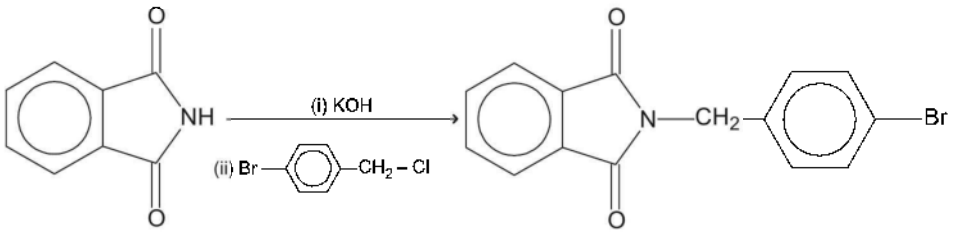
(C)



(D)



Sol. **(A)**



[**Reason:** Due to partial double bond character along C–Br bond prevents the attack of nucleophile at phenylic position]

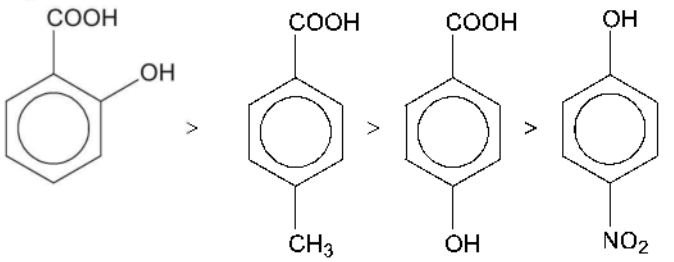
6. Among the following compounds, the most acidic is

(A) p-nitrophenol (B) p-hydroxybenzoic acid

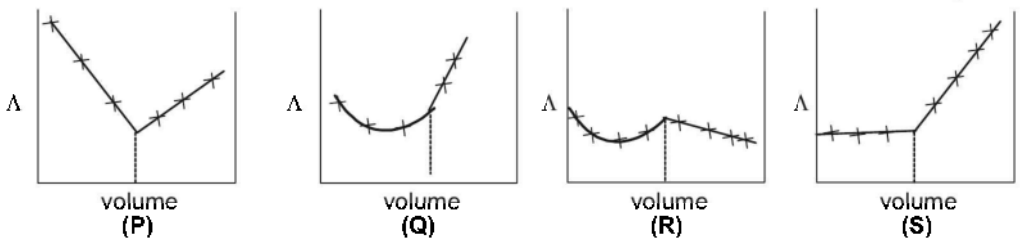
(C) o-hydroxybenzoic acid (D) p-toluic acid

Sol. **(C)**

Due to ortho effect o-hydroxy benzoic acid is strongest acid and correct order of decreasing K_a is



7. $AgNO_3$(aq.) was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. The plot of conductance (Λ) versus the volume of $AgNO_3$ is



(A) (P) (B) (Q)

(C) (R) (D) (S)

Sol. **(D)**

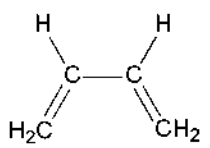
$AgNO_3 + KCl(aq) \rightarrow AgCl(s) + KNO_3(aq)$

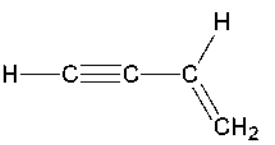
Initially there is aq. KCl solution now as solution of $AgNO_3$ is added, AgCl(s) is formed. Hence conductivity of solution is almost compensated (or slightly increase) by the formation of KNO_3. After end point conductivity increases more rapidly because addition of excess $AgNO_3$ solution.

SECTION–II
(Multiple Correct Answers Type)

This section contains **4 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

8. Amongst the given options, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are)

(A) 

(B) 

(C) $H_2C{=}C{=}O$

(D) $H_2C{=}C{=}CH_2$

Sol. **(B, C)**

Along C–C single bond conformations are possible in butadiene in which all the atoms may not lie in the same plane.

9. Extraction of metal from the ore **cassiterite** involves

(A) carbon reduction of an oxide ore
(B) self-reduction of a sulphide ore
(C) removal of copper impurity
(D) removal of iron impurity

Sol. **(A, C, D)**

$SnO_2 + 2C \rightarrow 2CO + Sn$

The ore cassiterite contains the impurity of Fe, Mn, W and traces of Cu.

10. According to kinetic theory of gases

(A) collisions are always elastic
(B) heavier molecules transfer more momentum to the wall of the container
(C) only a small number of molecules have very high velocity
(D) between collisions, the molecules move in straight lines with constant velocities

Sol. **(A, B, C, D)**

11. The correct statement(s) pertaining to the adsorption of a gas on a solid surface is (are)
(A) Adsorption is always exothermic
(B) Physisorption may transform into chemisorption at high temperature
(C) Physiosorption increases with increasing temperature but chemisorption decreases with increasing temperature
(D) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation

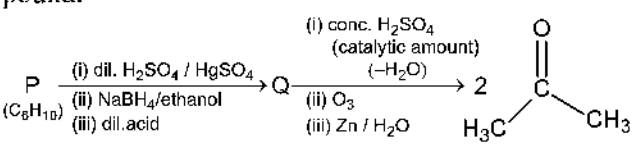
Sol. **(A, B, D)**

SECTION–III
(Paragraph Type)

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of **which ONLY ONE** is correct.

Paragraph for Question Nos. 12 and 13

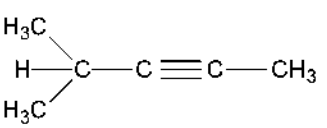
An acyclic hydrocarbon **P**, having molecular formula C_6H_{10}, gave acetone as the only organic product through the following sequence of reaction, in which **Q** is an intermediate organic compound.

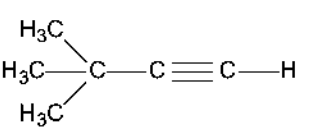


12. The structure of compound **P** is

(A) $CH_3CH_2CH_2CH_2 - C \equiv X - H$

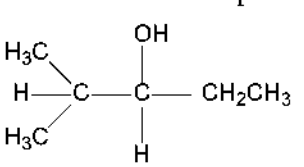
(B) $H_3CH_2C - C \equiv C - CH_2CH_3$

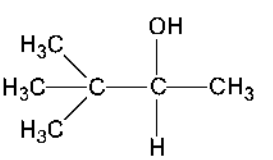
(C) 

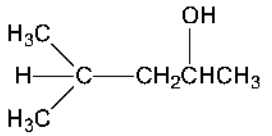
(D) 

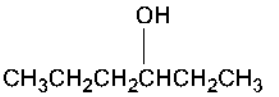
Sol. **(D)**

13. The structure of the compound **Q** is

(A) 

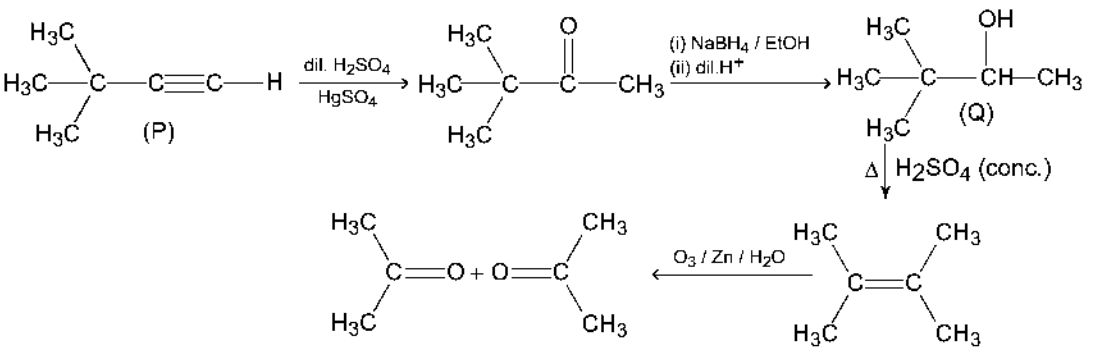
(B) 

(C) 

(D) 

Sol. **(B)**

Solution for the Q. No. 12 to 13



Paragraph for Question Nos. 14 to 16

When a metal rod **M** is dipped into an aqueous colourless concentrated solution of compound **N**, the solution turns light blue. Addition of aqueous NaCl to the blue solution gives a white precipitate **O**. Addition of aqueous NH_3 dissolves **O** and gives an intense blue solution.

14. The metal rod **M** is

(A) Fe (B) Cu

(C) Ni (D) Co

Sol. **(B)**

$$\underset{M}{Cu} + 2\underset{N}{AgNO_3} \rightarrow \underset{Blue}{Cu(NO_3)_2} + 2Ag$$

While Cu partially oxidizes to $Cu(NO_3)_2$ and remaining $AgNO_3$ reacts with NaCl.

15. The compound **N** is

(A) $AgNO_3$ (B) $Zn(NO_3)_2$

(C) $Al(NO_3)_3$ (D) $Pb(NO_3)_2$

Sol. **(A)**

16. The final solution contains

(A) $\left[Pb(NH_3)_4\right]^{2+}$ and $\left[CoCl_4\right]^{2-}$ (B) $\left[Al(NH_3)_4\right]^{3-}$ and $\left[Cu(NH_3)_4\right]^{2-}$

(C) $\left[Ag(NH_3)_2\right]^{1}$ and $\left[Cu(NH_3)_4\right]^{21}$ (D) $\left[Ag(NH_3)_2\right]$ and $\left[Ni(NH_3)_6\right]^{2}$

Sol. **(C)**

$$\underset{(X)}{AgNO_3} + NaCl \rightarrow \underset{(o)}{AgCl}\downarrow + NaNO_3$$

$$AgCl + 2NH_3 \rightarrow \left[Ag(NH_3)_2\right]^{+} Cl$$

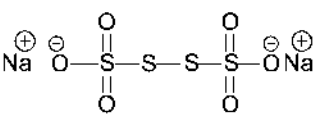
$$Cu(NO_3)_2 + 4NH_4OH \rightarrow \left[Cu(NH_3)_4\right]^{2+}$$

SECTION–IV
(Integer Answer Type)

This section contains **7** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.

17. The difference in the oxidation numbers of the two types of sulphur atoms in $Na_2S_4O_6$ is

Sol. **(5)**



S will have oxidation number = +5, 0
Difference in oxidation number = 5

18. A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0% to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is

Sol. **(6)**

For n-units of glycine,

$$\frac{n\times 75}{(796+9\times 18)}\times 100 = 47$$

$$\Rightarrow n = 6$$

19. The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
ϕ (eV)	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

Sol. **(4)**

The energy associated with incident photon $= \frac{hc}{\lambda}$

$\Rightarrow E = \frac{6.6\times10^{-34}\times3\times10^{8}}{300\times10^{-9}}$ J

E in eV $= \frac{6.6\times10^{-34}\times3\times10^{8}}{300\times10^{-9}\times1.6\times10^{-19}} = 4.16$ eV

So, number of metals showing photo-electric effects will be (4), i.e., Li, Na, K, Mg

20. The maximum number of electrons that can have principal quantum number, n 3, and spin quantum number, m_s $\frac{1}{2}$, is

Sol. **(9)**

For principal quantum number (n 3)

Number of orbitals n^2 9

So, number of electrons with m_s $\frac{1}{2}$ will be 9.

21. Reaction of Br_2 with Na_2CO_3 in aqueous solution gives sodium bromide and sodium bromate with evolution of CO_2 gas. The number of sodium bromide molecules involved in the balanced chemical equation is

Sol. **(5)**

$3Br_2 + 3Na_2CO_3 \longrightarrow 5NaBr + NaBrO_3 + 3CO_2$

So, number of NaBr molecules 5

22. To an evacuated vessel with movable piston under external pressure of 1 atm, 0.1 mol of He and 1.0 mol of an unknown compound (vapour pressure 0.68 atm. at 0°C) are introduced. Considering the ideal gas behaviour, the total volume (in litre) of the gases at 0°C is close to

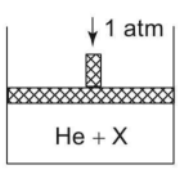
Sol. **(7)**

For any ideal gas, PV nRT

0.32 V 0.1 0.0821 273

V 7 litre

(unknown compound X will not follow ideal gas equation)

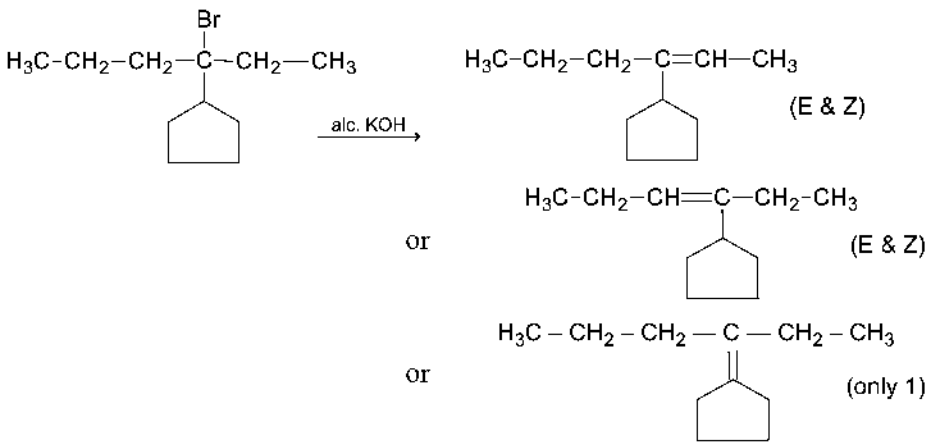


For He, n 0.1, P 0.32 atm., V ?, T 273

23. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentyl-hexane using alcoholic KOH is

Sol. **(5)**

Total no. of alkenes will be = 5



PAPER–II

SECTION–I

(Single Correct Answer Type)

This Section contains **8 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. The freezing point (in °C) of a solution containing 0.1 g of $K_3[Fe(CN)_6$ (Mol. Wt. 329) in 100 g of water ($K_f = 1.86$ K kg mol ^{1}) is

(A) -2.3×10^{2} (B) -5.7×10^{2}

(C) -5.7×10^{3} (D) -1.2×10^{2}

Sol. **(A)**

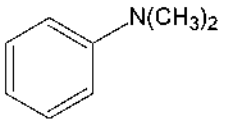
$$K_3\left[Fe(CN)_6\right] \rightarrow 3K^{1} + \left[Fe(CN)_6\right]^{3}$$

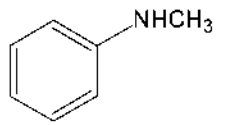
$i = 4$

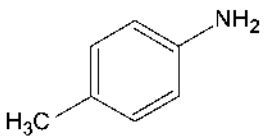
$$\Delta T_f = K_f \times i \times \frac{m}{M} \times \frac{1000}{W} = 1.86 \times 4 \times \frac{0.1}{329} \times \frac{1000}{100} = 2.3 \times 10^{2}$$

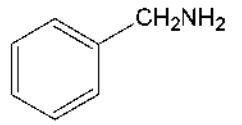
$$T_f^{1} = -2.3 \times 10^{2}$$

2. Amongst the compounds given, the one that would form a brilliant colored dye on treatment with $NaNO_2$ in dil. HCl followed by addition to an alkaline solution of β-naphthol is

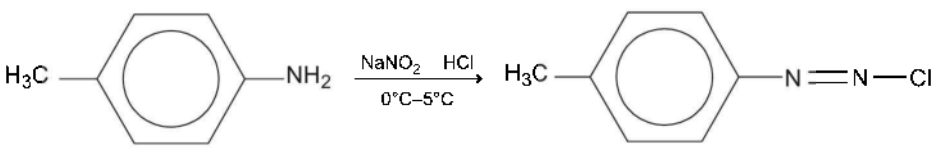
(A) 

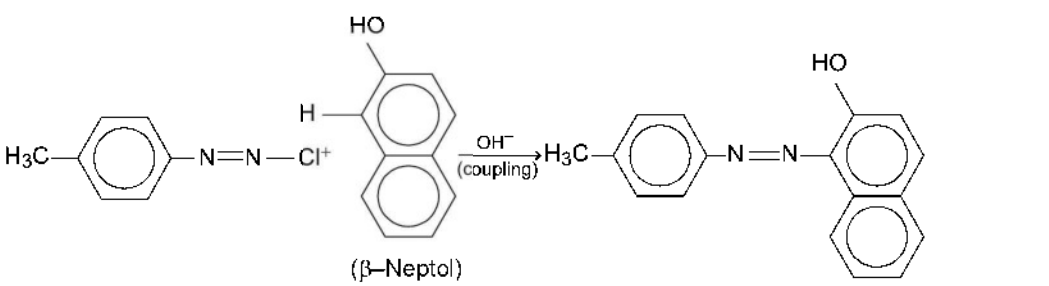
(B) 

(C) 

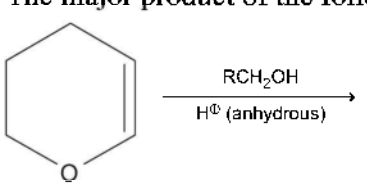
(D) 

Sol. **(C)**



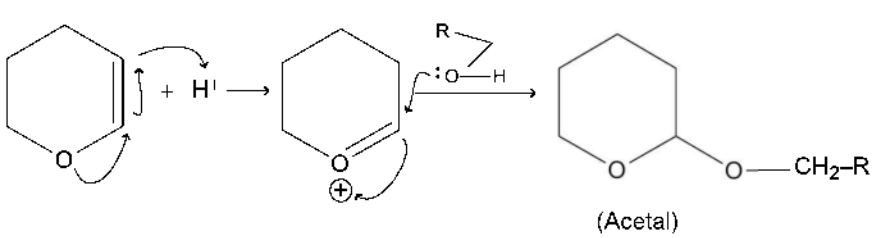


3. The major product of the following reaction is

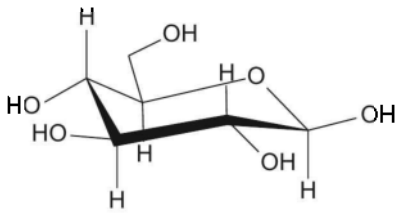


(A) a hemiacetal
(B) an acetal
(C) an ether
(D) an ester

Sol. **(B)**



4. The following carbohydrate is



(A) a ketohexose
(B) an aldohexose
(C) an α-furanose
(D) an α-pyranose

Sol. **(B)**

5. Oxidation states of the metal in the minerals haematite and magnetite, respectively, are
(A) II, III in haematite and III in magnetite
(B) II, III in haematite and II in magnetite
(C) II in haematite and II, III in magnetite
(D) III in haematite and II, III in magnetite

Sol. **(D)**
Haematite : Fe_2O_3 : $2x + 3 \times (-2) = 0$
$x = 3$
Magnetite: Fe_3O_4 [an equimolar mixture of FeO and Fe_2O_3]
Fe_2O_3: $x = 3$

6. Among the following complexes **(K–P)**
$K_3[Fe(CN)_6]$ **(K)**, $[Co(NH_3)_6]Cl_3$ **(L)**, $Na_3[Co(oxalate)_3]$ **(M)**, $[Ni(H_2O)_6]Cl_2$ **(N)**, $K_2[Pt(CN)_4]$ (O) and $[Zn(H_2O)_6](NO_3)_2$ **(P)**
(A) **K, L, M, N**
(B) **K, M, O, P**
(C) **L, M, 0, P**
(D) **L, M, N, 0**

7. (C)
Following compounds are diamagnetic.
L : $[Co(NH_3)_6]Cl_3$
M : $Na_3[Co(Ox)_3]$
O: $K_2[Pt(CN)_4]$
P : $[Zn(H_2O)_6](NO_3)_2$

8. Passing H_2S gas into a mixture of Mn^2 , Ni^{21}, Cu^2 and Hg^2 ions in an acidified aqueous solution precipitates
(A) CuS and HgS
(B) MnS and CuS
(C) MnS and NiS
(D) NiS and HgS

Sol. **(A)**

H_2S in presence of aqueous acidified solution precipitates as sulphide of Cu and Hg apart from Pb ^{2}, Bi^{13}, Cd^{12}, As^{13}, Sb+^{3} and Sn ^{2}.

9. Consider the following cell reaction:

$$2Fc_{(s)} + O_{2(g)} + 4H_{(aq)} \rightarrow 2Fc^{2}_{(aq)} + 2H_2O_{(l)} \quad E° = 1.67\ V$$

At [Fc^{21}] 10 ^{3} M, P(O_2) 0.1 atm and pH 3, the cell potential at 25°C is

(A) 1.47 V (B) 1.77 V

(C) 1.87 V (D) 1.57 V

Sol. **(D)**

$$2Fc(s) + O_2(g) + 4H^{1}(aq) \longrightarrow 2Fc^{2}(aq) + 2H_2O(l)$$

N 4 (no. of moles of electron involved)

From Nernst's equation,

$$E_{cell} = E°_{cell} - \frac{0.0591}{n}\log Q$$

$$= 1.67 - \frac{0.0591}{4}\log\frac{\left(10^{-3}\right)^{2}}{0.1\times\left(10^{-3}\right)^{4}} \quad \left\{\because \left[H^{+}\right] = 10^{-pH}\right\}$$

$$= 1.67 - 0.106$$

$$= 1.57\ V$$

SECTION II

(Multiple Correct Answer(s) Type)

This section contains **4 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

10. Reduction of the metal centre in aqueous permanganate ion involves

(A) 3 electrons in neutral medium (B) 5 electrons in neutral medium

(C) 3 electrons in alkaline medium (D) 5 electrons in acidic medium

Sol. **(A, D)**

In acidic medium

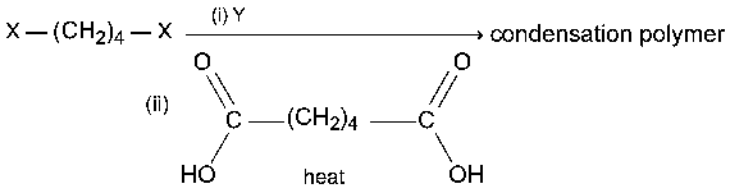
$$MnO_4^{-} + 8H + 5e \longrightarrow Mn^{21} + 4H_2O$$

In neutral medium

$$MnO_4^{-} + 2H_2O + 3e \longrightarrow MnO_2 + 4OH$$

Hence, number of electron loose in acidic and neutral medium 5 and 3 electrons respectively.

11. The correct functional group X and the reagent/reaction conditions Y in the following scheme are

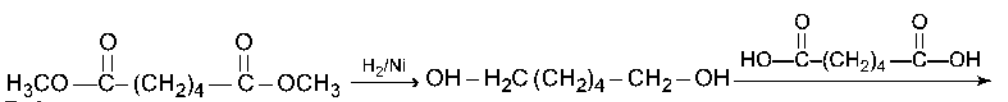


(A) X = $COOCH_3$, Y = H_2/Ni/heat
(B) X = $CONH_2$, Y = H_2/Ni/heat
(C) X = $CONH_2$, Y = Br_2/NaOH
(D) X = CN, Y = H_2/Ni/heat

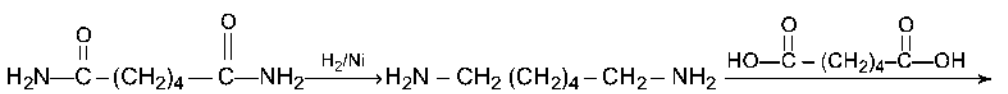
Sol. **(A, B, C, D)**

Condensation polymers are formed by condensation of a diols or diamine with dicarboxylic acids.

Hence, X may be $-\overset{O}{\overset{\|}{C}}-OR$ or $-\overset{O}{\overset{|}{C}}-NH_2$ or $-C \equiv N$

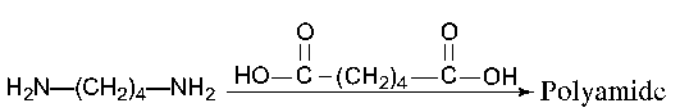


Polyester

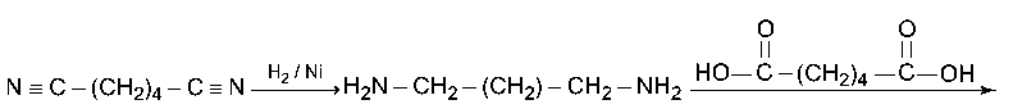


Polyamide

$Br_2 / OH^-, \Delta$
Hoffmann Bromamide Reaction



Polyamide



Polyamide

12. For the first order reaction

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

(A) the concentration of the reactant decreases exponentially with time
(B) the half-life of the reaction decreases with increasing temperature
(C) the half-life of the reaction depends on the initial concentration of the reactant
(D) the reaction proceeds to 99.6% completion in eight half-life duration

Sol. **(A, B, D)**
For first order reaction
$[A] = [A]_0 e^{-kt}$
Hence concentration of $|NO_2|$ decreases exponentially.

Also, $t_{\frac{1}{2}} = \frac{0.693}{K}$. Which is independent of concentration and $t_{\frac{1}{2}}$ decreases with the increase of temperature.

$$t_{99.6} = \frac{2.303}{K} \log\left(\frac{100}{0.4}\right)$$

$$t_{99.6} = \frac{2.303}{K}(2.4) = 8 \times \frac{0.693}{K} = 8\, t_{\frac{1}{2}}$$

13. The equilibrium

$$2Cu^{I} \rightleftharpoons Cu^{\circ} + Cu^{II}$$

in aqueous medium at 25°C shifts towards the left in the presence of

(A) NO_3 (B) Cl
(C) SCN (D) CN

Sol. **(B, C, D)**
Cu^{2+} ions will react with CN and SCN forming $[Cu(CN)_4]^3$ and $[Cu(SCN)_4]^3$ leading the reaction in the backward direction.

$$Cu^{2+} + 2CN^- \rightarrow Cu(CN)_2$$
$$2Cu(CN)_2 \rightarrow 2CuCN + (CN)_2$$
$$CuCN + 3CN^- \rightarrow \left[Cu(CN)_4\right]^3$$
$$Cu^{2+} + 4SCN^- \rightarrow \left[Cu(SCN)_4\right]^3$$

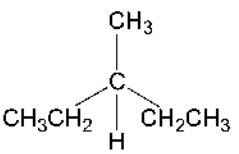
Cu^{2+} also combines with $CuCl_2$ which reacts with Cu to produce CuCl pushing the reaction in the backward direction.

$$CuCl_2 + Cu \rightarrow 2CuCl\downarrow$$

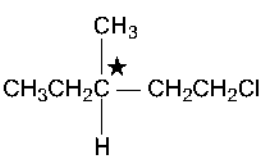
SECTION–III
(Integer Answer Type)

This section contains **6 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the **ORS**.

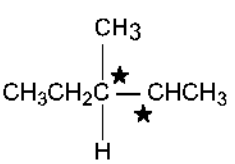
14. The maximum number of isomers (including stereoisomers) that are possible on monochlorination of the following compound is



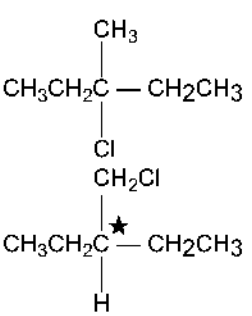
Sol. **(8)**



Enantiomeric pair = 2

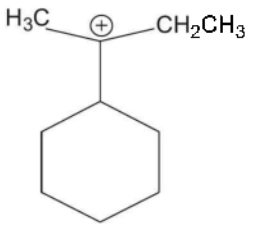


Two Enantiomeric pair = 2



Total = 2 + 4 + 1 + 1 = 8

15. The total number of contributing structure showing hyperconjugation (involving C–H bonds) for the following carbocation is



Sol. **(6)**

6 × H—atoms are there

16. Among the following, the number of compounds than can react with PCl_5 to give $POCl_3$ is

O_2, CO_2, SO_2, H_2O, H_2SO_4, P_4O_{10}

Sol. **(5)**

17. The volume (in mL) of 0.1 M $AgNO_3$ required for complete precipitation of chloride ions present in 30 mL of 0.01 M solution of $[Cr(H_2O)_5Cl]Cl_2$, as silver chloride is close to

Sol. **(6)**

Number of ionisable Cl^- in $[Cr(H_2O)_5Cl]Cl_2$ is 2

$\therefore$ Millimoles of $Cl^- = 30 \times 0.01 \times 2 = 0.6$

$\therefore$ Millimoles of Ag^- required = 0.6

$\therefore$ 0.6 = 0.1 V

$\therefore$ V = 6 ml

18. In 1 L saturated solution of AgCl [$K_{sp}(AgCl) = 1.6 \times 10^{-10}$], 0.1 mol of CuCl [$K_{sp}(CuCl) = 1.0 \times 10^{-6}$] is added. The resultant concentration of Ag^+ in the solution is 1.6×10^{-x}. The value of "x" is

Sol. **(7)**

Let the solubility of AgCl is x mollitre^{-1} and that of CuCl is y mollitre^{-1}

$$AgCl \rightleftharpoons \underset{x}{Ag^+} + \underset{x}{Cl^-}$$

$$CuCl \rightleftharpoons \underset{y}{Cu^+} + \underset{y}{Cl^-}$$

$\therefore$ K_{sp} of $AgCl = [Ag^+][Cl^-]$

$1.6 \times 10^{-10} = x(x + y)$ ... (i)

Similarly K_{sp} of $CuCl = [Cu^+][Cl^-]$

$1.6 \times 10^{-6} = y(x + y)$ ... (ii)

On solving (i) and (ii)

$[Ag^+] = 1.6 \times 10^{-7}$

$\therefore$ x = 7

19. The number of hexagonal faces that are present in a truncated octahedron is

Sol. **(8)**

SECTION–IV

(Matrix-Match Type)

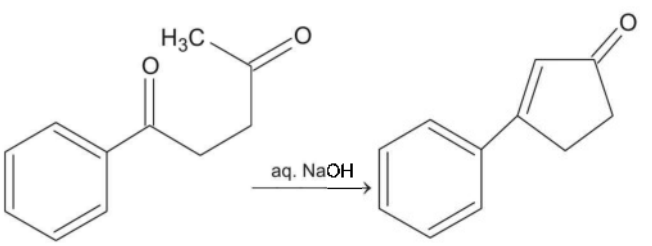
This section contains **2 questions.** Each question has four statements (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II.** Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.

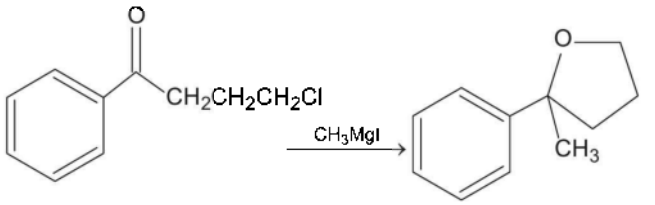
20. Match the transformations in **column I** with appropriate options in **column II**

Column I	**Column II**
(A) $CO_2(s) \rightarrow CO_2(g)$	(p) phase transition
(B) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$	(q) allotropic change
(C) $2H \bullet\!\!\rightarrow H_2(g)$	(r) Δ H is positive
(D) $P_{(white,\ solid)} \rightarrow P_{(red\ solid)}$	(s) Δ S is positive
	(t) Δ S is negative

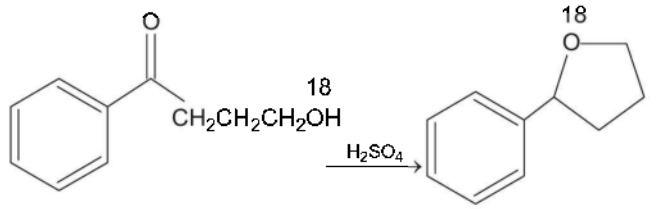
Sol. **(A) → (p, r, s) (B) → (r, s) (C) → (t) (D) → (p, q, t)**

21. Match the reactions in **column I** with appropriate types of steps/reactive intermediate involved in these reactions as given in **column II**

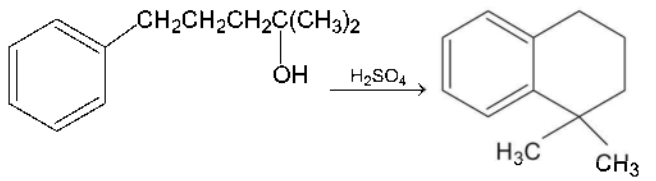
Column I	**Column II**
(A) 	(p) Nucleophilic substitution

(B) 

(q) Electrophilic substitution

(C) 

(r) Dehydration

(D) 

(s) Nucleophilic additio

(t) Carbanion

Sol. $(A) \rightarrow (r,s,t)$ $(B) \rightarrow (p,s)$ $(C) \rightarrow (r,s)$ $(D) \rightarrow (q,r)$

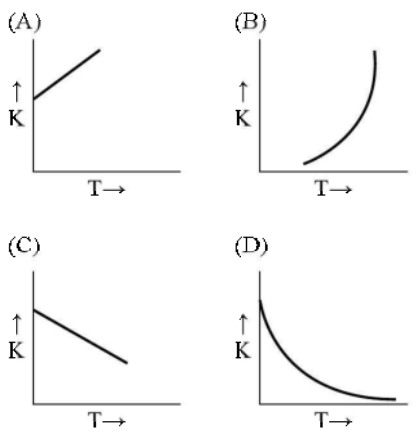
Solved Question Papers 2010

PHYSICAL CHEMISTRY

Section-I

Single Correct Choice Type

1. Plots showing the variation of the rate constant (κ) with temperature (T) are given below. The plot that follows Arrhenius equation is



Ans (B) [chemical. Equilibrium]

As according to the Arrhenius equation ($\kappa = Ae^{-Ea/RT}$), the rate constant increases exponentially with temperature so the correct choice is (A)

2. The bond energy (in kcal mol^{-1}) of a C C single bond is approximately

(A) 1 (B) 10
(C) 100 (D) 1000
Ans (C) [energetic]

C C bond energy = 348 J/mol $\frac{348}{4.2} kcal/mol = 82.85$ kcal/ mol ≈ 100 kcal/mol.

3. The species which by definition has ZERO standard molar enthalpy of formation at 298 K is

(A) Br_2 (g) (B) Cl_2 (g)
(C) H_3O (g) (D) CH_4 (g)
Ans (B) [energetic]

As we know that the species in its elemental form has zero standard molar enthalpy of formation at 298 K.

$Br_{2(l)} \rightarrow Br_{2(g)}$; $\Delta^\circ_f \neq 0$

$C_{(s)} \quad 2H_{2(g)} \rightarrow CH_{4(g)}$; $\Delta H^\circ_f \neq 0$

$H_{2(g)} + \frac{1}{2} O_{2(g)} \rightarrow H_2O_{(g)}$; $\Delta H^\circ_f \neq 0$

So correct choice is (B)

Multiple Correct Choice Type

4. Among the following, the intensive property is (properties are)

(A) molar conductivity
(B) electromotive force
(C) resistance
(D) heat capacity
Ans (A, B) [energetics]

As mass independent properties are known as intensive properties here resistance and heat capacity are extensive properties.

5. Aqueous solutions of HNO_3, KOH, CH_3COOH and CH_3COONa of identical concentrations are provided. The pair (s) of solutions which form a buffer upon mixing is (are)

(A) HNO_3 and CH_3COOH
(B) KOH and CH_3COONa
(C) HNO_3 and CH_3COONa
(D) CH_3COOH and CH_3COONa
Ans (C, D) [ionic equilibrium]

As any solution of a weak acid and its salt with strong base acts as an acidic buffer solution so if volume of HNO_3 solution added is less as compared to that of CH_3OONa

solution. It results in the formation of an acidic buffer solution.

$$CH_3COONa + HNO_3 \longrightarrow CH_3COOH + NaNO_3$$

Excess	limiting Reagent		
MV	MV'	–	–
M(V – V')	0	MV'	MV' (V' < V)

SECTION–III

Paragraph Type

Inorganic

Comprehension-1

Copper is the most noble of the first row transition meals and occurs in small deposits in several countries. Ores of copper include chalcanthite ($CuSO_4.5H_2O$), atacamite ($Cu_2Cl(OH)_3$), cuprite (Cu_2O), copper glance (Cu_2S) and malachite ($Cu_2(OH)_2CO_3$). However, 80 % of the world copper production comes from the ore chalcopyrite ($CuFeS_2$). The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self reduction.

[Metallurgy]

6. Partial roasting of chalcopyrite produces

(A) Cu_2S and FeO
(B) Cu_2O and FeO
(C) CuS and Fe_2O_3
(D) Cu_2O and Fe_2O_3
Ans (B)

$$2CuFeS_2 + 4O_2 \xrightarrow{1400-1450^\circ C} Cu_2S + 3SO_2 + 2FeO$$

The partial roasting of FeS and Cu_2S gives FeO and Cu_2O respectively

$$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$$

$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$$

7. Iron is removed from chalcopyrite as

(A) FeO (B) FeS
(C) Fe_2O_3 (D) $FeSiO_3$
Ans (D)

$$FeO + SiO_2 \longrightarrow \underset{\text{Slag}}{FeSiO_3\downarrow}$$

8. In self-reduction, the reducing species is

(A) S (B) O^{2-}
(C) S^{2-} (D) SO_2
Ans (C)

$$Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$$

As the reducing species is the one which gets oxidized. So, here S^{2-} ion getting oxidized to S^{4+} is the reductant.

Comprehension-2

The concentration of potassium ions inside a biological cell is atleast twenty times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal M is

$$M(s) \mid M^+(aq;\ 0.05\ molar) \parallel M^+(aq;\ 1\ molar) \mid M(s)$$

For the above electrolytic cell the magnitude of the cell potential $|E_{cell}| = 70$ mV.

[electrochemistry]

9. For the above cell

(A) $E_{cell} < 0;\ \Delta G > 0$
(B) $E_{cell} > 0;\ \Delta G < 0$
(C) $E_{cell} < 0;\ \Delta G^\circ > 0$
(D) $E_{cell} > 0;\ \Delta G^\circ < 0$
Ans (B)

(A) : $M(s) \longrightarrow M_A^+(aq) + e$

(C) : $M_c(aq) + e^- \longrightarrow M(s)$

$$M_c(aq) \longrightarrow M_A(aq)$$

$$E_{cell} = E^\circ_{cell} - \frac{RT}{F} \ln \frac{[M_A^+]}{[M_c^+]} - \frac{RT}{F} \ln \frac{[M_c^+]}{[M_A^+]} \quad (e^\circ_{cell} = 0)$$

$$E_{cell} = 0.059 \log \frac{[M_c^+]}{0.05}$$

$$0.07\ V = 0.059 \log \frac{[M_c^+]}{0.05}$$

$$= 0.059 \log 20[M_c^+]$$

$$= 0.059 (\log 20 + \log[M_c^+])$$

As $[M_c^+]$ is greater than 0.05 M, so E_{cell} is –ve.

$E_{cell} > 0$ & $\Delta G < 0$

10. If the 0.05 molar solution of M^+ is replaced by a 0.0025 molar M^- solution, then the magnitude of the cell potential would be

(A) 35 mV (B) 70 mV
(C) 140 mV (D) 700 mV

Ans (C)

$$E_{cell} = 0.059 \log \frac{[M_c^-]}{0.0025} = 0.059 \log 400\ [M_c^+]$$

$$= 0.059 \log (20)^2\ [M_c^+]$$

$$= 0.059\ (2 \log 20 + \log\ [M_c^-])$$

$$= 0.059 \left(2 \times \frac{0.07}{0.59}\right) = 0.14V = 140\ mV$$

SECTION IV
Integer Answer Type

11. Amongst the following, the total number of compounds whose aqueous solution turns red litmus paper blue is

KCN K_2SO_4 $(NH_4)_2C_2O_4$ NaCL $ZN(NO_3)_2$
$FeCl_3$ K_2CO_3 NH_4NO_3 LiCN

Solution[ion. Eq.]

As KCN, K_2CO_3 and LiCN are the salts of weak acid and strong base. So, their aqueous solutions turns red litmus paper blue.

So the answer is 3.

12. Based on VSEPR theory, the number of 90 degree F–Br–F angles in BrF5 is

Solution [ch. bonding]

According to VSEPR theory, number of electron pairs around central atom (Br) are 6.

$$\frac{N}{2} = \frac{7-5}{2} = 6$$

(Five are bond pairs and one is lone pair)

Its geometry is octahedral but because of lone pair bond pair repulsion, the four fluorine atoms at corner are forced towards the upper fluorine atom thus reducing F–Br–F angle from 90° to 84.8°.

Hence the answer is 0.

13. A student performs a titration with difference burettes and finds titre values of 25.2 ml, 25.25 mL, and 25.0 mL. The number of significant figure in the average titre value is

Solution [mole concept]

The least significant figure in titre values is 3.

$$\text{Average titre value} = \frac{25.2 \quad 25.25 \quad 25}{3} = \frac{75.4}{3} = 25.1$$

The number of significant figures in average titre value will also be 3.

The answer is 3.

14. The concentration of R in the reaction R→P was measured as a function of time and the following data is obtained.

[R](molar)	1.0	0.75	0.40	0.10
t(min.)	0.0	0.05	0.12	0.18

The order of reaction is

Solution [kinetics]

The integrated form of a zero-order reaction is

$$[A_0] \quad [A_t] = k_0 t$$

$$1.0 \quad 0.75 = k_0 \times 0.05,\ k_0 = 5$$

$$1.0 - 0.4 = k_0 \quad 0.12,\ k_0 = 5$$

The answer is 0.

15. The number of neutrons emitted when $^{235}_{92}U$ undergoes controlled nuclear fission to $^{142}_{54}Xe$ and $^{90}_{38}Sr$ is

Solution [nuclear]

$$_{92}U^{235} + {}_0n^1 \rightarrow {}_{54}Xe^{142} + {}_{38}Sr^{90} + y\ {}_0n^1$$

$235 - 1 = 142 + 0 - y$

$y = 4$. The number of neutrons emitted are $(4 - 1) = 3$.

The answer is 3.

PAPER–II

PART-1
Single Correct Choice type

1. Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is

(A) 1 and diamagnetic
(B) 0 and diamagnetic
(C) 1 and paramagnetic
(D) 0 and paramagnetic
Ans (A) [atomic structure]

Solution

Molecular orbital configuration of Br_2 as per the condition will be

$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_y^2$

Bond order of $B_2 - \frac{6-4}{1} - 1$

B_2 will be diamagnetic

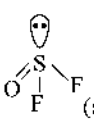
Correct choice (A)

2. The species having pyramidal shape is

(A) SO_3 (B) BrF_3
(C) SiO_3^{2-} (D) OSF_2
Ans (D) [ch bonding]

$OSF_2 : \frac{N}{2} - \frac{6+2}{2} - 4$

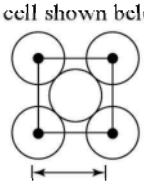
It has 1 lone pair



(shape is trigonal pyramidal)

The shapes of SO_3, BrF_3 and SiO_3^{2-} are triangular planar, T-shaped and triangular planar respectively.

3. The packing efficiency of the two-dimensional square unit cell shown below is



L

(A) 39.27 % (B) 68.02 %
(C) 74.05 % (D) 78.54 %
Ans (D) [solid state]

As Packing efficiency

$$= \frac{\text{Area occupied by effective circles}}{\text{Area of square}}$$

$$= \frac{2\pi r^2}{L^2} \times 100$$

$$= \frac{2\pi r^2}{(2\sqrt{2}r)^2} \times 100$$

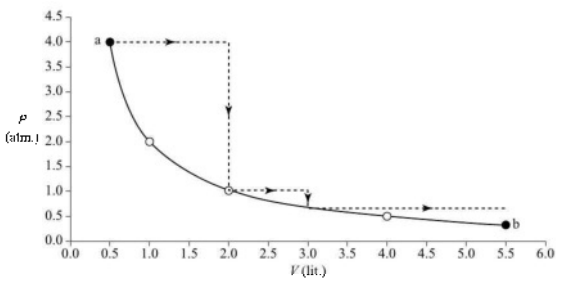
$$= \frac{\pi}{4} \times 100 = 78.54\%$$

Hence the correct choice is D

SECTION-II

Integer Type

4. One mole of an ideal gas is taken from a to b along two paths denoted by the solid and the dashed lines as shown in the graphs below. If the work done along the solid line path is w_s and that along the dotted line path is w_d, then the integer closest to the ratio w_d/w_s is



Solution [energetics]

$$w_d = -\left(-4 \times \frac{3}{2}\right) + (-1 \quad 1) - \left(-\frac{2}{3} \times \frac{5}{2}\right)$$

$$= -\left(6 + 1 + \frac{5}{3}\right)$$

$$w_d = -\frac{26}{3} Latm$$

$$w_s = -2.303 \text{ RT log } \frac{5.5}{1/2} = -2.303 \text{ PV log } 11$$

$$w_s = -4.606 \times 1.04 = -4.8 \text{ Latm}$$

$$\frac{w_d}{w_s} = \frac{-26/3}{-4.8} = 1.80 \simeq 2.0$$

The answer is 2.

5. Among the following, the number of elements showing only one non-zero oxidation state is

O, Cl, F, N, P, Sn. Tl, Na, Ti

Solution [redox and electrochemistry]

Fluorine generally shows O and –1 oxidation states while sodium shows 0 and +1 oxidation state.

So the answer is 2

6. Silver (atomic weight = 108 g mol^{-1}) has a density of 10.5 g cm^{-3}. The number of silver atoms on a

surface of area 10^{-12} m^2 can be expressed in scientific notation as y 10^x. The value of x is

Solution [solid state]

Silver has cubic close packed structure. It has fcc unit cell of edge length a cm.

$$10.5 - \frac{4 \times 108}{6.023 \quad 10^{23} (a)^3}$$

$$\text{So, } a - \left[\frac{4 \times 108}{6.023 \quad 10^{23} \times 10.5}\right]^{1/3}$$

$$- 4.087 \times 10^{-8} \text{ cm.}$$

(i) Assuming that surface area given is the area of any extended face of a FCC unit cell.

Area of square face $-$ $(4.087 \times 10^{-8})^2$ cm^2

Number of Ag atoms per unit area $- \frac{2}{a^2}$

Surface area $- 10^{-12}$ m$^2 - 10^{-8}$ cm^2.

Number of Ag atoms in the given surface area

$$= \frac{2 \times 10^{-8}}{(4.087 \times 10^{-8})^2} = 1.197 \times 10^7.$$

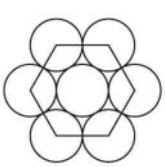
x $-$ 7

(ii) Assuming that surface area given is the area corresponding o closest packed layer

$$\sqrt{2}a - 4r, \; r - \frac{\sqrt{2}a}{4} - \frac{4.087 \times 10^{-8} \quad \sqrt{2}}{4}$$

$$- 1.44 \times 10^{-8}$$

In a closest packed layer, the arrangement appears like



$$6 \quad \frac{\sqrt{3}}{4}(2r)^2 - \frac{6\sqrt{3}}{4}(2 \times 1.44 \quad 10^{-8})^2$$

$$- \frac{6\sqrt{3}}{4} \times 4 \quad 2.07 \times 10^{-16}$$

$$- 21.5 \times 10^{-16} \text{ cm}^2$$

Number of silver atoms on surface

$$- \frac{3 \quad 10^{-8}}{21.5 \quad 10^{-16}} - 0.139 \times 10^8$$

$$- 1.39 \times 10^7$$

x $-$ 7

The answer is 7.

Comprehension 2

The hydrogen–like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

Atomic structure

7. The state S_1 is

(A) 1s (B) 2s

(C) 2p (D) 3s

Ans (B)

The spherically symmetric state S_1 of Li^{2+} with one radial node is 2s. Upon absorbing light, the ion gets excited to state S_2, which also has one radial node. The energy of electron is S_2 is same as that of H-atom in its ground state

$E_n - \frac{Z^2}{n^2} E_1$ where E_1 is the energy of H-atom in the ground state

$$- \frac{(3)^2 E_1}{n^2} \text{ for } Li^{2+}$$

$E_n - E_1$

n $-$ 3

State of S_2 of Li^{2+} having one radial node is 3p.

Orbital angular momentum quantum number of 3p is 1.

Energy of state $S_1 - \frac{(3)^2}{(2)^2} E_1 - 2.25 E_1$

8. Energy of the state S_1 in units of the hydrogen atom ground state energy is

(A) 0.75 (B) 1.50

(C) 2.25 (D) 4.50

Ans (C)

9. The orbital angular momentum quantum number of the state energy is

(A) 0 (B) 1

(C) 2 (D) 3

Ans (B)

Inorganic Chemistry
Paper I

SINGLE CHOICE TYPE

16. The ionization isomer of $[Cr(H_2O)_4Cl(NO_2)]Cl$ is

(A) $[Cr(H_2O)_4(O_2N)]Cl_2$
(B) $[Cr(H_2O)_4Cl_2](NO_2)$
(C) $[Cr(H_2O)_4Cl(ONO)]Cl$
(D) $[Cr(H_2O)_4Cl_2(NO_2)]H_2O$

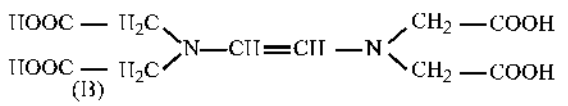
Ans (B) [co-ordination]

Ionization isomer of $[Cr(H_2O)_4Cl(NO_2)]Cl$ is $[Cr(H_2O)_4Cl_2]NO_2$.

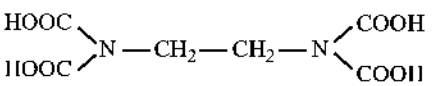
So the correct choice is (B)

17. The correct structure of ethylenediaminetetraacetic acid (EDTA) is

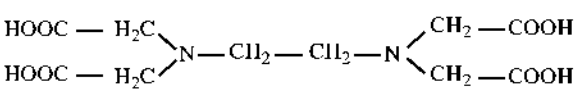
(A)



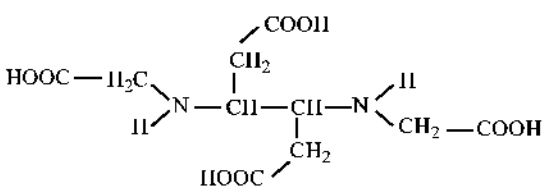
(B)



(C)

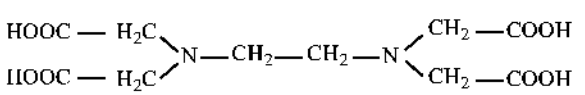


(D)



Ans (C) [coordination]

As the correct structure of EDTA is



Multiple Correct Type

18. The reagent (s) used for softening the temporary hardness of water is (are)

(A) $Ca_3(PO_4)_2$ (B) $Ca(OH)_2$
(C) Na_2CO_3 (D) NaOCl

Ans (B, C, D) [s- block]

As the temporary hardness is due to bicarbonates of calcium and magnesium. It can be removed by Clark's process, which involves the addition of slaked lime, $Ca(OH)_2$. Washing soda (Na_2CO_3) removes both the temporary and permanent hardness by converting soluble calcium and magnesium compounds into insoluble carbonates.

$$Ca(HCO_3)_2 \quad Ca(OH)_2 \rightarrow 2CaCO_3\downarrow \quad 2H_2O$$

$$Ca(HCO_3)_2 \quad Na_2CO_3 \rightarrow CaCO_3\downarrow \quad 2NaHCO_3$$

$$2OCl^- + 2H_2O \rightleftharpoons 2HOCl - 2OH$$

$$Ca(HCO_3)_2 - 2OH \rightarrow CaCO_3\downarrow + CO_3^{2-} + 2H_2O$$

Integer Type

19. The value of n in the molecular formula $Be_nAl_2Si_6O_{18}$ is

Solution [p-block]

It is the formula of Beryl (which has BeO, Al_2O_3 and SiO_2).

The formula of Beryl should be
$3BeO.Al_2O_3.6SiO$
Or
Total cationic charge = total anionic charge

$$2n + 6 = 24 = 36$$
$$n = 3$$

Hence the answer is 3.

Paper II

SINGLE CHOICE TYPE

10. The complex showing a spin only magnetic moment of 2.82 B.M. is

(A) $Ni(CO)_4$ (B) $[NiCl_4]^{2-}$
(C) $Ni(PPh_3)_4$ (D) $[Ni(CN)_4]^{2-}$

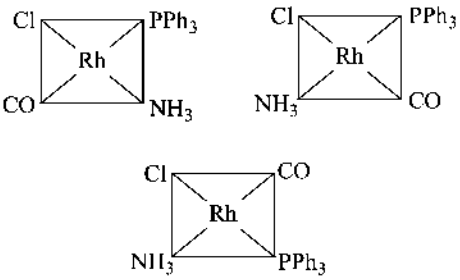
Ans (B) [coordination]

$Ni(CO)_4$ and $Ni(PPh_3)_4$ have Ni in zero oxidation state, while CO and Ph_3P are strong field ligands. Thus, both these molecules have sp^3 hybridization and are diamagnetic. $[Ni(CN)_4]^{2-}$ has electronic configuration of Ni^{2+} as $3d^8$ and hybridization is dsp^2. It is diamagnetic. $[NiCl_4]^{2-}$ has sp^3 hybridization with 2 unpaired electrons having spin only magnetic moment as $\sqrt{8}$ or 2.82 B.M.

Integer Type

11. Total number of geometrical isomers for the complex $[RhCl(CO)(PPh_3)(NH_3)]$ is

Solution [coordination]



The number of geometrical isomers is 3.

The answer is 3

12. The total number of diprotic acids among the following is

H_3PO_4 H_2SO_4 H_3PO_3 H_2CO_3 $H_2S_2O_7$
H_3BO_3 H_3PO_2 H_2CrO_4 H_2SO_3

Solution [p- block]

Diprotic acids are H_2SO_4, H_3PO_3, H_2CO_3, $H_2S_2O_7$, H_2CrO_4 and H_2SO_3.

The answer is 6

Matrix Match Type

13. Match the following

Column I

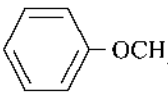
(A) $(CH_3)_2SiCl_2$
(B) XeF_4
(C) Cl_2
(D) VCl_5

Column II

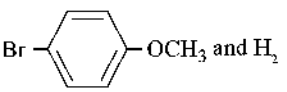
(p) Hydrogen halide formation
(q) Redox reaction
(r) Reacts with glass
(s) Polymerization
(t) O_2 formation

Ans A – (p, s); B – (p, q, r, t); C – (p, q); D – (p, q) [p- block I]

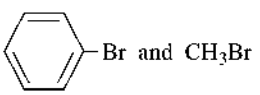
Organic Chemistry
Single Correct Choice
Type Paper I

20. In the reaction  the products are

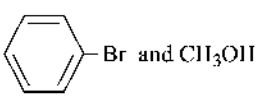
(A)



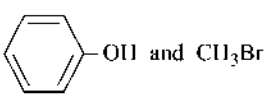
(B)



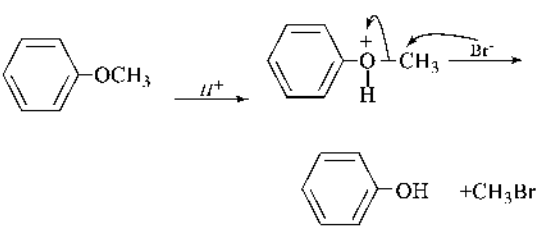
(C)



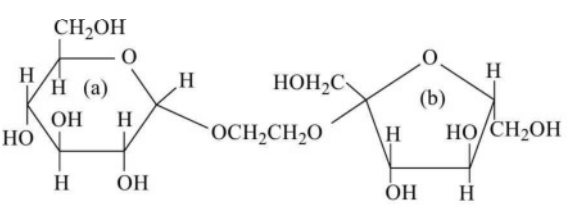
(D)



Ans (D) [alcohol]



21. The correct statement about the following disaccharide is



(A) Ring (a) is pyranose with α-glycosidic link.
(B) Ring (a) is furanose with α-glycosidic link.
(C) Ring (b) is furanose with α-glycosidic link.
(D) Ring (b) is pyranose with β-glycosidic link.

Ans (A) [biomolecule]

Correct choice is (A) see text part.

22. The synthesis of 3- octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. The bromoalkane and alkyne respectively are

(A) $BrCH_2CH_2CH_3$ and $CH_3CH_2CH_2C{-}CH$

(B) $BrCH_2CH_2CH_2CH_2CH_3$ and $CH_3CH_2C{=}CH$

(C) $BrCH_2CH_2CH_2CH_3$ and $CH_3CH_2C{=}CH$

(D) $BrCH_2CH_2CH_2CH_2CH_3$ and $CH_3C{=}CH$

Ans (C) [hydrocarbon]

$$CH_3CH_2C{=}CH \xrightarrow[-NH_3]{NaNH_2} CH_3CH_2C{=}C{-}Na^+$$

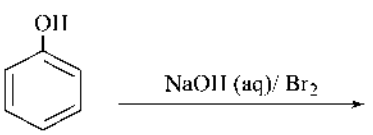
$$\xrightarrow[(S_N2)]{CH_3CH_2CH_2CH_2Br}$$

$$CH_3CH_2C{=}CCH_2CH_2CH_2CH_3 - NaBr$$

Hence the correct choice is (C)

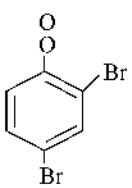
Multiple Choice Type

23. In the reaction

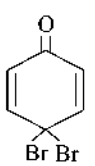


The intermediate(s) is/are

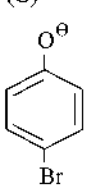
(A)



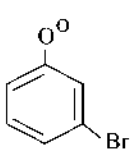
(B)



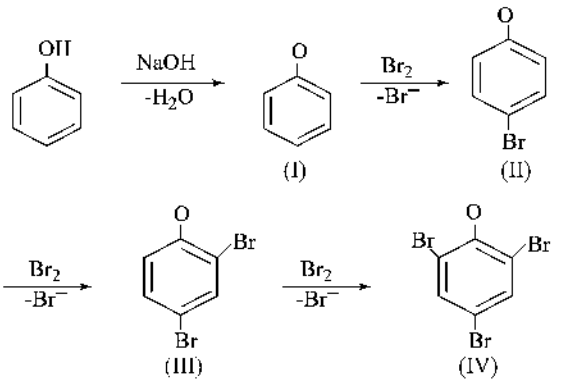
(C)



(D)

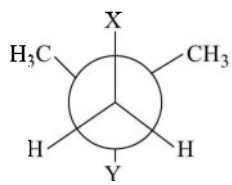


Ans (A, C) [alcohol]



As the product of reaction of phenol with NaOH /Br_2 is sodium salt of 2, 4, 6-tribromophenol. So the species (I), (II), (III) are formed as intermediate here. So the correct choice is (A, C)

24. In the Newman projection for 2, 2 dimethylbutane



X and Y can respectively be

(A) H and H

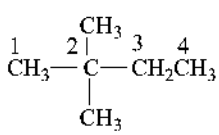
(B) H and C_2H_5

(C) C_2H_5 and H

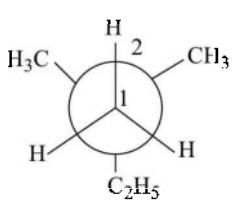
(D) CH_3 and CH_3

Ans (B, D) [GOC- I]

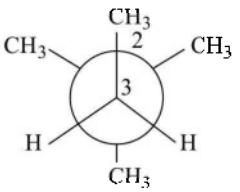
The Structural formula of 2, 2 –dimethylbutane can be given as



(I) Newman projection using $C_1 - C_2$ bond

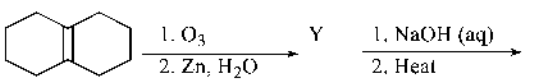


(II) Newman projection using $C_3 - C_2$ bond



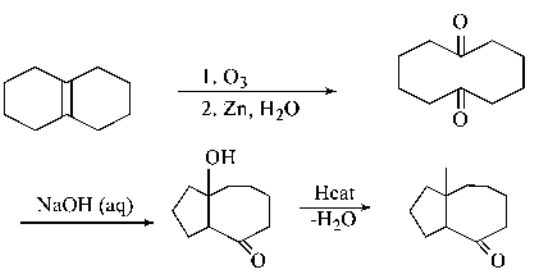
Integer Type

25. In the scheme given below, the total number of intramolecular aldol condensation products formed from 'Y' is



[carbonyl compound]

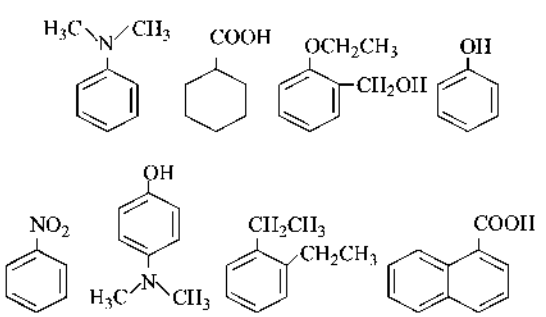
Solution



The number of intramolecular aldol condensation products (α, β-unsaturated carbonyl compound) formed from (Y) is 1.

The answer is 1.

26. Amongst the following, the total number of compounds soluble in aqueous NaOH is

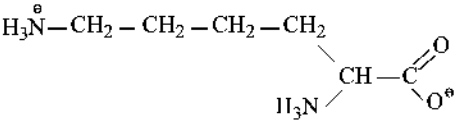


Solution [acids]

As all carboxylic acid and phenols are soluble in aqueous NaOH. So here four compounds are soluble in aqueous NaOH

The answer is 4.

27. The total number of basic groups in the following form of lysine is



Solution [GOC. II]

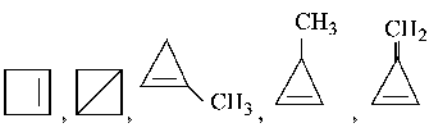
As the basic groups in the given form of lysine is NH_2 and CO_2. The answer is 2.

28. The total number of cyclic isomers possible for a hydrocarbon with the molecular formula C_4H_6 is

Solution [GOC I]

The number of cyclic isomers for a hydrocarbon with molecular formula C_4H_6 is 5.

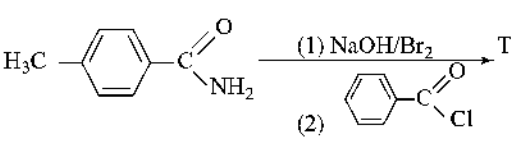
The structures are



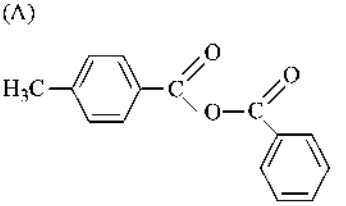
Paper II

SINGLE CHOICE TYPE

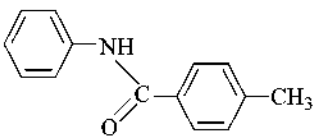
14. In the reaction



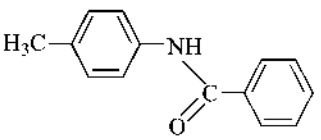
(A)



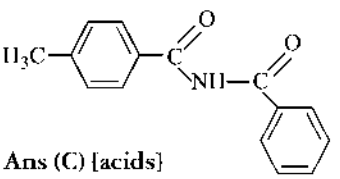
(B)



(C)

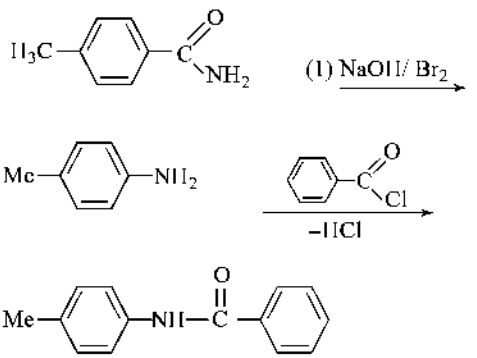


(D)



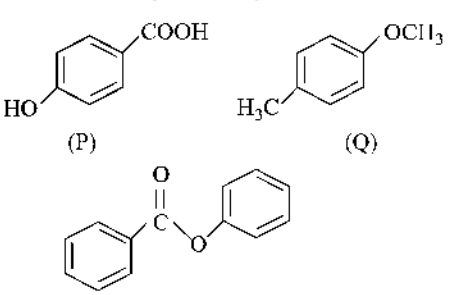
Ans (C) [acids]

Solution



Correct choice is (C)

15. The compound P, Q and S

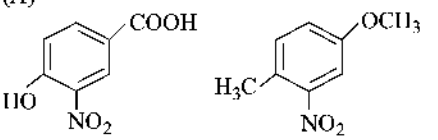


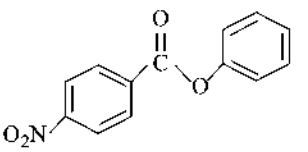
(P) (Q)

(S)

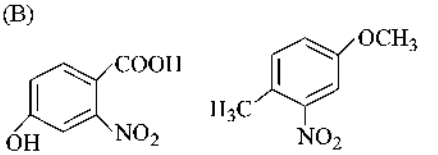
Were separately subjected to nitration using HNO_3 / H_2SO_4 mixture. The major product formed in each case respectively, is

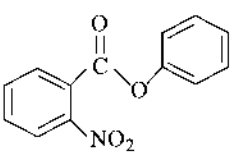
(A)



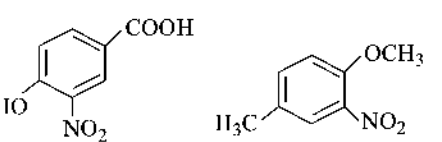


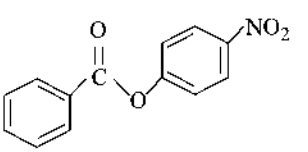
(B)



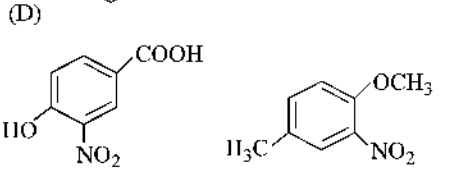


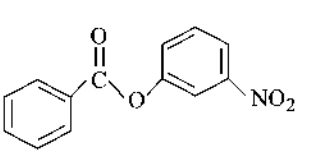
(C)



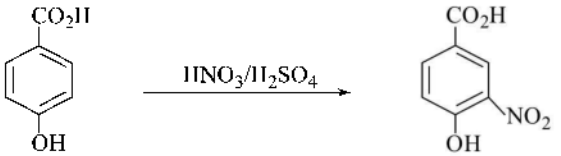


(D)

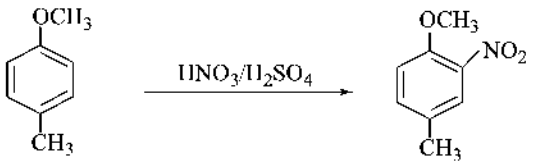




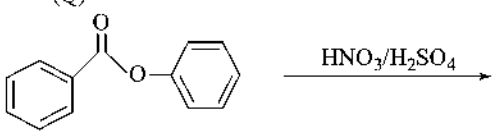
Ans (C) [acids]



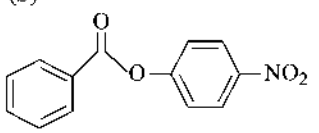
(P)



(Q)

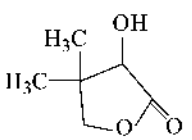


(S)



SECTION–III
Paragraph Type
Comprehension-I

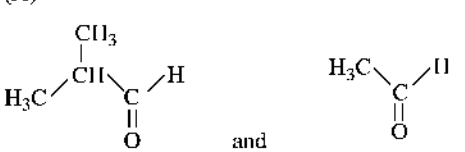
Two aliphatic aldehydes P and Q react in the presence of aqueous K_2CO_3 is give compound R, which upon treatment with HCN provides compound S. On acidification and heating, S gives the product shown below.



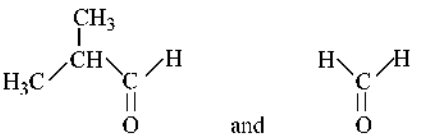
Carbonyl compound

16. The compound P and Q respectively are

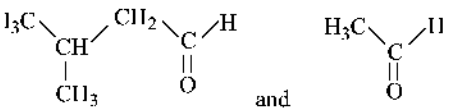
(A)



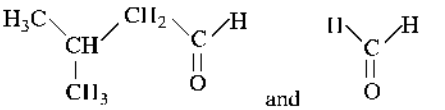
(B)



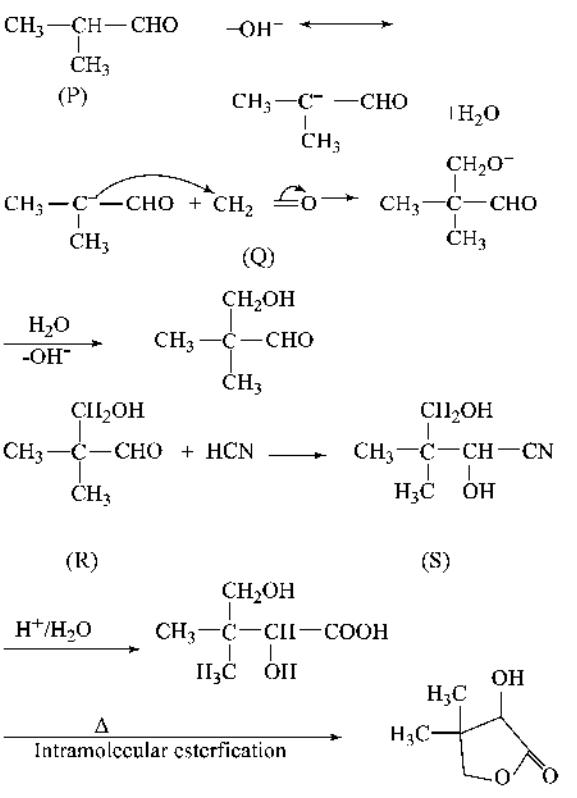
(C)



(D)

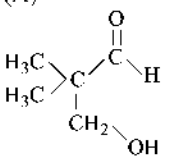


Ans (B)

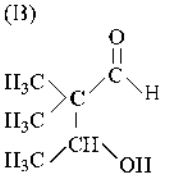


17. The compound R is

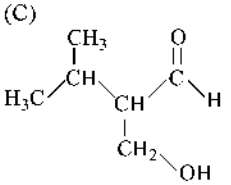
(A)



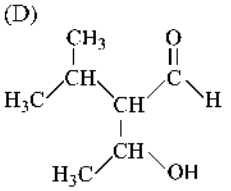
(B)



(C)



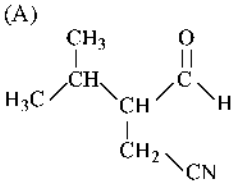
(D)



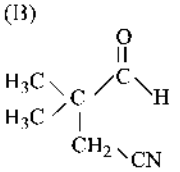
Ans (A)

18. The Compound S is

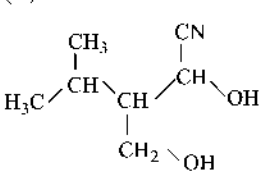
(A)



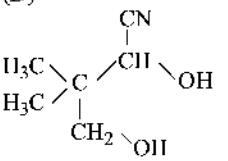
(B)



(C)



(D)



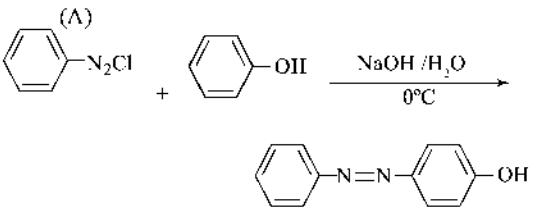
Ans (D)

MATRIX MATCH TYPE

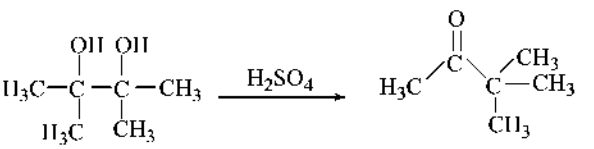
19. Match the following

Column I

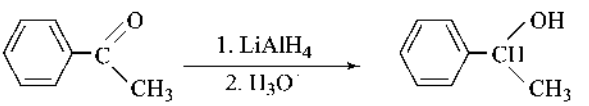
(A)



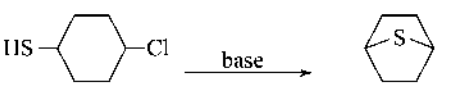
(B)



(C)



(D)



Column II

(p) Racemic mixture

(q) Addition reaction

(r) Substitution reaction

(s) Coupling reaction

(t) Carbocation intermediate

Ans A – (r, s); B – (t); C – (p, q); D – (r) [GOC III]